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47.3A: HABITAT LOSS AND SUSTAINABILITY
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47.3D: CLIMATE CHANGE AND BIODIVERSITY
47.4: PRESERVING BIODIVERSITY

47.4A: MEASURING BIODIVERSITY
47.4B: CHANGING HUMAN BEHAVIOR IN RESPONSE TO BIODIVERSITY LOSS
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5.2C: DIFFUSION
5.2D: FACILITATED TRANSPORT
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5.4: BULK TRANSPORT

5.4A: ENDOCYTOSIS
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6: METABOLISM

6.1: ENERGY AND METABOLISM

6.1A: THE ROLE OF ENERGY AND METABOLISM
6.1B: TYPES OF ENERGY
6.1C: METABOLIC PATHWAYS
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6.2: POTENTIAL, KINETIC, FREE, AND ACTIVATION ENERGY

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6.2D: ACTIVATION ENERGY
6.3: ATP: ADENOSINE TRIPHOSPHATE

6.3A: ATP: ADENOSINE TRIPHOSPHATE
6.4: ENZYMES

6.4A: ENZYME ACTIVE SITE AND SUBSTRATE SPECIFICITY
6.4B: CONTROL OF METABOLISM THROUGH ENZYME REGULATION

7: CELLULAR RESPIRATION

Cellular respiration is a set of metabolic reactions and processes that take place in the cells of organisms to convert biochemical energy from nutrients into adenosine triphosphate (ATP), and then release waste products.

7.1: ENERGY IN LIVING SYSTEMS

7.1A: TRANSFORMING CHEMICAL ENERGY
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7.1C: ATP IN METABOLISM
7.2: GLYCOLYSIS

7.2A: IMPORTANCE OF GLYCOLYSIS
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7.2D: OUTCOMES OF GLYCOLYSIS
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7.3A: BREAKDOWN OF PYRUVATE
7.3B: ACETYL COA TO CO2
7.3C: CITRIC ACID CYCLE
7.4: OXIDATIVE PHOSPHORYLATION

7.4A: ELECTRON TRANSPORT CHAIN
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7.6: CONNECTIONS OF CARBOHYDRATE, PROTEIN, AND LIPID METABOLIC PATHWAYS

7.6A: CONNECTING OTHER SUGARS TO GLUCOSE METABOLISM
7.6B: CONNECTING PROTEINS TO GLUCOSE METABOLISM
7.6C: CONNECTING LIPIDS TO GLUCOSE METABOLISM
7.7: REGULATION OF CELLULAR RESPIRATION

7.7A: REGULATORY MECHANISMS FOR CELLULAR RESPIRATION
7.7B: CONTROL OF CATABOLIC PATHWAYS

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8.1: OVERVIEW OF PHOTOSYNTHESIS

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The Science of Biology

Topic hierarchy

- 1.1A: Introduction to the Study of Biology
- 1.1B: Scientific Reasoning

Science is knowledge obtained from logical inferences and deductive experimentation that attempts to comprehend nature.

- 1.1C: The Scientific Method
- 1.1D: Basic and Applied Science
- 1.1E: Publishing Scientific Work
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1.1A: Introduction to the Study of Biology

Biology is the study of life and living things through rigorously-tested and peer-reviewed scientific research methods.

Learning Objectives

- Describe the field of biological science

Key Points

- Biology has evolved as a field of science since it was first studied in ancient civilizations, although modern biology is a relatively recent field.
- Science is a process that requires the testing of ideas using evidence gathered from the natural world. Science is iterative in nature and involves critical thinking, careful data collection, rigorous peer review, and the communication of results.
- Science also refers to the body of knowledge produced by scientific investigation.
- Pseudoscience is a belief presented as scientific although it is not a product of scientific investigation.

Key Terms

- **pseudoscience**: Any belief purported to be scientific or supported by science that is not a product of scientific investigation.
- **science**: A process for learning about the natural world that tests ideas using evidence gathered from nature.
- **Biology**: A natural science concerned with the study of life and living organisms.

The Study of Life

Biology is a natural science concerned with the study of life and living organisms. Modern biology is a vast and eclectic field composed of many specialized disciplines that study the structure, function, growth, distribution, evolution, or other features of living organisms. However, despite the broad scope of biology, there are certain general and unifying concepts that govern all study and research:

- the cell is the basic unit of life
- genes (consisting of DNA or RNA) are the basic unit of heredity
- evolution accounts for the unity and diversity seen among living organisms
- all organisms survive by consuming and transforming energy
- all organisms maintain a stable internal environment

Figure 1.1*A*.1: **Biology: The Study of Life**: A collection of organisms clockwise from top left: bacteria, koala, fern, toadstool, tree frog, tarantula.

Biological research indicates the first forms of life on Earth were microorganisms that existed for billions of years before the evolution of larger organisms. The mammals, birds, and flowers so familiar to us are all relatively recent, originating within the last 200 million years. Modern-appearing humans, *Homo sapiens*, are a relatively new species, having inhabited this planet for only the last 200,000 years (approximately).

Figure 1.1*A*.1: **Stromatolites**: Stromatolites, sedimentary accretions formed by the actions of cyanobacteria, provide fossil evidence of life on Earth about 3.5 billion years ago.

History of Biological Science

Although modern biology is a relatively recent development, sciences related to and included within it have been studied since ancient times. Natural philosophy was studied as early as the ancient civilizations of Mesopotamia, Egypt, the Indian subcontinent, and China. However, the origins of modern biology and its approach to the study of nature are most often traced

back to ancient Greece. (Biology is derived from the Greek word “bio” meaning “life” and the suffix “ology” meaning “study of.”)

Advances in microscopy also had a profound impact on biological thinking. In the early 19th century, a number of biologists pointed to the central importance of the cell and in 1838, Schleiden and Schwann began promoting the now universal ideas of the cell theory. Jean-Baptiste Lamarck was the first to present a coherent theory of evolution, although it was the British naturalist Charles Darwin who spread the theory of natural selection throughout the scientific community. In 1953, the discovery of the double helical structure of DNA marked the transition to the era of molecular genetics.

Figure 1.1*A*.1: **Phrenology: Dr. Spurzheim’s divisions of the organs of phrenology marked externally**: Phrenology is a pseudoscience that attempted to determine brain function and personality by analyzing an individual’s skull.

Science and Pseudoscience

Science is a process for learning about the natural world. Most scientific investigations involve the testing of potential answers to important research questions. For example, oncologists (cancer doctors) are interested in finding out why some cancers respond well to chemotherapy while others are unaffected. Based on their growing knowledge of molecular biology, some doctors suspect a connection between a patient’s genetics and their response to chemotherapy. Many years of research have produced numerous scientific papers documenting the evidence for a connection between cancer, genetics, and treatment response. Once published, scientific information is available for anyone to read, learn from, or even question/dispute. This makes science an iterative, or cumulative, process, where previous research is used as the foundation for new research. Our current understanding of any issue in the sciences is the culmination of all previous work.

Pseudoscience is a belief presented as scientific although it is not a product of scientific investigation. Pseudoscience is often known as fringe or alternative science. It usually lacks the carefully-controlled and thoughtfully-interpreted experiments which provide the foundation of the natural sciences and which contribute to their advancement.

1.1B: Scientific Reasoning

Learning Objectives

- Compare and contrast theories and hypotheses

The Process of Science

Science (from the Latin scientia, meaning “knowledge”) can be defined as knowledge that covers general truths or the operation of general laws, especially when acquired and tested by the scientific method. The steps of the scientific method will be examined in detail later, but one of the most important aspects of this method is the testing of hypotheses (testable statements) by means of repeatable experiments. Although using the scientific method is inherent to science, it is inadequate in determining what science is. This is because it is relatively easy to apply the scientific method to disciplines such as physics and chemistry, but when it comes to disciplines like archaeology, paleoanthropology, psychology, and geology, the scientific method becomes less applicable as it becomes more difficult to repeat experiments.

These areas of study are still sciences, however. Consider archaeology: even though one cannot perform repeatable experiments, hypotheses may still be supported. For instance, an archaeologist can hypothesize that an ancient culture existed based on finding a piece of pottery. Further hypotheses could be made about various characteristics of this culture. These hypotheses may be found to be plausible (supported by data) and tentatively accepted, or may be falsified and rejected altogether (due to contradictions from data and other findings). A group of related hypotheses, that have not been disproven, may eventually lead to the development of a verified theory. A theory is a tested and confirmed explanation for observations or phenomena that is supported by a large body of evidence. Science may be better defined as fields of study that attempt to comprehend the nature of the universe.

Scientific Reasoning

One thing is common to all forms of science: an ultimate goal “to know.” Curiosity and inquiry are the driving forces for the development of science. Scientists seek to understand the world and the way it operates. To do this, they use two methods of logical thinking: inductive reasoning and deductive reasoning.

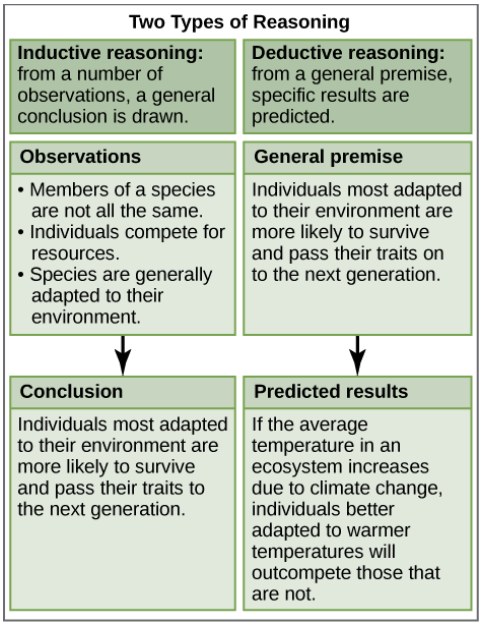

Figure 1.1B.1: **Scientific Reasoning**: Scientists use two types of reasoning, inductive and deductive, to advance scientific knowledge.

Inductive reasoning is a form of logical thinking that uses related observations to arrive at a general conclusion. This type of reasoning is common in descriptive science. A life scientist such as a biologist makes observations and records them. These data can be qualitative or quantitative and the raw data can be supplemented with drawings, pictures, photos, or

videos. From many observations, the scientist can infer conclusions (inductions) based on evidence. Inductive reasoning involves formulating generalizations inferred from careful observation and the analysis of a large amount of data. Brain studies provide an example. In this type of research, many live brains are observed while people are doing a specific activity, such as viewing images of food. The part of the brain that "lights up" during this activity is then predicted to be the part controlling the response to the selected stimulus; in this case, images of food. The "lighting up" of the various areas of the brain is caused by excess absorption of radioactive sugar derivatives by active areas of the brain. The resultant increase in radioactivity is observed by a scanner. Then researchers can stimulate that part of the brain to see if similar responses result.

Deductive reasoning or deduction is the type of logic used in hypothesis-based science. In deductive reason, the pattern of thinking moves in the opposite direction as compared to inductive reasoning. Deductive reasoning is a form of logical thinking that uses a general principle or law to forecast specific results. From those general principles, a scientist can extrapolate and predict the specific results that would be valid as long as the general principles are valid. Studies in climate change can illustrate this type of reasoning. For example, scientists may predict that if the climate becomes warmer in a particular region, then the distribution of plants and animals should change. These predictions have been written and tested, and many such predicted changes have been observed, such as the modification of arable areas for agriculture correlated with changes in the average temperatures.

Both types of logical thinking are related to the two main pathways of scientific study: descriptive science and hypothesis-based science. Descriptive (or discovery) science, which is usually inductive, aims to observe, explore, and discover, while hypothesis-based science, which is usually deductive, begins with a specific question or problem and a potential answer or solution that can be tested. The boundary between these two forms of study is often blurred and most scientific endeavors combine both approaches. The fuzzy boundary becomes apparent when thinking about how easily observation can lead to specific questions. For example, a gentleman in the 1940s observed that the burr seeds that stuck to his clothes and his dog's fur had a tiny hook structure. Upon closer inspection, he discovered that the burrs' gripping device was more reliable than a zipper. He eventually developed a company and produced the hook-and-loop fastener popularly known today as Velcro. Descriptive science and hypothesis-based science are in continuous dialogue.

Figure 1.1$B.1$: **A Burr**: This fruit attaches to animal fur via the hooks on its surface to improve distribution. Velcro is an example of a biomimetic invention which has copied burrs and uses small flexible hooks to reversibly attach to fluffy surfaces.

Key Points

- A hypothesis is a statement/prediction that can be tested by experimentation.
- A theory is an explanation for a set of observations or phenomena that is supported by extensive research and that can be used as the basis for further research.
- Inductive reasoning draws on observations to infer logical conclusions based on the evidence.
- Deductive reasoning is hypothesis-based logical reasoning that deduces conclusions from test results.

Key Terms

- **theory**: a well-substantiated explanation of some aspect of the natural world based on knowledge that has been repeatedly confirmed through observation and experimentation
- **hypothesis**: a tentative conjecture explaining an observation, phenomenon, or scientific problem that can be tested by further observation, investigation, and/or experimentation

1.1C: The Scientific Method

The scientific method is a process by which observations are questioned; hypotheses are created and tested; and the results are analyzed.

Learning Objectives

- Discuss hypotheses and the components of a scientific experiment as part of the scientific method

Key Points

- In the scientific method, observations lead to questions that require answers.
- In the scientific method, the hypothesis is a testable statement proposed to answer a question.
- In the scientific method, experiments (often with controls and variables) are devised to test hypotheses.
- In the scientific method, analysis of the results of an experiment will lead to the hypothesis being accepted or rejected.

Key Terms

- **scientific method**: a way of discovering knowledge based on making falsifiable predictions (hypotheses), testing them, and developing theories based on collected data
- **hypothesis**: an educated guess that usually is found in an "if…then…" format
- **control group**: a group that contains every feature of the experimental group except it is not given the manipulation that is hypothesized

The Scientific Method

Biologists study the living world by posing questions about it and seeking science -based responses. This approach is common to other sciences as well and is often referred to as the scientific method. The scientific method was used even in ancient times, but it was first documented by England's Sir Francis Bacon (1561–1626) who set up inductive methods for scientific inquiry. The scientific method can be applied to almost all fields of study as a logical, rational, problem-solving method.

Figure 1.1C.1: **Sir Francis Bacon**: Sir Francis Bacon (1561–1626) is credited with being the first to define the scientific method.

The scientific process typically starts with an observation (often a problem to be solved) that leads to a question. Let's think about a simple problem that starts with an observation and apply the scientific method to solve the problem. A teenager notices that his friend is really tall and wonders why. So his question might be, "Why is my friend so tall? "

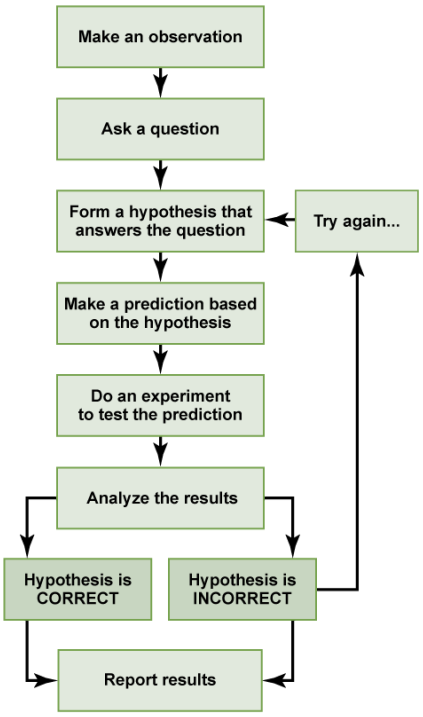

Figure 1.1C.1: **The Scientific Method**: The scientific method consists of a series of well-defined steps. If a hypothesis is not supported by experimental data, a new hypothesis can be proposed.

Proposing a Hypothesis

Recall that a hypothesis is an educated guess that can be tested. Hypotheses often also include an explanation for the educated guess. To solve one problem, several hypotheses may be proposed. For example, the student might believe that his friend is tall because he drinks a lot of milk. So his hypothesis might be "If a person drinks a lot of milk, then they will grow to be very tall because milk is good for your bones." Generally, hypotheses have the format "If…then…" Keep in mind that there could be other responses to the question; therefore, other hypotheses may be proposed. A second hypothesis might be, "If a person has tall parents, then they will also be tall, because they have the genes to be tall. "

Once a hypothesis has been selected, the student can make a prediction. A prediction is similar to a hypothesis but it is truly a guess. For instance, they might predict that their friend is tall because he drinks a lot of milk.

Testing a Hypothesis

A valid hypothesis must be testable. It should also be falsifiable, meaning that it can be disproven by experimental results. Importantly, science does not claim to "prove" anything because scientific understandings are always subject to modification with further information. This step—openness to disproving ideas—is what distinguishes sciences from non-sciences. The presence of the supernatural, for instance, is neither testable nor falsifiable. To test a hypothesis, a researcher will conduct one or more experiments designed to eliminate one or more of the hypotheses. Each experiment will have one or more variables and one or more controls. A variable is any part of the experiment that can vary or change during the experiment. The control group contains every feature of the experimental group except it is not given the manipulation that is hypothesized. For example, a control group could be a group of varied teenagers that did not drink milk and they could be compared to the experimental group, a group of varied teenagers that did drink milk. Thus, if the results of the experimental group differ from the control group, the difference must be due to the hypothesized manipulation rather than some outside factor. To test the first hypothesis, the student would find out if drinking milk affects height. If drinking milk has no affect on height, then there must be another reason for the height of the friend. To test the second hypothesis, the student could check whether or not his friend has tall parents. Each hypothesis should be tested by carrying out appropriate experiments. Be aware that rejecting one hypothesis does not determine whether or not the other hypotheses can be accepted. It simply eliminates one hypothesis that is not valid. Using the scientific method, the hypotheses that are inconsistent with experimental data are rejected.

While this “tallness” example is based on observational results, other hypotheses and experiments might have clearer controls. For instance, a student might attend class on Monday and realize she had difficulty concentrating on the lecture. One hypothesis to explain this occurrence might be, “If I eat breakfast before class, then I am better able to pay attention.” The student could then design an experiment with a control to test this hypothesis.

The scientific method may seem too rigid and structured. It is important to keep in mind that although scientists often follow this sequence, there is flexibility. Many times, science does not operate in a linear fashion. Instead, scientists continually draw inferences and make generalizations, finding patterns as their research proceeds. Scientific reasoning is more complex than the scientific method alone suggests.

1.1D: Basic and Applied Science

Basic science increases the knowledge base of a field of research while applied science uses that knowledge to solve specific problems.

Learning Objectives

- Differentiate between basic and applied science

Key Points

- The only goal of basic science research is to increase the knowledge base of a particular field of study.
- Applied science uses the knowledge base supplied by basic science to devise solutions, often technological, to specific problems.
- The basic science involved in mapping the human genome is leading to applied science techniques that will diagnose and treat genetic diseases.

Key Terms

- **basic science**: research done solely to expand the knowledge base
- **applied science**: The discipline dealing with the art or science of applying scientific knowledge to practical problems.

Two Types of Science: Basic Science and Applied Science

The scientific community has been debating for the last few decades about the value of different types of science. Is it valuable to pursue science for the sake of simply gaining knowledge, or does scientific knowledge only have worth if we can apply it to solving a specific problem or to bettering our lives? This question focuses on the differences between two types of science: basic science and applied science.

Basic science or "pure" science seeks to expand knowledge regardless of the short-term application of that knowledge. It is not focused on developing a product or a service of immediate public or commercial value. The goal of basic science is knowledge for knowledge's sake; though this does not mean that, in the end, it may not result in a practical application.

In contrast, applied science or "technology" aims to use science to solve real-world problems such as improving crop yields, finding a cure for a particular disease, or saving animals threatened by a natural disaster. In applied science, the problem is usually defined for the researcher.

Figure 1.1*D*.1: **Example of Applied Science**: After Hurricane Ike struck the Gulf Coast in 2008, the U.S. Fish and Wildlife Service rescued this brown pelican. Thanks to applied science, scientists knew how to rehabilitate the bird.

Some individuals may perceive applied science as "useful" and basic science as "useless." A question these people might pose to a scientist advocating knowledge acquisition would be, "What for?" A careful look at the history of science, however, reveals that basic knowledge has resulted in many remarkable applications of great value. Many scientists think that a basic understanding of science is necessary before an application is developed; therefore, applied science relies on the results generated through basic science. Other scientists think that it is time to move on from basic science and instead to find

solutions to actual problems. Both approaches are valid. It is true that there are problems that demand immediate attention; however, few solutions would be found without the help of the wide knowledge foundation generated through basic science.

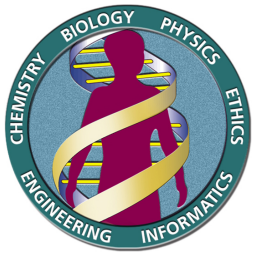

Figure 1.1*D*.1: **A Link Between Basic and Applied Science**: The Human Genome Project was a 13-year collaborative effort among researchers working in several different fields of science. The project, which sequenced the entire human genome, was completed in 2003.

One example of how basic and applied science can work together to solve practical problems occurred after the discovery of DNA structure led to an understanding of the molecular mechanisms governing DNA replication. Strands of DNA, unique in every human, are found in our cells where they provide the instructions necessary for life. During DNA replication, DNA makes new copies of itself shortly before a cell divides. Understanding the mechanisms of DNA replication enabled scientists to develop laboratory techniques that are now used to identify genetic diseases, pinpoint individuals who were at a crime scene, and determine paternity. Without basic science, it is unlikely that applied science would exist.

Another example of the link between basic and applied research is the Human Genome Project, a study in which each human chromosome was analyzed and mapped to determine the precise sequence of DNA subunits and the exact location of each gene. (The gene is the basic unit of heredity; an individual's complete collection of genes is his or her genome.) Other less complex organisms have also been studied as part of this project in order to gain a better understanding of human chromosomes. The Human Genome Project relied on basic research carried out with simple organisms and, later, with the human genome. An important end goal eventually became using the data for applied research to seek cures and early diagnoses for genetically-related diseases.

While research efforts in both basic science and applied science are usually carefully planned, it is important to note that some discoveries are made by serendipity; that is, by means of a fortunate accident or a lucky surprise. Penicillin was discovered when biologist Alexander Fleming accidentally left a petri dish of *Staphylococcus* bacteria open. An unwanted mold grew on the dish, killing the bacteria. The mold turned out to be Penicillium and a new antibiotic was discovered. Even in the highly organized world of science, luck, when combined with an observant, curious mind, can lead to unexpected breakthroughs.

: Publishing Scientific Work

Publication of scientific research in a peer-reviewed journal allows other scientists access to the research.

Learning Objectives

- Describe the role played by peer-reviewed scientific articles

Key Points

- The body of scientific knowledge is recorded in peer-reviewed science journals which allow other scientists to determine what has been done previously and where their own research fits in the larger field of study.
- A scientific article generally follows the steps of the scientific method: introduction (background, observations, question), materials and methods (hypothesis and experimental plan), results (analysis of collected data), and discussion (conclusions drawn from analysis).
- Peer reviewers are other researchers in that field of study who carefully dissect, analyze, and critique a research article submitted for publication.
- Review articles (summaries and commentaries on prior research in a field of study) also go through the peer-review process.

Key Terms

- **peer review**: The scholarly process whereby manuscripts intended to be published in an academic journal are reviewed by independent researchers to evaluate the contribution, importance, and accuracy of the manuscript's contents.

Reporting Scientific Work

Scientists must share their findings in order for other researchers to expand and build upon their discoveries. Collaboration with other scientists—when planning, conducting, and analyzing results—are all important for scientific research. For this reason, a major aspect of a scientist's work is communicating with peers and disseminating results to peers. Scientists can share results by presenting them at a scientific meeting or conference, but this approach can reach only the select few who are present. Instead, most scientists present their results in peer-reviewed manuscripts that are published in scientific journals. Peer-reviewed manuscripts are scientific papers that are reviewed by a scientist's colleagues or peers. These colleagues are qualified individuals, often experts in the same research area, who judge whether or not the scientist's work is suitable for publication. The process of peer review helps to ensure that the research described in a scientific paper or grant proposal is original, significant, logical, and thorough. Grant proposals, which are requests for research funding, are also subject to peer review. Scientists publish their work so other scientists can reproduce their experiments under similar or different conditions to expand on the findings. The experimental results must be consistent with the findings of other scientists.

Figure 1.1E.1: **Scientific Journal**: Scientific research is published in peer-reviewed scientific journals.

A scientific paper is very different from creative writing. Although creativity is required to design experiments, there are fixed guidelines when it comes to presenting scientific results. Scientific writing must be brief, concise, and accurate. It needs to be succinct but detailed-enough to allow peers to reproduce the experiments.

The scientific paper consists of several specific sections: introduction, materials and methods, results, and discussion. This structure is sometimes called the "IMRaD" format. There are usually acknowledgment and reference sections, as well as

tract (a concise summary) at the beginning of the paper. There might be additional sections depending on the type of and the journal where it will be published; for example, some review papers require an outline.

The introduction starts with brief, but broad, background information about what is known in the field. A good introduction also gives the rationale and justification for the work. The introduction refers to the published scientific work of others and, therefore, requires citations following the style of the journal. Using the work or ideas of others without proper citation is considered plagiarism.

The materials and methods section includes a complete and accurate description of the substances and the techniques used by the researchers to gather data. The description should be thorough, yet concise, while providing enough information to allow another researcher to repeat the experiment and obtain similar results. This section will also include information on how measurements were made and what types of calculations and statistical analyses were used to examine raw data. Although the materials and methods section gives an accurate description of the experiments, it does not discuss them.

Journals may require separate results and discussion sections, or it may combine them in one section. If the journal does not allow the combination of both sections, the results section simply narrates the findings without any further interpretation. The results are presented by means of tables or graphs, but no duplicate information should be presented. In the discussion section, the researcher will interpret the results, describe how variables may be related, and attempt to explain the observations. It is indispensable to conduct an extensive literature search to put the results in the context of previously-published scientific research. Therefore, proper citations are included in this section as well.

Finally, the conclusion section summarizes the importance of the experimental findings. While the scientific paper almost certainly answered one or more scientific questions that were stated, any good research should lead to more questions. A well-written scientific paper leaves doors open for the researcher and others to continue and expand on the findings.

Review articles do not follow the IMRAD format because they do not present original scientific findings or primary literature. Instead, they summarize and comment on findings that were published as primary literature. They typically include extensive reference sections.

1.1F: Branches and Subdisciplines of Biology

The field of biology can be divided into various branches and subdisciplines, which leads to careers that result in more focused fields.

Learning Objectives

- Recognize the various subfields of biology; e.g. microbiology, genetics, evolutionary, etc.

Key Points

- Biology is broad and focuses on the study of life from various perspectives.
- The branches and subdisciplines of biology, which are highly focused areas, have resulted in the development of careers that are specific to these branches and subdisciplines.
- Branches of biological study include microbiology, physiology, ecology and genetics; subdisciplines within these branches can include: microbial physiology, microbial ecology and microbial genetics.

Key Terms

- **genetic engineering**: the deliberate modification of the genetic structure of an organism
- **forensic**: Relating to the use of science and technology in the investigation and establishment of facts or evidence in a court of law.

Branches of Biological Study

The scope of biology is broad and therefore contains many branches and subdisciplines. Biologists may pursue one of those subdisciplines and work in a more focused field. The biological branches are divided according to the focus of the discipline and can even be divided based on the types of techniques and tools used to study that specific focus. However, with the increasing amount of basic biological information growing due to advances in technology and databases, there is often cross-discipline and collaboration between branches. For instance, molecular biology and biochemistry study biological processes at the molecular and chemical level, respectively, including interactions among molecules such as DNA, RNA, and proteins, as well as the way they are regulated. Microbiology, the study of microorganisms, is the study of the structure and function of single-celled organisms. It is quite a broad branch itself, and depending on the subject of study, there are also microbial physiologists, ecologists, and geneticists, among others.

Biological Disciplines and Careers

Forensic science is the application of science to answer questions related to the law. Biologists as well as chemists and biochemists can be forensic scientists. Forensic scientists provide scientific evidence for use in courts, and their job involves examining trace materials associated with crimes.Their job activities are primarily related to crimes against people such as murder, rape, and assault. Their work involves analyzing samples such as hair, blood, and other body fluids, including the processing of DNA found in many different environments and materials associated with the crime scenes.

Figure 1.1F.1: **Forensic Science**: This forensic scientist works in a DNA extraction room at the U.S. Army Criminal Investigation Laboratory at Fort Gillem, GA.

Another field of biological study, neurobiology, is the study of the nervous system, and although it is considered a branch of biology, it is also recognized as an interdisciplinary field of study known as neuroscience. Because of its interdisciplinary nature, this subdiscipline focuses on different functions of the nervous system using molecular, cellular, developmental, medical, and computational approaches.

Additional branches of biology include paleontology, which uses fossils to study life's history; zoology, which studies animals; and botany, which studies plants. Biologists can also specialize as biotechnologists, ecologists, or physiologists. This is just a small sample of the many fields that biologists can pursue.

Figure 1.1F.1: **Paleontology**: Researchers work on excavating dinosaur fossils at a site in Castellón, Spain.

Biology is the culmination of the achievements of the natural sciences from their inception to today. Excitingly, it is the cradle of emerging sciences such as the biology of brain activity, genetic engineering of custom organisms, and the biology of evolution that uses the laboratory tools of molecular biology to retrace the earliest stages of life on earth. A scan of news headlines—whether reporting on immunizations, a newly discovered species, sports doping, or a genetically-modified food—demonstrates the way biology is active in and important to our everyday world.

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1.2: Themes and Concepts of Biology

Topic hierarchy

1.2A: Properties of Life

Key characteristics or functions of living beings are order, stimuli, reproduction, growth/development, regulation, homeostasis, and energy.

1.2B: Levels of Organization of Living Things

The biological levels of organization range from a single organelle all the way up to the biosphere in a highly structured hierarchy.

1.2C: The Diversity of Life

The diversity of life can be classified within the three major domains (Bacteria, Eukarya and Archaea) using phylogenetic trees.

1.2A: Properties of Life

LEARNING OBJECTIVES

- Describe the properties of life

All living organisms share several key characteristics or functions: order, sensitivity or response to the environment, reproduction, growth and development, regulation, homeostasis, and energy processing. When viewed together, these eight characteristics serve to define life.

Figure 1.2A.1: **Multicellular Organisms**: A toad represents a highly organized structure consisting of cells, tissues, organs, and organ systems.

Order

Organisms are highly organized, coordinated structures that consist of one or more cells. Even very simple, single-celled organisms are remarkably complex: inside each cell, atoms make up molecules; these in turn make up cell organelles and other cellular inclusions. In multicellular organisms, similar cells form tissues. Tissues, in turn, collaborate to create organs (body structures with a distinct function). Organs work together to form organ systems.

Figure 1.2A.1: **Response to Stimuli**: The leaves of this sensitive plant (Mimosa pudica) will instantly droop and fold when touched. After a few minutes, the plant returns to normal.

Sensitivity or Response to Stimuli

Organisms can respond to diverse stimuli. For example, plants can grow toward a source of light, climb on fences and walls, or respond to touch. Even tiny bacteria can move toward or away from chemicals (a process called chemotaxis) or light (phototaxis). Movement toward a stimulus is considered a positive response, while movement away from a stimulus is considered a negative response.

Reproduction

Single-celled organisms reproduce by first duplicating their DNA. They then divide it equally as the cell prepares to divide to form two new cells. Multicellular organisms often produce specialized reproductive germline cells that will form new individuals. When reproduction occurs, genes containing DNA are passed along to an organism's offspring. These genes ensure that the offspring will belong to the same species and will have similar characteristics, such as size and shape.

Figure 1.2A.1: **Reproduction**: Although no two look alike, these kittens have inherited genes from both parents and share many of the same characteristics.

Growth and Development

All organisms grow and develop following specific instructions coded for by their genes. These genes provide instructions that will direct cellular growth and development, ensuring that a species' young will grow up to exhibit many of the same characteristics as its parents.

Regulation

Even the smallest organisms are complex and require multiple regulatory mechanisms to coordinate internal functions, respond to stimuli, and cope with environmental stresses. Two examples of internal functions regulated in an organism are nutrient transport and blood flow. Organs (groups of tissues working together) perform specific functions, such as carrying oxygen throughout the body, removing wastes, delivering nutrients to every cell, and cooling the body.

Figure 1.2A.1: **Homeostasis**: Polar bears (Ursus maritimus) and other mammals living in ice-covered regions maintain their body temperature by generating heat and reducing heat loss through thick fur and a dense layer of fat under their skin.

Homeostasis

In order to function properly, cells need to have appropriate conditions such as proper temperature, pH, and appropriate concentration of diverse chemicals. These conditions may, however, change from one moment to the next. Organisms are able to maintain internal conditions within a narrow range almost constantly, despite environmental changes, through homeostasis (literally, "steady state")—the ability of an organism to maintain constant internal conditions. For example, an organism needs to regulate body temperature through a process known as thermoregulation. Organisms that live in cold climates, such as the polar bear, have body structures that help them withstand low temperatures and conserve body heat. Structures that aid in this type of insulation include fur, feathers, blubber, and fat. In hot climates, organisms have methods (such as perspiration in humans or panting in dogs) that help them to shed excess body heat.

Figure 1.2A.1: **Energy Processing**: The California condor (Gymnogyps californianus) uses chemical energy derived from food to power flight.

Energy Processing

All organisms use a source of energy for their metabolic activities. Some organisms capture energy from the sun and convert it into chemical energy in food; others use chemical energy in molecules they take in as food.

Figure 1.2A.1: **Adaptation in the flat-tailed horned lizard**: This lizard exhibits a flattened body and coloring that helps camouflage it, both of which are adaptive traits that help it avoid predators.

Evolution

As a population of organisms interacts with the environment, individuals with traits that contribute to reproduction and survival in that particular environment will leave more offspring. Over time those advantageous traits (called adaptations) will become more common in the population. This process, change over time, is called evolution, and it is one of the processes that explain the diverse species seen in biology. Adaptations help organisms survive in their ecological niches, and adaptive traits may be structural, behavioral, or physiological; as such, adaptations frequently involve other properties of organisms such as homeostasis, reproduction, and growth and development.

Key Points

- Order can include highly organized structures such as cells, tissues, organs, and organ systems.
- Interaction with the environment is shown by response to stimuli.
- The ability to reproduce, grow and develop are defining features of life.
- The concepts of biological regulation and maintenance of homeostasis are key to survival and define major properties of life.
- Organisms use energy to maintain their metabolic processes.
- Populations of organisms evolve to produce individuals that are adapted to their specific environment.

Key Terms

- **phototaxis**: The movement of an organism either towards or away from a source of light
- **gene**: a unit of heredity; the functional units of chromosomes that determine specific characteristics by coding for specific proteins
- **chemotaxis**: the movement of a cell or an organism in response to a chemical stimulant

1.2B: Levels of Organization of Living Things

Learning Objectives

- Describe the biological levels of organization from the smallest to highest level

Living things are highly organized and structured, following a hierarchy that can be examined on a scale from small to large. The atom is the smallest and most fundamental unit of matter. It consists of a nucleus surrounded by electrons. Atoms form molecules which are chemical structures consisting of at least two atoms held together by one or more chemical bonds. Many molecules that are biologically important are macromolecules, large molecules that are typically formed by polymerization (a polymer is a large molecule that is made by combining smaller units called monomers, which are simpler than macromolecules). An example of a macromolecule is deoxyribonucleic acid (DNA), which contains the instructions for the structure and functioning of all living organisms.

Figure 1.2B.1: **DNA**: All molecules, including this DNA molecule, are composed of atoms.

From Organelles to Biospheres

Macromolecules can form aggregates within a cell that are surrounded by membranes; these are called organelles. Organelles are small structures that exist within cells. Examples of these include: mitochondria and chloroplasts, which carry out indispensable functions. Mitochondria produce energy to power the cell while chloroplasts enable green plants to utilize the energy in sunlight to make sugars. All living things are made of cells, and the cell itself is the smallest fundamental unit of structure and function in living organisms. (This requirement is why viruses are not considered living: they are not made of cells. To make new viruses, they have to invade and hijack the reproductive mechanism of a living cell; only then can they obtain the materials they need to reproduce.) Some organisms consist of a single cell and others are multicellular. Cells are classified as prokaryotic or eukaryotic. Prokaryotes are single-celled or colonial organisms that do not have membrane-bound nuclei; in contrast, the cells of eukaryotes do have membrane-bound organelles and a membrane-bound nucleus.

In larger organisms, cells combine to make tissues, which are groups of similar cells carrying out similar or related functions. Organs are collections of tissues grouped together performing a common function. Organs are present not only in animals but also in plants. An organ system is a higher level of organization that consists of functionally related organs. Mammals have many organ systems. For instance, the circulatory system transports blood through the body and to and from the lungs; it includes organs such as the heart and blood vessels. Furthermore, organisms are individual living entities. For example, each tree in a forest is an organism. Single-celled prokaryotes and single-celled eukaryotes are also considered organisms and are typically referred to as microorganisms.

All the individuals of a species living within a specific area are collectively called a population. For example, a forest may include many pine trees. All of these pine trees represent the population of pine trees in this forest. Different populations may live in the same specific area. For example, the forest with the pine trees includes populations of flowering plants and also insects and microbial populations. A community is the sum of populations inhabiting a particular area. For instance, all of the trees, flowers, insects, and other populations in a forest form the forest's community. The forest itself is an ecosystem. An ecosystem consists of all the living things in a particular area together with the abiotic, non-living parts of that environment such as nitrogen in the soil or rain water. At the highest level of organization, the biosphere is the collection of all ecosystems, and it represents the zones of life on earth. It includes land, water, and even the atmosphere to a certain extent. Taken together, all of these levels comprise the biological levels of organization, which range from organelles to the biosphere.

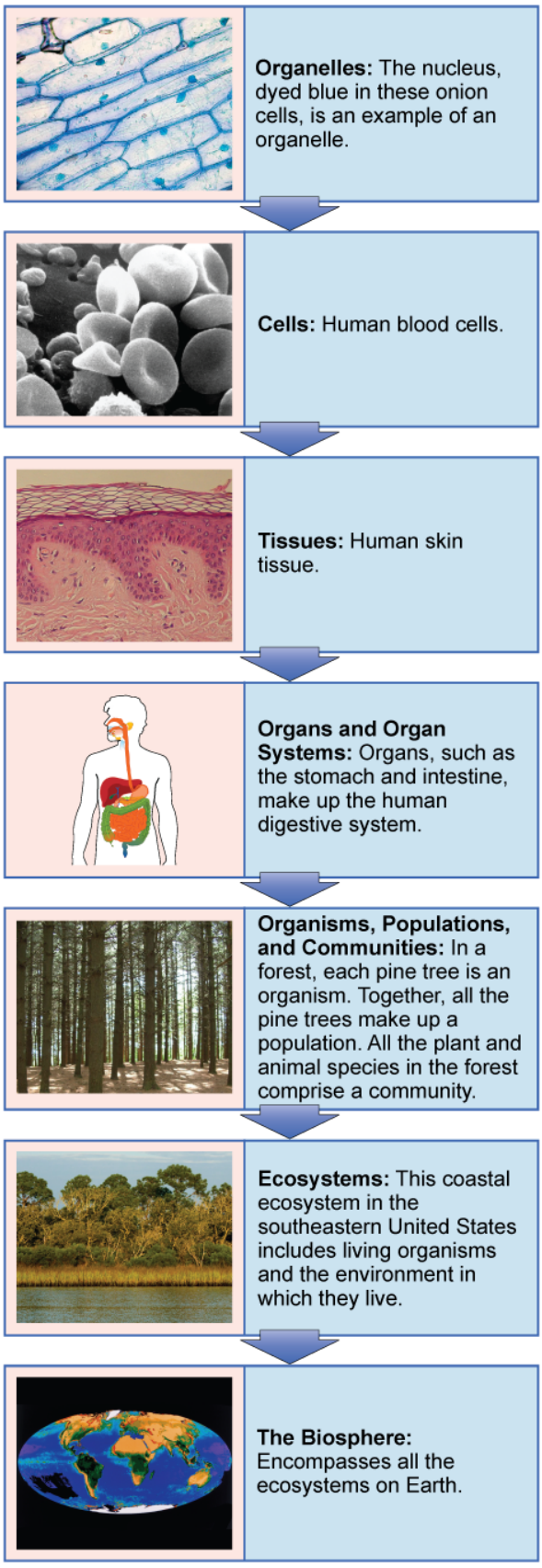

Figure 1.2B.1: **Biological Levels of Organization**: The biological levels of organization of living things follow a hierarchy, such as the one shown. From a single organelle to the entire biosphere, living organisms are part of a highly

structured hierarchy.

Key Points

- The atom is the smallest and most fundamental unit of matter. The bonding of at least two atoms or more form molecules.
- The simplest level of organization for living things is a single organelle, which is composed of aggregates of macromolecules.
- The highest level of organization for living things is the biosphere; it encompasses all other levels.
- The biological levels of organization of living things arranged from the simplest to most complex are: organelle, cells, tissues, organs, organ systems, organisms, populations, communities, ecosystem, and biosphere.

Key Terms

- **molecule**: The smallest particle of a specific compound that retains the chemical properties of that compound; two or more atoms held together by chemical bonds.
- **macromolecule**: a very large molecule, especially used in reference to large biological polymers (e.g. nucleic acids and proteins)
- **polymerization**: The chemical process, normally with the aid of a catalyst, to form a polymer by bonding together multiple identical units (monomers).

Learning Objectives

- Recognize the three major domains used for classification

The fact that biology has such a broad scope as a science has to do with the tremendous diversity of life on Earth. The source of this diversity is evolution, the process of gradual change during which new species arise from older species. Evolutionary biologists study the evolution of living things in everything from the microscopic world to ecosystems.

The evolution of various life forms on Earth can be summarized in a phylogenetic tree using phylogeny. A phylogenetic tree is a diagram showing the evolutionary relationships among biological species based on similarities and differences in genetic or physical traits or both. A phylogenetic tree is composed of nodes and branches. The internal nodes represent ancestors and are points in evolution when, based on scientific evidence, an ancestor is thought to have diverged to form two new species. The length of each branch is proportional to the time elapsed since the split.

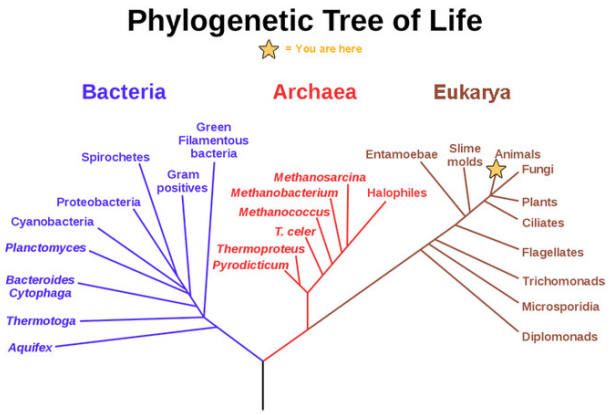

Figure 1.2C.1: **Phylogenetic Tree of Life**: This phylogenetic tree was constructed by microbiologist Carl Woese using data obtained from sequencing ribosomal RNA genes. The tree shows the separation of living organisms into three domains: Bacteria, Archaea, and Eukarya. Bacteria and Archaea are prokaryotes, single-celled organisms lacking intracellular organelles.

Carl Woese and the Phylogenetic Tree

In the past, biologists grouped living organisms into five kingdoms: animals, plants, fungi, protists, and bacteria. The organizational scheme was based mainly on physical features, as opposed to physiology, biochemistry, or molecular biology, all of which are used by modern systematics. The pioneering work of American microbiologist Carl Woese in the early 1970s has shown, however, that life on Earth has evolved along three lineages, now called domains—Bacteria, Archaea, and Eukarya. The first two are prokaryotic cells with microbes that lack membrane-enclosed nuclei and organelles. The third domain contains the eukaryotes and includes unicellular microorganisms together with the four original kingdoms (excluding bacteria). Woese defined Archaea as a new domain, and this resulted in a new taxonomic tree. Many organisms belonging to the Archaea domain live under extreme conditions and are called extremophiles. To construct his tree, Woese used genetic relationships rather than similarities based on morphology (shape).

Woese's tree was constructed from comparative sequencing of the genes that are universally distributed, present in every organism, and conserved (meaning that these genes have remained essentially unchanged throughout evolution). Woese's approach was revolutionary because comparisons of physical features are insufficient to differentiate between the prokaryotes that appear fairly similar in spite of their tremendous biochemical diversity and genetic variability. The comparison of homologous DNA and RNA sequences provided Woese with a sensitive device that revealed the extensive variability of prokaryotes, and which justified the separation of the prokaryotes into two domains: bacteria and archaea. DNA, the universal genetic material, contains the instructions for the structure and function of all living organisms and can be divided into genes whose expression varies between organisms. The RNA, which is transcribed from DNA, varies between organisms as well based on the expression of specific genes. Thus, to examine differences at this molecular level provides a more accurate depiction of the diversity which exists.

Points

- The three major Domains of Life include: Domain Bacteria, Domain Eukarya and Domain Archaea.
- Domain Bacteria and Domain Archaea include prokaryotic cells that lack membrane-enclosed nuclei and organelles.
- Domain Eukarya include eukaryotes and more complex organisms that contain membrane-bound nuclei and organelles.
- Carl Woese defined Archaea as a new domain and constructed the phylogentic tree of life which shows separation of all living organisms.
- The phylogenetic tree of life was constructed by Carl Woese using sequencing data of ribosomal RNA genes. Therefore, genetics classification surpassed morphological cataloguing, which was the traditional way of organizing living beings.

Key Terms

- **phylogeny**: the evolutionary history of an organism
- **extremophile**: an organism that lives under extreme conditions of temperature, salinity, etc; commercially important as a source of enzymes that operate under similar conditions
- **DNA**: a biopolymer of deoxyribonucleic acids (a type of nucleic acid) that has four different chemical groups, called bases: adenine, guanine, cytosine, and thymine

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CHAPTER OVERVIEW

10: CELL REPRODUCTION

10.1: CELL DIVISION

10.1A: THE ROLE OF THE CELL CYCLE
10.1B: GENOMIC DNA AND CHROMOSOMES
10.1C: EUKARYOTIC CHROMOSOMAL STRUCTURE AND COMPACTION
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10.2A: INTERPHASE
10.2B: THE MITOTIC PHASE AND THE G0 PHASE
10.3: CONTROL OF THE CELL CYCLE

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10.4: CANCER AND THE CELL CYCLE

10.4A: PROTO-ONCOGENES
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10.5: PROKARYOTIC CELL DIVISION

10.5A: BINARY FISSION

10.1: Cell Division

Topic hierarchy

- 10.1A: The Role of the Cell Cycle
- 10.1B: Genomic DNA and Chromosomes
- 10.1C: Eukaryotic Chromosomal Structure and Compaction

10.1A: The Role of the Cell Cycle

The cell cycle allows multiicellular organisms to grow and divide and single-celled organisms to reproduce.

Learning Objectives

- Explain the role of the cell cycle in carrying out the cell's essential functions

Key Points

- All multicellular organisms use cell division for growth and the maintenance and repair of cells and tissues.
- Single-celled organisms use cell division as their method of reproduction.
- Somatic cells divide regularly; all human cells (except for the cells that produce eggs and sperm) are somatic cells.
- Somatic cells contain two copies of each of their chromosomes (one copy from each parent).
- The cell cycle has two major phases: interphase and the mitotic phase.
- During interphase, the cell grows and DNA is replicated; during the mitotic phase, the replicated DNA and cytoplasmic contents are separated and the cell divides.

Key Terms

- **somatic cell**: any normal body cell of an organism that is not involved in reproduction; a cell that is not on the germline
- **interphase**: the stage in the life cycle of a cell where the cell grows and DNA is replicated
- **mitotic phase**: replicated DNA and the cytoplasmic material are divided into two identical cells

Introduction: Cell Division and Reproduction

A human, as well as every sexually-reproducing organism, begins life as a fertilized egg or zygote. Trillions of cell divisions subsequently occur in a controlled manner to produce a complex, multicellular human. In other words, that original single cell is the ancestor of every other cell in the body. Once a being is fully grown, cell reproduction is still necessary to repair or regenerate tissues. For example, new blood and skin cells are constantly being produced. All multicellular organisms use cell division for growth and the maintenance and repair of cells and tissues. Cell division is tightly regulated because the occasional failure of regulation can have life-threatening consequences. Single-celled organisms use cell division as their method of reproduction.

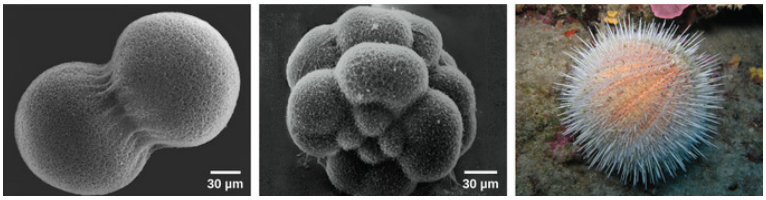

Figure 10.1*A*. 1: **Cell Division and Growth**: A sea urchin begins life as a single cell that (a) divides to form two cells, visible by scanning electron microscopy. After four rounds of cell division, (b) there are 16 cells, as seen in this SEM image. After many rounds of cell division, the individual develops into a complex, multicellular organism, as seen in this (c) mature sea urchin.

While there are a few cells in the body that do not undergo cell division, most somatic cells divide regularly. A somatic cell is a general term for a body cell: all human cells, except for the cells that produce eggs and sperm (which are referred to as germ cells), are somatic cells. Somatic cells contain two copies of each of their chromosomes (one copy received from each parent). Cells in the body replace themselves over the lifetime of a person. For example, the cells lining the gastrointestinal tract must be frequently replaced when constantly "worn off" by the movement of food through the gut. But what triggers a cell to divide and how does it prepare for and complete cell division?

The cell cycle is an ordered series of events involving cell growth and cell division that produces two new daughter cells. Cells on the path to cell division proceed through a series of precisely timed and carefully regulated stages of growth, DNA replication, and division that produces two identical (clone) cells. The cell cycle has two major phases: interphase and the mitotic phase. During interphase, the cell grows and DNA is replicated. During the mitotic phase, the replicated DNA and cytoplasmic contents are separated and the cell divides.

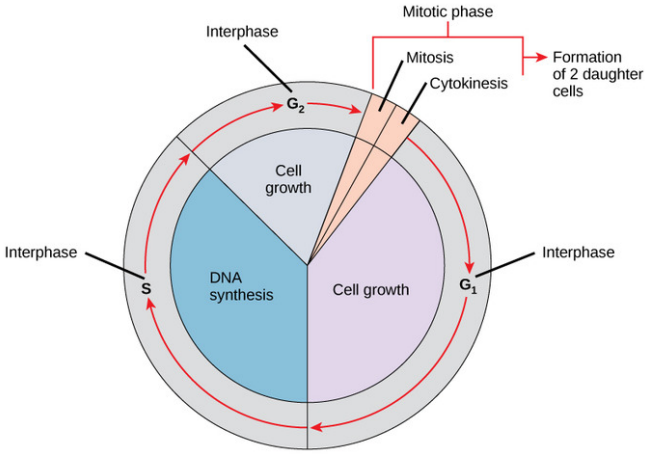

Figure 10.1A.1: **The Cell Cycle**: The cell cycle consists of interphase and the mitotic phase. During interphase, the cell grows and the nuclear DNA is duplicated. Interphase is followed by the mitotic phase. During the mitotic phase, the duplicated chromosomes are segregated and distributed into daughter nuclei. The cytoplasm is usually divided as well, resulting in two daughter cells

10.1B: Genomic DNA and Chromosomes

The genome of an organism consists of its entire complement of DNA, which encodes the genes that control the organism's characteristics.

Learning Objectives

- Explain the importance of a genome to an organism

Key Points

- A cell 's DNA, packaged as a double-stranded DNA molecule, is called its genome.
- In prokaryotes, the genome is composed of a single, double-stranded DNA molecule in the form of a loop or circle; the region in the cell containing this genetic material is called a nucleoid.
- In eukaryotes, the genome consists of several double-stranded linear DNA molecules; each species of eukaryotes has a characteristic number of chromosomes in the nuclei of its cells.
- Matched pairs of chromosomes in a diploid organism are called homologous chromosomes, which are the same length and have specific nucleotide segments called genes in exactly the same location, or locus.
- Each copy of a homologous pair of chromosomes originates from a different parent, so the genes themselves are not identical.
- The difference between the DNA sequences in pairs of homologous chromosomes is less than one percent; the sex chromosomes, X and Y, are the single exception to this rule since their genes are different.

Key Terms

- **genome**: the cell's complete genetic information packaged as a double-stranded DNA molecule
- **nucleoid**: the irregularly-shaped region within a prokaryote cell where the genetic material is localized
- **gene**: a unit of heredity; the functional units of chromosomes that determine specific characteristics by coding for specific proteins
- **chromosome**: a structure in the cell nucleus that contains DNA, histone protein, and other structural proteins
- **locus**: a fixed position on a chromosome that may be occupied by one or more genes

Genomic DNA

Before discussing the steps a cell must undertake to replicate, a deeper understanding of the structure and function of a cell's genetic information is necessary. A cell's DNA, packaged as a double-stranded DNA molecule, is called its genome. In prokaryotes, the genome is composed of a single, double-stranded DNA molecule in the form of a loop or circle. The region in the cell containing this genetic material is called a nucleoid. Some prokaryotes also have smaller loops of DNA called plasmids that are not essential for normal growth. Bacteria can exchange these plasmids with other bacteria, sometimes receiving beneficial new genes that the recipient can add to their chromosomal DNA. Antibiotic resistance is one trait that often spreads through a bacterial colony through plasmid exchange.

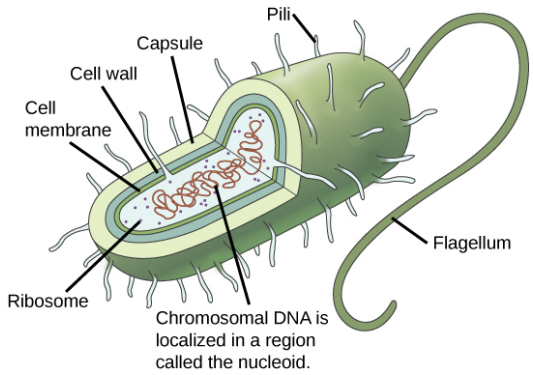

Figure 10.1B.1: **Prokaryotic Genome**: Prokaryotes, including bacteria and archaea, have a single, circular chromosome located in a central region called the nucleoid.

In eukaryotes, the genome consists of several double-stranded linear DNA molecules packaged into chromosomes. Each species of eukaryotes has a characteristic number of chromosomes in the nuclei of its cells. Human body cells have 46 chromosomes, while human gametes (sperm or eggs) have 23 chromosomes each. A typical body cell, or somatic cell, contains two matched sets of chromosomes, a configuration known as diploid. The letter n is used to represent a single set of chromosomes; therefore, a diploid organism is designated 2n. Human cells that contain one set of chromosomes are called gametes, or sex cells; these are eggs and sperm, and are designated 1n, or haploid.

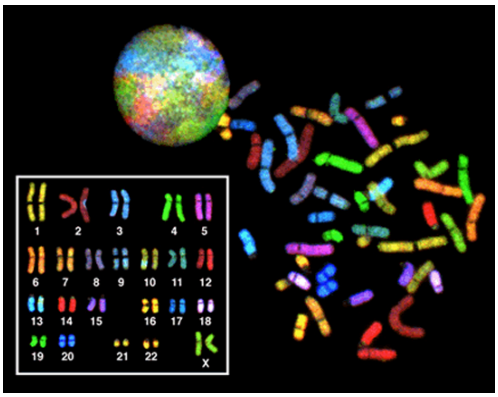

Figure 10.1B.1: **Eukaryotic Genome**: There are 23 pairs of homologous chromosomes in a female human somatic cell. The condensed chromosomes are viewed within the nucleus (top), removed from a cell in mitosis and spread out on a slide (right), and artificially arranged according to length (left); an arrangement like this is called a karyotype. In this image, the chromosomes were exposed to fluorescent stains for differentiation of the different chromosomes. A method of staining called "chromosome painting" employs fluorescent dyes that highlight chromosomes in different colors.

Matched pairs of chromosomes in a diploid organism are called homologous ("same knowledge") chromosomes. Homologous chromosomes are the same length and have specific nucleotide segments called genes in exactly the same location, or locus. Genes, the functional units of chromosomes, determine specific characteristics, or traits, by coding for specific proteins. For example, hair color is a trait that can be blonde, brown, or black.

Each copy of a homologous pair of chromosomes originates from a different parent; therefore, the genes themselves are not identical. The variation of individuals within a species is due to the specific combination of the genes inherited from both parents. Even a slightly altered sequence of nucleotides within a gene can result in an alternative trait. For example, there are three possible gene sequences on the human chromosome that code for blood type: sequence A, sequence B, and sequence O. Because all diploid human cells have two copies of the chromosome that determines blood type, the blood type (the trait) is determined by which two versions of the marker gene are inherited. It is possible to have two copies of the same gene sequence on both homologous chromosomes, with one on each (for example, AA, BB, or OO), or two different sequences, such as AB, AO, or BO.

Minor variations of traits, such as blood type, eye color, and handedness, contribute to the natural variation found within a species. However, if the entire DNA sequence from any pair of human homologous chromosomes is compared, the difference is less than one percent. The sex chromosomes, X and Y, are the single exception to the rule of homologous chromosome uniformity. Other than a small amount of homology that is necessary to accurately produce gametes, the genes found on the X and Y chromosomes are different.

10.1C: Eukaryotic Chromosomal Structure and Compaction

Chromosomes must coil to pack DNA into the cell during cell division, a process involving 3 levels of compaction.

Learning Objectives

- Describe the levels of chromsomal structure and compaction

Key Points

- During some stages of the cell cycle, the long strands of DNA are condensed into compact chromosomes to fit in the cell's nucleus.
- In the first level of compaction, short stretches of the DNA double helix wrap around a core of eight histone proteins at regular intervals along the entire length of the chromosome.
- The DNA surrouding the histone core is called a nucleosome; the DNA-histone complex is called chromatin.
- The second level of compaction occurs as the nucleosomes and the linker DNA between them are coiled into a 30-nm chromatin fiber, which shortens the chromosome so it's about 50 times shorter than the extended form.
- After replication, the chromosomes are composed of two linked sister chromatids; when fully compact, the pairs of identically-packed chromosomes are bound to each other by cohesin proteins.
- The connection between the sister chromatids is closest in a region called the centromere; this region is highly condensed.

Key Terms

- **nucleosome**: any of the subunits that repeat in chromatin; a coil of DNA surrounding a histone core
- **histone**: any of various simple water-soluble proteins that are rich in the basic amino acids lysine and arginine and are complexed with DNA in the nucleosomes of eukaryotic chromatin
- **chromatin**: a complex of DNA, RNA, and proteins within the cell nucleus out of which chromosomes condense during cell division

Eukaryotic Chromosomal Structure and Compaction

If the DNA from all 46 chromosomes in a human cell nucleus was laid out end to end, it would measure approximately two meters. However, the diameter would be only 2 nm. Considering that the size of a typical human cell is about 10 µm (100,000 cells lined up to equal one meter), DNA must be tightly packaged to fit in the cell's nucleus. At the same time, it must also be readily accessible for the genes to be expressed. During some stages of the cell cycle, the long strands of DNA are condensed into compact chromosomes. There are a number of ways that chromosomes are compacted to fit in the cell's nucleus and be accessible for gene expression.

In the first level of compaction, short stretches of the DNA double helix wrap around a core of eight histone proteins at regular intervals along the entire length of the chromosome. The DNA-histone complex is called chromatin. The beadlike, histone DNA complex is called a nucleosome. DNA connecting the nucleosomes is called linker DNA. A DNA molecule in this form is about seven times shorter than the double helix without the histones. The beads are about 10 nm in diameter, in contrast with the 2-nm diameter of a DNA double helix. The next level of compaction occurs as the nucleosomes and the linker DNA between them are coiled into a 30-nm chromatin fiber. This coiling further shortens the chromosome so that it is now about 50 times shorter than the extended form. In the third level of packing, a variety of fibrous proteins is used to pack the chromatin. These fibrous proteins also ensure that each chromosome in a non-dividing cell occupies a particular area of the nucleus that does not overlap with that of any other chromosome.

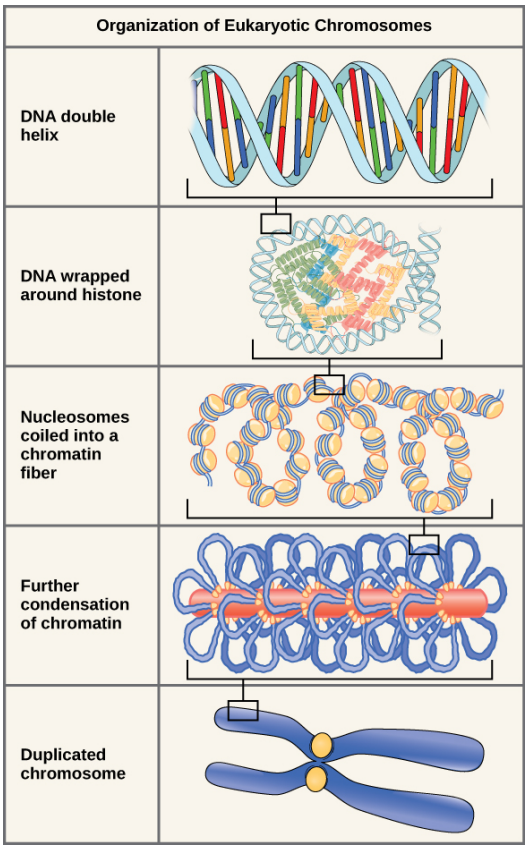

Figure 10.1C.1: **Levels of DNA Compaction**: Double-stranded DNA wraps around histone proteins to form nucleosomes that have the appearance of "beads on a string." The nucleosomes are coiled into a 30-nm chromatin fiber. When a cell undergoes mitosis, the chromosomes condense even further.

DNA replicates in the S phase of interphase. After replication, the chromosomes are composed of two linked sister chromatids. When fully compact, the pairs of identically-packed chromosomes are bound to each other by cohesin proteins. The connection between the sister chromatids is closest in a region called the centromere. The conjoined sister chromatids, with a diameter of about 1 µm, are visible under a light microscope. The centromeric region is highly condensed and will appear as a constricted area.

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Topic hierarchy

- 10.2A: Interphase
- 10.2B: The Mitotic Phase and the G0 Phase

10.2A: Interphase

Cells must grow and duplicate their internal structures during interphase before they can divide during mitosis.

Learning Objectives

- Describe the events that occur during Interphase

Key Points

- There are three stages of interphase: G_1 (first gap), S (synthesis of new DNA), and G_2 (second gap).
- Cells spend most of their lives in interphase, specifically in the S phase where genetic material must be copied.
- The cell grows and carries out biochemical functions, such as protein synthesis, in the G_1 phase.
- During the S phase, DNA is duplicated into two sister chromatids, and centrosomes, which give rise to the mitotic spindle, are also replicated.
- In the G_2 phase, energy is replenished, new proteins are synthesized, the cytoskeleton is dismantled, and additional growth occurs.

Key Terms

- **interphase**: the stage in the life cycle of a cell where the cell grows and DNA is replicated
- **sister chromatid**: either of the two identical strands of a chromosome (DNA material) that separate during mitosis
- **mitotic spindle**: the apparatus that orchestrates the movement of chromosomes during mitosis

Interphase

During interphase, the cell undergoes normal growth processes while also preparing for cell division. In order for a cell to move from interphase into the mitotic phase, many internal and external conditions must be met. The three stages of interphase are called G_1, S, and G_2 .

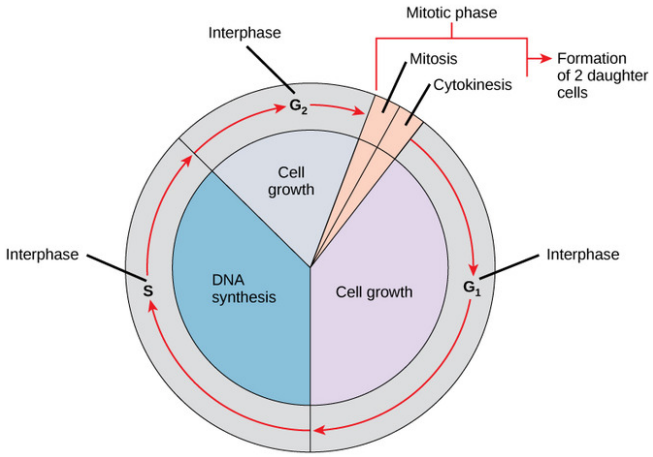

Figure 10.2*A*.1: **The Stages of Interphase and the Cell Cycle**: The cell cycle consists of interphase and the mitotic phase. During interphase, the cell grows and the nuclear DNA is duplicated. Interphase is followed by the mitotic phase. During the mitotic phase, the duplicated chromosomes are segregated and distributed into daughter nuclei. The cytoplasm is usually divided as well, resulting in two daughter cells.

G_1 Phase (First Gap)

The first stage of interphase is called the G_1 phase (first gap) because, from a microscopic aspect, little change is visible. However, during the G_1 stage, the cell is quite active at the biochemical level. The cell grows and accumulates the building blocks of chromosomal DNA and the associated proteins as well as sufficient energy reserves to complete the task of replicating each chromosome in the nucleus.

S Phase (Synthesis of DNA)

The synthesis phase of interphase takes the longest because of the complexity of the genetic material being duplicated. Throughout interphase, nuclear DNA remains in a semi-condensed chromatin configuration. In the S phase, DNA replication results in the formation of identical pairs of DNA molecules, sister chromatids, that are firmly attached to the centromeric region. The centrosome is duplicated during the S phase. The two centrosomes will give rise to the mitotic spindle, the apparatus that orchestrates the movement of chromosomes during mitosis. At the center of each animal cell, the centrosomes of animal cells are associated with a pair of rod-like objects, the centrioles, which are at right angles to each other. Centrioles help organize cell division. Centrioles are not present in the centrosomes of other eukaryotic species, such as plants and most fungi.

G_2 Phase (Second Gap)

In the G_2 phase, the cell replenishes its energy stores and synthesizes proteins necessary for chromosome manipulation. Some cell organelles are duplicated, and the cytoskeleton is dismantled to provide resources for the mitotic phase. There may be additional cell growth during G_2. The final preparations for the mitotic phase must be completed before the cell is able to enter the first stage of mitosis.

10.2B: The Mitotic Phase and the G0 Phase

During the multistep mitotic phase, the cell nucleus divides, and the cell components split into two identical daughter cells.

Learning Objectives

- Describe the events that occur at the different stages of mitosis

Key Points

- During prophase, the nucleus disappears, spindle fibers form, and DNA condenses into chromosomes (sister chromatids).
- During metaphase, the sister chromatids align along the equator of the cell by attaching their centromeres to the spindle fibers.
- During anaphase, sister chromatids are separated at the centromere and are pulled towards opposite poles of the cell by the mitotic spindle.
- During telophase, chromosomes arrive at opposite poles and unwind into thin strands of DNA, the spindle fibers disappear, and the nuclear membrane reappears.
- Cytokinesis is the actual splitting of the cell membrane; animal cells pinch apart, while plant cells form a cell plate that becomes the new cell wall.
- Cells enter the G_0 (inactive) phase after they exit the cell cycle when they are not actively preparing to divide; some cells remain in G_0 phase permanently.

Key Terms

- **karyokinesis**: (mitosis) the first portion of mitotic phase in which division of the cell nucleus takes place
- **centrosome**: an organelle near the nucleus in the cytoplasm of most organisms that controls the organization of its microtubules and gives rise to the mitotic spindle
- **cytokinesis**: the second portion of the mitotic phase in which the cytoplasm of a cell divides following the division of the nucleus

The Mitotic Phase

The mitotic phase is a multistep process during which the duplicated chromosomes are aligned, separated, and move into two new, identical daughter cells. The first portion of the mitotic phase is called karyokinesis or nuclear division. The second portion of the mitotic phase, called cytokinesis, is the physical separation of the cytoplasmic components into the two daughter cells.

Karyokinesis (Mitosis)

Karyokinesis, also known as mitosis, is divided into a series of phases (prophase, prometaphase, metaphase, anaphase, and telophase) that result in the division of the cell nucleus.

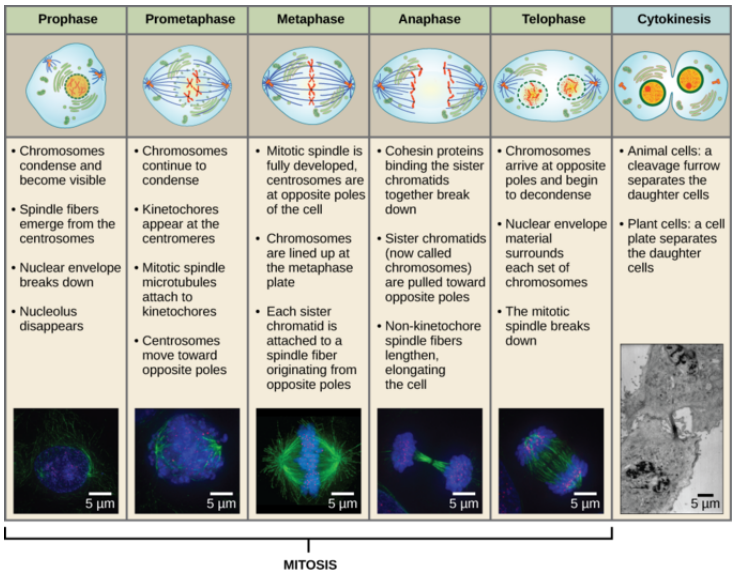

Figure 10.2B.1: **Stages of the Cell Cycle**: Karyokinesis (or mitosis) is divided into five stages: prophase, prometaphase, metaphase, anaphase, and telophase. The images at the bottom were taken by fluorescence microscopy (hence, the black background) of cells artificially stained by fluorescent dyes: blue fluorescence indicates DNA (chromosomes) and green fluorescence indicates microtubules (spindle apparatus).

During prophase, the "first phase," the nuclear envelope starts to dissociate into small vesicles. The membranous organelles (such as the Golgi apparatus and endoplasmic reticulum) fragment and disperse toward the periphery of the cell. The nucleolus disappears and the centrosomes begin to move to opposite poles of the cell. Microtubules that will eventually form the mitotic spindle extend between the centrosomes, pushing them farther apart as the microtubule fibers lengthen. The sister chromatids begin to coil more tightly with the aid of condensin proteins and become visible under a light microscope.

During prometaphase, the "first change phase," many processes that began in prophase continue to advance. The remnants of the nuclear envelope fragment. The mitotic spindle continues to develop as more microtubules assemble and stretch across the length of the former nuclear area. Chromosomes become more condensed and discrete. Each sister chromatid develops a protein structure called a kinetochore in the centromeric region. The proteins of the kinetochore attract and bind mitotic spindle microtubules.

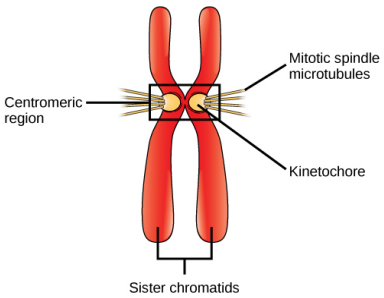

Figure 10.2B.1: **Kinetochore and Mitotic Spindle**: During prometaphase, mitotic spindle microtubules from opposite poles attach to each sister chromatid at the kinetochore. In anaphase, the connection between the sister chromatids breaks down and the microtubules pull the chromosomes toward opposite poles.

During metaphase, the "change phase," all the chromosomes are aligned on a plane called the metaphase plate, or the equatorial plane, midway between the two poles of the cell. The sister chromatids are still tightly attached to each other by cohesin proteins. At this time, the chromosomes are maximally condensed.

During anaphase, the "upward phase," the cohesin proteins degrade, and the sister chromatids separate at the centromere. Each chromatid, now called a chromosome, is pulled rapidly toward the centrosome to which its microtubule is attached. The cell becomes visibly elongated (oval shaped) as the polar microtubules slide against each other at the metaphase plate where they overlap.

telophase, the “distance phase,” the chromosomes reach the opposite poles and begin to decondense (unravel), g into a chromatin configuration. The mitotic spindles are depolymerized into tubulin monomers that will be used to assemble cytoskeletal components for each daughter cell. Nuclear envelopes form around the chromosomes and nucleosomes appear within the nuclear area.

Cytokinesis

Cytokinesis, or “cell motion,” is the second main stage of the mitotic phase during which cell division is completed via the physical separation of the cytoplasmic components into two daughter cells. Division is not complete until the cell components have been apportioned and completely separated into the two daughter cells. Although the stages of mitosis are similar for most eukaryotes, the process of cytokinesis is quite different for eukaryotes that have cell walls, such as plant cells.

In cells such as animal cells, which lack cell walls, cytokinesis follows the onset of anaphase. A contractile ring composed of actin filaments forms just inside the plasma membrane at the former metaphase plate. The actin filaments pull the equator of the cell inward, forming a fissure. This fissure or “crack” is called the cleavage furrow. The furrow deepens as the actin ring contracts; eventually the membrane is cleaved in two.

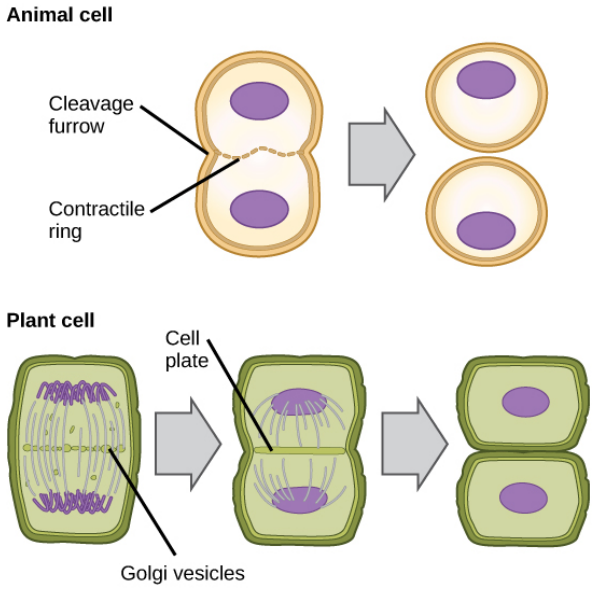

Figure 10.2B.1: **Cytokinesis**: During cytokinesis in animal cells, a ring of actin filaments forms at the metaphase plate. The ring contracts, forming a cleavage furrow, which divides the cell in two. In plant cells, Golgi vesicles coalesce at the former metaphase plate, forming a phragmoplast. A cell plate formed by the fusion of the vesicles of the phragmoplast grows from the center toward the cell walls and the membranes of the vesicles fuse to form a plasma membrane that divides the cell in two.

In plant cells, a new cell wall must form between the daughter cells. During interphase, the Golgi apparatus accumulates enzymes, structural proteins, and glucose molecules prior to breaking into vesicles and dispersing throughout the dividing cell. During telophase, these Golgi vesicles are transported on microtubules to form a phragmoplast (a vesicular structure) at the metaphase plate. There, the vesicles fuse and coalesce from the center toward the cell walls; this structure is called a cell plate. As more vesicles fuse, the cell plate enlarges until it merges with the cell walls at the periphery of the cell. Enzymes use the glucose that has accumulated between the membrane layers to build a new cell wall. The Golgi membranes become parts of the plasma membrane on either side of the new cell wall.

G_0 Phase

Not all cells adhere to the classic cell cycle pattern in which a newly-formed daughter cell immediately enters the preparatory phases of interphase, closely followed by the mitotic phase. Cells in G_0 phase are not actively preparing to divide. The cell is in a quiescent (inactive) stage that occurs when cells exit the cell cycle. Some cells enter G_0 temporarily until an external signal triggers the onset of G_1. Other cells that never or rarely divide, such as mature cardiac muscle and nerve cells, remain in G_0 permanently.

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10.3: Control of the Cell Cycle

Topic hierarchy

- 10.3A: Regulation of the Cell Cycle by External Events
- 10.3B: Regulation of the Cell Cycle at Internal Checkpoints
- 10.3C: Regulator Molecules of the Cell Cycle

10.3A: Regulation of the Cell Cycle by External Events

External factors can influence the cell cycle by inhibiting or initiating cell division.

Learning Objectives

- Describe external events that can affect cell cycle regulation

Key Points

- The death of nearby cells and the presence or absence of certain hormones can impact the cell cycle.
- The release of growth-promoting hormones, such as HGH, can initiate cell division, and a lack of these hormones can inhibit cell division.
- Cell growth initiates cell division because cells must divide as the surface-to-volume ratio decreases; cell crowding inhibits cell division.
- Key conditions must be met before the cell can move into interphase.

Key Terms

- **gigantism**: a condition caused by an over-production of growth hormone, resulting in excessive bone growth
- **growth hormone**: any polypeptide hormone secreted by the pituitary gland that promotes growth and regulates the metabolism of carbohydrates, proteins, and lipids
- **dwarfism**: a condition caused by a lack of growth hormone, resulting in short stature and limbs that are disproportionately small in relation to the body

Regulation of the Cell Cycle by External Events

Unlike the life of organisms, which is a straight progression from birth to death, the life of a cell takes place in a cyclical pattern. Each cell is produced as part of its parent cell. When a daughter cell divides, it turns into two new cells, which would lead to the assumption that each cell is capable of being immortal as long as its descendants can continue to divide. However, all cells in the body only live as long as the organism lives. Some cells do live longer than others, but eventually all cells die when their vital functions cease. Most cells in the body exist in the state of interphase, the non-dividing stage of the cell life cycle. When this stage ends, cells move into the dividing part of their lives called mitosis.

Both the initiation and inhibition of cell division are triggered by events external to the cell when it is about to begin the replication process. An event may be as simple as the death of a nearby cell or as sweeping as the release of growth-promoting hormones, such as human growth hormone (HGH). A lack of HGH can inhibit cell division, resulting in dwarfism, whereas too much HGH can result in gigantism. Crowding of cells can also inhibit cell division. Another factor that can initiate cell division is the size of the cell; as a cell grows, it becomes inefficient due to its decreasing surface-to-volume ratio. The solution to this problem is to divide.

Figure 10.3*A*. 1: **Dwarfism**: Commodore Nut (right) was a famous circus performer afflicted with dwarfism. This was a result of a lack of Human Growth Hormone.

Whatever the source of the message, the cell receives the signal, and a series of events within the cell allows it to proceed into interphase. Moving forward from this initiation point, every parameter required during each cell cycle phase must be met or the cycle cannot progress.

10.3B: Regulation of the Cell Cycle at Internal Checkpoints

The cell cycle is controlled by three internal checkpoints that evaluate the condition of the genetic information.

Learning Objectives

- Explain the effects of internal checkpoints on the regulation of the cell cycle

Key Points

- A checkpoint is one of several points in the eukaryotic cell cycle at which the progression of a cell to the next stage in the cycle can be halted until conditions are favorable.
- Damage to DNA and other external factors are evaluated at the G1 checkpoint; if conditions are inadequate, the cell will not be allowed to continue to the S phase of interphase.
- The G2 checkpoint ensures all of the chromosomes have been replicated and that the replicated DNA is not damaged before cell enters mitosis.
- The M checkpoint determines whether all the sister chromatids are correctly attached to the spindle microtubules before the cell enters the irreversible anaphase stage.

Key Terms

- **restriction point**: (G1 checkpoint) a point in the animal cell cycle at which the cell becomes “committed” to the cell cycle, which is determined by external factors and signals
- **spindle checkpoint**: (M checkpoint) prevents separation of the duplicated chromosomes until each chromosome is properly attached to the spindle apparatus
- **cyclin**: any of a group of proteins that regulates the cell cycle by forming a complex with kinases
- **G2 checkpoint**: ensures all of the chromosomes have been replicated and that the replicated DNA is not damaged

Regulation at Internal Checkpoints

It is essential that the daughter cells are exact duplicates of the parent cell. Mistakes in the duplication or distribution of the chromosomes lead to mutations that may be passed forward to every new cell produced from an abnormal cell. To prevent a compromised cell from continuing to divide, internal control mechanisms operate at three main cell cycle checkpoints. A checkpoint is one of several points in the eukaryotic cell cycle at which the progression of a cell to the next stage in the cycle can be halted until conditions are favorable (e.g. the DNA is repaired). These checkpoints occur near the end of G_1, at the G_2/M transition, and during metaphase.

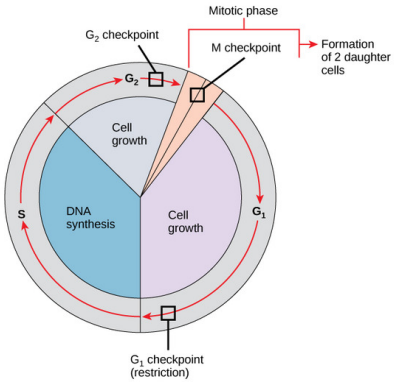

Figure 10.3B.1: **Internal Checkpoints During the Cell Cycle**: The cell cycle is controlled at three checkpoints. The integrity of the DNA is assessed at the G1 checkpoint. Proper chromosome duplication is assessed at the G2 checkpoint. Attachment of each kinetochore to a spindle fiber is assessed at the M checkpoint.

The G_1 Checkpoint

The G_1 checkpoint determines whether all conditions are favorable for cell division to proceed. The G_1 checkpoint, also called the restriction point (in yeast), is a point at which the cell irreversibly commits to the cell division process. External influences, such as growth factors, play a large role in carrying the cell past the G_1 checkpoint. The cell will only pass the

oint if it is an appropriate size and has adequate energy reserves. At this point, the cell also checks for DNA e. A cell that does not meet all the requirements will not progress to the S phase. The cell can halt the cycle and attempt to remedy the problematic condition, or the cell can advance into G_0 (inactive) phase and await further signals when conditions improve.

If a cell meets the requirements for the G_1 checkpoint, the cell will enter S phase and begin DNA replication. This transition, as with all of the major checkpoint transitions in the cell cycle, is signaled by cyclins and cyclin dependent kinases (CDKs). Cyclins are cell-signaling molecules that regulate the cell cycle.

The G_2 Checkpoint

The G_2 checkpoint bars entry into the mitotic phase if certain conditions are not met. As with the G_1 checkpoint, cell size and protein reserves are assessed. However, the most important role of the G_2 checkpoint is to ensure that all of the chromosomes have been accurately replicated without mistakes or damage. If the checkpoint mechanisms detect problems with the DNA, the cell cycle is halted and the cell attempts to either complete DNA replication or repair the damaged DNA. If the DNA has been correctly replicated, cyclin dependent kinases (CDKs) signal the beginning of mitotic cell division.

The M Checkpoint

The M checkpoint occurs near the end of the metaphase stage of mitosis. The M checkpoint is also known as the spindle checkpoint because it determines whether all the sister chromatids are correctly attached to the spindle microtubules. Because the separation of the sister chromatids during anaphase is an irreversible step, the cycle will not proceed until the kinetochores of each pair of sister chromatids are firmly anchored to at least two spindle fibers arising from opposite poles of the cell.

10.3C: Regulator Molecules of the Cell Cycle

The cell cycle is controlled by regulator molecules that either promote the process or stop it from progressing.

Learning Objectives

- Differentiate among the molecules that regulate the cell cycle

Key Points

- Two groups of proteins, cyclins and cyclin-dependent kinases (Cdks), are responsible for promoting the cell cycle.
- Cyclins regulate the cell cycle only when they are bound to Cdks; to be fully active, the Cdk/cyclin complex must be phosphorylated, which allows it to phosphorylate other proteins that advance the cell cycle.
- Negative regulator molecules (Rb, p53, and p21) act primarily at the G_1 checkpoint and prevent the cell from moving forward to division until damaged DNA is repaired.
- p53 halts the cell cycle and recruits enzymes to repair damaged DNA; if DNA cannot be repaired, p53 triggers apoptosis to prevent duplication.
- Production of p21 is triggered by p53; p21 halts the cycle by binding to and inhibiting the activity of the Cdk/cyclin complex.
- Dephosphorylated Rb binds to E2F, which halts the cell cycle; when the cell grows, Rb is phosphorylated and releases E2F, which advances the cell cycle.

Key Terms

- **cyclin**: any of a group of proteins that regulates the cell cycle by forming a complex with kinases
- **cyclin-dependent kinase**: (CDK) a member of a family of protein kinases first discovered for its role in regulating the cell cycle through phosphorylation
- **retinoblastoma protein**: (Rb) a group of tumor-suppressor proteins that regulates the cell cycle by monitoring cell size

Regulator Molecules of the Cell Cycle

In addition to the internally controlled checkpoints, there are two groups of intracellular molecules that regulate the cell cycle. These regulatory molecules either promote progress of the cell to the next phase (positive regulation) or halt the cycle (negative regulation). Regulator molecules may act individually or they can influence the activity or production of other regulatory proteins. Therefore, the failure of a single regulator may have almost no effect on the cell cycle, especially if more than one mechanism controls the same event. Conversely, the effect of a deficient or non-functioning regulator can be wide-ranging and possibly fatal to the cell if multiple processes are affected.

Positive Regulation of the Cell Cycle

Two groups of proteins, called cyclins and cyclin-dependent kinases (Cdks), are responsible for the progress of the cell through the various checkpoints. The levels of the four cyclin proteins fluctuate throughout the cell cycle in a predictable pattern. Increases in the concentration of cyclin proteins are triggered by both external and internal signals. After the cell moves to the next stage of the cell cycle, the cyclins that were active in the previous stage are degraded.

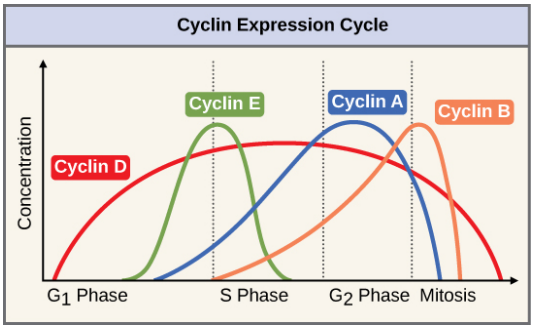

Figure 10.3C.1: **Cyclin Concentrations at Checkpoints**: The concentrations of cyclin proteins change throughout the cell cycle. There is a direct correlation between cyclin accumulation and the three major cell cycle checkpoints. Also, note the sharp decline of cyclin levels following each checkpoint (the transition between phases of the cell cycle) as cyclin is degraded by cytoplasmic enzymes.

Cyclins regulate the cell cycle only when they are tightly bound to Cdks. To be fully active, the Cdk/cyclin complex must also be phosphorylated in specific locations. Like all kinases, Cdks are enzymes (kinases) that phosphorylate other proteins. Phosphorylation activates the protein by changing its shape. The proteins phosphorylated by Cdks are involved in advancing the cell to the next phase.. The levels of Cdk proteins are relatively stable throughout the cell cycle; however, the concentrations of cyclin fluctuate and determine when Cdk/cyclin complexes form. The different cyclins and Cdks bind at specific points in the cell cycle and thus regulate different checkpoints.

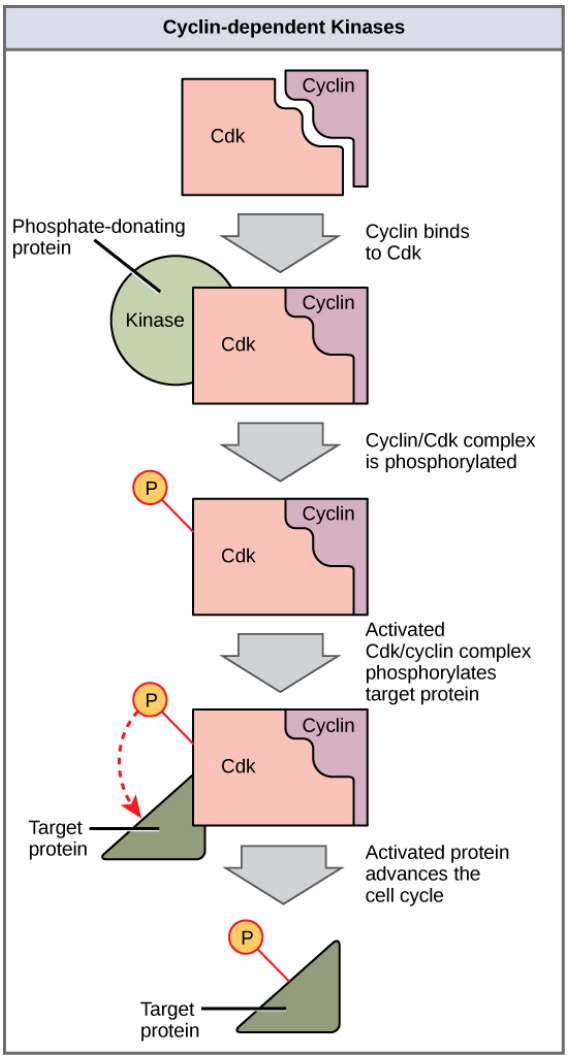

Figure 10.3C.1: **Activation of Cdks**: Cyclin-dependent kinases (Cdks) are protein kinases that, when fully activated, can phosphorylate and activate other proteins that advance the cell cycle past a checkpoint. To become fully activated, a Cdk must bind to a cyclin protein and then be phosphorylated by another kinase.

Although the cyclins are the main regulatory molecules that determine the forward momentum of the cell cycle, there are several other mechanisms that fine tune the progress of the cycle with negative, rather than positive, effects. These mechanisms essentially block the progression of the cell cycle until problematic conditions are resolved. Molecules that prevent the full activation of Cdks are called Cdk inhibitors. Many of these inhibitor molecules directly or indirectly monitor a particular cell cycle event. The block placed on Cdks by inhibitor molecules will not be removed until the specific event being monitored is completed.

Negative Regulation of the Cell Cycle

The second group of cell cycle regulatory molecules are negative regulators. Negative regulators halt the cell cycle. Remember that in positive regulation, active molecules cause the cycle to progress.

The best understood negative regulatory molecules are retinoblastoma protein (Rb), p53, and p21. Retinoblastoma proteins are a group of tumor-suppressor proteins common in many cells. Much of what is known about cell cycle regulation comes from research conducted with cells that have lost regulatory control. All three of these regulatory proteins were discovered to be damaged or non-functional in cells that had begun to replicate uncontrollably (became cancerous). In each case, the main cause of the unchecked progress through the cell cycle was a faulty copy of the regulatory protein.

Rb, p53, and p21 act primarily at the G_1 checkpoint. p53 is a multi-functional protein that has a major impact on the cell's commitment to division; it acts when there is damaged DNA in cells that are undergoing the preparatory processes during

damaged DNA is detected, p53 halts the cell cycle and recruits enzymes to repair the DNA. If the DNA cannot be d, p53 can trigger apoptosis (cell suicide) to prevent the duplication of damaged chromosomes. As p53 levels rise, the production of p21 is triggered. p21 enforces the halt in the cycle dictated by p53 by binding to and inhibiting the activity of the Cdk/cyclin complexes. As a cell is exposed to more stress, higher levels of p53 and p21 accumulate, making it less likely that the cell will move into the S phase.

Rb exerts its regulatory influence on other positive regulator proteins. Rb monitors cell size. In the active, dephosphorylated state, Rb binds to proteins called transcription factors, most commonly to E2F. Transcription factors "turn on" specific genes, allowing the production of proteins encoded by that gene. When Rb is bound to E2F, production of proteins necessary for the G_1/S transition is blocked. As the cell increases in size, Rb is slowly phosphorylated until it becomes inactivated. Rb releases E2F, which can now turn on the gene that produces the transition protein and this particular block is removed. For the cell to move past each of the checkpoints, all positive regulators must be "turned on" and all negative regulators must be "turned off."

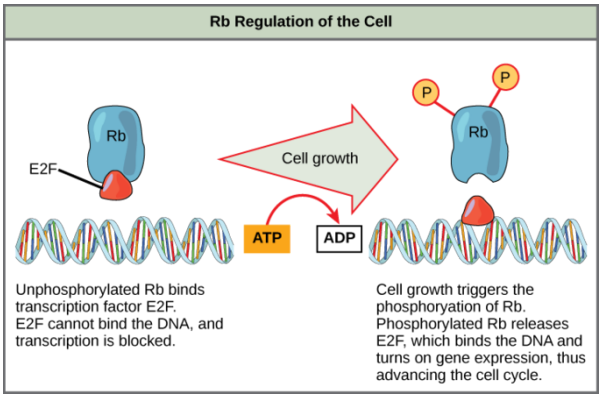

Figure 10.3C.1: **Function of the Rb Regulator Molecule**: Rb halts the cell cycle by binding E2F. Rb releases its hold on E2F in response to cell growth to advance the cell cycle.

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Topic hierarchy

10.4A: Proto-oncogenes

10.4B: Tumor Suppressor Genes

10.4A: Proto-oncogenes

Proto-oncogenes normally regulate cell division, but can be changed into oncogenes through mutation, which may cause cancers to form.

Learning Objectives

- Explain regulation of the cell cycle by proto-oncogenes

Key Points

- Proto- oncogenes positively regulate the cell cycle.
- Mutations may cause proto-oncogenes to become oncogenes, disrupting normal cell division and causing cancers to form.
- Some mutations prevent the cell from reproducing, which keeps the mutations from being passed on.
- If a mutated cell is able to reproduce because the cell division regulators are damaged, then the mutation will be passed on, possibly accumulating more mutations with successive divisions.

Key Terms

- **proto-oncogene**: a gene that promotes the specialization and division of normal cells that becomes an oncogene following mutation
- **mutation**: any heritable change of the base-pair sequence of genetic material
- **oncogene**: any gene that contributes to the conversion of a normal cell into a cancerous cell when mutated or expressed at high levels

Proto-oncogenes

The genes that code for the positive cell cycle regulators are called proto-oncogenes. Proto-oncogenes are normal genes that, when mutated in certain ways, become oncogenes: genes that cause a cell to become cancerous. There are several ways by which a proto-oncogene can be converted into an oncogene. Consider what might happen to the cell cycle in a cell with a recently-acquired oncogene. In most instances, the alteration of the DNA sequence will result in a less functional (or non-functional) protein. The result is detrimental to the cell and will likely prevent the cell from completing the cell cycle; however, the organism is not harmed because the mutation will not be carried forward. If a cell cannot reproduce, the mutation is not propagated and the damage is minimal.

Occasionally, however, a gene mutation causes a change that increases the activity of a positive regulator. For example, a mutation that allows the Cdk gene to be activated without being partnered with cyclin could push the cell cycle past a checkpoint before all of the required conditions are met. If the resulting daughter cells are too damaged to undergo further cell divisions, the mutation would not be propagated and no harm would come to the organism. However, if the atypical daughter cells are able to undergo further cell divisions, subsequent generations of cells will probably accumulate even more mutations, some possibly in additional genes that regulate the cell cycle.

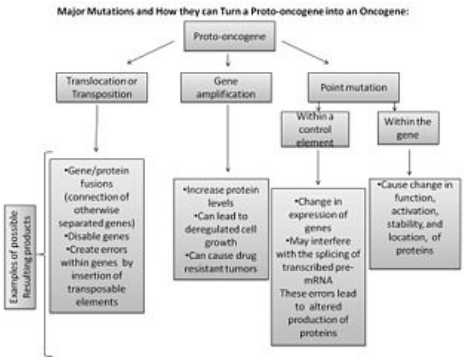

Figure 10.4A.1: **Proto-oncogene Conversion to Oncogene**: Examples of ways to convert proto-oncogenes into cancer-causing genes (oncogenes).

The Cdk gene in the above example is only one of many genes that are considered proto-oncogenes. In addition to the cell cycle regulatory proteins, any protein that influences the cycle can be altered in such a way as to override cell cycle checkpoints. An oncogene is any gene that, when altered, leads to an increase in the rate of cell cycle progression.

10.4B: Tumor Suppressor Genes

Tumor-suppressor genes keep regulatory mechanisms of cell division under control and prevent abnormal cell growth.

Learning Objectives

- Describe the role played by tumor suppressor genes in the cell cycle

Key Points

- Tumor suppressor genes are segments of DNA that code for negative regulator proteins, which keep the cell from undergoing uncontrolled division.
- Mutated p53 genes are believed to be responsible for causing tumor growth because they turn off the regulatory mechanisms that keep cells from dividing out of control.
- Sometimes cells with negative regulators can halt their transmission by inducing pre-programmed cell death called apoptosis.
- Without a fully functional p53, the G1 checkpoint of interphase is severely compromised and the cell proceeds directly from G1 to S; this creates two daughter cells that have inherited the mutated p53 gene.

Key Terms

- **apoptosis**: a process of programmed cell death

Tumor Suppressor Genes

Like proto- oncogenes, many of the negative cell cycle regulatory proteins were discovered in cells that had become cancerous. Tumor suppressor genes are segments of DNA that code for negative regulator proteins: the type of regulators that, when activated, can prevent the cell from undergoing uncontrolled division. The collective function of the best-understood tumor suppressor gene proteins, Rb, p53, and p21, is to put up a roadblock to cell cycle progression until certain events are completed. A cell that carries a mutated form of a negative regulator might not be able to halt the cell cycle if there is a problem. Tumor suppressors are similar to brakes in a vehicle: malfunctioning brakes can contribute to a car crash.

Mutated p53 genes have been identified in more than one-half of all human tumor cells. This discovery is not surprising in light of the multiple roles that the p53 protein plays at the G_1 checkpoint. A cell with a faulty p53 may fail to detect errors present in the genomic DNA. Even if a partially-functional p53 does identify the mutations, it may no longer be able to signal the necessary DNA repair enzymes. Either way, damaged DNA will remain uncorrected. At this point, a functional p53 will deem the cell unsalvageable and trigger programmed cell death (apoptosis). The damaged version of p53 found in cancer cells, however, cannot trigger apoptosis.

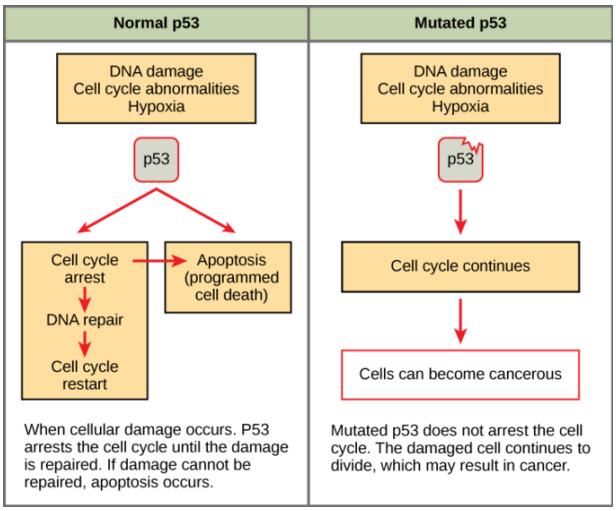

Figure 10.4B.1: **Function of Normal and Mutated p53 genes**: The role of normal p53 is to monitor DNA and the supply of oxygen (hypoxia is a condition of reduced oxygen supply). If damage is detected, p53 triggers repair mechanisms. If repairs are unsuccessful, p53 signals apoptosis. A cell with an abnormal p53 protein cannot repair damaged DNA and cannot signal apoptosis. Cells with abnormal p53 can become cancerous.

The loss of p53 function has other repercussions for the cell cycle. Mutated p53 might lose its ability to trigger p21 production. Without adequate levels of p21, there is no effective block on Cdk activation. Essentially, without a fully functional p53, the G_1 checkpoint is severely compromised and the cell proceeds directly from G_1 to S regardless of internal and external conditions. At the completion of this shortened cell cycle, two daughter cells are produced that have inherited the mutated p53 gene. Given the non-optimal conditions under which the parent cell reproduced, it is likely that the daughter cells will have acquired other mutations in addition to the faulty tumor suppressor gene. Cells such as these daughter cells quickly accumulate both oncogenes and non-functional tumor suppressor genes. Again, the result is tumor growth.

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10.5: Prokaryotic Cell Division

Topic hierarchy

- 10.5A: Binary Fission

10.5A: Binary Fission

Binary fission is the method by which prokaryotes produce new individuals that are genetically identical to the parent organism.

Learning Objectives

- Describe the process of binary fission in prokaryotes

Key Points

- In bacterial replication, the DNA is attached to the plasma membrane at about the midpoint of the cell.
- The origin, or starting point of bacterial replication, is close to the binding site of the DNA to the plasma membrane.
- Replication of the bacterial DNA is bidirectional, which means it moves away from the origin on both strands simultaneously.
- The formation of the FtsZ ring, a ring composed of repeating units of protein, triggers the accumulation of other proteins that work together to acquire and bring new membrane and cell wall materials to the site.
- When new cell walls are in place, due to the formation of a septum, the daughter cells separate to form individual cells.

Key Terms

- **mitotic spindle**: the apparatus that orchestrates the movement of DNA during mitosis
- **karyokinesis**: (mitosis) the first portion of mitotic phase where division of the cell nucleus takes place
- **binary fission**: the process whereby a cell divides asexually to produce two daughter cells

Binary Fission

Prokaryotes, such as bacteria, propagate by binary fission. For unicellular organisms, cell division is the only method used to produce new individuals. In both prokaryotic and eukaryotic cells, the outcome of cell reproduction is a pair of daughter cells that are genetically identical to the parent cell. In unicellular organisms, daughter cells are individuals.

Due to the relative simplicity of the prokaryotes, the cell division process, or binary fission, is a less complicated and much more rapid process than cell division in eukaryotes. The single, circular DNA chromosome of bacteria is not enclosed in a nucleus, but instead occupies a specific location, the nucleoid, within the cell. Although the DNA of the nucleoid is associated with proteins that aid in packaging the molecule into a compact size, there are no histone proteins and thus, no nucleosomes in prokaryotes. The packing proteins of bacteria are, however, related to the cohesin and condensin proteins involved in the chromosome compaction of eukaryotes.

The bacterial chromosome is attached to the plasma membrane at about the midpoint of the cell. The starting point of replication, the origin, is close to the binding site of the chromosome at the plasma membrane. Replication of the DNA is bidirectional, moving away from the origin on both strands of the loop simultaneously. As the new double strands are formed, each origin point moves away from the cell wall attachment toward the opposite ends of the cell. As the cell elongates, the growing membrane aids in the transport of the chromosomes. After the chromosomes have cleared the midpoint of the elongated cell, cytoplasmic separation begins. The formation of a ring composed of repeating units of a protein, FtsZ, directs the partition between the nucleoids. Formation of the FtsZ ring triggers the accumulation of other proteins that work together to recruit new membrane and cell wall materials to the site. A septum is formed between the nucleoids, extending gradually from the periphery toward the center of the cell. When the new cell walls are in place, the daughter cells separate.

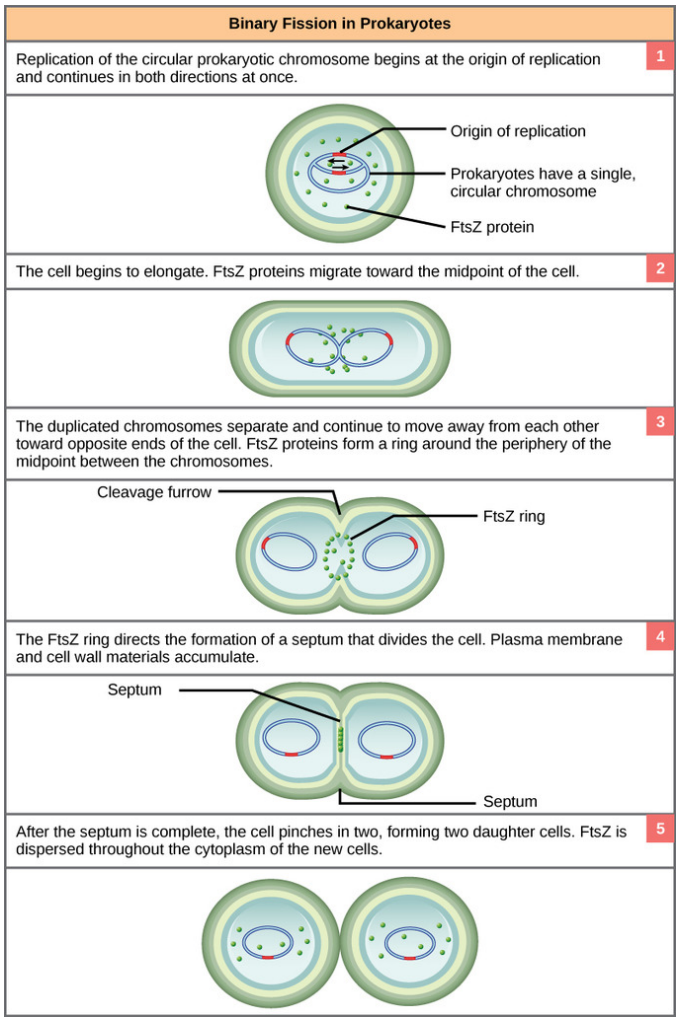

Figure 10.5.A.1: **Binary Fission**: These images show the steps of binary fission in prokaryotes.

Mitotic Spindle Apparatus

The precise timing and formation of the mitotic spindle is critical to the success of eukaryotic cell division. Prokaryotic cells, on the other hand, do not undergo karyokinesis and, therefore, have no need for a mitotic spindle. However, the FtsZ protein that plays such a vital role in prokaryotic cytokinesis is structurally and functionally very similar to tubulin, the building block of the microtubules that make up the mitotic spindle fibers that are necessary for eukaryotes. FtsZ proteins can form filaments, rings, and other three-dimensional structures that resemble the way tubulin forms microtubules, centrioles, and various cytoskeletal components. In addition, both FtsZ and tubulin employ the same energy source, GTP (guanosine triphosphate), to rapidly assemble and disassemble complex structures.

FtsZ and tubulin are homologous structures derived from common evolutionary origins. In this example, FtsZ is the ancestor protein to tubulin (a modern protein). While both proteins are found in extant organisms, tubulin function has evolved and diversified tremendously since evolving from its FtsZ prokaryotic origin. A survey of mitotic assembly components found in present-day unicellular eukaryotes reveals crucial intermediary steps to the complex membrane-enclosed genomes of multicellular eukaryotes.

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CHAPTER OVERVIEW

11: MEIOSIS AND SEXUAL REPRODUCTION

11.1: THE PROCESS OF MEIOSIS

11.1A: INTRODUCTION TO MEIOSIS
11.1B: MEIOSIS I
11.1C: MEIOSIS II
11.1D: COMPARING MEIOSIS AND MITOSIS
11.2: SEXUAL REPRODUCTION

11.2A: ADVANTAGES AND DISADVANTAGES OF SEXUAL REPRODUCTION
11.2B: LIFE CYCLES OF SEXUALLY REPRODUCING ORGANISMS

11.1: The Process of Meiosis

Topic hierarchy

- 11.1A: Introduction to Meiosis
- 11.1B: Meiosis I
- 11.1C: Meiosis II
- 11.1D: Comparing Meiosis and Mitosis

11.1A: Introduction to Meiosis

Meiosis is the nuclear division of diploid cells into haploid cells, which is a necessary step in sexual reproduction.

Learning Objectives

- Describe the importance of meiosis in sexual reproduction

Key Points

- Sexual reproduction is the production of haploid cells and the fusion of two of those cells to form a diploid cell.
- Before sexual reproduction can occur, the number of chromosomes in a diploid cell must decrease by half.
- Meiosis produces cells with half the number of chromosomes as the original cell.
- Haploid cells used in sexual reproduction, gametes, are formed during meiosis, which consists of one round of chromosome replication and two rounds of nuclear division.
- Meiosis I is the first round of meiotic division, while meiosis II is the second round.

Key Terms

- **haploid**: of a cell having a single set of unpaired chromosomes
- **gamete**: a reproductive cell, male (sperm) or female (egg), that has only half the usual number of chromosomes
- **diploid**: of a cell, having a pair of each type of chromosome, one of the pair being derived from the ovum and the other from the spermatozoon

Introduction: Meiosis and Sexual Reproduction

The ability to reproduce *in kind* is a basic characteristic of all living things. In kind means that the offspring of any organism closely resemble their parent or parents. Sexual reproduction requires fertilization: the union of two cells from two individual organisms. Haploid cells contain one set of chromosomes. Cells containing two sets of chromosomes are called diploid. The number of sets of chromosomes in a cell is called its ploidy level. If the reproductive cycle is to continue, then the diploid cell must somehow reduce its number of chromosome sets before fertilization can occur again or there will be a continual doubling in the number of chromosome sets in every generation. Therefore, sexual reproduction includes a nuclear division that reduces the number of chromosome sets.

Figure 11.1A.1: **Offspring Closely Resemble Their Parents**: *In kind* means that the offspring of any organism closely resemble their parent or parents. The hippopotamus gives birth to hippopotamus calves (a). Joshua trees produce seeds from which Joshua tree seedlings emerge (b). Adult flamingos lay eggs that hatch into flamingo chicks (c).

Sexual reproduction is the production of haploid cells (gametes) and the fusion (fertilization) of two gametes to form a single, unique diploid cell called a zygote. All animals and most plants produce these gametes, or eggs and sperm. In most plants and animals, through tens of rounds of mitotic cell division, this diploid cell will develop into an adult organism.

Haploid cells that are part of the sexual reproductive cycle are produced by a type of cell division called meiosis. Meiosis employs many of the same mechanisms as mitosis. However, the starting nucleus is always diploid and the nuclei that result at the end of a meiotic cell division are haploid, so the resulting cells have half the chromosomes as the original. To achieve this reduction in chromosomes, meiosis consists of one round of chromosome duplication and two rounds of nuclear division. Because the events that occur during each of the division stages are analogous to the events of mitosis, the same stage names are assigned. However, because there are two rounds of division, the major process and the stages are designated with a “I” or a “II.” Thus, meiosis I is the first round of meiotic division and consists of prophase I,

prometaphase I, and so on. Meiosis II, the second round of meiotic division, includes prophase II, prometaphase II, and so on.

11.1B: Meiosis I

In meiosis I, the first round of meiosis, homologous chromosomes exchange DNA and the diploid cell is divided into two haploid cells.

Learning Objectives

- Describe the stages and results of meiosis I

Key Points

- Meiosis is preceded by interphase which consists of the G_1 phase (growth), the S phase (DNA replication), and the G_2 phase.
- During prophase I, the homologous chromosomes condense and become visible as the x shape we know, pair up to form a tetrad, and exchange genetic material by crossing over.
- During prometaphase I, microtubules attach at the chromosomes' kinetochores and the nuclear envelope breaks down.
- In metaphase I, the tetrads line themselves up at the metaphase plate and homologous pairs orient themselves randomly.
- In anaphase I, centromeres break down and homologous chromosomes separate.
- In telophase I, chromosomes move to opposite poles; during cytokinesis the cell separates into two haploid cells.

Key Terms

- **crossing over**: the exchange of genetic material between homologous chromosomes that results in recombinant chromosomes
- **tetrad**: two pairs of sister chromatids (a dyad pair) aligned in a certain way and often on the equatorial plane during the meiosis process
- **chromatid**: either of the two strands of a chromosome that separate during meiosis

Meiosis I

Meiosis is preceded by an interphase consisting of three stages. The G_1 phase (also called the first gap phase) initiates this stage and is focused on cell growth. The S phase is next, during which the DNA of the chromosomes is replicated. This replication produces two identical copies, called sister chromatids, that are held together at the centromere by cohesin proteins. The centrosomes, which are the structures that organize the microtubules of the meiotic spindle, also replicate. Finally, during the G_2 phase (also called the second gap phase), the cell undergoes the final preparations for meiosis.

Prophase I

During prophase I, chromosomes condense and become visible inside the nucleus. As the nuclear envelope begins to break down, homologous chromosomes move closer together. The synaptonemal complex, a lattice of proteins between the homologous chromosomes, forms at specific locations, spreading to cover the entire length of the chromosomes. The tight pairing of the homologous chromosomes is called synapsis. In synapsis, the genes on the chromatids of the homologous chromosomes are aligned with each other. The synaptonemal complex also supports the exchange of chromosomal segments between non-sister homologous chromatids in a process called crossing over. The crossover events are the first source of genetic variation produced by meiosis. A single crossover event between homologous non-sister chromatids leads to an exchange of DNA between chromosomes. Following crossover, the synaptonemal complex breaks down and the cohesin connection between homologous pairs is also removed. At the end of prophase I, the pairs are held together only at the chiasmata; they are called tetrads because the four sister chromatids of each pair of homologous chromosomes are now visible.

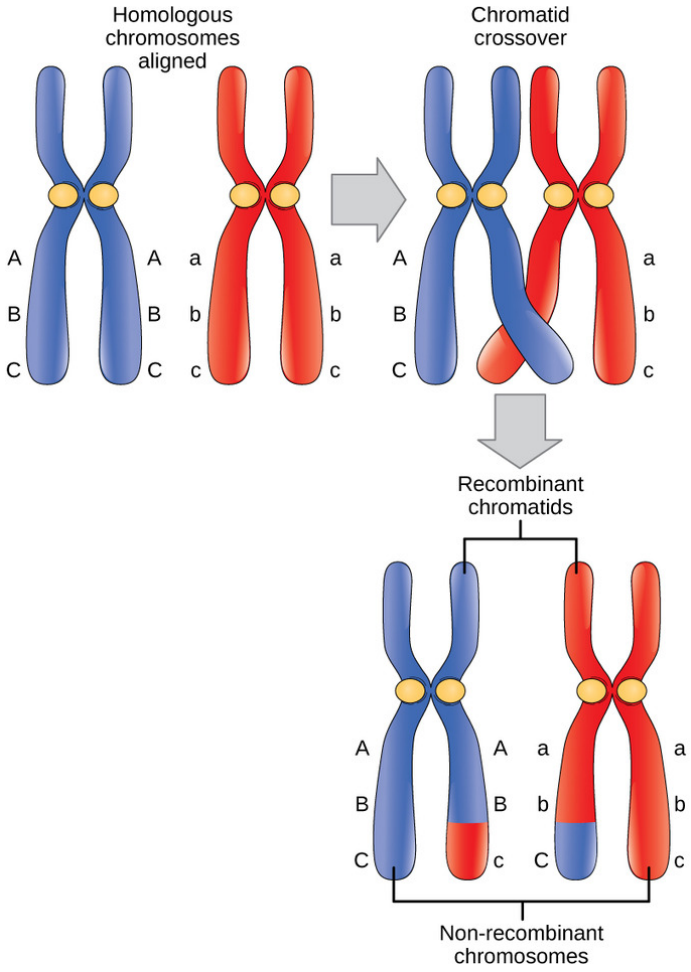


Figure 11.1B.1: **Crossover between homologous chromosomes**: Crossover occurs between non-sister chromatids of homologous chromosomes. The result is an exchange of genetic material between homologous chromosomes.

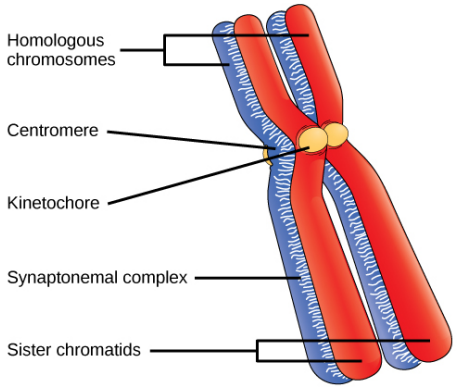


Figure 11.1B.1: **Synapsis holds pairs of homologous chromosomes together**: Early in prophase I, homologous chromosomes come together to form a synapse. The chromosomes are bound tightly together and in perfect alignment by a protein lattice called a synaptonemal complex and by cohesin proteins at the centromere.

Prometaphase I

The key event in prometaphase I is the formation of the spindle fiber apparatus where spindle fiber microtubules attach to the kinetochore proteins at the centromeres. Microtubules grow from centrosomes placed at opposite poles of the cell. The microtubules move toward the middle of the cell and attach to one of the two fused homologous chromosomes at the kinetochores. At the end of prometaphase I, each tetrad is attached to microtubules from both poles, with one homologous chromosome facing each pole. In addition, the nuclear membrane has broken down entirely.

Metaphase I

During metaphase I, the tetrads move to the metaphase plate with kinetochores facing opposite poles. The homologous pairs orient themselves randomly at the equator. This event is the second mechanism that introduces variation into the gametes or spores. In each cell that undergoes meiosis, the arrangement of the tetrads is different. The number of variations is dependent on the number of chromosomes making up a set. There are two possibilities for orientation at the metaphase plate. The possible number of alignments, therefore, equals 2n, where n is the number of chromosomes per set. Given these two mechanisms, it is highly unlikely that any two haploid cells resulting from meiosis will have the same genetic composition.

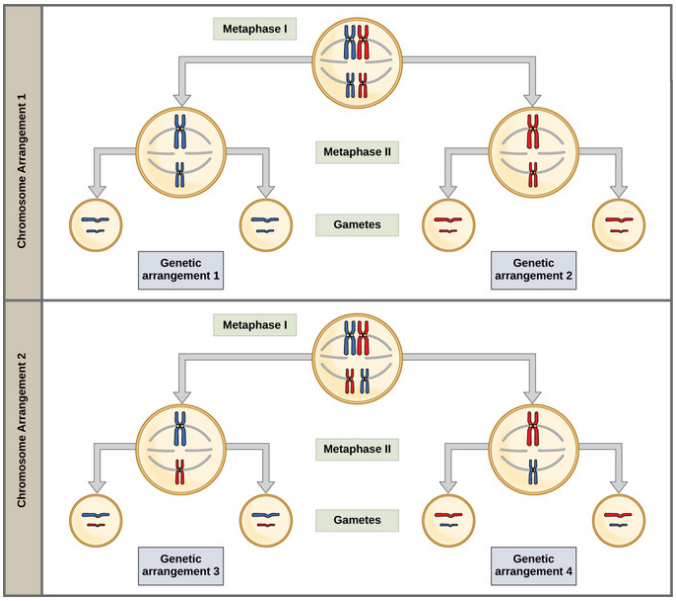


Figure 11.1B.1: **Meiosis I ensures unique gametes**: Random, independent assortment during metaphase I can be demonstrated by considering a cell with a set of two chromosomes (n = 2). In this case, there are two possible arrangements at the equatorial plane in metaphase I. The total possible number of different gametes is 2n, where n equals the number of chromosomes in a set. In this example, there are four possible genetic combinations for the gametes. With n = 23 in human cells, there are over 8 million possible combinations of paternal and maternal chromosomes.

Anaphase I

In anaphase I, the microtubules pull the attached chromosomes apart. The sister chromatids remain tightly bound together at the centromere. The chiasmata are broken in anaphase I as the microtubules attached to the fused kinetochores pull the homologous chromosomes apart.

Telophase I and Cytokinesis

In telophase I, the separated chromosomes arrive at opposite poles. In some organisms, the chromosomes decondense and nuclear envelopes form around the chromatids in telophase I. Then cytokinesis, the physical separation of the cytoplasmic components into two daughter cells, occurs without reformation of the nuclei. In nearly all species of animals and some fungi, cytokinesis separates the cell contents via a cleavage furrow (constriction of the actin ring that leads to cytoplasmic division). In plants, a cell plate is formed during cell cytokinesis by Golgi vesicles fusing at the metaphase plate. This cell plate will ultimately lead to the formation of cell walls that separate the two daughter cells.

Two haploid cells are the end result of the first meiotic division. The cells are haploid because at each pole there is just one of each pair of the homologous chromosomes. Therefore, only one full set of the chromosomes is present. Although there is only one chromosome set, each homolog still consists of two sister chromatids.

11.1C: Meiosis II

During meiosis II, the sister chromatids within the two daughter cells separate, forming four new haploid gametes.

LEARNING OBJECTIVES

Describe the stages and results of Meiosis II

KEY TAKEAWAYS

Key Points

- During prophase II, chromsomes condense again, centrosomes that were duplicated during interphase I move away from each other toward opposite poles, and new spindles are formed.
- During prometaphase II, the nuclear envelopes are completely broken down, and each sister chromatid forms an individual kinetochore that attaches to microtubules from opposite poles.
- During metaphase II, sister chromatids are condensed and aligned at the equator of the cell.
- During anaphase II sister chromatids are pulled apart by the kinetochore microtubules and move toward opposite poles.
- During telophase II and cytokinesis, chromosomes arrive at opposite poles and begin to decondense; the two cells divide into four unique haploid cells.

Key Terms

- **meiosis II**: the second part of the meiotic process; the end result is production of four haploid cells from the two haploid cells produced in meiosis I

Meiosis II

Meiosis II initiates immediately after cytokinesis, usually before the chromosomes have fully decondensed. In contrast to meiosis I, meiosis II resembles a normal mitosis. In some species, cells enter a brief interphase, or interkinesis, before entering meiosis II. Interkinesis lacks an S phase, so chromosomes are not duplicated. The two cells produced in meiosis I go through the events of meiosis II together. During meiosis II, the sister chromatids within the two daughter cells separate, forming four new haploid gametes. The mechanics of meiosis II is similar to mitosis, except that each dividing cell has only one set of homologous chromosomes.

Prophase II

If the chromosomes decondensed in telophase I, they condense again. If nuclear envelopes were formed, they fragment into vesicles. The centrosomes that were duplicated during interphase I move away from each other toward opposite poles and new spindles are formed.

Prometaphase II

The nuclear envelopes are completely broken down and the spindle is fully formed. Each sister chromatid forms an individual kinetochore that attaches to microtubules from opposite poles.

Metaphase II

The sister chromatids are maximally condensed and aligned at the equator of the cell.

Anaphase II

The sister chromatids are pulled apart by the kinetochore microtubules and move toward opposite poles. Non-kinetochore microtubules elongate the cell.

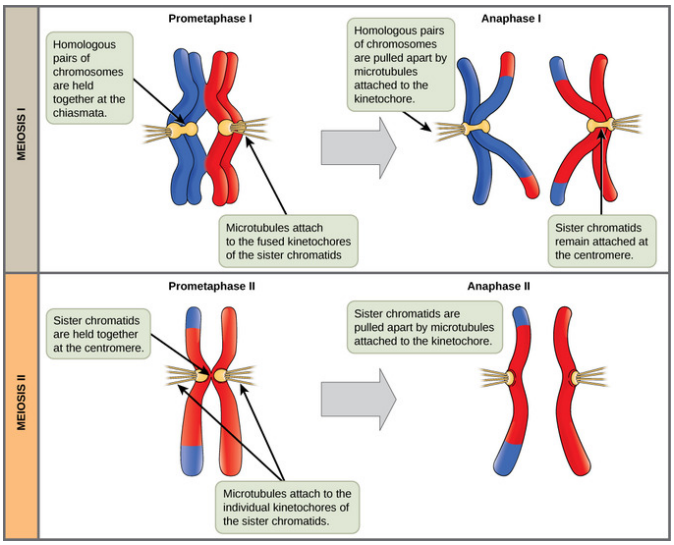


Figure 11.1C.1: **Meiosis I vs. Meiosis II**: The process of chromosome alignment differs between meiosis I and meiosis II. In prometaphase I, microtubules attach to the fused kinetochores of homologous chromosomes, and the homologous chromosomes are arranged at the midpoint of the cell in metaphase I. In anaphase I, the homologous chromosomes are separated. In prometaphase II, microtubules attach to the kinetochores of sister chromatids, and the sister chromatids are arranged at the midpoint of the cells in metaphase II. In anaphase II, the sister chromatids are separated.

Telophase II and Cytokinesis

The chromosomes arrive at opposite poles and begin to decondense. Nuclear envelopes form around the chromosomes. Cytokinesis separates the two cells into four unique haploid cells. At this point, the newly-formed nuclei are both haploid. The cells produced are genetically unique because of the random assortment of paternal and maternal homologs and because of the recombining of maternal and paternal segments of chromosomes (with their sets of genes) that occurs during crossover.

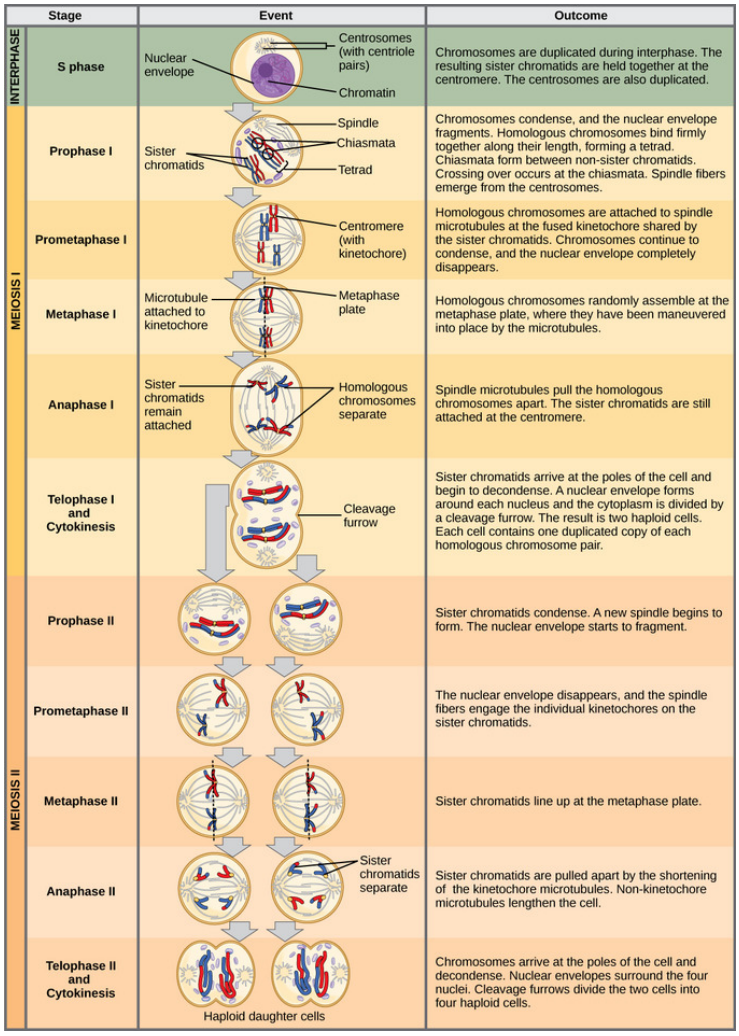


Figure 11.1C.1: **Complete Stages of Meiosis**: An animal cell with a diploid number of four (2n = 4) proceeds through the stages of meiosis to form four haploid daughter cells.

11.1D: Comparing Meiosis and Mitosis

Mitosis and meiosis share some similarities, but also some differences, most of which are observed during meiosis I.

Learning Objectives

- Compare and contrast mitosis and meiosis

Key Points

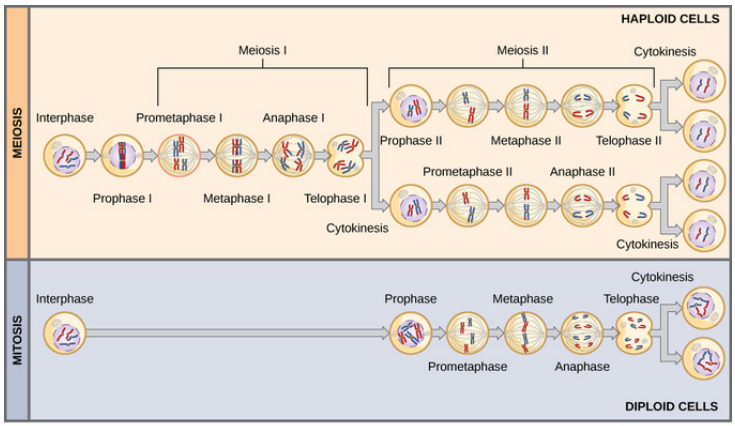
- For the most part, in mitosis, diploid cells are partitioned into two new diploid cells, while in meiosis, diploid cells are partitioned into four new haploid cells.
- In mitosis, the daughter cells have the same number of chromosomes as the parent cell, while in meiosis, the daughter cells have half the number of chromosomes as the parent.
- The daughter cells produced by mitosis are identical, whereas the daughter cells produced by meiosis are different because crossing over has occurred.
- The events that occur in meiosis but not mitosis include homologous chromosomes pairing up, crossing over, and lining up along the metaphase plate in tetrads.
- Meiosis II and mitosis are not reduction division like meiosis I because the number of chromosomes remains the same; therefore, meiosis II is referred to as equatorial division.
- When the homologous chromosomes separate and move to opposite poles during meiosis I, the ploidy level is reduced from two to one, which is referred to as a reduction division.

Key Terms

- **reduction division**: the first of the two divisions of meiosis, a type of cell division
- **ploidy**: the number of homologous sets of chromosomes in a cell
- **equatorial division**: a process of nuclear division in which each chromosome divides equally such that the number of chromosomes remains the same from parent to daughter cells

Comparing Meiosis and Mitosis

Mitosis and meiosis are both forms of division of the nucleus in eukaryotic cells. They share some similarities, but also exhibit distinct differences that lead to very different outcomes. The purpose of mitosis is cell regeneration, growth, and asexual reproduction,while the purpose of meiosis is the production of gametes for sexual reproduction. Mitosis is a single nuclear division that results in two nuclei that are usually partitioned into two new daughter cells. The nuclei resulting from a mitotic division are genetically identical to the original nucleus. They have the same number of sets of chromosomes, one set in the case of haploid cells and two sets in the case of diploid cells. In most plants and all animal species, it is typically diploid cells that undergo mitosis to form new diploid cells. In contrast, meiosis consists of two nuclear divisions resulting in four nuclei that are usually partitioned into four new haploid daughter cells. The nuclei resulting from meiosis are not genetically identical and they contain one chromosome set only. This is half the number of chromosome sets in the original cell, which is diploid.



						OUTCOME
PROCESS	DNA synthesis	Synapsis of homologous chromosomes	Crossover	Homologous chromosomes line up at metaphase plate	Sister chromatids line up at metaphase plate	Number and genetic composition of daughter cells
MEIOSIS	Occurs in S phase of interphase	During prophase I	During prophase I	During metaphase I	During metaphase II	Four haploid cells at the end of meiosis II
MITOSIS	Occurs in S phase of interphase	Does not occur in mitosis	Does not occur in mitosis	Does not occur in mitosis	During metaphase	Two diploid cells at the end of mitosis

Figure 11.1D.1: **Comparing Meiosis and Mitosis**: Meiosis and mitosis are both preceded by one round of DNA replication; however, meiosis includes two nuclear divisions. The four daughter cells resulting from meiosis are haploid and genetically distinct. The daughter cells resulting from mitosis are diploid and identical to the parent cell.

The main differences between mitosis and meiosis occur in meiosis I. In meiosis I, the homologous chromosome pairs become associated with each other and are bound together with the synaptonemal complex. Chiasmata develop and crossover occurs between homologous chromosomes, which then line up along the metaphase plate in tetrads with kinetochore fibers from opposite spindle poles attached to each kinetochore of a homolog in a tetrad. All of these events occur only in meiosis I.

When the tetrad is broken up and the homologous chromosomes move to opposite poles, the ploidy level is reduced from two to one. For this reason, meiosis I is referred to as a reduction division. There is no such reduction in ploidy level during mitosis.

Meiosis II is much more similar to a mitotic division. In this case, the duplicated chromosomes (only one set, as the homologous pairs have now been separated into two different cells) line up on the metaphase plate with divided kinetochores attached to kinetochore fibers from opposite poles. During anaphase II and mitotic anaphase, the kinetochores divide and sister chromatids, now referred to as chromosomes, are pulled to opposite poles. The two daughter cells of mitosis, however, are identical, unlike the daughter cells produced by meiosis. They are different because there has been at least one crossover per chromosome. Meiosis II is not a reduction division because, although there are fewer copies of the genome in the resulting cells, there is still one set of chromosomes, as there was at the end of meiosis I. Meiosis II is, therefore, referred to as equatorial division.

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: Sexual Reproduction

Topic hierarchy

- 11.2A: Advantages and Disadvantages of Sexual Reproduction
- 11.2B: Life Cycles of Sexually Reproducing Organisms

11.2A: Advantages and Disadvantages of Sexual Reproduction

The genetic diversity of sexual reproduction, observed in most eukaryotes, is thought to give species better chances of survival.

Learning Objectives

- Describe the benefits of sexual reproduction

Key Points

- The variation that sexual reproduction creates among offspring is very important to the survival and reproduction of the population.
- In sexual reproduction, different mutations are continually reshuffled from one generation to the next when different parents combine their unique genomes; this results in an increase of genetic diversity.
- On average, a sexually-reproducing population will leave more offspring than an otherwise similar asexually-reproducing population.

Key Terms

- **sexual reproduction**: Sexual reproduction is the creation of a new organism by combining the genetic material of two organisms. There are two main processes during sexual reproduction: meiosis, involving the halving of the number of chromosomes, and fertilization, involving the fusion of two gametes and the restoration of the original number of chromosomes.
- **asexual reproduction**: any form of reproduction that involves neither meiosis nor fusion of gametes

An Introduction to Sexual Reproduction

Sexual reproduction was an early evolutionary innovation after the appearance of eukaryotic cells. During sexual reproduction, the genetic material of two individuals is combined to produce genetically-diverse offspring that differ from their parents. The fact that most eukaryotes reproduce sexually is evidence of its evolutionary success. In many animals, it is actually the only mode of reproduction. The genetic diversity of sexually-produced offspring is thought to give species a better chance of surviving in an unpredictable or changing environment.

Scientists recognize some real disadvantages to sexual reproduction. On the surface, creating offspring that are genetic clones of the parent appears to be a better system. If the parent organism is successfully occupying a habitat, offspring with the same traits would be similarly successful. Species that reproduce sexually must maintain two different types of individuals, males and females, which can limit the ability to colonize new habitats as both sexes must be present. Therefore, there is an obvious benefit to an organism that can produce offspring whenever circumstances are favorable by asexual budding, fragmentation, or asexual eggs. These methods of asexual reproduction do not require another organism of the opposite sex. Indeed, some organisms that lead a solitary lifestyle have retained the ability to reproduce asexually. In addition, in asexual populations, every individual is capable of reproduction. In sexual populations, the males are not producing the offspring themselves. In theory, an asexual population could grow twice as fast.

Nevertheless, multicellular organisms that exclusively depend on asexual reproduction are exceedingly rare. Why is sexuality (and meiosis) so common? This is one of the important unanswered questions in biology and has been the focus of much research beginning in the latter half of the twentieth century. There are several possible explanations, one of which is that the variation that sexual reproduction creates among offspring is very important to the survival and reproduction of the population. Thus, on average, a sexually-reproducing population will leave more descendants than an otherwise similar asexually-reproducing population. The only source of variation in asexual organisms is mutation. This is the ultimate source of variation in sexual organisms, but, in addition, those different mutations are continually reshuffled from one generation to the next when different parents combine their unique genomes and the genes are mixed into different combinations by the process of meiosis. Meiosis is the division of the contents of the nucleus, dividing the chromosomes among gametes.

The process of meiosis produces unique reproductive cells called gametes, which have half the number of chromosomes as the parent cell. Fertilization, the fusion of haploid gametes from two individuals, restores the diploid condition. Thus,

sexually-reproducing organisms alternate between haploid and diploid stages. However, the ways in which reproductive cells are produced and the timing between meiosis and fertilization vary greatly. There are three main categories of sexual life cycles: diploid-dominant, demonstrated by most animals; haploid-dominant, demonstrated by all fungi and some algae; and the alternation of generations, demonstrated by plants and some algae.

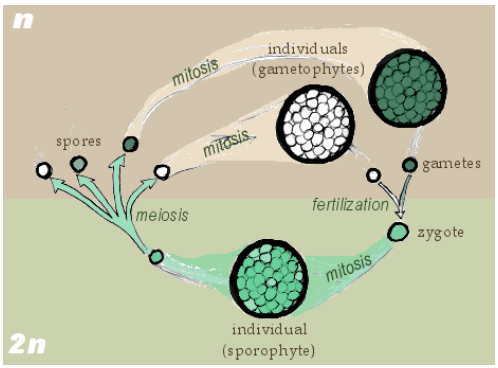

Figure 11.2A.1: **The Sexual Life Cycle**: In animals, sexually-reproducing adults form haploid gametes from diploid germ cells. Fusion of the gametes gives rise to a fertilized egg cell, or zygote. The zygote will undergo multiple rounds of mitosis to produce a multicellular offspring.

The Red Queen Hypothesis

It is not in dispute that sexual reproduction provides evolutionary advantages to organisms that employ this mechanism to produce offspring. But why, even in the face of fairly stable conditions, does sexual reproduction persist when it is more difficult and costly for individual organisms? Variation is the outcome of sexual reproduction, but why are ongoing variations necessary? Possible answers to these questions are explained in the Red Queen hypothesis, first proposed by Leigh Van Valen in 1973.

All species co-evolve with other organisms; for example, predators evolve with their prey and parasites evolve with their hosts. Each tiny advantage gained by favorable variation gives a species an edge over close competitors, predators, parasites, or even prey. The only method that will allow a co-evolving species to maintain its own share of the resources is to also continually improve its fitness. As one species gains an advantage, this increases selection on the other species; they must also develop an advantage or they will be out-competed. No single species progresses too far ahead because genetic variation among the progeny of sexual reproduction provides all species with a mechanism to improve rapidly. Species that cannot keep up become extinct. The Red Queen's catchphrase was, "It takes all the running you can do to stay in the same place." This is an apt description of co-evolution between competing species.

11.2B: Life Cycles of Sexually Reproducing Organisms

The main categories of sexual life cycles in eukaryotic organisms are: diploid-dominant, haploid-dominant, and alternation of generations.

Learning Objectives

- Explain the life cycles in sexual reproduction

Key Points

- In the diploid – dominant cycle, the multicellular diploid stage is the most obvious life stage; the only haploid cells produced by the organism are the gametes.
- Most fungi and algae employ a haploid-dominant life cycle type in which the “body” of the organism is haploid; specialized haploid cells from two individuals join to form a diploid zygote.
- Observed in all plants and some algae, species with alternation of generations have both haploid and diploid multicellular organisms as part of their life cycle.

Key Terms

- **zygote**: a diploid fertilized egg cell
- **gametophyte**: a plant (or the haploid phase in its life cycle) that produces gametes by mitosis in order to produce a zygote
- **sporophyte**: a plant (or the diploid phase in its life cycle) that produces spores by meiosis in order to produce gametophytes

Life Cycles of Sexually Reproducing Organisms

In sexual reproduction, the genetic material of two individuals is combined to produce genetically diverse offspring that differ from their parents. Fertilization and meiosis alternate in sexual life cycles. What happens between these two events depends upon the organism. The process of meiosis, the division of the contents of the nucleus that divides the chromosomes among gametes, reduces the chromosome number by half, while fertilization, the joining of two haploid gametes, restores the diploid condition. There are three main categories of life cycles in eukaryotic organisms: diploid-dominant, haploid-dominant, and alternation of generations.

Diploid-Dominant Life Cycle

In the diploid-dominant life cycle, the multicellular diploid stage is the most obvious life stage, as occurs with most animals, including humans. Nearly all animals employ a diploid-dominant life cycle strategy in which the only haploid cells produced by the organism are the gametes. Early in the development of the embryo, specialized diploid cells, called germ cells, are produced within the gonads (e.g. testes and ovaries). Germ cells are capable of mitosis to perpetuate the cell line and meiosis to produce gametes. Once the haploid gametes are formed, they lose the ability to divide again. There is no multicellular haploid life stage. Fertilization occurs with the fusion of two gametes, usually from different individuals, restoring the diploid state.

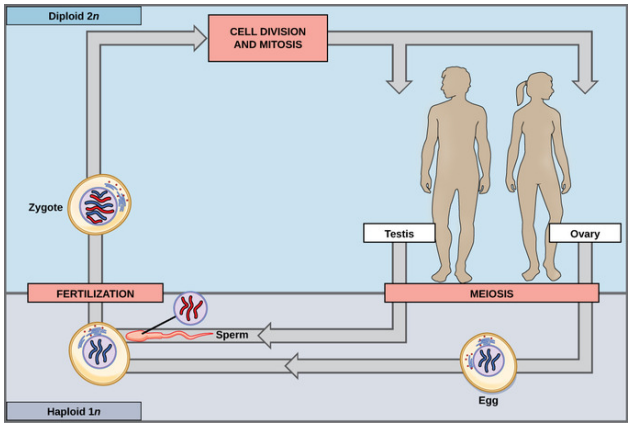

Figure 11.2B.1: **Diploid-Dominant Life Cycle**: In animals, sexually-reproducing adults form haploid gametes from diploid germ cells. Fusion of the gametes gives rise to a fertilized egg cell, or zygote. The zygote will undergo multiple rounds of mitosis to produce a multicellular offspring. The germ cells are generated early in the development of the zygote.

Haploid-Dominant Life Cycle

Within haploid-dominant life cycles, the multicellular haploid stage is the most obvious life stage. Most fungi and algae employ a life cycle type in which the "body" of the organism, the ecologically important part of the life cycle, is haploid. The haploid cells that make up the tissues of the dominant multicellular stage are formed by mitosis. During sexual reproduction, specialized haploid cells from two individuals, designated the (+) and (−) mating types, join to form a diploid zygote. The zygote immediately undergoes meiosis to form four haploid cells called spores. Although haploid like the "parents," these spores contain a new genetic combination from two parents. The spores can remain dormant for various time periods. Eventually, when conditions are conducive, the spores form multicellular haploid structures by many rounds of mitosis.

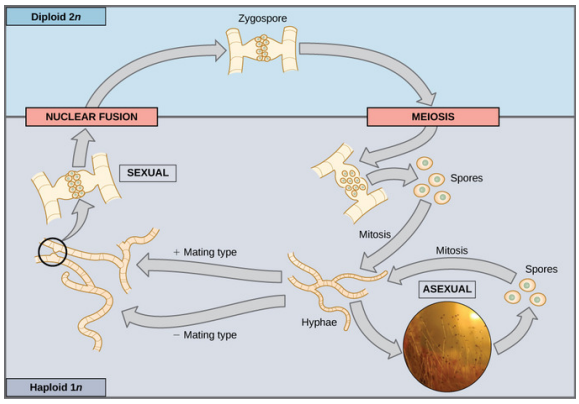

Figure 11.2B.1: **Haploid-Dominant Life Cycle**: Fungi, such as black bread mold (Rhizopus nigricans), have haploid-dominant life cycles. The haploid multicellular stage produces specialized haploid cells by mitosis that fuse to form a diploid zygote. The zygote undergoes meiosis to produce haploid spores. Each spore gives rise to a multicellular haploid organism by mitosis.

Alternation of Generations

The third life-cycle type, employed by some algae and all plants, is a blend of the haploid-dominant and diploid-dominant extremes. Species with alternation of generations have both haploid and diploid multicellular organisms as part of their life cycle. The haploid multicellular plants are called gametophytes because they produce gametes from specialized cells. Meiosis is not directly involved in the production of gametes because the organism that produces the gametes is already a haploid. Fertilization between the gametes forms a diploid zygote. The zygote will undergo many rounds of mitosis and give rise to a diploid multicellular plant called a sporophyte. Specialized cells of the sporophyte will undergo meiosis and produce haploid spores. The spores will subsequently develop into the gametophytes.

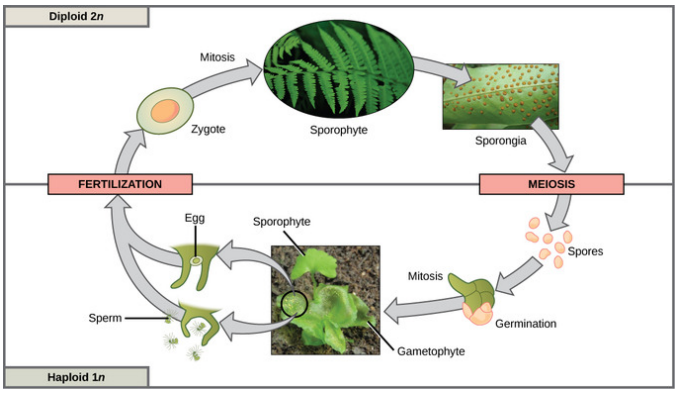

Figure 11.2B.1: **Alternation of Generations**: Plants have a life cycle that alternates between a multicellular haploid organism and a multicellular diploid organism. In some plants, such as ferns, both the haploid and diploid plant stages are free-living. The diploid plant is called a sporophyte because it produces haploid spores by meiosis. The spores develop into multicellular, haploid plants called gametophytes because they produce gametes. The gametes of two individuals will fuse to form a diploid zygote that becomes the sporophyte.

Although all plants utilize some version of the alternation of generations, the relative size of the sporophyte and the gametophyte and the relationship between them vary greatly. In plants such as moss, the gametophyte organism is the free-living plant, while the sporophyte is physically dependent on the gametophyte. In other plants, such as ferns, both the gametophyte and sporophyte plants are free-living; however, the sporophyte is much larger. In seed plants, such as magnolia trees and daisies, the gametophyte is composed of only a few cells and, in the case of the female gametophyte, is completely retained within the sporophyte.

Sexual reproduction takes many forms in multicellular organisms. However, at some point in each type of life cycle, meiosis produces haploid cells that will fuse with the haploid cell of another organism. The mechanisms of variation (crossover, random assortment of homologous chromosomes, and random fertilization) are present in all versions of sexual reproduction. The fact that nearly every multicellular organism on earth employs sexual reproduction is strong evidence for the benefits of producing offspring with unique gene combinations, although there are other possible benefits as well.

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CHAPTER OVERVIEW

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12.1: Mendels Experiments and the Laws of Probability

Topic hierarchy

- 12.1A: Introduction to Mendelian Inheritance
- 12.1B: Mendel’s Model System
- 12.1C: Mendelian Crosses
- 12.1D: Garden Pea Characteristics Revealed the Basics of Heredity
- 12.1E: Rules of Probability for Mendelian Inheritance

12.1A: Introduction to Mendelian Inheritance

While working with pea plants, Gregor Mendel noticed that offspring were similar to their parent plants, which led him to some of the earliest theories about genetics.

Learning Objectives

- Describe the traits of pea plants that were studied by Mendel

Key Points

- Mendel studied seven characteristics of the garden pea plants: flower color, seed texture, seed color, stem length, pod color, pod texture, and flower position to develop his Laws of Inheritance.
- Genetics is the study of genes passed from parents to offspring.
- Genes are the basic fundamental units of heredity.

Key Terms

- **genetics**: The branch of biology that deals with the transmission and variation of inherited characteristics, in particular chromosomes and DNA.

Gregor Mendel and the Study of Genetics

Genetics is the study of heredity, or the passing of traits from parents to offspring. Gregor Johann Mendel set the framework for genetics long before chromosomes or genes had been identified, at a time when meiosis was not well understood. For his work, Mendel is often referred to as the "father of modern genetics. " Mendel selected a simple biological system, garden peas, and conducted methodical, quantitative analyses using large sample sizes.

Figure 12.1*A*.1: **Gregor Mendel**: Gregor Johann Mendel was a German-speaking Moravian scientist and Augustinian friar who gained posthumous fame as the founder of the modern science of genetics.

Mendel entered the Augustinian St. Thomas's Abbey and began his training as a priest. He began studying heredity using mice, but his bishop did not like one of his friars studying animal sex, so he switched to plants. Mendel grew and studied around 29,000 garden pea plants in a monastery's garden, where he analyzed seven characteristics of the garden pea plants: flower color (purple or white), seed texture (wrinkled or round), seed color (yellow or green), stem length (long or short), pod color (yellow or green), pod texture (inflated or constricted), and flower position (axial or terminal). Based on the appearance, or phenotypes, of the seven traits, Mendel developed genotypes for those traits.

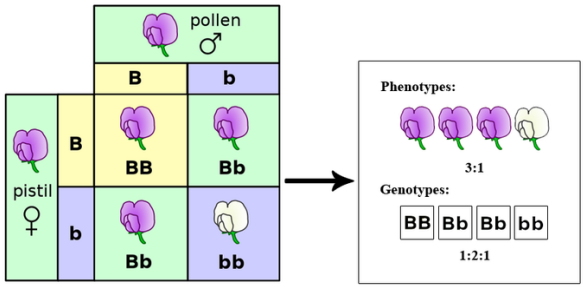

Figure 12.1*A*.1: **Appearance and genetic makeup of garden pea plant flowers**: Based on Mendel's experiments, the genotype of the pea flowers could be determined from the phenotypes of the flowers.

Because of Mendel's work, the fundamental principles of heredity were revealed, which are often referred to as Mendel's Laws of Inheritance. We now know that genes, carried on chromosomes, are the basic functional units of heredity with the capability to be replicated, expressed, or mutated. Today, the postulates put forth by Mendel form the basis of classical, or Mendelian, genetics. Not all genes are transmitted from parents to offspring according to Mendelian genetics, but Mendel's experiments serve as an excellent starting point for thinking about inheritance.

Mendel made all of his observations and findings crossing individual plants. We can now view a human karyotype of all of the chromosomes in an individual to visualize chromosomal abnormalities in offspring, even before birth. Shortly after Mendel proposed that traits were determined by what are now known as genes, other researchers observed that different traits were often inherited together, and thereby deduced that the genes were physically linked by being located on the same chromosome. Mendel's work was the beginning of many of the advances in molecular biology over the years.

B: Mendel's Model System

The garden pea has several advantageous characteristics that allowed Mendel to develop the laws of modern genetics.

Learning Objectives

- Describe the scientific reasons for the success of Mendel's experimental work

Key Points

- Mendel used true-breeding plants in his experiments. These plants, when self-fertilized, always produce offspring with the same phenotype.
- Pea plants are easily manipulated, grow in one season, and can be grown in large quantities; these qualities allowed Mendel to conduct methodical, quantitative analyses using large sample sizes.
- Based on his experiments with the garden peas, Mendel found that one phenotype was always dominant over another recessive phenotype for the same trait.

Key Terms

- **phenotype**: the observable characteristics of an organism, often resulting from its genetic information or a combination of genetic information and environmental factors
- **genotype**: the specific genetic information of a cell or organism, usually a description of the allele or alleles relating to a specific gene.
- **true-breeding plant**: a plant that always produces offspring of the same phenotype when self-fertilized; one that is homozygous for the trait being followed.

Mendel's Model System

Mendel's seminal work was accomplished using the garden pea, *Pisum sativum*, to study inheritance. Pea plant reproduction is easily manipulated; large quantities of garden peas could be cultivated simultaneously, allowing Mendel to conclude that his results did not occur simply by chance. The garden pea also grows to maturity within one season; several generations could be evaluated over a relatively short time.

Pea plants have both male and female parts and can easily be grown in large numbers. For this reason, garden pea plants can either self-pollinate or cross-pollinate with other pea plants. In the absence of outside manipulation, this species naturally self-fertilizes: ova (the eggs) within individual flowers are fertilized by pollen (containing the sperm cell) from the same flower. The sperm and the eggs that produce the next generation of plants both come from the same parent. What's more, the flower petals remain sealed tightly until after pollination, preventing pollination from other plants. The result is highly inbred, or "true-breeding," pea plants. These are plants that always produce offspring that look like the parent. Today, we know that these "true-breeding" plants are homozygous for most traits.

A gardener or researcher, such as Mendel, can cross-pollinate these same plants by manually applying sperm from one plant to the pistil (containing the ova) of another plant. Now the sperm and eggs come from different parent plants. When Mendel cross-pollinated a true-breeding plant that only produced yellow peas with a true-breeding plant that only produced green peas, he found that the first generation of offspring is always all yellow peas. The green pea trait did not show up. However, if this first generation of yellow pea plants were allowed to self-pollinate, the following or second generation had a ratio of 3:1 yellow to green peas.

In this and all the other pea plant traits Mendel followed, one form of the trait was "dominant" over another so it masked the presence of the other "recessive" form in the first generation after cross-breeding two homozygous plants.. Even if the phenotype (visible form) is hidden, the genotype (allele controlling that form of the trait) can be passed on to next generation and produce the recessive form in the second generation. By experimenting with true-breeding pea plants, Mendel avoided the appearance of unexpected (recombinant) traits in offspring that might occur if the plants were not true breeding.

Figure 12.1B.1: **Mendel's Experiments With Peas**: Experimenting with thousands of garden peas, Mendel uncovered the fundamentals of genetics.

12.1C: Mendelian Crosses

Mendel's crosses involved mating two true-breeding organisms that had different traits to produce new generations of pea plants.

Learning Objectives

- Identify Mendelian crosses

Key Points

- Mendel carefully controlled his experiments by removing the anthers from the pea plants before they matured.
- First generation pea plants were called parental generation, P_0, while the following generations were called filial, F_n, where n is the number of generations from P_0.
- The ratio of characteristics in the P_0–F_1–F_2 generations became the basis for Mendel's postulates.

Key Terms

- **filial**: of a generation or generations descending from a specific previous one
- **parental**: of the generation of organisms that produce a hybrid

Mendelian Crosses

Mendel performed crosses, which involved mating two true-breeding individuals that have different traits. In the pea, which is a naturally self-pollinating plant, this is done by manually transferring pollen from the anther of a mature pea plant of one variety to the stigma of a separate mature pea plant of the second variety. In plants, pollen carries the male gametes (sperm) to the stigma, a sticky organ that traps pollen and allows the sperm to move down the pistil to the female gametes (ova) below. To prevent the pea plant that was receiving pollen from self-fertilizing and confounding his results, Mendel painstakingly removed all of the anthers from the plant's flowers before they had a chance to mature.

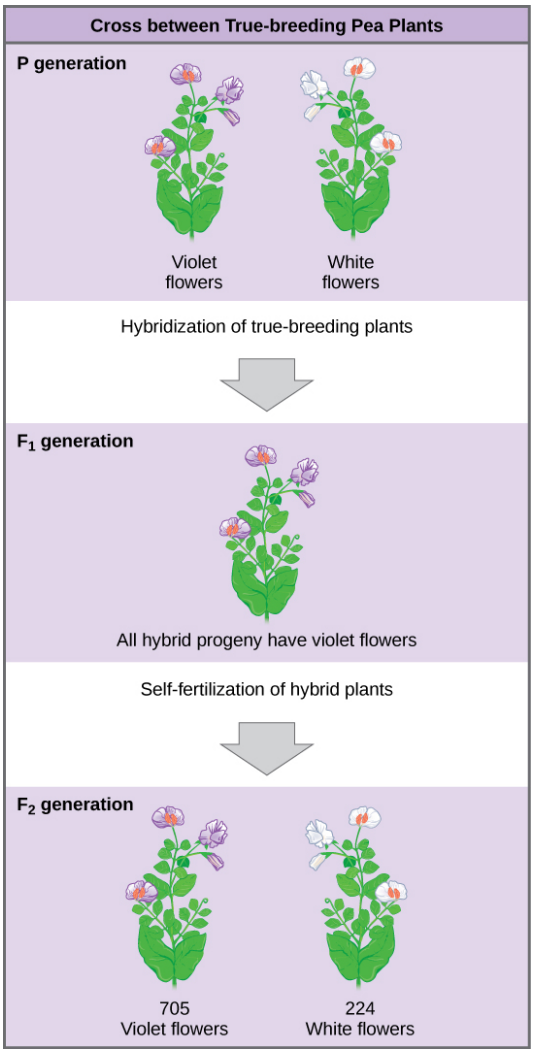

Figure 12.1C.1: **Mendelian Crosses**: In one of his experiments on inheritance patterns, Mendel crossed plants that were true-breeding for violet flower color with plants true-breeding for white flower color (the P generation). The resulting hybrids in the F1 generation all had violet flowers. In the F2 generation, approximately three-quarters of the plants had violet flowers, while one-quarter had white flowers.

Plants used in first-generation crosses were called P_0, or parental generation one, plants. Mendel collected the seeds belonging to the P_0 plants that resulted from each cross and grew them the following season. These offspring were called the F_1, or the first filial (filial = offspring, daughter or son), generation. Once Mendel examined the characteristics in the F_1generation of plants, he allowed them to self-fertilize naturally. He then collected and grew the seeds from the F_1 plants to produce the F_2, or second filial, generation. Mendel's experiments extended beyond the F_2 generation to the F_3 and F_4generations, and so on, but it was the ratio of characteristics in the P_0–F_1–F_2 generations that were the most intriguing and became the basis for Mendel's postulates.

12.1D: Garden Pea Characteristics Revealed the Basics of Heredity

Mendel's experiments with peas revealed the presence of dominant and recessive traits in the filial generations.

Learning Objectives

- Evaluate the results of F1 and F2 generations from Mendelian crosses of peas

Key Points

- Dominant traits are inherited unchanged from one generation to the next.
- Recessive traits disappear in the first filial generation, but reappear in the second filial generation at a ratio of 3:1, dominant:recessive.
- In the F1 generation, Mendel found that one of the two options for each trait had disappeared (all offspring were identical phenotypes), while in the F_2 generation, the trait reappeared in 1/4 of the offspring (a 3:1 ratio).

Key Terms

- **hybrid**: offspring resulting from cross-breeding different entities, e.g. two different species or two purebred parent strains
- **recessive**: able to be covered up by a dominant trait
- **dominant**: a relationship between alleles of a gene, in which one allele masks the expression (phenotype) of another allele at the same locus

Garden Pea Characteristics Revealed the Basics of Heredity

To fully examine each of the seven traits in garden peas, Mendel generated large numbers of F_1 and F_2 plants, reporting results from 19,959 F_2 plants alone. His findings were consistent.

What results did Mendel find in his crosses for flower color? First, Mendel confirmed that he had plants that bred true for white or violet flower color. Regardless of how many generations Mendel examined, all self-crossed offspring of parents with white flowers had white flowers, and all self-crossed offspring of parents with violet flowers had violet flowers. In addition, Mendel confirmed that, other than flower color, the pea plants were physically identical.

Once these validations were complete, Mendel applied the pollen from a plant with violet flowers to the stigma of a plant with white flowers. After gathering and sowing the seeds that resulted from this cross, Mendel found that 100 percent of the F_1hybrid generation had violet flowers. Conventional wisdom at that time would have predicted the hybrid flowers to be pale violet or for hybrid plants to have equal numbers of white and violet flowers. In other words, the contrasting parental traits were expected to blend in the offspring. Instead, Mendel's results demonstrated that the white flower trait in the F_1 generation had completely disappeared.

Importantly, Mendel did not stop his experimentation there. He allowed the F_1 plants to self-fertilize and found that, of F_2-generation plants, 705 had violet flowers and 224 had white flowers. This was a ratio of 3.15 violet flowers per one white flower, or approximately 3:1. When Mendel transferred pollen from a plant with violet flowers to the stigma of a plant with white flowers and vice versa, he obtained about the same ratio regardless of which parent, male or female, contributed which trait. This is called a reciprocal cross: a paired cross in which the respective traits of the male and female in one cross become the respective traits of the female and male in the other cross. For the other six characteristics Mendel examined, the F_1 and F_2generations behaved in the same way as they had for flower color. One of the two traits would disappear completely from the F_1 generation only to reappear in the F_2 generation at a ratio of approximately 3:1.

Upon compiling his results for many thousands of plants, Mendel concluded that the characteristics could be divided into expressed and latent traits. He called these, respectively, dominant and recessive traits. Dominant traits are those that are inherited unchanged in a hybridization. Recessive traits become latent, or disappear, in the offspring of a hybridization. The recessive trait does, however, reappear in the progeny of the hybrid offspring. An example of a dominant trait is the violet-flower trait. For this same characteristic (flower color), white-colored flowers are a recessive trait. The fact that the recessive trait reappeared in the F_2 generation meant that the traits remained separate (not blended) in the plants of the F_1 generation. Mendel also proposed that plants possessed two copies of the trait for the flower-color characteristic and that

each parent transmitted one of its two copies to its offspring, where they came together. Moreover, the physical observation of a dominant trait could mean that the genetic composition of the organism included two dominant versions of the characteristic or that it included one dominant and one recessive version. Conversely, the observation of a recessive trait meant that the organism lacked any dominant versions of this characteristic.

The Results of Mendel's Garden Pea Hybridizations

Characteristic	Contrasting P0 Traits	F1 Offspring Traits	F2 Offspring Traits	F2 Trait Ratios
Flower color	Violet vs. white	100 percent violet	705 violet 224 white	3.15:1
Flower position	Axial vs. terminal	100 percent axial	651 axial 207 terminal	3.14:1
Plant height	Tall vs. dwarf	100 percent tall	787 tall 277 dwarf	2.84:1
Seed texture	Round vs. wrinkled	100 percent round	5,474 round 1,850 wrinkled	2.96:1
Seed color	Yellow vs. green	100 percent yellow	6,022 yellow 2,001 green	3.01:1
Pea pod texture	Inflated vs. constricted	100 percent inflated	882 inflated 299 constricted	2.95:1
Pea pod color	Green vs. yellow	100 percent green	428 green 152 yellow	2.82:1

Figure 12.1D.1: **Results of Mendel's Garden Pea Hybridizations**: Mendel conducted thousands of experiments and found the same ratios of offspring every time, regardless of which trait he examined.

12.1E: Rules of Probability for Mendelian Inheritance

The rules of probability can be applied to Mendelian crosses to determine the expected phenotypes and genotypes of offspring.

Learning Objectives

- Calculate the probability of traits of pea plants using Mendelian crosses

Key Points

- The Product Rule is used to determine the outcome of an event with two independent events; the probability of the event is the product of the probabilities of each individual event.
- The Sum Rule is used to determine the outcome of an event with two mutually exclusive events from multiple pathways; the probability of the event is the sum of the probabilities of each individual event.
- The Product Rule of probability is used to determine the probability of having both dominant traits in the F_2progeny; it is the product of the probabilities of having the dominant trait for each characteristic.
- The Sum Rule of probability is used to determine the probability of having one dominant trait in the F2 generation of a dihybrid cross; it is the sum of the probabilities of each individual with that trait.

Key Terms

- **sum rule**: the probability of the occurrence of one event or the other event, of two mutually exclusive events, is the sum of their individual probabilities
- **product rule**: the probability of two independent events occurring together can be calculated by multiplying the individual probabilities of each event occurring alone
- **probability**: a number, between 0 and 1, expressing the precise likelihood of an event happening

Probability Basics

Probabilities are mathematical measures of likelihood. The empirical probability of an event is calculated by dividing the number of times the event occurs by the total number of opportunities for the event to occur. Empirical probabilities come from observations such as those of Mendel. An example of a genetic event is a round seed produced by a pea plant. Mendel demonstrated that the probability of the event "round seed" was guaranteed to occur in the F_1 offspring of true-breeding parents, one of which has round seeds and one of which has wrinkled seeds. When the F_1 plants were subsequently self-crossed, the probability of any given F_2 offspring having round seeds was now three out of four. In other words, in a large population of F_2 offspring chosen at random, 75 percent were expected to have round seeds, whereas 25 percent were expected to have wrinkled seeds. Using large numbers of crosses, Mendel was able to calculate probabilities and use these to predict the outcomes of other crosses.

The Product Rule

Mendel demonstrated that the pea-plant characteristics he studied were transmitted as discrete units from parent to offspring. Mendel also determined that different characteristics were transmitted independently of one another and could be considered in separate probability analyses. For instance, performing a cross between a plant with green, wrinkled seeds and a plant with yellow, round seeds produced offspring that had a 3:1 ratio of green:yellow seeds and a 3:1 ratio of round:wrinkled seeds. The characteristics of color and texture did not influence each other.

The product rule of probability can be applied to this phenomenon of the independent transmission of characteristics. It states that the probability of two independent events occurring together can be calculated by multiplying the individual probabilities of each event occurring alone. Imagine that you are rolling a six-sided die (D) and flipping a penny (P) at the same time. The die may roll any number from 1–6 ($D_{\#}$), whereas the penny may turn up heads (P_H) or tails (P_T). The outcome of rolling the die has no effect on the outcome of flipping the penny and vice versa. There are 12 possible outcomes, and each is expected to occur with equal probability: D_1P_H, D_1P_T, D_2P_H, D_2P_T, D_3P_H, D_3P_T, D_4P_H, D_4P_T, D_5P_H, D_5P_T, D_6P_H, D_6P_T.

Of the 12 possible outcomes, the die has a 2/12 (or 1/6) probability of rolling a two, and the penny has a 6/12 (or 1/2) probability of coming up heads. The probability that you will obtain the combined outcome 2 and heads is: (D_2) x (P_H) = (1/6) x (1/2) or 1/12. The word "and" is a signal to apply the product rule. Consider how the product rule is applied to a dihybrid: the probability of having both dominant traits in the F_2 progeny is the product of the probabilities of having the dominant trait for each characteristic.

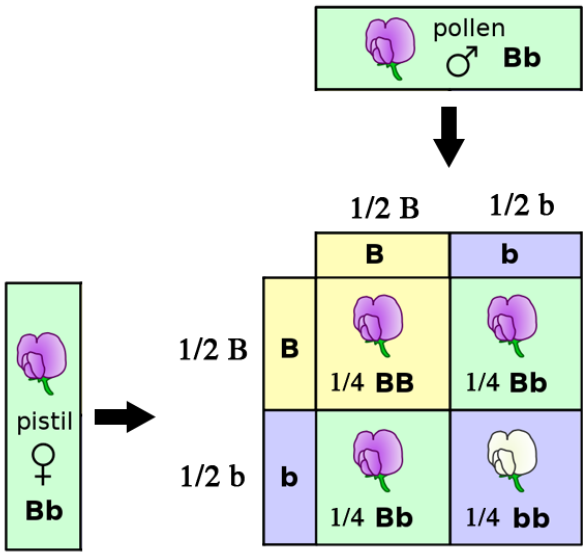

Figure 12.1*E*. 1: **Role of probability in segregation of alleles and fertilization**: In a genetic cross, the probability of the dominant trait being expressed is dependent upon its frequency. In this case, both parents possessed a dominant and a recessive gene for the trait of flower color. The dominant trait is expressed in 3/4 of the offspring and the recessive trait is expressed in 1/4.

The Sum Rule

The sum rule is applied when considering two mutually-exclusive outcomes that can result from more than one pathway. It states that the probability of the occurrence of one event or the other, of two mutually-exclusive events, is the sum of their individual probabilities. The word "or" indicates that you should apply the sum rule. Let's imagine you are flipping a penny (P) and a quarter (Q). What is the probability of one coin coming up heads and one coming up tails? This can be achieved by two cases: the penny is heads (P_H) and the quarter is tails (Q_T), or the quarter is heads (Q_H) and the penny is tails (P_T). Either case fulfills the outcome. We calculate the probability of obtaining one head and one tail as $[(P_H) \times (Q_T)] + [(Q_H) \times (P_T)] = [(1/2) \times (1/2)] + [(1/2) \times (1/2)] = 1/2$. You should also notice that we used the product rule to calculate the probability of P_H and Q_T and also the probability of P_T and Q_H, before we summed them. The sum rule can be applied to show the probability of having just one dominant trait in the F_2 generation of a dihybrid cross.

To use probability laws in practice, it is necessary to work with large sample sizes because small sample sizes are prone to deviations caused by chance. The large quantities of pea plants that Mendel examined allowed him to calculate the probabilities of the traits appearing in his F_2 generation. This discovery meant that when parental traits were known, the offspring's traits could be predicted accurately even before fertilization.

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12.2: Patterns of Inheritance

Topic hierarchy

- 12.2A: Genes as the Unit of Heredity
- 12.2B: Phenotypes and Genotypes
- 12.2C: The Punnett Square Approach for a Monohybrid Cross
- 12.2D: Alternatives to Dominance and Recessiveness
- 12.2E: Sex-Linked Traits
- 12.2F: Lethal Inheritance Patterns

12.2A: Genes as the Unit of Heredity

Genes exist in pairs within an organism, with one of each pair inherited from each parent.

Learning Objectives

- Describe the structure of a gene and how offspring inherit genes from each parent

Key Points

- A gene is a stretch of DNA that helps to control the development and function of all organs and working systems in the body.
- Genes are passed from parent to offspring; the combination of these genes affects all aspects of the human body, from eye and hair color to how well the liver can process toxins.
- A human will inherit 23 chromosomes from its mother and 23 from its father; together, these form 23 pairs of chromosomes that direct the inherited characteristics of the individual.
- If the two copies of a gene inherited from each parent are the same, that individual is said to be homozygous for the gene; if the two copies inherited from each parent are different, that individual is said to be heterozygous for the gene.

Key Terms

- **gene**: a unit of heredity; the functional units of chromosomes that determine specific characteristics by coding for specific proteins
- **chromosome**: a structure in the cell nucleus that contains DNA, histone protein, and other structural proteins
- **genetics**: the branch of biology that deals with the transmission and variation of inherited characteristics, in particular chromosomes and DNA

Pairs of Unit Factors, or Genes

Mendel proposed that paired unit factors of heredity were transmitted faithfully from generation to generation by the dissociation and reassociation of paired factors during gametogenesis and fertilization, respectively. After he crossed peas with contrasting traits and found that the recessive trait resurfaced in the F_2 generation, Mendel deduced that hereditary factors must be inherited as discrete units. This finding contradicted the belief at that time that parental traits were blended in the offspring.

A gene is made up of short sections of DNA that are contained on a chromosome within the nucleus of a cell. Genes control the development and function of all organs and all working systems in the body. A gene has a certain influence on how the cell works; the same gene in many different cells determines a certain physical or biochemical feature of the whole body (e.g., eye color or reproductive functions). All human cells hold approximately 21,000 different genes.

Genetics is the science of the way traits are passed from parent to offspring. For all forms of life, continuity of the species depends upon the genetic code being passed from parent to offspring. Evolution by natural selection is dependent on traits being heritable. Genetics is very important in human physiology because all attributes of the human body are affected by a person's genetic code. It can be as simple as eye color, height, or hair color. Or it can be as complex as how well your liver processes toxins, whether you will be prone to heart disease or breast cancer, and whether you will be color blind.

Genetic inheritance begins at the time of conception. You inherited 23 chromosomes from your mother and 23 from your father. Together they form 22 pairs of autosomal chromosomes and a pair of sex chromosomes (either XX if you are female, or XY if you are male). Homologous chromosomes have the same genes in the same positions, but may have different alleles (varieties) of those genes. There can be many alleles of a gene within a population, but an individual within that population only has two copies and can be homozygous (both copies the same) or heterozygous (the two copies are different) for any given gene.

Figure 12.2*A*.1: **Gene pairs enable genetic combinations**: A child will inherit half of its genes (one of each of its 23 pairs) from its mother and the other half from its father.

12.2B: Phenotypes and Genotypes

The observable traits expressed by an organism are referred to as its phenotype and its underlying genetic makeup is called its genotype.

Learning Objectives

- Distinguish between the phenotype and the genotype of an organism

Key Points

- Mendel used pea plants with seven distinct traits or phenotypes to determine the pattern of inheritance and the underlying genotypes.
- Mendel found that crossing two purebred pea plants which expressed different traits resulted in an F_1generation where all the pea plants expressed the same trait or phenotype.
- When Mendel allowed the F_1 plants to self-fertilize, the F_2 generation showed two different phenotypes, indicating that the F_1 plants had different genotypes.

Key Terms

- **phenotype**: the appearance of an organism based on a multifactorial combination of genetic traits and environmental factors, especially used in pedigrees
- **genotype**: the combination of alleles, situated on corresponding chromosomes, that determines a specific trait of an individual, such as “Aa” or “aa”

Phenotypes and Genotypes

The observable traits expressed by an organism are referred to as its phenotype. An organism’s underlying genetic makeup, consisting of both physically visible and non-expressed alleles, is called its genotype. Johann Gregor Mendel’s (1822–1884) hybridization experiments demonstrate the difference between phenotype and genotype.

Mendel crossed or mated two true-breeding (self-pollinating) garden peas, *Pisum saivum*, by manually transferring pollen from the anther of a mature pea plant of one variety to the stigma of a separate mature pea plant of the second variety. Plants used in first-generation crosses were called P_0, or parental generation one, plants. Mendel collected the seeds belonging to the P_0plants that resulted from each cross and grew them the following season. These offspring were called the F_1, or the first filial (filial = offspring, daughter or son), generation. Once Mendel examined the characteristics in the F_1 generation of plants, he allowed them to self-fertilize naturally. He then collected and grew the seeds from the F_1 plants to produce the F_2, or second filial, generation.

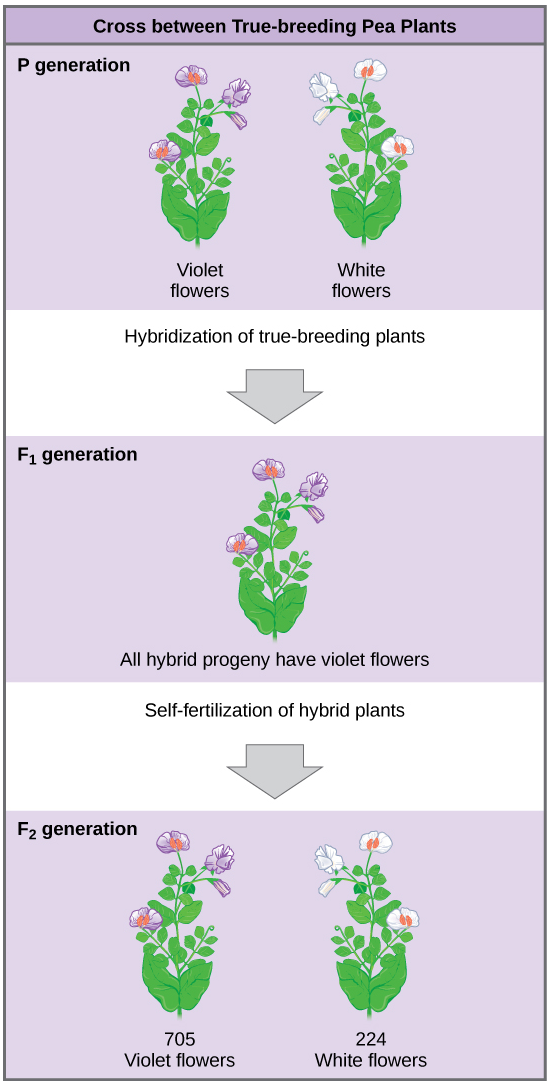

Figure 12.2B.1: **Mendelian crosses**: In one of his experiments on inheritance patterns, Mendel crossed plants that were true-breeding for violet flower color with plants true-breeding for white flower color (the P generation). The resulting hybrids in the F_1 generation all had violet flowers. In the F_2 generation, approximately three-quarters of the plants had violet flowers, and one-quarter had white flowers.

When true-breeding plants in which one parent had white flowers and one had violet flowers were cross-fertilized, all of the F1 hybrid offspring had violet flowers. That is, the hybrid offspring were phenotypically identical to the true-breeding parent with violet flowers. However, we know that the allele donated by the parent with white flowers was not simply lost because it reappeared in some of the F2 offspring. Therefore, the F1 plants must have been genotypically different from the parent with violet flowers.

In his 1865 publication, Mendel reported the results of his crosses involving seven different phenotypes, each with two contrasting traits. A trait is defined as a variation in the physical appearance of a heritable characteristic. The characteristics included plant height, seed texture, seed color, flower color, pea pod size, pea pod color, and flower position. To fully examine each characteristic, Mendel generated large numbers of F_1 and F_2 plants, reporting results from 19,959 F_2 plants alone. His findings were consistent. First, Mendel confirmed that he had plants that bred true for white or violet flower color. Regardless of how many generations Mendel examined, all self-crossed offspring of parents with white flowers had white flowers, and all self-crossed offspring of parents with violet flowers had violet flowers. In addition, Mendel confirmed that, other than flower color, the pea plants were physically identical.

12.2C: The Punnett Square Approach for a Monohybrid Cross

A Punnett square applies the rules of probability to predict the possible outcomes of a monohybrid cross and their expected frequencies.

Learning Objectives

- Describe the Punnett square approach to a monohybrid cross

Key Points

- Fertilization between two true-breeding parents that differ in only one characteristic is called a monohybrid cross.
- For a monohybrid cross of two true-breeding parents, each parent contributes one type of allele resulting in all of the offspring with the same genotype.
- A test cross is a way to determine whether an organism that expressed a dominant trait was a heterozygote or a homozygote.

Key Terms

- **monohybrid**: a hybrid between two species that only have a difference of one gene
- **homozygous**: of an organism in which both copies of a given gene have the same allele
- **heterozygous**: of an organism which has two different alleles of a given gene
- **Punnett square**: a graphical representation used to determine the probability of an offspring expressing a particular genotype

Punnett Square Approach to a Monohybrid Cross

When fertilization occurs between two true-breeding parents that differ in only one characteristic, the process is called a monohybrid cross, and the resulting offspring are monohybrids. Mendel performed seven monohybrid crosses involving contrasting traits for each characteristic. On the basis of his results in F_1 and F_2 generations, Mendel postulated that each parent in the monohybrid cross contributed one of two paired unit factors to each offspring and that every possible combination of unit factors was equally likely.

To demonstrate a monohybrid cross, consider the case of true-breeding pea plants with yellow versus green pea seeds. The dominant seed color is yellow; therefore, the parental genotypes were YY (homozygous dominant) for the plants with yellow seeds and yy (homozygous recessive) for the plants with green seeds, respectively. A Punnett square, devised by the British geneticist Reginald Punnett, can be drawn that applies the rules of probability to predict the possible outcomes of a genetic cross or mating and their expected frequencies.To prepare a Punnett square, all possible combinations of the parental alleles are listed along the top (for one parent) and side (for the other parent) of a grid, representing their meiotic segregation into haploid gametes. Then the combinations of egg and sperm are made in the boxes in the table to show which alleles are combining. Each box then represents the diploid genotype of a zygote, or fertilized egg, that could result from this mating. Because each possibility is equally likely, genotypic ratios can be determined from a Punnett square. If the pattern of inheritance (dominant or recessive) is known, the phenotypic ratios can be inferred as well. For a monohybrid cross of two true-breeding parents, each parent contributes one type of allele. In this case, only one genotype is possible. All offspring are Yy and have yellow seeds.

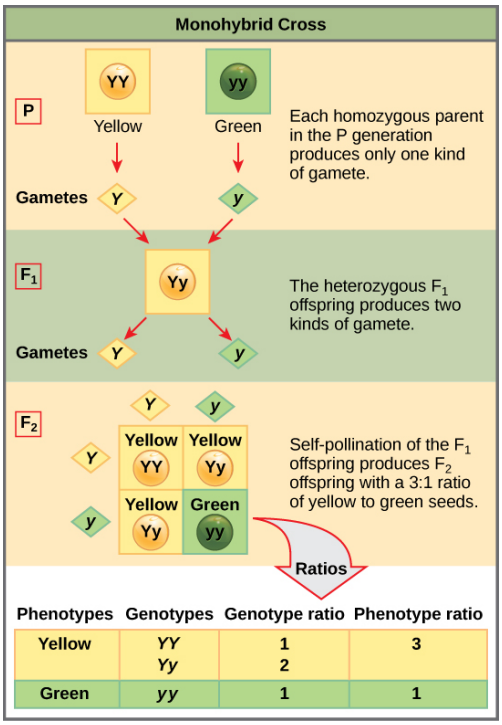

Figure 12.2C.1: **Punnett square analysis of a monohytbrid cross**: In the P generation, pea plants that are true-breeding for the dominant yellow phenotype are crossed with plants with the recessive green phenotype. This cross produces F1 heterozygotes with a yellow phenotype. Punnett square analysis can be used to predict the genotypes of the F2 generation.

A self-cross of one of the Yy heterozygous offspring can be represented in a 2 × 2 Punnett square because each parent can donate one of two different alleles. Therefore, the offspring can potentially have one of four allele combinations: YY, Yy, yY, or yy. Notice that there are two ways to obtain the Yy genotype: a Y from the egg and a y from the sperm, or a y from the egg and a Y from the sperm. Both of these possibilities must be counted. Recall that Mendel's pea-plant characteristics behaved in the same way in reciprocal crosses. Therefore, the two possible heterozygous combinations produce offspring that are genotypically and phenotypically identical despite their dominant and recessive alleles deriving from different parents. They are grouped together. Because fertilization is a random event, we expect each combination to be equally likely and for the offspring to exhibit a ratio of YY:Yy:yy genotypes of 1:2:1. Furthermore, because the YY and Yy offspring have yellow seeds and are phenotypically identical, applying the sum rule of probability, we expect the offspring to exhibit a phenotypic ratio of 3 yellow:1 green. Indeed, working with large sample sizes, Mendel observed approximately this ratio in every F_2 generation resulting from crosses for individual traits.

Beyond predicting the offspring of a cross between known homozygous or heterozygous parents, Mendel also developed a way to determine whether an organism that expressed a dominant trait was a heterozygote or a homozygote. Called the test cross, this technique is still used by plant and animal breeders. In a test cross, the dominant-expressing organism is crossed with an organism that is homozygous recessive for the same characteristic. If the dominant-expressing organism is a homozygote, then all F_1 offspring will be heterozygotes expressing the dominant trait. Alternatively, if the dominant expressing organism is a heterozygote, the F_1 offspring will exhibit a 1:1 ratio of heterozygotes and recessive homozygotes. The test cross further validates Mendel's postulate that pairs of unit factors segregate equally.

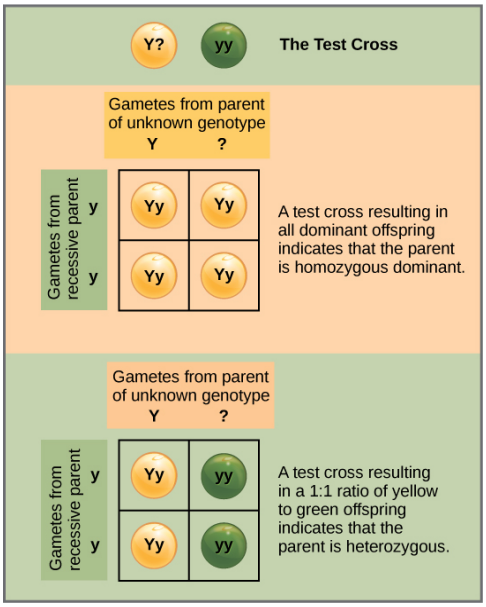

Figure 12.2C.1: **Example of a test cross**: A test cross can be performed to determine whether an organism expressing a dominant trait is a homozygote or a heterozygote.

12.2D: Alternatives to Dominance and Recessiveness

With the inclusion of incomplete dominance, codominance, multiple alleles, and mutant alleles, the inheritance of traits is complex process.

Learning Objectives

- Discuss incomplete dominance, codominance, and multiple alleles as alternatives to dominance and recessiveness

Key Points

- Incomplete dominance is the expression of two contrasting alleles such that the individual displays an intermediate phenotype.
- Codominance is a variation on incomplete dominance in which both alleles for the same characteristic are simultaneously expressed in the heterozygote.
- Diploid organisms can only have two alleles for a given gene; however, multiple alleles may exist at the population level such that many combinations of two alleles are observed.
- The complete dominance of a wild-type phenotype over all other mutants often occurs as an effect of "dosage" of a specific gene product: the wild-type allele supplies the correct amount of gene product whereas the mutant alleles cannot.
- One mutant allele can also be dominant over all other phenotypes, including the wild type.

Key Terms

- **allele**: one of a number of alternative forms of the same gene occupying a given position on a chromosome
- **incomplete dominance**: a condition in which the phenotype of the heterozygous genotype is distinct from and often intermediate to the phenotypes of the homozygous genotypes
- **codominance**: a condition in which both alleles of a gene pair in a heterozygote are fully expressed, with neither one being dominant or recessive to the other

Alternatives to Dominance and Recessiveness

Mendel's experiments with pea plants suggested that: (1) two "units" or alleles exist for every gene; (2) alleles maintain their integrity in each generation (no blending); and (3) in the presence of the dominant allele, the recessive allele is hidden and makes no contribution to the phenotype. Therefore, recessive alleles can be "carried" and not expressed by individuals. Such heterozygous individuals are sometimes referred to as "carriers." Further genetic studies in other plants and animals have shown that much more complexity exists, but that the fundamental principles of Mendelian genetics still hold true.

Mendel's results, that traits are inherited as dominant and recessive pairs, contradicted the view at that time that offspring exhibited a blend of their parents' traits. However, the heterozygote phenotype occasionally does appear to be intermediate between the two parents. For example, in the snapdragon, *Antirrhinum majus*, a cross between a homozygous parent with white flowers (C^WC^W) and a homozygous parent with red flowers (C^RC^R) will produce offspring with pink flowers (C^RC^W). This pattern of inheritance is described as incomplete dominance, denoting the expression of two contrasting alleles such that the individual displays an intermediate phenotype. The allele for red flowers is incompletely dominant over the allele for white flowers. However, the results of a heterozygote self-cross can still be predicted, just as with Mendelian dominant and recessive crosses. In this case, the genotypic ratio would be 1 C^RC^R:2 C^RC^W:1 C^WC^W, and the phenotypic ratio would be 1:2:1 for red:pink:white.

Figure 12.2D.1: **Example of incomplete dominance**: These pink flowers of a heterozygote snapdragon result from incomplete dominance.

A variation on incomplete dominance is codominance, in which both alleles for the same characteristic are simultaneously expressed in the heterozygote. An example of codominance is the MN blood groups of humans. The M and N alleles are expressed in the form of an M or N antigen present on the surface of red blood cells. Homozygotes ($L^ML^M$and L^NL^N) express either the M or the N allele, and heterozygotes (L^ML^N) express both alleles equally. In a self-cross between heterozygotes expressing a codominant trait, the three possible offspring genotypes are phenotypically distinct. However, the 1:2:1 genotypic ratio characteristic of a Mendelian monohybrid cross still applies.

Mendel implied that only two alleles, one dominant and one recessive, could exist for a given gene. We now know that this is an oversimplification. Although individual humans (and all diploid organisms) can only have two alleles for a given gene, multiple alleles may exist at the population level such that many combinations of two alleles are observed. Note that when many alleles exist for the same gene, the convention is to denote the most common phenotype or genotype among wild animals as the wild type (often abbreviated "+"); this is considered the standard or norm. All other phenotypes or genotypes are considered variants of this standard, meaning that they deviate from the wild type. The variant may be recessive or dominant to the wild-type allele. An example of multiple alleles is coat color in rabbits. Here, four alleles exist for the c gene. The wild-type version, C^+C^+, is expressed as brown fur. The chinchilla phenotype, $c^{ch}c^{ch}$, is expressed as black-tipped white fur. The Himalayan phenotype, c^hc^h, has black fur on the extremities and white fur elsewhere. Finally, the albino, or "colorless" phenotype, cc, is expressed as white fur. In cases of multiple alleles, dominance hierarchies can exist. In this case, the wild-type allele is dominant over all the others, chinchilla is incompletely dominant over Himalayan and albino, and Himalayan is dominant over albino. This hierarchy, or allelic series, was revealed by observing the phenotypes of each possible heterozygote offspring.

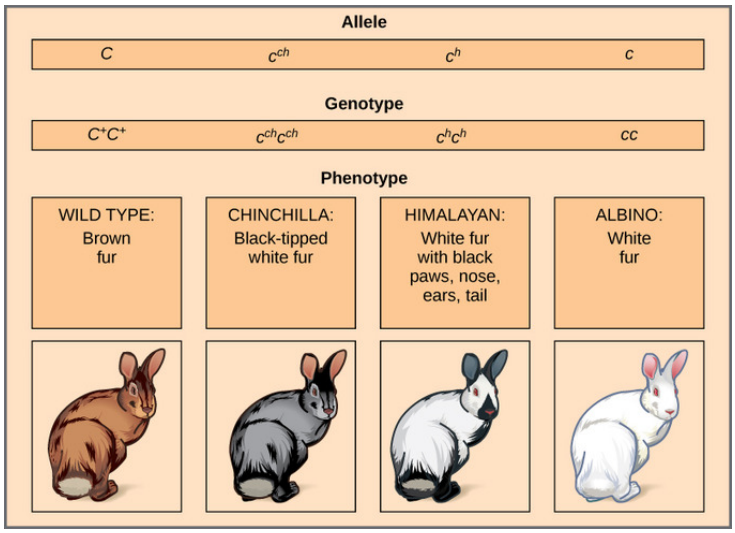

Figure 12.2D.1: **Example of multiple alleles for rabbit coat color**: Four different alleles exist for the rabbit coat color (C) gene.

The complete dominance of a wild-type phenotype over all other mutants often occurs as an effect of "dosage" of a specific gene product, such that the wild-type allele supplies the correct amount of gene product whereas the mutant alleles cannot. For the allelic series in rabbits, the wild-type allele may supply a given dosage of fur pigment, whereas the mutants supply a lesser dosage or none at all. Alternatively, one mutant allele can be dominant over all other phenotypes, including the wild type. This may occur when the mutant allele somehow interferes with the genetic message so that even a heterozygote with one wild-type allele copy expresses the mutant phenotype. One way in which the mutant allele can interfere is by enhancing the function of the wild-type gene product or changing its distribution in the body. One example of this is the Antennapedia mutation in *Drosophila*. In this case, the mutant allele expands the distribution of the gene product; as a result, the Antennapedia heterozygote develops legs on its head where its antennae should be.

Figure 12.2D.1: **Example of a mutant allele interfering with the function of a wild-type gene**: As seen in comparing the wild-type *Drosophila*(left) and the Antennapedia mutant (right), the Antennapedia mutant has legs on its head in place of antennae.

12.2E: Sex-Linked Traits

A gene present on one of the sex chromosomes (X or Y in mammals) is a sex-linked trait because its expression depends on the sex of the individual.

Learning Objectives

- Distinguish between sex-linked traits and other forms of inheritance

Key Points

- In mammals, females have a homologous pair of X chromosomes, whereas males have an XY chromosome pair.
- The Y chromosome contains a small region of similarity to the X chromosome so that they can pair during meiosis, but the Y is much shorter and contains fewer genes.
- Males are said to be hemizygous because they have only one allele for any X-linked characteristic; males will exhibit the trait of any gene on the X-chromosome regardless of dominance and recessiveness.
- Most sex-linked traits are actually X-linked, such as eye color in *Drosophila* or color blindness in humans.

Key Terms

- **hemizygous**: Having some single copies of genes in an otherwise diploid cell or organism.
- **X-linked**: Associated with the X chromosome.
- **carrier**: A person or animal that transmits a disease to others without itself contracting the disease.
- **sex chromosomes**: A chromosome involved with determining the sex of an organism, typically one of two kinds.

Sex Determination

In humans, as well as in many other animals and some plants, the sex of the individual is determined by sex chromosomes. However, there are other sex determination systems in nature. For example, temperature-dependent sex determination is relatively common,
and there are many other types of environmental sex determination. Some species, such as some snails, practice sex change adults start out male, then become female. In tropical clown fish, the dominant individual in a group becomes female while the others are male.

The sex chromosomes are one pair of non-homologous chromosomes. Until now, we have only considered inheritance patterns among non-sex chromosomes, or autosomes. In addition to 22 homologous pairs of autosomes, human females have a homologous pair of X chromosomes, whereas human males have an XY chromosome pair. Although the Y chromosome contains a small region of similarity to the X chromosome so that they can pair during meiosis, the Y chromosome is much shorter and contains many fewer genes. When a gene being examined is present on the X chromosome, but not on the Y chromosome, it is said to be X-linked.

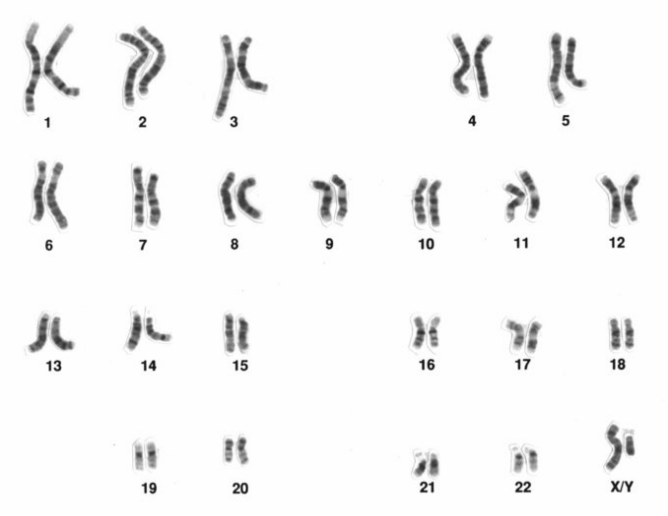

Figure 12.2E.1: **Human male karyotype**: A human males possesses XY chromosomes, as seen in the bottom left of this karyotype. The Y chromosome is much shorter than the X chromosome, unlike all of the other homologous chromosome pairs.

X-Linked Traits

Insects also follow an XY sex-determination pattern and like humans, *Drosophila* males have an XY chromosome pair and females are XX. Eye color in *Drosophila* was one of the first X-linked traits to be identified, and Thomas Hunt Morgan mapped this trait to the X chromosome in 1910.

In fruit flies, the wild-type eye color is red (X^W) and is dominant to white eye color (X^w). Because this eye-color gene is located on the X chromosome only, reciprocal crosses do not produce the same offspring ratios. Males are said to be hemizygous, because they have only one allele for any X-linked characteristic. Hemizygosity makes the descriptions of dominance and recessiveness irrelevant for XY males because each male only has one copy of the gene. *Drosophila* males lack a second allele copy on the Y chromosome; their genotype can only be X^WY or X^wY. In contrast, females have two allele copies of this gene and can be X^WX^W, X^WX^w, or X^wX^w.

Figure 12.2E.1: **Eye color in Drosophila is an example of a X-linked trait**: In *Drosophila*, the gene for eye color is located on the X chromosome. Clockwise from top left are brown, cinnabar, sepia, vermilion, white, and red. Red eye color is wild-type and is dominant to white eye color.

X-Linked Crosses

In an X-linked cross, the genotypes of F_1 and F_2 offspring depend on whether the recessive trait was expressed by the male or the female in the P_1 generation. With regard to *Drosophila* eye color, when the P_1 male expresses the white-eye phenotype and the female is homozygous red-eyed, all members of the F_1 generation exhibit red eyes. The F_1 females are heterozygous (X^WX^w), and the males are all X^WY, having received their X chromosome from the homozygous dominant P_1 female and their Y chromosome from the P_1 male.

A subsequent cross between the X^WX^w female and the X^WY male would produce only red-eyed females (with X^WX^W or X^WX^wgenotypes) and both red- and white-eyed males (with X^WY or X^wY genotypes). Now, consider a cross between a homozygous white-eyed female and a male with red eyes. The F_1 generation would exhibit only heterozygous red-eyed females (X^WX^w) and only white-eyed males (X^wY). Half of the F_2 females would be red-eyed (X^WX^w) and half would be white-eyed (X^wX^w). Similarly, half of the F_2 males would be red-eyed (X^WY) and half would be white-eyed (X^wY).

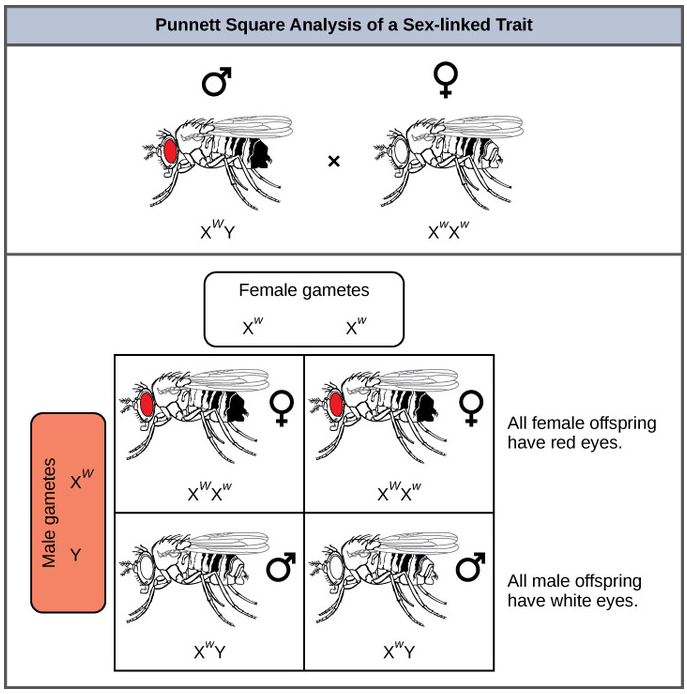

Figure 12.2E.1: **Punnett square analysis of *Drosophila* eye color**: Punnett square analysis is used to determine the ratio of offspring from a cross between a red-eyed male fruit fly (X^WY) and a white-eyed female fruit fly (X^wX^w).

X-Linked Recessive Disorders in Humans

Sex-linkage studies provided the fundamentals for understanding X-linked recessive disorders in humans, which include red-green color blindness and Types A and B hemophilia. Because human males need to inherit only one recessive mutant X allele to be affected, X-linked disorders are disproportionately observed in males. Females must inherit recessive X-linked alleles from both of their parents in order to express the trait.

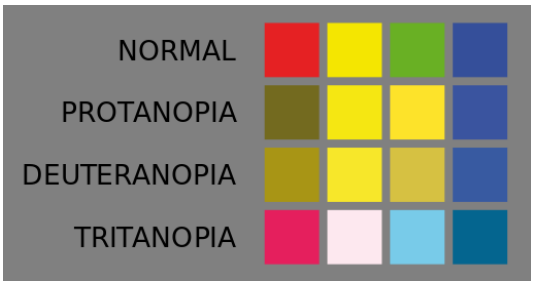

Figure 12.2E.1: **Color perception in different types of color blindness**: In this chart you can see what people with different types of color blindness can see versus the normal color vision line at top.

Recessive Carriers

When they inherit one recessive X-linked mutant allele and one dominant X-linked wild-type allele, they are carriers of the trait and are typically unaffected. Carrier females can manifest mild forms of the trait due to the inactivation of the dominant allele located on one of the X chromosomes. However, female carriers can contribute the trait to their sons, resulting in the son exhibiting the trait, or they can contribute the recessive allele to their daughters, resulting in the daughters being carriers of the trait. Although some Y-linked recessive disorders exist, typically they are associated with infertility in males and are, therefore, not transmitted to subsequent generations.

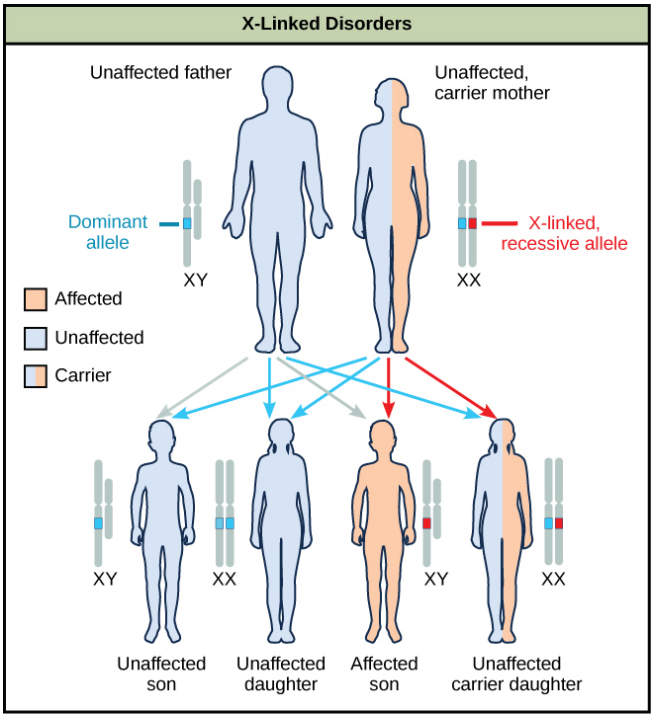

Figure 12.2E.1: **Inheritance of a recessive X-linked disorder**: The son of a woman who is a carrier of a recessive X-linked disorder will have a 50 percent chance of being affected. A daughter will not be affected, but she will have a 50 percent chance of being a carrier like her mother.

12.2F: Lethal Inheritance Patterns

Inheriting two copies of mutated genes that are nonfunctional can have lethal consequences.

Learning Objectives

- Describe recessive and dominant lethal inheritance patterns

Key Points

- An inheritance pattern in which an allele is only lethal in the homozygous form and in which the heterozygote may be normal or have some altered non-lethal phenotype is referred to as recessive lethal.
- The dominant lethal inheritance pattern is one in which an allele is lethal both in the homozygote and the heterozygote; this allele can only be transmitted if the lethality phenotype occurs after reproductive age.
- Dominant lethal alleles are very rare because the allele only lasts one generation and is, therefore, not usually transmitted.
- In the case where dominant lethal alleles might not be expressed until adulthood, the allele may be unknowingly passed on, resulting in a delayed death in both generations.

Key Terms

- **mutation**: any heritable change of the base-pair sequence of genetic material
- **recessive lethal**: an inheritance pattern in which an allele is only lethal in the homozygous form and in which the heterozygote may be normal or have some altered non-lethal phenotype
- **dominant lethal**: an inheritance pattern is one in which an allele is lethal both in the homozygote and the heterozygote; this allele can only be transmitted if the lethality phenotype occurs after reproductive age

Lethal Inheritance Patterns

A large proportion of genes in an individual's genome are essential for survival. Occasionally, a nonfunctional allele for an essential gene can arise by mutation and be transmitted in a population as long as individuals with this allele also have a wild-type, functional copy. The wild-type allele functions at a capacity sufficient to sustain life and is, therefore, considered to be dominant over the nonfunctional allele. However, consider two heterozygous parents that have a genotype of wild-type/nonfunctional mutant for a hypothetical essential gene. In one quarter of their offspring, we would expect to observe individuals that are homozygous recessive for the nonfunctional allele. Because the gene is essential, these individuals might fail to develop past fertilization, die in utero, or die later in life, depending on what life stage requires this gene. An inheritance pattern in which an allele is only lethal in the homozygous form and in which the heterozygote may be normal or have some altered non-lethal phenotype is referred to as recessive lethal.

For crosses between heterozygous individuals with a recessive lethal allele that causes death before birth when homozygous, only wild-type homozygotes and heterozygotes would be observed. The genotypic ratio would therefore be 2:1. In other instances, the recessive lethal allele might also exhibit a dominant (but not lethal) phenotype in the heterozygote. For instance, the recessive lethal Curly allele in Drosophila affects wing shape in the heterozygote form, but is lethal in the homozygote.

Dominant Lethal Alleles

A single copy of the wild-type allele is not always sufficient for normal functioning or even survival. The dominant lethal inheritance pattern is one in which an allele is lethal both in the homozygote and the heterozygote; this allele can only be transmitted if the lethality phenotype occurs after reproductive age. Individuals with mutations that result in dominant lethal alleles fail to survive even in the heterozygote form. Dominant lethal alleles are very rare because, as you might expect, the allele only lasts one generation and is not transmitted. However, just as the recessive lethal allele might not immediately manifest the phenotype of death, dominant lethal alleles also might not be expressed until adulthood. Once the individual reaches reproductive age, the allele may be unknowingly passed on, resulting in a delayed death in both generations. An example of this in humans is Huntington's disease in which the nervous system gradually wastes away. People who are heterozygous for the dominant Huntington allele (Hh) will inevitably develop the fatal disease. However,

the onset of Huntington's disease may not occur until age 40, at which point the afflicted persons may have already passed the allele to 50 percent of their offspring.

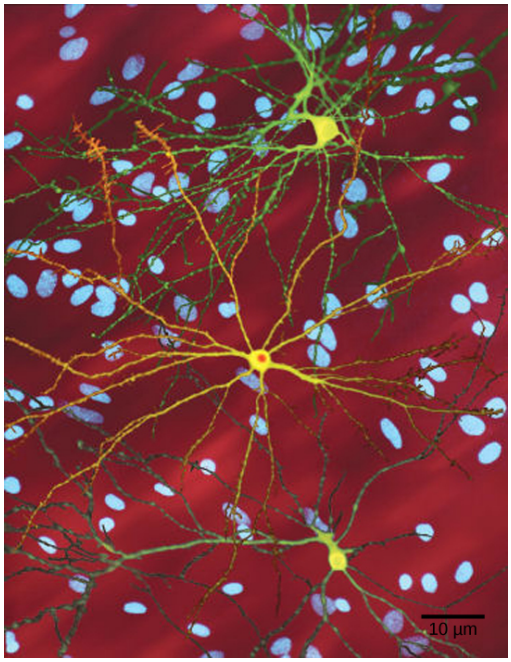

Figure 12.2F.1: **Effects of Huntington's disease on neurons**: The neuron in the center of this micrograph (yellow) has nuclear inclusions characteristic of Huntington's disease (orange area in the center of the neuron). Huntington's disease occurs when an abnormal dominant allele for the Huntington gene is present.

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12.3: Laws of Inheritance

Topic hierarchy

- 12.3A: Mendel’s Laws of Heredity
- 12.3B: Mendel’s Law of Dominance
- 12.3C: Mendel’s Law of Segregation
- 12.3D: Mendel’s Law of Independent Assortment
- 12.3E: Genetic Linkage and Violation of the Law of Independent Assortment
- 12.3F: Epistasis

12.3A: Mendel's Laws of Heredity

Mendel formed the Laws of Heredity (the Law of Segregation and the Law of Independent Assortment) from his pea plant experiments.

Learning Objectives

- Discuss the methods Mendel utilized in his research that led to his success in understanding the process of inheritance

Key Points

- By crossing purple and white pea plants, Mendel found the offspring were purple rather than mixed, indicating one color was dominant over the other.
- Mendel's Law of Segregation states individuals possess two alleles and a parent passes only one allele to his/her offspring.
- Mendel's Law of Independent Assortment states the inheritance of one pair of factors (genes) is independent of the inheritance of the other pair.
- If the two alleles are identical, the individual is called homozygous for the trait; if the two alleles are different, the individual is called heterozygous.
- Mendel cross-bred dihybrids and found that traits were inherited independently of each other.

Key Terms

- **homozygous**: of an organism in which both copies of a given gene have the same allele
- **heterozygous**: of an organism which has two different alleles of a given gene
- **allele**: one of a number of alternative forms of the same gene occupying a given position on a chromosome

Introduction

Mendelian inheritance (or Mendelian genetics or Mendelism) is a set of primary tenets relating to the transmission of hereditary characteristics from parent organisms to their children; it underlies much of genetics. The tenets were initially derived from the work of Gregor Mendel published in 1865 and 1866, which was "re-discovered" in 1900; they were initially very controversial, but they soon became the core of classical genetics.

The laws of inheritance were derived by Gregor Mendel, a 19th century monk conducting hybridization experiments in garden peas (*Pisum sativum*). Between 1856 and 1863, he cultivated and tested some 28,000 pea plants. From these experiments, he deduced two generalizations that later became known as Mendel's Laws of Heredity or Mendelian inheritance. He described these laws in a two part paper, "Experiments on Plant Hybridization", which was published in 1866.

Mendel's Laws

Mendel discovered that by crossing true-breeding white flower and true-breeding purple flower plants, the result was a hybrid offspring. Rather than being a mix of the two colors, the offspring was purple flowered. He then conceived the idea of heredity units, which he called "factors", one of which is a recessive characteristic and the other dominant. Mendel said that factors, later called genes, normally occur in pairs in ordinary body cells, yet segregate during the formation of sex cells. Each member of the pair becomes part of the separate sex cell. The dominant gene, such as the purple flower in Mendel's plants, will hide the recessive gene, the white flower. After Mendel self-fertilized the F1 generation and obtained an F2 generation with a 3:1 ratio, he correctly theorized that genes can be paired in three different ways for each trait: AA, aa, and Aa. The capital A represents the dominant factor while the lowercase a represents the recessive.

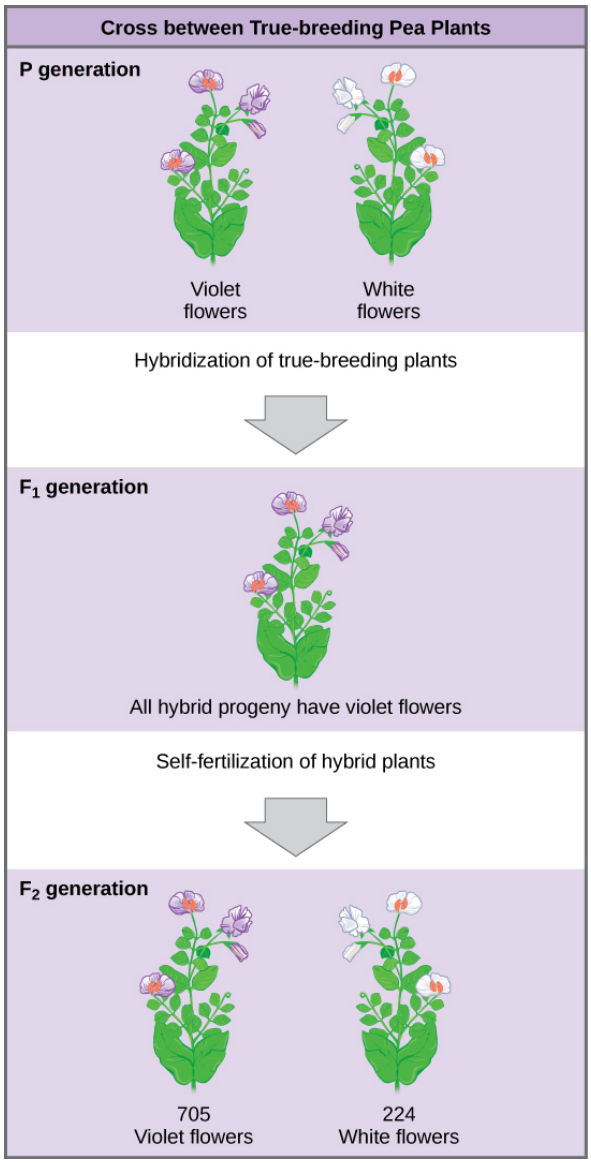

Figure 12.3A.1: **Mendel's Pea Plants**: In one of his experiments on inheritance patterns, Mendel crossed plants that were true-breeding for violet flower color with plants true-breeding for white flower color (the P generation). The resulting hybrids in the F1 generation all had violet flowers. In the F2 generation, approximately three-quarters of the plants had violet flowers, and one-quarter had white flowers.

Mendel stated that each individual has two alleles for each trait, one from each parent. Thus, he formed the "first rule", the Law of Segregation, which states individuals possess two alleles and a parent passes only one allele to his/her offspring. One allele is given by the female parent and the other is given by the male parent. The two factors may or may not contain the same information. If the two alleles are identical, the individual is called homozygous for the trait. If the two alleles are different, the individual is called heterozygous. The presence of an allele does not promise that the trait will be expressed in the individual that possesses it. In heterozygous individuals, the only allele that is expressed is the dominant. The recessive allele is present, but its expression is hidden. The genotype of an individual is made up of the many alleles it possesses. An individual's physical appearance, or phenotype, is determined by its alleles as well as by its environment.

Mendel also analyzed the pattern of inheritance of seven pairs of contrasting traits in the domestic pea plant. He did this by cross-breeding dihybrids; that is, plants that were heterozygous for the alleles controlling two different traits. Mendel then crossed these dihybrids. If it is inevitable that round seeds must always be yellow and wrinkled seeds must be green, then he would have expected that this would produce a typical monohybrid cross: 75 percent round-yellow; 25 percent wrinkled-green. But, in fact, his mating generated seeds that showed all possible combinations of the color and texture traits. He found 9/16 of the offspring were round-yellow, 3/16 were round-green, 3/16 were wrinkled-yellow, and 1/16

were wrinkled-green. Finding in every case that each of his seven traits was inherited independently of the others, he formed his “second rule”, the Law of Independent Assortment, which states the inheritance of one pair of factors (genes) is independent of the inheritance of the other pair. Today we know that this rule holds only if the genes are on separate chromosomes

12.3B: Mendel's Law of Dominance

In a heterozygote, the allele which masks the other is referred to as dominant, while the allele that is masked is referred to as recessive.

Learning Objectives

- Explain the concept of dominance versus recessiveness

Key Points

- Dominant alleles are expressed exclusively in a heterozygote, while recessive traits are expressed only if the organism is homozygous for the recessive allele.
- A single allele may be dominant over one allele, but recessive to another.
- Not all traits are controlled by simple dominance as a form of inheritance; more complex forms of inheritance have been found to exist.

Key Terms

- **dominant**: a relationship between alleles of a gene, in which one allele masks the expression (phenotype) of another allele at the same locus
- **recessive**: able to be covered up by a dominant trait

Alleles Can Be Dominant or Recessive

Most familiar animals and some plants have paired chromosomes and are described as diploid. They have two versions of each chromosome: one contributed by the female parent in her ovum and one by the male parent in his sperm. These are joined at fertilization. The ovum and sperm cells (the gametes) have only one copy of each chromosome and are described as haploid.

Figure $12.3B.1$: **Recessive traits are only visible if an individual inherits two copies of the recessive allele**: The child in the photo expresses albinism, a recessive trait.

Mendel's law of dominance states that in a heterozygote, one trait will conceal the presence of another trait for the same characteristic. Rather than both alleles contributing to a phenotype, the dominant allele will be expressed exclusively. The recessive allele will remain "latent," but will be transmitted to offspring by the same manner in which the dominant allele

is transmitted. The recessive trait will only be expressed by offspring that have two copies of this allele; these offspring will breed true when self-crossed.

By definition, the terms dominant and recessive refer to the genotypic interaction of alleles in producing the phenotype of the heterozygote. The key concept is genetic: which of the two alleles present in the heterozygote is expressed, such that the organism is phenotypically identical to one of the two homozygotes. It is sometimes convenient to talk about the trait corresponding to the dominant allele as the dominant trait and the trait corresponding to the hidden allele as the recessive trait. However, this can easily lead to confusion in understanding the concept as phenotypic. For example, to say that "green peas" dominate "yellow peas" confuses inherited genotypes and expressed phenotypes. This will subsequently confuse discussion of the molecular basis of the phenotypic difference. Dominance is not inherent. One allele can be dominant to a second allele, recessive to a third allele, and codominant to a fourth. If a genetic trait is recessive, a person needs to inherit two copies of the gene for the trait to be expressed. Thus, both parents have to be carriers of a recessive trait in order for a child to express that trait.

Since Mendel's experiments with pea plants, other researchers have found that the law of dominance does not always hold true. Instead, several different patterns of inheritance have been found to exist.

12.3C: Mendel's Law of Segregation

Mendel's Law of Segregation states that a diploid organism passes a randomly selected allele for a trait to its offspring, such that the offspring receives one allele from each parent.

Learning Objectives

- Apply the law of segregation to determine the chances of a particular genotype arising from a genetic cross

Key Points

- Each gamete acquires one of the two alleles as chromosomes separate into different gametes during meiosis.
- Heterozygotes, which posess one dominant and one recessive allele, can receive each allele from either parent and will look identical to homozygous dominant individuals; the Law of Segregation supports Mendel's observed 3:1 phenotypic ratio.
- Mendel proposed the Law of Segregation after observing that pea plants with two different traits produced offspring that all expressed the dominant trait, but the following generation expressed the dominant and recessive traits in a 3:1 ratio.

Key Terms

- **law of segregation**: a diploid individual possesses a pair of alleles for any particular trait and each parent passes one of these randomly to its offspring

Equal Segregation of Alleles

Observing that true-breeding pea plants with contrasting traits gave rise to F_1 generations that all expressed the dominant trait and F_2 generations that expressed the dominant and recessive traits in a 3:1 ratio, Mendel proposed the law of segregation. The law of segregation states that each individual that is a diploid has a pair of alleles (copy) for a particular trait. Each parent passes an allele at random to their offspring resulting in a diploid organism. The allele that contains the dominant trait determines the phenotype of the offspring. In essence, the law states that copies of genes separate or segregate so that each gamete receives only one allele.

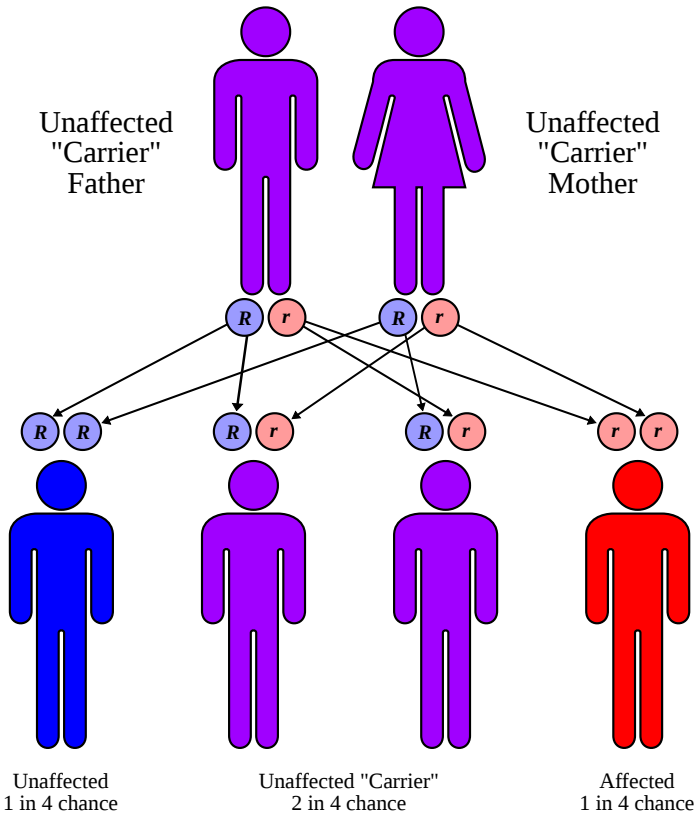

Figure 12.3C.1: **The Law of Segregation states that alleles segregate randomly into gametes**: When gametes are formed, each allele of one parent segregates randomly into the gametes, such that half of the parent's gametes carry each allele.

For the F_2 generation of a monohybrid cross, the following three possible combinations of genotypes could result: homozygous dominant, heterozygous, or homozygous recessive. Because heterozygotes could arise from two different pathways (receiving one dominant and one recessive allele from either parent), and because heterozygotes and homozygous dominant individuals are phenotypically identical, the law supports Mendel's observed 3:1 phenotypic ratio. The equal segregation of alleles is the reason we can apply the Punnett square to accurately predict the offspring of parents with known genotypes.

The physical basis of Mendel's law of segregation is the first division of meiosis in which the homologous chromosomes with their different versions of each gene are segregated into daughter nuclei. The behavior of homologous chromosomes during meiosis can account for the segregation of the alleles at each genetic locus to different gametes. As chromosomes separate into different gametes during meiosis, the two different alleles for a particular gene also segregate so that each gamete acquires one of the two alleles. In Mendel's experiments, the segregation and the independent assortment during meiosis in the F1 generation give rise to the F2 phenotypic ratios observed by Mendel. The role of the meiotic segregation of chromosomes in sexual reproduction was not understood by the scientific community during Mendel's lifetime.

12.3D: Mendel's Law of Independent Assortment

Independent assortment allows the calculation of genotypic and phenotypic ratios based on the probability of individual gene combinations.

Learning Objectives

- Use the probability or forked line method to calculate the chance of any particular genotype arising from a genetic cross

Key Points

- Mendel's law of independent assortment states that genes do not influence each other with regard to the sorting of alleles into gametes; every possible combination of alleles for every gene is equally likely to occur.
- The calculation of any particular genotypic combination of more than one gene is, therefore, the probability of the desired genotype at the first locus multiplied by the probability of the desired genotype at the other loci.
- The forked line method can be used to calculate the chances of all possible genotypic combinations from a cross, while the probability method can be used to calculate the chance of any one particular genotype that might result from that cross.

Key Terms

- **independent assortment**: separate genes for separate traits are passed independently of one another from parents to offspring

Independent Assortment

Mendel's law of independent assortment states that genes do not influence each other with regard to the sorting of alleles into gametes: every possible combination of alleles for every gene is equally likely to occur. The independent assortment of genes can be illustrated by the dihybrid cross: a cross between two true-breeding parents that express different traits for two characteristics. Consider the characteristics of seed color and seed texture for two pea plants: one that has green, wrinkled seeds (yyrr) and another that has yellow, round seeds (YYRR). Because each parent is homozygous, the law of segregation indicates that the gametes for the green/wrinkled plant all are yr, while the gametes for the yellow/round plant are all YR. Therefore, the F_1 generation of offspring all are YyRr.

For the F2 generation, the law of segregation requires that each gamete receive either an R allele or an r allele along with either a Y allele or a y allele. The law of independent assortment states that a gamete into which an r allele sorted would be equally likely to contain either a Y allele or a y allele. Thus, there are four equally likely gametes that can be formed when the YyRr heterozygote is self-crossed as follows: YR, Yr, yR, and yr. Arranging these gametes along the top and left of a 4 × 4 Punnett square gives us 16 equally likely genotypic combinations. From these genotypes, we infer a phenotypic ratio of 9 round/yellow:3 round/green:3 wrinkled/yellow:1 wrinkled/green. These are the offspring ratios we would expect, assuming we performed the crosses with a large enough sample size.

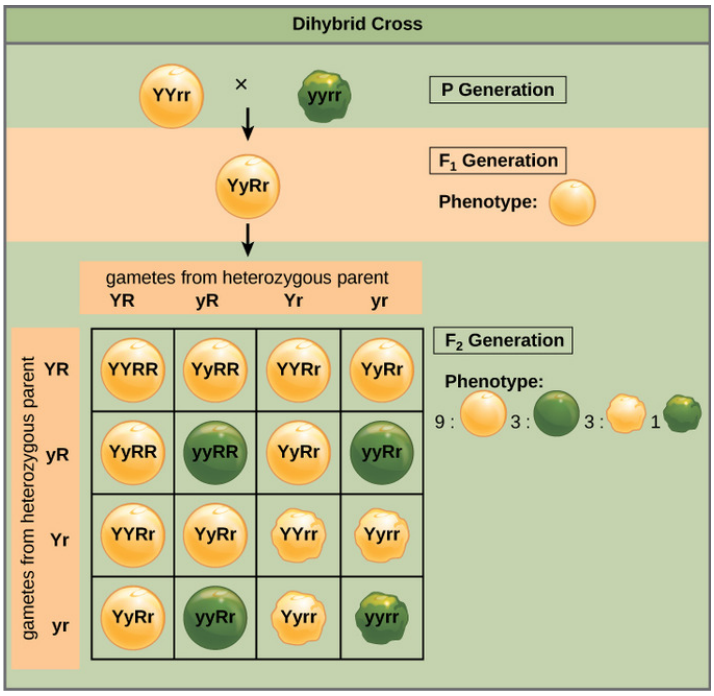

Figure 12.3D. 1: **Independent assortment of 2 genes**: This dihybrid cross of pea plants involves the genes for seed color and texture.

Because of independent assortment and dominance, the 9:3:3:1 dihybrid phenotypic ratio can be collapsed into two 3:1 ratios, characteristic of any monohybrid cross that follows a dominant and recessive pattern. Ignoring seed color and considering only seed texture in the above dihybrid cross, we would expect that three-quarters of the F_2 generation offspring would be round and one-quarter would be wrinkled. Similarly, isolating only seed color, we would assume that three-quarters of the F_2offspring would be yellow and one-quarter would be green. The sorting of alleles for texture and color are independent events, so we can apply the product rule. Therefore, the proportion of round and yellow F_2 offspring is expected to be (3/4) × (3/4) = 9/16, and the proportion of wrinkled and green offspring is expected to be (1/4) × (1/4) = 1/16. These proportions are identical to those obtained using a Punnett square. Round/green and wrinkled/yellow offspring can also be calculated using the product rule as each of these genotypes includes one dominant and one recessive phenotype. Therefore, the proportion of each is calculated as (3/4) × (1/4) = 3/16.

Forked-Line Method

When more than two genes are being considered, the Punnett-square method becomes unwieldy. For instance, examining a cross involving four genes would require a 16 × 16 grid containing 256 boxes. It would be extremely cumbersome to manually enter each genotype. For more complex crosses, the forked-line and probability methods are preferred.

To prepare a forked-line diagram for a cross between F_1 heterozygotes resulting from a cross between AABBCC and aabbcc parents, we first create rows equal to the number of genes being considered and then segregate the alleles in each row on forked lines according to the probabilities for individual monohybrid crosses. We then multiply the values along each forked path to obtain the F_2 offspring probabilities. Note that this process is a diagrammatic version of the product rule. The values along each forked pathway can be multiplied because each gene assorts independently. For a trihybrid cross, the F_2phenotypic ratio is 27:9:9:9:3:3:3:1.

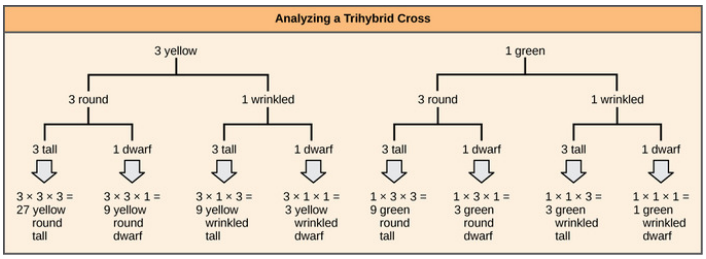

Figure 12.3D.1: **Independent assortment of 3 genes**: The forked-line method can be used to analyze a trihybrid cross. Here, the probability for color in the F2 generation occupies the top row (3 yellow:1 green). The probability for shape occupies the second row (3 round:1 wrinked), and the probability for height occupies the third row (3 tall:1 dwarf). The probability for each possible combination of traits is calculated by multiplying the probability for each individual trait. Thus, the probability of F2 offspring having yellow, round, and tall traits is 3 × 3 × 3, or 27.

Probability Method

While the forked-line method is a diagrammatic approach to keeping track of probabilities in a cross, the probability method gives the proportions of offspring expected to exhibit each phenotype (or genotype) without the added visual assistance.

To fully demonstrate the power of the probability method, however, we can consider specific genetic calculations. For instance, for a tetrahybrid cross between individuals that are heterozygotes for all four genes, and in which all four genes are sorting independently in a dominant and recessive pattern, what proportion of the offspring will be expected to be homozygous recessive for all four alleles? Rather than writing out every possible genotype, we can use the probability method. We know that for each gene the fraction of homozygous recessive offspring will be 1/4. Therefore, multiplying this fraction for each of the four genes, (1/4) × (1/4) × (1/4) × (1/4), we determine that 1/256 of the offspring will be quadruply homozygous recessive.

12.3E: Genetic Linkage and Violation of the Law of Independent Assortment

Genes that are on the same chromosome, or “linked”, do not assort independently, but can be separated by recombination.

Learning Objectives

- Describe how recombination can separate linked genes

Key Points

- Two genes close together on the same chromosome tend to be inherited together and are said to be linked.
- Linked genes can be separated by recombination in which homologous chromosomes exchange genetic information during meiosis; this results in parental, or nonrecombinant genotypes, as well as a smaller proportion of recombinant genotypes.
- Geneticists can use the amount of recombination between genes to estimate the distance between them on a chromosome.

Key Terms

- **linkage**: the property of genes of being inherited together
- **recombination**: the formation of genetic combinations in offspring that are not present in the parents

Linked Genes Violate the Law of Independent Assortment

Although all of Mendel’s pea characteristics behaved according to the law of independent assortment, we now know that some allele combinations are not inherited independently of each other. Genes that are located on separate non-homologous chromosomes will always sort independently. However, each chromosome contains hundreds or thousands of genes organized linearly on chromosomes like beads on a string. The segregation of alleles into gametes can be influenced by linkage, in which genes that are located physically close to each other on the same chromosome are more likely to be inherited as a pair. However, because of the process of recombination, or “crossover,” it is possible for two genes on the same chromosome to behave independently, or as if they are not linked. To understand this, let’s consider the biological basis of gene linkage and recombination.

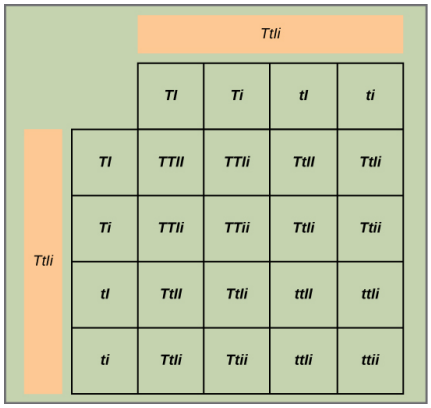

Figure 12.3E.1: **Unlinked genes assort independently**: This figure shows all possible combinations of offspring resulting from a dihybrid cross of pea plants that are heterozygous for the tall/dwarf and inflated/constricted alleles.

Homologous chromosomes possess the same genes in the same linear order. The alleles may differ on homologous chromosome pairs, but the genes to which they correspond do not. In preparation for the first division of meiosis, homologous chromosomes replicate and synapse. Like genes on the homologs align with each other. At this stage, segments of homologous chromosomes exchange linear segments of genetic material. This process is called recombination, or crossover, and it is a common genetic process. Because the genes are aligned during recombination, the gene order is not altered. Instead, the result of recombination is that maternal and paternal alleles are combined onto the same chromosome. Across a given chromosome, several recombination events may occur, causing extensive shuffling of alleles.

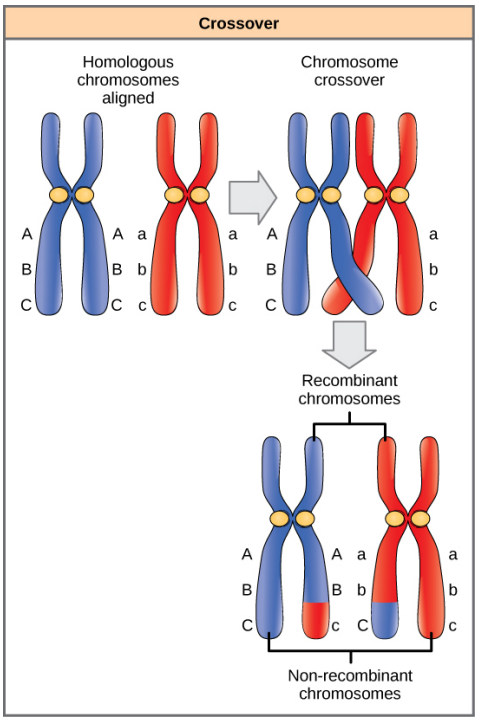

Figure 12.3E. 1: **Linked genes can be separated by recombination**: The process of crossover, or recombination, occurs when two homologous chromosomes align during meiosis and exchange a segment of genetic material. Here, the alleles for gene C were exchanged. The result is two recombinant and two non-recombinant chromosomes.

When two genes are located in close proximity on the same chromosome, they are considered linked, and their alleles tend to be transmitted through meiosis together. To exemplify this, imagine a dihybrid cross involving flower color and plant height in which the genes are next to each other on the chromosome. If one homologous chromosome has alleles for tall plants and red flowers, and the other chromosome has genes for short plants and yellow flowers, then when the gametes are formed, the tall and red alleles will go together into a gamete and the short and yellow alleles will go into other gametes. These are called the parental genotypes because they have been inherited intact from the parents of the individual producing gametes. But unlike if the genes were on different chromosomes, there will be no gametes with tall and yellow alleles and no gametes with short and red alleles. If you create the Punnett square with these gametes, you will see that the classical Mendelian prediction of a 9:3:3:1 outcome of a dihybrid cross would not apply. As the distance between two genes increases, the probability of one or more crossovers between them increases, and the genes behave more like they are on separate chromosomes. Geneticists have used the proportion of recombinant gametes (the ones not like the parents) as a measure of how far apart genes are on a chromosome. Using this information, they have constructed elaborate maps of genes on chromosomes for well-studied organisms, including humans.

Mendel's seminal publication makes no mention of linkage, and many researchers have questioned whether he encountered linkage, but chose not to publish those crosses out of concern that they would invalidate his independent assortment postulate. The garden pea has seven chromosomes and some have suggested that his choice of seven characteristics was not a coincidence. However, even if the genes he examined were not located on separate chromosomes, it is possible that he simply did not observe linkage because of the extensive shuffling effects of recombination.

12.3F: Epistasis

Epistasis occurs when one gene masks or interferes with the expression of another.

Learning Objectives

- Explain the phenotypic outcomes of epistatic effects between genes

Key Points

- In many cases, several genes may contribute to a particular phenotype; when the actions of one gene masks the effects of another, this gene is said to be epistatic to the second.
- Epistasis can occur when a recessive genotype masks the actions of another gene, or when a dominant allele masks the effects of another gene.
- Epistasis can be reciprocal: either gene, when present in the dominant (or recessive) form, expresses the same phenotype.
- Any single characteristic that results in a phenotypic ratio that totals 16 (such as 12:3:1, 9:3:4, or others) is typical of a two-gene interaction.

Key Terms

- **epistasis**: the modification of the expression of a gene by another unrelated one

Epistasis

Mendel's studies in pea plants implied that the sum of an individual's phenotype was controlled by genes (or as he called them, unit factors): every characteristic was distinctly and completely controlled by a single gene. In fact, single observable characteristics are almost always under the influence of multiple genes (each with two or more alleles) acting in unison. For example, at least eight genes contribute to eye color in humans.

In some cases, several genes can contribute to aspects of a common phenotype without their gene products ever directly interacting. In the case of organ development, for instance, genes may be expressed sequentially, with each gene adding to the complexity and specificity of the organ. Genes may function in complementary or synergistic fashions: two or more genes need to be expressed simultaneously to affect a phenotype. Genes may also oppose each other with one gene modifying the expression of another.

In epistasis, the interaction between genes is antagonistic: one gene masks or interferes with the expression of another. "Epistasis" is a word composed of Greek roots that mean "standing upon." The alleles that are being masked or silenced are said to be hypostatic to the epistatic alleles that are doing the masking. Often the biochemical basis of epistasis is a gene pathway in which the expression of one gene is dependent on the function of a gene that precedes or follows it in the pathway.

An example of epistasis is pigmentation in mice. The wild-type coat color, agouti (AA), is dominant to solid-colored fur (aa). However, a separate gene (C) is necessary for pigment production. A mouse with a recessive c allele at this locus is unable to produce pigment and is albino regardless of the allele present at locus A. Therefore, the genotypes AAcc, Aacc, and aacc all produce the same albino phenotype. A cross between heterozygotes for both genes (AaCc x AaCc) would generate offspring with a phenotypic ratio of 9 agouti:3 solid color:4 albino. In this case, the C gene is epistatic to the A gene.

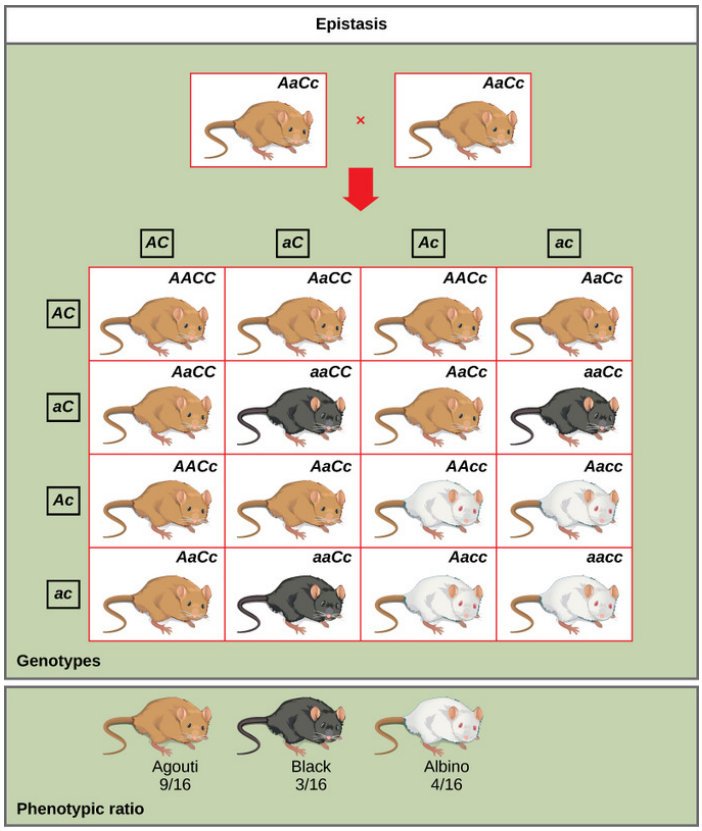

Figure 12.3F.1: **Epistasis in mouse coat color**: In mice, the mottled agouti coat color (A) is dominant to a solid coloration, such as black or gray. A gene at a separate locus (C) is responsible for pigment production. The recessive c allele does not produce pigmentnand a mouse with the homozygous recessive cc genotype is albino regardless of the allele present at the A locus. Thus, the C gene is epistatic to the A gene.

Epistasis can also occur when a dominant allele masks expression at a separate gene. Fruit color in summer squash is expressed in this way. Homozygous recessive expression of the W gene (ww) coupled with homozygous dominant or heterozygous expression of the Y gene (YY or Yy) generates yellow fruit, while the wwyy genotype produces green fruit. However, if a dominant copy of the W gene is present in the homozygous or heterozygous form, the summer squash will produce white fruit regardless of the Y alleles. A cross between white heterozygotes for both genes (WwYy × WwYy) would produce offspring with a phenotypic ratio of 12 white:3 yellow:1 green.

Finally, epistasis can be reciprocal: either gene, when present in the dominant (or recessive) form, expresses the same phenotype. In the shepherd's purse plant (*Capsella bursa-pastoris*), the characteristic of seed shape is controlled by two genes in a dominant epistatic relationship. When the genes A and B are both homozygous recessive (aabb), the seeds are ovoid. If the dominant allele for either of these genes is present, the result is triangular seeds. That is, every possible genotype other than aabb results in triangular seeds; a cross between heterozygotes for both genes (AaBb x AaBb) would yield offspring with a phenotypic ratio of 15 triangular:1 ovoid.

Keep in mind that any single characteristic that results in a phenotypic ratio that totals 16 is typical of a two-gene interaction. Recall the phenotypic inheritance pattern for Mendel's dihybrid cross, which considered two non-interacting genes: 9:3:3:1. Similarly, we would expect interacting gene pairs to also exhibit ratios expressed as 16 parts. Note that we are assuming the interacting genes are not linked; they are still assorting independently into gametes.

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CHAPTER OVERVIEW

13: MODERN UNDERSTANDINGS OF INHERITANCE

13.1: CHROMOSOMAL THEORY AND GENETIC LINKAGE

13.1A: CHROMOSOMAL THEORY OF INHERITANCE
13.1B: GENETIC LINKAGE AND DISTANCES
13.1C: IDENTIFICATION OF CHROMOSOMES AND KARYOTYPES
A karyotype depicts the number, size, and any abnormalities of the chromosomes in an organism.

13.2: CHROMOSOMAL BASIS OF INHERITED DISORDERS

13.2A: DISORDERS IN CHROMOSOME NUMBER
13.2B: CHROMOSOMAL STRUCTURAL REARRANGEMENTS
13.2C: X-INACTIVATION
The presence of extra X chromosomes in a cell is compensated for by X-inactivation in which all but one X chromosome are silenced.

13.1: Chromosomal Theory and Genetic Linkage

Topic hierarchy

- 13.1A: Chromosomal Theory of Inheritance
- 13.1B: Genetic Linkage and Distances
- 13.1C: Identification of Chromosomes and Karyotypes

A karyotype depicts the number, size, and any abnormalities of the chromosomes in an organism.

13.1A: Chromosomal Theory of Inheritance

The Chromosomal Theory of Inheritance identified chromosomes as the genetic material responsible for Mendelian inheritance.

Learning Objectives

- List the reasons that fruit flies are excellent model organisms for genetic research

Key Points

- Homologous chromosome pairs are independent of other chromosome pairs.
- Chromosomes from each homologous pair are sorted randomly into pre- gametes.
- Parents synthesize gametes that contain only half of their chromosomes; eggs and sperm have the same number of chromosomes.
- Gametic chromosomes combine during fertilization to produce offspring with the same chromosome number as their parents.
- Eye color in fruit flies was the first X-linked trait to be discovered; thus, Morgan's experiments with fruit flies solidified the Chromosomal Theory of Inheritance.

Key Terms

- **autosome**: any chromosome other than sex chromosomes
- **hemizygous**: having some single copies of genes in an otherwise diploid cell or organism
- **wild type**: the typical form of an organism, strain, gene or characteristic as it occurs in nature

Chromosomal Theory of Inheritance

The speculation that chromosomes might be the key to understanding heredity led several scientists to examine Mendel's publications and re-evaluate his model in terms of the behavior of chromosomes during mitosis and meiosis. In 1902, Theodor Boveri observed that proper embryonic development of sea urchins does not occur unless chromosomes are present. That same year, Walter Sutton observed the separation of chromosomes into daughter cells during meiosis. Together, these observations led to the development of the Chromosomal Theory of Inheritance, which identified chromosomes as the genetic material responsible for Mendelian inheritance.

(a) (b)

Figure 13.1*A*.1: **Sutton and Boveri**: (a) Walter Sutton and (b) Theodor Boveri are credited with developing the Chromosomal Theory of Inheritance, which states that chromosomes carry the unit of heredity (genes).

The Chromosomal Theory of Inheritance was consistent with Mendel's laws and was supported by the following observations:

- During meiosis, homologous chromosome pairs migrate as discrete structures that are independent of other chromosome pairs.

- The sorting of chromosomes from each homologous pair into pre-gametes appears to be random.
- Each parent synthesizes gametes that contain only half of their chromosomal complement.
- Even though male and female gametes (sperm and egg) differ in size and morphology, they have the same number of chromosomes, suggesting equal genetic contributions from each parent.
- The gametic chromosomes combine during fertilization to produce offspring with the same chromosome number as their parents.

Despite compelling correlations between the behavior of chromosomes during meiosis and Mendel's abstract laws, the Chromosomal Theory of Inheritance was proposed long before there was any direct evidence that traits were carried on chromosomes. Critics pointed out that individuals had far more independently segregating traits than they had chromosomes. It was only after several years of carrying out crosses with the fruit fly, *Drosophila melanogaster*, that Thomas Hunt Morgan provided experimental evidence to support the Chromosomal Theory of Inheritance.

In 1910, Thomas Hunt Morgan started his work with *Drosophila melanogaster*, a fruit fly. He chose fruit flies because they can be cultured easily, are present in large numbers, have a short generation time, and have only four pair of chromosomes that can be easily identified under the microscope. They have three pair of autosomes and a pair of sex chromosomes. At that time, he already knew that X and Y have to do with gender. He used normal flies with red eyes and mutated flies with white eyes and cross bred them. In flies, the wild type eye color is red (XW) and is dominant to white eye color (Xw). He was able to conclude that the gene for eye color was on the X chromosome. This trait was thus determined to be X-linked and was the first X-linked trait to be identified. Males are said to be hemizygous, in that they have only one allele for any X-linked characteristic.

Figure 13.1*A*. 1: **Eye Color in Fruit Flies**: In Drosophila, the gene for eye color is located on the X chromosome. Red eye color is wild type and is dominant to white eye color.

13.1B: Genetic Linkage and Distances

Linked genes can become unlinked during recombination; the probability of genes separating depends on their distance from each other.

Learning Objectives

- Discuss how linked genes can be inherited separately

Key Points

- Alleles positioned on the same chromosome are not always inherited together because during meiosis linked genes can became unlinked.
- Frans Janssen suggested chromosomes become unlinked during homologous recombination, a process where homologous chromosomes exchange segments of DNA.
- Alfred Sturtevant hypothesized that alleles that were closer together on a gene were more likely to be inherited together rather than alleles that were farther apart and used measurements of recombination between genes to create the first genetic map.
- When genes are perfectly linked, they have a recombination frequency of 0.
- When genes are unlinked, they have a recombination frequency of 0.5, which means 50 percent of offspring are recombinants and the other 50 percent are parental types.

Key Terms

- **homologous recombination**: a type of genetic recombination in which nucleotide sequences are exchanged between two similar or identical molecules of DNA
- **linkage**: the property of genes of being inherited together
- **synapsis**: the association of homologous maternal and paternal chromosomes during the initial part of meiosis

Genetic Linkage and Distances

Mendel's work suggested that traits are inherited independently of each other. Morgan identified a 1:1 ratio between a segregating trait and the X chromosome, suggesting that the random segregation of chromosomes was the physical basis of Mendel's model. This also demonstrated that linked genes disrupt Mendel's predicted outcomes. The fact that each chromosome can carry many linked genes explains how individuals can have many more traits than they have chromosomes. However, observations by researchers in Morgan's laboratory suggested that alleles positioned on the same chromosome were not always inherited together. During meiosis, linked genes somehow became unlinked.

Homologous Recombination

In 1909, Frans Janssen observed chiasmata (the point at which chromatids are in contact with each other and may exchange segments) prior to the first division of meiosis. He suggested that alleles become unlinked when chromosomes physically exchange segments. As chromosomes condensed and paired with their homologs, they appeared to interact at distinct points. Janssen suggested that these points corresponded to regions in which chromosome segments were exchanged. It is now known that the pairing and interaction between homologous chromosomes, known as synapsis, does more than simply organize the homologs for migration to separate daughter cells. When synapsed, homologous chromosomes undergo reciprocal physical exchanges of DNA at their arms in a process called homologous recombination, or more simply, "crossing over."

Genetic Maps

In 1913, Alfred Sturtevant, a student in Morgan's laboratory, created the first "chromosome map," a linear representation of gene order and relative distance on a chromosome.To construct a chromosome map, Sturtevant assumed that genes were ordered serially on threadlike chromosomes. He also assumed that the incidence of recombination between two homologous chromosomes could occur with equal likelihood anywhere along the length of the chromosome. Operating under these assumptions, Sturtevant hypothesized alleles that were far apart on a chromosome were more likely to dissociate during meiosis simply because there was a larger region over which recombination could occur. Conversely,

alleles that were close to each other on the chromosome were likely to be inherited together. The average number of crossovers between two alleles, or their recombination frequency, correlated with their genetic distance from each other, relative to the locations of other genes on that chromosome. Sturtevant divided his genetic map into map units, or centimorgans (cM), in which a recombination frequency of 0.01 corresponds to 1 cM.

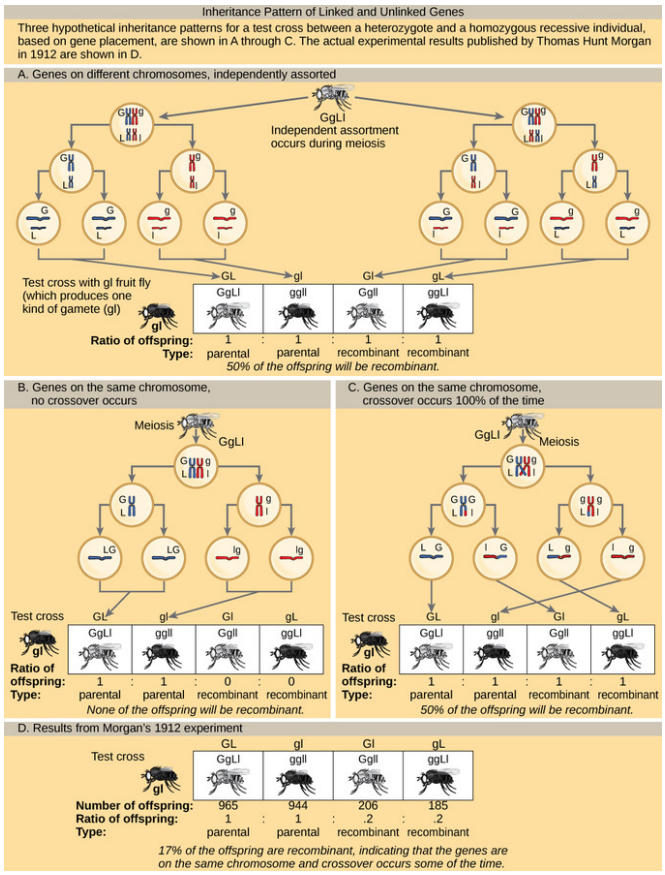

Figure 13.1B.1: **Inheritance Patterns of Unlinked and Linked Genes**: In (a), two genes are located on different chromosomes so independent assortment occurs during meiosis. The offspring have an equal chance of being the parental type (inheriting the same combination of traits as the parents) or a nonparental type (inheriting a different combination of traits than the parents). In (b), two genes are very close together on the same chromosome so that no crossing over occurs between them. The genes are, therefore, always inherited together and all of the offspring are the parental type. In (c), two genes are far apart on the chromosome such that crossing over occurs during every meiotic event. The recombination frequency will be the same as if the genes were on separate chromosomes. (d) The actual recombination frequency of fruit fly wing length and body color that Thomas Morgan observed in 1912 was 17 percent. A crossover frequency between 0 percent and 50 percent indicates that the genes are on the same chromosome and crossover occurs some of the time.

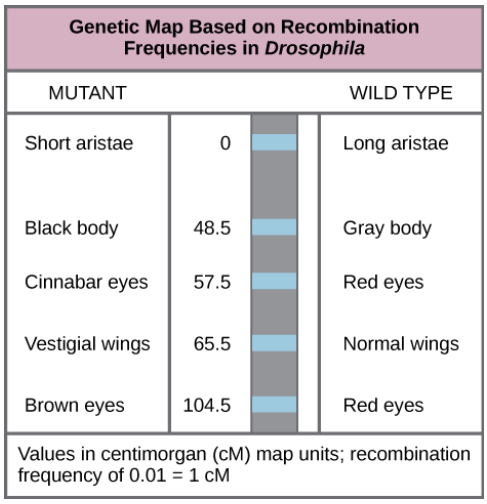

Figure 13.1B.1: **Genetic Maps**: This genetic map orders Drosophila genes on the basis of recombination frequency.

By representing alleles in a linear map, Sturtevant suggested that genes can range from being perfectly linked (recombination frequency = 0) to being perfectly unlinked (recombination frequency = 0.5) when genes are on different chromosomes or genes are separated very far apart on the same chromosome. Perfectly unlinked genes correspond to the frequencies predicted by Mendel to assort independently in a dihybrid cross. A recombination frequency of 0.5 indicates that 50 percent of offspring are recombinants and the other 50 percent are parental types. That is, every type of allele combination is represented with equal frequency. This allowed Sturtevant to calculate distances between several genes on the same chromosome.

13.1C: Identification of Chromosomes and Karyotypes

A karyotype depicts the number, size, and any abnormalities of the chromosomes in an organism.

Learning Objectives

- Describe a normal human karyotype and discuss the various abnormalities that can be detected using this technique

Key Points

- A normal human karyotype contains 23 pairs of chromosomes: 22 pairs of autosomes and 1 pair of sex chromosomes, generally arranged in order from largest to smallest.
- The short arm of a chromosome is referred to as the p arm, while the long arm is designated the q arm.
- To observe a karyotype, cells are collected from a blood or tissue sample and stimulated to begin dividing; the chromosomes are arrested in metaphase, preserved in a fixative and applied to a slide where they are stained with a dye to visualize the distinct banding patterns of each chromosome pair.
- A karyotype can be used to visualize abnormalities in the chromosomes, such as an incorrect number of chromosomes, deletions, insertions, or translocations of DNA.

Key Terms

- **autosome**: any chromosome other than sex chromosomes
- **karyotype**: the observed characteristics (number, type, shape etc) of the chromosomes of an individual or species
- **translocation**: a transfer of a chromosomal segment to a new position, especially on a nonhomologous chromosome

Identification of Chromosomes

The isolation and microscopic observation of chromosomes forms the basis of cytogenetics and is the primary method by which clinicians detect chromosomal abnormalities in humans. A karyotype is the number and appearance of chromosomes. To obtain a view of an individual's karyotype, cytologists photograph the chromosomes and then cut and paste each chromosome into a chart, or karyogram, also known as an ideogram.

In a given species, chromosomes can be identified by their number, size, centromere position, and banding pattern. In a human karyotype, autosomes or "body chromosomes" (all of the non–sex chromosomes) are generally organized in approximate order of size from largest (chromosome 1) to smallest (chromosome 22). However, chromosome 21 is actually shorter than chromosome 22. This was discovered after the naming of Down syndrome as trisomy 21, reflecting how this disease results from possessing one extra chromosome 21 (three total). Not wanting to change the name of this important disease, chromosome 21 retained its numbering, despite describing the shortest set of chromosomes. The X and Y chromosomes are not autosomes and are referred to as the sex chromosomes.

The chromosome "arms" projecting from either end of the centromere may be designated as short or long, depending on their relative lengths. The short arm is abbreviated p (for "petite"), whereas the long arm is abbreviated q (because it follows "p" alphabetically). Each arm is further subdivided and denoted by a number. Using this naming system, locations on chromosomes can be described consistently in the scientific literature.

Although Mendel is referred to as the "father of modern genetics," he performed his experiments with none of the tools that the geneticists of today routinely employ. One such powerful cytological technique is karyotyping, a method in which traits characterized by chromosomal abnormalities can be identified from a single cell. To observe an individual's karyotype, a person's cells (like white blood cells) are first collected from a blood sample or other tissue. In the laboratory, the isolated cells are stimulated to begin actively dividing. A chemical called colchicine is then applied to cells to arrest condensed chromosomes in metaphase. Cells are then made to swell using a hypotonic solution so the chromosomes spread apart. Finally, the sample is preserved in a fixative and applied to a slide.

The geneticist then stains chromosomes with one of several dyes to better visualize the distinct and reproducible banding patterns of each chromosome pair. Following staining, the chromosomes are viewed using bright-field microscopy. A common stain choice is the Giemsa stain. Giemsa staining results in approximately 400–800 bands (of tightly coiled DNA and condensed proteins) arranged along all of the 23 chromosome pairs. An experienced geneticist can identify each

osome based on its characteristic banding pattern. In addition to the banding patterns, chromosomes are further ied on the basis of size and centromere location. To obtain the classic depiction of the karyotype in which homologous pairs of chromosomes are aligned in numerical order from longest to shortest, the geneticist obtains a digital image, identifies each chromosome, and manually arranges the chromosomes into this pattern.

Figure 13.1C.1: **A human karyotype**: This karyotype is of a male human. Notice that homologous chromosomes are the same size, and have the same centromere positions and banding patterns. A human female would have an XX chromosome pair instead of the XY pair shown.

At its most basic, the karyotype may reveal genetic abnormalities in which an individual has too many or too few chromosomes per cell. Examples of this are Down Syndrome, which is identified by a third copy of chromosome 21, and Turner Syndrome, which is characterized by the presence of only one X chromosome in women instead of the normal two. Geneticists can also identify large deletions or insertions of DNA. For instance, Jacobsen Syndrome, which involves distinctive facial features as well as heart and bleeding defects, is identified by a deletion on chromosome 11. Finally, the karyotype can pinpoint translocations, which occur when a segment of genetic material breaks from one chromosome and reattaches to another chromosome or to a different part of the same chromosome. Translocations are implicated in certain cancers, including chronic myelogenous leukemia.

During Mendel's lifetime, inheritance was an abstract concept that could only be inferred by performing crosses and observing the traits expressed by offspring. By observing a karyotype, today's geneticists can actually visualize the chromosomal composition of an individual to confirm or predict genetic abnormalities in offspring, even before birth.

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13.2: Chromosomal Basis of Inherited Disorders

Topic hierarchy

- 13.2A: Disorders in Chromosome Number
- 13.2B: Chromosomal Structural Rearrangements
- 13.2C: X-Inactivation

The presence of extra X chromosomes in a cell is compensated for by X-inactivation in which all but one X chromosome are silenced.

13.2A: Disorders in Chromosome Number

Aneuploidy, an abnormal number of chromosomes in a cell, is caused by nondisjunction, or the failure of chromosomes to separate at meiosis.

Learning Objectives

- Define aneuploidy and explain how this condition results from nondisjunction

Key Points

- Aneuploidy is caused by nondisjunction, which occurs when pairs of homologous chromosomes or sister chromatids fail to separate during meiosis.
- The loss of a single chromosome from a diploid genome is called monosomy (2n-1), while the gain of one chromosome is called trisomy (2n+1).
- If homologous chromosomes fail to separate during meiosis I, the result is no gametes with the normal number (one) of chromosomes.
- If sister chromatids fail to separate during meiosis II, the result is two normal gametes each with one copy of the chromosome, and two abnormal gametes in which one carries two copies and the other carries none.
- Aneuploidy can be lethal or result in serious developmental disorders such as Turner Syndrome (X monosomy) or Downs Syndrome (trisomy 21).

Key Terms

- **aneuploidy**: the state of possessing a chromosome number that is not an exact multiple of the haploid number
- **nondisjunction**: the failure of chromosome pairs to separate properly during meiosis

Disorders in Chromosome Number

Of all of the chromosomal disorders, abnormalities in chromosome number are the most obviously identifiable from a karyotype and are referred to as aneuploidy. Aneuploidy is a condition in which one or more chromosomes are present in extra copies or are deficient in number, but not a complete set. To be more specific, the loss of a single chromosome from a diploid genome is called monosomy (2n-1). The gain of one chromosome is called trisomy (2n+1).They are caused by nondisjunction, which occurs when pairs of homologous chromosomes or sister chromatids fail to separate during meiosis. Misaligned or incomplete synapsis, or a dysfunction of the spindle apparatus that facilitates chromosome migration, can cause nondisjunction. The risk of nondisjunction occurring increases with the age of the parents.

Nondisjunction can occur during either meiosis I or II, with differing results. If homologous chromosomes fail to separate during meiosis I, the result is two gametes that lack that particular chromosome and two gametes with two copies of the chromosome. If sister chromatids fail to separate during meiosis II, the result is one gamete that lacks that chromosome, two normal gametes with one copy of the chromosome, and one gamete with two copies of the chromosome. If a gamete with two copies of the chromosome combines with a normal gamete during fertilization, the result is trisomy; if a gamete with no copies of the chromosomes combines with a normal gamete during fertilization, the result is monosomy.

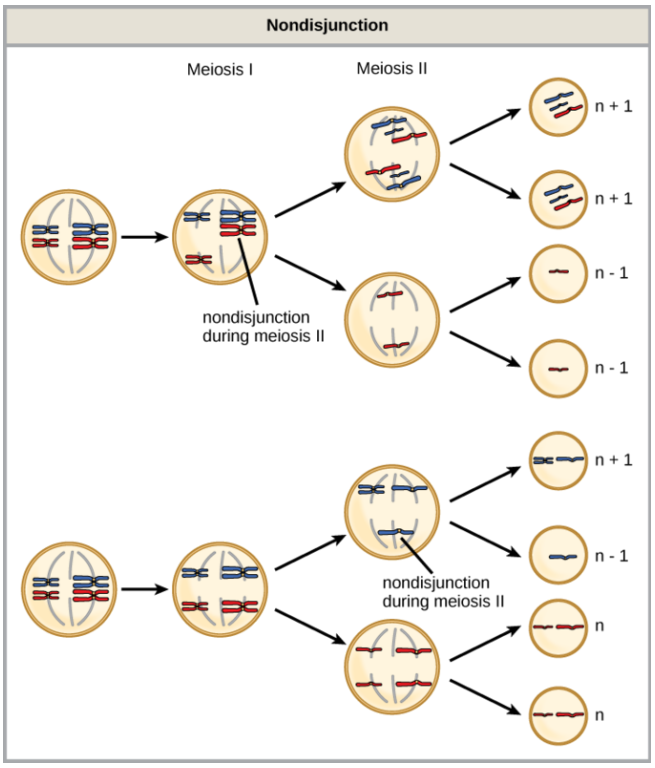

Figure 13.2*A*. 1: **Nondisjunction in Meiosis**: Nondisjunction occurs when homologous chromosomes or sister chromatids fail to separate during meiosis, resulting in an abnormal chromosome number. Nondisjunction may occur during meiosis I or meiosis II.

Aneuploidy often results in serious problems such as Turner syndrome, a monosomy in which females may contain all or part of an X chromosome. Monosomy for autosomes is usually lethal in humans and other animals. Klinefelter syndrome is a trisomy genetic disorder in males caused by the presence of one or more X chromosomes. The effects of trisomy are similar to those of monosomy. Down syndrome is the only autosomal trisomy in humans that has a substantial number of survivors one year after birth. Trisomy in chromosome 21 is the cause of Down syndrome; it affects 1 infant in every 800 live births.

13.2B: Chromosomal Structural Rearrangements

Structural rearrangements of chromosomes include both inversions and translocations, which may have detrimental effects on an organism.

Learning Objectives

- Describe the various types of structural rearrangements of chromosomes and how they can impact an organism

Key Points

- A chromosome inversion is the detachment, 180° rotation, and reinsertion of part of a chromosome; this may have no effect on the organism, but if the inversion occurs within a gene or moves a gene away from its regulatory elements it can have an adverse effect.
- Pericentric inversions include the centromere, while paracentric inversions occur outside of the centromere; a pericentric inversion can change the length of the chromosome arms above and below the centromere.
- A pericentric inversion on chromsome 18 appears to have been involved in the evolution of humans.
- A translocation occurs when a segment of a chromosome dissociates and reattaches to a different, nonhomologous chromosome and can be benign or detrimental; in reciprocal translocations, there is no gain or loss of genetic information, so these are usually benign.

Key Terms

- **inversion**: a segment of DNA in the context of a chromosome that is reversed in orientation relative to a reference karyotype or genome
- **translocation**: a transfer of a chromosomal segment to a new position, especially on a nonhomologous chromosome

Chromosomal Structural Rearrangements

Cytologists have characterized numerous structural rearrangements in chromosomes, but chromosome inversions and translocations are the most common. Both are identified during meiosis by the adaptive pairing of rearranged chromosomes with their former homologs to maintain appropriate gene alignment. If the genes carried on two homologs are not oriented correctly, a recombination event could result in the loss of genes from one chromosome and the gain of genes on the other. This would produce aneuploid gametes.

Chromosome Inversions

A chromosome inversion is the detachment, 180° rotation, and reinsertion of part of a chromosome. Inversions may occur in nature as a result of mechanical shear, or from the action of transposable elements (special DNA sequences capable of facilitating the rearrangement of chromosome segments with the help of enzymes that cut and paste DNA sequences). Unless they disrupt a gene sequence, inversions only change the orientation of genes and are likely to have milder effects than aneuploid errors. However, altered gene orientation can result in functional changes because regulators of gene expression could be moved out of position with respect to their targets, causing aberrant levels of gene products.

An inversion can be pericentric and include the centromere, or paracentric and occur outside of the centromere. A pericentric inversion that is asymmetric about the centromere can change the relative lengths of the chromosome arms, making these inversions easily identifiable.

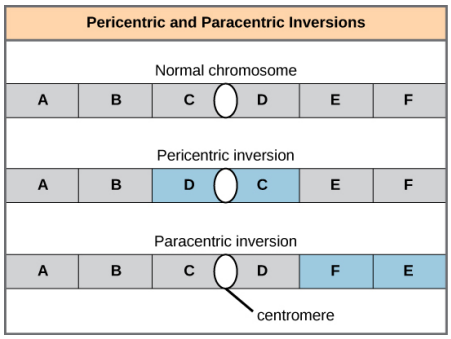

Figure 13.2B.1: **Inversions can be pericentric or paracentric**: Pericentric inversions include the centromere, and paracentric inversions do not. A pericentric inversion can change the relative lengths of the chromosome arms; a paracentric inversion cannot.

When one homologous chromosome undergoes an inversion, but the other does not, the individual is described as an inversion heterozygote. To maintain point-for-point synapsis during meiosis, one homolog must form a loop, and the other homolog must mold around it. Although this topology can ensure that the genes are correctly aligned, it also forces the homologs to stretch and can be associated with regions of imprecise synapsis.

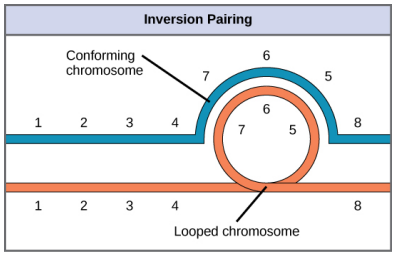

Figure 13.2B.1: **Inversion heterozygotes**: When one chromosome undergoes an inversion, but the other does not, one chromosome must form an inverted loop to retain point-for-point interaction during synapsis. This inversion pairing is essential to maintaining gene alignment during meiosis and to allow for recombination.

Not all structural rearrangements of chromosomes produce nonviable, impaired, or infertile individuals. In rare instances, such a change can result in the evolution of a new species. In fact, a pericentric inversion in chromosome 18 appears to have contributed to the evolution of humans. This inversion is not present in our closest genetic relatives, the chimpanzees. Humans and chimpanzees differ cytogenetically by pericentric inversions on several chromosomes and by the fusion of two separate chromosomes in chimpanzees that correspond to chromosome two in humans.

The pericentric chromosome 18 inversion is believed to have occurred in early humans following their divergence from a common ancestor with chimpanzees approximately five million years ago. Researchers characterizing this inversion have suggested that approximately 19,000 nucleotide bases were duplicated on 18p, and the duplicated region inverted and reinserted on chromosome 18 of an ancestral human.

A comparison of human and chimpanzee genes in the region of this inversion indicates that two genes—ROCK1 and USP14—that are adjacent on chimpanzee chromosome 17 (which corresponds to human chromosome 18) are more distantly positioned on human chromosome 18. This suggests that one of the inversion breakpoints occurred between these two genes. Interestingly, humans and chimpanzees express USP14 at distinct levels in specific cell types, including cortical cells and fibroblasts. Perhaps the chromosome 18 inversion in an ancestral human repositioned specific genes and reset their expression levels in a useful way. Because both ROCK1 and USP14 encode cellular enzymes, a change in their expression could alter cellular function. It is not known how this inversion contributed to hominid evolution, but it appears to be a significant factor in the divergence of humans from other primates.

Translocations

A translocation occurs when a segment of a chromosome dissociates and reattaches to a different, nonhomologous chromosome. Translocations can be benign or have devastating effects depending on how the positions of genes are altered with respect to regulatory sequences. Notably, specific translocations have been associated with several cancers and with schizophrenia. Reciprocal translocations result from the exchange of chromosome segments between two nonhomologous chromosomes such that there is no gain or loss of genetic information.

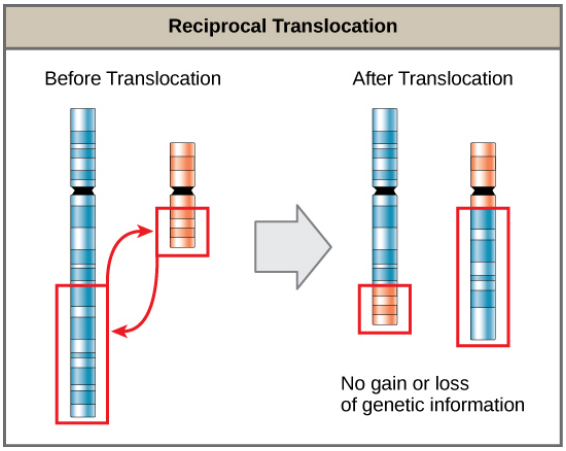

Figure 13.2B.1: **Reciprocal translocations do not involve loss of genetic information**: A reciprocal translocation occurs when a segment of DNA is transferred from one chromosome to another, nonhomologous chromosome.

13.2C: X-Inactivation

The presence of extra X chromosomes in a cell is compensated for by X-inactivation in which all but one X chromosome are silenced.

Learning Objectives

- Explain how and why X inactivation occurs in humans

Key Points

- Extra copies of the X chromosome are silenced by becoming Barr bodies.
- X chromosomal abnormalities are typically associated with mild mental and physical defects, as well as sterility.
- Conditions associated with aneuploidy of the sex chromosomes include individuals with three X chromosomes, called triplo-X; the XXY genotype, known as Klinefelter syndrome; and Turner syndrome, characterized as X monosomy.
- X-inactivation is a form of dosage compensation, in which an organism attempts to equalize the amount of X chromosome gene products in males and females.
- Since males only have one X chromosome, females inactivate one of theirs so that only one X chromosome is active in each gender.

Key Terms

- **dosage compensation**: a genetic regulatory mechanism that equalizes the phenotypic expression of characteristics determined by genes on the X chromosome so that they are equally expressed in males and females.
- **Barr body**: a sex chromosome inactivated by packing in heterochromatin
- **X inactivation**: a process by which one of the two copies of the X chromosome present in female mammals is inactivated

Sex Chromosome Nondisjunction in Humans

Humans display dramatic deleterious effects with autosomal trisomies and monosomies. Therefore, it may seem counterintuitive that human females and males can function normally, despite carrying different numbers of the X chromosome. Rather than a gain or loss of autosomes, variations in the number of X chromosomes are associated with relatively mild effects. In part, this occurs because of a molecular process called X inactivation. Early in development, when female mammalian embryos consist of just a few thousand cells (relative to trillions in the newborn), one X chromosome in each cell inactivates by tightly condensing into a quiescent (dormant) structure called a Barr body. The chance that an X chromosome (maternally or paternally derived) is inactivated in each cell is random, but once the inactivation occurs, all cells derived from that single cell will have the same inactive X chromosome or Barr body.

By this process, a phenomenon called dosage compensation is achieved. Females possess two X chromosomes, while males have only one; therefore, if both X chromosomes remained active in the female, they would produce twice as much product from the genes on the X chromosomes as males.

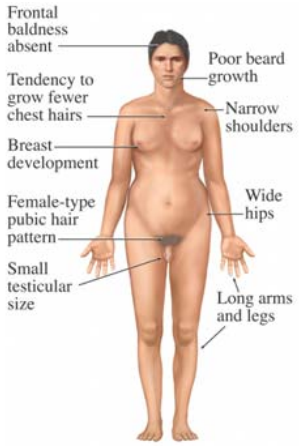

Figure 13.2C.1: **Sex Chromosome Nondisjunction**: The symptoms of Klinefelter's syndrome (XXY) in a human male.

So how does X-inactivation help alleviate the effects of extra X chromosomes? An individual carrying an abnormal number of X chromosomes will inactivate all but one X chromosome in each of her cells. If three X chromosomes are present, the cell will inactivate two of them. If four X chromosomes are present, three will be inactivated, and so on. This results in an individual that is relatively phenotypically normal. However, even inactivated X chromosomes continue to express a few genes, and X chromosomes must reactivate for the proper maturation of female ovaries. As a result, X-chromosomal abnormalities are typically associated with mild mental and physical defects, as well as sterility. If the X chromosome is absent altogether, the individual will not develop in utero.

Several errors in sex chromosome number have been characterized. Individuals with three X chromosomes, called triplo-X, are phenotypically female, but express developmental delays and reduced fertility. The XXY genotype, corresponding to one type of Klinefelter syndrome, corresponds to phenotypically male individuals with small testes, enlarged breasts, and reduced body hair. More complex types of Klinefelter syndrome exist in which the individual has as many as five X chromosomes. In all types, every X chromosome except one undergoes inactivation to compensate for the excess genetic dosage. This can be seen as several Barr bodies in each cell nucleus. Turner syndrome, characterized as an X0 genotype (i.e., only a single sex chromosome), corresponds to a phenotypically female individual with short stature, webbed skin in the neck region, hearing and cardiac impairments, and sterility.

Duplications and Deletions

In addition to the loss or gain of an entire chromosome, a chromosomal segment may be duplicated or lost. Duplications and deletions often produce offspring that survive but exhibit physical and mental abnormalities. Duplicated chromosomal segments may fuse to existing chromosomes or may be free in the nucleus. Cri-du-chat (from the French for "cry of the cat") is a syndrome associated with nervous system abnormalities and identifiable physical features that result from a deletion of most of 5p (the small arm of chromosome 5). Infants with this genotype emit a characteristic high-pitched cry on which the disorder's name is based.

Figure 13.2C.1: **Cri-du-chat Syndrome**: This individual with cri-du-chat syndrome is shown at two, four, nine, and 12 years of age.

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CHAPTER OVERVIEW

14: DNA STRUCTURE AND FUNCTION

14.1: HISTORICAL BASIS OF MODERN UNDERSTANDING

14.1A: DISCOVERY OF DNA
14.1B: MODERN APPLICATIONS OF DNA
14.2: DNA STRUCTURE AND SEQUENCING

14.2A: THE STRUCTURE AND SEQUENCE OF DNA
14.2B: DNA SEQUENCING TECHNIQUES
14.3: DNA REPLICATION

14.3A: BASICS OF DNA REPLICATION
14.3B: DNA REPLICATION IN PROKARYOTES
14.3C: DNA REPLICATION IN EUKARYOTES
14.3D: TELOMERE REPLICATION
14.4: DNA REPAIR

14.4A: DNA REPAIR

: Historical Basis of Modern Understanding

Topic hierarchy

- 14.1A: Discovery of DNA
- 14.1B: Modern Applications of DNA

14.1A: Discovery of DNA

The discovery of DNA is generally credited to Watson and Crick, but many other scientists contributed to the discovery.

Learning Objectives

- Explain the sequence of events leading up to the discovery of the structure of DNA

Key Points

- DNA is one of the most basic molecules of life; it carries genetic instructions for the development, functioning and reproduction of all known living organisms.
- The discovery of DNA's double-helix structure is largely credited to the scientists Watson and Crick, for which they won a Nobel Prize.
- However, the X-ray crystallography work of Rosalind Franklin and Erwin Chargaff's work in discovering the composition of DNA were instrumental to the discovery of DNA's structure.

Key Terms

- **double helix**: The structure formed by double-stranded molecules of nucleic acids such as DNA.

What is DNA?

Deoxyribonucleic acid (DNA) is a molecule that carries most of the genetic instructions used in the development, functioning and reproduction of all known living organisms and many viruses.

DNA is a nucleic acid; alongside proteins and carbohydrates, nucleic acids are one of the three major macromolecules essential for all known forms of life. DNA stores biological information and is involved in the expression of traits in all living organisms.

The Path to Discovery

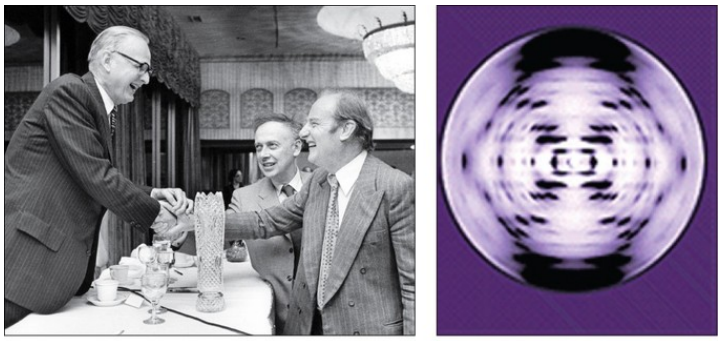
In the 1950s, Francis Crick and James Watson worked together to determine the structure of DNA at the University of Cambridge, England. At the time, other scientists like Linus Pauling and Maurice Wilkins were also actively exploring this field. Pauling had discovered the secondary structure of proteins using X-ray crystallography.

Chargaff's Rule

Erwin Chargaff met Francis Crick and James D. Watson at Cambridge in 1952, and, despite not getting along with them personally, he explained his findings to them. Chargaff's Rule showed that in natural DNA, the number of guanine units equals the number of cytosine units and the number of adenine units equals the number of thymine units. This strongly hinted towards the base pair makeup of the DNA. Chargaff's research would help the Watson and Crick laboratory team to deduce the double helical structure of DNA.

Franklin and X-Ray Diffraction

In Wilkins' lab, researcher Rosalind Franklin used X-ray diffraction methods to understand the structure of DNA. Watson and Crick were able to piece together the puzzle of the DNA molecule on the basis of Franklin's data, because Crick had also studied X-ray diffraction. In 1962, James Watson, Francis Crick, and Maurice Wilkins were awarded the Nobel Prize in Medicine.



(a) (b)

Figure 14.1*A*.1: **A team effort**: The work of pioneering scientists James Watson, Francis Crick, and Maclyn McCarty (pictured at left) led to our present day understanding of DNA. Scientist Rosalind Franklin discovered the X-ray diffraction pattern of DNA (pictured at right), which helped to elucidate its double helix structure. (credit a: modification of work by Marjorie McCarty, Public Library of Science)

Unfortunately by then, Franklin had died. Nobel prizes are not awarded posthumously, and though her work was crucial to the discovery of DNA, Franklin was never nominated for a Nobel Prize. Francis Crick, James Watson, and Maurice Wilkins were awarded a Nobel Prize for the discovery of the structure of DNA in 1962. There is still much controversy on how her image was given to Watson and Crick and why she was not given due credit.

14.1B: Modern Applications of DNA

DNA has many applications in a variety of fields including forensics and medicine.

Learning Objectives

- Explain why DNA is a practical tool in various fields, such as forensics and medicine

Key Points

- DNA is unique to each individual, and therefore can be used for identification purposes.
- The human genome consists of about 3 billion base pairs, corresponding to about 20,000 to 25,000 functional genes.
- Each person's DNA is inherited from their parents: 23 chromosomes from the mother, and 23 chromosomes from the father.

Key Terms

- **genotype**: The combination of alleles, situated on corresponding chromosomes, that determines a specific trait of an individual.
- **zygote**: The single cell that arises from the union of two gametes; in animals, the cell that arises from the union of sperm and ovum.
- **gene**: A unit of heredity; the functional units of chromosomes that determine specific characteristics by coding for specific RNAs or proteins.
- **phenotype**: The appearance of an organism based on a multifactorial combination of genetic traits and environmental factors.

The acronym "DNA" has become synonymous with solving crimes, testing for paternity, identifying human remains, and genetic testing. DNA can be retrieved from hair, blood, or saliva. Each person's DNA sequences are unique, and it is possible to detect differences between individuals within a species on the basis of these unique features. DNA testing can also be used to identify pathogens, identify biological remains in archaeological digs, trace disease outbreaks, and study human migration patterns. In the medical field, DNA is used in diagnostics, new vaccine development, and cancer therapy. It is now also possible to determine predispositions to some diseases by looking at genes.

Cloning

Reproductive cloning is a method used to make a clone or an identical copy of an entire multicellular organism.
In cloning both the original organism and the clone have identical DNA. Identical twins are, in one sense, clones of each other; they have identical DNA, having developed from the same fertilized egg.
Cloning became an issue in scientific ethics when a sheep became the first mammal cloned from an adult cell in 1996.
Since then several animals such as horses, bulls, and goats have been successfully cloned, although these individuals often exhibit facial, limb, and cardiac abnormalities.

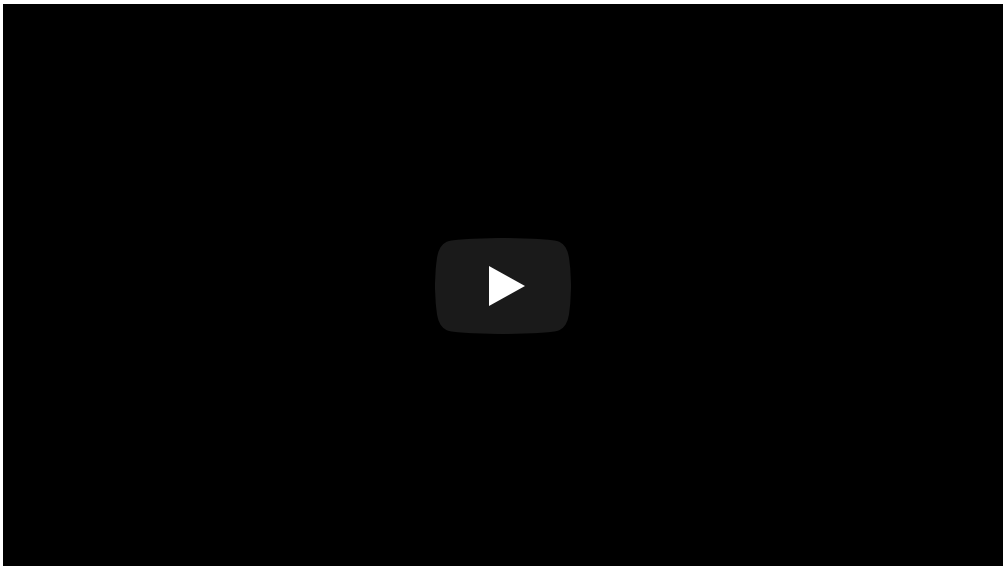
Figure 14.1B.1: **Modern understanding of DNA structure and function has led to cloning**: Dolly the sheep was the first large mammal to be cloned.

There have been attempts at producing cloned human embryos as sources of embryonic stem cells, sometimes referred to as 'cloning for therapeutic purposes'. Therapeutic cloning produces stem cells to attempt to remedy detrimental diseases or

defects (unlike reproductive cloning, which aims to reproduce an organism). Still, therapeutic cloning efforts have met with resistance because of bioethical considerations.

CRISPR

CRISPR (Clustered, Regularly-Interspaced Short Palindromic Repeats) allows scientists to edit genomes, far better than older techniques for gene splicing and editing. The CRISPR technique has enormous potential application, including altering the germline of humans, animals and other organisms, and modifying the genes of food crops.
Ethical concerns have surfaced about this biotechnology and the prospect of editing the human germline and making so-called 'designer babies'.



CRISPR Technique: This movie goes through the process of CRISPR.

Genetically Modified Organisms

A genetically modified organism (GMO) is any organism whose genetic material has been altered using genetic engineering techniques. GMOs are a source of medicines and genetically modified foods and are also widely used in scientific research, along with the production of other goods.
Genetic modification involves the mutation, insertion, or deletion of genes. Inserted genes usually come from a different species in a form of horizontal gene-transfer.

Figure 14.1$B.1$: **Glo Fish**: The GloFish is a patented and trademarked brand of genetically modified (GM) fluorescent fish. A gene that encodes the green fluorescent protein, originally extracted from a jellyfish, that naturally produced bright green fluorescence was inserted into a zebrafish embryo.

Bacteria, plants, and animals have been genetically modified since the early 1970s for academic, medical, agricultural, and industrial purposes. In the US, GMOs such as Roundup-Ready soybeans and borer-resistant corn are part of many common processed foods. As in many of these biotechnology areas there is considerable controversy in the use of GMOs.

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14.2: DNA Structure and Sequencing

Topic hierarchy

- 14.2A: The Structure and Sequence of DNA
- 14.2B: DNA Sequencing Techniques

14.2A: The Structure and Sequence of DNA

DNA is a double helix of two anti-parallel, complementary strands having a phosphate-sugar backbone with nitrogenous bases stacked inside.

Learning Objectives

- Describe the structure of DNA and summarize the importance of DNA sequencing

Key Points

- The two DNA strands are anti-parallel in nature; that is, the 3′ end of one strand faces the 5′ end of the other strand.
- The nucleotides that comprise DNA contain a nitrogenous base, a deoxyribose sugar, and a phosphate group which covalently link with other nucleotides to form phosphodiester bonds.
- Nucleotide bases can be classified as purines (containing a double-ring structure) or pyrimidines (containing a single-ring structure).
- Adenine (purine) and thymine (pyrimidine) are complementary base pairs as are guanine (purine) and cytosine (pyrimidine).
- DNA sequencing is the process of determining the precise order of nucleotides within a DNA molecule.

Key Terms

- **deoxyribose**: a derivative of the pentose sugar ribose in which the 2′ hydroxyl (-OH) is reduced to a hydrogen (H); a constituent of the nucleotides that comprise deoxyribonucleic acid, or DNA
- **hydrogen bond**: A weak bond in which a hydrogen atom already covalently bonded to a oxygen or nitrogen atom in one molecule is attracted to an electronegative atom (usually nitrogen or oxygen) in the same or different molecule.
- **nucleotide**: the monomer comprising DNA or RNA molecules; consists of a nitrogenous heterocyclic base that can be a purine or pyrimidine, a five-carbon pentose sugar, and a phosphate group

The monomeric building blocks of DNA are deoxyribomononucleotides (usually referred to as just nucleotides), and DNA is formed from linear chains, or polymers, of these nucleotides. The components of the nucleotide used in DNA synthesis are a nitrogenous base, a deoxyribose, and a phosphate group. The nucleotide is named depending on which nitrogenous base is present. The nitrogenous base can be a purine such as adenine (A) and guanine (G), characterized by double-ring structures, or a pyrimidine such as cytosine (C) and thymine (T), characterized by single-ring structures. In polynucleotides (the linear polymers of nucleotides) the nucleotides are connected to each other by covalent bonds known as phosphodiester bonds or phosphodiester linkages.

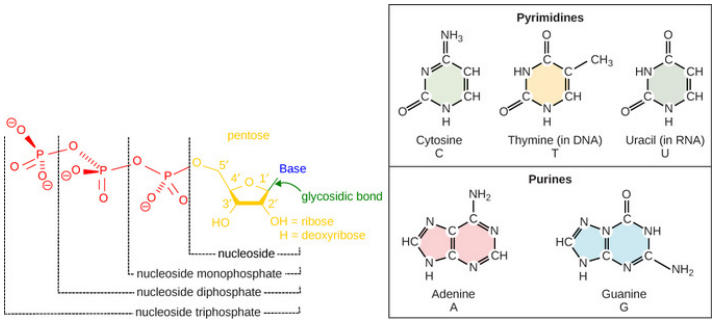

Figure 14.2A.1: **Nucleotide Structure**: Each nucleotide is made up of a sugar, a phosphate group, and a nitrogenous base. The sugar is deoxyribose in DNA and ribose in RNA. In their mononucleotide form, nucleotides can have one, two, or three phosphates attached to them. When linked together in polynucleotide chains, the nucleotides always have just one phosphate. A molecule with just a nitrogenous base and a sugar is known as a nucleoside. Once at least one phosphate is covalently attached, it is known as a nucleotide.

James Watson and Francis Crick, with some help from Rosalind Franklin and Maurice Wilkins, are credited with figuring out the structure of DNA. Watson and Crick proposed that DNA is made up of two polynucleotide strands that are twisted around each other to form a right-handed helix.

The two polynucleotide strands are anti-parallel in nature. That is, they run in opposite directions.

gars and phosphates of the nucleotides form the backbone of the structure, whereas the pairs of nitrogenous bases nted towards the interior of the molecule.

The twisting of the two strands around each other results in the formation of uniformly-spaced major and minor grooves bordered by the sugar-phosphate backbones of the two strands.

Figure 14.2A.1: **Three representations of DNA's double helical structure.**: A is a spacefill model of DNA, where every atom is represented as a sphere. The two anti-parallel polynucleotide strands are colored differently to illustrate how they coil around each other. B is a cartoon model of DNA, where the sugar-phosphate backbones are represented as violet strands and the nitrogenous bases are represented as color-coded rings. C is another spacefill model, with the sugar-phosphate atoms colored violet and all nitrogenous base atoms colored green. The major and minor grooves, which wrap around the entire molecule, are apparent as the spaces between the sugar-phosphate backbones.

The diameter of the DNA double helix is 2 nm and is uniform throughout. Only the pairing between a purine and pyrimidine can explain the uniform diameter. That is to say, at each point along the DNA molecule, the two sugar phosphate backbones are always separated by three rings, two from a purine and one from a pyrimidine.

The two strands are held together by base pairing between nitrogenous bases of one strand and nitrogenous bases from the other strand. Base pairing takes place between a purine and pyrimidine stabilized by hydrogen bonds: A pairs with T via two hydrogen bonds and G pairs with C via three hydrogen bonds.

The interior basepairs rotate with respect to one another, but are also stacked on top of each other when the molecule is viewed looking up or down its long axis.

Each base pair is separated from the previous base pair by a height of 0.34 nm and each 360^{o} turn of the helix travels 3.4 nm along the long axis of the molecule. Therefore, ten base pairs are present per turn of the helix.

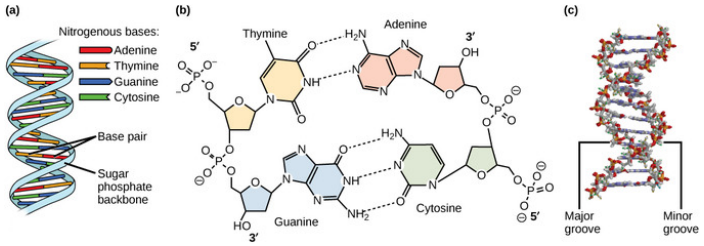

Figure 14.2A.1: **DNA Structure**: DNA has (a) a double helix structure and (b) phosphodiester bonds. The (c) major and minor grooves are binding sites for DNA binding proteins during processes such as transcription (the copying of RNA from DNA) and replication.

DNA sequencing is the process of determining the precise order of nucleotides within a DNA molecule. Rapid DNA sequencing methods has greatly accelerated biological and medical research and discovery. Knowledge of DNA sequences has become indispensable for basic biological research, and in numerous applied fields such as diagnostics, biotechnology, forensic biology, and biological systematics. The rapid speed of sequencing attained with modern technology has been instrumental in obtaining complete DNA sequences, or genomes, of numerous types and species of life, including the human genome and those of other animal, plant, and microbial species.

14.2B: DNA Sequencing Techniques

DNA sequencing techniques are used to determine the order of nucleotides (A,T,C,G) in a DNA molecule.

Learning Objectives

- Differentiate among the techniques used to sequence DNA

Key Points

- Genome sequencing will greatly advance our understanding of genetic biology and has vast potential for medical diagnosis and treatment.
- DNA sequencing technologies have gone through at least three “generations”: Sanger sequencing and Gilbert sequencing were first-generation, pyrosequencing was second-generation, and Illumina sequencing is next-generation.
- Sanger sequencing is based on the use of chain terminators, ddNTPs, that are added to growing DNA strands and terminate synthesis at different points.
- Illumina sequencing involves running up to 500,000,000 different sequencing reactions simultaneously on a single small slide. It makes use of a modified replication reaction and uses fluorescently-tagged nucleotides.
- Shotgun sequencing is a technique for determining the sequence of entire chromosomes and entire genomes based on producing random fragments of DNA that are then assembled by computers which order fragments by finding overlapping ends.

Key Terms

- **DNA sequencing**: a technique used in molecular biology that determines the sequence of nucleotides (A, C, G, and T) in a particular region of DNA
- **dideoxynucleotide**: any nucleotide formed from a deoxynucleotide by loss of an a second hydroxyl group from the deoxyribose group
- **in vitro**: any biochemical process done outside of its natural biological environment, such as in a test tube, petri dish, etc. (from the Latin for “in glass”)

DNA Sequencing Techniques

While techniques to sequence proteins have been around since the 1950s, techniques to sequence DNA were not developed until the mid-1970s, when two distinct sequencing methods were developed almost simultaneously, one by Walter Gilbert’s group at Harvard University, the other by Frederick Sanger’s group at Cambridge University. However, until the 1990s, the sequencing of DNA was a relatively expensive and long process. Using radiolabeled nucleotides also compounded the problem through safety concerns. With currently-available technology and automated machines, the process is cheaper, safer, and can be completed in a matter of hours. The Sanger sequencing method was used for the human genome sequencing project, which was finished its sequencing phase in 2003, but today both it and the Gilbert method have been largely replaced by better methods.

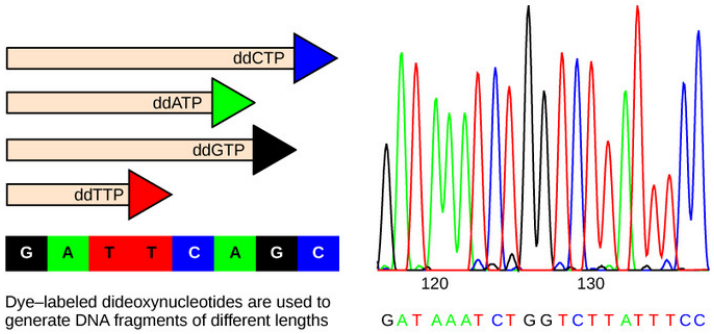

Figure 14.2*B*.1: **Sanger Method**: In Frederick Sanger's dideoxy chain termination method, fluorescent-labeled dideoxynucleotides are used to generate DNA fragments that terminate at each nucleotide along the template strand. The DNA is separated by capillary electrophoresis on the basis of size. From the order of fragments formed, the DNA sequence can be read. The smallest fragments were terminated earliest, and they come out of the column first, so the order in which different fluorescent tags exit the column is also the sequence of the strand. The DNA sequence readout is shown on an electropherogram that is generated by a laser scanner.

Sanger Sequencing

The Sanger method is also known as the dideoxy chain termination method. This sequencing method is based on the use of chain terminators, the dideoxynucleotides (ddNTPs). The dideoxynucleotides, or ddNTPSs, differ from deoxynucleotides by the lack of a free 3′ OH group on the five-carbon sugar. If a ddNTP is added to a growing DNA strand, the chain is not extended any further because the free 3′ OH group needed to add another nucleotide is not available. By using a predetermined ratio of deoxyribonucleotides to dideoxynucleotides, it is possible to generate DNA fragments of different sizes when replicating DNA in vitro.

A Sanger sequencing reaction is just a modified in vitro DNA replication reaction. As such the following components are needed: template DNA (which will the be DNA whose sequence will be determined), DNA Polymerase to catalyze the replication reactions, a primer that basepairs prior to the portion of the DNA you want to sequence, dNTPs, and ddNTPs. The ddNTPs are what distinguish a Sanger sequencing reaction from just a replication reaction. Most of the time in a Sanger sequencing reaction, DNA Polymerase will add a proper dNTP to the growing strand it is synthesizing in vitro. But at random locations, it will instead add a ddNTP. When it does, that strand will be terminated at the ddNTP just added. If enough template DNAs are included in the reaction mix, each one will have the ddNTP inserted at a different random location, and there will be at least one DNA terminated at each different nucleotide along its length for as long as the in vitro reaction can take place (about 900 nucleotides under optimal conditions.)

The ddNTPs which terminate the strands have fluorescent labels covalently attached to them. Each of the four ddNTPs carries a different label, so each different ddNTP will fluoresce a different color.

After the reaction is over, the reaction is subject to capillary electrophoresis. All the newly synthesized fragments, each terminated at a different nucleotide and so each a different length, are separated by size. As each differently-sized fragment exits the capillary column, a laser excites the flourescent tag on its terminal nucleotide. From the color of the resulting flouresence, a computer can keep track of which nucleotide was present as the terminating nucleotide. The computer also keeps track of the order in which the terminating nucleotides appeared, which is the sequence of the DNA used in the original reaction.

Second Generation and Next-generation Sequencing

The Sanger and Gilbert methods of sequencing DNA are often called "first-generation" sequencing because they were the first to be developed. In the late 1990s, new methods, called second-generation sequencing methods, that were faster and cheaper, began to be developed. The most popular, widely-used second-generation sequencing method was one called Pyrosequencing.

Today a number of newer sequencing methods are available and others are in the process of being developed. These are often called next-generation sequencing methods. The most widely-used sequencing method currently is one called Illumina sequencing (after the name of the company which commercialized the technique), but numerous competing methods are in the developmental pipeline and may supplant Illumina sequencing.

In Illumina sequencing, up to 500,000,000 separate sequencing reactions are run simultaneously on a single slide (the size of a microscope slide) put into a single machine. Each reaction is analyzed separately and the sequences generated from all 500 million DNAs are stored in an attached computer. Each sequencing reaction is a modified replication reaction involving flourescently-tagged nucleotides, but no chain-terminating dideoxy nucleotides are needed.

When the human genome was first sequenced using Sanger sequencing, it took several years, hundreds of labs working together, and a cost of around $100 million to sequence it to almost completion. Next generation sequencing can sequence a comparably-sized genome in a matter of days, using a single machine, at a cost of under $10,000. Many researchers have set a goal of improving sequencing methods even more until a single human genome can be sequenced for under $1000.

Shotgun Sequencing

Sanger sequence can only produce several hundred nucleotides of sequence per reaction. Most next-generation sequencing techniques generate even smaller blocks of sequence. Genomes are made up of chromosomes which are tens to hundreds of millions of basepairs long. They can only be sequenced in tiny fragments and the tiny fragments have to put in the correct order to generate the uninterrupted genome sequence. Most genomic sequencing projects today make use of an approach called whole genome shotgun sequencing.

Whole genome shotgun sequencing involves isolating many copies of the chromosomal DNA of interest. The chromosomes are all fragmented into sizes small enough to be sequenced (a few hundred basepairs) at random locations. As a result, each copy of the same chromosome is fragmented at different locations and the fragments from the same part of the chromosome will overlap each other. Each fragment is sequenced and sophisticated computer algorithms compare all the different fragments to find which overlaps with which. By lining up the overlapped regions, a process called tiling, the computer can find the largest possible continuous sequences that can be generated from the fragments. Ultimately, the sequence of entire chromosomes are assembled.

Figure 14.2B.1: **Whole genome shotgun sequencing.**: In shotgun sequencing, multiple copies of the same chromosome are isolated and then fragmented in random locations. The different copies of the chromosome end up generating different length fragments. When the complete collection of fragments has been sequenced, comparing the sequences of all the fragments will reveal which fragments have ends that overlap with other fragments. The complete sequence from one end of the original DNA to the other can be assembled by following the sequence from the first overlapping fragment to the last.

Genome sequencing will greatly advance our understanding of genetic biology. It has vast potential for medical diagnosis and treatment.

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14.3: DNA Replication

Topic hierarchy

- 14.3A: Basics of DNA Replication
- 14.3B: DNA Replication in Prokaryotes
- 14.3C: DNA Replication in Eukaryotes
- 14.3D: Telomere Replication

14.3A: Basics of DNA Replication

DNA replication uses a semi-conservative method that results in a double-stranded DNA with one parental strand and a new daughter strand.

Learning Objectives

- Explain how the Meselson and Stahl experiment conclusively established that DNA replication is semi-conservative.

Key Points

- There were three models suggested for DNA replication: conservative, semi-conservative, and dispersive.
- The conservative method of replication suggests that parental DNA remains together and newly-formed daughter strands are also together.
- The semi-conservative method of replication suggests that the two parental DNA strands serve as a template for new DNA and after replication, each double-stranded DNA contains one strand from the parental DNA and one new (daughter) strand.
- The dispersive method of replication suggests that, after replication, the two daughter DNAs have alternating segments of both parental and newly-synthesized DNA interspersed on both strands.
- Meselson and Stahl, using *E. coli* DNA made with two nitrogen istopes (^{14}N and ^{15}N) and density gradient centrifugation, determined that DNA replicated via the semi-conservative method of replication.

Key Terms

- **DNA replication**: a biological process occuring in all living organisms that is the basis for biological inheritance
- **isotope**: any of two or more forms of an element where the atoms have the same number of protons, but a different number of neutrons within their nuclei

Basics of DNA Replication

Watson and Crick's discovery that DNA was a two-stranded double helix provided a hint as to how DNA is replicated. During cell division, each DNA molecule has to be perfectly copied to ensure identical DNA molecules to move to each of the two daughter cells. The double-stranded structure of DNA suggested that the two strands might separate during replication with each strand serving as a template from which the new complementary strand for each is copied, generating two double-stranded molecules from one.

Models of Replication

There were three models of replication possible from such a scheme: conservative, semi-conservative, and dispersive. In conservative replication, the two original DNA strands, known as the parental strands, would re-basepair with each other after being used as templates to synthesize new strands; and the two newly-synthesized strands, known as the daughter strands, would also basepair with each other; one of the two DNA molecules after replication would be "all-old" and the other would be "all-new". In semi-conservative replication, each of the two parental DNA strands would act as a template for new DNA strands to be synthesized, but after replication, each parental DNA strand would basepair with the complementary newly-synthesized strand just synthesized, and both double-stranded DNAs would include one parental or "old" strand and one daughter or "new" strand. In dispersive replication, after replication both copies of the new DNAs would somehow have alternating segments of parental DNA and newly-synthesized DNA on each of their two strands.

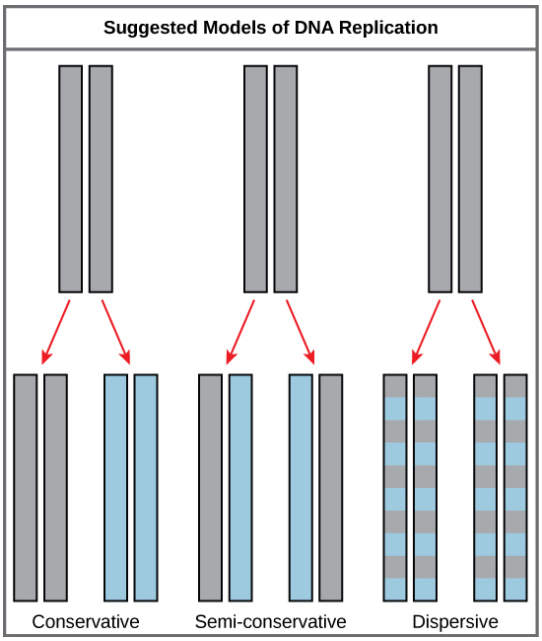


Figure 14.3A.1: **Suggested Models of DNA Replication**: The three suggested models of DNA replication. Grey indicates the original parental DNA strands or segments and blue indicates newly-synthesized daughter DNA strands or segments.

To determine which model of replication was accurate, a seminal experiment was performed in 1958 by two researchers: Matthew Meselson and Franklin Stahl.

Meselson and Stahl

Meselson and Stahl were interested in understanding how DNA replicates. They grew *E. coli* for several generations in a medium containing a "heavy" isotope of nitrogen (^{15}N) that is incorporated into nitrogenous bases and, eventually, into the DNA. The *E. coli* culture was then shifted into medium containing the common "light" isotope of nitrogen (^{14}N) and allowed to grow for one generation. The cells were harvested and the DNA was isolated. The DNA was centrifuged at high speeds in an ultracentrifuge in a tube in which a cesium chloride density gradient had been established. Some cells were allowed to grow for one more life cycle in ^{14}N and spun again.

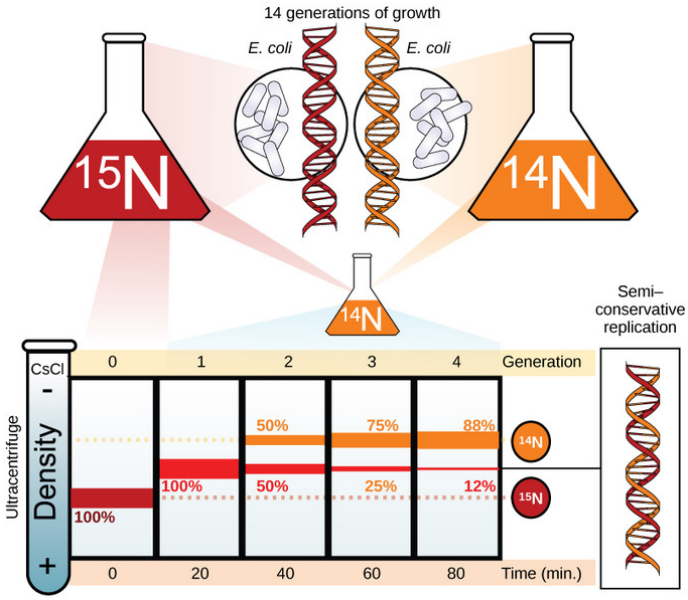


Figure 14.3A.1: **Meselson and Stahl**: Meselson and Stahl experimented with E. coli grown first in heavy nitrogen (^{15}N) then in ligher nitrogen (^{14}N.) DNA grown in ^{15}N (red band) is heavier than DNA grown in ^{14}N (orange band) and sediments to a lower level in the cesium chloride density gradient in an ultracentrifuge. When DNA grown in ^{15}N is switched to media containing ^{14}N, after one round of cell division the DNA sediments halfway between the ^{15}N and ^{14}N levels, indicating that it now contains fifty percent ^{14}N and fifty percent ^{15}N.. In subsequent cell divisions, an increasing amount of DNA contains ^{14}N only. These data support the semi-conservative replication model.

During the density gradient ultracentrifugation, the DNA was loaded into a gradient (Meselson and Stahl used a gradient of cesium chloride salt, although other materials such as sucrose can also be used to create a gradient) and spun at high speeds of 50,000 to 60,000 rpm. In the ultracentrifuge tube, the cesium chloride salt created a density gradient, with the cesium chloride solution being more dense the farther down the tube you went. Under these circumstances, during the spin the DNA was pulled down the ultracentrifuge tube by centrifugal force until it arrived at the spot in the salt gradient where the DNA molecules' density matched that of the surrounding salt solution. At the point, the molecules stopped sedimenting and formed a stable band. By looking at the relative positions of bands of molecules run in the same gradients, you can determine the relative densities of different molecules. The molecules that form the lowest bands have the highest densities.

DNA from cells grown exclusively in ^{15}N produced a lower band than DNA from cells grown exclusively in ^{14}N. So DNA grown in ^{15}N had a higher density, as would be expected of a molecule with a heavier isotope of nitrogen incorporated into its nitrogenous bases. Meselson and Stahl noted that after one generation of growth in ^{14}N (after cells had been shifted from ^{15}N), the DNA molecules produced only single band intermediate in position in between DNA of cells grown exclusively in ^{15}N and DNA of cells grown exclusively in ^{14}N. This suggested either a semi-conservative or dispersive mode of replication. Conservative replication would have resulted in two bands; one representing the parental DNA still with exclusively ^{15}N in its nitrogenous bases and the other representing the daughter DNA with exclusively ^{14}N in its nitrogenous bases. The single band actually seen indicated that all the DNA molecules contained equal amounts of both ^{15}N and ^{14}N.

The DNA harvested from cells grown for two generations in ^{14}N formed two bands: one DNA band was at the intermediate position between ^{15}N and ^{14}N and the other corresponded to the band of exclusively ^{14}N DNA. These results could only be explained if DNA replicates in a semi-conservative manner. Dispersive replication would have resulted in exclusively a single band in each new generation, with the band slowly moving up closer to the height of the ^{14}N DNA band. Therefore, dispersive replication could also be ruled out.

Meselson and Stahl's results established that during DNA replication, each of the two strands that make up the double helix serves as a template from which new strands are synthesized. The new strand will be complementary to the parental or "old" strand and the new strand will remain basepaired to the old strand. So each "daughter" DNA actually consists of one "old" DNA strand and one newly-synthesized strand. When two daughter DNA copies are formed, they have the identical sequences to one another and identical sequences to the original parental DNA, and the two daughter DNAs are divided equally into the two daughter cells, producing daughter cells that are genetically identical to one another and genetically identical to the parent cell.

14.3B: DNA Replication in Prokaryotes

Prokaryotic DNA is replicated by DNA polymerase III in the 5′ to 3′ direction at a rate of 1000 nucleotides per second.

Learning Objectives

- Explain the functions of the enzymes involved in prokaryotic DNA replication

Key Points

- Helicase separates the DNA to form a replication fork at the origin of replication where DNA replication begins.
- Replication forks extend bi-directionally as replication continues.
- Okazaki fragments are formed on the lagging strand, while the leading strand is replicated continuously.
- DNA ligase seals the gaps between the Okazaki fragments.
- Primase synthesizes an RNA primer with a free 3′-OH, which DNA polymerase III uses to synthesize the daughter strands.

Key Terms

- **DNA replication**: a biological process occuring in all living organisms that is the basis for biological inheritance
- **helicase**: an enzyme that unwinds the DNA helix ahead of the replication machinery
- **origin of replication**: a particular sequence in a genome at which replication is initiated

DNA Replication in Prokaryotes

DNA replication employs a large number of proteins and enzymes, each of which plays a critical role during the process. One of the key players is the enzyme DNA polymerase, which adds nucleotides one by one to the growing DNA chain that are complementary to the template strand. The addition of nucleotides requires energy; this energy is obtained from the nucleotides that have three phosphates attached to them, similar to ATP which has three phosphate groups attached. When the bond between the phosphates is broken, the energy released is used to form the phosphodiester bond between the incoming nucleotide and the growing chain. In prokaryotes, three main types of polymerases are known: DNA pol I, DNA pol II, and DNA pol III. DNA pol III is the enzyme required for DNA synthesis; DNA pol I and DNA pol II are primarily required for repair.

There are specific nucleotide sequences called origins of replication where replication begins. In *E. coli*, which has a single origin of replication on its one chromosome (as do most prokaryotes), it is approximately 245 base pairs long and is rich in AT sequences. The origin of replication is recognized by certain proteins that bind to this site. An enzyme called helicase unwinds the DNA by breaking the hydrogen bonds between the nitrogenous base pairs. ATP hydrolysis is required for this process. As the DNA opens up, Y-shaped structures called replication forks are formed. Two replication forks at the origin of replication are extended bi-directionally as replication proceeds. Single-strand binding proteins coat the strands of DNA near the replication fork to prevent the single-stranded DNA from winding back into a double helix. DNA polymerase is able to add nucleotides only in the 5′ to 3′ direction (a new DNA strand can be extended only in this direction). It also requires a free 3′-OH group to which it can add nucleotides by forming a phosphodiester bond between the 3′-OH end and the 5′ phosphate of the next nucleotide. This means that it cannot add nucleotides if a free 3′-OH group is not available. Another enzyme, RNA primase, synthesizes an RNA primer that is about five to ten nucleotides long and complementary to the DNA, priming DNA synthesis. A primer provides the free 3′-OH end to start replication. DNA polymerase then extends this RNA primer, adding nucleotides one by one that are complementary to the template strand.

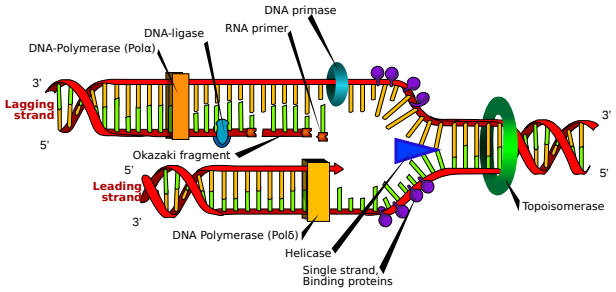


Figure 14.3$B.1$: **DNA Replication in Prokaryotes**: A replication fork is formed when helicase separates the DNA strands at the origin of replication. The DNA tends to become more highly coiled ahead of the replication fork. Topoisomerase breaks and reforms DNA's phosphate backbone ahead of the replication fork, thereby relieving the pressure that results from this supercoiling. Single-strand binding proteins bind to the single-stranded DNA to prevent the helix from re-forming. Primase synthesizes an RNA primer. DNA polymerase III uses this primer to synthesize the daughter DNA strand. On the leading strand, DNA is synthesized continuously, whereas on the lagging strand, DNA is synthesized in short stretches called Okazaki fragments. DNA polymerase I replaces the RNA primer with DNA. DNA ligase seals the gaps between the Okazaki fragments, joining the fragments into a single DNA molecule.

The replication fork moves at the rate of 1000 nucleotides per second. DNA polymerase can only extend in the 5′ to 3′ direction, which poses a slight problem at the replication fork. As we know, the DNA double helix is anti-parallel; that is, one strand is in the 5′ to 3′ direction and the other is oriented in the 3′ to 5′ direction. One strand (the leading strand), complementary to the 3′ to 5′ parental DNA strand, is synthesized continuously towards the replication fork because the polymerase can add nucleotides in this direction. The other strand (the lagging strand), complementary to the 5′ to 3′ parental DNA, is extended away from the replication fork in small fragments known as Okazaki fragments, each requiring a primer to start the synthesis. Okazaki fragments are named after the Japanese scientist who first discovered them.

The leading strand can be extended by one primer alone, whereas the lagging strand needs a new primer for each of the short Okazaki fragments. The overall direction of the lagging strand will be 3′ to 5′, while that of the leading strand will be 5′ to 3′. The sliding clamp (a ring-shaped protein that binds to the DNA) holds the DNA polymerase in place as it continues to add nucleotides. Topoisomerase prevents the over-winding of the DNA double helix ahead of the replication fork as the DNA is opening up; it does so by causing temporary nicks in the DNA helix and then resealing it. As synthesis proceeds, the RNA primers are replaced by DNA. The primers are removed by the exonuclease activity of DNA pol I, while the gaps are filled in by deoxyribonucleotides. The nicks that remain between the newly-synthesized DNA (that replaced the RNA primer) and the previously-synthesized DNA are sealed by the enzyme DNA ligase that catalyzes the formation of phosphodiester linkage between the 3′-OH end of one nucleotide and the 5′ phosphate end of the other fragment.

The table summarizes the enzymes involved in prokaryotic DNA replication and the functions of each.

Prokaryotic DNA Replication: Enzymes and Their Function	
Enzyme/protein	**Specific Function**
DNA pol I	Exonuclease activity removes RNA primer and replaces with newly synthesized DNA
DNA pol II	Repair function
DNA pol III	Main enzyme that adds nucleotides in the 5'-3' direction
Helicase	Opens the DNA helix by breaking hydrogen bonds between the nitrogenous bases
Ligase	Seals the gaps between the Okazaki fragments to create one continuous DNA strand
Primase	Synthesizes RNA primers needed to start replication
Sliding Clamp	Helps to hold the DNA polymerase in place when nucleotides are being added
Topoisomerase	Helps relieve the stress on DNA when unwinding by causing breaks and then resealing the DNA
Single-strand binding proteins (SSB)	Binds to single-stranded DNA to avoid DNA rewinding back.

Figure 14.3$B.1$: **Prokaryotic DNA Replication: Enzymes and Their Function**: The enzymes involved in prokaryotic DNA replication and their functions are summarized on this table.

14.3C: DNA Replication in Eukaryotes

DNA replication in eukaryotes occurs in three stages: initiation, elongation, and termination, which are aided by several enzymes.

Learning Objectives

- Describe how DNA is replicated in eukaryotes

Key Points

- During initiation, proteins bind to the origin of replication while helicase unwinds the DNA helix and two replication forks are formed at the origin of replication.
- During elongation, a primer sequence is added with complementary RNA nucleotides, which are then replaced by DNA nucleotides.
- During elongation the leading strand is made continuously, while the lagging strand is made in pieces called Okazaki fragments.
- During termination, primers are removed and replaced with new DNA nucleotides and the backbone is sealed by DNA ligase.

Key Terms

- **origin of replication**: a particular sequence in a genome at which replication is initiated
- **leading strand**: the template strand of the DNA double helix that is oriented so that the replication fork moves along it in the 3′ to 5′ direction
- **lagging strand**: the strand of the template DNA double helix that is oriented so that the replication fork moves along it in a 5′ to 3′ manner

Because eukaryotic genomes are quite complex, DNA replication is a very complicated process that involves several enzymes and other proteins. It occurs in three main stages: initiation, elongation, and termination.

Initiation

Eukaryotic DNA is bound to proteins known as histones to form structures called nucleosomes. During initiation, the DNA is made accessible to the proteins and enzymes involved in the replication process. There are specific chromosomal locations called origins of replication where replication begins. In some eukaryotes, like yeast, these locations are defined by having a specific sequence of basepairs to which the replication initiation proteins bind. In other eukaryotes, like humans, there does not appear to be a consensus sequence for their origins of replication. Instead, the replication initiation proteins might identify and bind to specific modifications to the nucleosomes in the origin region.

Certain proteins recognize and bind to the origin of replication and then allow the other proteins necessary for DNA replication to bind the same region. The first proteins to bind the DNA are said to “recruit” the other proteins. Two copies of an enzyme called helicase are among the proteins recruited to the origin. Each helicase unwinds and separates the DNA helix into single-stranded DNA. As the DNA opens up, Y-shaped structures called replication forks are formed. Because two helicases bind, two replication forks are formed at the origin of replication; these are extended in both directions as replication proceeds creating a replication bubble. There are multiple origins of replication on the eukaryotic chromosome which allow replication to occur simultaneously in hundreds to thousands of locations along each chromosome.

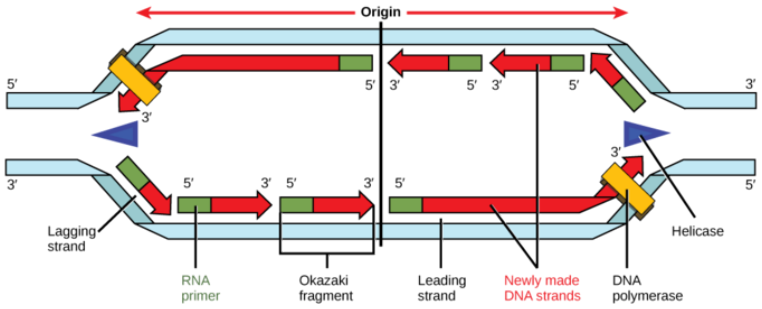


Figure 14.3C.1: **Replication Fork Formation**: A replication fork is formed by the opening of the origin of replication; helicase separates the DNA strands. An RNA primer is synthesized by primase and is elongated by the DNA polymerase. On the leading strand, only a single RNA primer is needed, and DNA is synthesized continuously, whereas on the lagging strand, DNA is synthesized in short stretches, each of which must start with its own RNA primer. The DNA fragments are joined by DNA ligase (not shown).

Elongation

During elongation, an enzyme called DNA polymerase adds DNA nucleotides to the 3′ end of the newly synthesized polynucleotide strand. The template strand specifies which of the four DNA nucleotides (A, T, C, or G) is added at each position along the new chain. Only the nucleotide complementary to the template nucleotide at that position is added to the new strand.

DNA polymerase contains a groove that allows it to bind to a single-stranded template DNA and travel one nucleotide at at time. For example, when DNA polymerase meets an adenosine nucleotide on the template strand, it adds a thymidine to the 3′ end of the newly synthesized strand, and then moves to the next nucleotide on the template strand. This process will continue until the DNA polymerase reaches the end of the template strand.

DNA polymerase cannot initiate new strand synthesis; it only adds new nucleotides at the 3′ end of an existing strand. All newly synthesized polynucleotide strands must be initiated by a specialized RNA polymerase called primase. Primase initiates polynucleotide synthesis and by creating a short RNA polynucleotide strand complementary to template DNA strand. This short stretch of RNA nucleotides is called the primer. Once RNA primer has been synthesized at the template DNA, primase exits, and DNA polymerase extends the new strand with nucleotides complementary to the template DNA.

Eventually, the RNA nucleotides in the primer are removed and replaced with DNA nucleotides. Once DNA replication is finished, the daughter molecules are made entirely of continuous DNA nucleotides, with no RNA portions.

The Leading and Lagging Strands

DNA polymerase can only synthesize new strands in the 5′ to 3′ direction. Therefore, the two newly-synthesized strands grow in opposite directions because the template strands at each replication fork are antiparallel. The “leading strand” is synthesized continuously toward the replication fork as helicase unwinds the template double-stranded DNA.

The “lagging strand” is synthesized in the direction away from the replication fork and away from the DNA helicase unwinds. This lagging strand is synthesized in pieces because the DNA polymerase can only synthesize in the 5′ to 3′ direction, and so it constantly encounters the previously-synthesized new strand. The pieces are called Okazaki fragments, and each fragment begins with its own RNA primer.

Termination

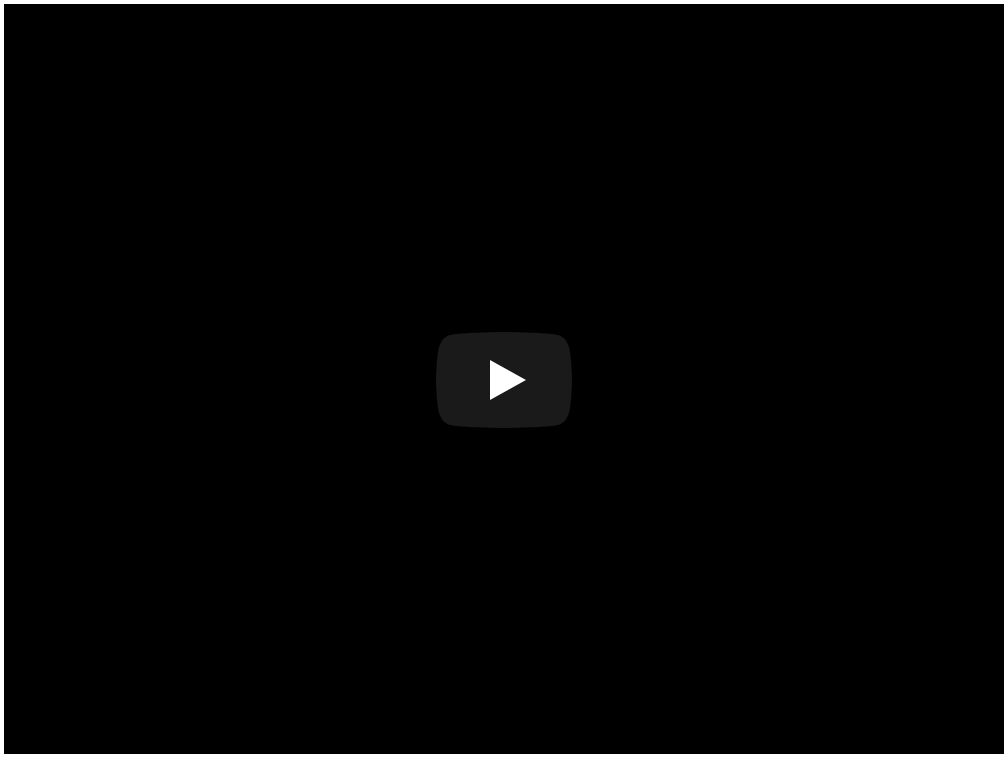
Eukaryotic chromosomes have multiple origins of replication, which initiate replication almost simultaneously. Each origin of replication forms a bubble of duplicated DNA on either side of the origin of replication. Eventually, the leading strand of one replication bubble reaches the lagging strand of another bubble, and the lagging strand will reach the 5′ end of the previous Okazaki fragment in the same bubble.

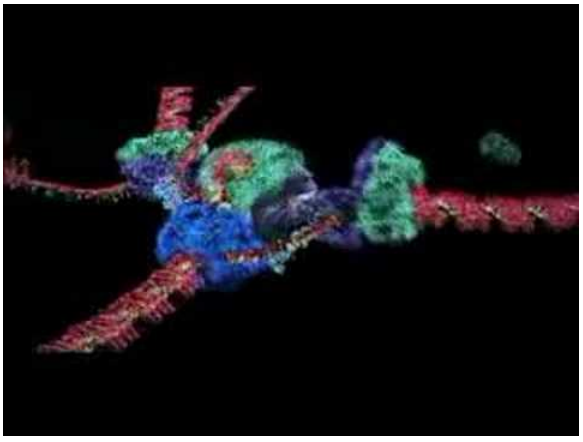
DNA polymerase halts when it reaches a section of DNA template that has already been replicated. However, DNA polymerase cannot catalyze the formation of a phosphodiester bond between the two segments of the new DNA strand, and it drops off. These unattached sections of the sugar-phosphate backbone in an otherwise full-replicated DNA strand are called nicks.

Once all the template nucleotides have been replicated, the replication process is not yet over. RNA primers need to be replaced with DNA, and nicks in the sugar-phosphate backbone need to be connected.

The group of cellular enzymes that remove RNA primers include the proteins FEN1 (flap endonulcease 1) and RNase H. The enzymes FEN1 and RNase H remove RNA primers at the start of each leading strand and at the start of each Okazaki fragment, leaving gaps of unreplicated template DNA. Once the primers are removed, a free-floating DNA polymerase lands at the 3′ end of the preceding DNA fragment and extends the DNA over the gap. However, this creates new nicks (unconnected sugar-phosphate backbone).

In the final stage of DNA replication, the enyzme ligase joins the sugar-phosphate backbones at each nick site. After ligase has connected all nicks, the new strand is one long continuous DNA strand, and the daughter DNA molecule is complete.





DNA Replication: This is a clip from a PBS production called “DNA: The Secret of Life.” It details the latest research (as of 2005) concerning the process of DNA replication.

14.3D: Telomere Replication

As DNA polymerase alone cannot replicate the ends of chromosomes, telomerase aids in their replication and prevents chromosome degradation.

Learning Objectives

- Describe the role played by telomerase in replication of telomeres

Key Points

- DNA polymerase cannot replicate and repair DNA molecules at the ends of linear chromosomes.
- The ends of linear chromosomes, called telomeres, protect genes from getting deleted as cells continue to divide.
- The telomerase enzyme attaches to the end of the chromosome; complementary bases to the RNA template are added on the 3′ end of the DNA strand.
- Once the lagging strand is elongated by telomerase, DNA polymerase can add the complementary nucleotides to the ends of the chromosomes and the telomeres can finally be replicated.
- Cells that undergo cell division continue to have their telomeres shortened because most somatic cells do not make telomerase; telomere shortening is associated with aging.
- Telomerase reactivation in telomerase-deficient mice causes extension of telomeres; this may have potential for treating age-related diseases in humans.

Key Terms

- **telomere**: either of the repetitive nucleotide sequences at each end of a eukaryotic chromosome, which protect the chromosome from degradation
- **telomerase**: an enzyme in eukaryotic cells that adds a specific sequence of DNA to the telomeres of chromosomes after they divide, giving the chromosomes stability over time

The End Problem of Linear DNA Replication

Linear chromosomes have an end problem. After DNA replication, each newly synthesized DNA strand is shorter at its 5′ end than at the parental DNA strand's 5′ end. This produces a 3′ overhang at one end (and one end only) of each daughter DNA strand, such that the two daughter DNAs have their 3′ overhangs at opposite ends

Figure 14.3D.1: **The telomere end problem**: A simplified schematic of DNA replication where the parental DNA (top) is replicated from three origins of replication, yielding three replication bubbles (middle) before giving rise to two daughter DNAs (bottom). Parental DNA strands are black, newly synthesized DNA strands are blue, and RNA primers are red. All RNA primers will be removed by Rnase H and FEN1, leaving gaps in the newly-synthesized DNA strands (not shown.) DNA Polymerase and Ligase will replace all the RNA primers with DNA except the RNA primer at the 5′ ends of each newly-synthesized (blue) strand. This means that each newly-synthesized DNA strand is shorter at its 5′ end than the equivalent strand in the parental DNA.

Every RNA primer synthesized during replication can be removed and replaced with DNA strands except the RNA primer at the 5′ end of the newly synthesized strand. This small section of RNA can only be removed, not replaced with DNA. Enzymes RNase H and FEN1 remove RNA primers, but DNA Polymerase will add new DNA only if the DNA Polymerase has an existing strand 5′ to it (“behind” it) to extend. However, there is no more DNA in the 5′ direction after the final RNA primer, so DNA polymerse cannot replace the RNA with DNA. Therefore, both daughter DNA strands have an incomplete 5′ strand with 3′ overhang.

In the absence of additional cellular processes, nucleases would digest these single-stranded 3′ overhangs. Each daughter DNA would become shorter than the parental DNA, and eventually entire DNA would be lost. To prevent this shortening, the ends of linear eukaryotic chromosomes have special structures called telomeres.

Telomere Replication

The ends of the linear chromosomes are known as telomeres: repetitive sequences that code for no particular gene. These telomeres protect the important genes from being deleted as cells divide and as DNA strands shorten during replication.

In humans, a six base pair sequence, TTAGGG, is repeated 100 to 1000 times. After each round of DNA replication, some telomeric sequences are lost at the 5′ end of the newly synthesized strand on each daughter DNA, but because these are noncoding sequences, their loss does not adversely affect the cell. However, even these sequences are not unlimited. After sufficient rounds of replication, all the telomeric repeats are lost, and the DNA risks losing coding sequences with subsequent rounds.

The discovery of the enzyme telomerase helped in the understanding of how chromosome ends are maintained. The telomerase enzyme attaches to the end of a chromosome and contains a catalytic part and a built-in RNA template. Telomerase adds complementary RNA bases to the 3′ end of the DNA strand. Once the 3′ end of the lagging strand template is sufficiently elongated, DNA polymerase adds the complementary nucleotides to the ends of the chromosomes; thus, the ends of the chromosomes are replicated.

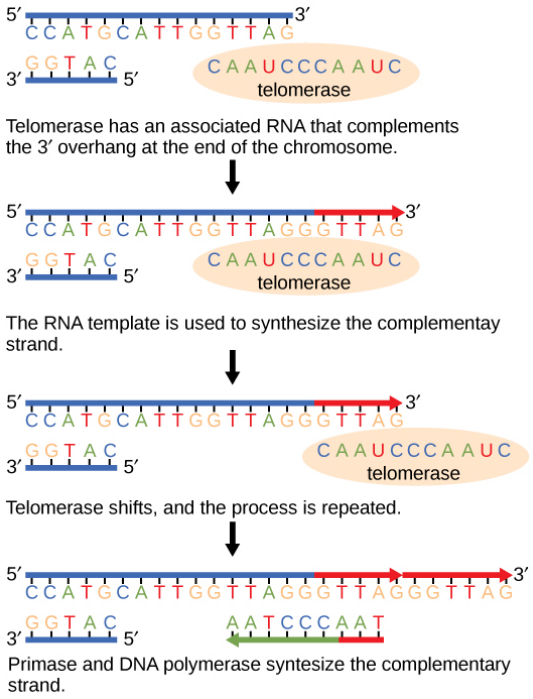


Figure 14.3D.1: **Telomerase is important for maintaining chromosome integrity**: The ends of linear chromosomes are maintained by the action of the telomerase enzyme.

Telomerase and Aging

Telomerase is typically active in germ cells and adult stem cells, but is not active in adult somatic cells. As a result, telomerase does not protect the DNA of adult somatic cells and their telomeres continually shorten as they undergo rounds of cell division.

In 2010, scientists found that telomerase can reverse some age-related conditions in mice. These findings may contribute to the future of regenerative medicine. In the studies, the scientists used telomerase-deficient mice with tissue atrophy, stem cell depletion, organ failure, and impaired tissue injury responses. Telomerase reactivation in these mice caused extension of telomeres, reduced DNA damage, reversed neurodegeneration, and improved the function of the testes, spleen, and intestines. Thus, telomere reactivation may have potential for treating age-related diseases in humans.

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14.4: DNA Repair

Topic hierarchy

- 14.4A: DNA Repair

14.4A: DNA Repair

Most mistakes during replication are corrected by DNA polymerase during replication or by post-replication repair mechanisms.

Learning Objectives

- Explain how errors during replication are repaired

Key Points

- Mismatch repair enzymes recognize mis-incorporated bases, remove them from DNA, and replace them with the correct bases.
- In nucleotide excision repair, enzymes remove incorrect bases with a few surrounding bases, which are replaced with the correct bases with the help of a DNA polymerase and the template DNA.
- When replication mistakes are not corrected, they may result in mutations, which sometimes can have serious consequences.
- Point mutations, one base substituted for another, can be silent (no effect) or may have effects ranging from mild to severe.
- Mutations may also involve insertions (addition of a base), deletion (loss of a base), or translocation (movement of a DNA section to a new location on the same or another chromosome).

Key Terms

- **mismatch repair**: a system for recognizing and repairing some forms of DNA damage and erroneous insertion, deletion, or mis-incorporation of bases that can arise during DNA replication and recombination
- **nucleotide excision repair**: a DNA repair mechanism that corrects damage done by UV radiation, including thymine dimers and 6,4 photoproducts that cause bulky distortions in the DNA

Errors during Replication

DNA replication is a highly accurate process, but mistakes can occasionally occur as when a DNA polymerase inserts a wrong base. Uncorrected mistakes may sometimes lead to serious consequences, such as cancer. Repair mechanisms can correct the mistakes, but in rare cases mistakes are not corrected, leading to mutations; in other cases, repair enzymes are themselves mutated or defective.

Mutations: In this interactive, you can “edit” a DNA strand and cause a mutation. Take a look at the effects!

Most of the mistakes during DNA replication are promptly corrected by DNA polymerase which proofreads the base that has just been added. In proofreading, the DNA pol reads the newly-added base before adding the next one so a correction can be made. The polymerase checks whether the newly-added base has paired correctly with the base in the template strand. If it is the correct base, the next nucleotide is added. If an incorrect base has been added, the enzyme makes a cut at the phosphodiester bond and releases the incorrect nucleotide. This is performed by the exonuclease action of DNA pol III. Once the incorrect nucleotide has been removed, a new one will be added again.

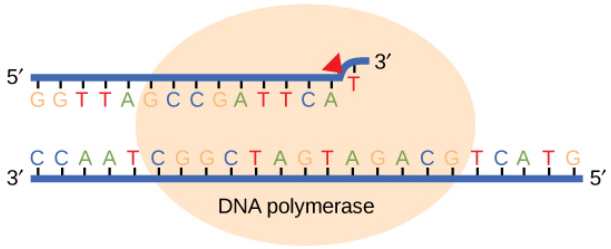

Figure 14.4*A*. 1: **DNA polymerase proofreading**: Proofreading by DNA polymerase corrects errors during replication.

Some errors are not corrected during replication, but are instead corrected after replication is completed; this type of repair is known as mismatch repair. The enzymes recognize the incorrectly-added nucleotide and excise it; this is then replaced by the correct base. If this remains uncorrected, it may lead to more permanent damage. How do mismatch repair enzymes

recognize which of the two bases is the incorrect one? In *E. coli*, after replication, the nitrogenous base adenine acquires a methyl group; the parental DNA strand will have methyl groups, whereas the newly-synthesized strand lacks them. Thus, DNA polymerase is able to remove the incorrectly-incorporated bases from the newly-synthesized, non-methylated strand. In eukaryotes, the mechanism is not very well understood, but it is believed to involve recognition of unsealed nicks in the new strand, as well as a short-term continuing association of some of the replication proteins with the new daughter strand after replication has been completed.

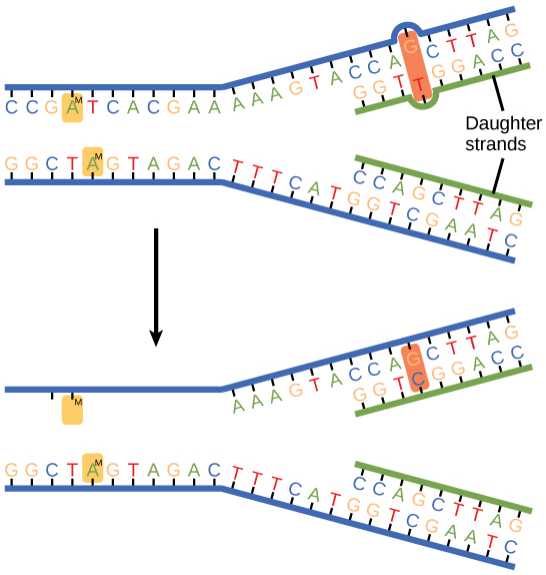

Figure 14.4*A*.1: **Mismatch Repair**: In mismatch repair, the incorrectly-added base is detected after replication. The mismatch-repair proteins detect this base and remove it from the newly-synthesized strand by nuclease action. The gap is now filled with the correctly-paired base.

In another type of repair mechanism, nucleotide excision repair, enzymes replace incorrect bases by making a cut on both the 3′ and 5′ ends of the incorrect base. The segment of DNA is removed and replaced with the correctly-paired nucleotides by the action of DNA pol. Once the bases are filled in, the remaining gap is sealed with a phosphodiester linkage catalyzed by DNA ligase. This repair mechanism is often employed when UV exposure causes the formation of pyrimidine dimers.

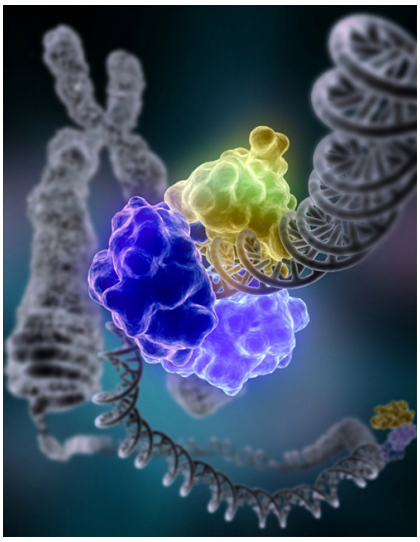

Figure 14.4A.1: **DNA Ligase I Repairing Chromosomal Damage**: DNA damage, due to environmental factors and normal metabolic processes inside the cell, occurs at a rate of 1,000 to 1,000,000 molecular lesions per cell per day. A special enzyme, DNA ligase (shown here in color), encircles the double helix to repair a broken strand of DNA. DNA ligase is responsible for repairing the millions of DNA breaks generated during the normal course of a cell's life. Without molecules that can mend such breaks, cells can malfunction, die, or become cancerous. DNA ligases catalyse the crucial step of joining breaks in duplex DNA during DNA repair, replication and recombination, and require either Adenosine triphosphate (ATP) or Nicotinamide adenine dinucleotide (NAD+) as a cofactor.

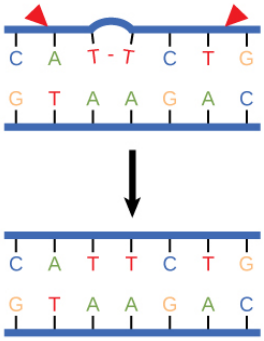

Figure 14.4A.1: **Nucleotide Excision Repairs**: Nucleotide excision repairs thymine dimers. When exposed to UV, thymines lying adjacent to each other can form thymine dimers. In normal cells, they are excised and replaced.

DNA Damage and Mutations

Errors during DNA replication are not the only reason why mutations arise in DNA. Mutations, variations in the nucleotide sequence of a genome, can also occur because of damage to DNA. Such mutations may be of two types: induced or spontaneous. Induced mutations are those that result from an exposure to chemicals, UV rays, X-rays, or some other environmental agent. Spontaneous mutations occur without any exposure to any environmental agent; they are a result of natural reactions taking place within the body.

Mutations may have a wide range of effects. Some mutations are not expressed; these are known as silent mutations. Point mutations are those mutations that affect a single base pair. The most common nucleotide mutations are substitutions, in which one base is replaced by another. These can be of two types: transitions or transversions. Transition substitution refers to a purine or pyrimidine being replaced by a base of the same kind; for example, a purine such as adenine may be replaced by the purine guanine. Transversion substitution refers to a purine being replaced by a pyrimidine or vice versa; for example, cytosine, a pyrimidine, is replaced by adenine, a purine. Mutations can also be the result of the addition of a base, known as an insertion, or the removal of a base, known as a deletion. Sometimes a piece of DNA from one chromosome may get translocated to another chromosome or to another region of the same chromosome.

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CHAPTER OVERVIEW

15: GENES AND PROTEINS

15.1: THE GENETIC CODE

15.1A: THE RELATIONSHIP BETWEEN GENES AND PROTEINS
15.1B: THE CENTRAL DOGMA: DNA ENCODES RNA AND RNA ENCODES PROTEIN
15.2: PROKARYOTIC TRANSCRIPTION

15.2A: TRANSCRIPTION IN PROKARYOTES
15.2B: INITIATION OF TRANSCRIPTION IN PROKARYOTES
15.2C: ELONGATION AND TERMINATION IN PROKARYOTES
15.3: EUKARYOTIC TRANSCRIPTION

15.3A: INITIATION OF TRANSCRIPTION IN EUKARYOTES
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15.4: RNA PROCESSING IN EUKARYOTES

15.4A: MRNA PROCESSING
15.4B: PROCESSING OF TRNAS AND RRNAS
15.5: RIBOSOMES AND PROTEIN SYNTHESIS

15.5A: THE PROTEIN SYNTHESIS MACHINERY
15.5B: THE MECHANISM OF PROTEIN SYNTHESIS
15.5C: PROTEIN FOLDING, MODIFICATION, AND TARGETING

15.1: The Genetic Code

Topic hierarchy

- 15.1A: The Relationship Between Genes and Proteins
- 15.1B: The Central Dogma: DNA Encodes RNA and RNA Encodes Protein

15.1A: The Relationship Between Genes and Proteins

Proteins, encoded by individual genes, orchestrate nearly every function of the cell.

Learning Objectives

- Describe transcription and translation

Key Points

- Genes are composed of DNA arranged on chromosomes.
- Some genes encode structural or regulatory RNAs. Other genes encode proteins.
- Replication copies DNA; transcription uses DNA to make complementary RNAs; translation uses mRNAs to make proteins.
- In eukaryotic cells, replication and transcription take place within the nucleus while translation takes place in the cytoplasm.
- In prokaryotic cells, replication, transcription, and translation occur in the cytoplasm.

Key Terms

- **DNA**: a biopolymer of deoxyribonucleic acids (a type of nucleic acid) that has four different chemical groups, called bases: adenine, guanine, cytosine, and thymine
- **messenger RNA**: Messenger RNA (mRNA) is a molecule of RNA that encodes a chemical “blueprint” for a protein product.
- **protein**: any of numerous large, complex naturally-produced molecules composed of one or more long chains of amino acids, in which the amino acid groups are held together by peptide bonds

Genes and Proteins

Since the rediscovery of Mendel’s work in 1900, the definition of the gene has progressed from an abstract unit of heredity to a tangible molecular entity capable of replication, transcription, translation, and mutation. Genes are composed of DNA and are linearly arranged on chromosomes. Some genes encode structural and regulatory RNAs. There is increasing evidence from research that profiles the transcriptome of cells (the complete set all RNA transcripts present in a cell) that these may be the largest classes of RNAs produced by eukaryotic cells, far outnumbering the protein-encoding messenger RNAs (mRNAs), but the 20,000 protein-encoding genes typically found in animal cells, and the 30,o00 protein-encoding genes typically found in plant cells, nonetheless have huge impacts on cellular functioning.

Protein-encoding genes specify the sequences of amino acids, which are the building blocks of proteins. In turn, proteins are responsible for orchestrating nearly every function of the cell. Both protein-encoding genes and the proteins that are their gene products are absolutely essential to life as we know it.

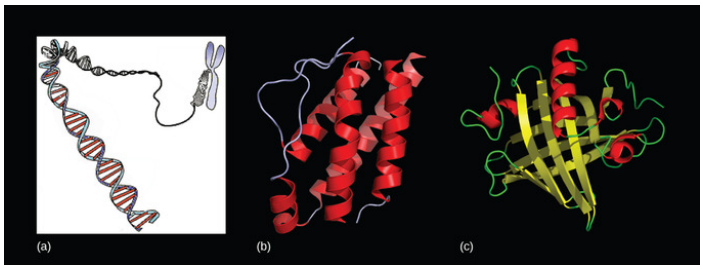


Figure 15.1A.1: **Genes Encode Proteins**: Genes, which are carried on (a) chromosomes, are linearly-organized instructions for making the RNA and protein molecules that are necessary for all of processes of life. The (b) interleukin-2 protein and (c) alpha-2u-globulin protein are just two examples of the array of different molecular structures that are encoded by genes.

Replication, Transcription, and Translation are the three main processes used by all cells to maintain their genetic information and to convert the genetic information encoded in DNA into gene products, which are either RNAs or proteins, depending on the gene. In eukaryotic cells, or those cells that have a nucleus, replication and transcription take

place within the nucleus while translation takes place outside of the nucleus in cytoplasm. In prokaryotic cells, or those cells that do not have a nucleus, all three processes occur in the cytoplasm.

Replication is the basis for biological inheritance. It copies a cell's DNA. The enzyme DNA polymerase copies a single parental double-stranded DNA molecule into two daughter double-stranded DNA molecules. Transcription makes RNA from DNA. The enzyme RNA polymerase creates an RNA molecule that is complementary to a gene-encoding stretch of DNA. Translation makes protein from mRNA. The ribosome generates a polypeptide chain of amino acids using mRNA as a template. The polypeptide chain folds up to become a protein.

15.1B: The Central Dogma: DNA Encodes RNA and RNA Encodes Protein

The central dogma describes the flow of genetic information from DNA to RNA to protein.

Learning Objectives

- Recall the central dogma of biology

Key Points

- The genetic code is degenerate because 64 codons encode only 22 amino acids.
- The genetic code is universal because it is the same among all organisms.
- Replication is the process of copying a molecule of DNA.
- Transcription is the process of converting a specific sequence of DNA into RNA.
- Translation is the process where a ribosome decodes mRNA into a protein.

Key Terms

- **codon**: a sequence of three adjacent nucleotides, which encode for a specific amino acid during protein synthesis or translation
- **ribosome**: protein/mRNA complexes found in all cells that are involved in the production of proteins by translating messenger RNA
- **degenerate**: the redundancy of the genetic code (more than one codon codes for each amino acid)

The Genetic Code Is Degenerate and Universal

The genetic code is degenerate as there are 64 possible nucleotide triplets (4^3), which is far more than the number of amino acids. These nucleotide triplets are called codons; they instruct the addition of a specific amino acid to a polypeptide chain. Sixty-one of the codons encode twenty different amino acids. Most of these amino acids can be encoded by more than one codon. Three of the 64 codons terminate protein synthesis and release the polypeptide from the translation machinery. These triplets are called stop codons. The stop codon UGA is sometimes used to encode a 21st amino acid called selenocysteine (Sec), but only if the mRNA additionally contains a specific sequence of nucleotides called a selenocysteine insertion sequence (SECIS). The stop codon UAG is sometimes used by a few species of microorganisms to encode a 22nd amino acid called pyrrolysine (Pyl). The codon AUG, also has a special function. In addition to specifying the amino acid methionine, it also serves as the start codon to initiate translation. The reading frame for translation is set by the AUG start codon.

The genetic code is universal. With a few exceptions, virtually all species use the same genetic code for protein synthesis. The universal nature of the genetic code is powerful evidence that all of life on Earth shares a common origin.

Figure 15.1$B.1$: **Codons and the universal genetic code.**: The genetic code for translating each nucleotide triplet (codon) in mRNA into an amino acid or a translation termination signal.

The Central Dogma: DNA Encodes RNA, RNA Encodes Protein

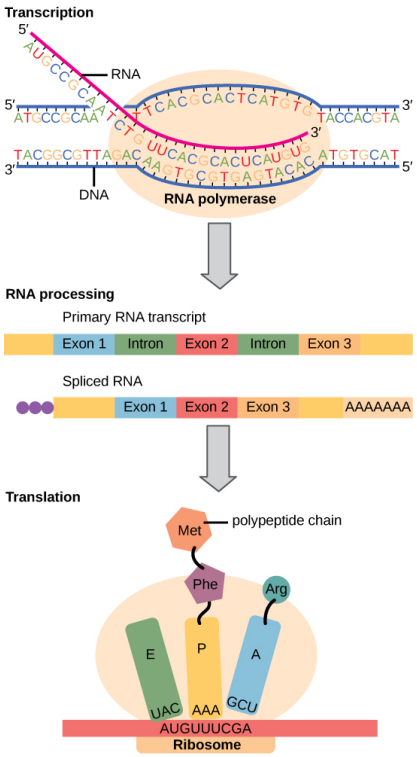


Figure 15.1*B*.1: **The central dogma**: Instructions on DNA are transcribed onto messenger RNA. Ribosomes are able to read the genetic information inscribed on a strand of messenger RNA and use this information to string amino acids together into a protein.

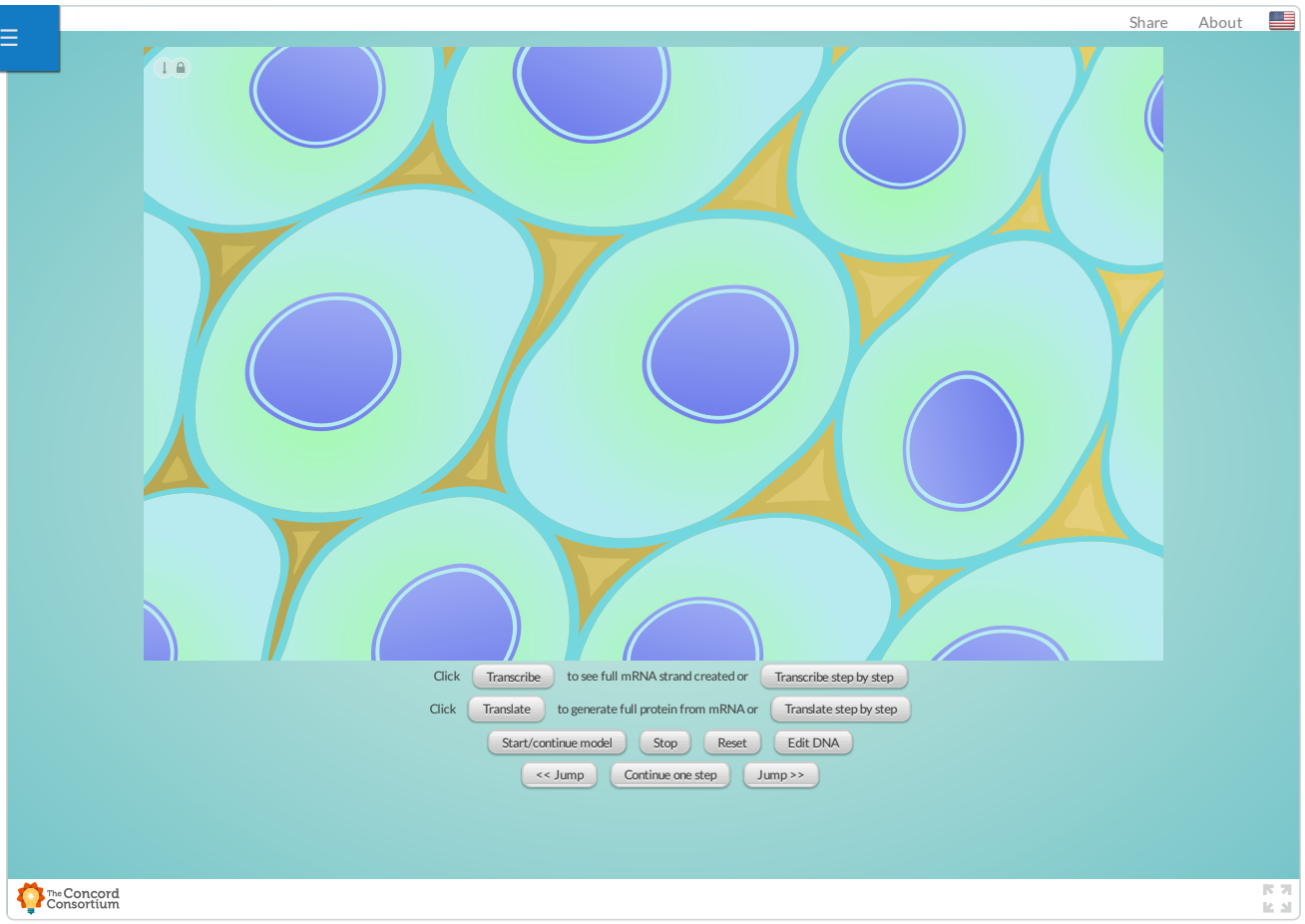
The central dogma of molecular biology describes the flow of genetic information in cells from DNA to messenger RNA (mRNA) to protein. It states that genes specify the sequence of mRNA molecules, which in turn specify the sequence of proteins. Because the information stored in DNA is so central to cellular function, the cell keeps the DNA protected and copies it in the form of RNA. An enzyme adds one nucleotide to the mRNA strand for every nucleotide it reads in the DNA strand. The translation of this information to a protein is more complex because three mRNA nucleotides correspond to one amino acid in the polypeptide sequence.

Transcription: DNA to RNA

Transcription is the process of creating a complementary RNA copy of a sequence of DNA. Both RNA and DNA are nucleic acids, which use base pairs of nucleotides as a complementary language that enzymes can convert back and forth from DNA to RNA. During transcription, a DNA sequence is read by RNA polymerase, which produces a complementary, antiparallel RNA strand. Unlike DNA replication, transcription results in an RNA complement that substitutes the RNA uracil (U) in all instances where the DNA thymine (T) would have occurred. Transcription is the first step in gene expression. The stretch of DNA transcribed into an RNA molecule is called a transcript. Some transcripts are used as structural or regulatory RNAs, and others encode one or more proteins. If the transcribed gene encodes a protein, the result of transcription is messenger RNA (mRNA), which will then be used to create that protein in the process of translation.

Translation: RNA to Protein

Translation is the process by which mRNA is decoded and translated to produce a polypeptide sequence, otherwise known as a protein. This method of synthesizing proteins is directed by the mRNA and accomplished with the help of a ribosome, a large complex of ribosomal RNAs (rRNAs) and proteins. In translation, a cell decodes the mRNA's genetic message and assembles the brand-new polypeptide chain. Transfer RNA, or tRNA, translates the sequence of codons on the mRNA strand. The main function of tRNA is to transfer a free amino acid from the cytoplasm to a ribosome, where it is attached to the growing polypeptide chain. tRNAs continue to add amino acids to the growing end of the polypeptide chain until they reach a stop codon on the mRNA. The ribosome then releases the completed protein into the cell.



DNA to protein: This interactive shows the process of DNA code being translated to a protein from start to finish!

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: Prokaryotic Transcription

Topic hierarchy

- 15.2A: Transcription in Prokaryotes
- 15.2B: Initiation of Transcription in Prokaryotes
- 15.2C: Elongation and Termination in Prokaryotes

15.2A: Transcription in Prokaryotes

The genetic code is a degenerate, non-overlapping set of 64 codons that encodes for 21 amino acids and 3 stop codons.

Learning Objectives

- Describe the genetic code and how the nucleotide sequence prescribes the amino acid and the protein sequence

Key Points

- The relationship between DNA base sequences and the amino acid sequence in proteins is called the genetic code.
- There are 61 codons that encode amino acids and 3 codons that code for chain termination for a total of 64 codons.
- Unlike, eukayrotes, a bacterial chromosome is a covalently-closed circle.
- The DNA double helix must partially unwind for transcription to occur; this unwound region is called a transcription bubble.

Key Terms

- **nucleotide**: the monomer comprising DNA or RNA molecules; consists of a nitrogenous heterocyclic base that can be a purine or pyrimidine, a five-carbon pentose sugar, and a phosphate group
- **amino acid**: Any of 20 naturally occurring α-amino acids (having the amino, and carboxylic acid groups on the same carbon atom), and a variety of side chains, that combine, via peptide bonds, to form proteins.
- **redundancy**: duplication of components, such as amino acid codons, to provide survival of the total system in case of failure of single components

The Genetic Code: Nucleotide sequences prescribe the amino acids

The genetic code is the relationship between DNA base sequences and the amino acid sequence in proteins. Features of the genetic code include:

- Amino acids are encoded by three nucleotides.
- It is non-overlapping.
- It is degenerate.

There are 21 genetically-encoded amino acids universally found in the species from all three domains of life. (There is a 22nd genetically-encooded amino acid, Pyl, but so far it has only been found in a handful of Archaea and Bacteria species.) Yet there are only four different nucleotides in DNA or RNA, so a minimum of three nucleotides are needed to code each of the 21 (or 22) amino acids. The set of three nucleotides that codes for a single amino acid is known as a codon. There are 64 codons in total, 61 that encode amino acids and 3 that code for chain termination. Two of the codons for chain termination can, under certain circumstances, instead code for amino acids.

Figure 15.2*A*.1: **Genetic Code Table.**: A codon is made of three nucleotides. Consequently there are 4^3 (=64) different codons. The 64 codons encode 22 different amino acids and three termination codons, also called stop codons.

Degeneracy is the redundancy of the genetic code. The genetic code has redundancy, but no ambiguity. For example, although codons GAA and GAG both specify glutamic acid (redundancy), neither of them specifies any other amino acid (no ambiguity). The codons encoding one amino acid may differ in any of their three positions. For example, the amino acid glutamic acid is specified by GAA and GAG codons (difference in the third position); the amino acid leucine is specified by UUA, UUG, CUU, CUC, CUA, CUG codons (difference in the first or third position); while the amino acid serine is specified by UCA, UCG, UCC, UCU, AGU, AGC (difference in the first, second or third position). These properties of the genetic code make it more fault-tolerant for point mutations.

Origin of transcription on prokaryotic organisms

Prokaryotes are mostly single-celled organisms that, by definition, lack membrane-bound nuclei and other organelles. The central region of the cell in which prokaryotic DNA resides is called the nucleoid region. Bacterial and Archaeal chromosomes are covalently-closed circles that are not as extensively compacted as eukaryotic chromosomes, but are compacted nonetheless as the diameter of a typical prokaryotic chromosome is larger than the diameter of a typical prokaryotic cell. Additionally, prokaryotes often have abundant plasmids, which are shorter, circular DNA molecules that may only contain one or a few genes and often carry traits such as antibiotic resistance.

Transcription in prokaryotes (as in eukaryotes) requires the DNA double helix to partially unwind in the region of RNA synthesis. The region of unwinding is called a transcription bubble. Transcription always proceeds from the same DNA strand for each gene, which is called the template strand. The RNA product is complementary to the template strand and is almost identical to the other (non-template) DNA strand, called the sense or coding strand. The only difference is that in RNA all of the T nucleotides are replaced with U nucleotides.

The nucleotide on the DNA template strand that corresponds to the site from which the first 5′ RNA nucleotide is transcribed is called the +1 nucleotide, or the initiation site. Nucleotides preceding, or 5′ to, the template strand initiation site are given negative numbers and are designated upstream. Conversely, nucleotides following, or 3′ to, the template strand initiation site are denoted with "+" numbering and are called downstream nucleotides.

15.2B: Initiation of Transcription in Prokaryotes

RNA polymerase initiates transcription at specific DNA sequences called promoters.

Learning Objectives

- Summarize the initial steps of transcription in prokaryotes

Key Points

- Transcription of mRNA begins at the initiation site.
- Two promoter consensus sequences are at the -10 and -35 regions upstream of the initiation site.
- The σ subunit of RNA polymerase recognizes and binds the -35 region.
- Five subunits (α, α, β, β', and σ) make up the complete RNA polymerase holoenzyme.

Key Terms

- **holoenzyme**: a fully functioning enzyme, composed of all its subunits
- **promoter**: the section of DNA that controls the initiation of RNA transcription

Prokaryotic RNA Polymerase

Prokaryotes use the same RNA polymerase to transcribe all of their genes. In *E. coli*, the polymerase is composed of five polypeptide subunits, two of which are identical. Four of these subunits, denoted α, α, β, and β', comprise the polymerase core enzyme. These subunits assemble each time a gene is transcribed; they disassemble once transcription is complete. Each subunit has a unique role: the two α-subunits are necessary to assemble the polymerase on the DNA; the β-subunit binds to the ribonucleoside triphosphate that will become part of the nascent "recently-born" mRNA molecule; and the β' binds the DNA template strand. The fifth subunit, σ, is involved only in transcription initiation. It confers transcriptional specificity such that the polymerase begins to synthesize mRNA from an appropriate initiation site. Without σ, the core enzyme would transcribe from random sites and would produce mRNA molecules that specified protein gibberish. The polymerase comprised of all five subunits is called the holoenzyme.

Prokaryotic Promoters and Initiation of Transcription

The nucleotide pair in the DNA double helix that corresponds to the site from which the first 5′ mRNA nucleotide is transcribed is called the +1 site, or the initiation site. Nucleotides preceding the initiation site are given negative numbers and are designated upstream. Conversely, nucleotides following the initiation site are denoted with "+" numbering and are called downstream nucleotides.

A promoter is a DNA sequence onto which the transcription machinery binds and initiates transcription. In most cases, promoters exist upstream of the genes they regulate. The specific sequence of a promoter is very important because it determines whether the corresponding gene is transcribed all the time, some of the time, or infrequently. Although promoters vary among prokaryotic genomes, a few elements are conserved. At the -10 and -35 regions upstream of the initiation site, there are two promoter consensus sequences, or regions that are similar across all promoters and across various bacterial species. The -10 consensus sequence, called the -10 region, is TATAAT. The -35 sequence, TTGACA, is recognized and bound by σ. Once this interaction is made, the subunits of the core enzyme bind to the site. The A–T-rich -10 region facilitates unwinding of the DNA template; several phosphodiester bonds are made. The transcription initiation phase ends with the production of abortive transcripts, which are polymers of approximately 10 nucleotides that are made and released.

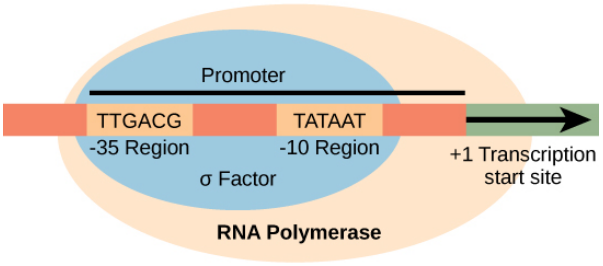

Figure 15.2B.1: **Promoter**: The σ subunit of prokaryotic RNA polymerase recognizes consensus sequences found in the promoter region upstream of the transcription start sight. The σ subunit dissociates from the polymerase after transcription has been initiated.

15.2C: Elongation and Termination in Prokaryotes

Transcription elongation begins with the release of the polymerase σ subunit and terminates via the rho protein or via a stable hairpin.

Learning Objectives

- Explain the process of elongation and termination in prokaryotes

Key Points

- The transcription elongation phase begins with the dissociation of the σ subunit, which allows the core RNA polymerase enzyme to proceed along the DNA template.
- Rho-dependent termination is caused by the rho protein colliding with the stalled polymerase at a stretch of G nucleotides on the DNA template near the end of the gene.
- Rho-independent termination is caused the polymerase stalling at a stable hairpin formed by a region of complementary C–G nucleotides at the end of the mRNA.

Key Terms

- **elongation**: the addition of nucleotides to the 3′-end of a growing RNA chain during transcription

Elongation in Prokaryotes

The transcription elongation phase begins with the release of the σ subunit from the polymerase. The dissociation of σ allows the core RNA polymerase enzyme to proceed along the DNA template, synthesizing mRNA in the 5′ to 3′ direction at a rate of approximately 40 nucleotides per second. As elongation proceeds, the DNA is continuously unwound ahead of the core enzyme and rewound behind it. Since the base pairing between DNA and RNA is not stable enough to maintain the stability of the mRNA synthesis components, RNA polymerase acts as a stable linker between the DNA template and the nascent RNA strands to ensure that elongation is not interrupted prematurely.

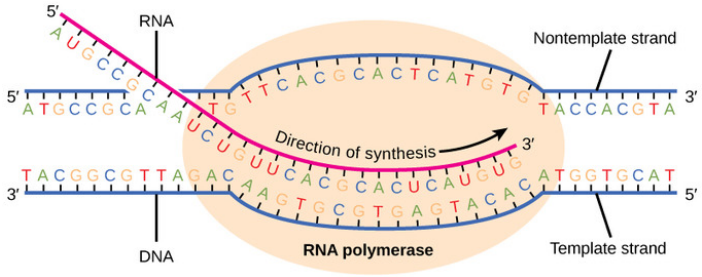

Figure 15.2C.1: **Elongation in prokaryotes**: During elongation, the prokaryotic RNA polymerase tracks along the DNA template, synthesizes mRNA in the 5′ to 3′ direction, and unwinds and rewinds the DNA as it is read.

Termination in Prokaryotes

Once a gene is transcribed, the prokaryotic polymerase needs to be instructed to dissociate from the DNA template and liberate the newly-made mRNA. Depending on the gene being transcribed, there are two kinds of termination signals: one is protein-based and the other is RNA-based.

Rho-dependent termination is controlled by the rho protein, which tracks along behind the polymerase on the growing mRNA chain. Near the end of the gene, the polymerase encounters a run of G nucleotides on the DNA template and it stalls. As a result, the rho protein collides with the polymerase. The interaction with rho releases the mRNA from the transcription bubble.

Rho-independent termination is controlled by specific sequences in the DNA template strand. As the polymerase nears the end of the gene being transcribed, it encounters a region rich in C–G nucleotides. The mRNA folds back on itself, and the complementary C–G nucleotides bind together. The result is a stable hairpin that causes the polymerase to stall as soon as it begins to transcribe a region rich in A–T nucleotides. The complementary U–A region of the mRNA transcript forms

weak interaction with the template DNA. This, coupled with the stalled polymerase, induces enough instability for e enzyme to break away and liberate the new mRNA transcript.

Upon termination, the process of transcription is complete. By the time termination occurs, the prokaryotic transcript would already have been used to begin synthesis of numerous copies of the encoded protein because these processes can occur concurrently in the cytoplasm. The unification of transcription, translation, and even mRNA degradation is possible because all of these processes occur in the same 5′ to 3′ direction and because there is no membranous compartmentalization in the prokaryotic cell. In contrast, the presence of a nucleus in eukaryotic cells prevents simultaneous transcription and translation.

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15.3: Eukaryotic Transcription

Topic hierarchy

- 15.3A: Initiation of Transcription in Eukaryotes
- 15.3B: Elongation and Termination in Eukaryotes

15.3A: Initiation of Transcription in Eukaryotes

Initiation is the first step of eukaryotic transcription and requires RNAP and several transcription factors to proceed.

Learning Objectives

- Describe how transcription is initiated and proceeds along the DNA strand

Key Points

- Eukaryotic transcription is carried out in the nucleus of the cell and proceeds in three sequential stages: initiation, elongation, and termination.
- Eukaryotes require transcription factors to first bind to the promoter region and then help recruit the appropriate polymerase.
- RNA Polymerase II is the polymerase responsible for transcribing mRNA.

Key Terms

- **repressor**: any protein that binds to DNA and thus regulates the expression of genes by decreasing the rate of transcription
- **activator**: any chemical or agent which regulates one or more genes by increasing the rate of transcription
- **polymerase**: any of various enzymes that catalyze the formation of polymers of DNA or RNA using an existing strand of DNA or RNA as a template

Steps in Eukaryotic Transcription

Eukaryotic transcription is carried out in the nucleus of the cell by one of three RNA polymerases, depending on the RNA being transcribed, and proceeds in three sequential stages:

1. Initiation
2. Elongation
3. Termination.

Initiation of Transcription in Eukaryotes

Unlike the prokaryotic RNA polymerase that can bind to a DNA template on its own, eukaryotes require several other proteins, called transcription factors, to first bind to the promoter region and then help recruit the appropriate polymerase. The completed assembly of transcription factors and RNA polymerase bind to the promoter, forming a transcription pre-initiation complex (PIC).

The most-extensively studied core promoter element in eukaryotes is a short DNA sequence known as a TATA box, found 25-30 base pairs upstream from the start site of transcription. Only about 10-15% of mammalian genes contain TATA boxes, while the rest contain other core promoter elements, but the mechanisms by which transcription is initiated at promoters with TATA boxes is well characterized.

The TATA box, as a core promoter element, is the binding site for a transcription factor known as TATA-binding protein (TBP), which is itself a subunit of another transcription factor: Transcription Factor II D (TFIID). After TFIID binds to the TATA box via the TBP, five more transcription factors and RNA polymerase combine around the TATA box in a series of stages to form a pre-initiation complex. One transcription factor, Transcription Factor II H (TFIIH), is involved in separating opposing strands of double-stranded DNA to provide the RNA Polymerase access to a single-stranded DNA template. However, only a low, or basal, rate of transcription is driven by the pre-initiation complex alone. Other proteins known as activators and repressors, along with any associated coactivators or corepressors, are responsible for modulating transcription rate. Activator proteins increase the transcription rate, and repressor proteins decrease the transcription rate.

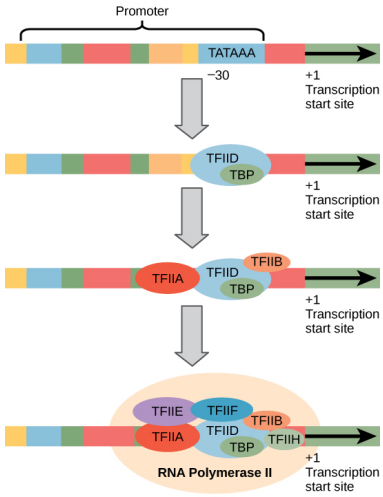


Figure 15.3A.1: **Eukaryotic Transcription Initiation**: A generalized promoter of a gene transcribed by RNA polymerase II is shown. Transcription factors recognize the promoter, RNA polymerase II then binds and forms the transcription initiation complex.

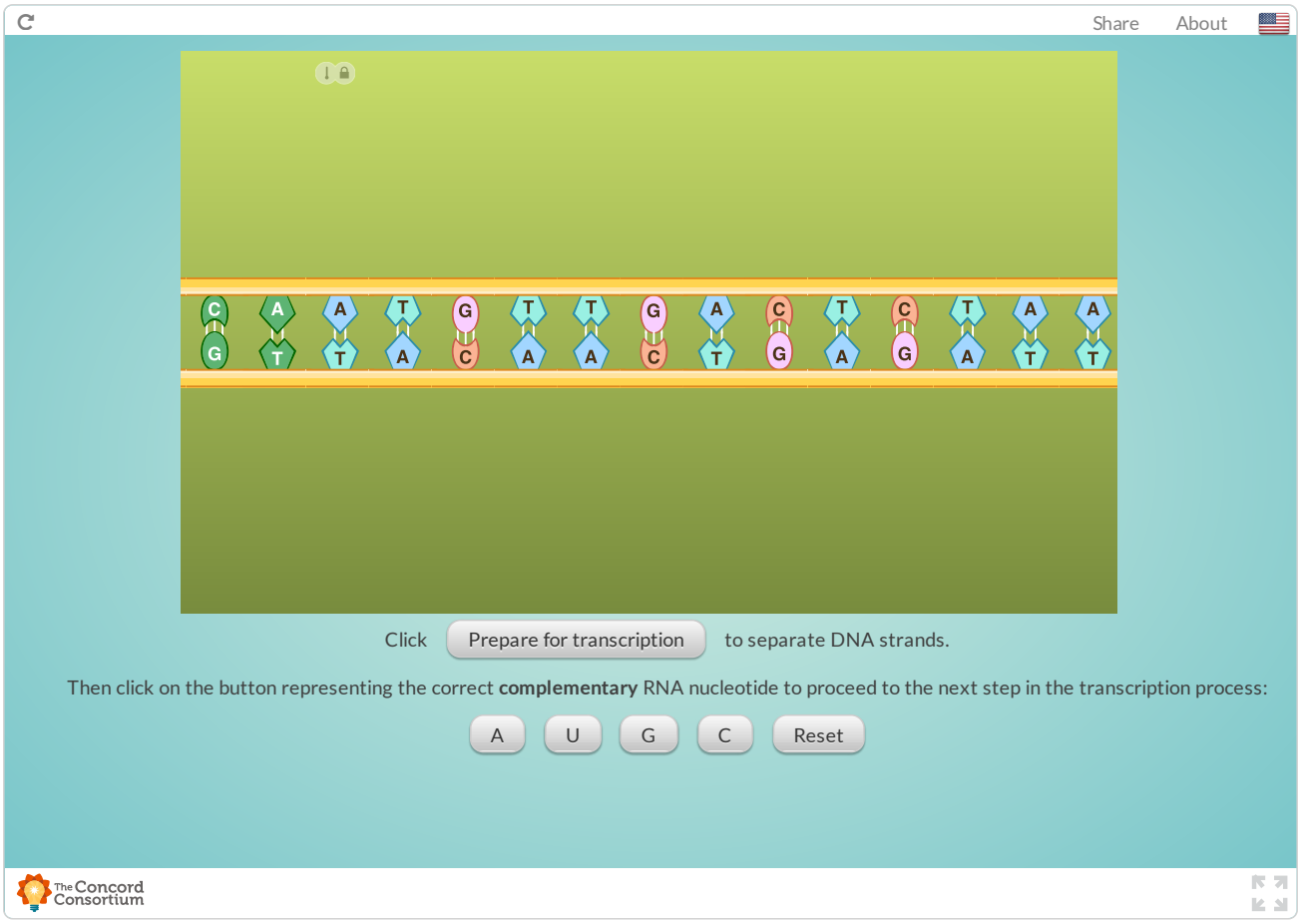
The Three Eukaryotic RNA Polymerases (RNAPs)

The features of eukaryotic mRNA synthesis are markedly more complex those of prokaryotes. Instead of a single polymerase comprising five subunits, the eukaryotes have three polymerases that are each made up of 10 subunits or more. Each eukaryotic polymerase also requires a distinct set of transcription factors to bring it to the DNA template.

RNA polymerase I is located in the nucleolus, a specialized nuclear substructure in which ribosomal RNA (rRNA) is transcribed, processed, and assembled into ribosomes. The rRNA molecules are considered structural RNAs because they have a cellular role but are not translated into protein. The rRNAs are components of the ribosome and are essential to the process of translation. RNA polymerase I synthesizes all of the rRNAs except for the 5S rRNA molecule.

RNA polymerase II is located in the nucleus and synthesizes all protein-coding nuclear pre-mRNAs. Eukaryotic pre-mRNAs undergo extensive processing after transcription, but before translation. RNA polymerase II is responsible for transcribing the overwhelming majority of eukaryotic genes, including all of the protein-encoding genes which ultimately are translated into proteins and genes for several types of regulatory RNAs, including microRNAs (miRNAs) and long-coding RNAs (lncRNAs).

RNA polymerase III is also located in the nucleus. This polymerase transcribes a variety of structural RNAs that includes the 5S pre-rRNA, transfer pre-RNAs (pre-tRNAs), and small nuclear pre-RNAs. The tRNAs have a critical role in translation: they serve as the adaptor molecules between the mRNA template and the growing polypeptide chain. Small nuclear RNAs have a variety of functions, including "splicing" pre-mRNAs and regulating transcription factors. Not all miRNAs are transcribed by RNA Polymerase II, RNA Polymerase III transcribes some of them.



Modeling transcription: This interactive models the process of DNA transcription in a eukaryotic cell.

15.3B: Elongation and Termination in Eukaryotes

Elongation synthesizes pre-mRNA in a 5′ to 3′ direction, and termination occurs in response to termination sequences and signals.

Learning Objectives

- Describe what is happening during transcription elongation and termination

Key Points

- RNA polymerase II (RNAPII) transcribes the major share of eukaryotic genes.
- During elongation, the transcription machinery needs to move histones out of the way every time it encounters a nucleosome.
- Transcription elongation occurs in a bubble of unwound DNA, where the RNA Polymerase uses one strand of DNA as a template to catalyze the synthesis of a new RNA strand in the 5′ to 3′ direction.
- RNA Polymerase I and RNA Polymerase III terminate transcription in response to specific termination sequences in either the DNA being transcribed (RNA Polymerase I) or in the newly-synthesized RNA (RNA Polymerase III).
- RNA Polymerase II terminates transcription at random locations past the end of the gene being transcribed. The newly-synthesized RNA is cleaved at a sequence-specified location and released before transcription terminates.

Key Terms

- **nucleosome**: any of the subunits that repeat in chromatin; a coil of DNA surrounding a histone core
- **histone**: any of various simple water-soluble proteins that are rich in the basic amino acids lysine and arginine and are complexed with DNA in the nucleosomes of eukaryotic chromatin
- **chromatin**: a complex of DNA, RNA, and proteins within the cell nucleus out of which chromosomes condense during cell division

Transcription through Nucleosomes

Following the formation of the pre-initiation complex, the polymerase is released from the other transcription factors, and elongation is allowed to proceed with the polymerase synthesizing RNA in the 5′ to 3′ direction. RNA Polymerase II (RNAPII) transcribes the major share of eukaryotic genes, so this section will mainly focus on how this specific polymerase accomplishes elongation and termination.

Although the enzymatic process of elongation is essentially the same in eukaryotes and prokaryotes, the eukaryotic DNA template is more complex. When eukaryotic cells are not dividing, their genes exist as a diffuse, but still extensively packaged and compacted mass of DNA and proteins called chromatin. The DNA is tightly packaged around charged histone proteins at repeated intervals. These DNA–histone complexes, collectively called nucleosomes, are regularly spaced and include 146 nucleotides of DNA wound twice around the eight histones in a nucleosome like thread around a spool.

For polynucleotide synthesis to occur, the transcription machinery needs to move histones out of the way every time it encounters a nucleosome. This is accomplished by a special protein dimer called FACT, which stands for “facilitates chromatin transcription.” FACT partially disassembles the nucleosome immediately ahead (upstream) of a transcribing RNA Polymerase II by removing two of the eight histones (a single dimer of H2A and H2B histones is removed.) This presumably sufficiently loosens the DNA wrapped around that nucleosome so that RNA Polymerase II can transcribe through it. FACT reassembles the nucleosome behind the RNA Polymerase II by returning the missing histones to it. RNA Polymerase II will continue to elongate the newly-synthesized RNA until transcription terminates.

Figure 15.3B.1: **The FACT protein dimer allows RNA Polymerase II to transcribe through packaged DNA**: DNA in eukaryotes is packaged in nucleosomes, which consist of an octomer of 4 different histone proteins. When DNA is tightly wound twice around a nucleosome, RNA Polymerase II cannot access it for transcription. FACT removes two of the histones from the nucleosome immediately ahead of RNA Polymerase, loosening the packaging so that RNA Polymerase II can continue transcription. FACT also reassembles the nucleosome immediately behindd the RNA Polymerase by returning the missing histones.

Elongation

RNA Polymerase II is a complex of 12 protein subunits. Specific subunits within the protein allow RNA Polymerase II to act as its own helicase, sliding clamp, single-stranded DNA binding protein, as well as carry out other functions. Consequently, RNA Polymerase II does not need as many accessory proteins to catalyze the synthesis of new RNA strands during transcription elongation as DNA Polymerase does to catalyze the synthesis of new DNA strands during replication elongation.

However, RNA Polymerase II does need a large collection of accessory proteins to initiate transcription at gene promoters, but once the double-stranded DNA in the transcription start region has been unwound, the RNA Polymerase II has been positioned at the +1 initiation nucleotide, and has started catalyzing new RNA strand synthesis, RNA Polymerase II clears or “escapes” the promoter region and leaves most of the transcription initiation proteins behind.

All RNA Polymerases travel along the template DNA strand in the 3′ to 5′ direction and catalyze the synthesis of new RNA strands in the 5′ to 3′ direction, adding new nucleotides to the 3′ end of the growing RNA strand.

RNA Polymerases unwind the double stranded DNA ahead of them and allow the unwound DNA behind them to rewind. As a result, RNA strand synthesis occurs in a transcription bubble of about 25 unwound DNA basebairs. Only about 8 nucleotides of newly-synthesized RNA remain basepaired to the template DNA. The rest of the RNA molecules falls off the template to allow the DNA behind it to rewind.

RNA Polymerases use the DNA strand below them as a template to direct which nucleotide to add to the 3′ end of the growing RNA strand at each point in the sequence. The RNA Polymerase travels along the template DNA one nucleotide at at time. Whichever RNA nucleotide is capable of basepairing to the template nucleotide below the RNA Polymerase is the next nucleotide to be added. Once the addition of a new nucleotide to the 3′ end of the growing strand has been catalyzed, the RNA Polymerase moves to the next DNA nucleotide on the template below it. This process continues until transcription termination occurs.

Termination

The termination of transcription is different for the three different eukaryotic RNA polymerases.

The ribosomal rRNA genes transcribed by RNA Polymerase I contain a specific sequence of basepairs (11 bp long in humans; 18 bp in mice) that is recognized by a termination protein called TTF-1 (Transcription Termination Factor for RNA Polymerase I.) This protein binds the DNA at its recognition sequence and blocks further transcription, causing the RNA Polymerase I to disengage from the template DNA strand and to release its newly-synthesized RNA.

The protein-encoding, structural RNA, and regulatory RNA genes transcribed by RNA Polymerse II lack any specific signals or sequences that direct RNA Polymerase II to terminate at specific locations. RNA Polymerase II can continue to transcribe RNA anywhere from a few bp to thousands of bp past the actual end of the gene. However, the transcript is cleaved at an internal site before RNA Polymerase II finishes transcribing. This releases the upstream portion of the transcript, which will serve as the initial RNA prior to further processing (the pre-mRNA in the case of protein-encoding genes.) This cleavage site is considered the “end” of the gene. The remainder of the transcript is digested by a 5′-

exonuclease (called Xrn2 in humans) while it is still being transcribed by the RNA Polymerase II. When the 5′-exonulease "catches up" to RNA Polymerase II by digesting away all the overhanging RNA, it helps disengage the polymerase from its DNA template strand, finally terminating that round of transcription.

In the case of protein-encoding genes, the cleavage site which determines the "end" of the emerging pre-mRNA occurs between an upstream AAUAAA sequence and a downstream GU-rich sequence separated by about 40-60 nucleotides in the emerging RNA. Once both of these sequences have been transcribed, a protein called CPSF in humans binds the AAUAAA sequence and a protein called CstF in humans binds the GU-rich sequence. These two proteins form the base of a complicated protein complex that forms in this region before CPSF cleaves the nascent pre-mRNA at a site 10-30 nucleotides downstream from the AAUAAA site. The Poly(A) Polymerase enzyme which catalyzes the addition of a 3′ poly-A tail on the pre-mRNA is part of the complex that forms with CPSF and CstF.

Figure 15.3B.1: **Transcription termination by RNA Polymerase II on a protein-encoding gene.**: RNA Polymerase II has no specific signals that terminate its transcription. In the case of protein-encoding genes, a protein complex will bind to two locations on the growing pre-mRNA once the RNA Polymerase has transcribed past the end of the gene. CPSF in the complex will bind a AAUAAA sequence, and CstF in the complex will bind a GU-rich sequence (top figure). CPSF in the complex will cleave the pre-mRNA at a site between the two bound sequences, releasing the pre-mRNA (middle figure). Poly(A) Polymerase is a part of the same complex and will begin to add a poly-A tail to the pre-mRNA. At the same time, Xrn2 protein, which is an exonuclease, attacks the 5′ end of the RNA strand still associated with the RNA Polymerase. Xrn2 will start digesting the non-released portion of the newly synthesized RNA until Xrn2 reaches the RNA Polymerase, where it aids in displacing the RNA Polymerase from the template DNA strand. This terminates transcription at some random location downstream from the true end of the gene (bottom figure).

The tRNA, 5S rRNA, and structural RNAs genes transcribed by RNA Polymerase III have a not-entirely-understood termination signal. The RNAs transcribed by RNA Polymerase III have a short stretch of four to seven U's at their 3′ end. This somehow triggers RNA Polymerase III to both release the nascent RNA and disengage from the template DNA strand.

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15.4: RNA Processing in Eukaryotes

Topic hierarchy

- 15.4A: mRNA Processing
- 15.4B: Processing of tRNAs and rRNAs

15.4A: mRNA Processing

Eukaryotic pre-mRNA receives a 5′ cap and a 3′ poly (A) tail before introns are removed and the mRNA is considered ready for translation.

Learning Objectives

- Outline the steps of pre-mRNA processing

Key Points

- A 7-methylguanosine cap is added to the 5′ end of the pre-mRNA while elongation is still in progress. The 5′ cap protects the nascent mRNA from degradation and assists in ribosome binding during translation.
- A poly (A) tail is added to the 3′ end of the pre-mRNA once elongation is complete. The poly (A) tail protects the mRNA from degradation, aids in the export of the mature mRNA to the cytoplasm, and is involved in binding proteins involved in initiating translation.
- Introns are removed from the pre-mRNA before the mRNA is exported to the cytoplasm.

Key Terms

- **intron**: a portion of a split gene that is included in pre-RNA transcripts but is removed during RNA processing and rapidly degraded
- **moiety**: a specific segment of a molecule
- **spliceosome**: a dynamic complex of RNA and protein subunits that removes introns from precursor mRNA

Pre-mRNA Processing

The eukaryotic pre-mRNA undergoes extensive processing before it is ready to be translated. The additional steps involved in eukaryotic mRNA maturation create a molecule with a much longer half-life than a prokaryotic mRNA. Eukaryotic mRNAs last for several hours, whereas the typical *E. coli* mRNA lasts no more than five seconds.

Pre-mRNAs are first coated in RNA-stabilizing proteins; these protect the pre-mRNA from degradation while it is processed and exported out of the nucleus. The three most important steps of pre-mRNA processing are the addition of stabilizing and signaling factors at the 5′ and 3′ ends of the molecule, and the removal of intervening sequences that do not specify the appropriate amino acids. In rare cases, the mRNA transcript can be “edited” after it is transcribed.

5′ Capping

While the pre-mRNA is still being synthesized, a 7-methylguanosine cap is added to the 5′ end of the growing transcript by a 5′-to-5′ phosphate linkage. This moiety protects the nascent mRNA from degradation. In addition, initiation factors involved in protein synthesis recognize the cap to help initiate translation by ribosomes.

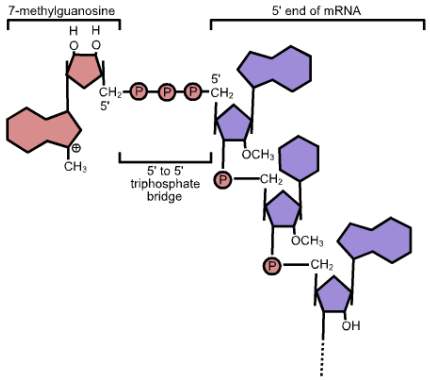

Figure 15.4*A*.1: **5′ cap structure**: Capping of the pre-mRNA involves the addition of 7-methylguanosine (m^7G) to the 5′ end. The cap protects the 5′ end of the primary RNA transcript from attack by ribonucleases and is recognized by eukaryotic initiation factors involved in assembling the ribosome on the mature mRNA prior to initiating translation.

3′ Poly-A Tail

While RNA Polymerase II is still transcribing downstream of the proper end of a gene, the pre-mRNA is cleaved by an endonuclease-containing protein complex between an AAUAAA consensus sequence and a GU-rich sequence. This releases the functional pre-mRNA from the rest of the transcript, which is still attached to the RNA Polymerase. An enzyme called poly (A) polymerase (PAP) is part of the same protein complex that cleaves the pre-mRNA and it immediately adds a string of approximately 200 A nucleotides, called the poly (A) tail, to the 3′ end of the just-cleaved pre-mRNA. The poly (A) tail protects the mRNA from degradation, aids in the export of the mature mRNA to the cytoplasm, and is involved in binding proteins involved in initiating translation.

Figure 15.4A.1: **Poly (A) Polymerase adds a 3′ poly (A) tail to the pre-mRNA.**: The pre-mRNA is cleaved off the rest of the growing transcript before RNA Polymerase II has stopped transcribing. This cleavage is done by an endonuclease-containing protein complex that binds to an AAUAAA sequence upstream of the cleavage site and to a GU-rich sequence downstream of the cut site. Immediately after the cleavage, Poly (A) Polymerase (PAP), which is also part of the protein complex, catalyzes the addition of up to 200 A nucleotides to the 3′ end of the just-cleaved pre-mRNA.

Pre-mRNA Splicing

Eukaryotic genes are composed of exons, which correspond to protein-coding sequences (*ex*-on signifies that they are *ex*pressed), and intervening sequences called introns (*int*-ron denotes their *int*ervening role), which may be involved in gene regulation, but are removed from the pre-mRNA during processing. Intron sequences in mRNA do not encode functional proteins.

Discovery of Introns

The discovery of introns came as a surprise to researchers in the 1970s who expected that pre-mRNAs would specify protein sequences without further processing, as they had observed in prokaryotes. The genes of higher eukaryotes very often contain one or more introns. While these regions may correspond to regulatory sequences, the biological significance

of having many introns or having very long introns in a gene is unclear. It is possible that introns slow down gene expression because it takes longer to transcribe pre-mRNAs with lots of introns. Alternatively, introns may be nonfunctional sequence remnants left over from the fusion of ancient genes throughout evolution. This is supported by the fact that separate exons often encode separate protein subunits or domains. For the most part, the sequences of introns can be mutated without ultimately affecting the protein product.

Intron Processing

All introns in a pre-mRNA must be completely and precisely removed before protein synthesis. If the process errs by even a single nucleotide, the reading frame of the rejoined exons would shift, and the resulting protein would be dysfunctional. The process of removing introns and reconnecting exons is called splicing. Introns are removed and degraded while the pre-mRNA is still in the nucleus. Splicing occurs by a sequence-specific mechanism that ensures introns will be removed and exons rejoined with the accuracy and precision of a single nucleotide. The splicing of pre-mRNAs is conducted by complexes of proteins and RNA molecules called spliceosomes.

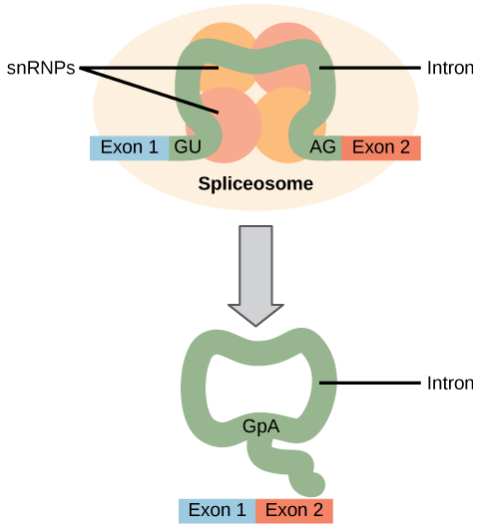

Figure 15.4A.1: **Pre-mRNA splicing**: Pre-mRNA splicing involves the precise removal of introns from the primary RNA transcript. The splicing process is catalyzed by large complexes called spliceosomes. Each spliceosome is composed of five subunits called snRNPs. The spliceseome's actions result in the splicing together of the two exons and the release of the intron in a lariat form.

Each spliceosome is composed of five subunits called snRNPs (for small nuclear ribonucleoparticles, and pronounced "snurps".) Each snRNP is itself a complex of proteins and a special type of RNA found only in the nucleus called snRNAs (small nuclear RNAs). Spliceosomes recognize sequences at the 5′ end of the intron because introns always start with the nucleotides GU and they recognize sequences at the 3′ end of the intron because they always end with the nucleotides AG. The spliceosome cleaves the pre-mRNA's sugar phosphate backbone at the G that starts the intron and then covalently attaches that G to an internal A nucleotide within the intron. Then the spliceosme connects the 3′ end of the first exon to the 5′ end of the following exon, cleaving the 3′ end of the intron in the process. This results in the splicing together of the two exons and the release of the intron in a lariat form.

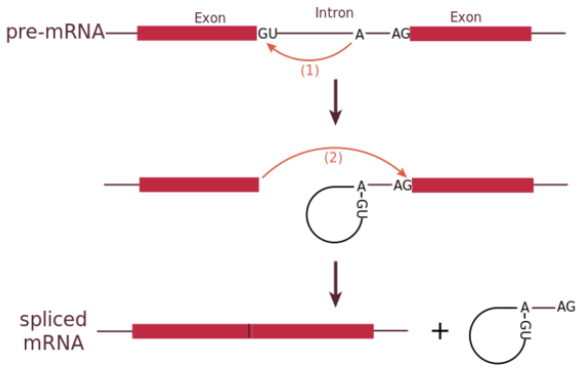

Figure 15.4A.1: **Mechanism of pre-mRNA splicing.**: The snRNPs of the spliceosome were left out of this figure, but it shows the sites within the intron whose interactions are catalyzed by the spliceosome. Initially, the conserved G which starts an intron is cleaved from the 3′ end of the exon upstream to it and the G is covalently attached to an internal A within the intron. Then the 3′ end of the just-released exon is joined to the 5′ end of the next exon, cleaving the bond that attaches the 3′ end of the intron to its adjacent exon. This both joins the two exons and removes the intron in lariat form.

15.4B: Processing of tRNAs and rRNAs

rRNA and tRNA are structural molecules that aid in protein synthesis but are not themselves translated into protein.

Learning Objectives

- Describe how pre-rRNAs and pre-tRNAs are processed into mature rRNAs and tRNAs.

Key Points

- Ribosomal RNA (rRNA) is a structural molecule that makes up over half of the mass of a ribosome and aids in protein synthesis.
- Transfer RNA (tRNA) recognizes a codon on mRNA and brings the appropriate amino acid to that site.
- rRNAs are processed from larger pre-rRNAs by trimming the larger rRNAs down and methylating some of the nucleotides.
- tRNAs are processed from pre-tRNAs by trimming both ends of the pre-tRNA, adding a CCA trinucleotide to the 3′ end, if needed, removing any introns present, and chemically modified 12 nucleotides on average per tRNA.

Key Terms

- **anticodon**: a sequence of three nucleotides in transfer RNA that binds to the complementary triplet (codon) in messenger RNA, specifying an amino acid during protein synthesis

Processing of tRNAs and rRNAs

The tRNAs and rRNAs are structural molecules that have roles in protein synthesis; however, these RNAs are not themselves translated. In eukaryotes, pre-rRNAs are transcribed, processed, and assembled into ribosomes in the nucleolus, while pre-tRNAs are transcribed and processed in the nucleus and then released into the cytoplasm where they are linked to free amino acids for protein synthesis.

Ribosomal RNA (rRNA)

The four rRNAs in eukaryotes are first transcribed as two long precursor molecules. One contains just the pre-rRNA that will be processed into the 5S rRNA; the other spans the 28S, 5.8S, and 18S rRNAs. Enzymes then cleave the precursors into subunits corresponding to each rRNA. In bacteria, there are only three rRNAs and all are transcribed in one long precursor molecule that is cleaved into the individual rRNAs. Some of the bases of pre-rRNAs are methylated for added stability. Mature rRNAs make up 50-60% of each ribosome. Some of a ribosome's RNA molecules are purely structural, whereas others have catalytic or binding activities.

The eukaryotic ribosome is composed of two subunits: a large subunit (60S) and a small subunit (40S). The 60S subunit is composed of the 28S rRNA, 5.8S rRNA, 5S rRNA, and 50 proteins. The 40S subunit is composed of the 18S rRNA and 33 proteins. The bacterial ribosome is composed of two similar subunits, with slightly different components. The bacterial large subunit is called the 50S subunit and is composed of the 23S rRNA, 5S rRNA, and 31 proteins, while the bacterial small subunit is called the 30S subunit and is composed of the 16S rRNA and 21 proteins.

The two subunits join to constitute a functioning ribosome that is capable of creating proteins.

Transfer RNA (tRNA)

Each different tRNA binds to a specific amino acid and transfers it to the ribosome. Mature tRNAs take on a three-dimensional structure through intramolecular basepairing to position the amino acid binding site at one end and the anticodon in an unbasepaired loop of nucleotides at the other end. The anticodon is a three-nucleotide sequence, unique to each different tRNA, that interacts with a messenger RNA (mRNA) codon through complementary base pairing.

There are different tRNAs for the 21 different amino acids. Most amino acids can be carried by more than one tRNA.

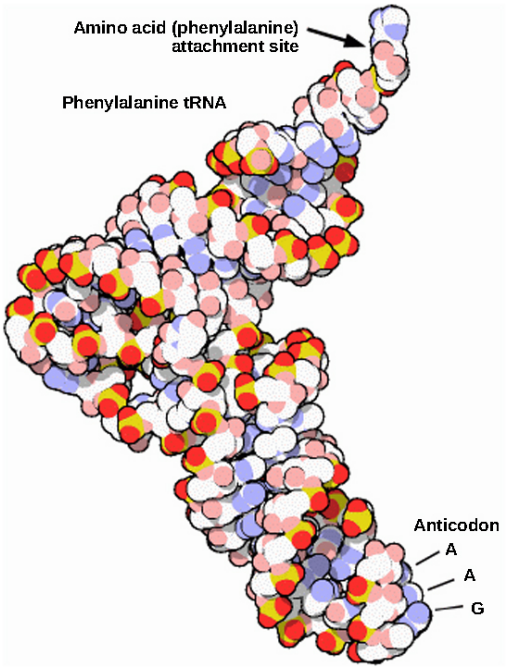

Figure 15.4B.1: **Structure of tRNA**: This is a space-filling model of a tRNA molecule that adds the amino acid phenylalanine to a growing polypeptide chain. The anticodon AAG binds the codon UUC on the mRNA. The amino acid phenylalanine is attached to the other end of the tRNA.

In all organisms, tRNAs are transcribed in a pre-tRNA form that requires multiple processing steps before the mature tRNA is ready for use in translation. In bacteria, multiple tRNAs are often transcribed as a single RNA. The first step in their processing is the digestion of the RNA to release individual pre-tRNAs. In archaea and eukaryotes, each pre-tRNA is transcribed as a separate transcript.

The processing to convert the pre-tRNA to a mature tRNA involves five steps.

1. The 5′ end of the pre-tRNA, called the 5′ leader sequence, is cleaved off.

2. The 3′ end of the pre-tRNA is cleaved off.

3. In all eukaryote pre-tRNAs, but in only some bacterial and archaeal pre-tRNAs, a CCA sequence of nucleotides is added to the 3′ end of the pre-tRNA after the original 3′ end is trimmed off. Some bacteria and archaea pre-tRNAs already have the CCA encoded in their transcript immediately upstream of the 3′ cleavage site, so they don't need to add one. The CCA at the 3′ end of the mature tRNA will be the site at which the tRNA's amino acid will be added.

4. Multiple nucleotides in the pre-tRNA are chemically modified, altering their nitorgen bases. On average about 12 nucleotides are modified per tRNA. The most common modifications are the conversion of adenine (A) to pseudouridine (ψ), the conversion of adenine to inosine (I), and the conversion of uridine to dihydrouridine (D). But over 100 other modifications can occur.

5. A significant number of eukaryotic and archaeal pre-tRNAs have introns that have to be spliced out. Introns are rarer in bacterial pre-tRNAs, but do occur occasionally and are spliced out.

After processing, the mature pre-tRNA is ready to have its cognate amino acid attached. The cognate amino acid for a tRNA is the one specified by its anticodon. Attaching this amino acid is called charging the tRNA. In eukaryotes, the mature tRNA is generated in the nucleus, and then exported to the cytoplasm for charging.

Figure 15.4B.1: **Processing of a pre-tRNA.**: A typical pre-tRNA undergoing processing steps to generate a mature tRNA ready to have its cognate amino acid attached. Nucleotides that are cleaved away are shown in green. Chemically-modified nucleotides are in yellow, as is the CAA trinucleotide that is added to the 3′ end of the pre-tRNA during processing. The anticodon nucleotides are shown in a lighter shade of red.

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15.5: Ribosomes and Protein Synthesis

Topic hierarchy

- 15.5A: The Protein Synthesis Machinery
- 15.5B: The Mechanism of Protein Synthesis
- 15.5C: Protein Folding, Modification, and Targeting

15.5A: The Protein Synthesis Machinery

Protein synthesis, or translation of mRNA into protein, occurs with the help of ribosomes, tRNAs, and aminoacyl tRNA synthetases.

Learning Objectives

- Explain the role played by ribosomes, tRNA, and aminoacyl tRNA synthetases in protein synthesis

Key Points

- Ribosomes, macromolecular structures composed of rRNA and polypeptide chains, are formed of two subunits (in bacteria and archaea, 30S and 50S; in eukaryotes, 40S and 60S), that bring together mRNA and tRNAs to catalyze protein synthesis.
- Fully assembled ribosomes have three tRNA binding sites: an A site for incoming aminoacyl-tRNAs, a P site for peptidyl-tRNAs, and an E site where empty tRNAs exit.
- tRNAs (transfer ribonucleic acids), which serve to deliver the appropriate amino acid to the growing peptide chain, consist of a modified RNA chain with the appropriate amino acid covalently attached.
- tRNAs have a loop of unbasepaired nucleotides at one end of the molecule that contains three nucleotides that act as the anticodon that basepairs to the mRNA codon.
- Aminoacyl tRNA synthetases are enzymes that load the individual amino acids onto the tRNAs.

Key Terms

- **ribosome**: protein/mRNA complexes found in all cells that are involved in the production of proteins by translating messenger RNA

The Protein Synthesis Machinery

In addition to the mRNA template, many molecules and macromolecules contribute to the process of translation. The composition of each component may vary across species. For instance, ribosomes may consist of different numbers of rRNAs and polypeptides depending on the organism. However, the general structures and functions of the protein synthesis machinery are comparable from bacteria to archaea to human cells. Translation requires the input of an mRNA template, ribosomes, tRNAs, and various enzymatic factors.

Ribosomes

A ribosome is a complex macromolecule composed of structural and catalytic rRNAs, and many distinct polypeptides. In eukaryotes, the synthesis and assembly of rRNAs occurs in the nucleolus.

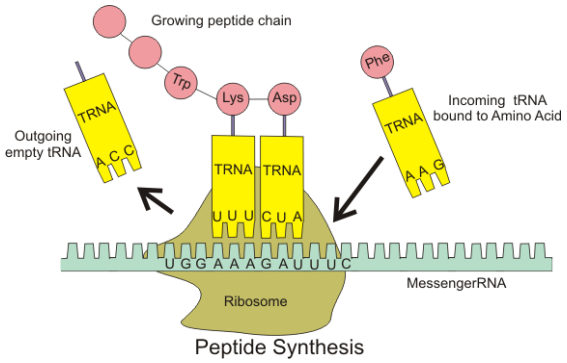


Figure 15.5A.1: **The ribosome in action**: Structure and role of ribosomes during translation

Ribosomes exist in the cytoplasm in prokaryotes and in the cytoplasm and on rough endoplasmic reticulum membranes in eukaryotes. Mitochondria and chloroplasts also have their own ribosomes, and these look more similar to prokaryotic ribosomes (and have similar drug sensitivities) than the cytoplasmic ribosomes. Ribosomes dissociate into large and small subunits when they are not synthesizing proteins and reassociate during the initiation of translation.*E. coli* have a 30S small subunit and a 50S large subunit, for a total of 70S when assembled (recall that Svedberg units are not additive). Mammalian

ribosomes have a small 40S subunit and a large 60S subunit, for a total of 80S. The small subunit is responsible for binding the mRNA template, whereas the large subunit sequentially binds tRNAs.

In bacteria, archaea, and eukaryotes, the intact ribosome has three binding sites that accomodate tRNAs: The A site, the P site, and the E site. Incoming aminoacy-tRNAs (a tRNA with an amino acid covalently attached is called an aminoacyl-tRNA) enter the ribosome at the A site. The peptidyl-tRNA carrying the growing polypeptide chain is held in the P site. The E site holds empty tRNAs just before they exit the ribosome.

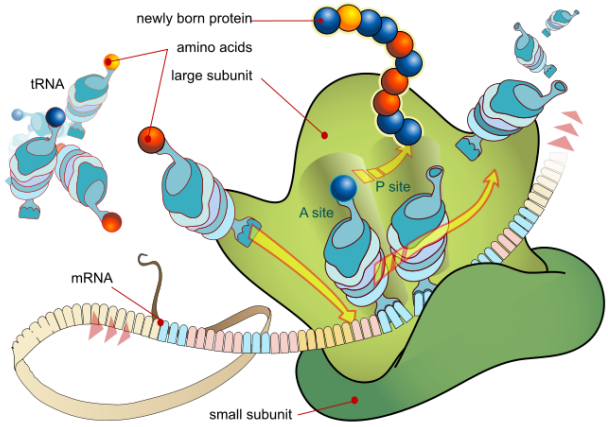


Figure 15.5A.1: **Ribosome structure**: The large ribosomal subunit sits atop the small ribosomal subunit and the mRNA is threaded through a groove near the interface of the two subunits. The intact ribosome has three tRNA binding sites: the A site for incoming aminoacyl-tRNAs; the P site for the peptidyl-tRNA carrying the growing polypeptide chain; and the E site where empty tRNAs exit (not shown in this figure but immediately adjacent to the P site.)

Each mRNA molecule is simultaneously translated by many ribosomes, all reading the mRNA from 5′ to 3′ and synthesizing the polypeptide from the N terminus to the C terminus. The complete mRNA/poly-ribosome structure is called a polysome.

tRNAs in eukaryotes

The tRNA molecules are transcribed by RNA polymerase III. Depending on the species, 40 to 60 types of tRNAs exist in the cytoplasm. Specific tRNAs bind to codons on the mRNA template and add the corresponding amino acid to the polypeptide chain. (More accurately, the growing polypeptide chain is added to each new amino acid bound in by a tRNA.)

The transfer RNAs (tRNAs) are structural RNA molecules. In eukaryotes, tRNA mole are transcribed from tRNA genes by RNA polymerase III. Depending on the species, 40 to 60 types of tRNAs exist in the cytoplasm. Serving as adaptors, specific tRNAs bind to sequences on the mRNA template and add the corresponding amino acid to the polypeptide chain. (More accurately, the growing polypeptide chain is added to each new amino acid brought in by a tRNA.) Therefore, tRNAs are the molecules that actually “translate” the language of RNA into the language of proteins.

Of the 64 possible mRNA codons (triplet combinations of A, U, G, and C) three specify the termination of protein synthesis and 61 specify the addition of amino acids to the polypeptide chain. Of the three termination codons, one (UGA) can also be used to encode the 21st amino acid, selenocysteine, but only if the mRNA contains a specific sequence of nucleotides known as a SECIS sequence. Of the 61 non-termination codons, one codon (AUG) also encodes the initiation of translation.

Each tRNA polynucleotide chain folds up so that some internal sections basepair with other internal sections. If just diagrammed in two dimensions, the regions where basepairing occurs are called stems, and the regions where no basepairs form are called loops, and the entire pattern of stems and loops that forms for a tRNA is called the “cloverleaf” structure. All tRNAs fold into very similar cloverleaf structures of four major stems and three major loops.

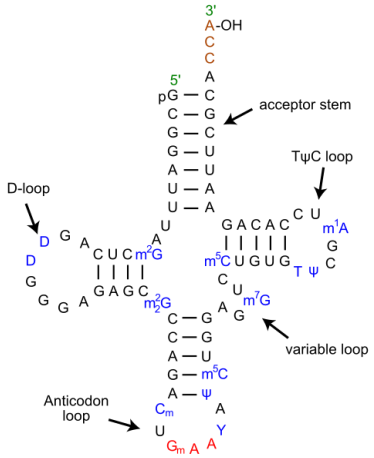


Figure 15.5A.1: **The two-dimensional cloverleaf structure of a typical tRNA.**: All tRNAs, regardless of the species they come from or the amino acid they carry, self-basepair to produce a cloverleaf structure of four main stems and three main loops. The amino acid carried by the tRNA is covalently attached to the nucleotide at the 3′ end of the tRNA, known as the tRNA's acceptor arm. The opposite end of the folded tRNA has the anticodon loop where the tRNA will basepair to the mRNA codon.

If viewed as a three-dimensional structure, all the basepaired regions of the tRNA are helical, and the tRNA folds into a L-shaped structure.

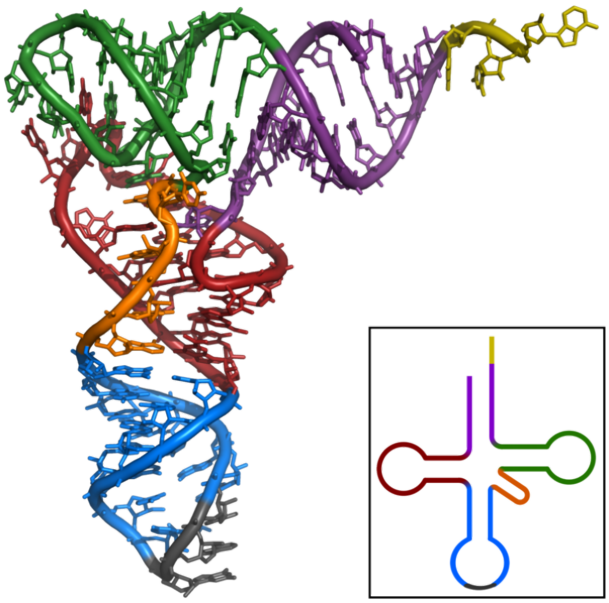


Figure 15.5A.1: **The three dimensional shape taken by tRNAs.**: If viewed as a three-dimensional structure, all tRNAs are partially helical molecules that are vaguely L-shaped. The anticodon-containing loop is at one end of the molecule (in grey here) and the amino acid acceptor arm is at the other end of the molecule (in yellow here) past the bend of the "L".

Each tRNA has a sequence of three nucleotides located in a loop at one end of the molecule that can basepair with an mRNA codon. This is called the tRNA's anticodon. Each different tRNA has a different anticodon. When the tRNA anticodon basepairs with one of the mRNA codons, the tRNA will add an amino acid to a growing polypeptide chain or terminate translation, according to the genetic code. For instance, if the sequence CUA occurred on a mRNA template in the proper reading frame, it would bind a tRNA with an anticodon expressing the complementary sequence, GAU. The tRNA with this anticodon would be linked to the amino acid leucine.

Aminoacyl tRNA Synthetases

The process of pre-tRNA synthesis by RNA polymerase III only creates the RNA portion of the adaptor molecule. The corresponding amino acid must be added later, once the tRNA is processed and exported to the cytoplasm. Through the process of tRNA "charging," each tRNA molecule is linked to its correct amino acid by a group of enzymes called aminoacyl

tRNA synthetases. When an amino acid is covalently linked to a tRNA, the resulting complex is known as an aminoacyl-tRNA. At least one type of aminoacyl tRNA synthetase exists for each of the 21 amino acids; the exact number of aminoacyl tRNA synthetases varies by species. These enzymes first bind and hydrolyze ATP to catalyze the formation of a covalent bond between an amino acid and adenosine monophosphate (AMP); a pyrophosphate molecule is expelled in this reaction. This is called “activating” the amino acid. The same enzyme then catalyzes the attachment of the activated amino acid to the tRNA and the simultaneous release of AMP. After the correct amino acid covalently attached to the tRNA, it is released by the enzyme. The tRNA is said to be charged with its cognate amino acid. (the amino acid specified by its anticodon is a tRNA’s cognate amino acid.)

B: The Mechanism of Protein Synthesis

Protein synthesis involves building a peptide chain using tRNAs to add amino acids and mRNA as a blueprint for the specific sequence.

Learning Objectives

- Describe the process of translation

Key Points

- Protein synthesis, or translation, begins with a process known as pre-initiation, when the small ribosmal subunit, the mRNA template, initiator factors, and a special initiator tRNA, come together.
- During translocation and elongation, the ribosome moves one codon 3′ down the mRNA, brings in a charged tRNA to the A site, transfers the growing polypeptide chain from the P-site tRNA to the carboxyl group of the A-site amino acid, and ejects the uncharged tRNA at the E site.
- When a stop or nonsense codon (UAA, UAG, or UGA) is reached on the mRNA, the ribosome terminates translation.

Key Terms

- **translation**: a process occurring in the ribosome in which a strand of messenger RNA (mRNA) guides assembly of a sequence of amino acids to make a protein

The Mechanism of Protein Synthesis

As with mRNA synthesis, protein synthesis can be divided into three phases: initiation, elongation, and termination.

Initiation of Translation

Protein synthesis begins with the formation of a pre-initiation complex. In *E. coli*, this complex involves the small 30S ribosome, the mRNA template, three initiation factors (IFs; IF-1, IF-2, and IF-3), and a special initiator tRNA, called fMet-tRNA. The initiator tRNA basepairs to the start codon AUG (or rarely, GUG) and is covalently linked to a formylated methionine called fMet. Methionine is one of the 21 amino acids used in protein synthesis; formylated methionine is a methione to which a formyl group (a one-carbon aldehyde) has been covalently attached at the amino nitrogen. Formylated methionine is inserted by fMet-tRNA at the beginning of every polypeptide chain synthesized by *E. coli*, and is usually clipped off after translation is complete. When an in-frame AUG is encountered during translation elongation, a non-formylated methionine is inserted by a regular Met-tRNA. In *E. coli* mRNA, a sequence upstream of the first AUG codon, called the Shine-Dalgarno sequence (AGGAGG), interacts with the rRNA molecules that compose the ribosome. This interaction anchors the 30S ribosomal subunit at the correct location on the mRNA template.

In eukaryotes, a pre-initiation complex forms when an initiation factor called eIF2 (eukaryotic initiation factor 2) binds GTP, and the GTP-eIF2 recruits the eukaryotic initiator tRNA to the 40s small ribosomal subunit. The initiator tRNA, called Met-$tRNA_i$, carries unmodified methionine in eukaryotes, not fMet, but it is distinct from other cellular Met-tRNAs in that it can bind eIFs and it can bind at the ribosome P site. The eukaryotic pre-initiation complex then recognizes the 7-methylguanosine cap at the 5′ end of a mRNA. Several other eIFs, specifically eIF1, eIF3, and eIF4, act as cap-binding proteins and assist the recruitment of the pre-initiation complex to the 5′ cap. Poly (A)-Binding Protein (PAB) binds both the poly (A) tail of the mRNA and the complex of proteins at the cap and also assists in the process. Once at the cap, the pre-initiation complex tracks along the mRNA in the 5′ to 3′ direction, searching for the AUG start codon. Many, but not all, eukaryotic mRNAs are translated from the first AUG sequence. The nucleotides around the AUG indicate whether it is the correct start codon.

Once the appropriate AUG is identified, eIF2 hydrolyzes GTP to GDP and powers the delivery of the $tRNA_i$-Met to the start codon, where the $tRNA_i$ anticodon basepairs to the AUG codon. After this, eIF2-GDP is released from the complex, and eIF5-GTP binds. The 60S ribosomal subunit is recruited to the pre-initiation complex by eIF5-GTP, which hydrolyzes its GTP to GDP to power the assembly of the full ribosome at the translation start site with the Met-tRNAi positioned in the ribosome P site. The remaining eIFs dissociate from the ribosome and translation is ready to begins.

In archaea, translation initiation is similar to that seen in eukaryotes, except that the initiation factors involved are called aIFs (archaeal inititiaion factors), not eIFs.

Figure 15.5B.1: **Translation initiation in eukaryotes.**: In eukaryotes, a preinitiation complex forms made of the small 40S subunit, the initiator Met-tRNAi, and eIF2-GTP. This preinitiation complex binds to the 5′-m^7G cap of the mRNA with the help of other eIFS and PAB, which binds the poly(A) tail of the mRNA, and loops the tail to the cap. Once at the cap, the preinitiation complex slides along the mRNA until it encounters the initiator AUG codon. There, GTP is hydrolyzed by eIF2 and the Met-$tRNA_i$ is loaded onto the AUG. Next, eIF5-GTP recruits the 60S large ribosomal subunit to the 40S subunit at the AUG and hydrolyzes GTP. This allows the large ribosomal subunit to assemble on top of the small subunit, generating the intact 80S ribosome, and places the Met-tRNAi in the P site of the intact ribosome. The ribosome A site is positioned over the second codon in the mRNA reading frame, and translation elongation can begin.

Translation Elongation

The basics of elongation are the same in prokaryotes and eukaryotes. The intact ribosome has three compartments: the A site binds incoming aminoacyl tRNAs; the P site binds tRNAs carrying the growing polypeptide chain; the E site releases dissociated tRNAs so that they can be recharged with amino acids. The initiator tRNA, rMet-tRNA in *E. coli* and Met-$tRNA_i$ in eukaryotes and archaea, binds directly to the P site. This creates an initiation complex with a free A site ready to accept the aminoacyl-tRNA corresponding to the first codon after the AUG.

The aminoacyl-tRNA with an anticodon complementary to the A site codon lands in the A site. A peptide bond is formed between the amino group of the A site amino acid and the carboxyl group of the most-recently attached amino acid in the growing polypeptide chain attached to the P-site tRNA.The formation of the peptide bond is catalyzed by peptidyl transferase, an RNA-based enzyme that is integrated into the large ribosomal subunit. The energy for the peptide bond formation is derived from GTP hydrolysis, which is catalyzed by a separate elongation factor.

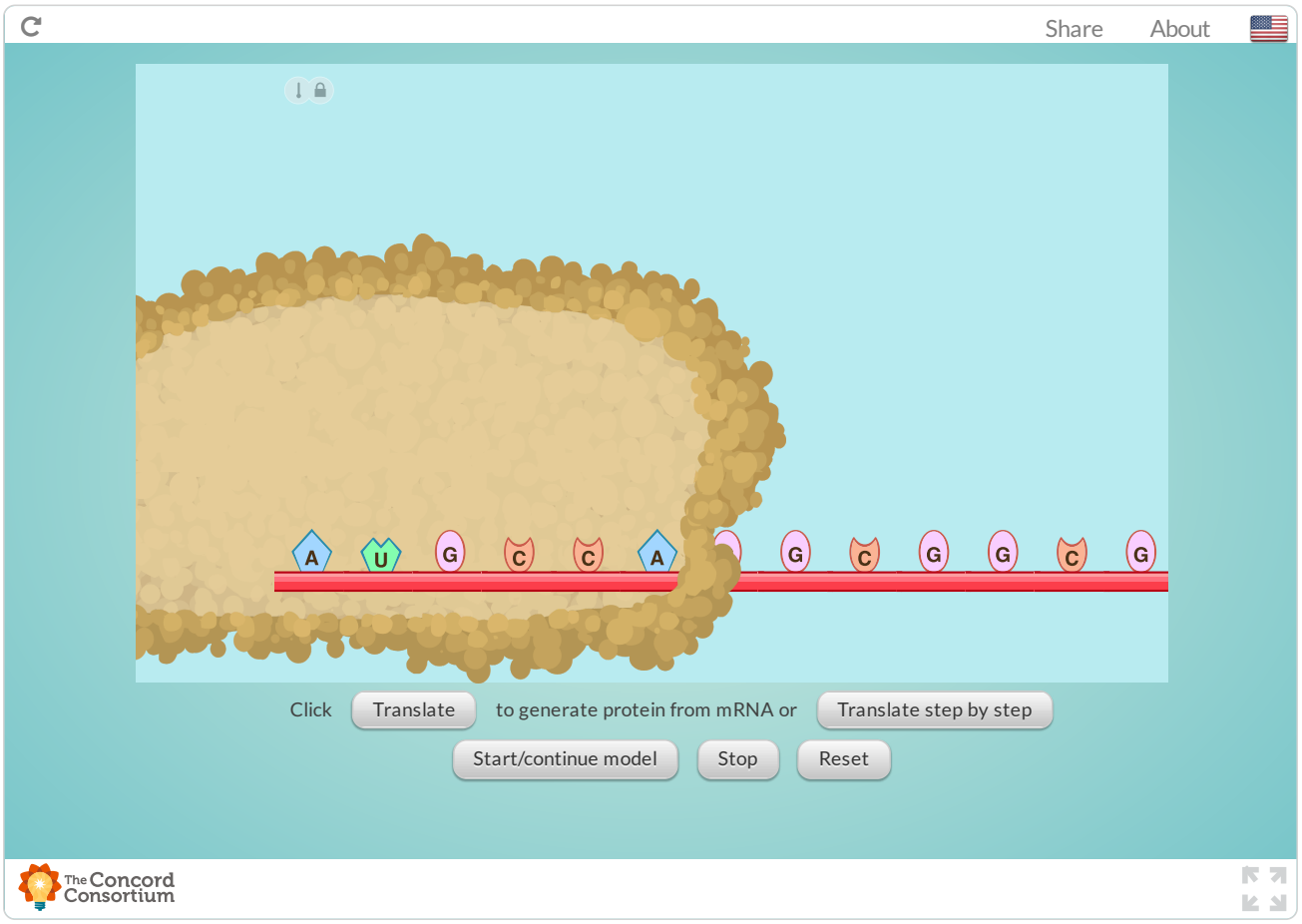
Catalyzing the formation of a peptide bond removes the bond holding the growing polypeptide chain to the P-site tRNA. The growing polypeptide chain is transferred to the amino end of the incoming amino acid, and the A-site tRNA temporarily holds the growing polypeptide chain, while the P-site tRNA is now empty or uncharged.

The ribosome moves three nucleotides down the mRNA. The tRNAs are basepaired to a codon on the mRNA, so as the ribosome moves over the mRNA, the tRNAs stay in place while the ribosome moves and each tRNA is moved into the next tRNA binding site. The E site moves over the former P-site tRNA, now empty or uncharged, the P site moves over the former A-site tRNA, now carrying the growing polypeptide chain, and the A site moves over a new codon. In the E site, the uncharged tRNA detaches from its anticodon and is expelled. A new aminoacyl-tRNA with an anticodon complementary to the new A-site codon enters the ribosome at the A site and the elongation process repeats itself. The energy for each step of the ribosome is donated by an elongation factor that hydrolyzes GTP.

Figure 15.5B.1: **Translation elongation in eukaryotes.**: During translation elongation, the incoming aminoacyl-tRNA enters the ribosome A site, where it binds if the tRNA anticodon is complementary to the A site mRNA codon. The elongation factor eEF1 assists in loading the aminoacyl-tRNA, powering the process through the hydrolysis of GTP. The growing polypeptide chain is attached to the tRNA in the ribosome P site. The ribosome's peptidyl transferase catalyses the transfer of the growing polypeptide chain from the P site tRNA to the amino group of the A site amino acid. This creates a peptide bond between the C terminus of the growing polypeptide chain and the A site amino acid. After the peptide bond is created, the growing polypeptide chain is attached to the A site tRNA, and the tRNA in the P site is empty. The ribosome translocates once codon on the mRNA. The elongation factor eEF2 assists in the translocation, powering the process through the hydrolysis of GTP. During translocation, the two tRNAs remain basepaired to their mRNA codons, so the ribosome moves over them, putting the empty tRNA in the E site (where it will be expelled from the ribosome) and the tRNA with the growing polypeptide chain in the P site. The A site moves over an empty codon, and the process repeats itself until a stop codon is reached.

Translation termination

Termination of translation occurs when the ribosome moves over a stop codon (UAA, UAG, or UGA). There are no tRNAs with anticodons complementary to stop codons, so no tRNAs enter the A site. Instead, in both prokaryotes and eukaryotes, a protein called a release factor enters the A site. The release factors cause the ribosome peptidyl transferase to add a water molecule to the carboxyl end of the most recently added amino acid in the growing polypeptide chain attached to the P-site tRNA. This causes the polypeptide chain to detach from its tRNA, and the newly-made polypeptide is released. The small and large ribosomal subunits dissociate from the mRNA and from each other; they are recruited almost immediately into another translation initiation complex. After many ribosomes have completed translation, the mRNA is degraded so the nucleotides can be reused in another transcription reaction.



Modeling translation: This interactive models the process of translation in eukaryotes.

15.5C: Protein Folding, Modification, and Targeting

In order to function, proteins must fold into the correct three-dimensional shape, and be targeted to the correct part of the cell.

Learning Objectives

- Discuss how post-translational events affect the proper function of a protein

Key Points

- Protein folding is a process in which a linear chain of amino acids attains a defined three-dimensional structure, but there is a possibility of forming misfolded or denatured proteins, which are often inactive.
- Proteins must also be located in the correct part of the cell in order to function correctly; therefore, a signal sequence is often attached to direct the protein to its proper location, which is removed after it attains its location.
- Protein misfolding is the cause of numerous diseases, such as mad cow disease, Creutzfeldt-Jakob disease, and cystic fibrosis.

Key Terms

- **prion**: a self-propagating misfolded conformer of a protein that is responsible for a number of diseases that affect the brain and other neural tissue
- **chaperone**: a protein that assists the non-covalent folding/unfolding of other proteins

Protein Folding

After being translated from mRNA, all proteins start out on a ribosome as a linear sequence of amino acids. This linear sequence must "fold" during and after the synthesis so that the protein can acquire what is known as its native conformation. The native conformation of a protein is a stable three-dimensional structure that strongly determines a protein's biological function. When a protein loses its biological function as a result of a loss of three-dimensional structure, we say that the protein has undergone denaturation. Proteins can be denatured not only by heat, but also by extremes of pH; these two conditions affect the weak interactions and the hydrogen bonds that are responsible for a protein's three-dimensional structure. Even if a protein is properly specified by its corresponding mRNA, it could take on a completely dysfunctional shape if abnormal temperature or pH conditions prevent it from folding correctly. The denatured state of the protein does not equate with the unfolding of the protein and randomization of conformation. Actually, denatured proteins exist in a set of partially-folded states that are currently poorly understood. Many proteins fold spontaneously, but some proteins require helper molecules, called chaperones, to prevent them from aggregating during the complicated process of folding.

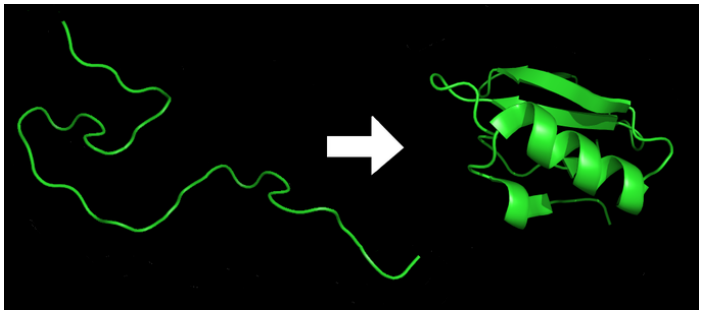


Figure $15.5C.1$: **Protein folding**: A protein starts as a linear sequence of amino acids, then folds into a 3-dimensional shape imbued with all the functional properties required inside the cell.

Protein Modification and Targeting

During and after translation, individual amino acids may be chemically modified and signal sequences may be appended to the protein. A signal sequence is a short tail of amino acids that directs a protein to a specific cellular compartment. These sequences at the amino end or the carboxyl end of the protein can be thought of as the protein's "train ticket" to its ultimate destination. Other cellular factors recognize each signal sequence and help transport the protein from the cytoplasm to its

correct compartment. For instance, a specific sequence at the amino terminus will direct a protein to the mitochondria or chloroplasts (in plants). Once the protein reaches its cellular destination, the signal sequence is usually clipped off.

Misfolding

It is very important for proteins to achieve their native conformation since failure to do so may lead to serious problems in the accomplishment of its biological function. Defects in protein folding may be the molecular cause of a range of human genetic disorders. For example, cystic fibrosis is caused by defects in a membrane-bound protein called cystic fibrosis transmembrane conductance regulator (CFTR). This protein serves as a channel for chloride ions. The most common cystic fibrosis-causing mutation is the deletion of a Phe residue at position 508 in CFTR, which causes improper folding of the protein. Many of the disease-related mutations in collagen also cause defective folding.

A misfolded protein, known as prion, appears to be the agent of a number of rare degenerative brain diseases in mammals, like the mad cow disease. Related diseases include kuru and Creutzfeldt-Jakob. The diseases are sometimes referred to as spongiform encephalopathies, so named because the brain becomes riddled with holes. Prion, the misfolded protein, is a normal constituent of brain tissue in all mammals, but its function is not yet known. Prions cannot reproduce independently and not considered living microoganisms. A complete understanding of prion diseases awaits new information about how prion protein affects brain function, as well as more detailed structural information about the protein. Therefore, improved understanding of protein folding may lead to new therapies for cystic fibrosis, Creutzfeldt-Jakob, and many other diseases.

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CHAPTER OVERVIEW

16: GENE EXPRESSION

16.1: REGULATION OF GENE EXPRESSION

16.1A: THE PROCESS AND PURPOSE OF GENE EXPRESSION REGULATION
16.2B: PROKARYOTIC VERSUS EUKARYOTIC GENE EXPRESSION
16.2: PROKARYOTIC GENE REGULATION

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16.2B: CATABOLITE ACTIVATOR PROTEIN (CAP): AN ACTIVATOR REGULATOR
16.2C: THE LAC OPERON: AN INDUCER OPERON
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16.1: Regulation of Gene Expression

Topic hierarchy

- 16.1A: The Process and Purpose of Gene Expression Regulation
- 16.2B: Prokaryotic versus Eukaryotic Gene Expression

16.1A: The Process and Purpose of Gene Expression Regulation

Gene expression is a highly complex, regulated process that begins with DNA transcribed into RNA, which is then translated into protein.

Learning Objectives

- Discuss how the genome and proteome contribute to the specialization of a cell

Key Points

- Every cell within an organism shares the same genome (with exceptions, i.e. mature red blood cells), but has variation between its proteomes.
- Gene expression involves the process of transcribing DNA into RNA and then translating RNA into proteins.
- Gene expression is a highly complex and tightly-regulated process.

Key Terms

- **somatic**: part of, or relating to the body of an organism
- **genome**: the cell's complete genetic information packaged as a double-stranded DNA molecule
- **proteome**: the complete set of proteins encoded by a particular genome

Each somatic cell in the body generally contains the same DNA. A few exceptions include red blood cells, which contain no DNA in their mature state, and some immune system cells that rearrange their DNA while producing antibodies. In general, however, the genes that determine whether you have green eyes, brown hair, and how fast you metabolize food are the same in the cells in your eyes and your liver, even though these organs function quite differently. If each cell has the same DNA, how is it that cells or organs are different? Why do cells in the eye differ so dramatically from cells in the liver ?

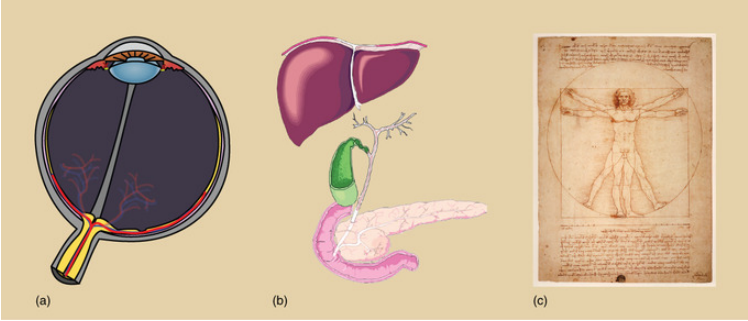

Figure 16.1A.1: **Gene Expression**: The genetic content of each somatic cell in an organism is the same, but not all genes are expressed in every cell. The control of which genes are expressed dictates whether a cell is (a) an eye cell or (b) a liver cell. It is the differential gene expression patterns that arise in different cells that give rise to (c) a complete organism.

Whereas each cell shares the same genome and DNA sequence, each cell does not turn on, or express, the same set of genes. Each cell type needs a different set of proteins to perform its function. Therefore, only a small subset of proteins is expressed in a cell that constitutes its proteome. For the proteins to be expressed, the DNA must be transcribed into RNA and the RNA must be translated into protein. In a given cell type, not all genes encoded in the DNA are transcribed into RNA or translated into protein because specific cells in our body have specific functions. Specialized proteins that make up the eye (iris, lens, and cornea) are only expressed in the eye, whereas the specialized proteins in the heart (pacemaker cells, heart muscle, and valves) are only expressed in the heart. At any given time, only a subset of all of the genes encoded by our DNA are expressed and translated into proteins. The expression of specific genes is a highly-regulated process with many levels and stages of control. This complexity ensures the proper expression in the proper cell at the proper time.

In this section, you will learn about the various methods of gene regulation and the mechanisms used to control gene expression, such as: epigenetic, transcriptional, post-transcriptional, translational, and post-translational controls in eukaryotic gene expression, and transcriptional control in prokaryotic gene expression.

16.2B: Prokaryotic versus Eukaryotic Gene Expression

Prokaryotes regulate gene expression by controlling the amount of transcription, whereas eukaryotic control is much more complex.

Learning Objectives

- Compare and contrast prokaryotic and eukaryotic gene expression

Key Points

- Prokaryotic gene expression is primarily controlled at the level of transcription.
- Eukaryotic gene expression is controlled at the levels of epigenetics, transcription, post-transcription, translation, and post-translation.
- Prokaryotic gene expression (both transcription and translation) occurs within the cytoplasm of a cell due to the lack of a defined nucleus; thus, the DNA is freely located within the cytoplasm.
- Eukaryotic gene expression occurs in both the nucleus (transcription) and cytoplasm (translation).

Key Terms

- **epigenetics**: the study of heritable changes caused by the activation and deactivation of genes without any change in DNA sequence
- **nucleosome**: any of the subunits that repeat in chromatin; a coil of DNA surrounding a histone core

Prokaryotic versus Eukaryotic Gene Expression

To understand how gene expression is regulated, we must first understand how a gene codes for a functional protein in a cell. The process occurs in both prokaryotic and eukaryotic cells, just in slightly different manners.

Prokaryotic organisms are single-celled organisms that lack a defined nucleus; therefore, their DNA floats freely within the cell cytoplasm. To synthesize a protein, the processes of transcription (DNA to RNA) and translation (RNA to protein) occur almost simultaneously. When the resulting protein is no longer needed, transcription stops. Thus, the regulation of transcription is the primary method to control what type of protein and how much of each protein is expressed in a prokaryotic cell. All of the subsequent steps occur automatically. When more protein is required, more transcription occurs. Therefore, in prokaryotic cells, the control of gene expression is mostly at the transcriptional level.

Eukaryotic cells, in contrast, have intracellular organelles that add to their complexity. In eukaryotic cells, the DNA is contained inside the cell's nucleus where it is transcribed into RNA. The newly-synthesized RNA is then transported out of the nucleus into the cytoplasm where ribosomes translate the RNA into protein. The processes of transcription and translation are physically separated by the nuclear membrane; transcription occurs only within the nucleus, and translation occurs only outside the nucleus within the cytoplasm. The regulation of gene expression can occur at all stages of the process. Regulation may occur when the DNA is uncoiled and loosened from nucleosomes to bind transcription factors (epigenetics), when the RNA is transcribed (transcriptional level), when the RNA is processed and exported to the cytoplasm after it is transcribed (post-transcriptional level), when the RNA is translated into protein (translational level), or after the protein has been made (post-translational level).

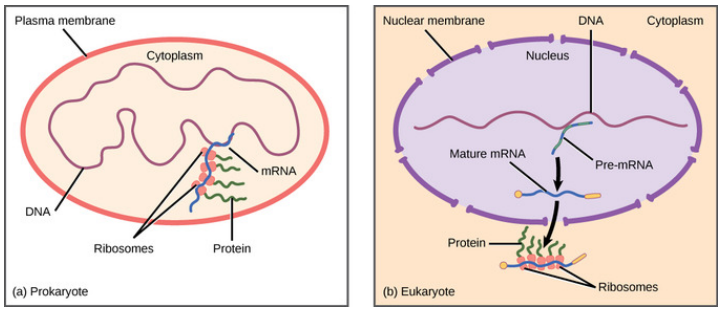

Figure 16.2*B*.1: **Prokaryotic vs Eukaryotic Gene Expression**: Prokaryotic transcription and translation occur simultaneously in the cytoplasm, and regulation occurs at the transcriptional level. Eukaryotic gene expression is regulated during transcription and RNA processing, which take place in the nucleus, and during protein translation, which takes place in the cytoplasm. Further regulation may occur through post-translational modifications of proteins.

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16.2: Prokaryotic Gene Regulation

Topic hierarchy

- 16.2A: The trp Operon: A Repressor Operon
- 16.2B: Catabolite Activator Protein (CAP): An Activator Regulator
- 16.2C: The lac Operon: An Inducer Operon

16.2A: The trp Operon: A Repressor Operon

The trp operon is a repressor operon that is either activated or repressed based on the levels of tryptophan in the environment.

Learning Objectives

- Explain the relationship between structure and function of an operon and the ways in which repressors regulate gene expression

Key Points

- The operator sequence is encoded between the promoter region and the first trp-coding gene.
- The trp operon is repressed when tryptophan levels are high by binding the repressor protein to the operator sequence via a corepressor which blocks RNA polymerase from transcribing the trp-related genes.
- The trp operon is activated when tryptophan levels are low by dissociation of the repressor protein to the operator sequence which allows RNA polymerase to transcribe the trp genes in the operon.

Key Terms

- **repressor**: any protein that binds to DNA and thus regulates the expression of genes by decreasing the rate of transcription
- **operon**: a unit of genetic material that functions in a coordinated manner by means of an operator, a promoter, and structural genes that are transcribed together

The trp Operon: A Repressor Operon

Bacteria such as *E. coli* need amino acids to survive. Tryptophan is one such amino acid that *E. coli* can ingest from the environment. *E. coli* can also synthesize tryptophan using enzymes that are encoded by five genes. These five genes are next to each other in what is called the tryptophan (trp) operon. If tryptophan is present in the environment, then *E. coli* does not need to synthesize it; the switch controlling the activation of the genes in the trp operon is turned off. However, when tryptophan availability is low, the switch controlling the operon is turned on, transcription is initiated, the genes are expressed, and tryptophan is synthesized.

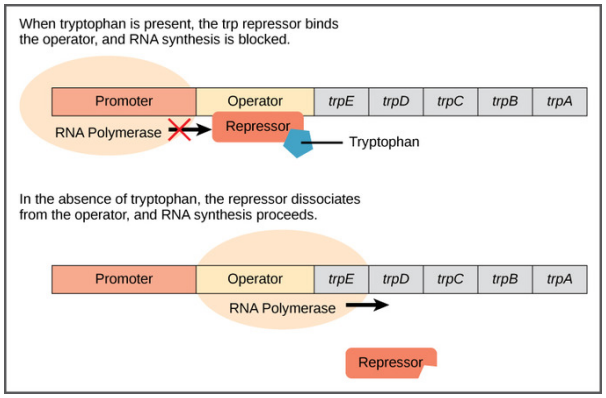

Figure 16.2*A*.1: **The trp operon**: The five genes that are needed to synthesize tryptophan in E. coli are located next to each other in the trp operon. When tryptophan is plentiful, two tryptophan molecules bind the repressor protein at the operator sequence. This physically blocks the RNA polymerase from transcribing the tryptophan genes. When tryptophan is absent, the repressor protein does not bind to the operator and the genes are transcribed.

A DNA sequence that codes for proteins is referred to as the coding region. The five coding regions for the tryptophan biosynthesis enzymes are arranged sequentially on the chromosome in the operon. Just before the coding region is the transcriptional start site. This is the region of DNA to which RNA polymerase binds to initiate transcription. The promoter sequence is upstream of the transcriptional start site. Each operon has a sequence within or near the promoter to which proteins (activators or repressors) can bind and regulate transcription.

A DNA sequence called the operator sequence is encoded between the promoter region and the first trp-coding gene. This operator contains the DNA code to which the repressor protein can bind. When tryptophan is present in the cell, two tryptophan molecules bind to the trp repressor, which changes shape to bind to the trp operator. Binding of the tryptophan–repressor complex at the operator physically prevents the RNA polymerase from binding and transcribing the downstream genes.

When tryptophan is not present in the cell, the repressor by itself does not bind to the operator; therefore, the operon is active and tryptophan is synthesized. Because the repressor protein actively binds to the operator to keep the genes turned off, the trp operon is negatively regulated and the proteins that bind to the operator to silence trp expression are negative regulators.

16.2B: Catabolite Activator Protein (CAP): An Activator Regulator

When glucose levels decline in *E. coli*, catabolite activator protein (CAP) is bound by cAMP to promote transcription of the lac operon.

Learning Objectives

- Explain how an activator works to increase transcription of a gene

Key Points

- Catabolite activator protein (CAP) must bind to cAMP to activate transcription of the lac operon by RNA polymerase.
- CAP is a transcriptional activator with a ligand-binding domain at the N-terminus and a DNA -binding domain at the C-terminus.
- cAMP molecules bind to CAP and function as allosteric effectors by increasing CAP's affinity to DNA.

Key Terms

- **RNA polymerase**: a DNA-dependent RNA polymerase, an enzyme, that produces RNA
- **operon**: a unit of genetic material that functions in a coordinated manner by means of an operator, a promoter, and structural genes that are transcribed together
- **promoter**: the section of DNA that controls the initiation of RNA transcription

Catabolite Activator Protein (CAP): An Activator Regulator

Just as the trp operon is negatively regulated by tryptophan molecules, there are proteins that bind to the operator sequences that act as a positive regulator to turn genes on and activate them. For example, when glucose is scarce, *E. coli* bacteria can turn to other sugar sources for fuel. To do this, new genes to process these alternate genes must be transcribed. This type of process can be seen in the lac operon which is turned on in the presence of lactose and absence of glucose.

When glucose levels drop, cyclic AMP (cAMP) begins to accumulate in the cell. The cAMP molecule is a signaling molecule that is involved in glucose and energy metabolism in *E. coli*. When glucose levels decline in the cell, accumulating cAMP binds to the positive regulator catabolite activator protein (CAP), a protein that binds to the promoters of operons that control the processing of alternative sugars, such as the lac operon. The CAP assists in production in the absence of glucose. CAP is a transcriptional activator that exists as a homodimer in solution, with each subunit comprising a ligand-binding domain at the N-terminus, which is also responsible for the dimerization of the protein and a DNA-binding domain at the C-terminus. Two cAMP molecules bind dimeric CAP with negative cooperativity and function as allosteric effectors by increasing the protein's affinity for DNA. CAP has a characteristic helix-turn-helix structure that allows it to bind to successive major grooves on DNA. This opens up the DNA molecule, allowing RNA polymerase to bind and transcribe the genes involved in lactose catabolism. When cAMP binds to CAP, the complex binds to the promoter region of the genes that are needed to use the alternate sugar sources. In these operons, a CAP-binding site is located upstream of the RNA-polymerase-binding site in the promoter. This increases the binding ability of RNA polymerase to the promoter region and the transcription of the genes. As cAMP-CAP is required for transcription of the lac operon, this requirement reflects the greater simplicity with which glucose may be metabolized in comparison to lactose.

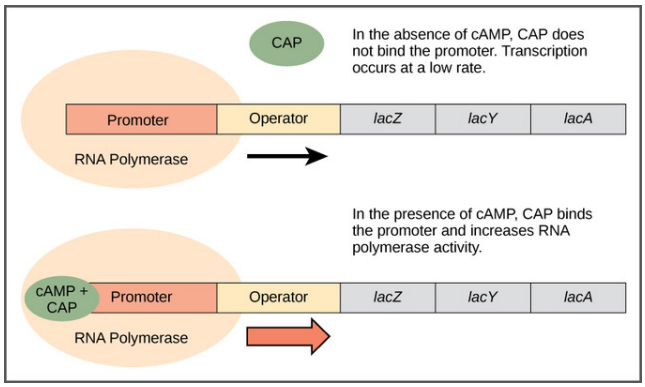

Figure 16.2B.1: **Catabolite Activator Protein (CAP) Regulation**: When glucose levels fall, E. coli may use other sugars for fuel, but must transcribe new genes to do so. As glucose supplies become limited, cAMP levels increase. This cAMP binds to the CAP protein, a positive regulator that binds to an operator region upstream of the genes required to use other sugar sources.

C: The lac Operon: An Inducer Operon

The lac operon is an inducible operon that utilizes lactose as an energy source and is activated when glucose is low and lactose is present.

Learning Objectives

- Describe the components of the lac operon and their role in its function

Key Points

- The lac operon contains an operator, promoter, and structural genes that are transcribed together and are under the control of the catabolite activator protein (CAP) or repressor.
- The lac operon is not activated and transcription remains off when the level of glucose is low or non-existent, but lactose is absent.
- The lac operon encodes for the genes needed to utilize lactose as an energy source.

Key Terms

- **operator**: a segment of DNA to which a transcription factor protein binds
- **repressor**: any protein that binds to DNA and thus regulates the expression of genes by decreasing the rate of transcription

The lac Operon: An Inducer Operon

A major type of gene regulation that occurs in prokaryotic cells utilizes and occurs through inducible operons. Inducible operons have proteins that can bind to either activate or repress transcription depending on the local environment and the needs of the cell. The lac operon is a typical inducible operon. As mentioned previously, *E. coli* is able to use other sugars as energy sources when glucose concentrations are low. To do so, the cAMP–CAP protein complex serves as a positive regulator to induce transcription. One such sugar source is lactose. The lac operon encodes the genes necessary to acquire and process the lactose from the local environment, which includes the structural genes lacZ, lacY, and lacA. *lacZ* encodes β-galactosidase (LacZ), an intracellular enzyme that cleaves the disaccharide lactose into glucose and galactose. *lacY* encodes β-galactoside permease (LacY), a membrane-bound transport protein that pumps lactose into the cell. *lacA* encodes β-galactoside transacetylase (LacA), an enzyme that transfers an acetyl group from acetyl-CoA to β-galactosides. Only lacZ and lacY appear to be necessary for lactose catabolism.

CAP binds to the operator sequence upstream of the promoter that initiates transcription of the lac operon. The *lac* operon uses a two-part control mechanism to ensure that the cell expends energy producing β-galactosidase, β-galactoside permease, and thiogalactoside transacetylase (also known as galactoside O-acetyltransferase) only when necessary. However, for the lac operon to be activated, two conditions must be met. First, the level of glucose must be very low or non-existent. Second, lactose must be present. If glucose is absent, then CAP can bind to the operator sequence to activate transcription. If lactose is absent, then the repressor binds to the operator to prevent transcription. If either of these requirements is met, then transcription remains off. The cell can use lactose as an energy source by producing the enzyme b-galactosidase to digest that lactose into glucose and galactose. Only when both conditions are satisfied is the lac operon transcribed, such as when glucose is absent and lactose is present. This process is beneficial and makes most sense for the cell as it would be energetically wasteful to create the proteins to process lactose if glucose were plentiful or if lactose were not available.

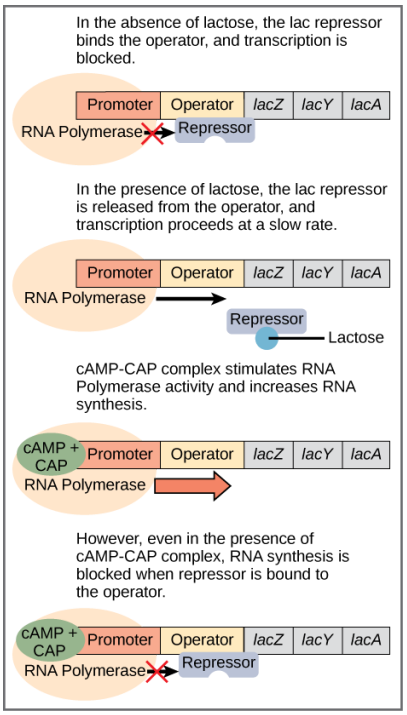

Figure 16.2C.1: **The lac Operon**: Transcription of the lac operon is carefully regulated so that its expression only occurs when glucose is limited and lactose is present to serve as an alternative fuel source.

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16.3: Eukaryotic Gene Regulation

Topic hierarchy

- 16.3A: The Promoter and the Transcription Machinery
- 16.3B: Transcriptional Enhancers and Repressors
- 16.3C: Epigenetic Control: Regulating Access to Genes within the Chromosome
- 16.3D: RNA Splicing
- 16.3E: The Initiation Complex and Translation Rate
- 16.3F: Regulating Protein Activity and Longevity

16.3A: The Promoter and the Transcription Machinery

When transcription factors bind to the promoter region, RNA polymerase is placed in an orientation that allows transcription to begin.

Learning Objectives

- Describe the role of promoters in RNA transcription

Key Points

- The purpose of the promoter is to bind transcription factors that control the initiation of transcription.
- The promoter region can be short or quite long; the longer the promoter is, the more available space for proteins to bind.
- To initiate transcription, a transcription factor (TFIID) binds to the TATA box, which causes other transcription factors to subsequently bind to the TATA box.
- Once the transcription initiation complex is assembled, RNA polymerase can bind to its upstream sequence and is then phosphorylated.
- Phosphorylation of RNA polymerase releases part of the protein from the DNA to activate the transcription initiation complex and places RNA polymerase in the correct orientation to begin transcription.
- Transcription factors respond to environmental stimuli that cause the proteins to find their binding sites and initiate transcription of the gene that is needed.

Key Terms

- **TATA box**: a DNA sequence (cis-regulatory element) found in the promoter region of genes in archaea and eukaryotes
- **transcription factor**: a protein that binds to specific DNA sequences, thereby controlling the flow (or transcription) of genetic information from DNA to mRNA
- **promoter**: the section of DNA that controls the initiation of RNA transcription

The Promoter and the Transcription Machinery

Genes are organized to make the control of gene expression easier. The promoter region is immediately upstream of the coding sequence. This region can be short (only a few nucleotides in length) or quite long (hundreds of nucleotides long). The longer the promoter, the more available space for proteins to bind. This also adds more control to the transcription process. The length of the promoter is gene-specific and can differ dramatically between genes. Consequently, the level of control of gene expression can also differ quite dramatically between genes. The purpose of the promoter is to bind transcription factors that control the initiation of transcription.

Within the promoter region, just upstream of the transcriptional start site, resides the TATA box. This box is simply a repeat of thymine and adenine dinucleotides (literally, TATA repeats). RNA polymerase binds to the transcription initiation complex, allowing transcription to occur. To initiate transcription, a transcription factor (TFIID) is the first to bind to the TATA box. Binding of TFIID recruits other transcription factors, including TFIIB, TFIIE, TFIIF, and TFIIH to the TATA box. Once this transcription initiation complex is assembled, RNA polymerase can bind to its upstream sequence. When bound along with the transcription factors, RNA polymerase is phosphorylated. This releases part of the protein from the DNA to activate the transcription initiation complex and places RNA polymerase in the correct orientation to begin transcription; DNA-bending protein brings the enhancer, which can be quite a distance from the gene, in contact with transcription factors and mediator proteins.

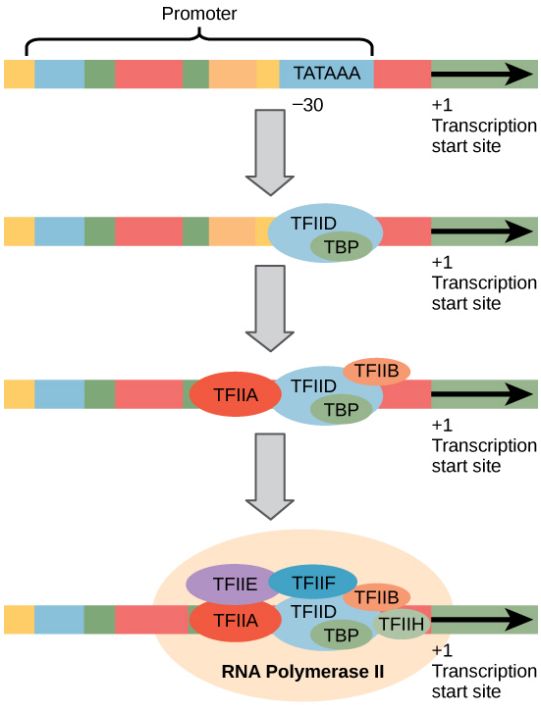

Figure 16.3A.1: **Promoters**: A generalized promoter of a gene transcribed by RNA polymerase II is shown. Transcription factors recognize the promoter. RNA polymerase II then binds and forms the transcription initiation complex.

In addition to the general transcription factors, other transcription factors can bind to the promoter to regulate gene transcription. These transcription factors bind to the promoters of a specific set of genes. They are not general transcription factors that bind to every promoter complex, but are recruited to a specific sequence on the promoter of a specific gene. There are hundreds of transcription factors in a cell that each bind specifically to a particular DNA sequence motif. When transcription factors bind to the promoter just upstream of the encoded gene, they are referred to as cis-acting elements because they are on the same chromosome, just next to the gene. The region that a particular transcription factor binds to is called the transcription factor binding site. Transcription factors respond to environmental stimuli that cause the proteins to find their binding sites and initiate transcription of the gene that is needed.

16.3B: Transcriptional Enhancers and Repressors

Enhancers increase the rate of transcription of genes, while repressors decrease the rate of transcription.

Learning Objectives

- Explain how enhancers and repressors regulate gene expression

Key Points

- Enhancers can be located upstream of a gene, within the coding region of the gene, downstream of a gene, or thousands of nucleotides away.
- When a DNA -bending protein binds to the enhancer, the shape of the DNA changes, which allows interactions between the activators and transcription factors to occur.
- Repressors respond to external stimuli to prevent the binding of activating transcription factors.
- Corepressors can repress transcriptional initiation by recruiting histone deacetylase.
- Histone deactylation increases the positive charge on histones, which strengthens the interaction between the histones and DNA, making the DNA less accessible to transcription.

Key Terms

- **enhancer**: a short region of DNA that can increase transcription of genes
- **repressor**: any protein that binds to DNA and thus regulates the expression of genes by decreasing the rate of transcription
- **activator**: any chemical or agent which regulates one or more genes by increasing the rate of transcription

Enhancers and Transcription

In some eukaryotic genes, there are regions that help increase or enhance transcription. These regions, called enhancers, are not necessarily close to the genes they enhance. They can be located upstream of a gene, within the coding region of the gene, downstream of a gene, or may be thousands of nucleotides away.

Enhancer regions are binding sequences, or sites, for transcription factors. When a DNA-bending protein binds to an enhancer, the shape of the DNA changes. This shape change allows the interaction between the activators bound to the enhancers and the transcription factors bound to the promoter region and the RNA polymerase to occur. Whereas DNA is generally depicted as a straight line in two dimensions, it is actually a three-dimensional object. Therefore, a nucleotide sequence thousands of nucleotides away can fold over and interact with a specific promoter.

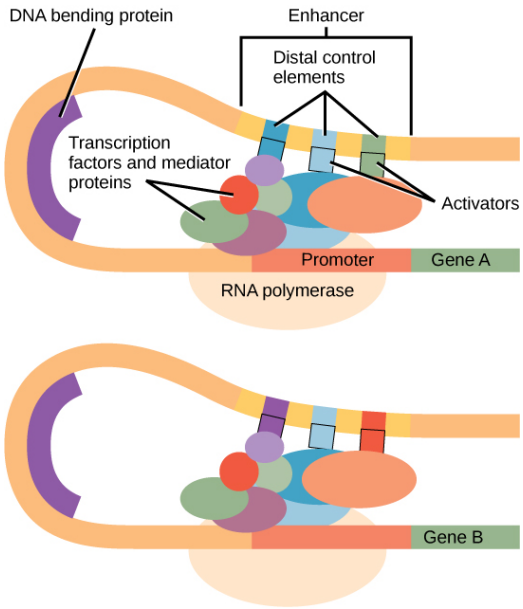

Figure 16.3B.1: **Enhancers**: An enhancer is a DNA sequence that promotes transcription. Each enhancer is made up of short DNA sequences called distal control elements. Activators bound to the distal control elements interact with mediator proteins and transcription factors.

Turning Genes Off: Transcriptional Repressors

Like prokaryotic cells, eukaryotic cells also have mechanisms to prevent transcription. Transcriptional repressors can bind to promoter or enhancer regions and block transcription. Like the transcriptional activators, repressors respond to external stimuli to prevent the binding of activating transcription factors.

A corepressor is a protein that decreases gene expression by binding to a transcription factor that contains a DNA-binding domain. The corepressor is unable to bind DNA by itself. The corepressor can repress transcriptional initiation by recruiting histone deacetylase, which catalyzes the removal of acetyl groups from lysine residues. This increases the positive charge on histones, which strengthens the interaction between the histones and DNA, making the DNA less accessible to the process of transcription.

16.3C: Epigenetic Control: Regulating Access to Genes within the Chromosome

Both the packaging of DNA around histone proteins, as well as chemical modifications to the DNA or proteins, can alter gene expression.

Learning Objectives

- Discuss how eukaryotic gene regulation occurs at the epigenetic level and the various epigenetic changes that can be made to DNA

Key Points

- DNA is packaged by wrapping around histone proteins into structures called nucleosomes, which resemble beads on a string.
- When DNA is to be transcribed, the nucleosomes can slide away from that region of DNA, opening it up to the transcription machinery of the cell.
- Chemical modifications to either the histone proteins or the DNA itself signals whether or not a particular region of the genome should be “open” or “closed” to the transcription machinery.
- Modifications such as acetylation or methylation of the histones can alter how tightly DNA is wrapped around them, while methylation of DNA changes how the DNA interacts with proteins, including the histone proteins that control access to the region.
- This type of genetic regulation is called epigenetic regulation (“above genetics”) as it does not change the nucleotide sequence of the DNA.

Key Terms

- **nucleosome**: any of the subunits that repeat in chromatin; a coil of DNA surrounding a histone core
- **epigenetics**: the study of heritable changes caused by the activation and deactivation of genes without any change in DNA sequence
- **histone**: any of various simple water-soluble proteins that are rich in the basic amino acids lysine and arginine and are complexed with DNA in the nucleosomes of eukaryotic chromatin

Epigenetic Control: Regulating Access to Genes within the Chromosome

The human genome encodes over 20,000 genes; each of the 23 pairs of human chromosomes encodes thousands of genes. The DNA in the nucleus is precisely wound, folded, and compacted into chromosomes so that it will fit into the nucleus. It is also organized so that specific segments can be accessed as needed by a specific cell type.

The first level of organization, or packing, is the winding of DNA strands around histone proteins. Histones package and order DNA into structural units called nucleosome complexes, which can control the access of proteins to the DNA regions. Under the electron microscope, this winding of DNA around histone proteins to form nucleosomes looks like small beads on a string. These beads (histone proteins) can move along the string (DNA) and change the structure of the molecule.

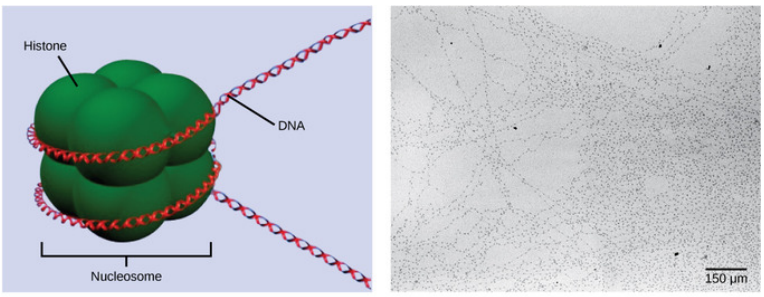

Figure 16.3C.1: **DNA Packaging**: DNA is folded around histone proteins to create (a) nucleosome complexes. These nucleosomes control the access of proteins to the underlying DNA. When viewed through an electron microscope (b), the nucleosomes look like beads on a string.

If DNA encoding a specific gene is to be transcribed into RNA, the nucleosomes surrounding that region of DNA can slide down the DNA to open that specific chromosomal region and allow for the transcriptional machinery (RNA polymerase) to initiate transcription. Nucleosomes can move to open the chromosome structure to expose a segment of DNA, but do so in a very controlled manner.

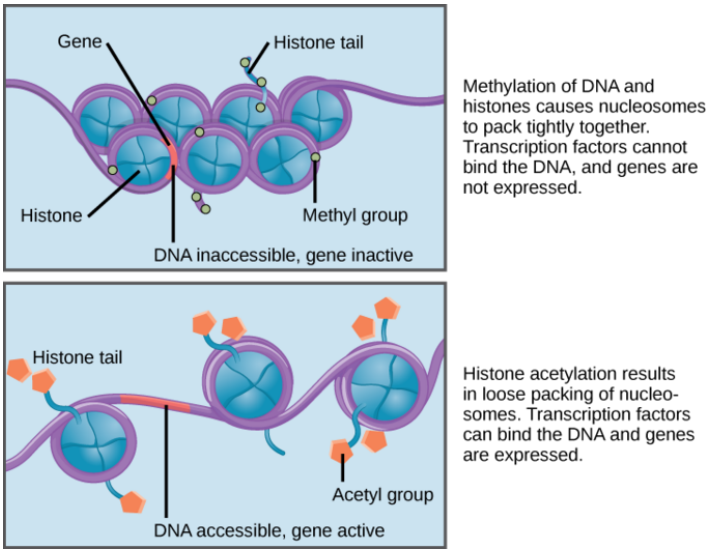

Figure 16.3C.1: **Nucleosomes can change position to allow transcription of genes**: Nucleosomes can slide along DNA. When nucleosomes are spaced closely together (top), transcription factors cannot bind and gene expression is turned off. When the nucleosomes are spaced far apart (bottom), the DNA is exposed. Transcription factors can bind, allowing gene expression to occur. Modifications to the histones and DNA affect nucleosome spacing.

How the histone proteins move is dependent on signals found on both the histone proteins and on the DNA. These signals are tags, or modifications, added to histone proteins and DNA that tell the histones if a chromosomal region should be open or closed. These tags are not permanent, but may be added or removed as needed. They are chemical modifications (phosphate, methyl, or acetyl groups) that are attached to specific amino acids in the protein or to the nucleotides of the DNA. The tags do not alter the DNA base sequence, but they do alter how tightly wound the DNA is around the histone proteins. DNA is a negatively-charged molecule; therefore, changes in the charge of the histone will change how tightly wound the DNA molecule will be. When unmodified, the histone proteins have a large positive charge; by adding chemical modifications, such as acetyl groups, the charge becomes less positive.

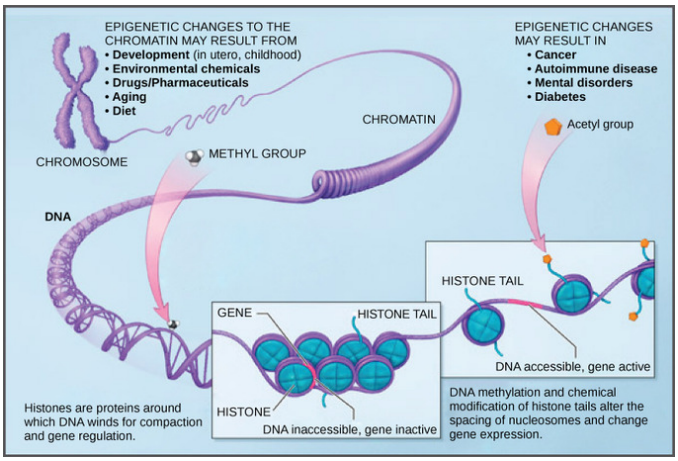

Figure 16.3C.1: **Modifications to histones and DNA can alter gene expression**: Histone proteins and DNA nucleotides can be modified chemically. Modifications affect nucleosome spacing and gene expression.

The DNA molecule itself can also be modified. This occurs within very specific regions called CpG islands. These are stretches with a high frequency of cytosine and guanine dinucleotide DNA pairs (CG) found in the promoter regions of genes. When this configuration exists, the cytosine member of the pair can be methylated (a methyl group is added). This modification changes how the DNA interacts with proteins, including the histone proteins that control access to the region. Highly-methylated (hypermethylated) DNA regions with deacetylated histones are tightly coiled and transcriptionally inactive. These changes to DNA are inherited from parent to offspring, such that while the DNA sequence is not altered, the pattern of gene expression is passed to the next generation.

This type of gene regulation is called epigenetic regulation. Epigenetics means “above genetics.” The changes that occur to the histone proteins and DNA do not alter the nucleotide sequence and are not permanent. Instead, these changes are temporary (although they often persist through multiple rounds of cell division) and alter the chromosomal structure (open or closed) as needed. A gene can be turned on or off depending upon the location and modifications to the histone proteins and DNA. If a gene is to be transcribed, the histone proteins and DNA are modified surrounding the chromosomal region encoding that gene. This opens the chromosomal region to allow access for RNA polymerase and other proteins, called transcription factors, to bind to the promoter region, located just upstream of the gene, and initiate transcription. If a gene is to remain turned off, or silenced, the histone proteins and DNA have different modifications that signal a closed chromosomal configuration. In this closed configuration, the RNA polymerase and transcription factors do not have access to the DNA and transcription cannot occur.

16.3D: RNA Splicing

RNA splicing allows for the production of multiple protein isoforms from a single gene by removing introns and combining different exons.

Learning Objectives

- Explain the role of RNA splicing in regulating gene expression

Key Points

- Introns are intervening sequences within a pre-mRNA molecule that do not code for proteins and are removed during RNA processing by a spliceosome.
- Exons are expressing sequences within a pre-mRNA molecule that are spliced together once introns are removed to form mature mRNA molecules that are translated into proteins.
- Alternative splicing allows for the production of various protein isoforms from one single gene coding.
- A spliceosome is a complex comprised of both RNA molecules and proteins which determine which introns to leave out and which exons to keep and bind together.

Key Terms

- **intron**: a portion of a split gene that is included in pre-RNA transcripts but is removed during RNA processing and rapidly degraded
- **exon**: a region of a transcribed gene present in the final functional RNA molecule
- **spliceosome**: a dynamic complex of RNA and protein subunits that removes introns from precursor mRNA

RNA splicing, the first stage of post-transcriptional control

Gene expression is the process that transfers genetic information from a gene made of DNA to a functional gene product made of RNA or protein. Genetic Information flows from DNA to RNA by the process of transcription and then from RNA to protein by the process of translation. In order to ensure that the proper products are produced, gene expression is regulated at many different stages during and in between transcription and translation. In eukaryotes, the gene contains extra sequences that do not code for protein. In these organisms, transcription of DNA produces pre-mRNA. These pre-mRNA transcripts often contain regions, called introns, that are intervening sequences which must be removed prior to translation by the process of splicing. The regions of RNA that code for protein are called exons. Splicing can be regulated so that different mRNAs can contain or lack exons, in a process called alternative splicing. Alternative splicing allows more than one protein to be produced from a gene and is an important regulatory step in determining which functional proteins are produced from gene expression. Thus, splicing is the first stage of post-transcriptional control.

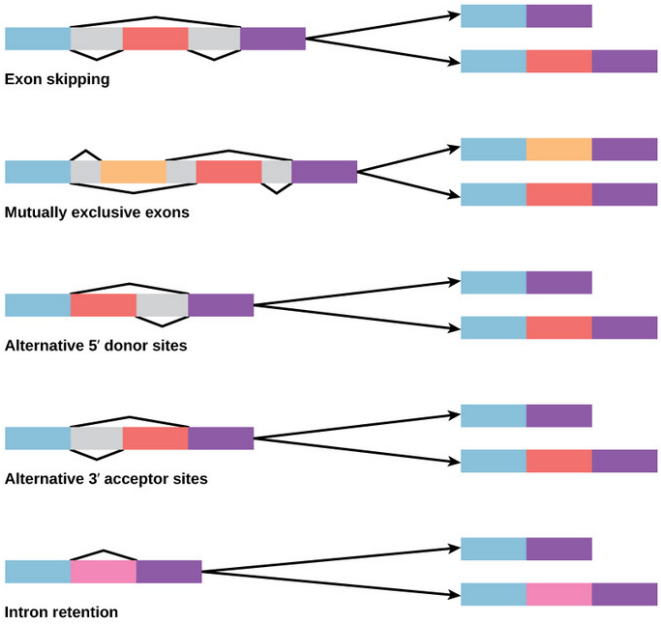

Figure 16.3D.1: **Alternative Splicing**: There are five basic modes of alternative splicing.

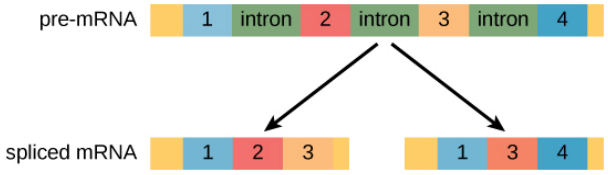

Figure 16.3D.1: **Alternative Splicing**: Pre-mRNA can be alternatively spliced to create different proteins.

Alternative Splicing

Alternative splicing is a process that occurs during gene expression and allows for the production of multiple proteins (protein isoforms) from a single gene coding. Alternative splicing can occur due to the different ways in which an exon can be excluded from or included in the messenger RNA. It can also occur if portions on an exon are excluded/included or if there is an inclusion of introns. For example, if a pre-mRNA has four exons (A, B, C, and D), these can be spliced and translated in a number of different combinations. Exons A, B, and C can be translated together or Exons A, C, and D can be translated. This results in what is called alternative splicing. The pattern of splicing and production of alternatively-spliced messenger RNA is controlled by the binding of regulatory proteins (trans-acting proteins that contain the genes) to cis-acting sites that are found on the pre-RNA. Some of these regulatory proteins include splicing activators (proteins that promote certain splicing sites) and splicing repressors (proteins that reduce the use of certain sites). Some common splicing repressors include: heterogeneous nuclear ribonucleoprotein (hnRNP) and polypyrimidine tract binding protein (PTB). Proteins that are translated from alternatively-spliced messenger RNAs differ in the sequence of their amino acids which results in altered function of the protein. This is one reason why the human genome can encode a wide diversity of proteins. Alternative splicing is a common process that occurs in eukaryotes; most of the multi-exonic genes in humans are spliced alternatively. Unfortunately, abnormal variations in splicing are also the reason why there are many genetic diseases and disorders.

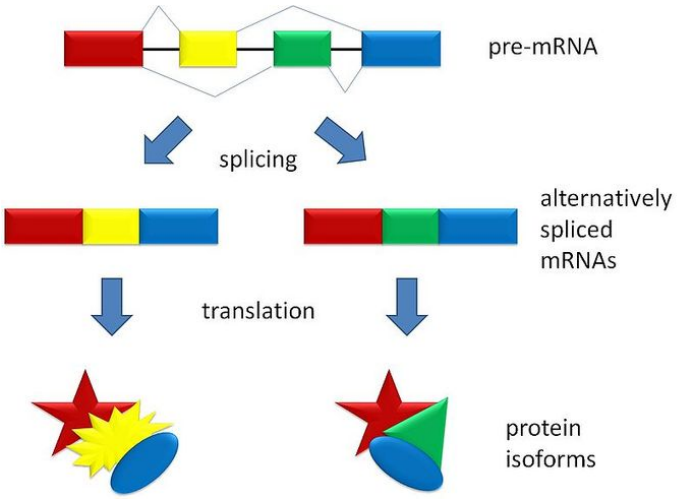

Figure 16.3D.1: **Mechanism of Splicing**: Alternative splicing can result in protein isoforms.

Spliceosome

The splicing of messenger RNA is accomplished and catalyzed by a macro-molecule complex known as the spliceosome. The areas for ligation and cleavage are determined by the many sub-units of the spliceosome which include the branch site (A) and the 5′ and 3′ splice sites. Interactions between these sub-units and the small nuclear ribonucleoproteins (snRNP) found in the spliceosome create a spliceosome A complex which helps determine which introns to leave out and which exons to keep and bind together. Once the introns are cleaved and removed, the exons are joined together by a phosphodiester bond.

Regulatory Proteins

As noted above, splicing is regulated by repressor proteins and activator proteins, which are are also known as trans-acting proteins. Equally as important are the silencers and enhancers that are found on the messenger RNAs, also known as cis-acting sites. These regulatory functions work together in order to create splicing code that determines alternative splicing.

E: The Initiation Complex and Translation Rate

The first step of translation is ribosome assembly, which requires initiation factors.

Learning Objectives

- Discuss how eukaryotes assemble ribosomes on the mRNA to begin translation

Key Points

- The components involved in ribosome assembly are brought together by the help of proteins called initiation factors which bind to the small ribosomal subunit.
- Initiator tRNA is used to locate the start codon AUG (the amino acid methionine) which establishes the reading frame for the mRNA strand.
- GTP carried by eIF2 is the energy source used for loading the initiator tRNA carried by the small ribosomal subunit on the correct start codon in the mRNA.
- GTP carried by eIF5 is the energy source for assembling the large and small ribosomal subunits together.

Key Terms

- **reading frame**: either of three possible triplets of codons in which a DNA sequence could be transcribed
- **phosphorylation**: the addition of a phosphate group to a compound; often catalyzed by enzymes

Ribosome Assembly and Translation Rate

Like transcription, translation is controlled by proteins that bind and initiate the process. In translation, before protein synthesis can begin, ribosome assembly has to be completed. This is a multi-step process.

In ribosome assembly, the large and small ribosomal subunits and an initiator tRNA ($tRNA_i$) containing the first amino acid of the final polypeptide chain all come together at the translation start codon on an mRNA to allow translation to begin. First, the small ribosomal subunit binds to the $tRNA_i$ which carries methionine in eukaryotes and archaea and carries N-formyl-methionine in bacteria. (Because the $tRNA_i$ is carrying an amino acid, it is said to be charged.) Next, the small ribosomal subunit with the charged $tRNA_i$ still bound scans along the mRNA strand until it reaches the start codon AUG, which indicates where translation will begin. The start codon also establishes the reading frame for the mRNA strand, which is crucial to synthesizing the correct sequence of amino acids. A shift in the reading frame results in mistranslation of the mRNA. The anticodon on the $tRNA_i$ then binds to the start codon via basepairing. The complex consisting of mRNA, charged $tRNA_i$, and the small ribosomal subunit attaches to the large ribosomal subunit, which completes ribosome assembly. These components are brought together by the help of proteins called initiation factors which bind to the small ribosomal subunit during initiation and are found in all three domains of life. In addition, the cell spends GTP energy to help form the initiation complex. Once ribosome assembly is complete, the charged $tRNA_i$ is positioned in the P site of the ribosome and the empty A site is ready for the next aminoacyl-tRNA. The polypeptide synthesis begins and always proceeds from the N-terminus to the C-terminus, called the N-to-C direction.

In eukaryotes, several eukaryotic initiation factor proteins (eIFs) assist in ribosome assembly. The eukaryotic initiation factor-2 (eIF-2) is active when it binds to guanosine triphosphate (GTP). With GTP bound to it, eIF-2 protein binds to the small 40S ribosomal subunit. Next, the initiator tRNA charged with methionine (Met-$tRNA_i$) associates with the GTP-eIF-2/40S ribosome complex, and once all these components are bound to each other, they are collectively called the 43S complex.

Eukaryotic initiation factors eIF1, eIF3, eIF4, and eIF5 help bring the 43S complex to the 5′-m^7G cap of an mRNA be translated. Once bound to the mRNA's 5′ m^7G cap, the 43S complex starts travelling down the mRNA until it reaches the initiation AUG codon at the start of the mRNA's reading frame. Sequences around the AUG may help ensure the correct AUG is used as the initiation codon in the mRNA.

Once the 43S complex is at the initiation AUG, the tRNAi-Met is positioned over the AUG. The anticodon on $tRNA_i$-Met basepairs with the AUG codon. At this point, the GTP bound to eIF2 in the 43S complexx is hydrolyzed to GDP +

hate, and energy is released. This energy is used to release the eIF2 (with GDP bound to it) from the 43S complex, g the 40S ribosomal subunit and the $tRNA_i$-Met at the translation start site of the mRNA.

Next, eIF5 with GTP bound binds to the 40S ribosomal subunit complexed to the mRNA and the $tRNA_i$-Met. The eIF5-GTP allows the 60S large ribosomal subunit to bind. Once the 60S ribosomal subunit arrives, eIF5 hydrolyzes its bound GTP to GDP + phosphate, and energy is released. This energy powers assembly of the two ribosomal subunits into the intact 80S ribosome, with tRNAi-Met in its P site while also basepaired to the initiation AUG codon on the mRNA. Translation is ready to begin.

The binding of eIF-2 to the 40S ribosomal subunit is controlled by phosphorylation. If eIF-2 is phosphorylated, it undergoes a conformational change and cannot bind to GTP. Therefore, the 43S complex cannot form properly and translation is impeded. When eIF-2 remains unphosphorylated, it binds the 40S ribosomal subunit and actively translates the protein.

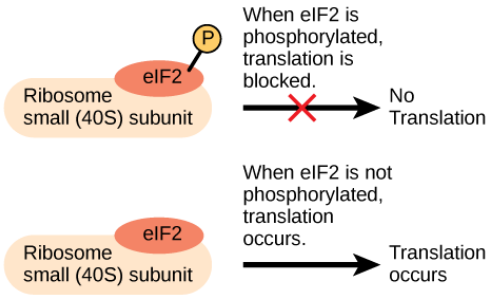

Figure 16.3E.1: **Translation Initiation Complex**: Gene expression can be controlled by factors that bind the translation initiation complex.

The ability to fully assemble the ribosome directly affects the rate at which translation occurs. But protein synthesis is regulated at various other levels as well, including mRNA synthesis, tRNA synthesis, rRNA synthesis, and eukaryotic initiation factor synthesis. Alteration in any of these components affects the rate at which translation can occur.

16.3F: Regulating Protein Activity and Longevity

A cell can rapidly change the levels of proteins in response to the environment by adding specific chemical groups to alter gene regulation.

Learning Objectives

- Explain how chemical modifications affect protein activity and longevity

Key Points

- Proteins can be chemically modified by adding methyl, phosphate, acetyl, and ubiquitin groups.
- Protein longevity can be affected by altering stages of gene regulation, including but not limited to altering: accessibility to chromosomal DNA for transcription, rate of translation, nuclear shuttling, RNA stability, and post-translational modifications.
- Ubiquitin is added to a protein to mark it for degradation by the proteasome.

Key Terms

- **ubiquitin**: a small polypeptide present in the cells of all eukaryotes; it plays a part in modifying and degrading proteins
- **proteasome**: a complex protein, found in bacterial, archaeal and eukaryotic cells, that breaks down other proteins via proteolysis

Chemical Modifications, Protein Activity, and Longevity

Proteins can be chemically modified with the addition of methyl, phosphate, acetyl, and ubiquitin groups. The addition or removal of these groups from proteins regulates their activity or the length of time they exist in the cell. Sometimes these modifications can regulate where a protein is found in the cell; for example, in the nucleus, the cytoplasm, or attached to the plasma membrane.

Chemical modifications occur in response to external stimuli such as stress, the lack of nutrients, heat, or ultraviolet light exposure. These changes can alter protein function, epigenetic accessibility, transcription, mRNA stability, or translation; all resulting in changes in expression of various genes. This is an efficient way for the cell to rapidly change the abundance levels of specific proteins in response to the environment. Because proteins are involved in every stage of gene regulation, the phosphorylation of a protein (depending on the protein that is modified) can alter accessibility to the chromosome, can alter translation (by altering transcription factor binding or function), can change nuclear shuttling (by influencing modifications to the nuclear pore complex), can alter RNA stability (by binding or not binding to the RNA to regulate its stability), can modify translation (increase or decrease), or can change post-translational modifications (add or remove phosphates or other chemical modifications). All of these protein activities are affected by the phosphorylation process. The enzymes which are responsible for phosphorylation are known as protein kinases. The addition of a phosphate group to a protein can result in either activation or deactivation; it is protein dependent.

Another example of chemical modifications affecting protein activity include the addition or removal of methyl groups. Methyl groups are added to proteins via the process of methylation; this is the most common form of post-translational modification. The addition of methyl groups to a protein can result in protein-protein interactions that allows for transcriptional regulation, response to stress, protein repair, nuclear transport, and even differentiation processes. Methylation on side chain nitrogens is considered largely irreversible while methylation of the carboxyl groups is potentially reversible. Methylation in the proteins negates the negative charge on it and increases the hydrophobicity of the protein. Methylation on carboxylate side chains covers up a negative charge and adds hydrophobicity. The addition of this chemical group changes the property of the protein and, thus, affects it activity.

The addition of an ubiquitin group to a protein marks that protein for degradation. Ubiquitin acts like a flag indicating that the protein lifespan is complete. These proteins are moved to the proteasome, an organelle that functions to remove proteins to be degraded. One way to control gene expression is to alter the longevity of the protein: ubiquitination shortens a protein's lifespan.

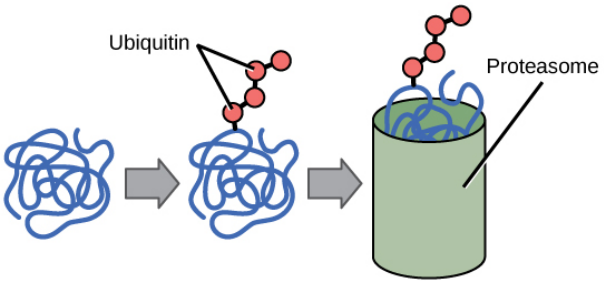

Figure 16.3F.1: **Ubiquitin Tags**: Proteins with ubiquitin tags are marked for degradation within the proteasome.

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16.4: Regulating Gene Expression in Cell Development

Topic hierarchy

- 16.4A: Gene Expression in Stem Cells
- 16.4B: Cellular Differentiation
- 16.4C: Mechanics of Cellular Differentation
- 16.4D: Establishing Body Axes during Development
- 16.4E: Gene Expression for Spatial Positioning
- 16.4F: Cell Migration in Multicellular Organisms
- 16.4G: Programmed Cell Death

16.4A: Gene Expression in Stem Cells

Symmetric division maintains stem cell lines and asymmetric division yields differentiated cells.

Learning Objectives

- Discuss the types of cell division that can occur to add cells during development

Key Points

- Symmetric cell division of stem cells ensures that a constant pool of stem cells is available by giving rise to two identical daughter cells both endowed with stem cell properties.
- Asymmetric division of stem cells results in the production of only one stem cell and a progenitor cell with limited self-renewal potential.
- Progenitor cells that are produced via asymmetric cell division will go through additional rounds of cell division until they are terminally differentiated into a mature, specialized cell.
- Asymmetric division can be controlled by both intrinsic and extrinsic factors.
- Intrinsic factors involve differing amounts of cell-fate determinants being distributed into each daughter cell, while extrinsic factors involve interactions with neighboring cells and the micro and macro environment of the precursor cell.

Key Terms

- **totipotency**: the ability of a cell to produce differentiated cells upon division
- **progenitor cell**: a biological cell that, like a stem cell, has a tendency to differentiate into a specific type of cell, but is already more specific than a stem cell and is pushed to differentiate into its “target” cell.
- **autologous**: derived from part of the same individual (i.e. from the recipient rather than the donor)
- **morula**: a spherical mass of blastomeres that forms following the splitting of a zygote; it becomes the blastula
- **pluripotent**: able to develop into more than one mature cell or tissue type, but not all

Adding cells through cellular division

Stem cells are undifferentiated biological cells found in multicellular organisms, that can differentiate into specialized cells (asymmetric division) or can divide to produce more stem cells (symmetric division). In mammals, there are two broad types of stem cells: embryonic stem cells, which are isolated from the inner cell mass of blastocysts, and adult stem cells, which are found in various tissues. In adult organisms, stem cells and progenitor cells act as a repair system for the body by replenishing adult tissues. In a developing embryo, stem cells can differentiate into all of the specialized cells (including ectoderm, endoderm and mesoderm cells) but also maintain the normal turnover of regenerative organs, such as blood, skin, or intestinal tissues. The pathway that is taken to produced specialized cells included: the embryonic cells develop from totipotent cells, to pluripotent cells which undergo differentiation and become more specialized. The key component however, in the ability to maintain tissues is the ability to maintain a key of stem cells.

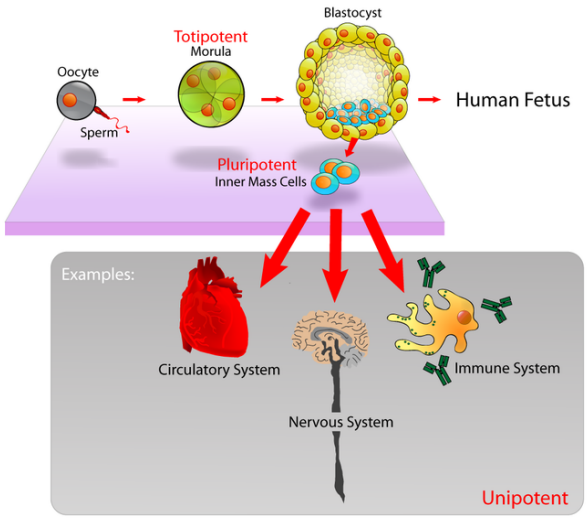


Figure 16.4*A*.1: **Stem Cells**: Pluripotent, embryonic stem cells originate as inner cell mass (ICM) cells within a blastocyst. These stem cells can become any tissue in the body, excluding a placenta. Only cells from an earlier stage of the embryo, known as the morula, are totipotent, able to become all tissues in the body and the extraembryonic placenta.

There are three accessible sources of autologous adult stem cells in humans: (1) bone marrow, which requires extraction by harvesting (i.e., drilling into bone); (2) adipose tissue (lipid cells), which requires extraction by liposuction; and (3) blood, which requires extraction through apheresis (wherein blood is drawn from the donor, passed through a machine that extracts the stem cells, and returned to the donor). Stem cells can also be taken from umbilical cord blood just after birth. Of all the stem cell types, autologous harvesting involves the least risk. By definition, autologous cells are obtained from one's own body, just as one may bank his or her own blood for elective surgical procedures. Highly plastic adult stem cells are routinely used in medical therapies, for example in bone marrow transplantation. Stem cells can now be artificially grown and differentiated into specialized cell types with characteristics consistent with muscle or nerve cells through cell culture. Embryonic cell lines and autologous embryonic stem cells generated through therapeutic cloning have also been proposed as promising candidates for future therapies.

Symmetric and asymmetric cell division

To ensure self-renewal, stem cells undergo two types of cell division: symmetric and asymmetric. Symmetric division gives rise to two identical daughter cells both endowed with stem cell properties. Asymmetric division, on the other hand, produces only one stem cell and a progenitor cell with limited self-renewal potential. Progenitors can go through several rounds of cell division themselves before terminally differentiating into a mature cell.. It is possible that the molecular distinction between symmetric and asymmetric division lies in differential segregation of cell membrane proteins between the daughter cells. An alternative theory is that stem cells remain undifferentiated due to environmental cues in their particular niche. Stem cells differentiate when they leave that niche or no longer receive those signals.

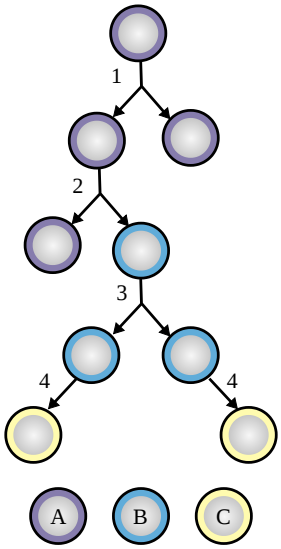


Figure 16.4A.1: **Symmetric and Asymmetric Division**: This diagram illustrates stem cell division and differentiation, through the processes of (1) symmetric stem cell division, (2) asymmetric stem cell division, (3) progenitor division, and (4) terminal differentiation. Stem cells are indicated by (A), progenitor cells by (B), and differentiated cells by (C).

An asymmetric cell division produces two daughter cells with different cellular fates. This is in contrast to normal symmetric cell divisions, which give rise to daughter cells of equivalent fates. Notably, stem cells divide asymmetrically to give rise to two distinct daughter cells: one copy of the original stem cell as well as a second daughter programmed to differentiate into a non-stem cell fate.

In principle, there are two mechanisms by which distinct properties may be conferred on the daughters of a dividing cell. In one, the daughter cells are initially equivalent but a difference is induced by signaling between the cells, from surrounding cells, or from the precursor cell. This mechanism is known as extrinsic asymmetric cell division. Extrinsic factors involve interactions with neighboring cells and the micro and macro environment of the precursor cell.

In the second mechanism, the prospective daughter cells are inherently different at the time of division of the mother cell. Because this latter mechanism does not depend on interactions of cells with each other or with their environment, it must rely on intrinsic asymmetry. The term asymmetric cell division usually refers to such intrinsic asymmetric divisions. Intrinsic factors generally involve differing amounts of cell-fate determinants being distributed into each daughter cell.

Animals are made up of a vast number of distinct cell types. During development, the zygote undergoes many cell divisions that give rise to various cell types, including embryonic stem cells. Asymmetric divisions of these embryonic cells gives rise to one cell of the same potency (self-renewal), and another that may be of the same potency or stimulated to further differentiate into specialized cell types such as neurons. Asymmetric division of stem cells plays a key role in development by allowing for the differentiation of a subset of daughter cells while maintaining stem cell pluripotency. Since it can be controlled by both intrinsic and extrinsic factors, upon delineating these particular factors it may be possible to use this knowledge in applications of tissue and whole organ generation.

16.4B: Cellular Differentiation

Cellular differentiation occurs so cells can specialize for different functions within an organism.

Learning Objectives

- Discuss how differentiated cells can serve different functions

Key Points

- The three major cell types in the mammalian body include germ cells (which develop into gametes), somatic cells (diploid cells that develop into a majority of the human body) and stem cells (cells that can divide indefinitely).
- In human development, the inner cell mass exhibits the ability to differentiate and form all tissues of the body; however, they cannot form an organism.
- The various types of stem and progenitor cells included in the body that will differentiate to develop more specialized cells includes: hematopoietic stem cells, mesenchymal stem cells, epithelial stem cells and muscle satellite cells.
- To develop a multicellular oragnisms, cells must differentiate to specialize for different functions.

Key Terms

- **blastocyst**: the mammalian blastula formed during development where the inner cell mass can be found which forms the embryo
- **inner cell mass**: a mass of cells within a primordial embryo that will eventually develop into the distinct form of a fetus in most eutherian mammals
- **proteome**: the complete set of proteins encoded by a particular genome
- **pluripotent**: able to develop into more than one mature cell or tissue type, but not all

Cellular Differentiation

To develop a multicellular organisms, cells must differentiate to specialize for different functions. Three basic categories of cells make up the mammalian body: germ cells, somatic cells, and stem cells. Each of the approximately 100 trillion cells in an adult human has its own copy or copies of the genome except certain cell types, such as red blood cells, that lack nuclei in their fully differentiated state. Most cells are diploid; they have two copies of each chromosome. The process of cellular differentiation is regulated by transcription factors and growth factors, and results in expression or inhibition of various genes between the cell types, thereby resulting in varying proteomes between cell types. The variation in proteomes between cell types is what drives differentiation and thus, specialization of cells. The ability of transcription factors to control whether a gene will be transcribed or not that contributes to specialization and growth factors to aid in the division process are key components of cell differentiation.

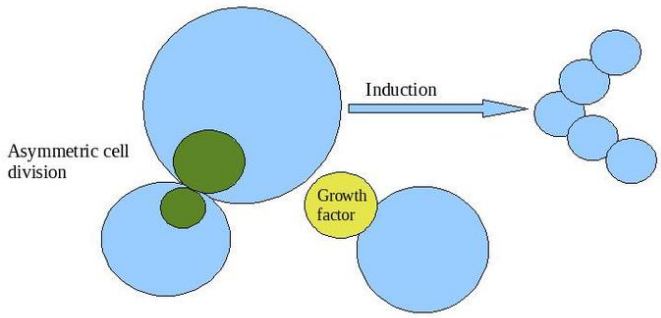


Figure 16.4B.1: **Cell Differentiation**: Mechanics of cellular differentiation can be controlled by growth factors which can induce cell division. In asymetric cell division the cell will be induced to differentiate into a specialized cell and the growth factors will work in tandem.

Somatic cells are diploid cells that make up most of the human body, such as the skin and muscle. Germ cells are any line of cells that give rise to gametes—eggs and sperm—and thus are continuous through the generations. Stem cells, on the

other hand, have the ability to divide for indefinite periods and to give rise to specialized cells. They are best described in the context of normal human development.

Embryonic Development

Development begins when a sperm fertilizes an egg and creates a single cell that has the potential to form an entire organism. In the first hours after fertilization, this cell divides into identical cells. In humans, approximately four days after fertilization and after several cycles of cell division, these cells begin to specialize, forming a hollow sphere of cells, called a blastocyst. The blastocyst has an outer layer of cells, and inside this hollow sphere, there is a cluster of cells called the inner cell mass. The cells of the inner cell mass go on to form virtually all of the tissues of the human body. Although the cells of the inner cell mass can form virtually every type of cell found in the human body, they cannot form an organism. These cells are referred to as pluripotent.

Pluripotent stem cells undergo further specialization into multipotent progenitor cells that then give rise to functional cells. Examples of stem and progenitor cells include:

1. Hematopoietic stem cells (adult stem cells) from the bone marrow that give rise to red blood cells, white blood cells, and platelets
2. Mesenchymal stem cells (adult stem cells) from the bone marrow that give rise to stromal cells, fat cells, and types of bone cells;
3. Epithelial stem cells (progenitor cells) that give rise to the various types of skin cells
4. Muscle satellite cells (progenitor cells) that contribute to differentiated muscle tissue

A pathway that is guided by the cell adhesion molecules is created as the cellular blastomere differentiates from the single-layered blastula to the three primary layers of germ cells in mammals, namely the ectoderm, mesoderm and endoderm (listed from most distal, or exterior, to the most proximal, or interior). The ectoderm ends up forming the skin and the nervous system, the mesoderm forms the bones and muscular tissue, and the endoderm forms the internal organ tissues.

16.4C: Mechanics of Cellular Differentation

Cellular differentiation, a necessary process in development and maintenance of multicellularity, is regulated by transcription factors.

Learning Objectives

- Summarize how a cell can differentiate into a specialized cell

Key Points

- Different types of stem cells exhibit varying abilities to differentiate into specialized cells (from the most unlimited stem cell to the most restricted): totipotent, pluripotent, multipotent to oligopotent.
- Totipotent cells have the potential to differentiate into any of the cells needed to enable an organism to grow and develop; pluripotent cells have the potential to differentiate into any type of human tissue but cannot support the full development of an organism.
- A multipotent stem cell has the potential to differentiate into different types of cells within a given cell lineage or small number of lineages, while an oligopotent stem cell is limited to becoming one of a few different cell types.
- The process of cellular differentiation is under strict regulation by transcription factors which can either activate or repress expression of genes that will affect the proteome of the cell and thus, provide the necessary components it needs to become a specialized cell.
- All cells contain the same complement of DNA, or genome, but once differentiation occurs, it is the changes in the proteome that will distinguish one cell type from another.

Key Terms

- **differentiate**: to produce distinct cells, organs or to achieve specific functions by a process of development
- **proteome**: the complete set of proteins encoded by a particular genome
- **transcription**: the synthesis of RNA under the direction of DNA

Cellular differentiation

How does a complex organism such as a human develop from a single cell—a fertilized egg—into the vast array of cell types such as nerve cells, muscle cells, and epithelial cells that characterize the adult? Throughout development and adulthood, the process of cellular differentiation leads cells to assume their final morphology and physiology. Differentiation is the process by which unspecialized cells become specialized to carry out distinct functions.

Stem Cells

A stem cell is an unspecialized cell that can divide without limit as needed and can, under specific conditions, differentiate into specialized cells. Stem cells are divided into several categories according to their potential to differentiate. The first embryonic cells that arise from the division of the zygote are the ultimate stem cells; these stems cells are described as totipotent because they have the potential to differentiate into any of the cells needed to enable an organism to grow and develop. The embryonic cells that develop from totipotent stem cells and are precursors to the fundamental tissue layers of the embryo are classified as pluripotent. A pluripotent stem cell is one that has the potential to differentiate into any type of human tissue but cannot support the full development of an organism. These cells then become slightly more specialized, and are referred to as multipotent cells. A multipotent stem cell has the potential to differentiate into different types of cells within a given cell lineage or small number of lineages, such as a red blood cell or white blood cell.

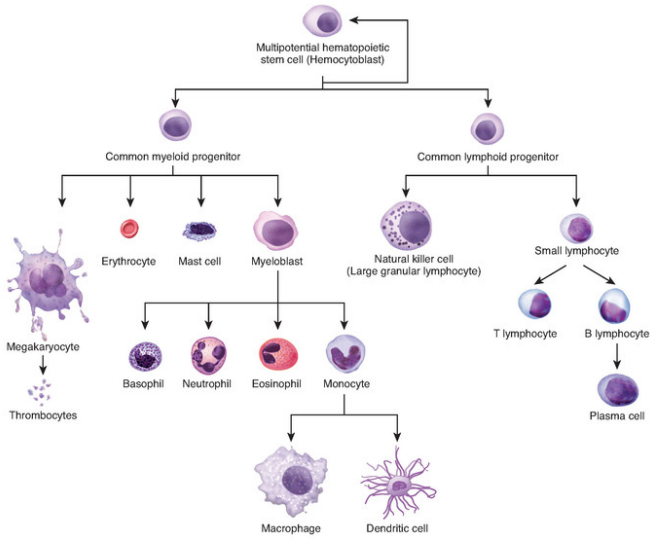


Figure 16.4C.1: **Hematopoiesis: the differentiation of multipotent cells**: The process of hematopoiesis involves the differentiation of multipotent cells into blood and immune cells. The multipotent hematopoietic stem cells give rise to many different cell types, including the cells of the immune system and red blood cells.

Finally, multipotent cells can become further specialized oligopotent cells. An oligopotent stem cell is limited to becoming one of a few different cell types. In contrast, a unipotent cell is fully specialized and can only reproduce to generate more of its own specific cell type. Stem cells are unique in that they can also continually divide and regenerate new stem cells instead of further specializing.

There are different stem cells present at different stages of a human's life, including the embryonic stem cells of the embryo, fetal stem cells of the fetus, and adult stem cells in the adult. One type of adult stem cell is the epithelial stem cell, which gives rise to the keratinocytes (cells that produce keratin, the primary protein in nails and hair) in the multiple layers of epithelial cells in the epidermis of skin. Adult bone marrow has three distinct types of stem cells: hematopoietic stem cells, which give rise to red blood cells, white blood cells, and platelets; endothelial stem cells, which give rise to the endothelial cell types that line blood and lymph vessels; and mesenchymal stem cells, which give rise to the different types of muscle cells.

Differentiation

When a cell differentiates (i.e., becomes more specialized), it may undertake major changes in its size, shape, metabolic activity, and overall function. Because all cells in the body, beginning with the fertilized egg, contain the same DNA, how do the different cell types come to be so different? The answer is analogous to a movie script. Different actors in a movie all read from the same script, but each one only reads their own part of the script. Similarly, all cells contain the same full complement of DNA, but each type of cell only "reads" the portions of DNA that are relevant to its own functioning. In other terms, each cell has the genome but will only express specific genes, thereby having unique proteomes. In biology, this is referred to as the unique genetic expression of each cell. In order for a cell to differentiate into its specialized form and function, it need only manipulate those genes (and thus those proteins) that will be expressed, and not those that will remain silent.

Mechanism

The primary mechanism by which genes are turned "on" or "off" is through transcription factors. A transcription factor is one of a class of proteins that bind to specific genes on the DNA molecule and either promote or inhibit their transcription. The primary mechanism that determines which genes will be expressed and which ones will not is through the use of different transcription factor proteins, which bind to DNA and promote or hinder the transcription of different genes. Through the action of these transcription factors, cells specialize into one of hundreds of different cell types in the human body.

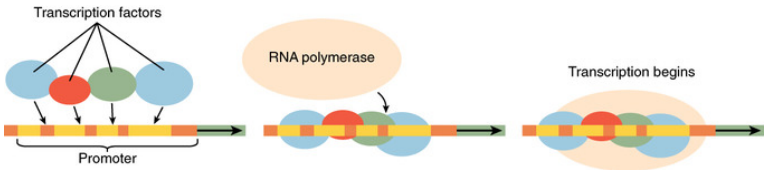


Figure 16.4C.1: **Transcription Factors Regulate Gene Expression**: While each body cell contains the organism's entire genome, different cells regulate gene expression with the use of various transcription factors. Transcription factors are proteins that affect the binding of RNA polymerase to a particular gene on the DNA molecule.

16.4D: Establishing Body Axes during Development

Animal bodies have three axes for symmetry (lateral-medial, dorsal-ventral and anterior-posterior) which are established in development.

Learning Objectives

- Summarize the mechanisms and cell types that establish the body axes

Key Points

- The three axes of the animal body are established in development via the expression of specific sets of genes that regulate which cells will develop into specific structures.
- During development, the dorsal cells are genetically programmed to develop into the notochord and define the axis.
- The neural tube can develop in two ways: primary or secondary neurulation, which are used by organisms in varying degrees to establish the neural tube that will develop into the central nervous system (brain and spinal cord).
- Specific patterns along the neural tube that are established via secretion and production of specific signaling molecules (such as Wnt, Shh, BMP and retinoic acid) play a key role in patterning the dorsal and ventral axes.

Key Terms

- **neural tube**: hollow longitudinal dorsal tube formed in the folding and subsequent fusion of the opposite ectodermal folds in the embryo that gives rise to the brain and spinal cord
- **neurulation**: the process by which the beginnings of the vertebrate nervous system is formed in embryos
- **anencephaly**: a lethal birth defect in which most of the brain and parts of the skull are missing; absence of the encephalon
- **notochord**: a flexible rodlike structure that forms the main support of the body in the lowest chordates; a primitive spine

Vertebrate axis formation

During development, as the germ layers form, the ball of cells still retains its spherical shape. However, animal bodies have lateral-medial (left-right), dorsal-ventral (back-belly), and anterior-posterior (head-feet) axes. How are these established? In one of the most seminal experiments ever to be carried out in developmental biology, Spemann and Mangold took dorsal cells from one embryo and transplanted them into the belly region of another embryo. They found that the transplanted embryo now had two notochords: one at the dorsal site from the original cells and another at the transplanted site. This suggested that the dorsal cells were genetically programmed to form the notochord and define the axis. Since then, researchers have identified many genes that are responsible for axis formation. Mutations in these genes leads to the loss of symmetry required for organism development. Animal bodies have externally visible symmetry. However, the internal organs are not symmetric. For example, the heart is on the left side and the liver on the right. The formation of the central left-right axis is an important process during development. This internal asymmetry is established very early during development and involves many genes. Research is still ongoing to fully understand the developmental implications of these genes.

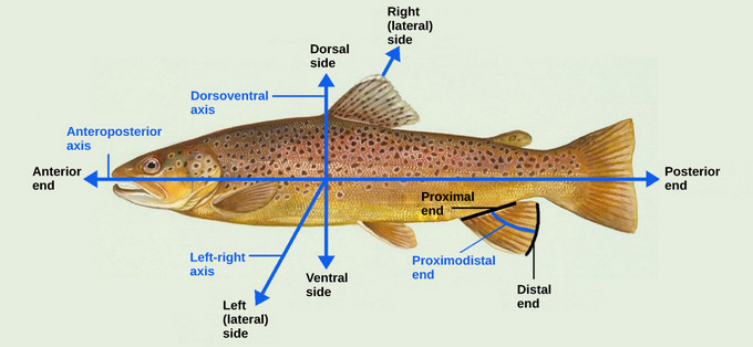


Figure 16.4D.1: **Vertebrate Axis Formation**: Animal bodies have three axes for symmetry:lateral-medial (left-right), dorsal-ventral (back-belly), and anterior-posterior (head-feet).

Neural tube

In the developing chordate (including vertebrates), the neural tube is the embryo's precursor to the central nervous system, which comprises the brain and spinal cord. The neural groove gradually deepens as the neural folds become elevated, and ultimately the folds meet and coalesce in the middle line and convert the groove into a closed tube, the neural tube or neural canal, the ectodermal wall of which forms the rudiment of the nervous system.

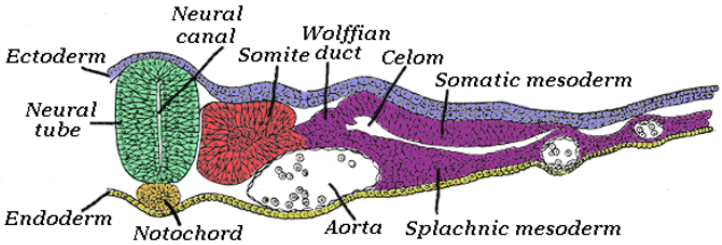


Figure 16.4D.1: **Neural Tube**: Transverse section of half of a chick embryo of forty-five hours' incubation. The dorsal (back) surface of the embryo is toward the top of this page, while the ventral (front) surface is toward the bottom. (Neural tube is in green.)

Primary and secondary neurulation

The neural tube develops in two ways: primary neurulation and secondary neurulation. Primary neurulation divides the ectoderm into three cell types: the internally located neural tube, the externally located epidermis, and the neural crest cells, which develop in the region between the neural tube and epidermis but then migrate to new locations. Primary neurulation begins after the neural plate forms. The edges of the neural plate start to thicken and lift upward, forming the neural folds. The center of the neural plate remains grounded, allowing a U-shaped neural groove to form. This neural groove sets the boundary between the right and left sides of the embryo. The neural folds pinch in towards the midline of the embryo and fuse together to form the neural tube. In secondary neurulation, the cells of the neural plate form a cord-like structure that migrates inside the embryo and hollows to form the tube.

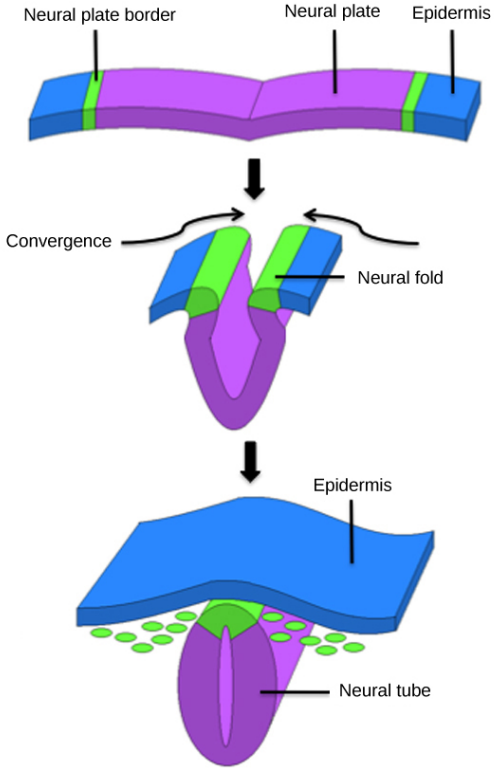


Figure 16.4D.1: **Neural Tube Formation**: The central region of the ectoderm forms the neural tube, which gives rise to the brain and the spinal cord.

Each organism uses primary and secondary neurulation to varying degrees. Neurulation in fish proceeds only via the secondary form. In avian species the posterior regions of the tube develop using secondary neurulation and the anterior regions develop by primary neurulation. In mammals, secondary neurulation begins around the 35th somite. Mammalian neural tubes close in the head in the opposite order that they close in the trunk. In the head, neural crest cells migrate, the neural tube closes, and the overlying ectoderm closes. In the trunk, overlying ectoderm closes, the neural tube closes and neural crest cells migrate.

Neural tube subdivisions

Four neural tube subdivisions eventually develop into distinct regions of the central nervous system by the division of neuroepithelial cells: the prosencephalon, the mesencephalon, the rhombencephalon and the spinal cord. The prosencephalon further goes on to develop into the telencephalon (the forebrain or cerebrum) and the diencephalon (the optic vesicles and hypothalamus). The mesencephalon develops into the midbrain. The rhombencephalon develops into the metencephalon (the pons and cerebellum) and the myelencephalon (the medulla oblongata).

For a short time, the neural tube is open both cranially and caudally. These openings, called neuropores, close during the fourth week in the human. Improper closure of the neuropores can result in neural tube defects such as anencephaly or spina bifida. The dorsal part of the neural tube contains the alar plate, which is primarily associated with sensation. The ventral part of the neural tube contains the basal plate, which is primarily associated with motor (i.e., muscle) control.

Signaling molecules and other factors

The neural tube patterns along the dorsal-ventral axis establish defined compartments of neural progenitor cells that lead to distinct classes of neurons. This patterning occurs early in development and results from the activity of several secreted signaling molecules. Sonic hedgehog (Shh) is a key player in patterning the ventral axis, while Bone morphogenic proteins (Bmp) and Wnt family members play an important role in patterning the dorsal axis. Other factors shown to provide positional information to the neural progenitor cells include Fibroblast growth factors (FGF) and Retinoic Acid. Retinoic acid is required ventrally along with Shh to induce Pax6 and Olig2 during differentiation of motor neurons. Three main ventral cell types are established during early neural tube development: the floor plate cells, which form at the ventral midline during the neural fold stage; as well as the more dorsally located motor neurons and interneurons. These cell types are specified by the secretion of Shh from the notochord (located ventrally to the neural tube), and later from the floor plate cells. Shh acts as a morphogen, meaning that it acts in a concentration-dependent manner to specify cell types as it moves further from its source. The different combinations of expression of transcription factors along the dorsal-ventral axis of the neural tube are responsible for creating the identity of the neuronal progenitor cells.

16.4E: Gene Expression for Spatial Positioning

During development it is critical that specific gene expression patterns are established to signal and differentiate the cells appropriately.

Learning Objectives

- Describe the role genes play in development and ensuring proper spatial positioning

Key Points

- Organogenesis results in the formation of the various organs in the body; however it will only occur if specific sets of genes are expressed to determine ultimate cell type.
- The ability of specific cells to migrate to the the edge of the ectoderm is highly regulated by specific gene expression and allows for differentiation into epidermal cells; in contrast, the cells which remain in the center will develop into the neural plate.
- The expression of specific sets of genes will also regulate the reorganization of the mesoderm into distinct groups of cells, called somites, which develop into the ribs, lungs, spine muscle and notochord.

Key Terms

- **gastrulation**: the stage of embryo development at which a gastrula is formed from the blastula by the inward migration of cells
- **organogenesis**: the formation and development of the organs of an organism from embryonic cells
- **somite**: one of the paired masses of mesoderm distributed along the sides of the neural tube that will eventually become dermis, skeletal muscle, or vertebrae

Genes provide positional information

Gastrulation leads to the formation of the three germ layers that give rise, during further development, to the different organs in the animal body. This process, known as organogenesis, is characterized by rapid and precise movements of the cells within the embryo.

Organogenesis

Organs form from the germ layers through the process of differentiation. During differentiation, the embryonic stem cells express specific sets of genes which will determine their ultimate cell type. For example, some cells in the ectoderm (the outer tissue layer of the embryo) will express the genes specific to skin cells. As a result, these cells will differentiate into epidermal cells. The process of differentiation is regulated by cellular signaling cascades. Scientists study organogenesis extensively in the lab in fruit flies (Drosophila) and the nematode Caenorhabditis elegans. Drosophila have segments along their bodies, and the patterning associated with the segment formation has allowed scientists to study which genes play important roles in organogenesis along the length of the embryo at different time points. The nematode C.elegans has roughly 1000 somatic cells and scientists have studied the fate of each of these cells during their development in the nematode life cycle. There is little variation in patterns of cell lineage between individuals, unlike in mammals where cell development from the embryo is dependent on cellular cues.

In vertebrates, one of the primary steps during organogenesis is the formation of the neural system. The ectoderm forms epithelial cells and tissues, as well as neuronal tissues. During the formation of the neural system, special signaling molecules called growth factors signal some cells at the edge of the ectoderm to become epidermis cells. The remaining cells in the center form the neural plate. If the signaling by growth factors were disrupted, then the entire ectoderm would differentiate into neural tissue. The neural plate undergoes a series of cell movements where it rolls up and forms a tube called the neural tube. In further development, the neural tube will give rise to the brain and the spinal cord. The mesoderm that lies on either side of the vertebrate neural tube will develop into the various connective tissues of the animal body. A spatial pattern of gene expression reorganizes the mesoderm into groups of cells called somites with spaces between them. The somites will further develop into the ribs, lungs, and segmental (spine) muscle. The mesoderm also forms a structure called the notochord, which is rod-shaped and forms the central axis of the animal body.

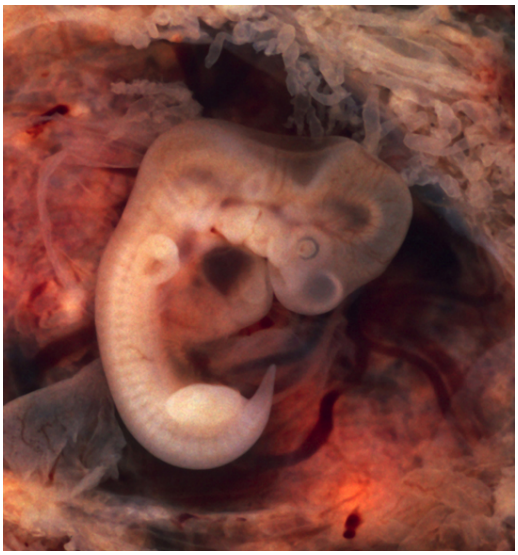


Figure 16.4E. 1: **Somites**: In this five-week old human embryo, somites are segments along the length of the body.

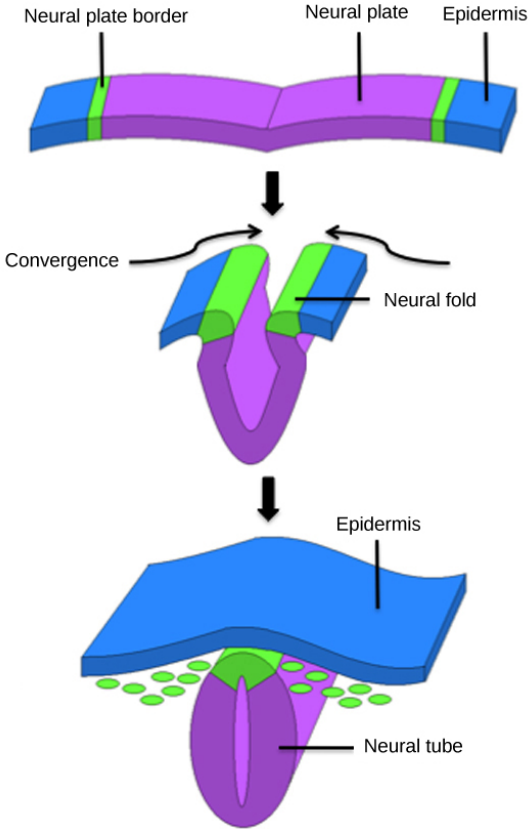


Figure 16.4E. 1: **Neural Tube Formation**: The central region of the ectoderm forms the neural tube, which gives rise to the brain and the spinal cord.

16.4F: Cell Migration in Multicellular Organisms

Cell migration is necessary for development and maintenance of multicellularity, and occurs through varying mechanisms.

Learning Objectives

- Describe how cells can migrate within an organism

Key Points

- The disruption or dysfunction of cell migration processes can lead to formation of various diseases such as metastasis, tumor formation and vascular disease.
- In prokaryotic organisms, and some eukaryotic cells such as sperm cells, cell migration occurs via the use of a cilia or flagella to propel forward.
- In eukaryotic organisms, cell migration is a much more complex process and can include, but is not excluded to, changes in the cytoskeleton, motor proteins, blebbing, and cytoplasmic displacement; it involves both external and internal signals that mediate these processes.

Key Terms

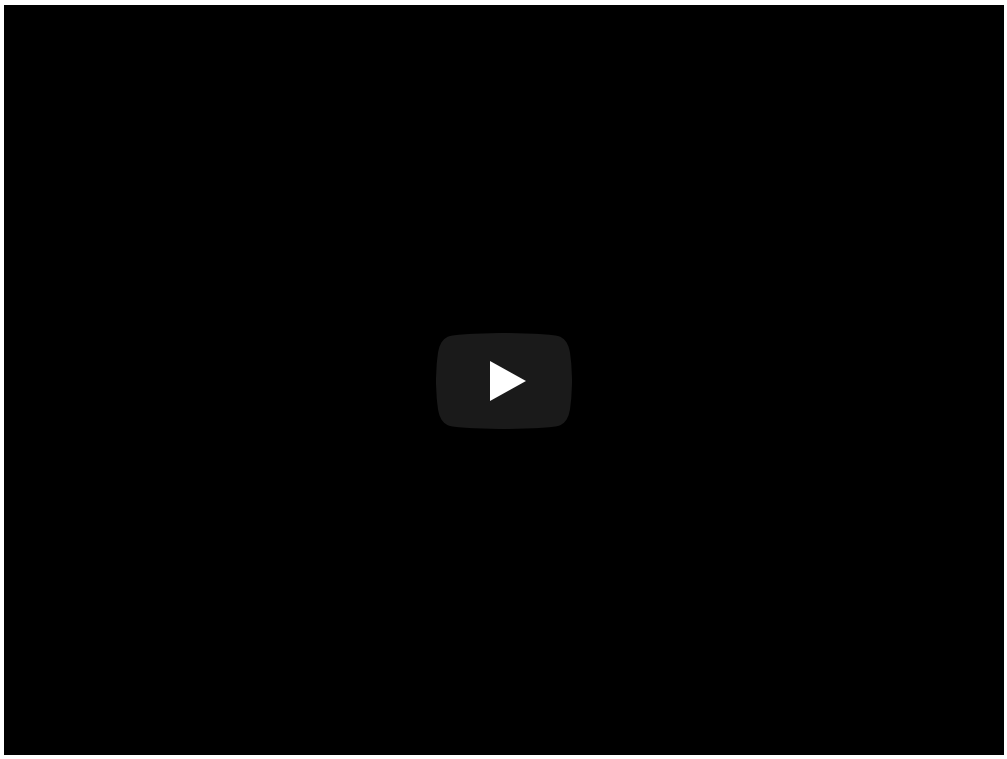
- **bleb**: an irregular bulge in the plasma membrane of a cell
- **chemotaxis**: the movement of a cell or an organism in response to a chemical stimulant
- **laminar**: of fluid motion, smooth and regular, flowing as though in different layers
- **metastasis**: the transference of a bodily function or disease to another part of the body; specifically the development of a secondary area of disease remote from the original site, as with some cancers

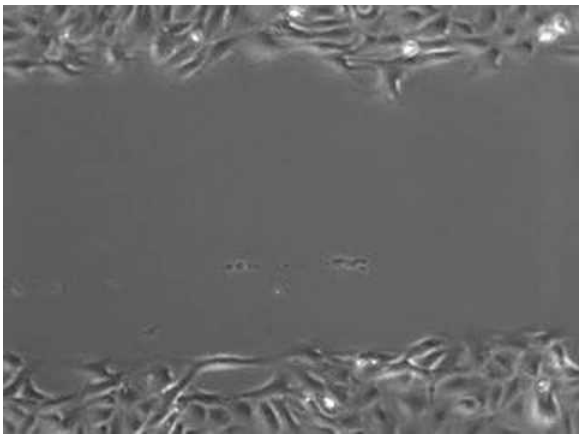
Cell migration

Cell migration is a central process in the development and maintenance of multicellular organisms. Processes such as tissue formation during embryonic development, wound healing, and immune responses, all require the orchestrated movement of cells in particular directions to specific locations. Errors during this process have serious consequences, including intellectual disability, vascular disease, tumor formation and metastasis. An understanding of the mechanism by which cells migrate may lead to the development of novel therapeutic strategies for controlling, for example, invasive tumor cells.

Cells often migrate in response to specific external signals, including chemical signals and mechanical signals. Due to a highly viscous environment, cells need to permanently produce forces in order to move. Cells achieve active movement by very different mechanisms. Many less complex prokaryotic organisms (and sperm cells) use flagella or cilia to propel themselves. Eukaryotic cell migration typically is far more complex and can consist of combinations of different migration mechanisms. It generally involves drastic changes in cell shape which are driven by the cytoskeleton, for instance a series of contractions and expansions due to cytoplasmic displacement. Two very distinct migration scenarios are crawling motion (most commonly studied) and blebbing motility.

The migration of cultured cells attached to a surface is commonly studied using microscopy. As cell movement is very slow (only a few µm/minute), time-lapse microscopy videos are recorded of the migrating cells to speed up the movement. Such videos reveal that the leading cell front is very active with a characteristic behavior of successive contractions and expansions. It is generally accepted that the leading front is the main motor that pulls the cell forward.





Cell Migration: Phase images of BSC 1 cells migrating in a scratch assay in the absence of serum over a period of 15 hours.

Common features of cell migration

The processes underlying mammalian cell migration are believed to be consistent with those of (non-spermatozoic) locomotion. Observations in common include cytoplasmic displacement at the leading front and laminar removal of dorsally-accumulated debris toward trailing end. The latter feature is most easily observed when aggregates of a surface molecule are cross-linked with a fluorescent antibody or when small beads become artificially bound to the front of the cell. Other eukaryotic cells are observed to migrate similarly. The amoeba *Dictyostelium discoideum* is useful to researchers because they consistently exhibit chemotaxis in response to cyclic AMP; they move more quickly than cultured mammalian cells; and they have a haploid genome that simplifies the process of connecting a particular gene product with its effect on cellular behavior.

16.4G: Programmed Cell Death

Programmed cell death describes the death of a cell through a highly regulated process, and serves many functions in an organism.

Learning Objectives

- Explain the importance of programmed cell death, including apoptosis and autophagy

Key Points

- Programmed cell death can provide an advantage to an organism during development, for instance by maintaining homeostasis and protection against potentially disruptive issues which may arise during the life of a cell.
- Apoptosis is a process of programmed cell death that is regulated by numerous biochemical events and appears to be genetically mediated.
- Autophagy is a process of programmed cell death that is characterized as a catabolic process via formation of an autophagolysosome which degrades damaged cellular contents.
- Necrosis occurs when cellular death is caused by external factors and is characterized as an alternate form of programmed cell death, called necroptosis.

Key Terms

- **apoptosis**: a process of programmed cell death
- **extracellular matrix**: All the connective tissues and fibres that are not part of a cell, but rather provide support.
- **autophagy**: a type of programmed cell death accomplished through self-digestion

Programmed cell death

Programmed cell-death (or PCD) is death of a cell in any form, mediated by an intracellular program. PCD is carried out in a regulated process, which usually confers advantage during an organism's life-cycle. For example, the differentiation of fingers and toes in a developing human embryo occurs because cells between the fingers apoptose, or die; the result is that the digits are separate. PCD serves fundamental functions during both plant and metazoa (multicellular animal) tissue development. Apoptosis and autophagy are both forms of PCD, but necrosis is a non-physiological process that occurs as a result of infection or injury.

Apoptosis

Apoptosis is the process of PCD that may occur in multicellular organisms. Biochemical events lead to characteristic cell changes (morphology) and death. These changes include blebbing, cell shrinkage, nuclear fragmentation, chromatin condensation, and chromosomal DNA fragmentation. There appears to be some variation in the morphology and indeed the biochemistry of these "suicide" pathways. Some tread the path of apoptosis, while others follow a more generalized pathway to deletion; however both are usually genetically and synthetically motivated. There is some evidence that certain symptoms of apoptosis, such as endonuclease activation, can be spuriously induced without engaging a genetic cascade; however, it is presumed that true apoptosis and PCD must be genetically mediated. It is also becoming clear that mitosis (the division of the cell nucleus) and apoptosis are linked in some way, and that the balance achieved depends on signals received from appropriate growth or survival factors.

When a cell is damaged, superfluous, or potentially dangerous to an organism, a cell can initiate a mechanism to trigger apoptosis. Apoptosis allows a cell to die in a controlled manner that prevents the release of potentially damaging molecules from inside the cell. There are many internal checkpoints that monitor a cell's health; if abnormalities are observed, a cell can spontaneously initiate the process of apoptosis. However, in some cases, such as a viral infection or uncontrolled cell division due to cancer, the cell's normal checks and balances fail.

Apoptosis can also be initiated via external signaling. For example, most normal animal cells have receptors that interact with the extracellular matrix, a network of glycoproteins that provides structural support for cells in an organism. The binding of cellular receptors to the extracellular matrix initiates a signaling cascade within the cell. However, if the cell

moves away from the extracellular matrix, the signaling ceases, and the cell undergoes apoptosis. This system keeps cells from traveling through the body and proliferating out of control, as happens with tumor cells that metastasize.

Figure 16.4G.1: **Programmed Cell Death**: This histological section of a foot of a 15-day-old mouse embryo, visualized using light microscopy, reveals areas of tissue between the toes, which apoptosis will eliminate before the mouse reaches its full gestational age at 27 days.

Another example of external signaling that leads to apoptosis occurs in T-cell development. T-cells are immune cells that bind to foreign macromolecules and particles, and target them for destruction by the immune system. Normally, T-cells do not target "self" proteins (those of their own organism), a process that can lead to autoimmune diseases. In order to develop the ability to discriminate between self and non-self, immature T-cells undergo screening to determine whether they bind to so-called self proteins. If the T-cell receptor binds to self proteins, the cell initiates apoptosis to remove the potentially dangerous cell.

Apoptosis is also essential for normal embryological development. In vertebrates, for example, early stages of development include the formation of web-like tissue between individual fingers and toes. During the course of normal development, these unneeded cells must be eliminated, enabling fully separated fingers and toes to form. A cell signaling mechanism triggers apoptosis, which destroys the cells between the developing digits.

Autophagy

Macroautophagy, often referred to as autophagy, is a type of programmed cell death accomplished through self-digestion. It is a catabolic process that results in the autophagosomic-lysosomal degradation of bulk cytoplasmic contents, abnormal protein aggregates, and excess or damaged organelles. Autophagy is generally activated by conditions of nutrient deprivation but has also been associated with physiological as well as pathological processes such as development, differentiation, neurodegenerative diseases, stress (physiology), infection and cancer.

Necrosis

Necrosis is the death of a cell caused by external factors such as trauma or infection and occurs in several different forms. Recently a form of programmed necrosis, called necroptosis, has been recognized as an alternate form of PCD. It is hypothesized that necroptosis can serve as a cell-death backup to apoptosis when the apoptosis signaling is blocked by endogenous or exogenous factors, such as viruses or mutations.

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16.5: Cancer and Gene Regulation

Topic hierarchy

- 16.5A: Altered Gene Expression in Cancer
- 16.5B: Epigenetic Alterations in Cancer
- 16.5C: Cancer and Transcriptional Control
- 16.5D: Cancer and Post-Transcriptional Control
- 16.5E: Cancer and Translational Control

16.5A: Altered Gene Expression in Cancer

Cancer, a disease of altered gene expression, is the result of gene mutations or dramatic changes in gene regulation.

Learning Objectives

- Describe how cancer is caused by uncontrolled cell growth

Key Points

- Cancer results from a gene that is not normally expressed in a cell, but is switched on and expressed at high levels due to mutations or alterations in gene regulation.
- Alterations in histone acetylation, activation of transcription factors, increased RNA stability, increased translational control, and protein modification are all observed in cancer cells.
- Tumor suppressor genes, active in normal cells, work to prevent uncontrolled cell growth.
- Proto- oncogenes, which are positive cell-cycle regulators, can become oncogenes and cause cancer when mutated.

Key Terms

- **oncogene**: any gene that contributes to the conversion of a normal cell into a cancerous cell when mutated or expressed at high levels
- **proto-oncogene**: a gene that promotes the specialization and division of normal cells that becomes an oncogene following mutation
- **cancer**: a disease in which the cells of a tissue undergo uncontrolled (and often rapid) proliferation

Cancer: Disease of Altered Gene Expression

Cancer can be described as a disease of altered gene expression. There are many proteins that are turned on or off (gene activation or gene silencing) that dramatically alter the overall activity of the cell. A gene that is not normally expressed in that cell can be switched on and expressed at high levels. This can be the result of gene mutation or changes in gene regulation (epigenetic, transcription, post-transcription, translation, or post-translation).

Changes in epigenetic regulation, transcription, RNA stability, protein translation, and post-translational control can be detected in cancer. While these changes do not occur simultaneously in one cancer, changes at each of these levels can be detected when observing cancer at different sites in different individuals. Therefore, changes in histone acetylation (epigenetic modification that leads to gene silencing), activation of transcription factors by phosphorylation, increased RNA stability, increased translational control, and protein modification can all be detected at some point in various cancer cells. Scientists are working to understand the common changes that give rise to certain types of cancer or how a modification might be exploited to destroy a tumor cell.

Tumor Suppressor Genes, Oncogenes, and Cancer

In normal cells, some genes function to prevent excess, inappropriate cell growth. These are tumor suppressor genes, which are active in normal cells to prevent uncontrolled cell growth. There are many tumor suppressor genes in cells, but the one most studied is p53, which is mutated in over 50 percent of all cancer types. The p53 protein itself functions as a transcription factor. It can bind to sites in the promoters of genes to initiate transcription. Therefore, the mutation of p53 in cancer will dramatically alter the transcriptional activity of its target genes.

Another type of gene often deregulated in cancers are proto-oncogenes which are positive cell-cycle regulators. When mutated, proto-oncogenes can become oncogenes and cause cancer. Overexpression of the oncogene can lead to uncontrolled cell growth because oncogenes can alter transcriptional activity, stability, or protein translation of another gene that directly or indirectly controls cell growth. An example of an oncogene involved in cancer is a protein called myc. Myc is a transcription factor that is aberrantly activated in Burkett's Lymphoma, a cancer of the lymph system. Overexpression of myc transforms normal B cells into cancerous cells that continue to grow uncontrollably. High B-cell numbers can result in tumors that can interfere with normal bodily function. Patients with Burkett's lymphoma can develop tumors on their jaw or in their mouth that interfere with the ability to eat.

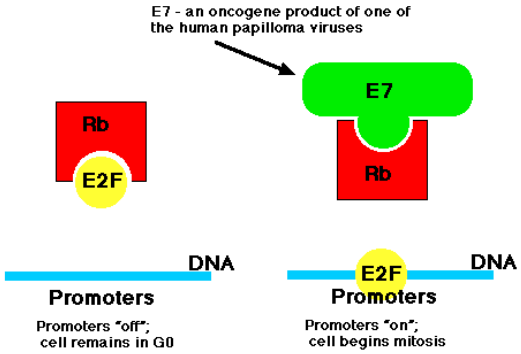

Figure 16.5A.1: **Proto-oncogenes Can Become Oncogenes**: When mutated, proto-oncogenes can become oncogenes and cause cancer due to uncontrolled cell growth.

16.5B: Epigenetic Alterations in Cancer

Common in cancer cells, silencing genes, which occur through epigenetic mechanisms, include modifications to histone proteins and DNA.

Learning Objectives

- Describe the role played by epigenetic alterations to gene expression in the development of cancer

Key Points

- The DNA in the promoter region of silenced genes in cancer cells is methylated on cytosine DNA residues in CpG islands.
- Histone proteins that surround the promoter region of silenced genes lack the acetylation modification that is present when the genes are expressed in normal cells.
- When the combination of DNA methylation and histone deacetylation occur within cancer cells, the gene present in that chromosomal region is silenced.
- Epigenetic changes that are altered in cancer can be reversed and may, therefore, be helpful in new drug and therapy design.

Key Terms

- **epigenetic**: the study of heritable changes in gene expression or cellular phenotype caused by mechanisms other than changes in the underlying DNA sequence
- **methylation**: the addition of a methyl group to cytosine and adenine residues in DNA that leads to the epigenetic modification of DNA and the reduction of gene expression and protein production
- **acetylation**: the reaction of a substance with acetic acid or one of its derivatives; the introduction of one or more acetyl groups into a substance

Cancer and Epigenetic Alterations

Cancer epigenetics is the study of epigenetic modifications to the genome of cancer cells that do not involve a change in the nucleotide sequence. Epigenetic alterations are as important as genetic mutations in a cell's transformation to cancer. Mechanisms of epigenetic silencing of tumor suppressor genes and activation of oncogenes include: alteration in CpG island methylation patterns, histone modifications, and dysregulation of DNA binding proteins.

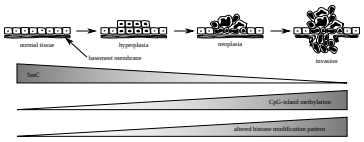

Figure 16.5B.1: **Epigenetic Alterations in Cancer Cells**: In cancer cells, silencing genes through epigenetic mechanisms is a common occurrence. Mechanisms can include modifications to histone proteins and DNA associated with these silencing genes.

Silencing genes through epigenetic mechanisms is very common in cancer cells and include modifications to histone proteins and DNA that are associated with silenced genes. In cancer cells, the DNA in the promoter region of silenced genes is methylated on cytosine DNA residues in CpG islands, genomic regions that contain a high frequency of CpG sites, where a cytosine nucleotide occurs next to a guanine nucleotide. Histone proteins that surround that region lack the acetylation modification (the addition of an acetyl group) that is present when the genes are expressed in normal cells. This combination of DNA methylation and histone deacetylation (epigenetic modifications that lead to gene silencing) is commonly found in cancer. When these modifications occur, the gene present in that chromosomal region is silenced. Increasingly, scientists are understanding how these epigenetic changes are altered in cancer. Because these changes are temporary and can be reversed (for example, by preventing the action of the histone deacetylase protein that removes acetyl groups, or by DNA methyl transferase enzymes that add methyl groups to cytosines in DNA) it is possible to design new drugs and new therapies to take advantage of the reversible nature of these processes. Indeed, many researchers are testing how a silenced gene can be switched back on in a cancer cell to help re-establish normal growth patterns.

Genes involved in the development of many other illnesses, ranging from allergies to inflammation to autism, are also thought to be regulated by epigenetic mechanisms. As our knowledge deepens of how genes are controlled, new ways to treat these diseases and cancer will emerge.

16.5C: Cancer and Transcriptional Control

Increased transcriptional activation of genes result in alterations of cell growth leading to abnormal gene expression, as seen in cancer.

Learning Objectives

- Explain the role of transcription factors in cancer

Key Points

- The mutations that activate transcription factors can increase the binding of a transcription factor to its binding site in a promoter leading to increased transcriptional activation of that gene and resulting in altered cell growth.
- A mutation in the DNA of a promoter or enhancer region may increase the binding ability of a transcription factor, which may then lead to the increased transcription and anomalous gene expression that is seen in cancer cells.
- Studying how to control the transcriptional activation of gene expression in cancer cells along with identifying how a transcription factor binds or a pathway activates where a gene can be turned off has led researchers to new drugs and novel ways of treating cancer.

Key Terms

- **transcription factor**: a protein that binds to specific DNA sequences, thereby controlling the flow (or transcription) of genetic information from DNA to mRNA

Cancer and Transcriptional Control

Many transcription factors, especially some that are proto-oncogenes or tumor suppressors, help regulate the cell cycle and, as such, determine how large a cell will get and when it can divide into two daughter cells. Alterations in cells that give rise to cancer can affect the transcriptional control of gene expression. Mutations that activate transcription factors, such as increased phosphorylation, can increase the binding of a transcription factor to its binding site in a promoter. This could lead to increased transcriptional activation of that gene that results in modified cell growth. Alternatively, a mutation in the DNA of a promoter or enhancer region can increase the binding ability of a transcription factor. This could also lead to the increased transcription and aberrant gene expression that is seen in cancer cells.

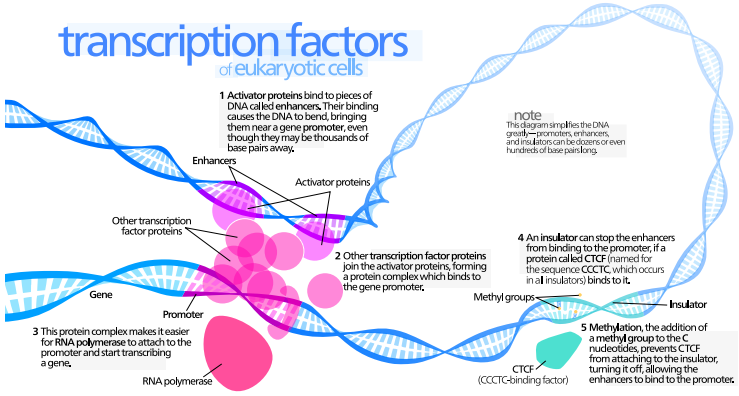

Figure 16.5C.1: **Transcription Factors**: Transcription factors, especially some that are proto-oncogenes or tumor suppressors, help regulate the cell cycle; however, when regulation gives rise to cancer cells, then transcriptional control of gene expression is affected.

Researchers have been investigating how to control the transcriptional activation of gene expression in cancer. Identifying how a transcription factor binds, or a pathway that activates where a gene can be turned off, has led to new drugs and new ways to treat cancer. In breast cancer, for example, many proteins are overexpressed. This can lead to increased phosphorylation of key transcription factors that increase transcription. One such example is the overexpression of the epidermal growth factor receptor (EGFR) in a subset of breast cancers. The EGFR pathway activates many protein kinases that, in turn, activate many transcription factors that control genes involved in cell growth. New drugs that prevent the activation of EGFR have been developed and are used to treat these cancers.

16.5D: Cancer and Post-Transcriptional Control

Modifications, such as the overexpression of miRNAs, in the post-transcriptional control of a gene can result in cancer.

Learning Objectives

- Explain how post-transcriptional control can result in cancer

Key Points

- Specific cancers have altered expression of miRNAs; changes in the miRNA population of particular cancers varies depending on the type of cancer.
- Having too many miRNAs can dramatically decrease the RNA population leading to a decrease in protein expression.
- Studies have found that some miRNAs are specifically expressed only in cancer cells.

Key Terms

- **microRNA**: a single-stranded, non-coding form of RNA, having only about 20-30 nucleotides, that has a number of functions including the regulation of gene expression
- **exosome**: a vesicle responsible for the selective removal of plasma membrane proteins

Cancer and Post-transcriptional Control

Post-transcriptional regulation is the control of gene expression at the RNA level; therefore, between the transcription and the translation of the gene. After being produced, the stability and distribution of the different transcripts is regulated (post-transcriptional regulation) by means of RNA-binding proteins (RBP) that control the various steps and rates of the transcripts: events such as alternative splicing, nuclear degradation (exosome), processing, nuclear export (three alternative pathways), sequestration in DCP2-bodies for storage or degradation, and, ultimately, translation.

Changes in the post-transcriptional control of a gene can result in cancer. Recently, several groups of researchers have shown that specific cancers have altered expression of microRNAs (miRNAs). miRNAs bind to the 3′ UTR or 5′ UTR of RNA molecules to degrade them. Overexpression of these miRNAs could be detrimental to normal cellular activity. An increase in many miRNAs could dramatically decrease the RNA population leading to a decrease in protein expression. Several studies have demonstrated a change in the miRNA population in specific cancer types. It appears that the subset of miRNAs expressed in breast cancer cells is quite different from the subset expressed in lung cancer cells or even from normal breast cells. This suggests that alterations in miRNA activity can contribute to the growth of breast cancer cells. These types of studies also suggest that if some miRNAs are specifically expressed only in cancer cells, they could be potential drug targets. It would, therefore, be conceivable that new drugs that turn off miRNA expression in cancer could be an effective method to treat cancer.

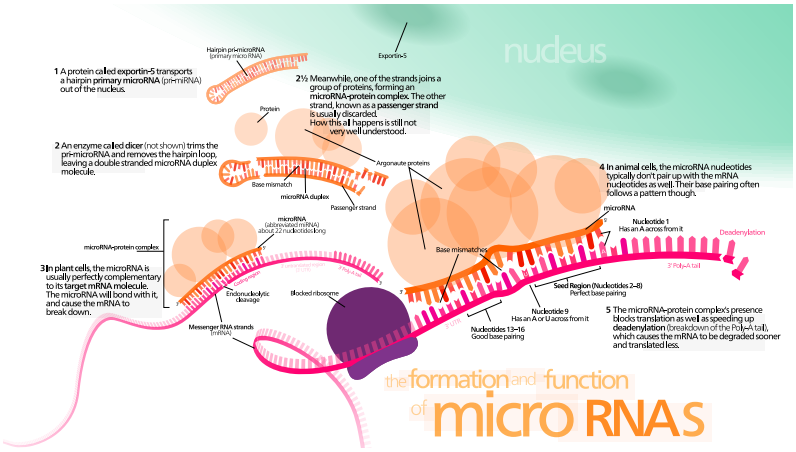

Figure 16.5D.1: **MicroRNA**: Overexpression of miRNAs could be detrimental to normal cellular activity because miRNAs bind to the 3′ UTR of RNA molecules to degrade them. Specific types of miRNAs are only found in cancer cells.

16.5E: Cancer and Translational Control

Cancer can arise from translational or post-translational modifications of proteins.

Learning Objectives

- Determine how translational or post-translational modifications can lead to cancer

Key Points

- Protein modifications from the increased translation of a protein to changes in protein phosphorylation to alternative splice variants of a protein are found in cancer cells.
- The expression of the wrong protein dramatically alters cell function and contributes to the progression of cancer.
- Gene regulation and gene function provide scientists with the opportunity to design medicines and therapies that specifically target diseased cells or exploit the overexpression of specific proteins as cancer treatment.

Key Terms

- **targeted therapy**: a type of medication that blocks the growth of cancer cells by interfering with specific targeted molecules rather than by interfering with rapidly dividing cells
- **cancer**: a disease in which the cells of a tissue undergo uncontrolled (and often rapid) proliferation
- **post-translational modification**: the chemical modification of a protein after its translation; one of the later steps in protein biosynthesis, and thus gene expression, for many proteins

Cancer and Translational/Post-translational Control

There are many examples of translational or post-translational modifications of proteins that arise in cancer. Modifications are found in cancer cells from the increased translation of a protein to changes in protein phosphorylation to alternative splice variants of a protein. An example of how the expression of an alternative form of a protein can have dramatically different outcomes is seen in colon cancer cells. The c-Flip protein, a protein involved in mediating the cell death pathway, comes in two forms: long (c-FLIPL) and short (c-FLIPS). Both forms appear to be involved in initiating controlled cell death mechanisms in normal cells. However, in colon cancer cells, expression of the long form results in increased cell growth instead of cell death. Clearly, the expression of the wrong protein dramatically alters cell function and contributes to the development of cancer.

New Drugs to Combat Cancer: Targeted Therapy

Scientists are using what is known about the regulation of gene expression in disease states, including cancer, to develop new ways to treat and prevent disease development. Many scientists are designing drugs on the basis of the gene expression patterns within individual tumors. This idea, that therapy and medicines can be tailored to an individual, has given rise to the field of personalized medicine. With an increased understanding of gene regulation and gene function, medicines can be designed to specifically target diseased cells without harming healthy cells. Some new medicines, called targeted therapies, have exploited the overexpression of a specific protein or the mutation of a gene to develop a new medication to treat disease. One such example is the use of anti-EGF receptor medications to treat the subset of breast cancer tumors that have very high levels of the EGF protein. Undoubtedly, more targeted therapies will be developed as scientists learn more about how gene expression changes can cause cancer.

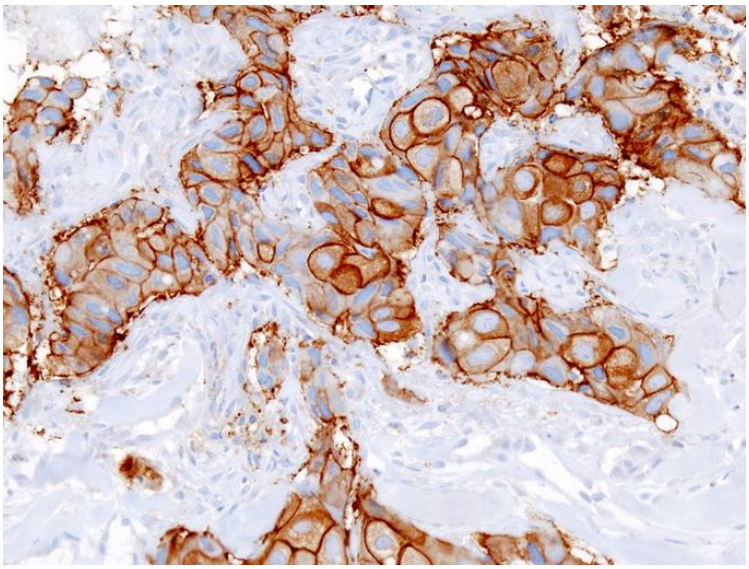

Figure 16.5E. 1: **Using Gene Expression in Targeted Therapy**: Scientists are using knowledge of the regulation of gene expression in individual cancers to develop new ways to treat target diseased cells and prevent the disease from occurring. Target therapies exploit the overexpression of a specific protein or gene mutation to develop new medications against the specific cancer.

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CHAPTER OVERVIEW

17: BIOTECHNOLOGY AND GENOMICS

17.1: BIOTECHNOLOGY

17.1A: BIOTECHNOLOGY
17.1B: BASIC TECHNIQUES TO MANIPULATE GENETIC MATERIAL (DNA AND RNA)
17.1C: MOLECULAR AND CELLULAR CLONING
17.1D: REPRODUCTIVE CLONING
17.1E: GENETIC ENGINEERING
17.1F: GENETICALLY MODIFIED ORGANISMS (GMOS)
17.1G: BIOTECHNOLOGY IN MEDICINE
17.1H: PRODUCTION OF VACCINES, ANTIBIOTICS, AND HORMONES
17.2: MAPPING GENOMES

17.2A: GENETIC MAPS
17.2B: PHYSICAL MAPS AND INTEGRATION WITH GENETIC MAPS
17.3: WHOLE-GENOME SEQUENCING

17.3A: STRATEGIES USED IN SEQUENCING PROJECTS
17.3B: USE OF WHOLE-GENOME SEQUENCES OF MODEL ORGANISMS
17.3C: USES OF GENOME SEQUENCES
17.4: APPLYING GENOMICS

17.4A: PREDICTING DISEASE RISK AT THE INDIVIDUAL LEVEL
17.4B: PHARMACOGENOMICS, TOXICOGENOMICS, AND METAGENOMICS
The effectiveness and safety of drugs can be determined through pharmacogenomics.

17.4C: GENOMICS AND BIOFUELS
17.5: GENOMICS AND PROTEOMICS

17.5A: GENOMICS AND PROTEOMICS
17.5B: BASIC TECHNIQUES IN PROTEIN ANALYSIS
17.5C: CANCER PROTEOMICS
Proteomics, the analysis of proteins, plays a prominent role in the study and treatment of cancer.

17.1: Biotechnology

Topic hierarchy

- 17.1A: Biotechnology
- 17.1B: Basic Techniques to Manipulate Genetic Material (DNA and RNA)
- 17.1C: Molecular and Cellular Cloning
- 17.1D: Reproductive Cloning
- 17.1E: Genetic Engineering
- 17.1F: Genetically Modified Organisms (GMOs)
- 17.1G: Biotechnology in Medicine
- 17.1H: Production of Vaccines, Antibiotics, and Hormones

17.1A: Biotechnology

Relying on the study of DNA, genomics analyzes entire genomes, while biotechnology uses biological agents for technological advancements.

Learning Objectives

- Justify an overview of the field of biotechnology

Key Points

- Genomics includes the study of a complete set of genes, their nucleotide sequence and organization, and their interactions within a species and with other species.
- Through DNA sequencing, genomic information is used to create maps of the DNA of different organisms.
- Biotechnology, or the use of biological agents for technological progression, has applications in medicine, agriculture, and in industry, which include processes such as fermentation and the production of biofuels.

Key Terms

- **genomics**: the study of the complete genome of an organism
- **sequencing**: the procedure of determining the order of amino acids in the polypeptide chain of a protein (protein sequencing) or of nucleotides in a DNA section comprising a gene (gene sequencing)
- **biotechnology**: the use of living organisms (especially microorganisms) in industrial, agricultural, medical, and other technological applications

The study of nucleic acids began with the discovery of DNA, progressed to the study of genes and small fragments, and has now exploded to the field of genomics. Genomics is the study of entire genomes, including the complete set of genes, their nucleotide sequence and organization, and their interactions within a species and with other species. The advances in genomics have been made possible by DNA sequencing technology. Just as information technology has led to Google maps that enable people to get detailed information about locations around the globe, genomic information is used to create similar maps of the DNA of different organisms. These findings have helped anthropologists to better understand human migration and have aided the field of medicine through the mapping of human genetic diseases. The ways in which genomic information can contribute to scientific understanding are varied and quickly growing.

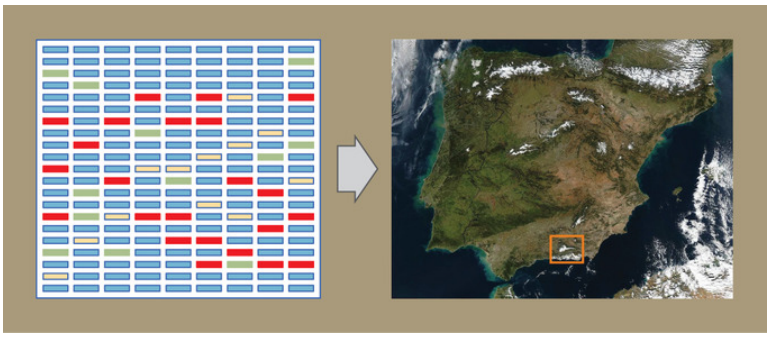

Figure 17.1*A*.1: **Genomics**: In genomics, the DNA of different organisms is compared, enabling scientists to create maps with which to navigate the DNA of different organisms.

Another rapidly-advancing field that utilizes DNA is biotechnology. This field involves the use of biological agents for technological advancement. Biotechnology was used for breeding livestock and crops long before the scientific basis of these techniques was understood. Since the discovery of the structure of DNA in 1953, the field of biotechnology has grown rapidly through both academic research and private companies. The primary applications of this technology are in medicine (production of vaccines and antibiotics) and agriculture (genetic modification of crops, such as to increase yields). Biotechnology also has many industrial applications, such as fermentation, the treatment of oil spills, and the production of biofuels.

17.1B: Basic Techniques to Manipulate Genetic Material (DNA and RNA)

Basic techniques used in genetic material manipulation include extraction, gel electrophoresis, PCR, and blotting methods.

Learning Objectives

- Distinguish among the basic techniques used to manipulate DNA and RNA

Key Points

- The first step to study or work with nucleic acids includes the isolation or extraction of DNA or RNA from cells.
- Gel electrophoresis depends on the negatively-charged ions present on nucleic acids at neutral or basic pH to separate molecules on the basis of size.
- Specific regions of DNA can be amplified through the use of polymerase chain reaction for further analysis.
- Southern blotting involves the transfer of DNA to a nylon membrane, while northern blotting is the transfer of RNA to a nylon membrane; these techniques allow samples to be probed for the presence of certain sequences.

Key Terms

- **denaturation**: the change of folding structure of a protein (and thus of physical properties) caused by heating, changes in pH, or exposure to certain chemicals
- **electrophoresis**: a method for the separation and analysis of large molecules, such as proteins or nucleic acids, by migrating a colloidal solution of them through a gel under the influence of an electric field
- **polymerase chain reaction**: a technique in molecular biology for creating multiple copies of DNA from a sample

Basic Techniques to Manipulate Genetic Material (DNA and RNA)

To understand the basic techniques used to work with nucleic acids, remember that nucleic acids are macromolecules made of nucleotides (a sugar, a phosphate, and a nitrogenous base) linked by phosphodiester bonds. The phosphate groups on these molecules each have a net negative charge. An entire set of DNA molecules in the nucleus is called the genome. DNA has two complementary strands linked by hydrogen bonds between the paired bases. The two strands can be separated by exposure to high temperatures (DNA denaturation) and can be reannealed by cooling. The DNA can be replicated by the DNA polymerase enzyme. Unlike DNA, which is located in the nucleus of eukaryotic cells, RNA molecules leave the nucleus. The most common type of RNA that is analyzed is the messenger RNA (mRNA) because it represents the protein -coding genes that are actively expressed.

DNA and RNA Extraction

To study or manipulate nucleic acids, the DNA or RNA must first be isolated or extracted from the cells. This can be done through various techniques. Most nucleic acid extraction techniques involve steps to break open the cell and use enzymatic reactions to destroy all macromolecules that are not desired (such as degradation of unwanted molecules and separation from the DNA sample). Cells are broken using a lysis buffer (a solution that is mostly a detergent); lysis means “to split.” These enzymes break apart lipid molecules in the membranes of the cell and the nucleus. Macromolecules are inactivated using enzymes such as proteases that break down proteins, and ribonucleases (RNAses) that break down RNA. The DNA is then precipitated using alcohol. Human genomic DNA is usually visible as a gelatinous, white mass. Samples can be stored at –80°C for years.

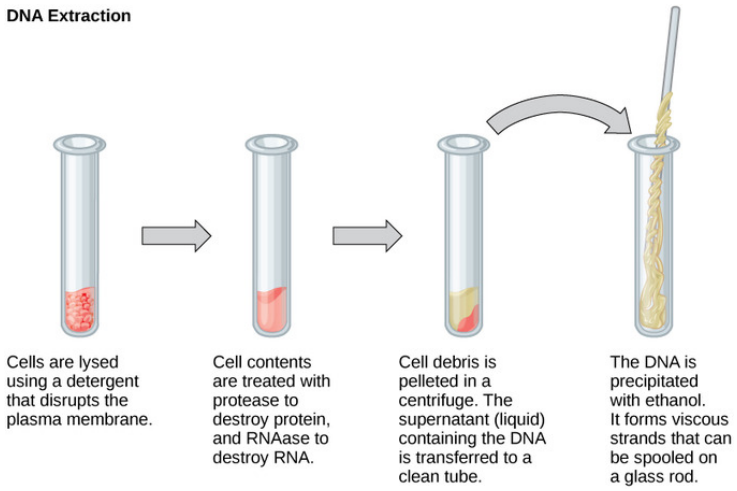

Figure 17.1B.1: **DNA Extraction**: This diagram shows the basic method used for extraction of DNA.

RNA analysis is performed to study gene expression patterns in cells. RNA is naturally very unstable because RNAses are commonly present in nature and very difficult to inactivate. Similar to DNA, RNA extraction involves the use of various buffers and enzymes to inactivate macromolecules and preserve the RNA.

Gel Electrophoresis

Because nucleic acids are negatively-charged ions at neutral or basic pH in an aqueous environment, they can be mobilized by an electric field. Gel electrophoresis is a technique used to separate molecules on the basis of size using this charge and may be separated as whole chromosomes or fragments. The nucleic acids are loaded into a slot near the negative electrode of a porous gel matrix and pulled toward the positive electrode at the opposite end of the gel. Smaller molecules move through the pores in the gel faster than larger molecules; this difference in the rate of migration separates the fragments on the basis of size. There are molecular-weight standard samples that can be run alongside the molecules to provide a size comparison. Nucleic acids in a gel matrix can be observed using various fluorescent or colored dyes. Distinct nucleic acid fragments appear as bands at specific distances from the top of the gel (the negative electrode end) on the basis of their size.

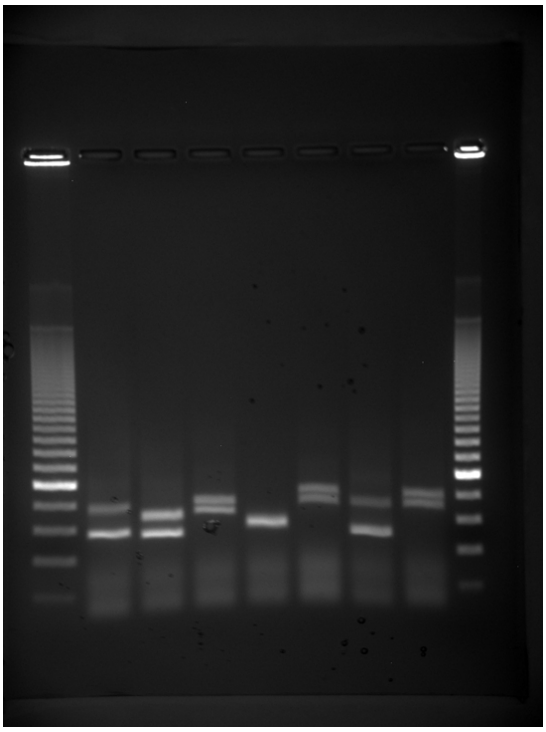

Figure 17.1$B.1$: **Gel Electrophoresis**: Shown are DNA fragments from seven samples run on a gel, stained with a fluorescent dye, and viewed under UV light.

Amplification of Nucleic Acid Fragments by Polymerase Chain Reaction

Polymerase chain reaction (PCR) is a technique used to amplify specific regions of DNA for further analysis. PCR is used for many purposes in laboratories, such as the cloning of gene fragments to analyze genetic diseases, identification of contaminant foreign DNA in a sample, and the amplification of DNA for sequencing. More practical applications include the determination of paternity and detection of genetic diseases.

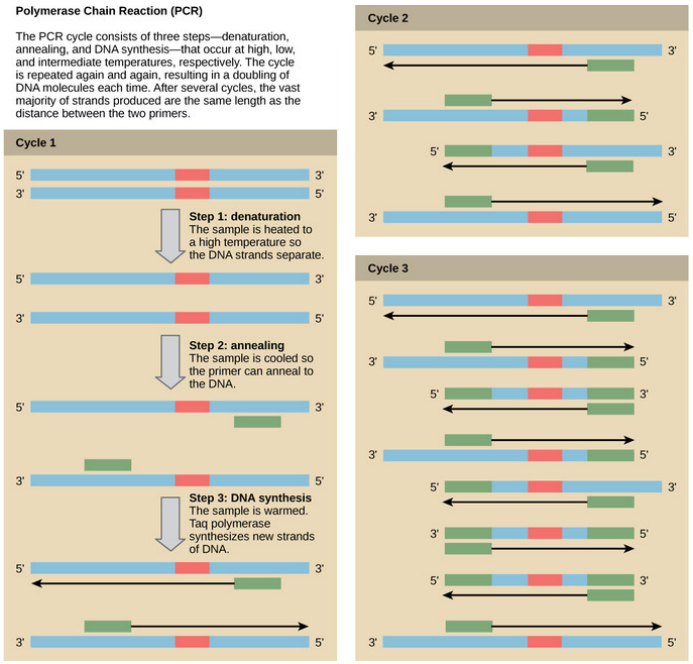

Figure 17.1B.1: **PCR Amplification**: Polymerase chain reaction, or PCR, is used to amplify a specific sequence of DNA. Primers—short pieces of DNA complementary to each end of the target sequence—are combined with genomic DNA, Taq polymerase, and deoxynucleotides. Taq polymerase is a DNA polymerase isolated from the thermostable bacterium Thermus aquaticus that is able to withstand the high temperatures used in PCR. Thermus aquaticus grows in the Lower Geyser Basin of Yellowstone National Park. Reverse transcriptase PCR (RT-PCR) is similar to PCR, but cDNA is made from an RNA template before PCR begins.

DNA fragments can also be amplified from an RNA template in a process called reverse transcriptase PCR (RT-PCR). The first step is to recreate the original DNA template strand (called cDNA) by applying DNA nucleotides to the mRNA. This process is called reverse transcription. This requires the presence of an enzyme called reverse transcriptase. After the cDNA is made, regular PCR can be used to amplify it.

Hybridization, Southern Blotting, and Northern Blotting

Nucleic acid samples, such as fragmented genomic DNA and RNA extracts, can be probed for the presence of certain sequences. Short DNA fragments called probes are designed and labeled with radioactive or fluorescent dyes to aid detection. Gel electrophoresis separates the nucleic acid fragments according to their size. The fragments in the gel are then transferred onto a nylon membrane in a procedure called blotting. The nucleic acid fragments that are bound to the surface of the membrane can then be probed with specific radioactively- or fluorescently-labeled probe sequences. When DNA is transferred to a nylon membrane, the technique is called Southern blotting; when RNA is transferred to a nylon membrane, it is called northern blotting. Southern blots are used to detect the presence of certain DNA sequences in a given genome, and northern blots are used to detect gene expression.

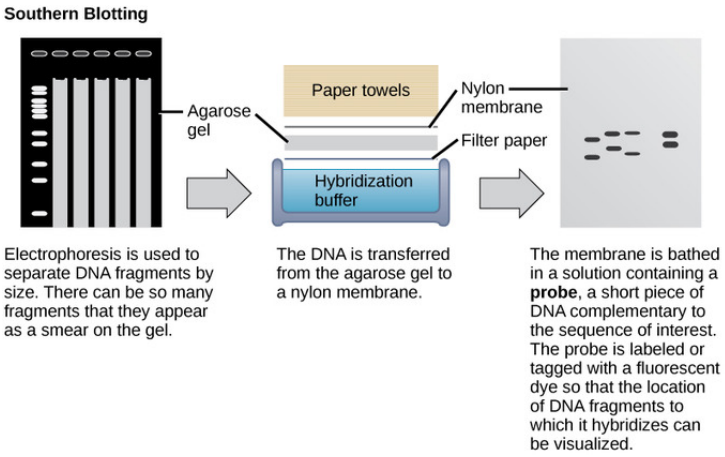

Figure 17.1B.1: **Blotting Techniques**: Southern blotting is used to find a particular sequence in a sample of DNA. DNA fragments are separated on a gel, transferred to a nylon membrane, and incubated with a DNA probe complementary to the sequence of interest. Northern blotting is similar to Southern blotting, but RNA is run on the gel instead of DNA. In western blotting, proteins are run on a gel and detected using antibodies.

17.1C: Molecular and Cellular Cloning

Molecular cloning reproduces the desired regions or fragments of a genome, enabling the manipulation and study of genes.

Learning Objectives

- Describe the process of molecular cloning

Key Points

- Cloning small fragments of a genome allows specific genes, their protein products, and non-coding regions to be studied in isolation.
- A plasmid, also known as a vector, is a small circular DNA molecule that replicates independently of the chromosomal DNA; it can be used to provide a “folder” in which to insert a desired DNA fragment.
- Recombinant DNA molecules are plasmids with foreign DNA inserted into them; they are created artificially as they do not occur in nature.
- Bacteria and yeast naturally produce clones of themselves when they replicate asexually through cellular cloning.

Key Terms

- **recombinant DNA**: DNA that has been engineered by splicing together fragments of DNA from multiple species and introduced into the cells of a host
- **molecular cloning**: a biological method that creates many identical DNA molecules and directs their replication within a host organism
- **plasmid**: a circle of double-stranded DNA that is separate from the chromosomes, which is found in bacteria and protozoa

Molecular Cloning

In general, the word “cloning” means the creation of a perfect replica; however, in biology, the re-creation of a whole organism is referred to as “reproductive cloning.” Long before attempts were made to clone an entire organism, researchers learned how to reproduce desired regions or fragments of the genome, a process that is referred to as molecular cloning.

Cloning small fragments of the genome allows for the manipulation and study of specific genes (and their protein products) or noncoding regions in isolation. A plasmid (also called a vector) is a small circular DNA molecule that replicates independently of the chromosomal DNA. In cloning, the plasmid molecules can be used to provide a “folder” in which to insert a desired DNA fragment. Plasmids are usually introduced into a bacterial host for proliferation. In the bacterial context, the fragment of DNA from the human genome (or the genome of another organism that is being studied) is referred to as foreign DNA (or a transgene) to differentiate it from the DNA of the bacterium, which is called the host DNA.

Plasmids occur naturally in bacterial populations (such as *Escherichia coli*) and have genes that can contribute favorable traits to the organism such as antibiotic resistance (the ability to be unaffected by antibiotics). Plasmids have been repurposed and engineered as vectors for molecular cloning and the large-scale production of important reagents such as insulin and human growth hormone. An important feature of plasmid vectors is the ease with which a foreign DNA fragment can be introduced via the multiple cloning site (MCS). The MCS is a short DNA sequence containing multiple sites that can be cut with different commonly-available restriction endonucleases. Restriction endonucleases recognize specific DNA sequences and cut them in a predictable manner; they are naturally produced by bacteria as a defense mechanism against foreign DNA. Many restriction endonucleases make staggered cuts in the two strands of DNA, such that the cut ends have a 2- or 4-base single-stranded overhang. Because these overhangs are capable of annealing with complementary overhangs, these are called “sticky ends.” Addition of an enzyme called DNA ligase permanently joins the DNA fragments via phosphodiester bonds. In this way, any DNA fragment generated by restriction endonuclease cleavage can be spliced between the two ends of a plasmid DNA that has been cut with the same restriction endonuclease.

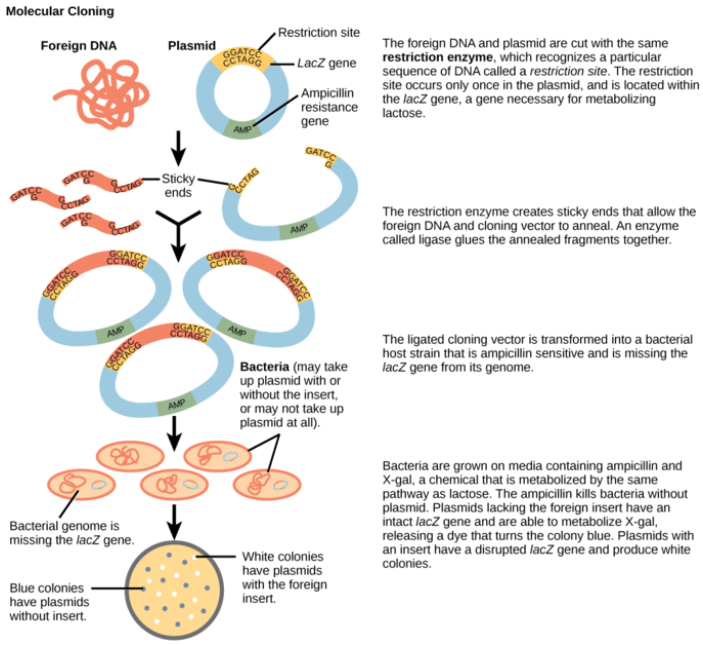

Figure 17.1C.1: **Molecular Cloning**: This diagram shows the steps involved in molecular cloning, where regions or fragments of a genome are reproduced to allow the study or manipulation of genes and their protein products.

Recombinant DNA Molecules

Plasmids with foreign DNA inserted into them are called recombinant DNA molecules because they are created artificially and do not occur in nature. They are also called chimeric molecules because the origin of different parts of the molecules can be traced back to different species of biological organisms or even to chemical synthesis. Proteins that are expressed from recombinant DNA molecules are called recombinant proteins. Not all recombinant plasmids are capable of expressing genes. The recombinant DNA may need to be moved into a different vector (or host) that is better designed for gene expression. Plasmids may also be engineered to express proteins only when stimulated by certain environmental factors so that scientists can control the expression of the recombinant proteins.

Cellular Cloning

Unicellular organisms, such as bacteria and yeast, naturally produce clones of themselves when they replicate asexually by binary fission; this is known as cellular cloning. The nuclear DNA duplicates by the process of mitosis, which creates an exact replica of the genetic material.

17.1D: Reproductive Cloning

Reproductive cloning, possible through artificially-induced asexual reproduction, is a method used to make a clone of an entire organism.

Learning Objectives

- Differentiate reproductive cloning from cellular and molecular cloning

Key Points

- A form of asexual reproduction, parthenogenesis, occurs when an embryo grows and develops without the fertilization of the egg.
- In reproductive cloning, if the haploid nucleus of an egg cell is replaced with a diploid nucleus from the cell of an individual of the same species, it will become a zygote that is genetically identical to the donor.
- Reproductive cloning has become successful, but still has limitations as cloned individuals often exhibit facial, limb, and cardiac abnormalities.
- Therapeutic cloning, the cloning of human embryos as a source of embryonic stem cells, has been attempted in order to produce cells that can be used to treat detrimental diseases or defects.

Key Terms

- **clone**: a living organism produced asexually from a single ancestor, to which it is genetically identical
- **stem cell**: a primal undifferentiated cell from which a variety of other cells can develop through the process of cellular differentiation
- **parthenogenesis**: a form of asexual reproduction where growth and development of embryos occur without fertilization

Reproductive Cloning

Reproductive cloning is a method used to make a clone or an identical copy of an entire multicellular organism. Most multicellular organisms undergo reproduction by sexual means, which involves genetic hybridization of two individuals (parents), making it impossible to generate an identical copy or clone of either parent. Recent advances in biotechnology have made it possible to artificially induce asexual reproduction of mammals in the laboratory.

Parthenogenesis, or “virgin birth,” occurs when an embryo grows and develops without the fertilization of the egg occurring; this is a form of asexual reproduction. An example of parthenogenesis occurs in species in which the female lays an egg. If the egg is fertilized, it is a diploid egg and the individual develops into a female; if the egg is not fertilized, it remains a haploid egg and develops into a male. The unfertilized egg is called a parthenogenic, or virgin, egg. Some insects and reptiles lay parthenogenic eggs that can develop into adults.

Sexual reproduction requires two cells; when the haploid egg and sperm cells fuse, a diploid zygote results. The zygote nucleus contains the genetic information to produce a new individual. However, early embryonic development requires the cytoplasmic material contained in the egg cell. This idea forms the basis for reproductive cloning. If the haploid nucleus of an egg cell is replaced with a diploid nucleus from the cell of any individual of the same species (called a donor), it will become a zygote that is genetically identical to the donor. Somatic cell nuclear transfer is the technique of transferring a diploid nucleus into an enucleated egg. It can be used for either therapeutic cloning or reproductive cloning.

The first cloned animal was Dolly, a sheep who was born in 1996. The success rate of reproductive cloning at the time was very low. Dolly lived for seven years and died of respiratory complications. There is speculation that because the cell DNA belongs to an older individual, the age of the DNA may affect the life expectancy of a cloned individual. Since Dolly, several animals (e.g. horses, bulls, and goats) have been successfully cloned, although these individuals often exhibit facial, limb, and cardiac abnormalities. There have been attempts at producing cloned human embryos as sources of embryonic stem cells. Sometimes referred to as cloning for therapeutic purposes, the technique produces stem cells that attempt to remedy detrimental diseases or defects (unlike reproductive cloning, which aims to reproduce an organism). Still, therapeutic cloning efforts have met with resistance because of bioethical considerations.

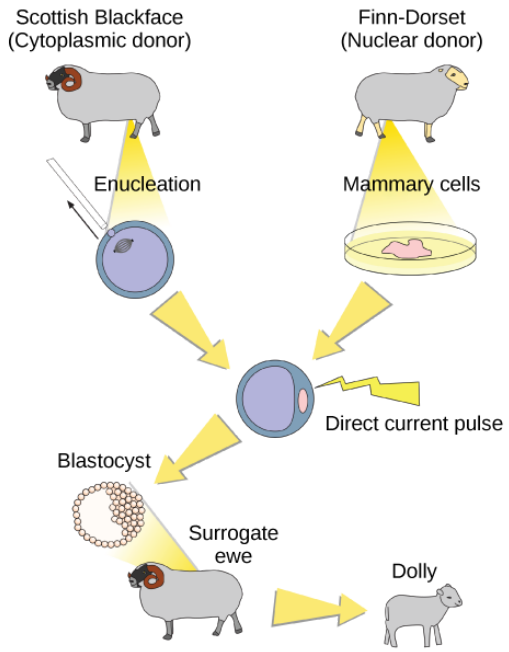

Figure 17.1D.1: **Reproductive Cloning of Dolly, the Sheep**: Dolly the sheep was the first mammal to be cloned. To create Dolly, the nucleus was removed from a donor egg cell. The nucleus from a second sheep was then introduced into the cell, which was allowed to divide to the blastocyst stage before being implanted in a surrogate mother.

17.1E: Genetic Engineering

In genetic engineering, an organism's genotype is altered using recombinant DNA, created by molecular cloning, to modify an organism's DNA.

Learning Objectives

- Discuss how genetic engineering leads to DNA modification.

Key Points

- A genetically modified organism receives recombinant DNA generated through molecular cloning.
- Transgenic host organisms receive their foreign DNA from a different species.
- The use of recombinant DNA vectors to alter the expression of a particular gene is known as gene targeting, which is done through the addition of mutations in a gene or the exclusion of the expression of a certain gene.
- Recombinant DNA technology involves transferring a DNA fragment of interest from one organism to another by inserting it into a vector.

Key Terms

- **recombinant DNA**: DNA that has been engineered by splicing together fragments of DNA from multiple species and introduced into the cells of a host
- **genetic engineering**: the deliberate modification of the genetic structure of an organism
- **genetically modified organism**: an organism whose genetic material has been altered using genetic engineering techniques

Genetic Engineering

Genetic engineering is the alteration of an organism's genotype using recombinant DNA technology to modify an organism's DNA to achieve desirable traits. Recombinant DNA technology, or DNA cloning, is the process of transferring a DNA fragment of interest from one organism to a self-replicating genetic element, such as a bacteria plasmid, which is called a vector. The DNA of interest can then be propagated in another organism. The addition of foreign DNA in the form of recombinant DNA vectors generated by molecular cloning is the most common method of genetic engineering. The organism that receives the recombinant DNA is called a genetically-modified organism (GMO). If the foreign DNA that is introduced comes from a different species, the host organism is called transgenic. Bacteria, plants, and animals have been genetically modified since the early 1970s for academic, medical, agricultural, and industrial purposes. In the US, GMOs such as Roundup-ready soybeans and borer-resistant corn are part of many common processed foods.

Figure 17.1E.1: **GMO Corn**: Borer-resistant corn is an example of a genetically- modified organism made possible through genetic engineering methods that allow scientists to alter an organism's DNA to achieve specific traits, such as herbicide resistance.

Gene Targeting

Although classical methods of studying the function of genes began with a given phenotype and determined the genetic basis of that phenotype, modern techniques allow researchers to start at the DNA sequence level and ask: “What does this gene or DNA element do? ” This technique, called reverse genetics, has resulted in reversing the classic genetic methodology. This method would be similar to damaging a body part to determine its function. An insect that loses a wing cannot fly, which means that the function of the wing is flight. The classical genetic method would compare insects that cannot fly with insects that can fly, and observe that the non-flying insects have lost wings. Similarly, mutating or deleting genes provides researchers with clues about gene function. The methods used to disable gene function are collectively called gene targeting. Gene targeting is the use of recombinant DNA vectors to alter the expression of a particular gene, either by introducing mutations in a gene, or by eliminating the expression of a certain gene by deleting a part or all of the gene sequence from the genome of an organism.

17.1F: Genetically Modified Organisms (GMOs)

Transgenic modification, adding recombinant DNA to a species, has led to the expression of desirable genes in plants and animals.

Learning Objectives

- Describe how research on transgenic plants and animals aids humans.

Key Points

- Transgenic animals are those that have been modified to express recombinant DNA from another species.
- Manipulation of transgenic plants, those that have received recombinant DNA from other species, has led to the creation of species that display disease resistance, herbicide and pesticide resistance, better nutritional value, and better shelf-life.
- The thickness of a plant's cell wall makes the artificial introduction of DNA into plant cells much more challenging than in animal cells.

Key Terms

- **transgenic**: of or pertaining to an organism whose genome has been changed by the addition of a gene from another species; genetically modified
- **genetically modified organism**: an organism whose genetic material has been altered using genetic engineering techniques

Transgenic Animals

Although several recombinant proteins used in medicine are successfully produced in bacteria, some proteins require a eukaryotic animal host for proper processing. For this reason, the desired genes are cloned and expressed in animals, such as sheep, goats, chickens, and mice. Animals that have been modified to express recombinant DNA are called transgenic animals. Several human proteins are expressed in the milk of transgenic sheep and goats, while others are expressed in the eggs of chickens. Mice have been used extensively for expressing and studying the effects of recombinant genes and mutations.

Transgenic Plants

Manipulating the DNA of plants (or creating genetically modified organisms called GMOs) has helped to create desirable traits, such as disease resistance, herbicide and pesticide resistance, better nutritional value, and better shelf-life. Plants are the most important source of food for the human population. Farmers developed ways to select for plant varieties with desirable traits long before modern-day biotechnology practices were established. Plants that have received recombinant DNA from other species are called transgenic plants. Because foreign genes can spread to other species in the environment, extensive testing is required to ensure ecological stability. Staples like corn, potatoes, and tomatoes were the first crop plants to be genetically engineered.

Figure 17.1F.1: **Transgenic Plants**: Corn, a major agricultural crop used to create products for a variety of industries, is often modified through plant biotechnology.

Transformation of Plants Using *Agrobacterium tumefaciens*

Gene transfer occurs naturally between species in microbial populations. Many viruses that cause human diseases, such as cancer, act by incorporating their DNA into the human genome. In plants, tumors caused by the bacterium *Agrobacterium tumefaciens* occur by transfer of DNA from the bacterium to the plant. Although the tumors do not kill the plants, they stunt the plants, which become more susceptible to harsh environmental conditions. Many plants, such as walnuts, grapes, nut trees, and beets, are affected by *A. tumefaciens*. The artificial introduction of DNA into plant cells is more challenging than in animal cells because of the thick plant cell wall.

Researchers used the natural transfer of DNA from *Agrobacterium* to a plant host to introduce DNA fragments of their choice into plant hosts. In nature, the disease-causing *A. tumefaciens* have a set of plasmids, called the Ti plasmids (tumor-inducing plasmids), that contain genes for the production of tumors in plants. DNA from the Ti plasmid integrates into the infected plant cell's genome. Researchers manipulate the Ti plasmids to remove the tumor-causing genes and insert the desired DNA fragment for transfer into the plant genome. The Ti plasmids carry antibiotic resistance genes to aid selection and can be propagated in *E. coli* cells as well.

The Organic Insecticide *Bacillus thuringiensis*

Bacillus thuringiensis (Bt) is a bacterium that produces protein crystals during sporulation that are toxic to many insect species that affect plants. Bt toxin has to be ingested by insects for the toxin to be activated. Insects that have eaten Bt toxin stop feeding on the plants within a few hours. After the toxin is activated in the intestines of the insects, death occurs within a couple of days. Modern biotechnology has allowed plants to encode their own crystal Bt toxin that acts against insects. The crystal toxin genes have been cloned from Bt and introduced into plants. Bt toxin has been found to be safe for the environment, non-toxic to humans and other mammals, and is approved for use by organic farmers as a natural insecticide.

Flavr Savr Tomato

The first GM crop to be introduced into the market was the Flavr Savr Tomato, produced in 1994. Antisense RNA technology was used to slow down the process of softening and rotting caused by fungal infections, which led to increased shelf life of the GM tomatoes. Additional genetic modification improved the flavor of this tomato. The Flavr Savr tomato did not successfully stay in the market because of problems maintaining and shipping the crop.

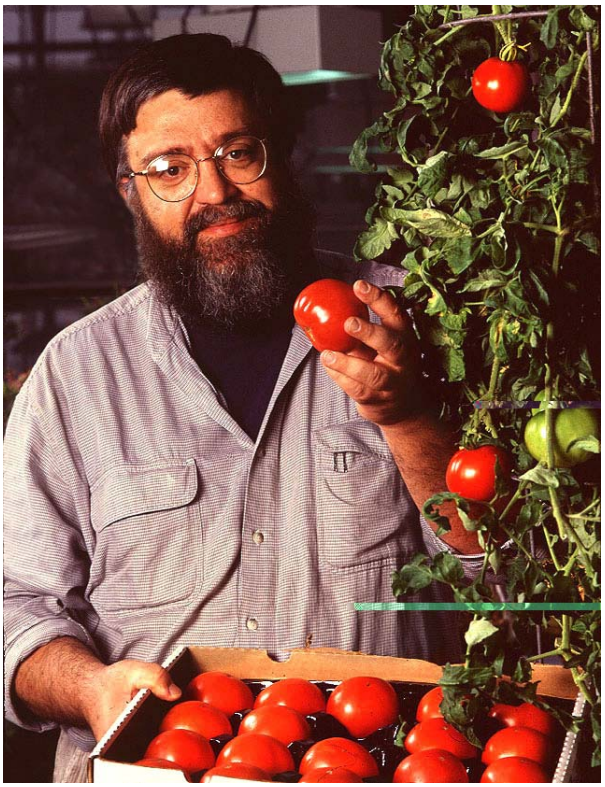

Figure 17.1$F.1$: **The Flavr Savr Tomato**: Plant physiologist Athanasios Theologis with tomatoes that contain the bioengineered ACC synthase gene (the Flavr Savr Tomato).

G: Biotechnology in Medicine

From manipulation of mutant genes to enhanced resistance to disease, biotechnology has allowed advances in medicine.

Learning Objectives

- Give examples of how biotechnology is used in medicine.

Key Points

- The study of pharmacogenomics can result in the development of tailor-made vaccines for people, more accurate means of determining drug dosages, improvements in drug discovery and approval, and the development of safer vaccines.
- Modern biotechnology can be used to manufacture drugs more easily and cheaply, as they can be produced in larger quantities from existing genetic sources.
- Genetic diagnosis involves the process of testing for suspected genetic defects before administering treatment through genetic testing.
- In gene therapy, a good gene is introduced at a random location in the genome to aid the cure of a disease that is caused by a mutated gene.

Key Terms

- **gene therapy**: any of several therapies involving the insertion of genes into a patient's cells in order to replace defective ones
- **pharmacogenomics**: the study of genes that code for enzymes that metabolize drugs, and the design of tailor-made drugs adapted to an individual's genetic make-up
- **immunodeficiency**: a depletion in the body's natural immune system, or in some component of it

Biotechnology in Medicine

It is easy to see how biotechnology can be used for medicinal purposes. Knowledge of the genetic makeup of our species, the genetic basis of heritable diseases, and the invention of technology to manipulate and fix mutant genes provides methods to treat the disease.

Pharmacogenomics is the study of how the genetic inheritance of an individual affects his/her body's response to drugs. It is a coined word derived from the words "pharmacology" and " genomics ". It is, therefore, the study of the relationship between pharmaceuticals and genetics. The vision of pharmacogenomics is to be able to design and produce drugs that are adapted to each person's genetic makeup. Pharmacogenomics results in the following benefits:

1. Development of tailor-made medicines. Using pharmacogenomics, pharmaceutical companies can create drugs based on the proteins, enzymes, and RNA molecules that are associated with specific genes and diseases. These tailor-made drugs promise not only to maximize therapeutic effects, but also to decrease damage to nearby healthy cells.

2. More accurate methods of determining appropriate drug dosages. Knowing a patient's genetics will enable doctors to determine how well the patient's body can process and metabolize a medicine. This will maximize the value of the medicine and decrease the likelihood of overdose.

3. Improvements in the drug discovery and approval process. The discovery of potential therapies will be made easier using genome targets. Genes have been associated with numerous diseases and disorders. With modern biotechnology, these genes can be used as targets for the development of effective new therapies, which could significantly shorten the drug discovery process.

4. Better vaccines. Safer vaccines can be designed and produced by organisms transformed by means of genetic engineering. These vaccines will elicit the immune response without the attendant risks of infection. They will be inexpensive, stable, easy to store, and capable of being engineered to carry several strains of pathogen at once.

Modern biotechnology can be used to manufacture existing drugs more easily and cheaply. The first genetically-engineered products were medicines designed to combat human diseases. In 1978, Genentech joined a gene for insulin with a plasmid vector and put the resulting gene into a bacterium called *Escherichia coli*. Insulin, widely used for the

ent of diabetes, was previously extracted from sheep and pigs. It was very expensive and often elicited unwanted c responses. The resulting genetically-engineered bacterium enabled the production of vast quantities of human insulin at low cost. Since then, modern biotechnology has made it possible to produce more easily and cheaply the human growth hormone, clotting factors for hemophiliacs, fertility drugs, erythropoietin, and other drugs. Genomic knowledge of the genes involved in diseases, disease pathways, and drug-response sites are expected to lead to the discovery of thousands more new targets.

Genetic Diagnosis and Gene Therapy

The process of testing for suspected genetic defects before administering treatment is called genetic diagnosis by genetic testing. Depending on the inheritance patterns of a disease-causing gene, family members are advised to undergo genetic testing. Treatment plans are based on the findings of genetic tests that determine the type of cancer. If the cancer is caused by inherited gene mutations, other female relatives are also advised to undergo genetic testing and periodic screening for breast cancer. Genetic testing is also offered for fetuses to determine the presence or absence of disease-causing genes in families with specific, debilitating diseases.

Genetic testing involves the direct examination of the DNA molecule itself. A scientist scans a patient's DNA sample for mutated sequences. There are two major types of gene tests. In the first type, a researcher may design short pieces of DNA whose sequences are complementary to the mutated sequences. These probes will seek their complement among the base pairs of an individual's genome. If the mutated sequence is present in the patient's genome, the probe will bind to it and flag the mutation. In the second type, a researcher may conduct the gene test by comparing the sequence of DNA bases in a patient's gene to a normal version of the gene.

Gene therapy is a genetic engineering technique used to cure disease. In its simplest form, it involves the introduction of a good gene at a random location in the genome to aid the cure of a disease that is caused by a mutated gene. The good gene is usually introduced into diseased cells as part of a vector transmitted by a virus that can infect the host cell and deliver the foreign DNA. More advanced forms of gene therapy try to correct the mutation at the original site in the genome, such as is the case with treatment of severe combined immunodeficiency (SCID).

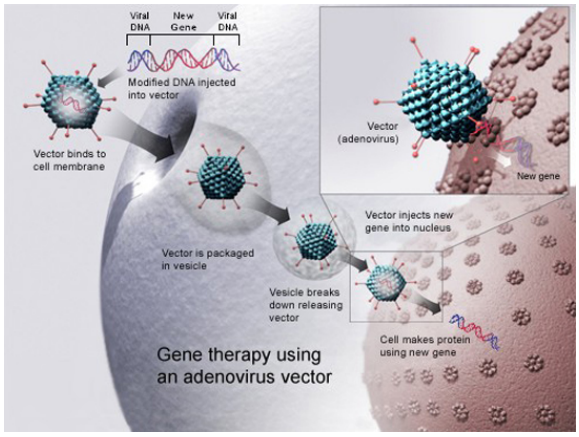

Figure 17.1G.1: **Gene Therapy**: Gene therapy using an adenovirus vector can be used to cure certain genetic diseases in which a person has a defective gene.

17.1H: Production of Vaccines, Antibiotics, and Hormones

Biotechnological advances in gene manipulation techniques have further resulted in the production of vaccines, antibiotics, and hormones.

Learning Objectives

- Discuss the methods by which biotechnology is used to produce vaccines, antibiotics, and hormones.

Key Points

- Vaccines use weakened or inactive forms of microorganisms to mount the initial immune response through the use of antigens, which are produced through use the genes of microbes that are cloned into vectors.
- Antibiotics, agents that inhibit bacterial growth or kill bacteria, are produced by cultivating and manipulating fungal cells.
- Hormones, such as the human growth hormone (HGH), can be formulated through recombinant DNA technology; for example, HGH can be cloned from a cDNA library and inserted into *E. coli* cells by cloning it into a bacterial vector.

Key Terms

- **bactericidal**: that which kills bacteria
- **bacteriostatic**: that which slows down or stalls bacterial growth
- **antigen**: a substance that binds to a specific antibody; may cause an immune response

Production of Vaccines, Antibiotics, and Hormones

Vaccines

Traditional vaccination strategies use weakened or inactive forms of microorganisms to mount the initial immune response. Modern techniques use the genes of microorganisms cloned into vectors to mass produce the desired antigen. The antigen is then introduced into the body to stimulate the primary immune response and trigger immune memory. Genes cloned from the influenza virus have been used to combat the constantly-changing strains of this virus.

Antibiotics

Antibiotics are biotechnological products that inhibit bacterial growth or kill bacteria. They are naturally produced by microorganisms, such as fungi, to attain an advantage over bacterial populations. Antibiotics are produced on a large scale by cultivating and manipulating fungal cells. Many antibacterial compounds are classified on the basis of their chemical or biosynthetic origin into natural, semisynthetic, and synthetic. Another classification system is based on biological activity. In this classification, antibiotics are divided into two broad groups according to their biological effect on microorganisms: bactericidal agents kill bacteria, and bacteriostatic agents slow down or stall bacterial growth.

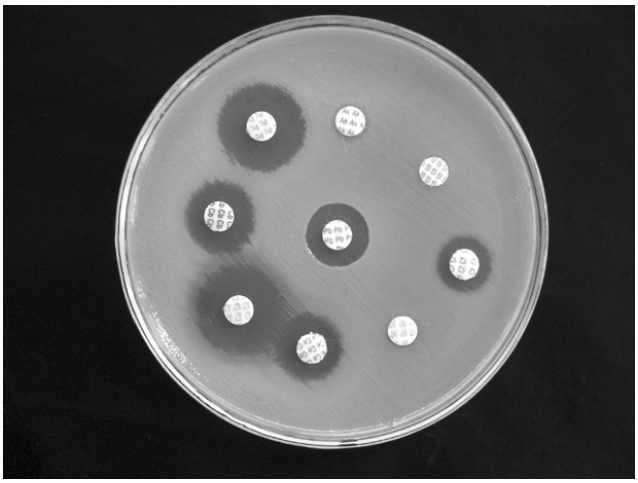

Figure 17.1H.1: **Antibiotic Treatment**: Assays such as the one shown help scientists understand the effects of antibiotics on bacterial species. Clear rings around the round inserts, which contain antibiotic, mean that bacteria on the plate are inhibited or killed by the compound.

Hormones

Recombinant DNA technology was used to produce large-scale quantities of human insulin (a hormone) in *E. coli* as early as 1978. Previously, it was only possible to treat diabetes with pig insulin, which caused allergic reactions in humans because of differences in the gene product. In recent times, human growth hormone (HGH) has been used to treat growth disorders in children. The HGH gene was cloned from a cDNA library and inserted into *E. coli* cells by cloning it into a bacterial vector. The bacteria was then grown and the hormone isolated, enabling large scale commercial production.

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Topic hierarchy

- 17.2A: Genetic Maps
- 17.2B: Physical Maps and Integration with Genetic Maps

17.2A: Genetic Maps

Genetic maps provide information about which chromosomes contain specific genes and precisely where the genes lie on that chromosome.

Learning Objectives

- Describe the different types of genetic markers that are used in generating genetic maps of DNA

Key Points

- Genetic mapping, often called linkage mapping, provides information about the location of a specific gene along a chromosome.
- Gene linkage describes the phenomenon that certain genes are physically linked by being located on the same chromosome and have a tendency to be inherited together.
- Genetic recombination involves the production of a novel set of genetic information by breaking and rejoining DNA fragments that have a great distance between them along the chromosome.
- The construction of genetic maps is reliant on the natural process of recombination which results in the ability to identify genetic markers with variability within a population.
- Genetic markers that can be used in generating genetic maps include restriction length polymorphisms (RFLP); variable number of tandem repeats (VNTRs); microsatellite polymorphisms; and single nucleotide polymorphisms (SNPs).

Key Terms

- **polymorphism**: the regular existence of two or more different genotypes within a given species or population
- **SNP**: single nucleotide polymorphism is single base pair of DNA which is polymorphic with respect to a population
- **microsatellite**: any of a group of polymorphic loci in DNA that consist of repeat units of just a few base pairs
- **RFLP**: restriction fragment length polymorphism is a section of DNA whose length varies among individuals and which is delimited by a base which does not occur within it

Genetic Maps

The study of genetic maps begins with linkage analysis, a procedure that analyzes the recombination frequency between genes to determine if they are linked or show independent assortment. The term linkage was used before the discovery of DNA. Early geneticists relied on the observation of phenotypic changes to understand the genotype of an organism. Shortly after Gregor Mendel (the father of modern genetics) proposed that traits were determined by what are now known as genes, other researchers observed that different traits were often inherited together and, thereby, deduced that the genes were physically linked by being located on the same chromosome. The mapping of genes relative to each other based on linkage analysis led to the development of the first genetic maps.

Observations that certain traits were always linked and certain others were not linked came from studying the offspring of crosses between parents with different traits. For example, in experiments performed on the garden pea, it was discovered that the color of the flower and shape of the plant’s pollen were linked traits; therefore, the genes encoding these traits were in close proximity on the same chromosome. The exchange of DNA between homologous pairs of chromosomes is called genetic recombination, which occurs by the crossing over of DNA between homologous strands of DNA, such as nonsister chromatids. Linkage analysis involves studying the recombination frequency between any two genes. The greater the distance between two genes, the higher the chance that a recombination event will occur between them, and the higher the recombination frequency between them. If the recombination frequency between two genes is less than 50 percent, they are said to be linked.

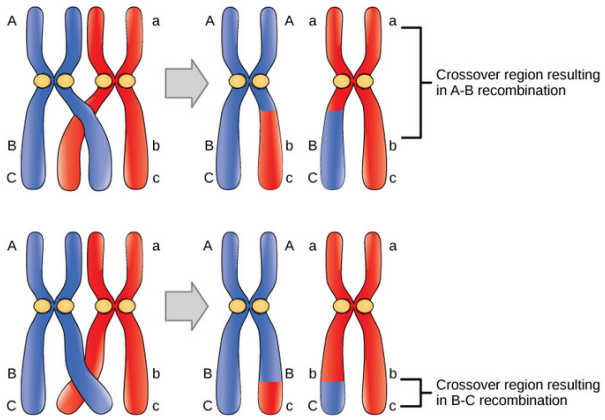

Figure 17.2*A*.1: **Crossovers and Recombination**: Crossover may occur at different locations on the chromosome. Recombination between genes A and B is more frequent than recombination between genes B and C because genes A and B are farther apart; a crossover is, therefore, more likely to occur between them.

The generation of genetic maps requires markers, just as a road map requires landmarks (such as rivers and mountains). Early genetic maps were based on the use of known genes as markers. More sophisticated markers, including those based on non-coding DNA, are now used to compare the genomes of individuals in a population. Although individuals of a given species are genetically similar, they are not identical; every individual has a unique set of traits. These minor differences in the genome between individuals in a population are useful for the purposes of genetic mapping. In general, a good genetic marker is a region on the chromosome that shows variability or polymorphism (multiple forms) in the population.

Some genetic markers used in generating genetic maps are restriction fragment length polymorphisms (RFLP), variable number of tandem repeats (VNTRs), microsatellite polymorphisms, and the single nucleotide polymorphisms (SNPs). RFLPs (sometimes pronounced “rif-lips”) are detected when the DNA of an individual is cut with a restriction endonuclease that recognizes specific sequences in the DNA to generate a series of DNA fragments, which are then analyzed by gel electrophoresis. The DNA of every individual will give rise to a unique pattern of bands when cut with a particular set of restriction endonucleases; this is sometimes referred to as an individual’s DNA “fingerprint.” Certain regions of the chromosome that are subject to polymorphism will lead to the generation of the unique banding pattern. VNTRs are repeated sets of nucleotides present in the non-coding regions of DNA. Non-coding DNA has no known biological function; however, research shows that much of this DNA is actually transcribed. While its function is uncertain, it is certainly active; it may be involved in the regulation of coding genes. The number of repeats may vary in individual organisms of a population. Microsatellite polymorphisms are similar to VNTRs, but the repeat unit is very small; thus, it is often referred to as short tandem repeats(STRs). SNPs are variations in a single nucleotide.

Because genetic maps rely completely on the natural process of recombination, mapping is affected by natural increases or decreases in the level of recombination in any given area of the genome. Some parts of the genome are recombination hotspots, whereas others do not show a propensity for recombination. For this reason, it is important to look at mapping information developed by multiple methods.

17.2B: Physical Maps and Integration with Genetic Maps

Physical maps display the physical distance between genes and can be constructed using cytogenetic, radiation hybrid, or sequence mapping.

Learning Objectives

- Describe the methods used to physically map genes: cytogenetic mapping, radiation hybrid mapping, and sequence mapping

Key Points

- Physical maps provide specified detail about the number of bases and physical distance that exists between genetic markers.
- Cytogenetic mapping is a method used to construct physical maps that uses stained sections of chromosomes to approximate the distance between genetic markers.
- Radiation hybrid mapping is a method used to construct physical maps that uses radiation or x-rays to break DNA into fragments to determine the distance between genetic markers and their order on the chromosome.
- Sequence mapping is a method used to construct physical maps that uses already-known locations of genetic markers to determine distances in number of base pairs.

Key Terms

- **cytogenetic**: of or pertaining to the origin and development of cells
- **physical map**: a map showing how much DNA separates two genes and is measured in base pairs
- **expressed sequence tag**: a short sub-sequence of a cDNA sequence that may be used to identify gene transcripts

Physical Maps

A physical map provides detail of the actual physical distance between genetic markers, as well as the number of nucleotides. There are three methods used to create a physical map: cytogenetic mapping, radiation hybrid mapping, and sequence mapping. Cytogenetic mapping uses information obtained by microscopic analysis of stained sections of the chromosome. It is possible to determine the approximate distance between genetic markers using cytogenetic mapping, but not the exact distance (number of base pairs). Radiation hybrid mapping uses radiation, such as x-rays, to break the DNA into fragments. The amount of radiation can be adjusted to create smaller or larger fragments. This technique overcomes the limitation of genetic mapping and is not affected by increased or decreased recombination frequency. Sequence mapping resulted from DNA sequencing technology that allowed for the creation of detailed physical maps with distances measured in terms of the number of base pairs. The creation of genomic libraries and complementary DNA (cDNA) libraries (collections of cloned sequences or all DNA from a genome) has sped up the process of physical mapping. A genetic site used to generate a physical map with sequencing technology (a sequence-tagged site, or STS) is a unique sequence in the genome with a known exact chromosomal location. An expressed sequence tag (EST) and a single sequence length polymorphism (SSLP) are common STSs. An EST is a short STS that is identified with cDNA libraries, while SSLPs are obtained from known genetic markers and provide a link between genetic maps and physical maps.

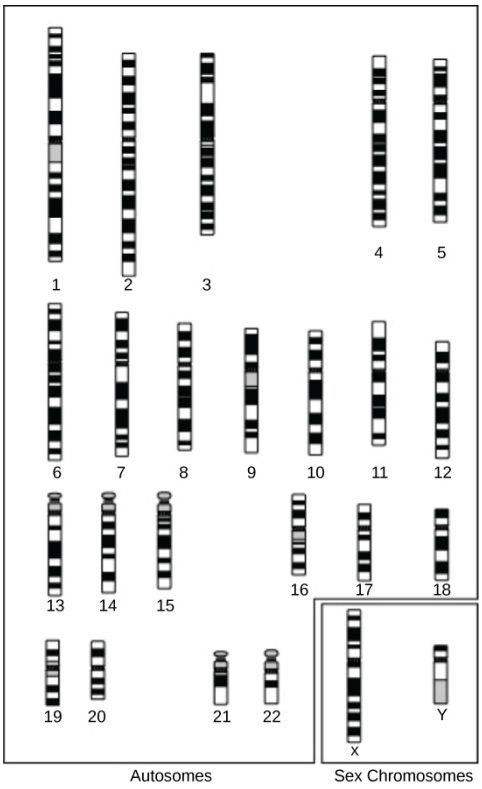

Figure 17.2B.1: **Cytogenetic Map**: A cytogenetic map shows the appearance of a chromosome after it is stained and examined under a microscope.

Integration of Genetic and Physical Maps

Genetic maps provide the outline and physical maps provide the details. It is easy to understand why both types of genome mapping techniques are important to show the big picture. Information obtained from each technique is used in combination to study the genome. Genomic mapping is being used with different model research organisms. Genome mapping is an-ongoing process; as better techniques are developed, more advances are expected. Genome mapping is similar to completing a complicated puzzle using every piece of available data. Mapping information generated in laboratories worldwide is entered into central databases, such as GenBank at the National Center for Biotechnology Information (NCBI). Efforts are being made to make the information more easily accessible to researchers and the general public. Just as we use global positioning systems instead of paper maps to navigate through roadways, NCBI has created a genome viewer tool to simplify the data-mining process.

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17.3: Whole-Genome Sequencing

Topic hierarchy

- 17.3A: Strategies Used in Sequencing Projects
- 17.3B: Use of Whole-Genome Sequences of Model Organisms
- 17.3C: Uses of Genome Sequences

17.3A: Strategies Used in Sequencing Projects

The strategies used for sequencing genomes include the Sanger method, shotgun sequencing, pairwise end, and next-generation sequencing.

Learning Objectives

- Compare the different strategies used for whole-genome sequencing: Sanger method, shotgun sequencing, pairwise-end sequencing, and next-generation sequencing

Key Points

- The Sanger method is a basic sequencing technique that uses fluorescently-labeled dideoxynucleotides (ddNTPs) during DNA replication which results in multiple short strands of replicated DNA that terminate at different points, based on where the ddNTP was incorporated.
- Shotgun sequencing is a method that randomly cuts DNA fragments into smaller pieces and then, with the help of a computer, takes the DNA fragments, analyzes them for overlapping sequences, and reassembles the entire DNA sequence.
- Pairwise-end sequencing is a type of shotgun sequencing which is used for larger genomes and analyzes both ends of the DNA fragments for overlap.
- Next-generation sequencing is a type of sequencing which is automated and relies on sophisticated software for rapid DNA sequencing.

Key Terms

- **fluorophore**: a molecule or functional group which is capable of fluorescence
- **contig**: a set of overlapping DNA segments, derived from a single source of genetic material, from which the complete sequence may be deduced
- **dideoxynucleotide**: any nucleotide formed from a deoxynucleotide by loss of an a second hydroxyl group from the deoxyribose group

Strategies Used in Sequencing Projects

The basic sequencing technique used in all modern day sequencing projects is the chain termination method (also known as the dideoxy method), which was developed by Fred Sanger in the 1970s. The chain termination method involves DNA replication of a single-stranded template with the use of a primer and a regular deoxynucleotide (dNTP), which is a monomer, or a single unit, of DNA. The primer and dNTP are mixed with a small proportion of fluorescently-labeled dideoxynucleotides (ddNTPs). The ddNTPs are monomers that are missing a hydroxyl group (–OH) at the site at which another nucleotide usually attaches to form a chain. Each ddNTP is labeled with a different color of fluorophore. Every time a ddNTP is incorporated in the growing complementary strand, it terminates the process of DNA replication, which results in multiple short strands of replicated DNA that are each terminated at a different point during replication. When the reaction mixture is processed by gel electrophoresis after being separated into single strands, the multiple, newly-replicated DNA strands form a ladder due to their differing sizes. Because the ddNTPs are fluorescently labeled, each band on the gel reflects the size of the DNA strand and the ddNTP that terminated the reaction. The different colors of the fluorophore-labeled ddNTPs help identify the ddNTP incorporated at that position. Reading the gel on the basis of the color of each band on the ladder produces the sequence of the template strand.

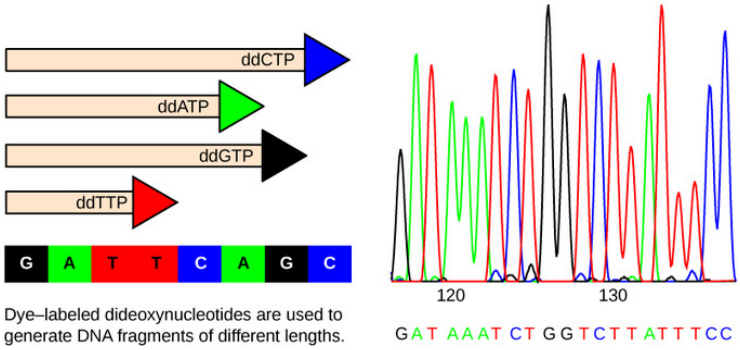
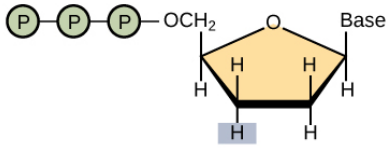
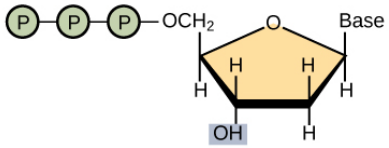


Figure 17.3A.1: **Sanger's Method**: Frederick Sanger's dideoxy chain termination method uses dideoxynucleotides, in which the DNA fragment can be terminated at different points. The DNA is separated on the basis of size, and these bands, based on the size of the fragments, can be read.



Dideoxynucleotide (ddNTP)



Deoxynucleotide (dNTP)

Figure 17.3A.1: **Structure of a Dideoxynucleotide**: A dideoxynucleotide is similar in structure to a deoxynucleotide, but is missing the 3′ hydroxyl group (indicated by the box). When a dideoxynucleotide is incorporated into a DNA strand, DNA synthesis stops.

Early Strategies: Shotgun Sequencing and Pair-Wise End Sequencing

In the shotgun sequencing method, several copies of a DNA fragment are cut randomly into many smaller pieces (somewhat like what happens to a round shot cartridge when fired from a shotgun). All of the segments are then sequenced using the chain-sequencing method. Then, with the help of a computer, the fragments are analyzed to see where their sequences overlap. By matching overlapping sequences at the end of each fragment, the entire DNA sequence can be reformed. A larger sequence that is assembled from overlapping shorter sequences is called a contig. As an analogy, consider that someone has four copies of a landscape photograph that you have never seen before and know nothing about how it should appear. The person then rips up each photograph with their hands, so that different size pieces are present from each copy. The person then mixes all of the pieces together and asks you to reconstruct the photograph. In one of the smaller pieces you see a mountain. In a larger piece, you see that the same mountain is behind a lake. A third fragment shows only the lake, but it reveals that there is a cabin on the shore of the lake. Therefore, from looking at the overlapping information in these three fragments, you know that the picture contains a mountain behind a lake that has a cabin on its shore. This is the principle behind reconstructing entire DNA sequences using shotgun sequencing.

Originally, shotgun sequencing only analyzed one end of each fragment for overlaps. This was sufficient for sequencing small genomes. However, the desire to sequence larger genomes, such as that of a human, led to the development of double-barrel shotgun sequencing, more formally known as pairwise-end sequencing. In pairwise-end sequencing, both ends of each fragment are analyzed for overlap. Pairwise-end sequencing is, therefore, more cumbersome than shotgun sequencing, but it is easier to reconstruct the sequence because there is more available information.

Next-generation Sequencing

Since 2005, automated sequencing techniques used by laboratories are under the umbrella of next-generation sequencing, which is a group of automated techniques used for rapid DNA sequencing. These automated, low-cost sequencers can

generate sequences of hundreds of thousands or millions of short fragments (25 to 500 base pairs) in the span of one day. Sophisticated software is used to manage the cumbersome process of putting all the fragments in order.

17.3B: Use of Whole-Genome Sequences of Model Organisms

Sequencing genomes of model organisms allows scientists to study homologous proteins in more complex eukaryotes, such as humans.

Learning Objectives

- Describe the model organisms used in whole-genome sequencing

Key Points

- The first genome to be completely sequenced was the bacterial virus, bacteriophage fx174, which is 5368 base pairs.
- Scientists utilize genome sequencing from model organisms to study homologous proteins and establish evolutionary relationships.
- Genome annotation is the process of attaching biological information to gene sequences identified using whole-genome sequencing.
- Model organisms include the fruit fly (*Drosophila melanogaster*), brewers yeast (*Saccharomyces cerevisiae*), the nematode, *Caenorhabditis elegans*, and the mouse (*Mus musculus*).

Key Terms

- **genome annotation**: the process of attaching biological information to gene sequences.
- **model organism**: any organism (e.g. the fruit fly) that has been extensively studied as an example of many others and from which general principles may be established

Use of Whole-Genome Sequences of Model Organisms

The first genome to be completely sequenced was of a bacterial virus, the bacteriophage fx174 (5368 base pairs). This was accomplished by Fred Sanger using shotgun sequencing. Several other organelle and viral genomes were later sequenced. The first organism whose genome was sequenced was the bacterium *Haemophilus influenzae*, which was accomplished by Craig Venter in the 1980s. Approximately 74 different laboratories collaborated on the sequencing of the genome of the yeast *Saccharomyces cerevisiae*, which began in 1989 and was completed in 1996. It took this long because it was 60 times bigger than any other genome that had been sequenced at that point. By 1997, the genome sequences of two important model organisms were available: the bacterium *Escherichia coli* K12 and the yeast *Saccharomyces cerevisiae*. Genomes of other model organisms, such as the mouse *Mus musculus*, the fruit fly *Drosophila melanogaster*, the nematode *Caenorhabditis elegans*, and the human *Homo sapiens* are now known. Much basic research is performed using model organisms because the information can be applied to the biological processes of genetically-similar organisms. Having entire genomes sequenced aids these research efforts.

The process of attaching biological information to gene sequences is called genome annotation. Annotation aids researchers doing basic experiments in molecular biology, such as designing PCR primers and RNA targets.

Sequencing genomes allows scientists to identify homologous proteins and establish evolutionary relationships. Furthermore, if a newly-discovered protein is homologous to a known protein, through homology, scientists can make an educated guess as to how the new protein functions.

Eukaryotes are organisms containing cells that enclose complex organelles within a well-defined cell membrane. The defining characteristic that sets eukaryotes and prokaryotes apart is the eukaryotes' nucleus, or nuclear envelope, in which an organism's genetic information is contained. The first eukaryotic genome to be sequenced was that of *S. cerevisiae*, which is the yeast used in baking and brewing. It is the most-studied eukaryotic model organism in molecular and cell biology, similar to *E. coli*'s role in the study of prokaryotic organisms. Research on many proteins that are important to humans is done by examining their homologs in yeasts. For example, signaling proteins and protein-processing enzymes were discovered through the help of yeast genome.

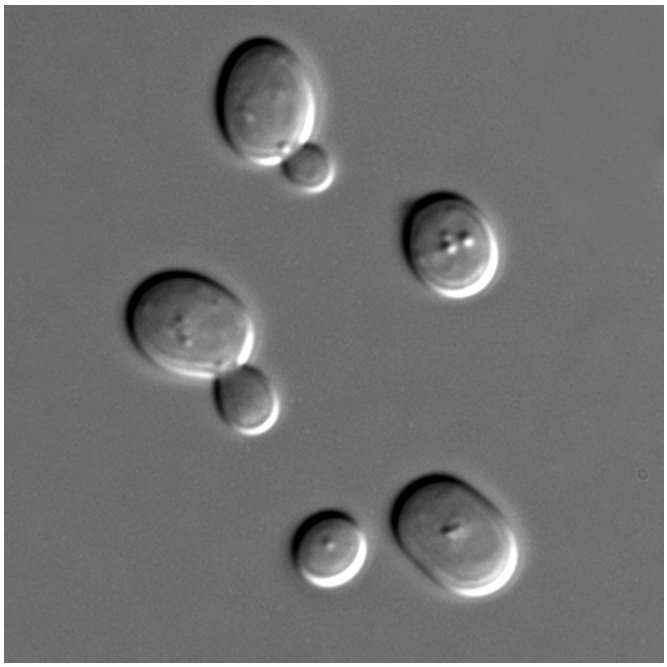


Figure 17.3B.1: **S. cerevisiae: A Model Organism for Eukaryotes**: Saccharomyces cerevisiae, a yeast, is used as a model organism for studying signaling proteins and protein-processing enzymes which have homologs in humans.

17.3C: Uses of Genome Sequences

Genome sequences and expression can be analyzed using DNA microarrays, which can contribute to detection of disease and genetic disorders.

Learning Objectives

- Describe the various uses of genome sequences

Key Points

- DNA microarrays can be used to detect gene expression within specific samples by analyzing active genes and sequences using an array of DNA fragments fixed to a slide.
- Genome sequences can be used to discover the possibility of disease and genetic disorders prior to onset.
- Genome sequences can also be used to develop agrochemicals and pharmaceuticals.

Key Terms

- **microarray**: any of several devices containing a two-dimensional array of small quantities of biological material used for various types of assays
- **genomics**: the study of the complete genome of an organism

Uses of Genome Sequences

DNA microarrays are methods used to detect gene expression by analyzing an array of DNA fragments that are fixed to a glass slide or a silicon chip to identify active genes and identify sequences. Almost one million genotypic abnormalities can be discovered using microarrays, whereas whole- genome sequencing can provide information about all six billion base pairs in the human genome. Although the study of medical applications of genome sequencing is interesting, this discipline tends to dwell on abnormal gene function. Knowledge of the entire genome will allow future onset diseases and other genetic disorders to be discovered early, which will allow for more informed decisions to be made about lifestyle, medication, and having children. Genomics is still in its infancy, although someday it may become routine to use whole-genome sequencing to screen every newborn to detect genetic abnormalities.

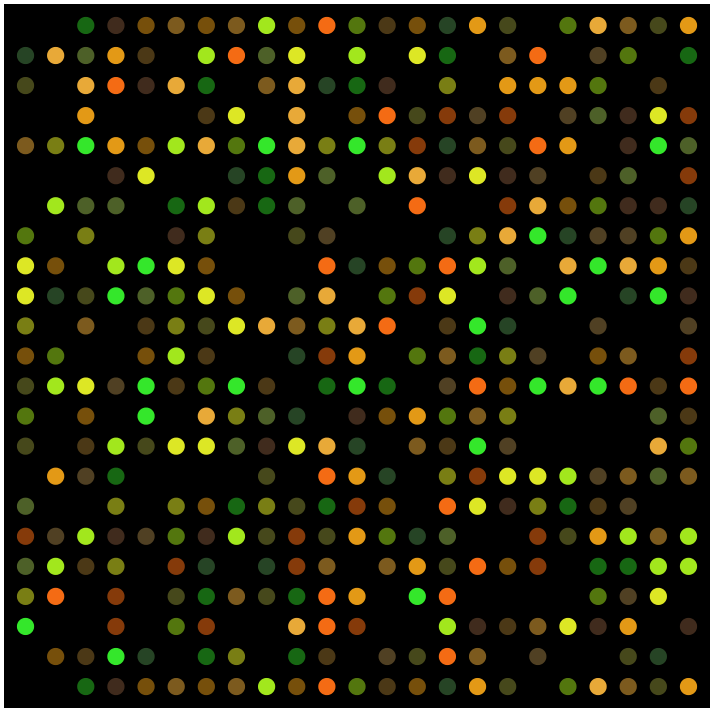


Figure 17.3C.1: **DNA Microarray**: DNA microarrays can be used to analyze gene expression within the genome.

In addition to disease and medicine, genomics can contribute to the development of novel enzymes that convert biomass to biofuel, which results in higher crop and fuel production, and lower cost to the consumer. This knowledge should allow better methods of control over the microbes that are used in the production of biofuels. Genomics could also improve the methods used to monitor the impact of pollutants on ecosystems and help clean up environmental contaminants. In addition, genomics has allowed for the development of agrochemicals and pharmaceuticals that could benefit medical science and agriculture.

It sounds great to have all the knowledge we can get from whole-genome sequencing; however, humans have a responsibility to use this knowledge wisely. Otherwise, it could be easy to misuse the power of such knowledge, leading to discrimination based on a person's genetics, human genetic engineering, and other ethical concerns. This information could also lead to legal issues regarding health and privacy.

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Topic hierarchy

- 17.4A: Predicting Disease Risk at the Individual Level
- 17.4B: Pharmacogenomics, Toxicogenomics, and Metagenomics

The effectiveness and safety of drugs can be determined through pharmacogenomics.

- 17.4C: Genomics and Biofuels

17.4A: Predicting Disease Risk at the Individual Level

Genome analysis is used to predict the level of disease risk in healthy individuals.

Learning Objectives

- Explain how analysis of an individual's genome can aid in predicting disease risk

Key Points

- Genome sequencing can predict the risk of developing diseases brought on by a single gene defect, but these defects only account for five percent of common diseases.
- Most diseases are polygenic or are brought on by environmental factors; genome sequencing cannot predict the risk of acquiring these diseases.
- Genome sequencing is becoming more reliable, but many scientists still question if it reduces the risk of death from certain diseases such as prostate cancer.

Key Terms

- **Human Genome Project**: an organized international scientific endeavor to determine the complete structure of human genetic material (DNA) to identify all the genes and understand their function
- **genome sequencing**: a laboratory process that determines the complete DNA sequence of an organism's genome at a single time
- **polygenic**: a phenotypic characteristic controlled by the interaction of two or more genes with the environment

Predicting Disease Risk at the Individual Level

The introduction of DNA sequencing and whole genome sequencing projects, particularly the Human Genome project, has expanded the applicability of DNA sequence information. Genomics is now being used in a wide variety of fields, such as metagenomics, pharmacogenomics, and mitochondrial genomics. The most commonly-known application of genomics is to understand and find cures for diseases.

Predicting the risk of disease involves screening currently-healthy individuals by genome analysis at the individual level. Intervention with lifestyle changes and drugs can be recommended before disease onset. However, this approach is most applicable when the problem resides within a single gene defect. Such defects only account for approximately five percent of diseases in developed countries. Most of the common diseases, such as heart disease, are multi-factored or polygenic, which is a phenotypic characteristic that involves two or more genes interacting with environmental factors such as diet. In April 2010, scientists at Stanford University published the genome analysis of a healthy individual (Stephen Quake, a scientist at Stanford University, who had his genome sequenced); the analysis predicted his propensity to acquire various diseases. A risk assessment was performed to analyze Quake's percentage of risk for 55 different medical conditions. A rare genetic mutation was found, which showed him to be at risk for sudden heart attack. He was also predicted to have a 23 percent risk of developing prostate cancer and a 1.4 percent risk of developing Alzheimer's. The scientists used databases and several publications to analyze the genomic data. Even though genomic sequencing is becoming more affordable and analytical tools are becoming more reliable, ethical issues surrounding genomic analysis at a population level remain to be addressed.

In 2011, the United States Preventative Services Task Force recommended against using the PSA test to screen healthy men for prostate cancer. Their recommendation was based on evidence that screening does not reduce the risk of death from prostate cancer. Prostate cancer often develops very slowly and does not cause problems, while the cancer treatment can have severe side effects. The PCA3 test is considered to be more accurate, but screening may still result in men suffering side effects from treatment who would not have been harmed by the cancer itself.

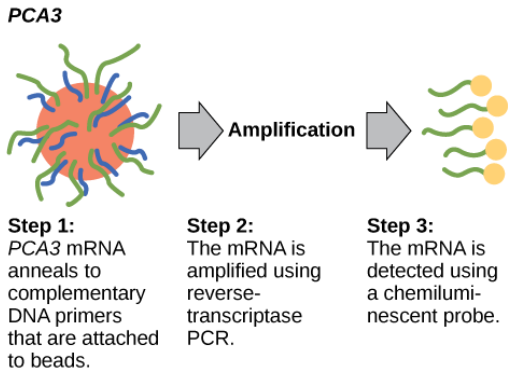

Figure 17.4A.1: **PCA3**: PCA3 is a gene that is expressed in prostate epithelial cells and overexpressed in cancerous cells. A high concentration of PCA3 in urine is indicative of prostate cancer.

17.4B: Pharmacogenomics, Toxicogenomics, and Metagenomics

LEARNING OBJECTIVES

- Explain how microbial metagenomics can give researchers a more complete picture of a particular environment

Pharmacogenomics, also called toxicogenomics, involves evaluating the effectiveness and safety of drugs on the basis of information from an individual's genomic sequence. Genomic responses to drugs can be studied using experimental animals (such as laboratory rats or mice) or live cells in the laboratory before embarking on studies with humans. Studying changes in gene expression could provide information about the transcription profile in the presence of the drug, which can be used as an early indicator of the potential for toxic effects. For example, genes involved in cellular growth and controlled cell death, when disturbed, could lead to the growth of cancerous cells. Genome-wide studies can also help to find new genes involved in drug toxicity. Personal genome sequence information can be used to prescribe medications that will be most effective and least toxic on the basis of the individual patient's genotype. The gene signatures may not be completely accurate, but can be tested further before pathologic symptoms arise.

Microbial Genomics: Metagenomics

Traditionally, microbiology has been taught with the view that microorganisms are best studied under pure culture conditions, which involves isolating a single type of cell and culturing it in the laboratory. Because microorganisms can go through several generations in a matter of hours, their gene expression profiles adapt to the new laboratory environment very quickly. In addition, the vast majority of bacterial species resist being cultured in isolation. Most microorganisms do not live as isolated entities, but in microbial communities known as biofilms. For all of these reasons, pure culture is not always the best way to study microorganisms. Metagenomics is the study of the collective genomes of multiple species that grow and interact in an environmental niche. Metagenomics can be used to identify new species more rapidly and to analyze the effect of pollutants on the environment.

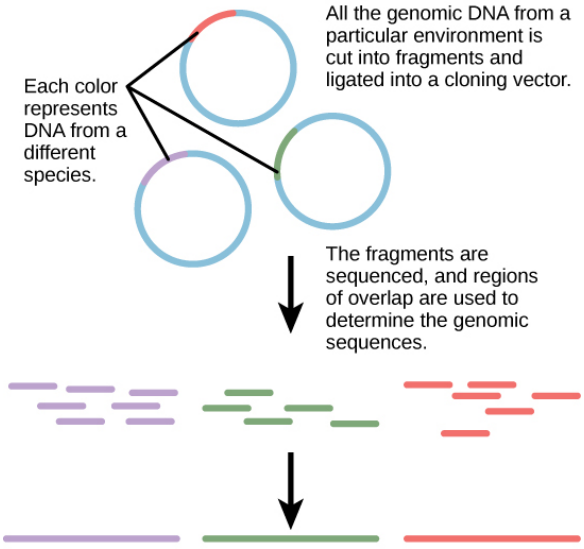

Figure 17.4B.1: **Metagenomics**: Metagenomics involves isolating DNA from multiple species within an environmental niche.

Key Points

- Pharmacogenomics involves evaluating the effectiveness and safety of drugs on the basis of information from an individual's genomic sequence.
- Genomic responses to drugs can be studied using experimental animals or live cells in the laboratory, which help indicate the potentially toxic effects of a drug.
- Personal genome sequence information can be used to prescribe medications that will be most effective and least toxic on an individual level.

- Metagenomics, the study of the collective genomes of multiple species that grow and interact in an environmental niche, is often a better way to study microrganisms rather than pure culture.

Key Terms

- **pharmacogenomics**: the study of genes that code for enzymes that metabolize drugs, and the design of tailor-made drugs adapted to an individual's genetic make-up
- **metagenomics**: the study of the collective genomes of multiple species that grow and interact in an environmental niche

17.4C: Genomics and Biofuels

Microbial genomics can be used to create new biofuels.

Learning Objectives

- Explain the process of creating new biofuels by using microbial genomics

Key Points

- Microorganisms can encode new enzymes and produce new organic compounds that can be used as biofuels.
- Genomic analysis of the fungus *Pichia* will allow optimization of its use in fermenting ethanol fuels.
- Analysis of the microbes in the hindgut of termites have found 500 genes that may be useful in enzymatic destruction of cellulose.
- Genetic markers have been used in forensic analysis, like in 2001 when the FBI used microbial genomics to determine a specific strain of anthrax that was found in several pieces of mail.
- Genomics is used in agriculture to develop plants with more desirable traits, such as drought and disease resistance.

Key Terms

- **renewable resource**: a natural resource such that it is replenished by natural processes at a rate comparable to its rate of consumption by humans or other users
- **biofuel**: any fuel that is obtained from a renewable biological resource

Knowledge of the genomics of microorganisms is being used to find better ways to harness biofuels from algae and cyanobacteria. The primary sources of fuel today are coal, oil, wood, and other plant products, such as ethanol. Although plants are renewable resources, there is still a need to find more alternative renewable sources of energy to meet our population ‘s energy demands. The microbial world is one of the largest resources for genes that encode new enzymes and produce new organic compounds, and it remains largely untapped.

For microbial biomass breakdown, many candidates have already been identified. These include Clostridia species for their ability to degrade cellulose, and fungi that express genes associated with the decomposition of the most recalcitrant features of the plant cell wall, lignin, the phenolic “glue” that imbues the plant with structural integrity and pest resistance. The white rot fungus *Phanerochaete chrysosporium* produces unique extracellular oxidative enzymes that effectively degrade lignin by gaining access through the protective matrix surrounding the cellulose microfibrils of plant cell walls.

Another fungus, the yeast *Pichia stipitis*, ferments the five-carbon “wood sugar” xylose abundant in hardwoods and agricultural harvest residue. *Pichia*‘s recently-sequenced genome has revealed insights into the metabolic pathways responsible for this process, guiding efforts to optimize this capability in commercial production strains. Pathway engineering promises to produce a wider variety of organisms able to ferment the full repertoire of sugars derived from cellulose and hemicellulose and tolerate higher ethanol concentrations to optimize fuel yields. For instance, the hindgut contents of nature’s own bioreactor, the termite, has yielded more than 500 genes related to the enzymatic deconstruction of cellulose and hemicellulose.

Figure 17.4C.1: **Termites: Nature's Bioreactors**: The hindgut of the termite has yielded more than 500 genes of microbes related to the enzymatic deconstruction of cellulose.

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17.5: Genomics and Proteomics

Topic hierarchy

- 17.5A: Genomics and Proteomics
- 17.5B: Basic Techniques in Protein Analysis
- 17.5C: Cancer Proteomics

Proteomics, the analysis of proteins, plays a prominent role in the study and treatment of cancer.

17.5A: Genomics and Proteomics

Proteomics is the study of the entire set of proteins produced by a cell type in order to understand its structure and function.

Learning Objectives

- Explain how the field of genomics led to the development of proteomics

Key Points

- Proteomics investigates how proteins affect and are affected by cell processes or the external environment.
- Within an individual organism, the genome is constant, but the proteome varies and is dynamic.
- Every cell in an individual organism has the same set of genes, but the set of proteins produced in different tissues differ from one another and are dependent on gene expression.

Key Terms

- **proteomics**: the branch of molecular biology that studies the set of proteins expressed by the genome of an organism
- **proteome**: the complete set of proteins encoded by a particular genome
- **genomics**: the study of the complete genome of an organism

Proteomics is a relatively-recent field; the term was coined in 1994 while the science itself had its origins in electrophoresis techniques of the 1970's and 1980's. The study of proteins, however, has been a scientific focus for a much longer time. Studying proteins generates insight into how they affect cell processes. Conversely, this study also investigates how proteins themselves are affected by cell processes or the external environment. Proteins provide intricate control of cellular machinery; they are, in many cases, components of that same machinery. They serve a variety of functions within the cell; there are thousands of distinct proteins and peptides in almost every organism. The goal of proteomics is to analyze the varying proteomes of an organism at different times in order to highlight differences between them. Put more simply, proteomics analyzes the structure and function of biological systems. For example, the protein content of a cancerous cell is often different from that of a healthy cell. Certain proteins in the cancerous cell may not be present in the healthy cell, making these unique proteins good targets for anti-cancer drugs. The realization of this goal is difficult; both purification and identification of proteins in any organism can be hindered by a multitude of biological and environmental factors.

The study of the function of proteomes is called proteomics. A proteome is the entire set of proteins produced by a cell type. Genomics led to proteomics (via transcriptomics) as a logical step. Proteomes can be studied using the knowledge of genomes because genes code for mRNAs and the mRNAs encode proteins. Although mRNA analysis is a step in the right direction, not all mRNAs are translated into proteins. Proteomics complements genomics and is useful when scientists want to test their hypotheses that were based on genes. Even though all cells of a multicellular organism have the same set of genes, the set of proteins produced in different tissues is different and dependent on gene expression. Thus, the genome is constant, but the proteome varies and is dynamic within an organism. In addition, RNAs can be alternately spliced (cut and pasted to create novel combinations and novel proteins) and many proteins are modified after translation by processes such as proteolytic cleavage, phosphorylation, glycosylation, and ubiquitination. There are also protein-protein interactions, which complicate the study of proteomes. Although the genome provides a blueprint, the final architecture depends on several factors that can change the progression of events that generate the proteome.

Figure 17.5A. 1: **Large-scale proteomics machinery**: This machine is preparing to do a proteomic pattern analysis to identify specific cancers so that an accurate cancer prognosis can be made.

17.5B: Basic Techniques in Protein Analysis

The basic techniques used to analyze proteins are mass spectrometry, x-ray crystallography, NMR, and protein microarrays.

Learning Objectives

- Describe the techniques used in proteomics to analyze proteins

Key Points

- Mass Spectrometry is a technique that is useful for determining the size of a protein or protein complex.
- X-ray crystallography and NMR are techniques useful for determining the 3-D structure of a protein or protein complex.
- Protein microarrays are useful for determining protein-protein interactions.

Key Terms

- **microarray**: any of several devices containing a two-dimensional array of small quantities of biological material used for various types of assays
- **reporter gene**: a gene that researchers attach to a regulatory sequence of another gene of interest and whose product is easily identifiable in assays

Basic Techniques in Protein Analysis

The ultimate goal of proteomics is to identify or compare the proteins expressed in a given genome under specific conditions, study the interactions between the proteins, and use the information to predict cell behavior or develop drug targets. Just as the genome is analyzed using the basic technique of DNA sequencing, proteomics requires techniques for protein analysis. The basic technique for protein analysis, analogous to DNA sequencing, is mass spectrometry.

Figure $17.5B.1$: **Mass Spectrometer**: Matrix-Assisted Laser Desorbtion Ionisation – Time Of Flight (MALDI-TOF) Mass Spectrometer. Mass spectrometry can be used in protein analysis.

Mass Spectrometry

Mass spectrometry is used to identify and determine the characteristics of a molecule. It is a technique in which gas phase molecules are ionized and their mass-to-charge ratio is measured by observing acceleration differences of ions when an electric field is applied. Lighter ions will accelerate faster and be detected first. If the mass is measured with precision, then the composition of the molecule can be identified. In the case of proteins, the sequence can be identified. The challenge of techniques used for proteomic analyses is the difficulty in detecting small quantities of proteins, but advances in spectrometry have allowed researchers to analyze very small samples of protein. Variations in protein expression in diseased states, however, can be difficult to discern. Proteins are naturally-unstable molecules, which makes proteomic analysis much more difficult than genomic analysis.

X-ray crystallography and Nuclear Magnetic Resonance

X-ray crystallography enables scientists to determine the three-dimensional structure of a protein crystal at atomic resolution. Crystallographers aim high-powered X-rays at a tiny crystal containing trillions of identical molecules. The crystal scatters the X-rays onto an electronic detector that is the same type used to capture images in a digital camera. After each blast of X-rays, lasting from a few seconds to several hours, the researchers precisely rotate the crystal by entering its desired orientation into the computer that controls the X-ray apparatus. This enables the scientists to capture in three dimensions how the crystal scatters, or diffracts, X-rays. The intensity of each diffracted ray is fed into a computer, which uses a mathematical equation to calculate the position of every atom in the crystallized molecule. The result is a three-dimensional digital image of the molecule.

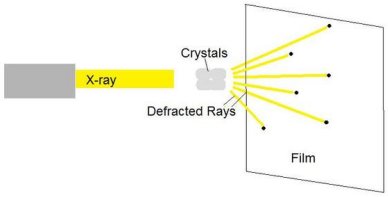

Figure $17.5B.1$: **X-ray crystallography**: X-rays that hit atomic nuclei are diffracted onto a detector.

Another protein imaging technique, nuclear magnetic resonance (NMR), uses the magnetic properties of atoms to determine the three-dimensional structure of proteins. NMR spectroscopy is unique in being able to reveal the atomic structure of macromolecules in solution, provided that highly-concentrated solution can be obtained. This technique depends on the fact that certain atomic nuclei are intrinsically magnetic. The chemical shift of nuclei depends on their local environment. The spins of neighboring nuclei interact with each other in ways that provide definitive structural information that can be used to determine complete three-dimensional structures of proteins.

Protein Microarrays and Two- Hybrid Screening

Protein microarrays have also been used to study interactions between proteins. These are large-scale adaptations of the basic two-hybrid screen. The premise behind the two-hybrid screen is that most eukaryotic transcription factors have modular activating and binding domains that can still activate transcription even when split into two separate fragments, as long as the fragments are brought within close proximity to each other. Generally, the transcription factor is split into a DNA-binding domain (BD) and an activation domain (AD). One protein of interest is genetically fused to the BD and another protein is fused to the AD. If the two proteins of interest bind each other, then the BD and AD will also come together and activate a reporter gene that signals interaction of the two hybrid proteins.

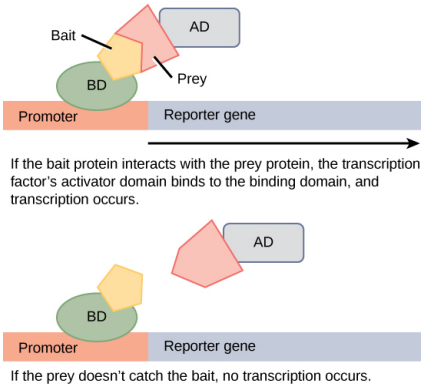

Figure $17.5B.1$: **Two-hybrid screening**: Two-hybrid screening is used to determine whether two proteins interact. In this method, a transcription factor is split into a DNA-binding domain (BD) and an activation domain (AD). The binding domain is able to bind the promoter in the absence of the activator domain, but it does not turn on transcription. A protein called the bait is attached to the BD, and a protein called the prey is attached to the AD. Transcription occurs only if the prey "catches" the bait.

Western Blot

The western blot, or protein immunoblot, is a technique that combines protein electrophoresis and antibodies to detect proteins in a sample. A western blot is fairly quick and simple compared to the above techniques and, thus, can serve as an assay to validate results from other experiments. The protein sample is first separated by gel electrophoresis, then transferred to a nitrocellulose or other type of membrane, and finally stained with a primary antibody that specifically binds the protein of interest. A fluorescent or radioactive-labeled secondary antibody binds to the primary antibody and provides a means of detection via either photography or x-ray film, respectively.

17.5C: Cancer Proteomics

LEARNING OBJECTIVES

- Explain the ways in which cancer proteomics may lead to better treatments

Genomes and proteomes of patients suffering from specific diseases are being studied to understand the genetic basis of diseases. The most prominent set of diseases being studied with proteomic approaches is cancer. Proteomic approaches are being used to improve screening and early detection of cancer, which is achieved by identifying proteins whose expression is affected by the disease process.

An individual protein that indicates disease is called a biomarker, whereas a set of proteins with altered expression levels is called a protein signature. For a biomarker or protein signature to be useful as a candidate for early screening and detection of a cancer, it must be secreted in body fluids (e.g. sweat, blood, or urine) such that large-scale screenings can be performed in a non-invasive fashion. The current problem with using biomarkers for the early detection of cancer is the high rate of false-negative results. A false-negative is an incorrect test result that should have been positive. In other words, many cases of cancer go undetected, which makes biomarkers unreliable. Some examples of protein biomarkers used in cancer detection are CA-125 for ovarian cancer and PSA for prostate cancer. Protein signatures may be more reliable than biomarkers to detect cancer cells.

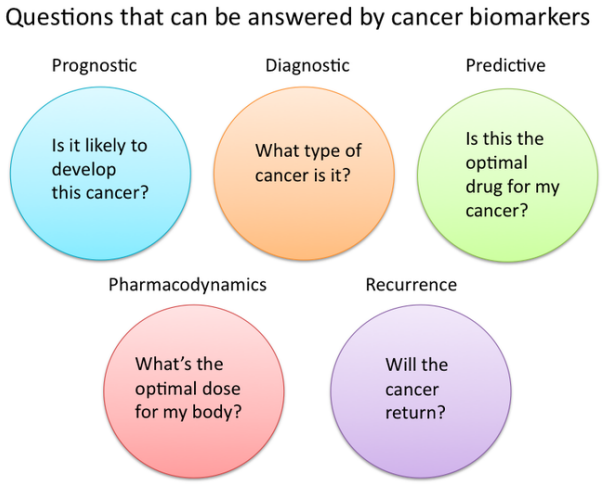

Figure 17.5C.1: **Questions that can be answered by biomarkers**: In cancer research and medicine, biomarkers are used in three primary ways: (A) Diagnostic – To help diagnose conditions, as in the case of identifying early stage cancers. (B) Prognostic – To forecast how aggressive a condition is, as in the case of determining a patient's ability to fare in the absence of treatment. (C) Predictive – To predict how well a patient will respond to treatment.

Proteomics is also being used to develop individualized treatment plans, which involves the prediction of whether or not an individual will respond to specific drugs and the side effects that the individual may experience. In addition, proteomics can be used to predict the possibility of disease recurrence. The National Cancer Institute has developed programs to improve the detection and treatment of cancer. The Clinical Proteomic Technologies for Cancer and the Early Detection Research Network are efforts to identify protein signatures specific to different types of cancers. The Biomedical Proteomics Program is designed to identify protein signatures and design effective therapies for cancer patients.

Key Points

- Identifying those proteins whose expression is affected by disease processes can be used to improve screening and early detection of cancer.
- Different biomarkers and protein signatures are being used to analyze each type of cancer.
- A future goal of cancer proteomics is to have a personalized treatment plan for each individual.

Key Terms

- **biomarker**: a substance used as an indicator of a biological state, most commonly disease

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CHAPTER OVERVIEW

18: EVOLUTION AND THE ORIGIN OF SPECIES

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18.1A: WHAT IS EVOLUTION?
18.1B: CHARLES DARWIN AND NATURAL SELECTION
18.1C: THE GALAPAGOS FINCHES AND NATURAL SELECTION
The differences in shape and size of beaks in Darwin's finches illustrate ongoing evolutionary change.

18.1D: PROCESSES AND PATTERNS OF EVOLUTION
18.1E: EVIDENCE OF EVOLUTION
18.1F: MISCONCEPTIONS OF EVOLUTION
18.2: FORMATION OF NEW SPECIES

18.2A: THE BIOLOGICAL SPECIES CONCEPT
18.2B: REPRODUCTIVE ISOLATION
18.2C: SPECIATION
18.2D: ALLOPATRIC SPECIATION
18.2E: SYMPATRIC SPECIATION
18.3: HYBRID ZONES AND RATES OF SPECIATION

18.3A: HYBRID ZONES
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18.4: EVOLUTION OF GENOMES

18.4A: GENOMIC SIMILIARITIES BETWEEN DISTANT SPECIES
18.4B: GENOME EVOLUTION
18.4C: WHOLE-GENOME DUPLICATION
18.4D: GENE DUPLICATIONS AND DIVERGENCE
Gene duplications create genetic redudancy and can have various effects, including detrimental mutations or divergent evolution.

18.4E: NONCODING DNA
18.4F: VARIATIONS IN SIZE AND NUMBER OF GENES
18.5: EVIDENCE OF EVOLUTION

18.5A: THE FOSSIL RECORD AS EVIDENCE FOR EVOLUTION
18.5B: FOSSIL FORMATION
18.5C: GAPS IN THE FOSSIL RECORD
18.5D: CARBON DATING AND ESTIMATING FOSSIL AGE
18.5E: THE FOSSIL RECORD AND THE EVOLUTION OF THE MODERN HORSE
The detailed fossil record of horses has provided insight into their evolutionary progress.

18.5F: HOMOLOGOUS STRUCTURES
18.5G: CONVERGENT EVOLUTION
18.5H: VESTIGIAL STRUCTURES
18.5I: BIOGEOGRAPHY AND THE DISTRIBUTION OF SPECIES

18.1: Understanding Evolution

Topic hierarchy

- 18.1A: What is Evolution?
- 18.1B: Charles Darwin and Natural Selection
- 18.1C: The Galapagos Finches and Natural Selection

The differences in shape and size of beaks in Darwin’s finches illustrate ongoing evolutionary change.

- 18.1D: Processes and Patterns of Evolution
- 18.1E: Evidence of Evolution
- 18.1F: Misconceptions of Evolution

18.1A: What is Evolution?

Evolution, the unifying theory of biology, describes a mechanism for the change and diversification of species over time.

Learning Objectives

- Describe the historical influences on Darwin's theory of evolution

Key Points

- Ancient Greeks expressed ideas about evolution, which were reintroduced in the eighteenth century by Georges-Louis Leclerc Comte de Buffon who observed different environments had different plant and animal populations.
- James Hutton proposed that geological changes occur gradually over time via the accumulation of small changes rather than through large catastrophic events.
- Charles Lyell popularized James Hutton's theory; this theory of incremental change influenced Darwin's theory of evolution.
- Jean-Baptiste Lamarck proposed the theory of the inheritance of acquired characterstics; this theory has now been discredited, but it served as an important influence on the theory of evolution.

Key Terms

- **evolution**: the change in the genetic composition of a population over successive generations
- **inheritance of acquired characteristics**: hypothesis that physiological changes acquired over the life of an organism may be transmitted to its offspring

Introduction: Evolution

All species of living organisms, including bacteria and chimpanzees, evolved at some point from a different species. Although it may seem that living things today stay the same, this is not the case: evolution is a gradual and ongoing process.

The theory of evolution is the unifying theory of biology, meaning it is the framework within which biologists ask questions about the living world. The Ukrainian-born American geneticist Theodosius Dobzhansky famously wrote that "nothing makes sense in biology except in the light of evolution." The tenet that all species have evolved and diversified from a common ancestor is the foundation from which we approach all questions in biology. It provides a direction for predictions about living things, which has been validated through extensive scientific experimentation.

Evolution by natural selection describes a mechanism for the change of species over time. Well before Darwin began to explore the concept of evolution, the idea that species change over time had already been suggested and debated. The view that species are static and unchanging was grounded in the writings of Plato, yet there were also ancient Greeks who expressed ideas about evolution. During the eighteenth century, ideas about the evolution of animals were reintroduced by the naturalist Georges-Louis Leclerc Comte de Buffon who observed that various geographic regions have different plant and animal populations, even when the environments are similar. It was also accepted that there are extinct species.

Figure 18.1A.1: **Evolution by Natural Selection**: All organisms are products of evolution adapted to their environment. (a) Saguaro (Carnegiea gigantea) can soak up 750 liters of water in a single rain storm, enabling these cacti to survive the dry conditions of the Sonoran desert in Mexico and the Southwestern United States. (b) The Andean semiaquatic lizard (Potamites montanicola), discovered in Peru in 2010, lives between 1,570 to 2,100 meters in elevation and, unlike most lizards, is nocturnal and swims. Scientists still do not know how these cold-blood animals are able to move in the cold (10 to 15°C) temperatures of the Andean night.

During this time, a Scottish naturalist named James Hutton proposed that geological change occurs gradually by the accumulation of small changes over long periods of time. This theory contrasted with the predominant view of the time: that the geology of the planet is a consequence of catastrophic events that occurred during a relatively brief past. During the nineteenth century, Hutton's views were popularized by the geologist Charles Lyell, who was a friend of Charles Darwin. Lyell's ideas, in turn, influenced Darwin's concept of evolution. The greater age of the earth proposed by Lyell supported the gradual evolution that Darwin proposed, and the slow process of geological change provided an analogy for the gradual change in species.

In the early nineteenth century, Jean-Baptiste Lamarck published a book that detailed a different mechanism for evolutionary change. This mechanism is now referred to as an inheritance of acquired characteristics. This idea states that modifications in an individual are caused by its environment, or the use or disuse of a structure during its lifetime, and that these changes can be inherited by its offspring, bringing about change in a species. While this mechanism for evolutionary change was discredited, Lamarck's ideas were an important influence on the concept of evolution.

18.1B: Charles Darwin and Natural Selection

Charles Darwin and Alfred Wallace independently developed the theories of evolution and its main operating principle: natural selection.

Learning Objectives

- Explain how natural selection can lead to evolution

Key Points

- Wallace traveled to Brazil to collect and observe insects from the Amazon rainforest.
- Darwin observed that finches in the Galápagos Islands had different beaks than finches in South America; these adaptations equiped the birds to acquire specific food sources.
- Wallace and Darwin observed similar patterns in the variation of organisms and independently developed the same explanation for how such variations could occur over time, a mechanism Darwin called natural selection.
- According to natural selection, also known as “survival of the fittest,” individuals with traits that enable them to survive are more reproductively successful; this leads to those traits becoming predominant within a population.
- Natural selection is an inevitable outcome of three principles: most characteristics are inherited, more offspring are produced than are able to survive, and offspring with more favorable characteristics will survive and have more offspring than those individuals with less favorable traits.

Key Terms

- **natural selection**: a process in which individual organisms or phenotypes that possess favorable traits are more likely to survive and reproduce
- **descent with modification**: change in populations over generations

Charles Darwin and Natural Selection

In the mid-nineteenth century, the mechanism for evolution was independently conceived of and described by two naturalists: Charles Darwin and Alfred Russel Wallace. Importantly, each naturalist spent time exploring the natural world on expeditions to the tropics. From 1831 to 1836, Darwin traveled around the world to places like South America, Australia, and the southern tip of Africa. Wallace traveled to Brazil to collect insects in the Amazon rainforest from 1848 to 1852 and to the Malay Archipelago from 1854 to 1862. Darwin’s journey, as with Wallace’s later journeys to the Malay Archipelago, included stops at several island chains, the last being the Galápagos Islands west of Ecuador. On these islands, Darwin observed that species of organisms on different islands were clearly similar, yet had distinct differences. For example, the ground finches inhabiting the Galápagos Islands comprised several species with a unique beak shape. The species on the islands had a graded series of beak sizes and shapes with very small differences between the most similar. He observed that these finches closely resembled another finch species on the mainland of South America. Darwin imagined that the island species might be modified from one of the original mainland species. Upon further study, he realized that the varied beaks of each finch helped the birds acquire a specific type of food. For example, seed-eating finches had stronger, thicker beaks for breaking seeds, while insect-eating finches had spear-like beaks for stabbing their prey.

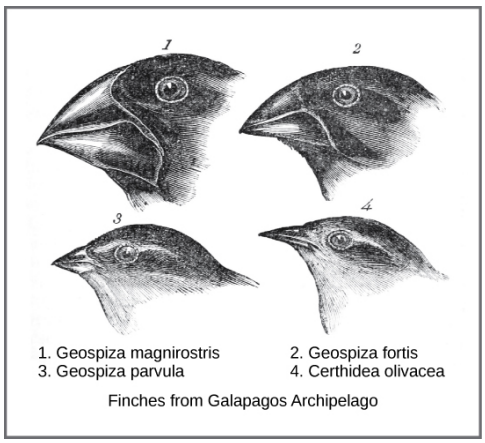


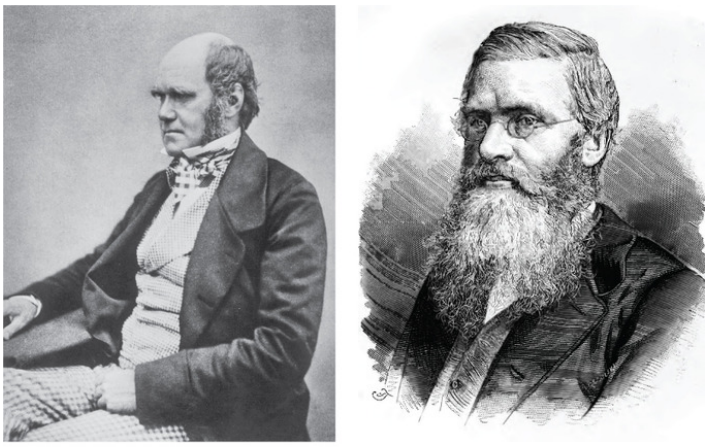
Figure 18.1B.1: **Beak Shape Among Finch Species**: Darwin observed that beak shape varies among finch species. He postulated that the beak of an ancestral species had adapted over time to equip the finches to acquire different food sources.

Natural Selection

Wallace and Darwin observed similar patterns in other organisms and independently developed the same explanation for how and why such changes could take place. Darwin called this mechanism natural selection. Natural selection, also known as "survival of the fittest," is the more prolific reproduction of individuals with favorable traits that survive environmental change because of those traits. This leads to evolutionary change, the trait becoming predominant within a population. For example, Darwin observed that a population of giant tortoises found in the Galapagos Archipelago have longer necks than those that lived on other islands with dry lowlands. These tortoises were "selected" because they could reach more leaves and access more food than those with short necks. In times of drought, when fewer leaves would be available, those that could reach more leaves had a better chance to eat and survive than those that could not reach the food source. Consequently, long-necked tortoises would more probably be reproductively successful and pass the long-necked trait to their offspring. Over time, only long-necked tortoises would be present in the population.

Natural selection, Darwin argued, was an inevitable outcome of three principles that operated in nature. First, most characteristics of organisms are inherited, or passed from parent to offspring, although *how* traits were inherited was unknown. Second, more offspring are produced than are able to survive. The capacity for reproduction in all organisms outstrips the availability of resources to support their numbers. Thus, there is competition for those resources in each generation. Both Darwin and Wallace were influenced by an essay written by economist Thomas Malthus who discussed this principle in relation to human populations. Third, Darwin and Wallace reasoned that offspring with the inherited characteristics that allow them to best compete for limited resources will survive and have more offspring than those individuals with variations that are less able to compete. Because characteristics are inherited, these traits will be better represented in the next generation. This will lead to change in populations over successive generations in a process that Darwin called descent with modification. Ultimately, natural selection leads to greater adaptation of the population to its local environment; it is the only mechanism known for adaptive evolution.

Papers by Darwin and Wallace presenting the idea of natural selection were read together in 1858 before the Linnean Society in London. The following year, Darwin's book, *On the Origin of Species*, was published. His book outlined his arguments for evolution by natural selection.



(a) (b)

Figure 18.1$B.1$: **Charles Darwin and Alfred Wallace**: Both (a) Charles Darwin and (b) Alfred Wallace wrote scientific papers on natural selection that were presented together before the Linnean Society in 1858.

18.1C: The Galapagos Finches and Natural Selection

The differences in shape and size of beaks in Darwin's finches illustrate ongoing evolutionary change.

Learning Objectives

- Describe how finches provide visible evidence of evolution

Key Points

- Darwin observed the Galapagos finches had a graded series of beak sizes and shapes and predicted these species were modified from one original mainland species.
- Darwin called differences among species natural selection, which is caused by the inheritance of traits, competition between individuals, and the variation of traits.
- Offspring with inherited characteristics that allow them to best compete will survive and have more offspring than those individuals with variations that are less able to compete.
- Large-billed finches feed more efficiently on large, hard seeds, whereas smaller billed finches feed more efficiently on small, soft seeds.
- When small, soft seeds become rare, large-billed finches will survive better, and there will be more larger-billed birds in the following generation; when large, hard seeds become rare, the opposite will occur.

Key Terms

- **natural selection**: a process in which individual organisms or phenotypes that possess favorable traits are more likely to survive and reproduce
- **evolution**: the change in the genetic composition of a population over successive generations

Visible Evidence of Ongoing Evolution: Darwin's Finches

From 1831 to 1836, Darwin traveled around the world, observing animals on different continents and islands. On the Galapagos Islands, Darwin observed several species of finches with unique beak shapes. He observed these finches closely resembled another finch species on the mainland of South America and that the group of species in the Galápagos formed a graded series of beak sizes and shapes, with very small differences between the most similar. Darwin imagined that the island species might be all species modified from one original mainland species. In 1860, he wrote, "seeing this gradation and diversity of structure in one small, intimately related group of birds, one might really fancy that from an original paucity of birds in this archipelago, one species had been taken and modified for different ends."

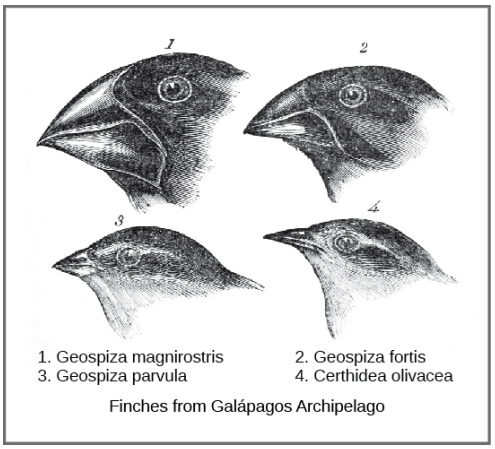


Figure 18.1C.1: **Darwin's Finches**: Darwin observed that beak shape varies among finch species. He postulated that the beak of an ancestral species had adapted over time to equip the finches to acquire different food sources. This illustration shows the beak shapes for four species of ground finch: 1. *Geospiza magnirostris* (the large ground finch), 2. *G. fortis* (the medium ground finch), 3. *G. parvula* (the small tree finch), and 4. *Certhidea olivacea* (the green-warbler finch).

Natural Selection

Darwin called this mechanism of change natural selection. Natural selection, Darwin argued, was an inevitable outcome of three principles that operated in nature. First, the characteristics of organisms are inherited, or passed from parent to offspring. Second, more offspring are produced than are able to survive; in other words, resources for survival and reproduction are limited. The capacity for reproduction in all organisms exceeds the availability of resources to support their numbers. Thus, there is a competition for those resources in each generation. Third, offspring vary among each other in regard to their characteristics and those variations are inherited. Out of these three principles, Darwin reasoned that offspring with inherited characteristics that allow them to best compete for limited resources will survive and have more offspring than those individuals with variations that are less able to compete. Because characteristics are inherited, these traits will be better represented in the next generation. This will lead to change in populations over generations in a process that Darwin called “descent with modification,” or evolution.

Studies of Natural Selection After Darwin

Demonstrations of evolution by natural selection can be time consuming. Peter and Rosemary Grant and their colleagues have studied Galápagos finch populations every year since 1976 and have provided important demonstrations of the operation of natural selection. The Grants found changes from one generation to the next in the beak shapes of the medium ground finches on the Galápagos island of Daphne Major.

The medium ground finch feeds on seeds. The birds have inherited variation in the bill shape with some individuals having wide, deep bills and others having thinner bills. Large-billed birds feed more efficiently on large, hard seeds, whereas smaller billed birds feed more efficiently on small, soft seeds. During 1977, a drought period altered vegetation on the island. After this period, the number of seeds declined dramatically; the decline in small, soft seeds was greater than the decline in large, hard seeds. The large-billed birds were able to survive better than the small-billed birds the following year.

The year following the drought when the Grants measured beak sizes in the much-reduced population, they found that the average bill size was larger. This was clear evidence for natural selection of bill size caused by the availability of seeds. The Grants had studied the inheritance of bill sizes and knew that the surviving large-billed birds would tend to produce offspring with larger bills, so the selection would lead to evolution of bill size. Subsequent studies by the Grants have demonstrated selection on and evolution of bill size in this species in response to other changing conditions on the island. The evolution has occurred both to larger bills, as in this case, and to smaller bills when large seeds became rare.

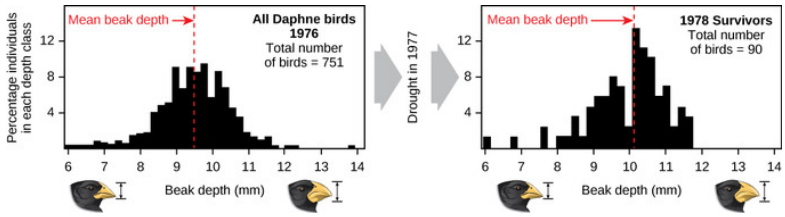


Figure 18.1C.1: **Finches of Daphne Major**: A drought on the Galápagos island of Daphne Major in 1977 reduced the number of small seeds available to finches, causing many of the small-beaked finches to die. This caused an increase in the finches’ average beak size between 1976 and 1978.

18.1D: Processes and Patterns of Evolution

Natural selection can only occur in the presence of genetic variation; environmental conditions determine which traits are selected.

Learning Objectives

- Explain why only heritable variation can be acted upon by natural selection

Key Points

- Genetic variation within a population is a result of mutations and sexual reproduction.
- A mutation may be neutral, reduce an organism's fitness, or increase an organism's fitness.
- An adaptation is a heritable trait that increases the survival and rate of reproduction of an organism in its present environment.
- Divergent evolution describes the process in which two species evolve in diverse directions from a common point.
- Convergent evolution is the process in which similar traits evolve independently in species that do not share a recent common ancestry.

Key Terms

- **adaptation**: modification of something or its parts that makes it more fit for existence under the conditions of its current environment
- **divergent evolution**: the process by which a species with similar traits become groups that are tremendously different from each other over many generations
- **convergent evolution**: a trait of evolution in which species not of similar recent origin acquire similar properties due to natural selection

Processes and Patterns of Evolution

Variation

Natural selection can only take place if there is variation, or differences, among individuals in a population. Importantly, these differences must have some genetic basis; otherwise, the selection will not lead to change in the next generation. This is critical because variation among individuals can be caused by non-genetic reasons, such as an individual being taller due to better nutrition rather than different genes.

Genetic diversity within a population comes from two main mechanisms: mutation and sexual reproduction. Mutation, a change in the DNA sequence, is the ultimate source of new alleles, or new genetic variation in any population. The genetic changes caused by mutation can have one of three outcomes:

- Many mutations will have no effect on the fitness of the phenotype; these are called neutral mutations.
- A mutation may affect the phenotype of the organism in a way that gives it reduced fitness (a lower likelihood of survival or fewer offspring).
- A mutation may produce a phenotype with a beneficial effect on fitness. Different mutations will have a range of effects on the fitness of an organism that expresses them in their phenotype, from a small effect to a great effect.

Sexual reproduction also leads to genetic diversity: when two parents reproduce, unique combinations of alleles assemble to produce the unique genotypes and thus phenotypes in each of the offspring. However, sexual reproduction can not lead to new genes, but rather provides a new combination of genes in a given individual.

Adaptations

A heritable trait that aids the survival and reproduction of an organism in its present environment is called an adaptation. Scientists describe groups of organisms becoming adapted to their environment when a change in the range of genetic variation occurs over time that increases or maintains the "fitness" of the population to its environment. The webbed feet of platypuses are an adaptation for swimming. The snow leopards' thick fur is an adaptation for living in the cold. The cheetahs' fast speed is an adaptation for catching prey.

Whether or not a trait is favorable depends on the environmental conditions at the time. The same traits are not always selected because environmental conditions can change. For example, consider a species of plant that grew in a moist climate and did not need to conserve water. Large leaves were selected because they allowed the plant to obtain more energy from the sun. Large leaves require more water to maintain than small leaves, and the moist environment provided favorable conditions to support large leaves. After thousands of years, the climate changed and the area no longer had excess water. The direction of natural selection shifted so that plants with small leaves were selected because those populations were able to conserve water to survive the new environmental conditions.

The evolution of species has resulted in enormous variation in form and function. Sometimes, evolution gives rise to groups of organisms that become tremendously different from each other. When two species evolve in diverse directions from a common point, it is called divergent evolution. Such divergent evolution can be seen in the forms of the reproductive organs of flowering plants which share the same basic anatomies; however, they can look very different as a result of selection in different physical environments and adaptation to different kinds of pollinators.

Figure 18.1D.1: **Flowering Plants**: Flowering plants evolved from a common ancestor. Notice that the (a) dense blazing star (Liatrus spicata) and the (b) purple coneflower (Echinacea purpurea) vary in appearance, yet both share a similar basic morphology.

In other cases, similar phenotypes evolve independently in distantly-related species. For example, flight has evolved in both bats and insects; they both have structures we refer to as wings, which are adaptations to flight. However, the wings of bats and insects have evolved from very different original structures. This phenomenon is called convergent evolution, where similar traits evolve independently in species that do not share a recent common ancestry. The two species came to the same function, flying, but did so separately from each other.

These physical changes occur over enormous spans of time and help explain how evolution occurs. Natural selection acts on individual organisms, which in turn can shape an entire species. Although natural selection may work in a single generation on an individual, it can take thousands or even millions of years for the genotype of an entire species to evolve. It is over these large time spans that life on earth has changed and continues to change.

18.1E: Evidence of Evolution

Evidence for evolution has been obtained through fossil records, embryology, geography, and molecular biology.

Learning Objectives

- Explain the development of the theory of evolution

Key Points

- Fossils serve to highlight the differences and similarities between current and extinct species, showing the evolution of form over time.
- Similar anatomy across different species highlights their common origin and can be seen in homologous and vestigial structures.
- Embryology provides evidence for evolution since the embryonic forms of divergent groups are extremely similar.
- The natural distribution of species across different continents supports evolution; species that evolved before the breakup of the supercontinent are distributed worldwide, whereas species that evolved more recently are more localized.
- Molecular biology indicates that the molecular basis for life evolved very early and has been maintained with little variation across all life on the planet.

Key Terms

- **homologous structure**: the traits of organisms that result from sharing a common ancestor; such traits often have similar embryological origins and development
- **biogeography**: the study of the geographical distribution of living things
- **vestigial structure**: genetically determined structures or attributes that have apparently lost most or all of their ancestral function in a given species

Evidence of Evolution

The evidence for evolution is compelling and extensive. Looking at every level of organization in living systems, biologists see the signature of past and present evolution. Darwin dedicated a large portion of his book, *On the Origin of Species*, to identifying patterns in nature that were consistent with evolution. Since Darwin, our understanding has become clearer and broader.

Fossils, Anatomy, and Embryology

Fossils provide solid evidence that organisms from the past are not the same as those found today; they show a progression of evolution. Scientists calculate the age of fossils and categorize them to determine when the organisms lived relative to each other. The resulting fossil record tells the story of the past and shows the evolution of form over millions of years. For example, scientists have recovered highly-detailed records showing the evolution of humans and horses. The whale flipper shares a similar morphology to appendages of birds and mammals, indicating that these species share a common ancestor. Over time, evolution led to changes in the shapes and sizes of these bones in different species, but they have maintained the same overall layout. Scientists call these synonymous parts homologous structures.

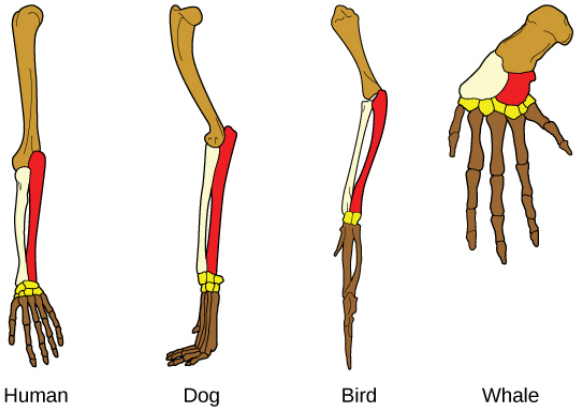


Figure 18.1E.1: **Common Ancestors**: The similar construction of these appendages indicates that these organisms share a common ancestor.

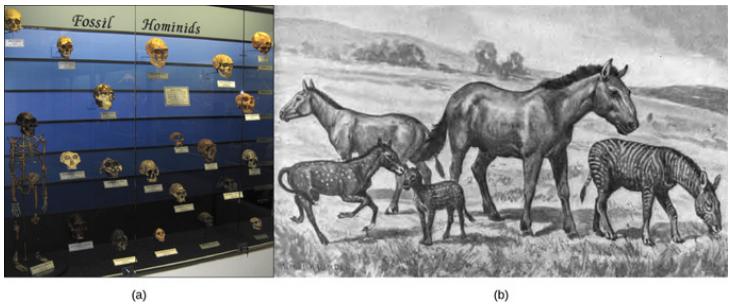


Figure 18.1E.1: **Evolution of Humans and Horses**: (a) In this display, fossil hominids are arranged from oldest (bottom) to newest (top). As hominids evolved, the shape of the skull changed. (b) An artist's rendition of extinct species of the genus Equus reveals that these ancient species resembled the modern horse (Equus ferus), but varied in size.

Some structures exist in organisms that have no apparent function at all, appearing to be residual parts from a common ancestor. These unused structures (such as wings on flightless birds, leaves on some cacti, and hind leg bones in whales) are vestigial.

Embryology, the study of the development of the anatomy of an organism to its adult form, provides evidence for evolution as embryo formation in widely-divergent groups of organisms tends to be conserved. Structures that are absent in the adults of some groups often appear in their embryonic forms, disappearing by the time the adult or juvenile form is reached. For example, all vertebrate embryos, including humans, exhibit gill slits and tails at some point in their early development. These disappear in the adults of terrestrial groups, but are maintained in adults of aquatic groups, such as fish and some amphibians. Great ape embryos, including humans, have a tail structure during their development that is lost by birth.

Another form of evidence of evolution is the convergence of form in organisms that share similar environments. For example, species of unrelated animals, such as the arctic fox and ptarmigan living in the arctic region, have been selected for seasonal white phenotypes during winter to blend with the snow and ice. These similarities occur not because of common ancestry, but because of similar selection pressures: the benefits of not being seen by predators.

(a) (b)

Figure 18.1E.1: **Adaptations: Winter Coats**: The white winter coat of the (a) arctic fox and the (b) ptarmigan's plumage are adaptations to their environments.

Biogeography

The geographic distribution of organisms on the planet follows patterns that are best explained by evolution in conjunction with the movement of tectonic plates over geological time. Broad groups that evolved before the breakup of the supercontinent Pangaea (about 200 million years ago) are distributed worldwide. Groups that evolved since the breakup appear uniquely in regions of the planet, such as the unique flora and fauna of northern continents that formed from the supercontinent Laurasia compared to that of the southern continents that formed from the supercontinent Gondwana.

The great diversification of marsupials in Australia and the absence of other mammals reflect Australia's long isolation. Australia has an abundance of endemic species (those found nowhere else) which is typical of islands whose isolation by expanses of water prevents species from migrating. Over time, these species diverge evolutionarily into new species that look very different from their ancestors that may exist on the mainland. The marsupials of Australia, the finches on the Galápagos, and many species on the Hawaiian Islands are all unique to their one point of origin, yet they display distant relationships to ancestral species on mainlands.

Molecular Biology

Like anatomical structures, the structures of the molecules of life reflect descent with modification. Evidence of a common ancestor for all of life is reflected in the universality of DNA as the genetic material, in the near universality of the genetic code, and in the machinery of DNA replication and expression. In general, the relatedness of groups of organisms is reflected in the similarity of their DNA sequences. This is exactly the pattern that would be expected from descent and diversification from a common ancestor.

DNA sequences have also shed light on some of the mechanisms of evolution. For example, it is clear that the evolution of new functions for proteins commonly occurs after gene duplications that allow the free modification of one copy by mutation, selection, or drift (changes in a population 's gene pool resulting from chance), while the second copy continues to produce a functional protein.

18.1F: Misconceptions of Evolution

There are many misconceptions about evolution, including the meaning of the word *theory*, the way populations change, and the origin of life.

Learning Objectives

- Discuss misconceptions about the theory of evolution

Key Points

- Attacks on the theory of evolution sometimes take issue with the word “theory”, which in the vernacular means a guess or suggested explanation. In scientific language, “theory” indicates a body of thoroughly-tested and verified explanations for a set of observations of the natural world.
- Evolution does not take place on an individual level; evolution is the average change of a characteristic within an entire population.
- Evolution does not explain the origin of life; the theory of evolution instead explains how populations change over time and how traits are selected in order to increase the fitness of a population.
- Favorable traits do not arise as a result of the environment as these traits are already present; individuals with favorable traits are more likely to survive and, thus, will have greater fitness than individuals with less desirable traits.
- Evolution and natural selection are not synonymous. Natural selection is just one mechanism by which evolution occurs.

Key Terms

- **theory**: a well-substantiated explanation of some aspect of the natural world based on knowledge that has been repeatedly confirmed through observation and experimentation

Misconceptions of Evolution

Although the theory of evolution generated controversy when it was first proposed, it was almost universally accepted by biologists within 20 years of the publication of *On the Origin of Species*. Nevertheless, the theory of evolution is a difficult concept and misconceptions about it abound.

Evolution is Just a Theory

Critics of the theory of evolution dismiss its importance by purposefully confounding the everyday usage of the word “theory” with the way scientists use the word. In science, a “theory” is understood to be a body of thoroughly-tested and verified explanations for a set of observations of the natural world. Scientists have a theory of the atom, a theory of gravity, and the theory of relativity, each of which describes understood facts about the world. In the same way, the theory of evolution describes facts about the living world. A theory in science has also survived significant efforts to discredit it by scientists. In contrast, a “theory” in common vernacular is a word meaning a guess or suggested explanation; this meaning is more akin to the scientific concept of “hypothesis. ” When critics of evolution say evolution is “just a theory,” they are implying that there is little evidence supporting it and that it is still in the process of being rigorously tested. This is a mis-characterization.

Individuals Evolve

Evolution is the change in genetic composition of a population over time, specifically over generations, resulting from differential reproduction of individuals with certain alleles. Individuals do change over their lifetime, obviously, but this is called development and involves changes programmed by the set of genes the individual acquired at birth in coordination with the individual’s environment. When thinking about the evolution of a characteristic, it is probably best to think about the change of the average value of the characteristic in the population over time. For example, when natural selection leads to bill-size change in medium-ground finches in the Galápagos, this does not mean that individual bills on the finches are changing. If one measures the average bill size among all individuals in the population at one time and then measures the average bill size in that population several years later, this average value *of the population* will be different as a result of evolution.

Evolution Explains the Origin of Life

It is a common misunderstanding that evolution includes an explanation of life's origins. The theory of evolution explains how populations change over time. It does not shed light on the beginnings of life, including the origins of the first cells, which is how life is defined. The mechanisms of the origin of life on earth are a particularly difficult problem because it occurred a very long time ago and, presumably, it occurred just once. However, while evolution does not explain the origin of life, it may have something to say about some of the processes operating once pre-living entities acquired certain properties. Once a mechanism of inheritance was in place in the form of a molecule like DNA, either within a cell or pre-cell, these entities would be subject to the principle of natural selection. More effective reproducers would increase in frequency at the expense of inefficient reproducers.

Organisms Evolve on Purpose

Statements such as "organisms evolve in response to a change in an environment" may lead to the misunderstanding that evolution is somehow intentional. A changed environment results in some individuals in the population, those with particular phenotypes, benefiting and, therefore, producing proportionately more offspring than other phenotypes. This results in change in the population if the characteristics are genetically determined.

It is important to understand that the variation that natural selection works on is already present in a population and does not arise in response to an environmental change. For example, applying antibiotics to a population of bacteria will, over time, select a population of bacteria that are resistant to antibiotics. The resistance, which is caused by a gene, did not arise by mutation because of the application of the antibiotic. The gene for resistance was already present in the gene pool of the bacteria, probably at a low frequency. The antibiotic, which kills the bacterial cells without the resistance gene, strongly selects individuals that are resistant, since these would be the only ones that survived and divided. Experiments have demonstrated that mutations for antibiotic resistance do not arise as a result of antibiotics.

In a larger sense, evolution is not goal directed. Species do not become "better" over time; they track their changing environment with adaptations that maximize their reproduction. The characteristics that evolve in a species are a function of preexisting variation and the environment, both of which are constantly changing non-directionally. A trait that is fit in one environment at one time may also be fatal at some point in the future.

Evolution = Natural Selection

The terms "evolution" and "natural selection" are often conflated, as the two concepts are closely related. They are not, however, synonymous. Natural selection refers to the process by which organisms better suited for their environment are more likely to survive and produce offspring, thereby proliferating those favorable genetics in a population. Evolution is defined more broadly as any change in the genetic makeup of a population over time. As expounded by Darwin, natural selection is a major driving force of evolution, but it is not the only one.

Genetic drift, for example, is another mechanism by which evolution may occurs. Genetic drift occurs when allelic frequency is altered due to random sampling. It is evolution by chance, and the smaller the population, the more significant the effects on genetic distribution due to sampling error. For example, a population bottleneck, which occurs when an event such as a natural disaster dramatically reduces the size of a population, can result in the elimination or significant reduction of a trait within a population, regardless of how beneficial that trait may be to survival or reproduction. Thus evolution can occur without natural selection.

Figure 18.1F.1: **Misconceptions of Evolution: Lamarckian evolution and giraffes**: Lamarck's hypothesis for how evolution worked stated that organisms could pass on traits they had acquired during their lifetime to the progeny, such as giraffes gaining a long neck because their parents needed to reach high food.

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18.2: Formation of New Species

Topic hierarchy

- 18.2A: The Biological Species Concept
- 18.2B: Reproductive Isolation
- 18.2C: Speciation
- 18.2D: Allopatric Speciation
- 18.2E: Sympatric Speciation

18.2A: The Biological Species Concept

A species is defined as a group of individuals that, in nature, are able to mate and produce viable, fertile offspring.

Learning Objectives

- Explain the biological species concept

Key Points

- Members of the same species are similar both in their external appearance and their internal physiology; the closer the relationship between two organisms, the more similar they will be in these features.
- Some species can look very dissimilar, such as two very different breeds of dogs, but can still mate and produce viable offspring, which signifies that they belong to the same species.
- Some species may look very similar externally, but can be dissimilar enough in their genetic makeup that they cannot produce viable offspring and are, therefore, different species.
- Mutations can occur in any cell of the body, but if a change does not occur in a sperm or egg cell, it cannot be passed on to the organism's offspring.

Key Terms

- **species**: a group of organsms that, in nature, are capable of mating and producing viable, fertile offspring
- **hybrid**: offspring resulting from cross-breeding different entities, e.g. two different species or two purebred parent strains
- **gene pool**: the complete set of unique alleles that would be found by inspecting the genetic material of every living member of a species or population

Species and the Ability to Reproduce

A species is a group of individual organisms that interbreed and produce fertile, viable offspring. According to this definition, one species is distinguished from another when, in nature, it is not possible for matings between individuals from each species to produce fertile offspring.

Members of the same species share both external and internal characteristics which develop from their DNA. The closer relationship two organisms share, the more DNA they have in common, just like people and their families. People's DNA is likely to be more like their father or mother's DNA than their cousin's or grandparent's DNA. Organisms of the same species have the highest level of DNA alignment and, therefore, share characteristics and behaviors that lead to successful reproduction.

Species' appearance can be misleading in suggesting an ability or inability to mate. For example, even though domestic dogs (*Canis lupus familiaris*) display phenotypic differences, such as size, build, and coat, most dogs can interbreed and produce viable puppies that can mature and sexually reproduce.

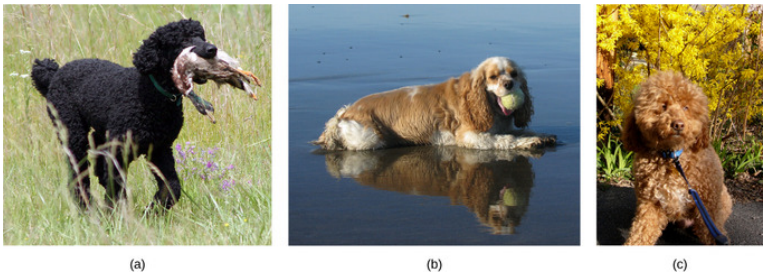

Figure 18.2*A*.1: **Interbreeding in Dogs**: Dogs of different breeds still have the ability to reproduce. The (a) poodle and (b) cocker spaniel can reproduce to produce a breed known as (c) the cockapoo.

In other cases, individuals may appear similar although they are not members of the same species. For example, even though bald eagles (*Haliaeetus leucocephalus*) and African fish eagles (*Haliaeetus vocifer*) are both birds and eagles, each belongs to a separate species group. If humans were to artificially intervene and fertilize the egg of a bald eagle with the sperm of an African fish eagle and a chick did hatch, that offspring, called a hybrid (a cross between two species), would

probably be infertile: unable to successfully reproduce after it reached maturity. Different species may have different genes that are active in development; therefore, it may not be possible to develop a viable offspring with two different sets of directions. Thus, even though hybridization may take place, the two species still remain separate.

(a) (b)

Figure 18.2*A*.1: **Species Similarity & Reproduction**: Species that appear similar may not be able to reproduce. The (a) African fish eagle is similar in appearance to the (b) bald eagle, but the two birds are members of different species.

Populations of species share a gene pool: a collection of all the variants of genes in the species. Again, the basis to any changes in a group or population of organisms must be genetic for this is the only way to share and pass on traits. When variations occur within a species, they can only be passed to the next generation along two main pathways: asexual reproduction or sexual reproduction. The change will be passed on asexually simply if the reproducing cell possesses the changed trait. For the changed trait to be passed on by sexual reproduction, a gamete, such as a sperm or egg cell, must possess the changed trait. In other words, sexually-reproducing organisms can experience several genetic changes in their body cells, but if these changes do not occur in a sperm or egg cell, the changed trait will never reach the next generation. Only heritable traits can evolve. Therefore, reproduction plays a paramount role for genetic change to take root in a population or species. In short, organisms must be able to reproduce with each other to pass new traits to offspring.

18.2B: Reproductive Isolation

Reproductive isolation, through mechanical, behavioral, and physiological barriers, is an important component of speciation.

Learning Objectives

- Explain how reproductive isolation can result in speciation

Key Points

- Reproductive isolation can be either prezygotic (barriers that prevent fertilization) or postzygotic (barriers that occur after zygote formation such as organisms that die as embryos or those that are born sterile).
- Some species may be prevented from mating with each other by the incompatibility of their anatomical mating structures, or a resulting offspring may be prevented by the incompatibility of their gametes.
- Postzygotic barriers include the creation of hybrid individuals that do not survive past the embryonic stages (hybrid inviability) or the creation of a hybrid that is sterile and unable to produce offspring (hybrid sterility).
- Temporal isolation can result in species that are physically similar and may even live in the same habitat, but if their breeding schedules do not overlap then interbreeding will never occur.
- Behavioral isolation, in which the behaviors involved in mating are so unique as to prevent mating, is a prezygotic barrier that can cause two otherwise-compatible species to be uninterested in mating with each other.
- Behavioral isolation, in which the behaviors involved in mating are so unique as to prevent mating, is a prezygotic barrier that can cause two otherwise compatible species to be uninterested in mating with each other.

Key Terms

- **reproductive isolation**: a collection of mechanisms, behaviors, and physiological processes that prevent two different species that mate from producing offspring, or which ensure that any offspring produced is not fertile
- **temporal isolation**: factors that prevent potentially fertile individuals from meeting that reproductively isolate the members of distinct species
- **behavioral isolation**: the presence or absence of a specific behavior that prevents reproduction between two species from taking place
- **prezygotic barrier**: a mechanism that blocks reproduction from taking place by preventing fertilization
- **postzygotic barrier**: a mechanism that blocks reproduction after fertilization and zygote formation
- **hybrid inviability**: a situation in which a mating between two individuals creates a hybrid that does not survive past the embryonic stages
- **hybrid sterility**: a situation in which a mating between two individuals creates a hybrid that is sterile

Reproductive Isolation

Given enough time, the genetic and phenotypic divergence between populations will affect characters that influence reproduction: if individuals of the two populations were to be brought together, mating would be improbable, but if mating did occur, offspring would be non-viable or infertile. Many types of diverging characters may affect reproductive isolation, the ability to interbreed, of the two populations. Reproductive isolation is a collection of mechanisms, behaviors, and physiological processes that prevent the members of two different species that cross or mate from producing offspring, or which ensure that any offspring that may be produced is not fertile.

Scientists classify reproductive isolation in two groups: prezygotic barriers and postzygotic barriers. Recall that a zygote is a fertilized egg: the first cell of the development of an organism that reproduces sexually. Therefore, a prezygotic barrier is a mechanism that blocks reproduction from taking place; this includes barriers that prevent fertilization when organisms attempt reproduction. A postzygotic barrier occurs after zygote formation; this includes organisms that don't survive the embryonic stage and those that are born sterile.

Some types of prezygotic barriers prevent reproduction entirely. Many organisms only reproduce at certain times of the year, often just annually. Differences in breeding schedules, called temporal isolation, can act as a form of reproductive

isolation. For example, two species of frogs inhabit the same area, but one reproduces from January to March, whereas the other reproduces from March to May.

(a) (b)

Figure 18.2B.1: **Temporal isolation**: These two related frog species exhibit temporal reproductive isolation. (a) Rana aurora breeds earlier in the year than (b) Rana boylii.

In some cases, populations of a species move to a new habitat and take up residence in a place that no longer overlaps with other populations of the same species; this is called habitat isolation. Reproduction with the parent species ceases and a new group exists that is now reproductively and genetically independent. For example, a cricket population that was divided after a flood could no longer interact with each other. Over time, the forces of natural selection, mutation, and genetic drift will likely result in the divergence of the two groups.

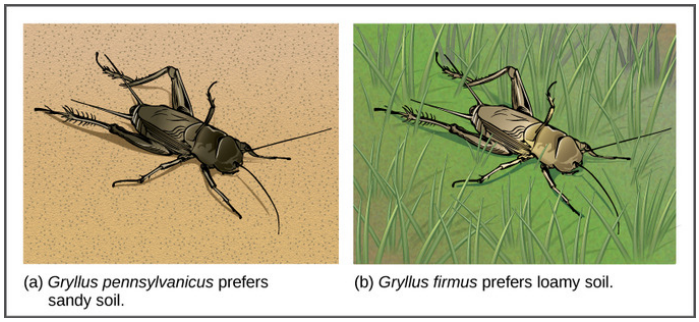

Figure 18.2B.1: **Habitat isolation**: Speciation can occur when two populations occupy different habitats. The habitats need not be far apart. The cricket (a) Gryllus pennsylvanicus prefers sandy soil, while the cricket (b) Gryllus firmus prefers loamy soil. The two species can live in close proximity, but because of their different soil preferences, they became genetically isolated.

Behavioral isolation occurs when the presence or absence of a specific behavior prevents reproduction from taking place. For example, male fireflies use specific light patterns to attract females. Various species display their lights differently; if a male of one species tried to attract the female of another, she would not recognize the light pattern and would not mate with the male.

Other prezygotic barriers work when differences in their gamete cells prevent fertilization from taking place; this is called a gametic barrier. Similarly, in some cases, closely-related organisms try to mate, but their reproductive structures simply do not fit together. For example, damselfly males of different species have differently-shaped reproductive organs. If one species tries to mate with the female of another, their body parts simply do not fit together..

Figure 18.2B.1: **Differences in reproductive structures in male damselflies**: The shape of the male reproductive organ varies among male damselfly species and is only compatible with the female of that species. Reproductive organ incompatibility keeps the species reproductively isolated.

In plants, certain structures aimed to attract one type of pollinator simultaneously prevent a different pollinator from accessing the pollen. The tunnel through which an animal must access nectar can vary in length and diameter, which prevents the plant from being cross-pollinated with a different species.

(a) Honeybee drinking nectar from a foxglove flower

(b) Ruby-throated hummingbird drinking nectar from a trumpet creeper flower

Figure 18.2B.1: **Reproductive isolation in plants**: Some flowers have evolved to attract certain pollinators. The (a) wide foxglove flower is adapted for pollination by bees, while the (b) long, tube-shaped trumpet creeper flower is adapted for pollination by humming birds.

When fertilization takes place and a zygote forms, postzygotic barriers can prevent reproduction. Hybrid individuals in many cases cannot form normally in the womb and simply do not survive past the embryonic stages; this is called hybrid inviability. In another postzygotic situation, reproduction leads to the birth and growth of a hybrid that is sterile and unable to reproduce offspring of their own; this is called hybrid sterility.

18.2C: Speciation

Speciation is an event in which a single species may branch to form two or more new species.

Learning Objectives

- Define speciation and discuss the ways in which it may occur

Key Points

- For the majority of species, the definition of a species is a group of animals that can potentially interbreed, although some different species are capable of producing hybrid offspring.
- Darwin was the first to envision speciation as the branching of two or more new species from one ancestral species; indicated by a diagram he made that bears a striking resemblance to modern-day phylogenetic diagrams.
- For a new species to be formed from an old species, certain events or changes must occur such that the new population is no longer capable of interbreeding with the old one.
- Speciation can occur either through allopatric speciation, when a population is geographically separated from one another, or through sympatric speciation, in which the two new species are not geographically separated.
- Speciation, the formation of two species from one original species, occurs as one species changes over time and branches to form more than one new species.

Key Terms

- **sympatric**: living in the same territory without interbreeding
- **allopatric**: not living in the same territory; geographically isolated and thus unable to crossbreed
- **speciation**: the process by which new distinct species evolve

Speciation

The biological definition of species, which works for sexually-reproducing organisms, is a group of actually or potentially interbreeding individuals. There are exceptions to this rule. Many species are similar enough that hybrid offspring are possible and may often occur in nature, but for the majority of species this rule generally holds. In fact, the presence in nature of hybrids between similar species suggests that they may have descended from a single interbreeding species: the speciation process may not yet be completed.

Given the extraordinary diversity of life on the planet, there must be mechanisms for speciation: the formation of two species from one original species. Darwin envisioned this process as a branching event and diagrammed the process in the only illustration found in *On the Origin of Species*, which bears some resemblance to the more modern phylogenetic diagram of elephant evolution. The diagram shows that as one species changes over time, it branches repeatedly to form more than one new species as long as the population survives or until the organism becomes extinct.

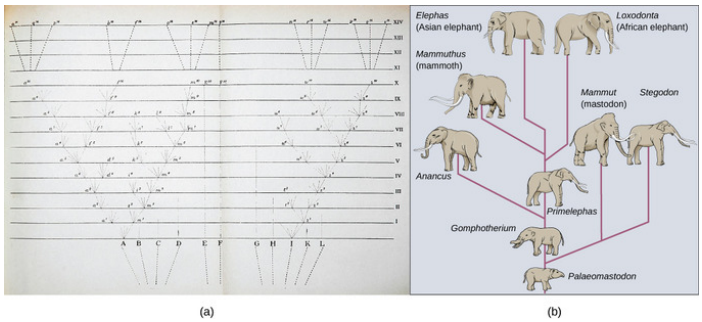

Figure 18.2C.1: **The Evolution of Species**: The only illustration in Darwin's On the Origin of Species is (a) a diagram showing speciation events leading to biological diversity. The diagram shows similarities to phylogenetic charts that are drawn today to illustrate the relationships of species. (b) Modern elephants evolved from the Palaeomastodon, a species that lived in Egypt 35–50 million years ago.

For speciation to occur, two new populations must be formed from one original population; they must evolve in such a way that it becomes impossible for individuals from the two new populations to interbreed. Biologists have proposed

mechanisms by which this could occur that fall into two broad categories: allopatric speciation and sympatric speciation. Allopatric speciation (allo- = “other”; -patric = “homeland”) involves geographic separation of populations from a parent species and subsequent evolution. Sympatric speciation (sym- = “same”; -patric = “homeland”) involves speciation occurring within a parent species remaining in one location.

Biologists think of speciation events as the splitting of one ancestral species into two descendant species. There is no reason why there might not be more than two species formed at one time except that it is less likely; multiple events can be conceptualized as single splits occurring close in time.

18.2D: Allopatric Speciation

Allopatric speciation occurs when a single species becomes geographically separated; each group evolves new and distinctive traits.

Learning Objectives

- Give examples of allopatric speciation

Key Points

- When a population is geographically continuous, the allele frequencies among its members are similar; however, when a population becomes separated, the allele frequencies between the two groups can begin to vary.
- If the separation between groups continues for a long period of time, the differences between their alleles can become more and more pronounced due to differences in climate, predation, food sources, and other factors, eventually leading to the formation of a new species.
- Geographic separation between populations can occur in many ways; the severity of the separation depends on the travel capabilities of the species.
- Allopatric speciation events can occur either by dispersal, when a few members of a species move to a new geographical area, or by vicariance, when a natural situation, such as the formation of a river or valley, physically divide organisms.
- When a population disperses throughout an area, into new, different and often isolated habitats, multiple speciation events can occur in which the single original species gives rise to many new species; this phenomenon is called adaptive radiation.

Key Terms

- **vicariance**: the separation of a group of organisms by a geographic barrier, resulting in differentiation of the original group into new varieties or species
- **adaptive radiation**: the diversification of species into separate forms that each adapt to occupy a specific environmental niche
- **dispersal**: the movement of a few members of a species to a new geographical area, resulting in differentiation of the original group into new varieties or species

Allopatric Speciation

A geographically-continuous population has a gene pool that is relatively homogeneous. Gene flow, the movement of alleles across the range of the species, is relatively free because individuals can move and then mate with individuals in their new location. Thus, the frequency of an allele at one end of a distribution will be similar to the frequency of the allele at the other end. When populations become geographically discontinuous, that free-flow of alleles is prevented. When that separation continues for a period of time, the two populations are able to evolve along different trajectories. This is known as allopatric speciation. Thus, their allele frequencies at numerous genetic loci gradually become more and more different as new alleles independently arise by mutation in each population. Typically, environmental conditions, such as climate, resources, predators, and competitors for the two populations will differ causing natural selection to favor divergent adaptations in each group.

Isolation of populations leading to allopatric speciation can occur in a variety of ways: a river forming a new branch, erosion forming a new valley, a group of organisms traveling to a new location without the ability to return, or seeds floating over the ocean to an island. The nature of the geographic separation necessary to isolate populations depends entirely on the biology of the organism and its potential for dispersal. If two flying insect populations took up residence in separate nearby valleys, chances are individuals from each population would fly back and forth, continuing gene flow. However, if two rodent populations became divided by the formation of a new lake, continued gene flow would be improbable; therefore, speciation would be probably occur.

Biologists group allopatric processes into two categories: dispersal and vicariance. Dispersal occurs when a few members of a species move to a new geographical area, while vicariance occurs when a natural situation arises to physically divide

organisms.

Scientists have documented numerous cases of allopatric speciation. For example, along the west coast of the United States, two separate sub-species of spotted owls exist. The northern spotted owl has genetic and phenotypic differences from its close relative, the Mexican spotted owl, which lives in the south.

Figure $18.2D.1$: **Allopatric speciation due to geographic separation**: The northern spotted owl and the Mexican spotted owl inhabit geographically separate locations with different climates and ecosystems. The owl is an example of allopatric speciation.

Additionally, scientists have found that the further the distance between two groups that once were the same species, the more probable it is that speciation will occur. This seems logical because as the distance increases, the various environmental factors would generally have less in common than locations in close proximity. Consider the two owls: in the north, the climate is cooler than in the south causing the types of organisms in each ecosystem differ, as do their behaviors and habits. Also, the hunting habits and prey choices of the southern owls vary from the northern owls. These variances can lead to evolved differences in the owls, resulting in speciation.

Adaptive Radiation

In some cases, a population of one species disperses throughout an area with each finding a distinct niche or isolated habitat. Over time, the varied demands of their new lifestyles lead to multiple speciation events originating from a single species. This is called adaptive radiation because many adaptations evolve from a single point of origin, causing the species to radiate into several new ones. Island archipelagos like the Hawaiian Islands provide an ideal context for adaptive radiation events because water surrounds each island which leads to geographical isolation for many organisms. The Hawaiian honeycreeper illustrates one example of adaptive radiation. From a single species, called the founder species, numerous species have evolved.

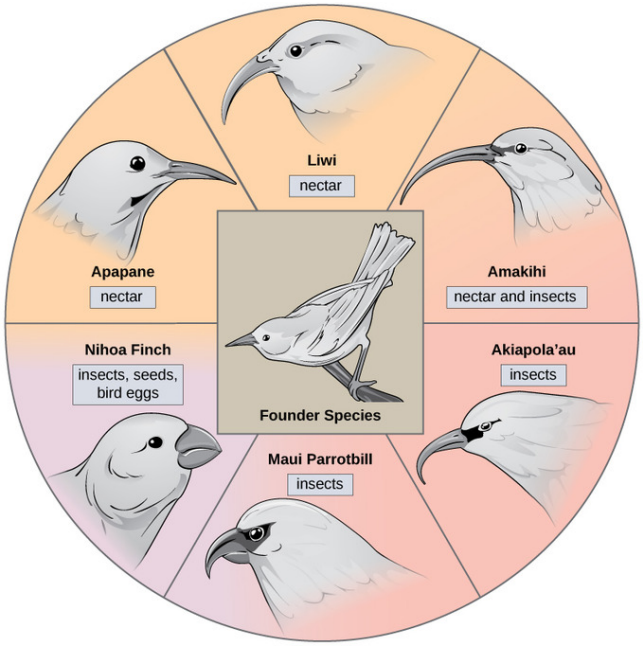

Figure 18.2D.1: **Adaptive Radiation**: The honeycreeper birds illustrate adaptive radiation. From one original species of bird, multiple others evolved, each with its own distinctive characteristics.

In Hawaiian honeycreepers, the response to natural selection based on specific food sources in each new habitat led to the evolution of a different beak suited to the specific food source. The seed-eating birds have a thicker, stronger beak which is suited to break hard nuts. The nectar-eating birds have long beaks to dip into flowers to reach the nectar. The insect-eating birds have beaks like swords, appropriate for stabbing and impaling insects.

E: Sympatric Speciation

Sympatric speciation occurs when two individual populations diverge from an ancestral species without being separated geographically.

Learning Objectives

- Give examples of sympatric speciation

Key Points

- Sympatric speciation can occur when one individual develops an abnormal number of chromosomes, either extra chromosomes (polyploidy) or fewer, such that viable interbreeding can no longer occur.
- When the extra sets of chromosomes in a polyploid originate with the individual because their own gametes do not undergo cytokinesis after meiosis, the result is autopolyploidy.
- When individuals of two different species reproduce to form a viable offspring, such that the extra chromosomes come from two different species, the result is an allopolyploid.
- Once a species develops an abnormal number of chromosomes, it can then only interbreed with members of the population that have the same abnormal number, which can lead to the development of a new species.

Key Terms

- **sympatric speciation**: the process through which new species evolve from a single ancestral species while inhabiting the same geographic region
- **autopolyploid**: having more than two sets of chromosomes, derived from the same species, as a result of redoubling
- **allopolyploid**: having multiple complete sets of chromosomes derived from different species

Sympatric Speciation

Can divergence occur if no physical barriers are in place to separate individuals who continue to live and reproduce in the same habitat? The answer is yes. The process of speciation within the same space is called sympatric speciation. The prefix “sym” means same, so “sympatric” means “same homeland” in contrast to “allopatric” meaning “other homeland.” A number of mechanisms for sympatric speciation have been proposed and studied.

One form of sympatric speciation can begin with a serious chromosomal error during cell division. In a normal cell division event, chromosomes replicate, pair up, and then separate so that each new cell has the same number of chromosomes. However, sometimes the pairs separate and the end cell product has too many or too few individual chromosomes in a condition called aneuploidy.

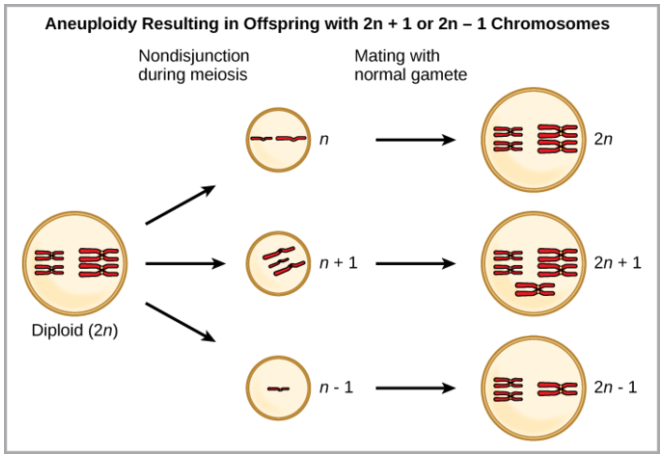

Figure 18.2E.1: **Aneuploidy of chromosomes**: Aneuploidy results when the gametes have too many or too few chromosomes due to nondisjunction during meiosis. In the example shown here, the resulting offspring will have 2n+1 or 2n-1 chromosomes

oidy is a condition in which a cell or organism has an extra set, or sets, of chromosomes. Scientists have identified ain types of polyploidy that can lead to reproductive isolation, or the inability to interbreed with normal individuals, of an individual in the polyploidy state. In some cases, a polyploid individual will have two or more complete sets of chromosomes from its own species in a condition called autopolyploidy. The prefix “auto-” means “self,” so the term means multiple chromosomes from one’s own species. Polyploidy results from an error in meiosis in which all of the chromosomes move into one cell instead of separating.

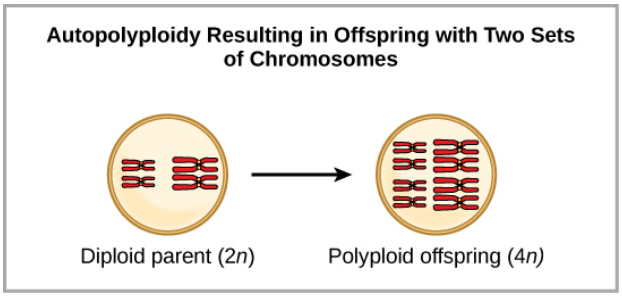

Figure 18.2E.1: **The generation of autopolyploidy**: Autopolyploidy results when meiosis is not followed by cytokinesis.

For example, if a plant species with 2n = 6 produces autopolyploid gametes that are also diploid (2n = 6, when they should be n = 3), the gametes now have twice as many chromosomes as they should have. These new gametes will be incompatible with the normal gametes produced by this plant species. However, they could either self-pollinate or reproduce with other autopolyploid plants with gametes having the same diploid number. In this way, sympatric speciation can occur quickly by forming offspring with 4n: a tetraploid. These individuals would immediately be able to reproduce only with those of this new kind and not those of the ancestral species.

The other form of polyploidy occurs when individuals of two different species reproduce to form a viable offspring called an allopolyploid. The prefix “allo-” means “other” (recall from allopatric). Therefore, an allopolyploid occurs when gametes from two different species combine. Notice how it takes two generations, or two reproductive acts, before the viable fertile hybrid results.

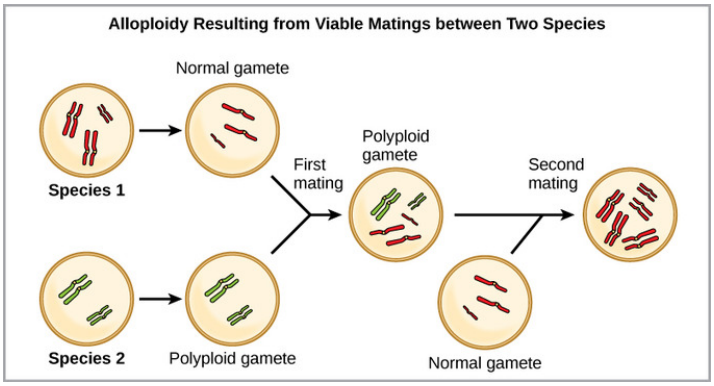

Figure 18.2E.1: **The generation of allopolyploidy**: Alloploidy results when two species mate to produce viable offspring. In the example shown, a normal gamete from one species fuses with a polyploidy gamete from another. Two matings are necessary to produce viable offspring.

The cultivated forms of wheat, cotton, and tobacco plants are all allopolyploids. Although polyploidy occurs occasionally in animals, it takes place most commonly in plants. (Animals with any of the types of chromosomal aberrations described here are unlikely to survive and produce normal offspring.) Scientists have discovered more than half of all plant species studied relate back to a species evolved through polyploidy. With such a high rate of polyploidy in plants, some scientists hypothesize that this mechanism takes place more as an adaptation than as an error.

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18.3: Hybrid Zones and Rates of Speciation

Topic hierarchy

- 18.3A: Hybrid Zones
- 18.3B: Varying Rates of Speciation

18.3A: Hybrid Zones

Over time, two species may further diverge or reconnect, depending on the fitness strength and the reproductive barriers of the hybrids.

Learning Objectives

- Discuss how the fitness of a hybrid will lead to changes in the hybrid zone over time

Key Points

- After speciation, or sufficient evolutionary change for one species to become two distinct species, the two species may continue to co-habitate and interact.
- The area in which two closely-related species interact and reproduce is known as the hybrid zone; their offspring are known as hybrids.
- Depending on the fitness of the hybrid offspring relative to the parents, the two species may either stay as two distinct species (reinforcement), or become one species again (reconnection).

Key Terms

- **hybrid zone**: an area where the ranges of two interbreeding species meet and interbreed
- **hybrid speciation**: the formation of a new species as the direct result of mating between members of two existing species
- **reconnection**: a convergence of two species over time

Reconnection After Speciation

Speciation occurs over a span of evolutionary time. When a new species arises, there is a transition period during which the closely-related species continue to interact.

After speciation, two species may recombine or even continue interacting indefinitely. Individual organisms will mate with any nearby individual with which they are capable of breeding. An area where two closely-related species continue to interact and reproduce, forming hybrids, is called a hybrid zone. Over time, the hybrid zone may change depending on the fitness strength and the reproductive barriers of the hybrids.

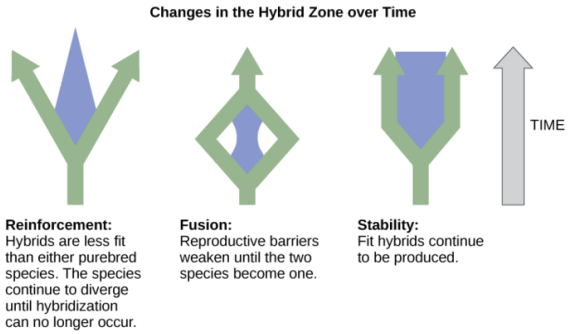

Figure 18.3A.1: **Speciation and the Hybrid Zone**: After speciation has occurred, the two separate-but-closely-related species may continue to produce offspring in an area called the hybrid zone. Reinforcement, fusion, or stability may result, depending on reproductive barriers and the relative fitness of the hybrids.

Hybrids can have less fitness, more fitness, or about the same fitness level as the purebred parents. Usually, hybrids tend to be less fit; therefore, reproduction to produce hybrids will diminish over time, which nudges the two species to diverge further in a process called reinforcement. This term is used because the low success of the hybrids reinforces the original speciation. If the hybrids are less fit than the parents, reinforcement of speciation occurs, and the species will continue to diverge until they can no longer mate and produce viable offspring.

If the hybrids are as fit or more fit than the parents, or the reproductive barriers weaken, the two species may fuse back into one species (reconnection). For a hybrid form to persist, it will generally have to be able to exploit the available resources better than either parent species, with which, in most cases, it will have to compete.

Over time, via a process called hybrid speciation, the hybrids themselves can become a separate species. Reproductive isolation between hybrids and their parents was once thought to be particularly difficult to achieve; thus, hybrid species were thought to be extremely rare. With DNA analysis becoming more accessible in the 1990s, hybrid speciation has been shown to be a fairly common phenomenon, particularly in plants.

Scientists have also observed that sometimes two species will remain separate, but continue to interact to produce some hybrid individuals; this is classified as stability because no real net change is taking place. For a hybrid zone to be stable, the offspring produced by the hybrids have to be less fit than members of the parent species.

18.3B: Varying Rates of Speciation

Two patterns are currently observed in the rates of speciation: gradual speciation and punctuated equilibrium.

Learning Objectives

- Explain how the interaction of an organism's population size in association with environmental changes can lead to different rates of speciation

Key Points

- In the gradual speciation model, species diverge slowly over time in small steps while in the punctuated equilibrium model, a new species diverges rapidly from the parent species.
- The two key influencing factors on the change in speciation rate are the environmental conditions and the population size.
- Gradual speciation is most likely to occur in large populations that live in a stable environment, while the punctuation equilibrium model is more likely to occur in a small population with rapid environmental change.

Key Terms

- **punctuated equilibrium**: a theory of evolution holding that evolutionary change tends to be characterized by long periods of stability, with infrequent episodes of very fast development
- **gradualism**: in evolutionary biology, belief that evolution proceeds at a steady pace, without the sudden development of new species or biological features from one generation to the next

Varying Rates of Speciation

Scientists around the world study speciation, documenting observations both of living organisms and those found in the fossil record. As their ideas take shape and as research reveals new details about how life evolves, they develop models to help explain rates of speciation. In terms of how quickly speciation occurs, two patterns are currently observed: the gradual speciation model and the punctuated equilibrium model.

In the gradual speciation model, species diverge gradually over time in small steps. In the punctuated equilibrium model, a new species changes quickly from the parent species and then remains largely unchanged for long periods of time afterward. This early change model is called punctuated equilibrium, because it begins with a punctuated or periodic change and then remains in balance afterward. While punctuated equilibrium suggests a faster tempo, it does not necessarily exclude gradualism.

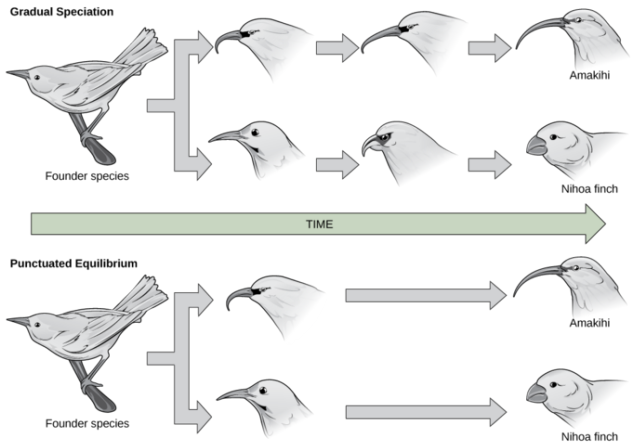

Figure 18.3*B*.1: **Graduated Speciation vs Punctuated Equilibrium**: In (a) gradual speciation, species diverge at a slow, steady pace as traits change incrementally. In (b) punctuated equilibrium, species diverge quickly and then remain unchanged for long periods of time.

The primary influencing factor on changes in speciation rate is environmental conditions. Under some conditions, selection occurs quickly or radically. Consider a species of snails that had been living with the same basic form for many thousands of years. Layers of their fossils would appear similar for a long time. When a change in the environment takes place, such as a

drop in the water level, a small number of organisms are separated from the rest in a brief period of time, essentially forming one large and one tiny population. The tiny population faces new environmental conditions. Because its gene pool quickly became so small, any variation that surfaces and that aids in surviving the new conditions becomes the predominant form.

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Topic hierarchy

18.4A: Genomic Similiarities between Distant Species

18.4B: Genome Evolution

18.4C: Whole-Genome Duplication

18.4D: Gene Duplications and Divergence

Gene duplications create genetic redudancy and can have various effects, including detrimental mutations or divergent evolution.

18.4E: Noncoding DNA

18.4F: Variations in Size and Number of Genes

18.4A: Genomic Similiarities between Distant Species

Genomic similarities between distant species can be established via analysis of genomes using advanced technology.

Learning Objectives

- Discuss the evolutionary implications of observed genome similarities between distant species

Key Points

- Genomic similarities between distant species can be explained by the theory that all organisms share a common ancestor.
- Genomic similarities between distant species can be analysed using genomic analysis tools to create phylogenetic trees that explain these relationships.
- Genetic distance is used to explain the genetic divergence between species or between populations within a species and can indicate how closely related they are and whether they have a recent common ancestor or recent interbreeding has taken place.
- Horizontal gene transfer (HGT) occurs when two unrelated species exchange genes, usually two prokaryotes, although HGT occurs in some eurokaryotes as well.

Key Terms

- **conjugation**: the temporary fusion of organisms, especially as part of sexual reproduction
- **phylogeny**: the evolutionary history of an organism
- **horizontal gene transfer**: the transfer of genetic material from one organism to another one that is not its offspring; especially common among bacteria
- **transformation**: the alteration of a bacterial cell caused by the transfer of DNA from another, especially if pathogenic
- **transduction**: horizontal gene transfer mechanism in prokaryotes where genes are transferred using a virus

Genomic Similarities Between Distant Species

Genetic distance refers to the genetic divergence between species or between populations within a species. Smaller genetic distances indicate that the populations have more similar genes, which indicates they are closely related; they have a recent common ancestor, or recent interbreeding has taken place. Genetic distance is useful in reconstructing the history of populations. For example, evidence from genetic distance suggests that humans arrived in America about 30,000 years ago. By examining the difference between allele frequencies between the populations, genetic distance can estimate how long ago the two populations were together.

Phylogenetic Relationships

Phylogeny describes the relationships of an organism, such as the relationship with its ancestors and the species it is most closely related. Phylogenetic relationships provide information on shared ancestry but not necessarily on how organisms are similar or different. The use of advanced genomic analysis has allowed us to establish phylogenetic trees, which map the relationship between species at a genetic and molecular level. The ability to use these technologies has established previously unknown relationships and has contributed to a more complex evolutionary history. These technologies have established genomic similarities between distant species by establishing genetic distances. In addition, the mechanisms by which genomic similarities between distant species occur can include horizontal gene transfer.

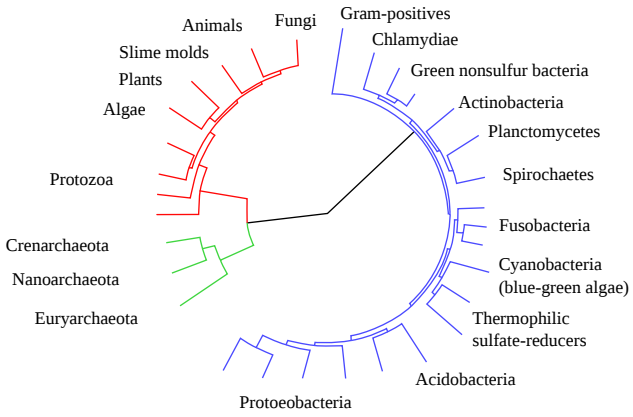

Figure 18.4*A*.1: **Tree of Life**: Diagrammatic representation of the divergence of modern taxonomic groups from their common ancestor. This shows the genomic similarities that can exist between distant species based on their relationship with this ancestor.

Horizontal Gene Transfer

Horizontal gene transfer (HGT), also known as lateral gene transfer, is the transfer of genes between unrelated species. Genes have been shown to be passed between species which are only distantly related using standard phylogeny, thus adding a layer of complexity to the understanding of phylogenetic relationships. The various ways that HGT occurs in prokaryotes is important to understanding phylogenies. Although at present, HGT is not viewed as important to eukaryotic evolution, HGT does occur in this domain as well. Finally, as an example of the ultimate gene transfer, theories of genome fusion between symbiotic or endosymbiotic organisms have been proposed to explain an event of great importance—the evolution of the first eukaryotic cell, without which humans could not have come into existence.

The mechanism of HGT has been shown to be quite common in the prokaryotic domains of Bacteria and Archaea, significantly changing the way their evolution is viewed. The majority of evolutionary models, such as in the endosymbiont theory, propose that eukaryotes descended from multiple prokaryotes, which makes HGT all the more important to understanding the phylogenetic relationships of all extant and extinct species. These gene transfers between species are the major mechanism whereby bacteria acquire resistance to antibiotics. Classically, this type of transfer has been thought to occur by three different mechanisms: transformation, transduction and conjugation.

Although it is easy to see how prokaryotes exchange genetic material by HGT, it was initially thought that this process was absent in eukaryotes, followed by the idea that the gene transfers between multicellular eukaryotes should be more difficult. In spite of this fact, HGT between distantly related organisms has been demonstrated in several eukaryotic species.

In animals, a particularly interesting example of HGT occurs within the aphid species. Aphids are insects that vary in color based on carotenoid content. Carotenoids are pigments made by a variety of plants, fungi, and microbes, and they serve a variety of functions in animals, who obtain these chemicals from their food. Humans require carotenoids to synthesize vitamin A, and we obtain them by eating orange fruits and vegetables: carrots, apricots, mangoes, and sweet potatoes. On the other hand, aphids have acquired the ability to make the carotenoids on their own. According to DNA analysis, this ability is due to the transfer of fungal genes into the insect by HGT, presumably as the insect consumed fungi for food.

(a) (b)

Figure 18.4*A*.1: **Horizontal Gene Transfer in Animals**: (a) Red aphids get their color from red carotenoid pigment. Genes necessary to make this pigment are present in certain fungi, and scientists speculate that aphids acquired these genes through HGT after consuming fungi for food. If genes for making carotenoids are inactivated by mutation, the aphids revert back to (b) their green color. Red coloration makes the aphids a lot more conspicuous to predators, but evidence suggests that red aphids are more resistant to insecticides than green ones. Thus, red aphids may be more fit to survive in some environments than green ones.

18.4B: Genome Evolution

Processes such as mutations, duplications, exon shuffling, transposable elements and pseudogenes have contributed to genomic evolution.

Learning Objectives

- Explain the importance of genomic changes in an evolutionary context

Key Points

- Gene and whole genome duplications have contributed accumulations that have contributed to genome evolution.
- Mutations are constantly occurring in an organism's genome and can cause either a negative effect, positive effect or no effect at all; however, it will still result in changes to the genome.
- Transposable elements are regions of DNA that can be inserted into the genetic code and will causes changes within the genome.
- Pseudogenes are dysfunctional genes derived from previously functional gene relatives and will become a pseudogene by deletion or insertion of one or multiple nucleotides.
- Exon shuffling occurs when two or more exons from different genes are combined together or when exons are duplicated, and will result in new genes.
- Species can also exhibit genome reduction when subsets of their genes are not needed anymore.

Key Terms

- **intron**: a portion of a split gene that is included in pre-RNA transcripts but is removed during RNA processing and rapidly degraded
- **exon**: a region of a transcribed gene present in the final functional RNA molecule
- **pseudogene**: a segment of DNA that is part of the genome of an organism, and which is similar to a gene but does not code for a gene product

Accumulating Changes Over Time

The evolution of the genome is characterized by the accumulation of changes. The analaysis of genomes and their changes in sequence or size over time involves various fields. There are various mechanisms that have contributed to genome evolution and these include gene and genome duplications, polyploidy, mutation rates, transposable elements, pseudogenes, exon shuffling and genomic reduction and gene loss. The concepts of gene and whole-genome duplication are discussed as their own independent concepts, thus, the focus will be on other mechanisms.

Mutation Rates

Mutation rates differ between species and even between different regions of the genome of a single species. Spontaneous mutations often occur which can cause various changes in the genome. Mutations can result in the addition or deletion of one or more nucleotide bases. A change in the code can result in a frameshift mutation which causes the entire code to be read in the wrong order and thus often results in a protein becoming non-functional. A mutation in a promoter region, enhancer region or a region coding for transcription factors can also result in either a loss of function or and upregulation or downregulation in transcription of that gene. Mutations are constantly occurring in an organism's genome and can cause either a negative effect, positive effect or no effect at all.

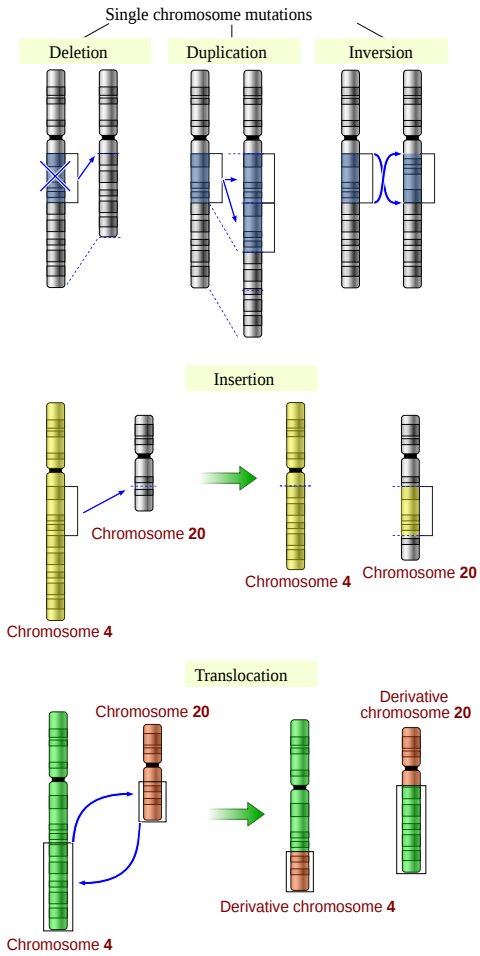

Figure 18.4B.1: **Chromosomal Mutations**: Chromosomal mutations over time can accumulate and promote diversity and evolution if a produced trait is favorable.

Transposable Elements

Transposable elements are regions of DNA that can be inserted into the genetic code through one of two mechanisms. These mechanisms work similarly to “cut-and-paste” and “copy-and-paste” functionalities in word processing programs. The “cut-and-paste” mechanism works by excising DNA from one place in the genome and inserting itself into another location in the code. The “copy-and-paste” mechanism works by making a genetic copy or copies of a specific region of DNA and inserting these copies elsewhere in the code. The most common transposable element in the human genome is the Alu sequence, which is present in the genome over one million times.

Pseudogenes

Often a result of spontaneous mutation, pseudogenes are dysfunctional genes derived from previously functional gene relatives. There are many mechanisms by which a functional gene can become a pseudogene including the deletion or insertion of one or multiple nucleotides. This can result in a shift of reading frame, causing the gene to longer code for the expected protein, a premature stop codon or a mutation in the promoter region. Often cited examples of pseudogenes within the human genome include the once functional olfactory gene families. Over time, many olfactory genes in the human genome became pseudogenes and were no longer able to produce functional proteins, explaining the poor sense of smell humans possess in comparison to their mammalian relatives.

Exon Shuffling

Exon shuffling is a mechanism by which new genes are created. This can occur when two or more exons from different genes are combined together or when exons are duplicated. Exon shuffling results in new genes by altering the current intron-exon structure. This can occur by any of the following processes: transposon mediated shuffling, sexual

bination or illegitimate recombination. Exon shuffling may introduce new genes into the genome that can be either d against and deleted or selectively favored and conserved.

Genome Reduction and Gene Loss

Many species exhibit genome reduction when subsets of their genes are not needed anymore. This typically happens when organisms adapt to a parasitic life style, e.g. when their nutrients are supplied by a host. As a consequence, they lose the genes need to produce these nutrients. In many cases, there are both free living and parasitic species that can be compared and their lost genes identified. Good examples are the genomes of Mycobacterium tuberculosis and Mycobacterium leprae, the latter of which has a dramatically reduced genome. Another beautiful example are endosymbiont species. For instance, Polynucleobacter necessarius was first described as a cytoplasmic endosymbiont of the ciliate Euplotes aediculatus. The latter species dies soon after being cured of the endosymbiont. In the few cases in which P. necessarius is not present, a different and rarer bacterium apparently supplies the same function. No attempt to grow symbiotic P. necessarius outside their hosts has yet been successful, strongly suggesting that the relationship is obligate for both partners. Yet, closely related free-living relatives of P. necessarius have been identified. The endosymbionts have a significantly reduced genome when compared to their free-living relatives (1.56 Mbp vs. 2.16 Mbp).

C: Whole-Genome Duplication

Whole-genome duplication is characterized by an organisms entire genetic information being copied once or multiple times.

Learning Objectives

- State the evolutionary implications of whole-genome duplication

Key Points

- Whole- genome duplication can provide an evolutionary advantage by providing the organism with multiple copies of a gene that is considered favorable.
- Whole-genome duplication can result in divergence and formation of new species over time.
- Whole-genome duplication can result in mutation and cause disease if the genes are rendered non-functional.

Key Terms

- **polyploidy**: having more than the usual two homologous sets of chromosomes
- **palaeopolyploidization**: the development of polyploid organisms in the geologic past
- **sympatric speciation**: the process through which new species evolve from a single ancestral species while inhabiting the same geographic region

Whole-Genome Duplication

Gene duplication is the process by which a region of DNA coding for a gene creates additional copies of the gene. Similar to gene duplication, whole-genome duplication is the process by which an organism's entire genetic information is copied, once or multiple times, which is known as polyploidy. This may provide an evolutionary benefit to the organism by supplying it with multiple copies of a gene, thus, creating a greater possibility of functional and selectively favored genes.

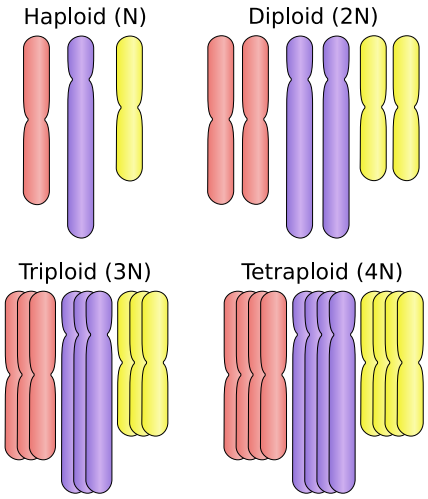

Figure $18.4C.1$: **Polyploidy**: This image shows haploid (single), diploid (double), triploid (triple), and tetraploid (quadruple) sets of chromosomes. Triploid and tetraploid chromosomes are examples of polyploidy.

Evolutionary importance

Paleopolyploidization events lead to massive cellular changes, including doubling of the genetic material, changes in gene expression and increased cell size. Gene loss during diploidization is not completely random, but heavily selected. Genes from large gene families are duplicated. On the other hand, individual genes are not duplicated. Overall, paleopolyploidy can have both short-term and long-term evolutionary effects on an organism's fitness in the natural environment.

ome diversity

e doubling provides organisms with redundant alleles that can evolve freely with little selection pressure. The duplicated genes can undergo neofunctionalization or subfunctionalization which could help the organism adapt to the new environment or survive different stress conditions.

Speciation

Sympatric speciation can begin with a chromosomal error during meiosis or the formation of a hybrid individual with too many chromosomes, such as polyploidy which can occur during whole-genome duplication. Scientists have identified types of polyploidy that can lead to reproductive isolation of an individual in the polyploid state. In some cases a polyploid individual will have two or more complete sets of chromosomes from its own species in a condition called autopolyploidy. The other form of polyploidy occurs when individuals of two different species reproduce to form a viable offspring called an allopolyploid. The prefix “allo” means “other” (recall from allopatric); therefore, an allopolyploid occurs when gametes from two different species combine.

It has been suggested that many polyploidization events created new species, via a gain of adaptive traits, or by sexual incompatibility with their diploid counterparts. An example would be the recent speciation of allopolyploid Spartina — S. anglica; the polyploid plant is so successful that it is listed as an invasive species in many regions.

Evidence of Whole-Genome Duplication

In 1997, Wolfe & Shields gave evidence for an ancient duplication of the *Saccharomyces cerevisiae* (Yeast) genome. It was initially noted that this yeast genome contained many individual gene duplications. Wolfe & Shields hypothesized that this was actually the result of an entire genome duplication in the yeast’s distant evolutionary history. They found 32 pairs of homologous chromosomal regions, accounting for over half of the yeast’s genome. They also noted that although homologs were present, they were often located on different chromosomes. Based on these observations, they determined that *Saccharomyces cerevisiae* underwent a whole-genome duplication soon after its evolutionary split from *Kluyveromyces*, a genus of ascomycetous yeasts. Over time, many of the duplicate genes were deleted and rendered non-functional. A number of chromosomal rearrangements broke the original duplicate chromosomes into the current manifestation of homologous chromosomal regions.

18.4D: Gene Duplications and Divergence

Learning Objectives

- Explain the mechanisms of gene duplication and divergence

Gene Duplication

Gene duplication is the process by which a region of DNA coding for a gene is copied. Gene duplication can occur as the result of an error in recombination or through a retrotransposition event. Duplicate genes are often immune to the selective pressure under which genes normally exist. This can result in a large number of mutations accumulating in the duplicate gene code. This may render the gene non-functional or in some cases confer some benefit to the organism. There are multiple mechanisms by which gene duplication can occur.

Ectopic Recombination

Duplications can arise from unequal crossing-over that occurs during meiosis between misaligned homologous chromosomes. The product of this recombination is a duplication at the site of the exchange and a reciprocal deletion. Ectopic recombination is typically mediated by sequence similarity at the duplicate breakpoints, which form direct repeats. Repetitive genetic elements, such as transposable elements, offer one source of repetitive DNA that can facilitate recombination, and they are often found at duplication breakpoints in plants and mammals.

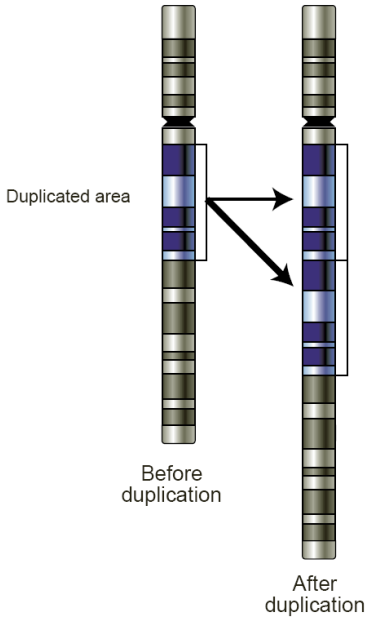

Figure 18.4*D*.1: **Gene Duplication**: This figure indicates a schematic of a region of a chromosome before and after a duplication event. Ectopic recombination is typically mediated by sequence similarity at the duplicate breakpoints, which form direct repeats.

Replication Slippage

Replication slippage is an error in DNA replication, which can produce duplications of short genetic sequences. During replication, DNA polymerase begins to copy the DNA, and at some point during the replication process, the polymerase dissociates from the DNA and replication stalls. When the polymerase reattaches to the DNA strand, it aligns the replicating strand to an incorrect position and incidentally copies the same section more than once. Replication slippage is also often facilitated by repetitive sequence but requires only a few bases of similarity.

Retrotransposition

During cellular invasion by a replicating retroelement or retrovirus, viral proteins copy their genome by reverse transcribing RNA to DNA. If viral proteins attach irregularly to cellular mRNA, they can reverse-transcribe copies of genes to create retrogenes. Retrogenes usually lack intronic sequence and often contain poly A sequences that are also integrated into the genome. Many retrogenes display changes in gene regulation in comparison to their parental gene sequences, which sometimes results in novel functions.

Aneuploidy

Aneuploidy occurs when nondisjunction at a single chromosome results in an abnormal number of chromosomes. Aneuploidy is often harmful and in mammals regularly leads to spontaneous abortions. Some aneuploid individuals are viable. For example, trisomy 21 in humans leads to Down syndrome, but it is not fatal. Aneuploidy often alters gene dosage in ways that are detrimental to the organism and therefore, will not likely spread through populations.

Gene duplication as an evolutionary event

Gene duplications are an essential source of genetic novelty that can lead to evolutionary innovation. Duplication creates genetic redundancy and if one copy of a gene experiences a mutation that affects its original function, the second copy can serve as a ‘spare part’ and continue to function correctly. Thus, duplicate genes accumulate mutations faster than a functional single-copy gene, over generations of organisms, and it is possible for one of the two copies to develop a new and different function. This is an examples of neofunctionalization.

Gene duplication is believed to play a major role in evolution; this stance has been held by members of the scientific community for over 100 years. It has been argued that gene duplication is the most important evolutionary force since the emergence of the universal common ancestor.

Another possible fate for duplicate genes is that both copies are equally free to accumulate degenerative mutations, so long as any defects are complemented by the other copy. This leads to a neutral “subfunctionalization” model, in which the functionality of the original gene is distributed among the two copies. Neither gene can be lost, as both now perform important non-redundant functions, but ultimately neither is able to achieve novel functionality. Subfunctionalization can occur through neutral processes in which mutations accumulate with no detrimental or beneficial effects. However, in some cases subfunctionalization can occur with clear adaptive benefits. If an ancestral gene is pleiotropic and performs two functions, often times neither one of these two functions can be changed without affecting the other function. In this way, partitioning the ancestral functions into two separate genes can allow for adaptive specialization of subfunctions, thereby providing an adaptive benefit.

Divergence

Genetic divergence is the process in which two or more populations of an ancestral species accumulate independent genetic changes through time, often after the populations have become reproductively isolated for some period of time. In some cases, subpopulations living in ecologically distinct peripheral environments can exhibit genetic divergence from the remainder of a population, especially where the range of a population is very large. The genetic differences among divergent populations can involve silent mutations (that have no effect on the phenotype) or give rise to significant morphological and/or physiological changes. Genetic divergence will always accompany reproductive isolation, either due to novel adaptations via selection and/or due to genetic drift, and is the principal mechanism underlying speciation.

Genetic drift or allelic drift is the change in the frequency of a gene variant (allele) in a population due to random sampling. The alleles in the offspring are a sample of those in the parents, and chance has a role in determining whether a given individual survives and reproduces. A population’s allele frequency is the fraction of the copies of one gene that share a particular form. Genetic drift may cause gene variants to disappear completely and thereby reduce genetic variation. When there are few copies of an allele, the effect of genetic drift is larger, and when there are many copies the effect is smaller. These changes in gene frequency can contribute to divergence.

Divergent evolution is usually a result of diffusion of the same species to different and isolated environments, which blocks the gene flow among the distinct populations allowing differentiated fixation of characteristics through genetic drift and natural selection.Divergent evolution can also be applied to molecular biology characteristics. This could apply to a

pathway in two or more organisms or cell types. This can apply to genes and proteins, such as nucleotide sequences or protein sequences that are derived from two or more homologous genes. Both orthologous genes (resulting from a speciation event) and paralogous genes (resulting from gene duplication within a population) can be said to display divergent evolution.

Key Points

- Ectopic recombination occurs when there is an unequal crossing-over and the product of this recombination are a duplication at the site of the exchange and a reciprocal deletion.
- Gene duplications do not always result in detrimental mutations; they can contribute to divergent evolution, which causes genetic differences between groups to develop and eventually form new species.
- Replication slippage can occur when there is an error during DNA replication and duplications of short genetic sequences are produced.
- Retrotranspositions occur when a retrovirus copies their genome by reverse transcribing RNA to DNA and aberrantly attach to cellular mRNA and reverse transcribe copies of genes to create retrogenes.
- Aneuploidy can occur when there is a nondisjunction even at a single chromosome thus, the result is an abnormal number of chromosomes.
- Genetic divergence can occur by mechanisms such as genetic drift which contibute to the accumulation of independent genetic changes of two or more populations derived from a common ancestor.

Key Terms

- **paralogous**: having a similar structure indicating divergence from a common ancestral gene
- **nondisjunction**: the failure of chromosome pairs to separate properly during meiosis
- **retrogene**: a DNA gene copied back from RNA by reverse transcription
- **genetic drift**: an overall shift of allele distribution in an isolated population, due to random fluctuations in the frequencies of individual alleles of the genes

E: Noncoding DNA

Noncoding DNA are sequences of DNA that do not encode protein sequences but can be transcribed to produce important regulatory molecules.

Learning Objectives

- Summarize the importance of noncoding DNA

Key Points

- In the human genome, over 98% of DNA is classified as noncoding DNA and can be transcribed to regulatory noncoding RNAs (i.e. tRNAs, rRNAs), origins of DNA replication, centromeres, telomeres and scaffold attachment regions (SARs).
- Noncoding regions are most commonly referred to as 'junk DNA', however, this term is misleading as noncoding DNA does have functional importance.
- The proportion of coding and noncoding DNA within organisms varies and the amount of noncoding DNA typically correlates with organism complexity, though there are many notable exceptions.

Key Terms

- **intergenic**: describing the noncoding sections of nucleic acid between genes
- **noncoding**: DNA which does not code for protein
- **intron**: a portion of a split gene that is included in pre-RNA transcripts but is removed during RNA processing and rapidly degraded

Noncoding DNA

In genomics and related disciplines, noncoding DNA sequences are components of an organism's DNA that do not encode protein sequences. Some noncoding DNA is transcribed into functional noncoding RNA molecules (e.g. transfer RNA, ribosomal RNA, and regulatory RNAs), while others are not transcribed or give rise to RNA transcripts of unknown function. The amount of noncoding DNA varies greatly among species. For example, over 98% of the human genome is noncoding DNA, while only about 2% of a typical bacterial genome is noncoding DNA.

Initially, a large proportion of noncoding DNA had no known biological function and was therefore sometimes referred to as "junk DNA", particularly in the lay press. However, many types of noncoding DNA sequences do have important biological functions, including the transcriptional and translational regulation of protein-coding sequences, origins of DNA replication, centromeres, telomeres, scaffold attachment regions (SARs), genes for functional RNAs, and many others. Other noncoding sequences have likely, but as-yet undetermined, functions. Some sequences may have no biological function for the organism, such as endogenous retroviruses.

Genomic Variation between Organisms

The amount of total genomic DNA varies widely between organisms, and the proportion of coding and noncoding DNA within these genomes varies greatly as well. More than 98% of the human genome does not encode protein sequences, including most sequences within introns and most intergenic DNA. While overall genome size, and by extension the amount of noncoding DNA, are correlated to organism complexity, there are many exceptions. For example, the genome of the unicellular Polychaos dubium (formerly known as Amoeba dubia) has been reported to contain more than 200 times the amount of DNA in humans. The pufferfish Takifugu rubripes genome is only about one eighth the size of the human genome, yet seems to have a comparable number of genes; approximately 90% of the Takifugu genome is noncoding DNA.

In 2013, a new "record" for most efficient genome was discovered. Utricularia gibba, a bladderwort plant, has only 3% noncoding DNA. The extensive variation in nuclear genome size among eukaryotic species is known as the C-value enigma or C-value paradox. Most of the genome size difference appears to lie in the noncoding DNA. About 80 percent of the nucleotide bases in the human genome may be transcribed, but transcription does not necessarily imply function.

Figure 18.4E. 1: **Utricularia gibba flower**: Utricularia gibba has 3% noncoding DNA, which is low for flowering plants. This 3% has given this plant the title the ‘most efficient’ genome.

18.4F: Variations in Size and Number of Genes

The genome size does not always correlate with the complexity of the organism and, in fact, shows great variation in size and gene number.

Learning Objectives

- Describe how variations in the size and number of genes can arise through evolutionary mechanisms

Key Points

- Harmless mutations and sexual recombination of chromosomes may allow the evolution of new characteristics.
- Genome size can be affected by various events, including duplication, insertion, recombination, deletion and polyploidization events.
- Genome size can be affected by evolution of an organism and result is an increased or decreased need for specific genes for survival based on behavior.
- The human genome exemplifies the concept that complexity does not always correlate with an increase in genome size; there are fewer protein coding genes present than expected relative to the genome size.

Key Terms

- **polyploidization**: hybridization that leads to polyploidy
- **pseudogene**: a segment of DNA that is part of the genome of an organism, and which is similar to a gene but does not code for a gene product
- **genome**: the cell's complete genetic information packaged as a double-stranded DNA molecule

Variations in Size and Number of Genes

Genetic diversity refers to any variation in the nucleotides, genes, chromosomes, or whole genomes of organisms. Genetic diversity at its most elementary level is represented by differences in the sequences of nucleotides (adenine, cytosine, guanine, and thymine) that form the DNA (deoxyribonucleic acid) within the cells of the organism. The DNA is contained in the chromosomes present within the cell; some chromosomes are contained within specific organelles in the cell (for example, the chromosomes of mitochondria and chloroplast). Nucleotide variation is measured for discrete sections of the chromosomes, called genes. Thus, each gene compromises a hereditary section of DNA that occupies a specific place of the chromosome, and controls a particular characteristic of an organism.

Chromosomes

Most organisms are diploid, having two sets of chromosomes, and therefore two copies (called alleles) of each gene. However, some organisms can be haploid, triploid, or tetraploid (having one, three, or four sets of chromosomes respectively). Within any single organism, there may be variation between the two (or more) alleles for each gene. This variation is introduced either through mutation of one of the alleles, or as a result of sexual reproduction.

During sexual reproduction, offspring inherit alleles from both parents and these alleles might be slightly different, especially if there has been migration or hybridization of organisms, so that the parents may come from different populations and gene pools. Also, when the offspring's chromosomes are copied after fertilization, genes can be exchanged in a process called sexual recombination. Harmless mutations and sexual recombination may allow the evolution of new characteristics.

Genome Size and Number

Genome size is usually measured in base pairs (or bases in single-stranded DNA or RNA). The C-value is another measure of genome size. The C-value refers to the amount, in picograms, of DNA contained within a haploid nucleus (e.g. a gamete) or one half the amount in a diploid somatic cell of a eukaryotic organism. In some cases (notably among diploid organisms), the terms C-value and genome size are used interchangeably, however in polyploids the C-value may represent two or more genomes contained within the same nucleus.

Different species can have different numbers of genes within the entire DNA or genome of the organism. However, a greater total number of genes might not correspond with a greater observable complexity in the anatomy and physiology of the

organism (i.e. greater phenotypic complexity). For example, the predicted size of the human genome is not much larger than the genomes of some invertebrates and plants, and may even be smaller than the Indian rice genome. In humans, more proteins are encoded per gene than in other species. In prokaryotic genomes, research has shown that there is a significant positive correlation between the C-value of prokaryotes and the amount of genes that compose the genome. This indicates that gene number is the main factor influencing the size of the prokaryotic genome.

Genes vs Genome Size

In eukaryotic organisms, there is a paradox observed, namely that the number of genes that make up the genome does not correlate with genome size. In other words, the genome size is much larger than would be expected given the total number of protein coding genes. Genome size can increase by duplication, insertion, or polyploidization and the process of recombination can lead to both DNA loss or gain. It is also possible that genomes can shrink due to deletions.

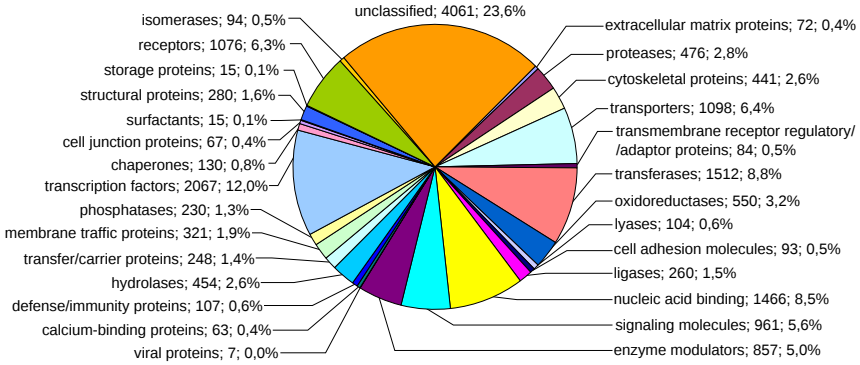

Figure 18.4F.1: **Gene variation in the Genome**: This figure represents the human genome, categorized by function of each gene product, given both as number of genes and as percentage of all genes. Importantly, genome size does not necessarily correlate with complexity.

A famous example for such gene decay is the genome of Mycobacterium leprae, the causative agent of leprosy. M.leprae has lost many once-functional genes over time due to the formation of pseudogenes. This is evident in looking at its closest ancestor Mycobacterium tuberculosis. M. leprae lives inside and replicates inside of a host and due to this arrangement it does not have a need for many of the genes it once carried which allowed it to live and prosper outside of the host. Thus over time these genes have lost their function through mechanisms such as mutation causing them to become pseudogenes. It is beneficial to an organism to rid itself of non-essential genes because it makes replicating its DNA much faster and more energy-efficient.

An example of increasing genome size over time is seen in filamentous plant pathogens. These plant pathogen genomes have been growing larger over the years due to repeat-driven expansion. The repeat-rich regions contain genes coding for host interaction proteins. With the addition of more and more repeats to these regions the plants increase the possibility of developing new virulence factors through mutation and other forms of genetic recombination. In this way it is beneficial for these plant pathogens to have larger genomes.

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18.5: Evidence of Evolution

Topic hierarchy

- 18.5A: The Fossil Record as Evidence for Evolution
- 18.5B: Fossil Formation
- 18.5C: Gaps in the Fossil Record
- 18.5D: Carbon Dating and Estimating Fossil Age
- 18.5E: The Fossil Record and the Evolution of the Modern Horse

The detailed fossil record of horses has provided insight into their evolutionary progress.

- 18.5F: Homologous Structures
- 18.5G: Convergent Evolution
- 18.5H: Vestigial Structures
- 18.5I: Biogeography and the Distribution of Species

18.5A: The Fossil Record as Evidence for Evolution

Fossils tell us when organisms lived, as well as provide evidence for the progression and evolution of life on earth over millions of years.

Learning Objectives

- Synthesize the contributions of the fossil record to our understanding of evolution

Key Points

- Fossils are the preserved remains or traces of animals, plants, and other organisms from the past.
- Fossils are important evidence for evolution because they show that life on earth was once different from life found on earth today.
- Usually only a portion of an organism is preserved as a fossil, such as body fossils (bones and exoskeletons), trace fossils (feces and footprints), and chemofossils (biochemical signals).
- Paleontologists can determine the age of fossils using methods like radiometric dating and categorize them to determine the evolutionary relationships between organisms.

Key Terms

- **biomarker**: A substance used as an indicator of a biological state, most commonly disease.
- **trace fossil**: A type of fossil reflecting the reworking of sediments and hard substrates by organisms including structures like burrows, trails, and impressions.
- **fossil record**: All discovered and undiscovered fossils and their placement in rock formations and sedimentary layers.
- **strata**: Layers of sedimentary rock.
- **fossiliferous**: Containing fossils.

What Fossils Tell Us

Fossils are the preserved remains or traces of animals, plants, and other organisms from the past. Fossils range in age from 10,000 to 3.48 billion years old. The observation that certain fossils were associated with certain rock strata led 19th century geologists to recognize a geological timescale. Like extant organisms, fossils vary in size from microscopic, like single-celled bacteria, to gigantic, like dinosaurs and trees.

Figure 18.5A.1: **"Sue" T-rex skeleton**: The bones of this Tyrannosaurus rex were preserved through the process of permineralization, which suggests that this organism was covered by sediment soon after death.

Permineralization

Permineralization is a process of fossilization that occurs when an organism is buried. The empty spaces within an organism (spaces filled with liquid or gas during life) become filled with mineral-rich groundwater. Minerals precipitate from the groundwater, occupying the empty spaces. This process can occur in very small spaces, such as within the cell

wall of a plant cell. Small-scale permineralization can produce very detailed fossils. For permineralization to occur, the organism must be covered by sediment soon after death, or soon after the initial decay process.

The degree to which the remains are decayed when covered determines the later details of the fossil. Fossils usually consist of the portion of the organisms that was partially mineralized during life, such as the bones and teeth of vertebrates or the chitinous or calcareous exoskeletons of invertebrates. However, other fossils contain traces of skin, feathers or even soft tissues.

Trace Fossils

Fossils may also consist of the marks left behind by the organism while it was alive, such as footprints or feces. These types of fossils are called trace fossils, or ichnofossils, as opposed to body fossils. Past life may also leave some markers that cannot be seen but can be detected in the form of biochemical signals; these are known as chemofossils or biomarkers.

Figure 18.5A.1: **Dinosaur footprints**: Footprints are examples of trace fossils, which contribute to the fossil record.

The Fossil Record

The totality of fossils, both discovered and undiscovered, and their placement in fossiliferous (fossil-containing) rock formations and sedimentary layers (strata) is known as the fossil record. The fossil record was one of the early sources of data underlying the study of evolution and continues to be relevant to the history of life on Earth. The development of radiometric dating techniques in the early 20th century allowed geologists to determine the numerical or “absolute” age of various strata and their included fossils.

Evidence for Evolution

Fossils provide solid evidence that organisms from the past are not the same as those found today; fossils show a progression of evolution. Fossils, along with the comparative anatomy of present-day organisms, constitute the morphological, or anatomical, record. By comparing the anatomies of both modern and extinct species, paleontologists can infer the lineages of those species. This approach is most successful for organisms that had hard body parts, such as shells, bones or teeth. The resulting fossil record tells the story of the past and shows the evolution of form over millions of years.

18.5B: Fossil Formation

Fossils can form under ideal conditions by preservation, permineralization, molding (casting), replacement, or compression.

Learning Objectives

- Predict the conditions suitable to fossil formation

Key Points

- Preservation of remains in amber or other substances is the rarest from of fossilization; this mechanism allows scientists to study the skin, hair, and organs of ancient creatures.
- Permineralization, where minerals like silica fill the empty spaces of shells, is the most common form of fossilization.
- Molds form when shells or bones dissolve, leaving behind an empty depression; a cast is then formed when the depression is filled by sediment.
- Replacement occurs when the original shell or bone dissolves away and is replaced by a different mineral; when this occurs with permineralization, it is called petrification.
- In compression, the most common form of fossilization of leaves and ferns, a dark imprint of the fossil remains.
- Decay, chemical weathering, erosion, and predators are factors that deter fossilization.
- Fossilization of soft body parts is rare, and hard parts are better preserved when buried.

Key Terms

- **amber**: a hard, generally yellow to brown translucent fossil resin
- **permineralization**: form of fossilization in which minerals are deposited in the pores of bone and similar hard animal parts
- **petrification**: process by which organic material is converted into stone through the replacement of the original material and the filling of the original pore spaces with minerals

Fossil Formation

The process of a once living organism becoming a fossil is called fossilization. Fossilization is a very rare process, and of all the organisms that have lived on Earth, only a tiny percentage of them ever become fossils. To see why, imagine an antelope that dies on the African plain. Most of its body is quickly eaten by scavengers, and the remaining flesh is soon eaten by insects and bacteria, leaving behind only scattered bones. As the years go by, the bones are scattered and fragmented into small pieces, eventually turning into dust and returning their nutrients to the soil. As a result, it would be rare for any of the antelope's remains to actually be preserved as a fossil.

Fossilization can occur in many ways. Most fossils are preserved in one of five processes:

- preserved remains
- permineralization
- molds and casts
- replacement
- compression

Preserved Remains

The rarest form of fossilization is the preservation of original skeletal material and even soft tissue. For example, some insects have been preserved perfectly in amber, which is ancient tree sap. In addition, several mammoths and even a Neanderthal hunter have been discovered frozen in glaciers. These preserved remains allow scientists the rare opportunity to examine the skin, hair, and organs of ancient creatures. Scientists have collected DNA from these remains and compared the DNA sequences to those of modern creatures.

Figure 18.5B.1: **Amber**: The image depicts a gnat preserved in amber. A lot of insects have been found to be perfectly maintained in this ancient tree sap.

Permineralization

The most common method of fossilization is permineralization. After a bone, wood fragment, or shell is buried in sediment, it may be exposed to mineral-rich water that moves through the sediment. This water will deposit minerals, typically silica, into empty spaces, producing a fossil. Fossilized dinosaur bones, petrified wood, and many marine fossils were formed by permineralization.

Figure 18.5B.1: **Permineralization**: These fossils from the Road Canyon Formation (Middle Permian of Texas) have been silicified (replaced with silica), which is a form of permineralization.

Molds and Casts

In some cases, the original bone or shell dissolves away, leaving behind an empty space in the shape of the shell or bone. This depression is called a mold. Later, the space may be filled with other sediments to form a matching cast in the shape of the original organism. Many mollusks (bivalves, snails, and squid) are commonly found as molds and casts because their shells dissolve easily.

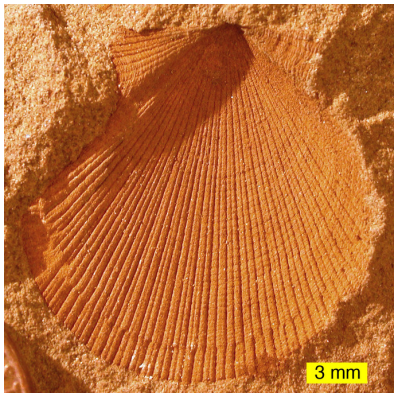

Figure 18.5B.1: **Molds**: The depression in the image is an external mold of a bivalve from the Logan Formation, Lower Carboniferous, Ohio

Replacement

In some cases, the original shell or bone dissolves away and is replaced by a different mineral. For example, shells that were originally calcite may be replaced by dolomite, quartz, or pyrite. If quartz fossils are surrounded by a calcite matrix, the calcite can be dissolved away by acid, leaving behind an exquisitely preserved quartz fossil. When permineralization and replacement occur together, the organism is said to have undergone petrification, the process of turning organic material into stone. However, replacement can occur without permineralization and vice versa.

Compression

Some fossils form when their remains are compressed by high pressure. This can leave behind a dark imprint of the fossil. Compression is most common for fossils of leaves and ferns but also can occur with other organisms.

Conditions for Fossilization

Following the death of an organism, several forces contribute to the dissolution of its remains. Decay, predators, or scavengers will typically rapidly remove the flesh. The hard parts, if they are separable at all, can be dispersed by predators, scavengers, or currents. The individual hard parts are subject to chemical weathering and erosion, as well as to splintering by predators or scavengers, which will crunch up bones for marrow and shells to extract the flesh inside. Also, an animal swallowed whole by a predator, such as a mouse swallowed by a snake, will have not just its flesh but some, and perhaps all, its bones destroyed by the gastric juices of the predator.

It would not be an exaggeration to say that the typical vertebrate fossil consists of a single bone, or tooth, or fish scale. The preservation of an intact skeleton with the bones in the relative positions they had in life requires a remarkable circumstances, such as burial in volcanic ash, burial in aeolian sand due to the sudden slumping of a sand dune, burial in a mudslide, burial by a turbidity current, and so forth. The mineralization of soft parts is even less common and is seen only in exceptionally rare chemical and biological conditions.

18.5C: Gaps in the Fossil Record

Because not all animals have bodies which fossilize easily, the fossil record is considered incomplete.

Learning Objectives

- Explain the gap in the fossil record

Key Points

- The number of species known about through fossils is less than 1% of all species that have ever lived.
- Because hard body parts are more easily preserved than soft body parts, there are more fossils of animals with hard body parts, such as vertebrates, echinoderms, brachiopods, and some groups of arthropods.
- Very few fossils have been found in the period from 360 to 345 million years ago, known as Romer's gap. Theories to explain this include the period's geochemistry, errors in excavation, and limited vertebrate diversity.

Key Terms

- **transitional fossil**: Fossilized remains of a life form that exhibits traits common to both an ancestral group and its derived descendant group.
- **Romer's gap**: A period in the tetrapod fossil record (360 to 345 million years ago) from which excavators have not yet found relevant fossils.

Incompleteness of the Fossil Record

Each fossil discovery represents a snapshot of the process of evolution. Because of the specialized and rare conditions required for a biological structure to fossilize, many important species or groups may never leave fossils at all. Even if they do leave fossils, humans may never find them—for example, if they are buried under hundreds of feet of ice in Antarctica. The number of species known about through the fossil record is less than 5% of the number of species alive today. Fossilized species may represent less than 1% of all the species that have ever lived.

Types of Fossils in the Fossil Record

The fossil record is very uneven and is mostly comprised of fossils of organisms with hard body parts, leaving most groups of soft-bodied organisms with little to no fossil record. Groups considered to have a good fossil record, including transitional fossils between these groups, are the vertebrates, the echinoderms, the brachiopods, and some groups of arthropods. Their hard bones and shells fossilize easily, unlike the bodies of organisms like cephalopods or jellyfish.

Romer's Gap

Romer's gap is an example of an apparent gap in the tetrapod fossil record used in the study of evolutionary biology. These gaps represent periods from which no relevant fossils have been found. Romer's gap is named after paleontologist Alfred Romer, who first recognized it. Romer's gap spanned from approximately 360 to 345 million years ago, corresponding to the first 15 million years of the Carboniferous Period.

Figure 18.5C.1: **Romer's Gap**: The bank of the Whiteadder Water in Scotland is one of the few known localities bearing fossils of tetrapods from Romer's gap.

There has been much debate over why there are so few fossils from this time period. Some scientists have suggested that the geochemistry of the time period caused bad conditions for fossil formation, so few organisms were fossilized. Another theory suggests that scientists have simply not yet discovered an excavation site for these fossils, due to inaccessibility or random chance.

18.5D: Carbon Dating and Estimating Fossil Age

The age of fossils can be determined using stratigraphy, biostratigraphy, and radiocarbon dating.

Learning Objectives

- Summarize the available methods for dating fossils

Key Points

- Determining the ages of fossils is an important step in mapping out how life evolved across geologic time.
- The study of stratigraphy enables scientists to determine the age of a fossil if they know the age of layers of rock that surround it.
- Biostratigraphy enables scientists to match rocks with particular fossils to other rocks with those fossils to determine age.
- Paleontology seeks to map out how life evolved across geologic time. A substantial hurdle is the difficulty of working out fossil ages.
- Scientists use carbon dating when determining the age of fossils that are less than 60,000 years old, and that are composed of organic materials such as wood or leather.

Key Terms

- **half-life**: The time required for half of the nuclei in a sample of a specific isotope to undergo radioactive decay.
- **stratigraphy**: The study of rock layers and the layering process.
- **radiocarbon dating**: A method of estimating the age of an artifact or biological vestige based on the relative amounts of various isotopes of carbon present in a sample.

Determining Fossil Ages

Paleontology seeks to map out how life evolved across geologic time. A substantial hurdle is the difficulty of working out fossil ages. There are several different methods for estimating the ages of fossils, including:

1. stratigraphy
2. biostratigraphy
3. carbon dating

Stratigraphy

Paleontologists rely on stratigraphy to date fossils. Stratigraphy is the science of understanding the strata, or layers, that form the sedimentary record. Strata are differentiated from each other by their different colors or compositions and are exposed in cliffs, quarries, and river banks. These rocks normally form relatively horizontal, parallel layers, with younger layers forming on top.

If a fossil is found between two layers of rock whose ages are known, the fossil's age is thought to be between those two known ages. Because rock sequences are not continuous, but may be broken up by faults or periods of erosion, it is difficult to match up rock beds that are not directly adjacent.

Figure 18.5D.1: **Sedimentary layers**: The layers of sedimentary rock, or strata, can be seen as horizontal bands of differently colored or differently structured materials exposed in this cliff. The deeper layers are older than the layers found at the top, which aids in determining the relative age of fossils found within the strata.

Biostratigraphy

Fossils of species that survived for a relatively short time can be used to match isolated rocks: this technique is called biostratigraphy. For instance, the extinct chordate *Eoplacognathus pseudoplanus* is thought to have existed during a short range in the Middle Ordovician period. If rocks of unknown age have traces of *E. pseudoplanus*, they have a mid-Ordovician age. Such index fossils must be distinctive, globally distributed, and occupy a short time range to be useful. Misleading results can occur if the index fossils are incorrectly dated.

Relative Dating

Stratigraphy and biostratigraphy can in general provide only relative dating (A was before B), which is often sufficient for studying evolution. This is difficult for some time periods, however, because of the barriers involved in matching rocks of the same age across continents. Family-tree relationships can help to narrow down the date when lineages first appeared. For example, if fossils of B date to X million years ago and the calculated "family tree" says A was an ancestor of B, then A must have evolved earlier.

It is also possible to estimate how long ago two living branches of a family tree diverged by assuming that DNA mutations accumulate at a constant rate. However, these "molecular clocks" are sometimes inaccurate and provide only approximate timing. For example, they are not sufficiently precise and reliable for estimating when the groups that feature in the Cambrian explosion first evolved, and estimates produced by different approaches to this method may vary as well.

Carbon Dating

Together with stratigraphic principles, radiometric dating methods are used in geochronology to establish the geological time scale. Beds that preserve fossils typically lack the radioactive elements needed for radiometric dating (" radiocarbon dating " or simply "carbon dating"). The principle of radiocarbon dating is simple: the rates at which various radioactive elements decay are known, and the ratio of the radioactive element to its decay products shows how long the radioactive element has existed in the rock. This rate is represented by the half-life, which is the time it takes for half of a sample to decay.

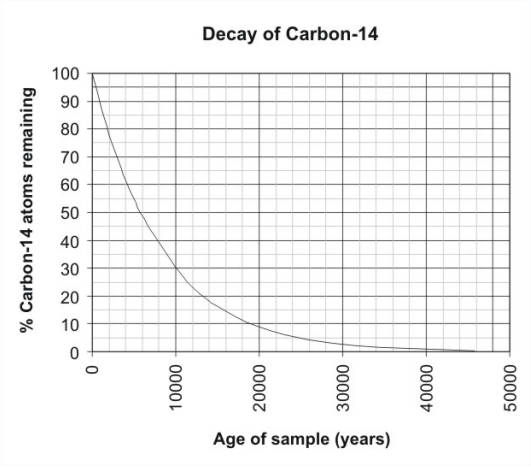

Figure 18.5D.1: **Half-life of Carbon-14**: Radiometric dating is a technique used to date materials such as rocks or carbon, usually based on a comparison between the observed abundance of a naturally occurring radioactive isotope and its decay products, using known decay rates.

The half-life of carbon-14 is 5,730 years, so carbon dating is only relevant for dating fossils less than 60,000 years old. Radioactive elements are common only in rocks with a volcanic origin, so the only fossil-bearing rocks that can be dated radiometrically are volcanic ash layers. Carbon dating uses the decay of carbon-14 to estimate the age of organic materials, such as wood and leather.

18.5E: The Fossil Record and the Evolution of the Modern Horse

LEARNING OBJECTIVES

- Analyze the fossil record to understand the evolution of horses

Fossils provide evidence that organisms from the past are not the same as those found today, and demonstrate a progression of evolution. Scientists date and categorize fossils to determine when the organisms lived relative to each other. The resulting fossil record tells the story of the past and shows the evolution of forms over millions of years.

Case Study: Evolution of the Modern Horse

Highly detailed fossil records have been recovered for sequences in the evolution of modern horses. The fossil record of horses in North America is especially rich and contains transition fossils: fossils that show intermediate stages between earlier and later forms. The fossil record extends back to a dog-like ancestor some 55 million years ago, which gave rise to the first horse-like species 55 to 42 million years ago in the genus *Eohippus*.

The first equid fossil was found in the gypsum quarries in Montmartre, Paris in the 1820s. The tooth was sent to the Paris Conservatory, where Georges Cuvier identified it as a browsing equine related to the tapir. His sketch of the entire animal matched later skeletons found at the site. During the H.M.S. Beagle survey expedition, Charles Darwin had remarkable success with fossil hunting in Patagonia. In 1833 in Santa Fe, Argentina, he was “filled with astonishment” when he found a horse’s tooth in the same stratum as fossils of giant armadillos and wondered if it might have been washed down from a later layer, but concluded this was “not very probable.” In 1836, the anatomist Richard Owen confirmed the tooth was from an extinct species, which he subsequently named *Equus curvidens*.

The original sequence of species believed to have evolved into the horse was based on fossils discovered in North America in the 1870s by paleontologist Othniel Charles Marsh. The sequence, from *Eohippus* to the modern horse (*Equus*), was popularized by Thomas Huxley and became one of the most widely known examples of a clear evolutionary progression. The sequence of transitional fossils was assembled by the American Museum of Natural History into an exhibit that emphasized the gradual, “straight-line” evolution of the horse.

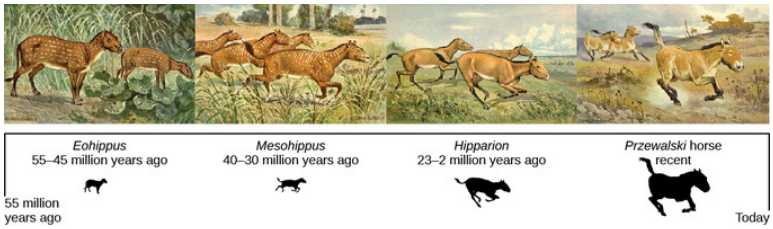

Figure 18.5E.1: **Horse evolution**: This illustration shows an artist’s renderings of species derived from fossils of the evolutionary history of the horse and its ancestors. The species depicted are only four from a very diverse lineage that contains many branches, dead ends, and adaptive radiations. One of the trends, depicted here, is the evolutionary tracking of a drying climate and increase in prairie versus forest habitat reflected in forms that are more adapted to grazing and predator escape through running.

Since then, as the number of equid fossils has increased, the actual evolutionary progression from *Eohippus* to *Equus* has been discovered to be much more complex and multibranched than was initially supposed. Detailed fossil information on the rate and distribution of new equid species has also revealed that the progression between species was not as smooth and consistent as was once believed.

Although some transitions were indeed gradual progressions, a number of others were relatively abrupt in geologic time, taking place over only a few million years. Both anagenesis, a gradual change in an entire population ‘s gene frequency, and cladogenesis, a population “splitting” into two distinct evolutionary branches, occurred, and many species coexisted with “ancestor” species at various times.

Adaptation for Grazing

The series of fossils tracks the change in anatomy resulting from a gradual drying trend that changed the landscape from a forested habitat to a prairie habitat. Early horse ancestors were originally specialized for tropical forests, while modern

horses are now adapted to life on drier land. Successive fossils show the evolution of teeth shapes and foot and leg anatomy to a grazing habit with adaptations for escaping predators.

The horse belongs to the order Perissodactyla (odd-toed ungulates), the members of which all share hoofed feet and an odd number of toes on each foot, as well as mobile upper lips and a similar tooth structure. This means that horses share a common ancestry with tapirs and rhinoceroses. Later species showed gains in size, such as those of *Hipparion*, which existed from about 23 to 2 million years ago. The fossil record shows several adaptive radiations in the horse lineage, which is now much reduced to only one genus, *Equus*, with several species. Paleozoologists have been able to piece together a more complete outline of the modern horse's evolutionary lineage than that of any other animal.

Key Points

- A dog-like organism gave rise to the first horse ancestors 55-42 million years ago.
- The fossil record shows modern horses moved from tropical forests to prairie habitats, developed teeth, and grew in size.
- The first equid fossil was a tooth from the extinct species *Equus curvidens* found in Paris in the 1820s.
- Thomas Huxley popularized the evolutionary sequence of horses, which became one of the most common examples of clear evolutionary progression.
- Horse evolution was previously believed to be a linear progress, but after more fossils were discovered, it was determined the evolution of horses was more complex and multi-branched.
- Horses have evolved from gradual change (anagenesis) as well as abrupt progression and division (cladogenesis).

Key Terms

- **cladogenesis**: An evolutionary splitting event in which each branch and its smaller branches forms a clade.
- **equid**: A member of the horse family.
- **anagenesis**: Evolution of a new species through a large scale change in gene frequency so that the new species replaces the old, rather than branching to produce an additional species.

18.5F: Homologous Structures

Homologous structures are similar structures that evolved from a common ancestor.

Learning Objectives

- Describe the connection between evolution and the appearance of homologous structures

Key Points

- Homology is a relationship defined between structures or DNA derived from a common ancestor and illustrates descent from a common ancestor.
- Analogous structures are physically (but not genetically) similar structures that were not present the last common ancestor.
- Homology can also be partial; new structures can evolve through the combination or parts of developmental pathways.
- Analogy may also be referred to as homoplasy, which is further divided into parallelism, reversal, and convergence.

Key Terms

- **homology**: A correspondence of structures in two life forms with a common evolutionary origin, such as flippers and hands.
- **analogy**: The relationship between characteristics that are apparently similar but did not develop from the same structure
- **homoplasy**: A correspondence between the parts or organs of different species acquired as the result of parallel evolution or convergence.

Homologous Structures

Homology is the relationship between structures or DNA derived from the most recent common ancestor. A common example of homologous structures in evolutionary biology are the wings of bats and the arms of primates. Although these two structures do not look similar or have the same function, genetically, they come from the same structure of the last common ancestor. Homologous traits of organisms are therefore explained by descent from a common ancestor.

It's important to note that defining two structures as homologous depends on what ancestor is being described as the common ancestor. If we go all the way back to the beginning of life, all structures are homologous!

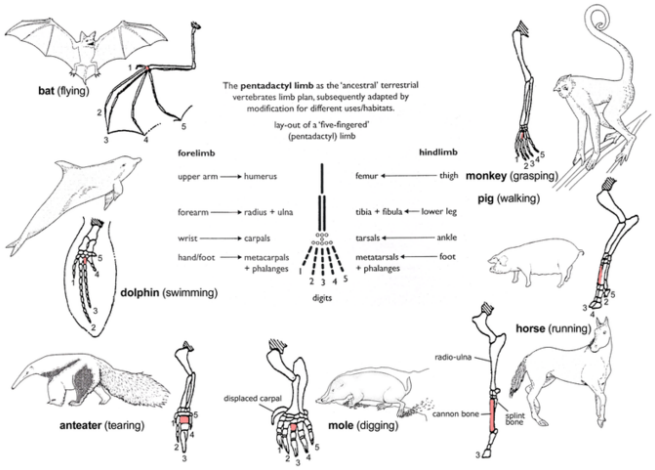

Figure 18.5F.1: **Homology in the forelimbs of vertebrates**: The principle of homology illustrated by the adaptive radiation of the forelimb of mammals. All conform to the basic pentadactyl pattern but are modified for different usages. The third metacarpal is shaded throughout; the shoulder is crossed-hatched.

In genetics, homology is measured by comparing protein or DNA sequences. Homologous gene sequences share a high similarity, supporting the hypothesis that they share a common ancestor.

ogy can also be partial: new structures can evolve through the combination of developmental pathways or parts of As a result, hybrid or mosaic structures can evolve that exhibit partial homologies. For example, certain compound leaves of flowering plants are partially homologous both to leaves and shoots because they combine some traits of leaves and some of shoots.

Paralogous Structures

Homologous sequences are considered paralogous if they were separated by a gene duplication event; if a gene in an organism is duplicated to occupy two different positions in the same genome, then the two copies are paralogous.

A set of sequences that are paralogous are called paralogs of each other. Paralogs typically have the same or similar function, but sometimes do not. It is considered that due to lack of the original selective pressure upon one copy of the duplicated gene, this copy is free to mutate and acquire new functions.

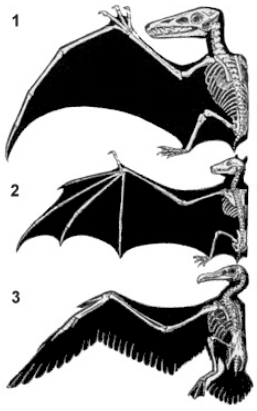

Figure 18.5F.1: **Homology vs. analogy**: The wings of pterosaurs (1), bats (2), and birds (3) are analogous as wings, but homologous as forelimbs. This is because they are similar characteristically and even functionally, but evolved from different ancestral roots.

Paralogous genes often belong to the same species, but not always. For example, the hemoglobin gene of humans and the myoglobin gene of chimpanzees are considered paralogs. This is a common problem in bioinformatics; when genomes of different species have been sequenced and homologous genes have been found, one can not immediately conclude that these genes have the same or similar function, as they could be paralogs whose function has diverged.

Analogous Structures

The opposite of homologous structures are analogous structures, which are physically similar structures between two taxa that evolved separately (rather than being present in the last common ancestor). Bat wings and bird wings evolved independently and are considered analogous structures. Genetically, a bat wing and a bird wing have very little in common; the last common ancestor of bats and birds did not have wings like either bats or birds. Wings evolved independently in each lineage after diverging from ancestors with forelimbs that were not used as wings (terrestrial mammals and theropod dinosaurs, respectively).

It is important to distinguish between different hierarchical levels of homology in order to make informative biological comparisons. In the above example, the bird and bat wings are analogous as wings, but homologous as forelimbs because the organ served as a forearm (not a wing) in the last common ancestor of tetrapods.

Analogy is different than homology. Although analogous characteristics are superficially similar, they are not homologous because they are phylogenetically independent. The wings of a maple seed and the wings of an albatross are analogous but not homologous (they both allow the organism to travel on the wind, but they didn't both develop from the same structure). Analogy is commonly also referred to as homoplasy.

18.5G: Convergent Evolution

Convergent evolution occurs in different species that have evolved similar traits independently of each other.

Learning Objectives

- Predict the circumstances supporting convergent evolution of two species

Key Points

- Examples of convergent evolution include the relationship between bat and insect wings, shark and dolphin bodies, and vertebrate and cephalopod eyes.
- Analogous structures arise from convergent evolution, but homologous structures do not.
- Convergent evolution is the opposite of divergent evolution, in which related species evolve different traits.
- Convergent evolution is similar to parallel evolution, in which two similar but independent species evolve in the same direction and independently acquire similar characteristics.

Key Terms

- **parallel evolution**: the development of a similar trait in related, but distinct, species descending from the same ancestor, but from different clades
- **convergent evolution**: a trait of evolution in which species not of similar recent origin acquire similar properties due to natural selection
- **divergent evolution**: the process by which a species with similar traits become groups that are tremendously different from each other over many generations
- **morphology**: the form and structure of an organism

Convergent Evolution

Sometimes, similar phenotypes evolve independently in distantly related species. For example, flight has evolved in both bats and insects, and they both have wings, which are adaptations to flight. However, the wings of bats and insects have evolved from very different original structures. This phenomenon is called convergent evolution, where similar traits evolve independently in species that do not share a recent common ancestry.

Examples of Convergent Evolution

Convergent evolution describes the independent evolution of similar features in species of different lineages. The two species came to the same function, flying, but did so separately from each other. They have “converged” on this useful trait. Both sharks and dolphins have similar body forms, yet are only distantly related: sharks are fish and dolphins are mammals. Such similarities are a result of both populations being exposed to the same selective pressures. Within both groups, changes that aid swimming have been favored. Thus, over time, they developed similar appearances (morphology), even though they are not closely related.

One of the most well-known examples of convergent evolution is the camera eye of cephalopods (e.g., octopus), vertebrates (e.g., mammals), and cnidaria (e.g., box jellies). Their last common ancestor had at most a very simple photoreceptive spot, but a range of processes led to the progressive refinement of this structure to the advanced camera eye. There is, however, one subtle difference: the cephalopod eye is “wired” in the opposite direction, with blood and nerve vessels entering from the back of the retina, rather than the front as in vertebrates.

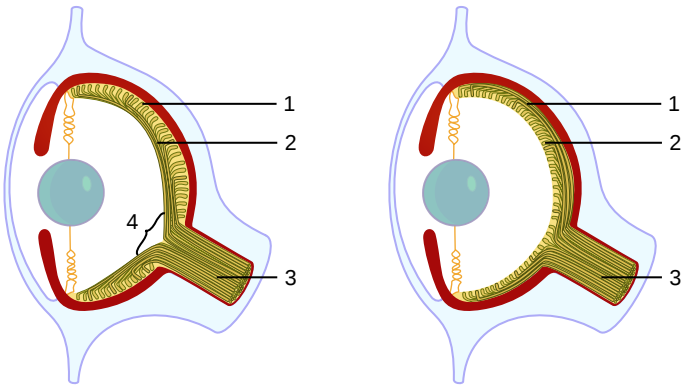

Figure 18.5G.1: **Eye evolution**: Vertebrates and octopi developed the camera eye independently. In the vertebrate version the nerve fibers pass in front of the retina, and there is a blind spot (4) where the nerves pass through the retina. In the octopus version, the eye is constructed the "right way out," with the nerves attached to the rear of the retina. This means that octopi do not have a blind spot.

Convergent evolution is similar to, but distinguishable from, the phenomenon of parallel evolution. Parallel evolution occurs when two independent but similar species evolve in the same direction and thus independently acquire similar characteristics; for example, gliding frogs have evolved in parallel from multiple types of tree frog.

Analogous Structures

Traits arising through convergent evolution are analogous structures, in contrast to homologous structures, which have a common origin, but not necessarily similar function. The British anatomist Richard Owen was the first scientist to recognize the fundamental difference between analogies and homologies. Bat and pterosaur wings are an example of analogous structures, while the bat wing is homologous to human and other mammal forearms, sharing an ancestral state despite serving different functions.

Divergent Evolution

The opposite of convergent evolution is divergent evolution, whereby related species evolve different traits. On a molecular level, this can happen due to random mutation unrelated to adaptive changes.

18.5H: Vestigial Structures

Vestigial structures have no function but may still be inherited to maintain fitness.

Learning Objectives

- Discuss the connection between evolution and the existence of vestigial structures

Key Points

- Structures that have no apparent function and appear to be residual parts from a past ancestor are called vestigial structures.
- Examples of vestigial structures include the human appendix, the pelvic bone of a snake, and the wings of flightless birds.
- Vestigial structures can become detrimental, but in most cases these structures are harmless; however, these structures, like any other structure, require extra energy and are at risk for disease.
- Vestigial structures, especially non-harmful ones, take a long time to be phased out since eliminating them would require major alterations that could result in negative side effects.

Key Terms

- **vestigial structure**: Genetically determined structures or attributes that have lost most or all of their ancestral function in a given species.
- **adaptation**: A modification of something or its parts that makes it more fit for existence under the conditions of its current environment.

What Are Vestigial Structures?

Some organisms possess structures with no apparent function which appear to be residual parts from a past ancestor. For example, some snakes have pelvic bones despite having no legs because they descended from reptiles that did have legs. Another example of a structure with no function is the human vermiform appendix. These unused structures without function are called vestigial structures. Other examples of vestigial structures are wings (which may have other functions) on flightless birds like the ostrich, leaves on some cacti, traces of pelvic bones in whales, and the sightless eyes of cave animals.

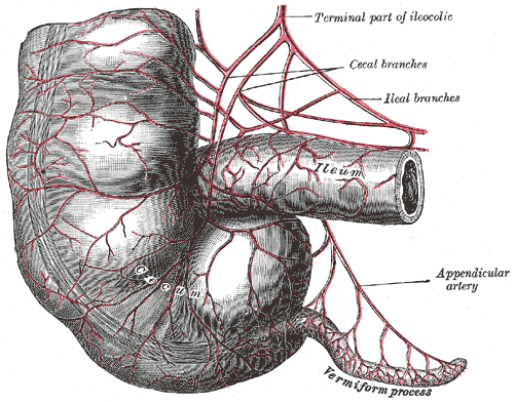

Figure 18.5H.1: **Vestigial appendix**: In humans the vermiform appendix is a vestigial structure; it has lost much of its ancestral function.

There are also several reflexes and behaviors that are considered to be vestigial. The formation of goose bumps in humans under stress is a vestigial reflex its function in human ancestors was to raise the body's hair, making the ancestor appear larger and scaring off predators. The arrector pili muscle, which is a band of smooth muscle that connects the hair follicle to connective tissue, contracts and creates the goose bumps on skin.

Vestigial Structures in Evolution

Vestigial structures are often homologous to structures that function normally in other species. Therefore, vestigial structures can be considered evidence for evolution, the process by which beneficial heritable traits arise in populations over an extended period of time. The existence of vestigial traits can be attributed to changes in the environment and behavior patterns of the organism in question. As the function of the trait is no longer beneficial for survival, the likelihood that future offspring will inherit the "normal" form of it decreases. In some cases the structure becomes detrimental to the organism.

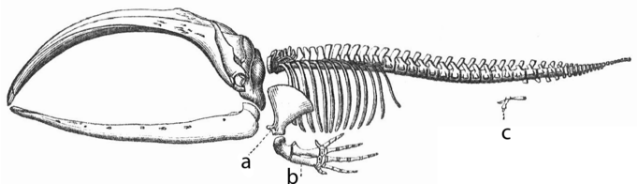

Figure 18.5H.1: **Whale Skeleton**: The pelvic bones in whales are also a good example of vestigial evolution (whales evolved from four-legged land mammals and secondarily lost their hind legs). Letter c in the picture indicates the undeveloped hind legs of a baleen whale.

If there are no selection pressures actively lowering the fitness of the individual, the trait will persist in future generations unless the trait is eliminated through genetic drift or other random events.

Although in many cases the vestigial structure is of no direct harm, all structures require extra energy in terms of development, maintenance, and weight and are also a risk in terms of disease (e.g., infection, cancer). This provides some selective pressure for the removal of parts that do not contribute to an organism's fitness, but a structure that is not directly harmful will take longer to be 'phased out' than one that is. Some vestigial structures persist due to limitations in development, such that complete loss of the structure could not occur without major alterations of the organism's developmental pattern, and such alterations would likely produce numerous negative side-effects.

The vestigial versions of a structure can be compared to the original version of the structure in other species in order to determine the homology of the structure. Homologous structures indicate common ancestry with those organisms that have a functional version of the structure. Vestigial traits can still be considered adaptations because an adaptation is often defined as a trait that has been favored by natural selection. Adaptations, therefore, need not be adaptive, as long as they were at some point.

18.5I: Biogeography and the Distribution of Species

The biological distribution of species is based on the movement of tectonic plates over a period of time.

Learning Objectives

- Relate biogeography and the distribution of species

Key Points

- Biogeography is the study of geological species distribution, which is influenced by both biotic and abiotic factors.
- Some species are endemic and are only found in a particular region, while others are generalists and are distributed worldwide.
- Species that evolved before the breakup of continents are distributed worldwide.
- Species that evolved after the breakup of continents are found in only certain regions of the planet.

Key Terms

- **endemic**: unique to a particular area or region; not found in other places
- **generalist**: species which can thrive in a wide variety of environmental conditions
- **Pangaea**: supercontinent that included all the landmasses of the earth before the Triassic period and that broke up into Laurasia and Gondwana

Distribution of Species

Biogeography is the study of the geographic distribution of living things and the abiotic factors that affect their distribution. Abiotic factors, such as temperature and rainfall, vary based on latitude and elevation, primarily. As these abiotic factors change, the composition of plant and animal communities also changes.

Patterns of Species Distribution

Ecologists who study biogeography examine patterns of species distribution. No species exists everywhere; for example, the Venus flytrap is endemic to a small area in North and South Carolina. An endemic species is one which is naturally found only in a specific geographic area that is usually restricted in size. Other species are generalists: species which live in a wide variety of geographic areas; the raccoon, for example, is native to most of North and Central America.

Since species distribution patterns are based on biotic and abiotic factors and their influences during the very long periods of time required for species evolution, early studies of biogeography were closely linked to the emergence of evolutionary thinking in the eighteenth century. Some of the most distinctive assemblages of plants and animals occur in regions that have been physically separated for millions of years by geographic barriers. Biologists estimate that Australia, for example, has between 600,000 and 700,000 species of plants and animals. Approximately 3/4 of living plant and mammal species are endemic species found solely in Australia.

Figure 18.5I.1: **Australia**: Australia is home to many endemic species. The (a) wallaby (*Wallabia bicolor*), a medium-sized member of the kangaroo family, is a pouched mammal, or marsupial. The (b) echidna (*Tachyglossus aculeatus*) is an egg-laying mammal.

The geographic distribution of organisms on the planet follows patterns that are best explained by evolution in conjunction with the movement of tectonic plates over geological time. Broad groups that evolved before the breakup of the supercontinent Pangaea (about 200 million years ago) are distributed worldwide. Groups that evolved since the breakup appear uniquely in regions of the planet, such as the unique flora and fauna of northern continents that formed from the supercontinent Laurasia and of the southern continents that formed from the supercontinent Gondwana. The presence of Proteaceae in Australia, southern Africa, and South America is best explained by the plant family's presence there prior to the southern supercontinent Gondwana breaking up.

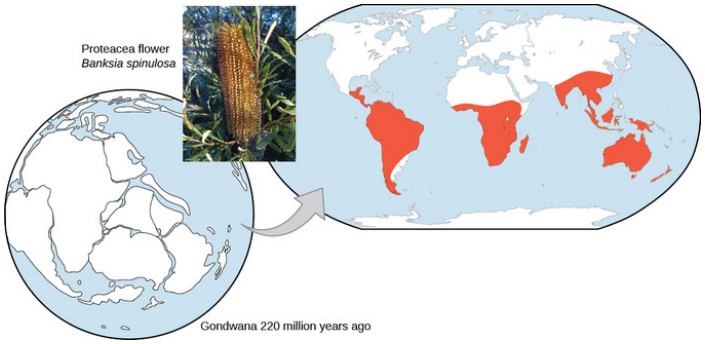

Figure 18.5*I*.1: **Biogeography**: The Proteacea family of plants evolved before the supercontinent Gondwana broke up. Today, members of this plant family are found throughout the southern hemisphere (shown in red).

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CHAPTER OVERVIEW

19: THE EVOLUTION OF POPULATIONS

19.1: POPULATION EVOLUTION

19.1A: DEFINING POPULATION EVOLUTION
19.1B: POPULATION GENETICS
19.1C: HARDY-WEINBERG PRINCIPLE OF EQUILIBRIUM

The Hardy-Weinberg principle can be used to estimate the frequency of alleles and genotypes in a population.

19.2: POPULATION GENETICS

19.2A: GENETIC VARIATION
19.2B: GENETIC DRIFT
19.2C: GENE FLOW AND MUTATION
19.2D: NONRANDOM MATING AND ENVIRONMENTAL VARIANCE
19.3: ADAPTIVE EVOLUTION

19.3A: NATURAL SELECTION AND ADAPTIVE EVOLUTION
19.3B: STABILIZING, DIRECTIONAL, AND DIVERSIFYING SELECTION
19.3C: FREQUENCY-DEPENDENT SELECTION
19.3D: SEXUAL SELECTION
19.3E: NO PERFECT ORGANISM

: Population Evolution

Topic hierarchy

- 19.1A: Defining Population Evolution
- 19.1B: Population Genetics
- 19.1C: Hardy-Weinberg Principle of Equilibrium

The Hardy-Weinberg principle can be used to estimate the frequency of alleles and genotypes in a population.

19.1A: Defining Population Evolution

Genetic variation in a population is determined by mutations, natural selection, genetic drift, genetic hitchhiking, and gene flow.

Learning Objectives

- Describe how the forces of genetic drift, genetic hitchhiking, gene flow, and mutation can lead to differences in population variation

Key Points

- The theory of evolution gives us a unifying theory to explain the similarities and differences within life's organisms and processes.
- Populations (or gene pools) evolve as gene frequencies change; individual organisms cannot evolve.
- Variation in populations is determined by the genes present in the population's gene pool, which may be directly altered by mutation.
- Natural selection is the gradual process that increases the frequency of advantageous inherited traits (allowing it to survive and reproduce) and decreases the frequency of detrimental inherited traits within a population.
- A population's genetic makeup can also be affected by random chance events like genetic drift, or when genes are inherited together in genetic hitchhiking.

Key Terms

- **gene flow**: the transfer of alleles or genes from one population to another
- **genetic hitchhiking**: a phenomenon in which a gene increases in a population because it lies near genes on the same chromosome that are advantageous to an organism
- **genetic drift**: an overall shift of allele distribution in an isolated population, due to random fluctuations in the frequencies of individual alleles of the genes
- **fitness**: an individual's ability to propagate its genes
- **natural selection**: a process in which individual organisms or phenotypes that possess favorable traits are more likely to survive and reproduce
- **mutation**: any heritable change of the base-pair sequence of genetic material

The Evolution of Populations

According to evolutionary theory, every organism from humans to beetles to plants to bacteria share a common ancestor. Millions of years of evolutionary pressure caused some organisms to died while others survived, leaving earth with the diverse life forms we have today. Within this diversity is unity; for example, all organisms are composed of cells and use DNA. The theory of evolution gives us a unifying theory to explain the similarities and differences within life's organisms and processes.

Figure 19.1A.1: **Evolution on earth**: Evolution has resulted in living things that may be single-celled or complex, multicellular organisms. They may be plants, animals, fungi, bacteria, or archaea. This diversity results from evolution.

Genetic Variation in Populations

A population is a group of individuals that can all interbreed, often distinguished as a species. Because these individuals can share genes and pass on combinations of genes to the next generation, the collection of these genes is called a gene pool. The process of evolution occurs *only* in populations and not in individuals. A single individual cannot evolve alone; evolution is the process of changing the gene frequencies within a gene pool. Five forces can cause genetic variation and evolution in a population: mutations, natural selection, genetic drift, genetic hitchhiking, and gene flow.

Mutations

Why do some organisms survive while others die? These surviving organisms generally possess traits or characteristics that bestow benefits that help them survive (e.g., better camouflage, faster swimming, or more efficient digestion). Each of these characteristics is the result of a mutation, or a change in the genetic code. Mutations occur spontaneously, but not all mutations are heritable; they are passed down to offspring only if the mutations occur in the gametes. These heritable mutations are responsible for the rise of new traits in a population.

Natural Selection

Just as mutations cause new traits in a population, natural selection acts on the frequency of those traits. Because there are more organisms than resources, all organisms are in a constant struggle for existence. In natural selection, those individuals with superior traits will be able to produce more offspring. The more offspring an organism can produce, the higher its fitness. As novel traits and behaviors arise from mutation, natural selection perpetuates the traits that confer a benefit.

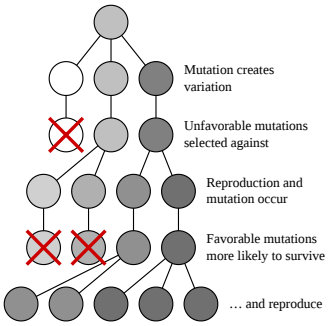

Figure 19.1A.1: **Mutation and natural selection**: As mutations create variation, natural selection affects the frequency of that trait in a population. Mutations that confer a benefit (such as running faster or digesting food more efficiently) can help that organism survive and reproduce, carrying the mutation to the next generation.

Genetic Drift

When selective forces are absent or relatively weak, gene frequencies tend to “drift” due to random events. This drift halts when the variation of the gene becomes “fixed” by either disappearing from the population or replacing the other variations completely. Even in the absence of selective forces, genetic drift can cause two separate populations that began with the same genetic structure to drift apart into two divergent populations.

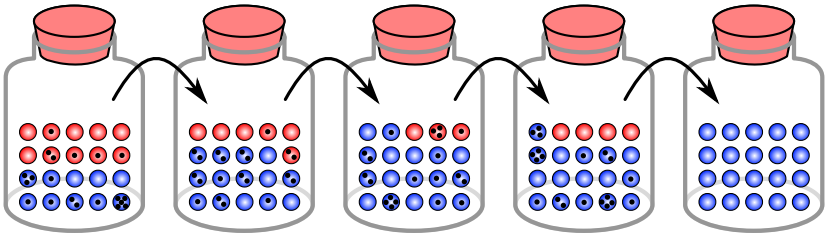

Figure 19.1A.1: **Genetic drift and gene fixation**: In this simulation, there is fixation in the blue gene variation within five generations. Images these dots are beetles and some of them are destroyed by a wildfire. As the surviving population changes over time, some traits (red) may be completely eliminated from the population, leaving only the beetles with other traits (blue).

Genetic Hitchhiking

When recombination occurs during sexual reproduction, genes are usually shuffled so that each parent gives its offspring a random assortment of its genetic variation. However, genes that are close together on the same chromosome are often assorted together. Therefore, the frequency of a gene may increase in a population through genetic hitchhiking if its proximal genes confer a benefit.

Gene Flow

Gene flow is the exchange of genes between populations or between species.If the gene pools between two populations are different, the exchange of genes can introduce variation that is advantageous or disadvantageous to one of the populations. If advantageous, this gene variation may replace all the other variations until the entire population exhibits that trait.

19.1B: Population Genetics

Population genetics is the study of the distributions and changes of allele frequency in a population.

Learning Objectives

- Define a population gene pool and explain how the size of the gene pool can affect the evolutionary success of a population

Key Points

- A gene pool is the sum of all the alleles (variants of a gene) in a population.
- Allele frequencies range from 0 (present in no individuals) to 1 (present in all individuals); all allele frequencies for a given gene add up to 100 percent in a population.
- The smaller a population, the more susceptible it is to mechanisms like natural selection and genetic drift, as the effects of such mechanisms are magnified when the gene pool is small.
- The founder effect occurs when part of an original population establishes a new population with a separate gene pool, leading to less genetic variation in the new population.

Key Terms

- **allele**: one of a number of alternative forms of the same gene occupying a given position on a chromosome
- **gene pool**: the complete set of unique alleles that would be found by inspecting the genetic material of every living member of a species or population
- **founder effect**: a decrease in genetic variation that occurs when an entire population descends from a small number of founders

Population Genetics

A gene for a particular characteristic may have several variations called alleles. These variations code for different traits associated with that characteristic. For example, in the ABO blood type system in humans, three alleles (I^A, I^B, or i) determine the particular blood-type protein on the surface of red blood cells. A human with a type I^A allele will display A-type proteins (antigens) on the surface of their red blood cells. Individuals with the phenotype of type A blood have the genotype I^AI^A or I^Ai, type B have I^BI^B or I^Bi, type AB have I^AI^B, and type O have ii.

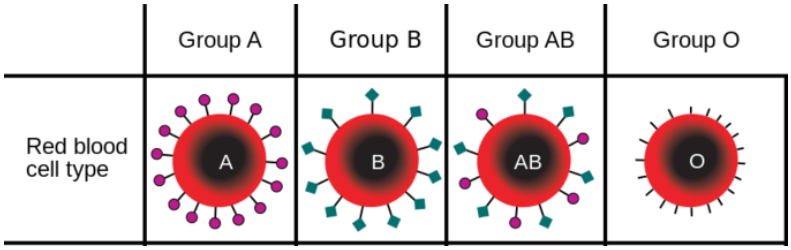

Figure 19.1*B*.1: **ABO blood type in humans**: In humans, each blood type corresponds to a combination of two alleles, which represent a the type of antigens displayed on the outside of a red blood cell. Human blood types are A, B, AB, and O.

A diploid organism can only carry two alleles for a particular gene. In human blood type, the combinations are composed of two alleles such as I^AI^A or I^AI^B. Although each organism can only carry two alleles, more than those two alleles may be present in the larger population. For example, in a population of fifty people where all the blood types are represented, there may be more I^A alleles than i alleles. Population genetics is the study of how selective forces change a population through changes in allele and genotypic frequencies.

Allele Frequency

The allele frequency (or gene frequency) is the rate at which a specific allele appears within a population. In population genetics, the term evolution is defined as a change in the frequency of an allele in a population. Frequencies range from 0,

present in no individuals, to 1, present in all individuals. The gene pool is the sum of all the alleles at all genes in a population.

Using the ABO blood type system as an example, the frequency of one of the alleles, for example I^A, is the number of copies of that allele divided by all the copies of the ABO gene in the population, i.e. all the alleles. Allele frequencies can be expressed as a decimal or as a percent and always add up to 1, or 100 percent, of the total population. For example, in a sample population of humans, the frequency of the I^A allele might be 0.26, which would mean that 26% of the chromosomes in that population carry the I^A allele. If we also know that the frequency of the I^B allele in this population is 0.14, then the frequency of the i allele is 0.6, which we obtain by subtracting all the known allele frequencies from 1 (thus: 1 – 0.26 – 0.14 = 0.6). A change in any of these allele frequencies over time would constitute evolution in the population.

Population Size and Evolution

When allele frequencies within a population change randomly with no advantage to the population over existing allele frequencies, the phenomenon is called genetic drift. The smaller a population, the more susceptible it is to mechanisms such as genetic drift as alleles are more likely to become fixed at 0 (absent) or 1 (universally present). Random events that alter allele frequencies will have a much larger effect when the gene pool is small. Genetic drift and natural selection usually occur simultaneously in populations, but the cause of the frequency change is often impossible to determine.

Natural selection also affects allele frequency. If an allele confers a phenotype that enables an individual to better survive or have more offspring, the frequency of that allele will increase. Because many of those offspring will also carry the beneficial allele and, therefore, the phenotype, they will have more offspring of their own that also carry the allele. Over time, the allele will spread throughout the population and may become fixed: every individual in the population carries the allele. If an allele is dominant but detrimental, it may be swiftly eliminated from the gene pool when the individual with the allele does not reproduce. However, a detrimental recessive allele can linger for generations in a population, hidden by the dominant allele in heterozygotes. In such cases, the only individuals to be eliminated from the population are those unlucky enough to inherit two copies of such an allele.

The Founder Effect

The founder effect occurs when part of a population becomes isolated and establishes a separate gene pool with its own allele frequencies. When a small number of individuals become the basis of a new population, this new population can be very different genetically from the original population if the founders are not representative of the original. Therefore, many different populations, with very different and uniform gene pools, can all originate from the same, larger population. Together, the forces of natural selection, genetic drift, and founder effect can lead to significant changes in the gene pool of a population.

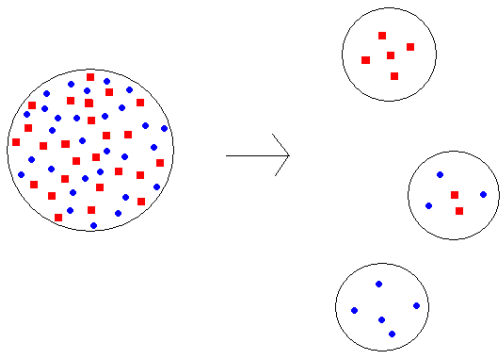

Figure 19.1*B*.1: **The Founder Effect**: Here are three possible outcomes of the founder effect, each with gene pools separate from the original populations.

19.1C: Hardy-Weinberg Principle of Equilibrium

LEARNING OBJECTIVES

- Use the Hardy Weinberg equation to calculate allelic and genotypic frequencies in a population

The Hardy-Weinberg principle states that a population's allele and genotype frequencies will remain constant in the absence of evolutionary mechanisms. Ultimately, the Hardy-Weinberg principle models a population without evolution under the following conditions:

1. no mutations
2. no immigration/emigration
3. no natural selection
4. no sexual selection
5. a large population

Although no real-world population can satisfy all of these conditions, the principle still offers a useful model for population analysis.

Hardy-Weinberg Equations and Analysis

According to the Hardy-Weinberg principle, the variable *p* often represents the frequency of a particular allele, usually a dominant one. For example, assume that *p* represents the frequency of the dominant allele, Y, for yellow pea pods. The variable *q* represents the frequency of the recessive allele, y, for green pea pods. If p and q are the only two possible alleles for this characteristic, then the sum of the frequencies must add up to 1, or 100 percent. We can also write this as p + q = 1.If the frequency of the Y allele in the population is 0.6, then we know that the frequency of the y allele is 0.4.

From the Hardy-Weinberg principle and the known allele frequencies, we can also infer the frequencies of the genotypes. Since each individual carries two alleles per gene (Y or y), we can predict the frequencies of these genotypes with a chi square. If two alleles are drawn at random from the gene pool, we can determine the probability of each genotype.

In the example, our three genotype possibilities are: pp (YY), producing yellow peas; pq (Yy), also yellow; or qq (yy), producing green peas. The frequency of homozygous pp individuals is p^2; the frequency of hereozygous pq individuals is 2pq; and the frequency of homozygous qq individuals is q^2. If p and q are the only two possible alleles for a given trait in the population, these genotypes frequencies will sum to one: $p^2 + 2pq + q^2 = 1$.

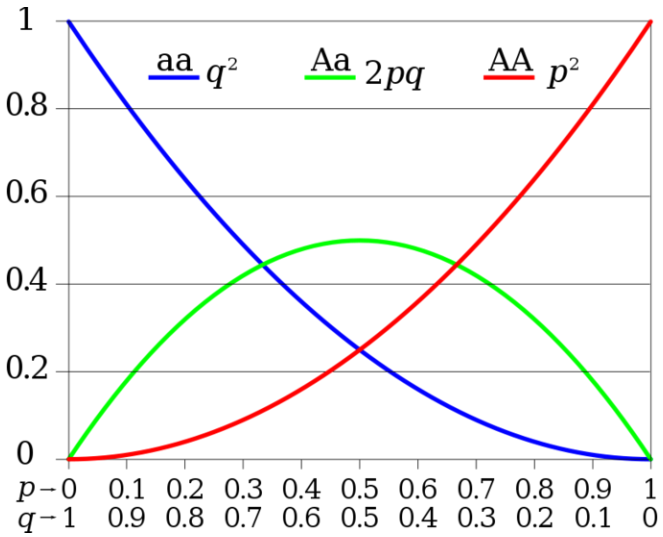

Figure 19.1C.1: **Hardy-Weinberg proportions for two alleles**: The horizontal axis shows the two allele frequencies *p* and *q* and the vertical axis shows the expected genotype frequencies.Each line shows one of the three possible genotypes.

In our example, the possible genotypes are homozygous dominant (YY), heterozygous (Yy), and homozygous recessive (yy). If we can only observe the phenotypes in the population, then we know only the recessive phenotype (yy). For

le, in a garden of 100 pea plants, 86 might have yellow peas and 16 have green peas. We do not know how many are ygous dominant (Yy) or heterozygous (Yy), but we do know that 16 of them are homozygous recessive (yy).

Therefore, by knowing the recessive phenotype and, thereby, the frequency of that genotype (16 out of 100 individuals or 0.16), we can calculate the number of other genotypes. If q^2 represents the frequency of homozygous recessive plants, then $q^2 = 0.16$. Therefore, q = 0.4.Because p + q = 1, then 1 – 0.4 = p, and we know that p = 0.6. The frequency of homozygous dominant plants (p^2) is $(0.6)^2 = 0.36$. Out of 100 individuals, there are 36 homozygous dominant (YY) plants. The frequency of heterozygous plants (2pq) is 2(0.6)(0.4) = 0.48. Therefore, 48 out of 100 plants are heterozygous yellow (Yy).

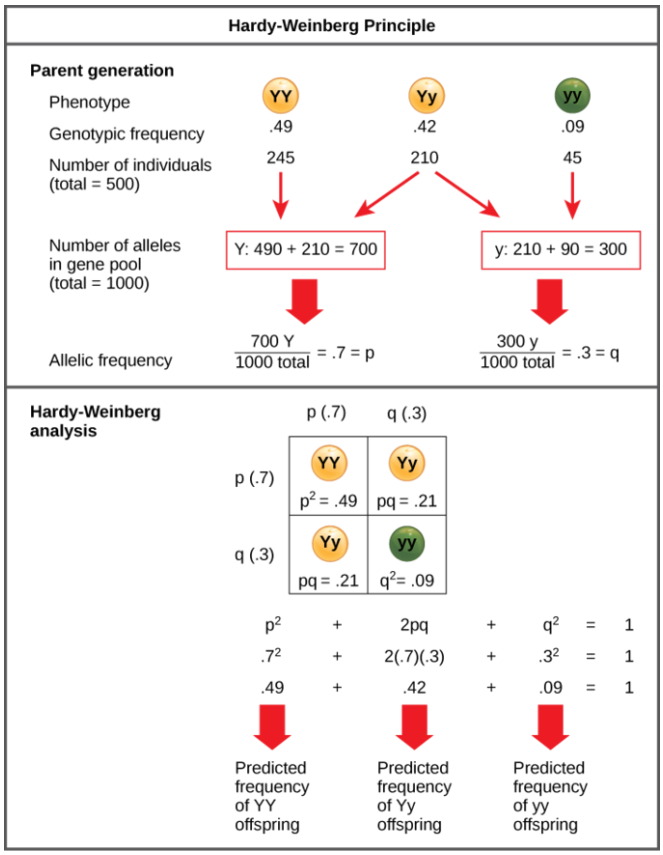

Figure 19.1C. 1: **The Hardy-Weinberg Principle**: When populations are in the Hardy-Weinberg equilibrium, the allelic frequency is stable from generation to generation and the distribution of alleles can be determined. If the allelic frequency measured in the field differs from the predicted value, scientists can make inferences about what evolutionary forces are at play.

Applications of Hardy-Weinberg

The genetic variation of natural populations is constantly changing from genetic drift, mutation, migration, and natural and sexual selection. The Hardy-Weinberg principle gives scientists a mathematical baseline of a non-evolving population to which they can compare evolving populations. If scientists record allele frequencies over time and then calculate the expected frequencies based on Hardy-Weinberg values, the scientists can hypothesize the mechanisms driving the population's evolution.

Key Points

- The Hardy-Weinberg principle assumes that in a given population, the population is large and is not experiencing mutation, migration, natural selection, or sexual selection.
- The frequency of alleles in a population can be represented by p + q = 1, with p equal to the frequency of the dominant allele and q equal to the frequency of the recessive allele.
- The frequency of genotypes in a population can be represented by $p^2+2pq+q^2= 1$, with p^2 equal to the frequency of the homozygous dominant genotype, 2pq equal to the frequency of the heterozygous genotype, and q^2 equal to the

quency of the recessive genotype.
e frequency of alleles can be estimated by calculating the frequency of the recessive genotype, then calculating the square root of that frequency in order to determine the frequency of the recessive allele.

Key Terms

- **genotype**: the combination of alleles, situated on corresponding chromosomes, that determines a specific trait of an individual, such as “Aa” or “aa”
- **phenotype**: the appearance of an organism based on a multifactorial combination of genetic traits and environmental factors, especially used in pedigrees

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19.2: Population Genetics

Topic hierarchy

- 19.2A: Genetic Variation
- 19.2B: Genetic Drift
- 19.2C: Gene Flow and Mutation
- 19.2D: Nonrandom Mating and Environmental Variance

19.2A: Genetic Variation

Genetic variation is a measure of the variation that exists in the genetic makeup of individuals within population.

Learning Objectives

- Assess the ways in which genetic variance affects the evolution of populations

Key Points

- Genetic variation is an important force in evolution as it allows natural selection to increase or decrease frequency of alleles already in the population.
- Genetic variation can be caused by mutation (which can create entirely new alleles in a population), random mating, random fertilization, and recombination between homologous chromosomes during meiosis (which reshuffles alleles within an organism's offspring).
- Genetic variation is advantageous to a population because it enables some individuals to adapt to the environment while maintaining the survival of the population.

Key Terms

- **genetic diversity**: the level of biodiversity, refers to the total number of genetic characteristics in the genetic makeup of a species
- **crossing over**: the exchange of genetic material between homologous chromosomes that results in recombinant chromosomes
- **phenotypic variation**: variation (due to underlying heritable genetic variation); a fundamental prerequisite for evolution by natural selection
- **genetic variation**: variation in alleles of genes that occurs both within and among populations

Genetic Variation

Genetic variation is a measure of the genetic differences that exist within a population. The genetic variation of an entire species is often called genetic diversity. Genetic variations are the differences in DNA segments or genes between individuals and each variation of a gene is called an allele.For example, a population with many different alleles at a single chromosome locus has a high amount of genetic variation. Genetic variation is essential for natural selection because natural selection can only increase or decrease frequency of alleles that already exist in the population.

Genetic variation is caused by:

- mutation
- random mating between organisms
- random fertilization
- crossing over (or recombination) between chromatids of homologous chromosomes during meiosis

The last three of these factors reshuffle alleles within a population, giving offspring combinations which differ from their parents and from others.

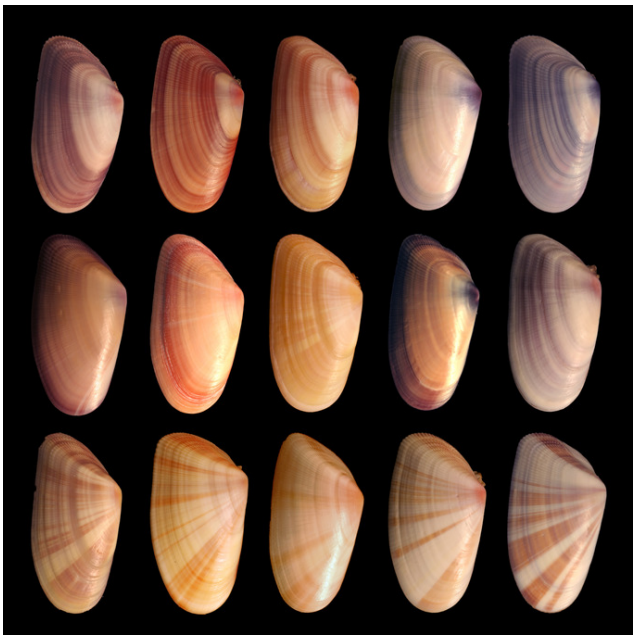

Figure 19.2A.1: **Genetic variation in the shells of Donax variabilis**: An enormous amount of phenotypic variation exists in the shells of Donax varabilis, otherwise known as the coquina mollusc. This phenotypic variation is due at least partly to genetic variation within the coquina population.

Evolution and Adaptation to the Environment

Variation allows some individuals within a population to adapt to the changing environment. Because natural selection acts directly only on phenotypes, more genetic variation within a population usually enables more phenotypic variation. Some new alleles increase an organism's ability to survive and reproduce, which then ensures the survival of the allele in the population. Other new alleles may be immediately detrimental (such as a malformed oxygen-carrying protein) and organisms carrying these new mutations will die out. Neutral alleles are neither selected for nor against and usually remain in the population. Genetic variation is advantageous because it enables some individuals and, therefore, a population, to survive despite a changing environment.

Figure 19.2A.1: **Low genetic diversity in the wild cheetah population**: Populations of wild cheetahs have very low genetic variation. Because wild cheetahs are threatened, their species has a very low genetic diversity. This low genetic diversity means they are often susceptible to disease and often pass on lethal recessive mutations; only about 5% of cheetahs survive to adulthood.

Geographic Variation

Some species display geographic variation as well as variation within a population. Geographic variation, or the distinctions in the genetic makeup of different populations, often occurs when populations are geographically separated by environmental barriers or when they are under selection pressures from a different environment. One example of geographic variation are clines: graded changes in a character down a geographic axis.

Sources of Genetic Variation

Gene duplication, mutation, or other processes can produce new genes and alleles and increase genetic variation. New genetic variation can be created within generations in a population, so a population with rapid reproduction rates will probably have high genetic variation. However, existing genes can be arranged in new ways from chromosomal crossing over and recombination in sexual reproduction. Overall, the main sources of genetic variation are the formation of new alleles, the altering of gene number or position, rapid reproduction, and sexual reproduction.

19.2B: Genetic Drift

Genetic drift is the change in allele frequencies of a population due to random chance events, such as natural disasters.

Learning Objectives

- Distinguish between selection and genetic drift

Key Points

- Genetic drift is the change in the frequency of an allele in a population due to random sampling and the random events that influence the survival and reproduction of those individuals.
- The bottleneck effect occurs when a natural disaster or similar event randomly kills a large portion (i.e. random sample) of the population, leaving survivors that have allele frequencies that were very different from the previous population.
- The founder effect occurs when a portion of the population (i.e. “founders”) separates from the old population to start a new population with different allele frequencies.
- Small populations are more susceptible genetic drift than large populations, whose larger numbers can buffer the population against chance events.

Key Terms

- **genetic drift**: an overall shift of allele distribution in an isolated population, due to random sampling
- **founder effect**: a decrease in genetic variation that occurs when an entire population descends from a small number of founders
- **random sampling**: a subset of individuals (a sample) chosen from a larger set (a population) by chance

Genetic Drift vs. Natural Selection

Genetic drift is the converse of natural selection. The theory of natural selection maintains that some individuals in a population have traits that enable to survive and produce more offspring, while other individuals have traits that are detrimental and may cause them to die before reproducing. Over successive generation, these selection pressures can change the gene pool and the traits within the population. For example, a big, powerful male gorilla will mate with more females than a small, weak male and therefore more of his genes will be passed on to the next generation. His offspring may continue to dominate the troop and pass on their genes as well. Over time, the selection pressure will cause the allele frequencies in the gorilla population to shift toward large, strong males.

Unlike natural selection, genetic drift describes the effect of chance on populations in the absence of positive or negative selection pressure. Through random sampling, or the survival or and reproduction of a random sample of individuals within a population, allele frequencies within a population may change. Rather than a male gorilla producing more offspring because he is stronger, he may be the only male available when a female is ready to mate. His genes are passed on to future generation because of chance, not because he was the biggest or the strongest. Genetic drift is the shift of alleles within a population due to chance events that cause random samples of the population to reproduce or not.

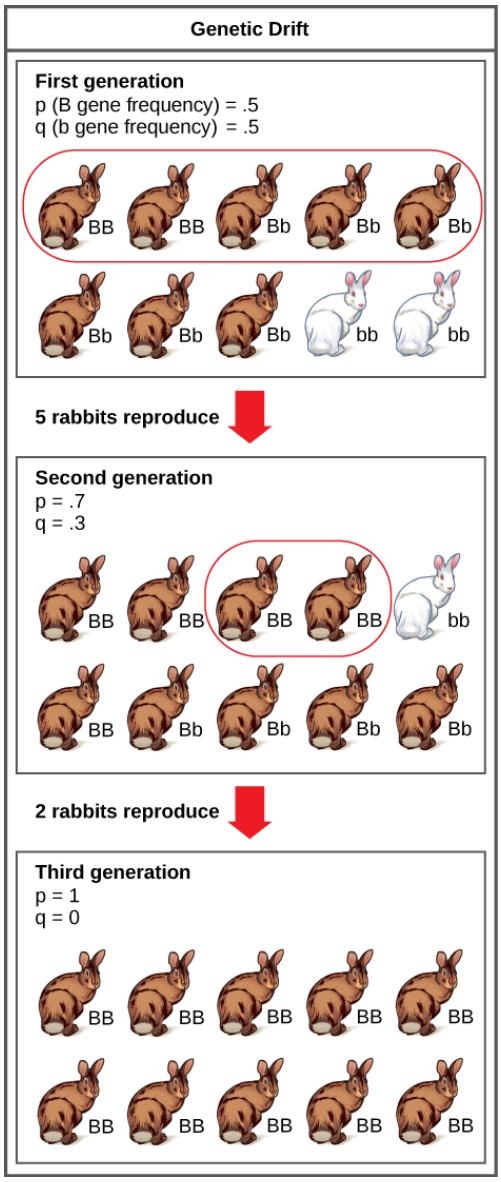

Figure 19.2B.1: **Effect of genetic drift**: Genetic drift in a population can lead to the elimination of an allele from that population by chance. In this example, the brown coat color allele (B) is dominant over the white coat color allele (b). In the first generation, the two alleles occur with equal frequency in the population, resulting in p and q values of.5. Only half of the individuals reproduce, resulting in a second generation with p and q values of.7 and.3, respectively. Only two individuals in the second generation reproduce and, by chance, these individuals are homozygous dominant for brown coat color. As a result, in the third generation the recessive b allele is lost.

Small populations are more susceptible to the forces of genetic drift. Large populations, on the other hand, are buffered against the effects of chance. If one individual of a population of 10 individuals happens to die at a young age before leaving any offspring to the next generation, all of its genes (1/10 of the population's gene pool) will be suddenly lost. In a population of 100, that individual represents only 1 percent of the overall gene pool; therefore, genetic drift has much less impact on the larger population's genetic structure.

The Bottleneck Effect

Genetic drift can also be magnified by natural events, such as a natural disaster that kills a large portion of the population at random. The bottleneck effect occurs when only a few individuals survive and reduces variation in the gene pool of a population. The genetic structure of the survivors becomes the genetic structure of the entire population, which may be very different from the pre-disaster population.

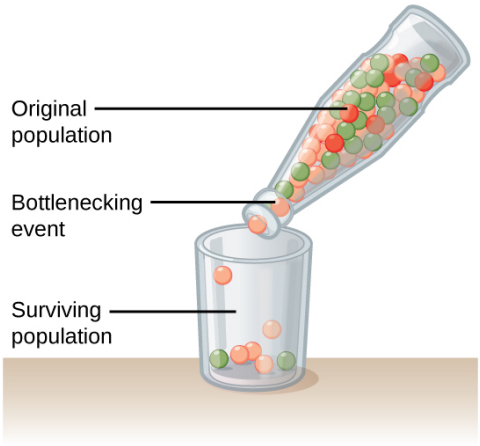

Figure 19.2B.1: **Effect of a bottleneck on a population**: A chance event or catastrophe can reduce the genetic variability within a population.

The Founder Effect

Another scenario in which populations might experience a strong influence of genetic drift is if some portion of the population leaves to start a new population in a new location or if a population gets divided by a physical barrier of some kind. In this situation, it is improbable that those individuals are representative of the entire population, which results in the founder effect. The founder effect occurs when the genetic structure changes to match that of the new population's founding fathers and mothers.

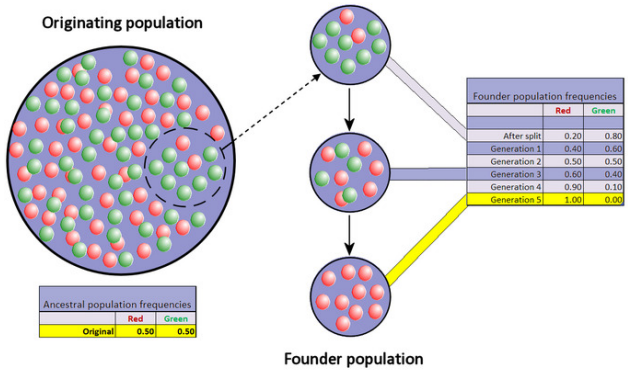

Figure 19.2B.1: **The Founder Effect**: The founder effect occurs when a portion of the population (i.e. "founders") separates from the old population to start a new population with different allele frequencies.

The founder effect is believed to have been a key factor in the genetic history of the Afrikaner population of Dutch settlers in South Africa, as evidenced by mutations that are common in Afrikaners, but rare in most other populations. This was probably due to the fact that a higher-than-normal proportion of the founding colonists carried these mutations. As a result, the population expresses unusually high incidences of Huntington's disease (HD) and Fanconi anemia (FA), a genetic disorder known to cause blood marrow and congenital abnormalities, even cancer.

Drift and fixation

The Hardy–Weinberg principle states that within sufficiently large populations, the allele frequencies remain constant from one generation to the next unless the equilibrium is disturbed by migration, genetic mutation, or selection.

Because the random sampling can remove, but not replace, an allele, and because random declines or increases in allele frequency influence expected allele distributions for the next generation, genetic drift drives a population towards genetic uniformity over time. When an allele reaches a frequency of 1 (100%) it is said to be "fixed" in the population and when an allele reaches a frequency of 0 (0%) it is lost. Once an allele becomes fixed, genetic drift for that allele comes to a halt, and the allele frequency cannot change unless a new allele is introduced in the population via mutation or gene flow. Thus even while genetic drift is a random, directionless process, it acts to eliminate genetic variation over time.

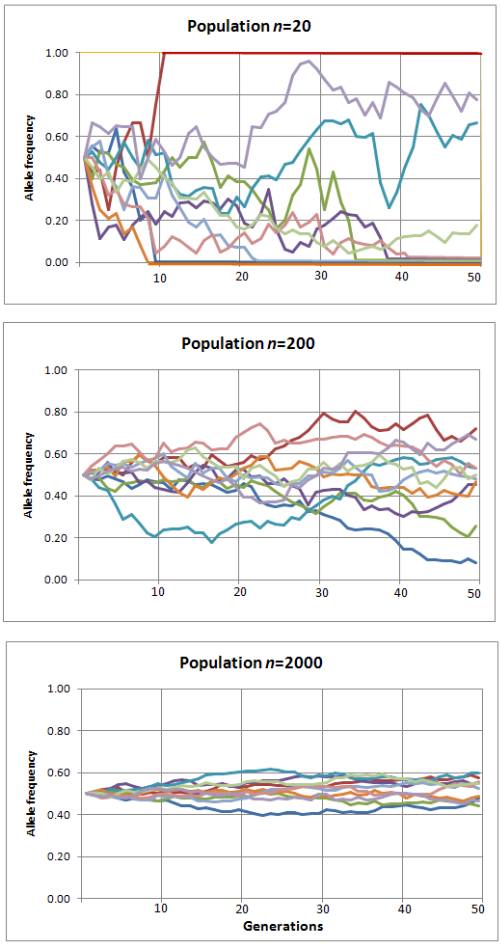

Figure 19.2B.1: **Genetic drift over time**: Ten simulations of random genetic drift of a single given allele with an initial frequency distribution 0.5 measured over the course of 50 generations, repeated in three reproductively synchronous populations of different sizes. In these simulations, alleles drift to loss or fixation (frequency of 0.0 or 1.0) only in the smallest population.Effect of population size on genetic drift: Ten simulations each of random change in the frequency distribution of a single hypothetical allele over 50 generations for different sized populations; first population size n=20, second population n=200, and third population n=2000.

19.2C: Gene Flow and Mutation

A population's genetic variation changes as individuals migrate into or out of a population and when mutations introduce new alleles.

Learning Objectives

- Explain how gene flow and mutations can influence the allele frequencies of a population

Key Points

- Plant populations experience gene flow by spreading their pollen long distances.
- Animals experience gene flow when individuals leave a family group or herd to join other populations.
- The flow of individuals in and out of a population introduces new alleles and increases genetic variation within that population.
- Mutations are changes to an organism's DNA that create diversity within a population by introducing new alleles.
- Some mutations are harmful and are quickly eliminated from the population by natural selection; harmful mutations prevent organisms from reaching sexual maturity and reproducing.
- Other mutations are beneficial and can increase in a population if they help organisms reach sexual maturity and reproduce.

Key Terms

- **gene flow**: the transfer of alleles or genes from one population to another
- **mutation**: any heritable change of the base-pair sequence of genetic material

Gene Flow

An important evolutionary force is gene flow: the flow of alleles in and out of a population due to the migration of individuals or gametes. While some populations are fairly stable, others experience more movement and fluctuation. Many plants, for example, send their pollen by wind, insects, or birds to pollinate other populations of the same species some distance away. Even a population that may initially appear to be stable, such as a pride of lions, can receive new genetic variation as developing males leave their mothers to form new prides with genetically-unrelated females. This variable flow of individuals in and out of the group not only changes the gene structure of the population, but can also introduce new genetic variation to populations in different geological locations and habitats.

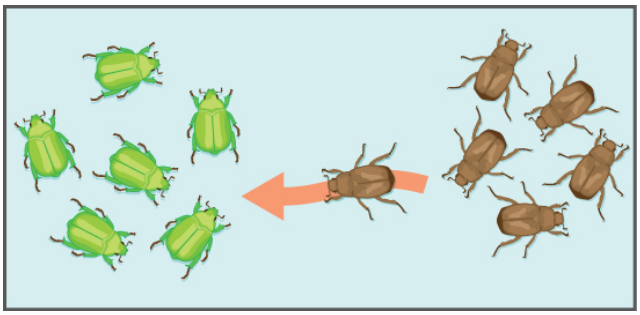

Figure 19.2C.1: **Gene flow**: Gene flow can occur when an individual travels from one geographic location to another.

Maintained gene flow between two populations can also lead to a combination of the two gene pools, reducing the genetic variation between the two groups. Gene flow strongly acts against speciation, by recombining the gene pools of the groups, and thus, repairing the developing differences in genetic variation that would have led to full speciation and creation of daughter species.

For example, if a species of grass grows on both sides of a highway, pollen is likely to be transported from one side to the other and vice versa. If this pollen is able to fertilize the plant where it ends up and produce viable offspring, then the alleles in the pollen have effectively linked the population on one side of the highway with the other.

Mutation

Mutations are changes to an organism's DNA and are an important driver of diversity in populations. Species evolve because of the accumulation of mutations that occur over time. The appearance of new mutations is the most common way to introduce novel genotypic and phenotypic variance. Some mutations are unfavorable or harmful and are quickly eliminated from the population by natural selection. Others are beneficial and will spread through the population. Whether or not a mutation is beneficial or harmful is determined by whether it helps an organism survive to sexual maturity and reproduce. Some mutations have no effect on an organism and can linger, unaffected by natural selection, in the genome while others can have a dramatic effect on a gene and the resulting phenotype.

Figure 19.2C.1: **Mutation in a garden rose**: A mutation has caused this garden moss rose to produce flowers of different colors. This mutation has introduce a new allele into the population that increases genetic variation and may be passed on to the next generation.

19.2D: Nonrandom Mating and Environmental Variance

Population structure can be altered by nonrandom mating (the preference of certain individuals for mates) as well as the environment.

Learning Objectives

- Explain how environmental variance and nonrandom mating can change gene frequencies in a population

Key Points

- Nonrandom mating can occur when individuals prefer mates with particular superior physical characteristics or by the preference of individuals to mate with individuals similar to themselves.
- Nonrandom mating can also occur when mates are chosen based on physical accessibility; that is, the availability of some mates over others.
- Phenotypes of individuals can also be influenced by the environment in which they live, such as temperature, terrain, or other factors.
- A cline occurs when populations of a given species vary gradually across an ecological gradient.

Key Terms

- **cline**: a gradation in a character or phenotype within a species or other group
- **sexual selection**: a mode of natural selection in which some individuals out-reproduce others of a population because they are better at securing mates
- **assortative mating**: between males and females of a species, the mutual attraction or selection, for reproductive purposes, of individuals with similar characteristics

Nonrandom Mating

If individuals nonrandomly mate with other individuals in the population, i.e. they choose their mate, choices can drive evolution within a population. There are many reasons nonrandom mating occurs. One reason is simple mate choice or sexual selection; for example, female peahens may prefer peacocks with bigger, brighter tails. Traits that lead to more matings for an individual lead to more offspring and through natural selection, eventually lead to a higher frequency of that trait in the population. One common form of mate choice, called positive assortative mating, is an individual's preference to mate with partners that are phenotypically similar to themselves.

Figure $19.2D.1$: **Assortative mating in the American Robin**: The American Robin may practice assortative mating on plumage color, a melanin based trait, and mate with other robins who have the most similar shade of color. However, there may also be some sexual selection for more vibrant plumage which indicates health and reproductive performance.

Another cause of nonrandom mating is physical location. This is especially true in large populations spread over large geographic distances where not all individuals will have equal access to one another. Some might be miles apart through

woods or over rough terrain, while others might live immediately nearby.

Environmental Variance

Genes are not the only players involved in determining population variation. Phenotypes are also influenced by other factors, such as the environment. A beachgoer is likely to have darker skin than a city dweller, for example, due to regular exposure to the sun, an environmental factor. Some major characteristics, such as gender, are determined by the environment for some species. For example, some turtles and other reptiles have temperature-dependent sex determination (TSD). TSD means that individuals develop into males if their eggs are incubated within a certain temperature range, or females at a different temperature range.

Figure 19.2D.1: **Temperature-dependent sex determination**: The sex of the American alligator (Alligator mississippiensis) is determined by the temperature at which the eggs are incubated. Eggs incubated at 30 degrees C produce females, and eggs incubated at 33 degrees C produce males.

Geographic separation between populations can lead to differences in the phenotypic variation between those populations. Such geographical variation is seen between most populations and can be significant. One type of geographic variation, called a cline, can be seen as populations of a given species vary gradually across an ecological gradient.

image

Geographic variation in moose: This graph shows geographical variation in moose; body mass increase positively with latitude. Bergmann's Rule is an ecologic principle which states that as latitude increases the body mass of a particular species increases. The data are taken from a Swedish study investigating the size of moose as latitude increases as shows the positive relationship between the two, supporting Bergmann's Rule.

Species of warm-blooded animals, for example, tend to have larger bodies in the cooler climates closer to the earth's poles, allowing them to better conserve heat. This is considered a latitudinal cline. Alternatively, flowering plants tend to bloom at different times depending on where they are along the slope of a mountain, known as an altitudinal cline.

If there is gene flow between the populations, the individuals will likely show gradual differences in phenotype along the cline. Restricted gene flow, on the other hand, can lead to abrupt differences, even speciation.

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19.3: Adaptive Evolution

Topic hierarchy

- 19.3A: Natural Selection and Adaptive Evolution
- 19.3B: Stabilizing, Directional, and Diversifying Selection
- 19.3C: Frequency-Dependent Selection
- 19.3D: Sexual Selection
- 19.3E: No Perfect Organism

19.3A: Natural Selection and Adaptive Evolution

Natural selection drives adaptive evolution by selecting for and increasing the occurrence of beneficial traits in a population.

Learning Objectives

- Explain how natural selection leads to adaptive evolution

Key Points

- Natural selection increases or decreases biological traits within a population, thereby selecting for individuals with greater evolutionary fitness.
- An individual with a high evolutionary fitness will provide more beneficial contributions to the gene pool of the next generation.
- Relative fitness, which compares an organism's fitness to the others in the population, allows researchers to establish how a population may evolve by determining which individuals are contributing additional offspring to the next generation.
- Stabilizing selection, directional selection, diversifying selection, frequency -dependent selection, and sexual selection all contribute to the way natural selection can affect variation within a population.

Key Terms

- **natural selection**: a process in which individual organisms or phenotypes that possess favorable traits are more likely to survive and reproduce
- **fecundity**: number, rate, or capacity of offspring production
- **Darwinian fitness**: the average contribution to the gene pool of the next generation that is made by an average individual of the specified genotype or phenotype

An Introduction to Adaptive Evolution

Natural selection only acts on the population's heritable traits: selecting for beneficial alleles and, thus, increasing their frequency in the population, while selecting against deleterious alleles and, thereby, decreasing their frequency. This process is known as adaptive evolution. Natural selection does not act on individual alleles, however, but on entire organisms. An individual may carry a very beneficial genotype with a resulting phenotype that, for example, increases the ability to reproduce (fecundity), but if that same individual also carries an allele that results in a fatal childhood disease, that fecundity phenotype will not be passed on to the next generation because the individual will not live to reach reproductive age. Natural selection acts at the level of the individual; it selects for individuals with greater contributions to the gene pool of the next generation, known as an organism's evolutionary fitness (or Darwinian fitness).

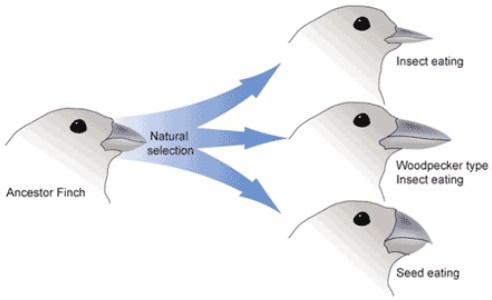


Figure 19.3A.1: **Adaptive evolution in finches**: Through natural selection, a population of finches evolved into three separate species by adapting to several difference selection pressures. Each of the three modern finches has a beak adapted to its life history and diet.

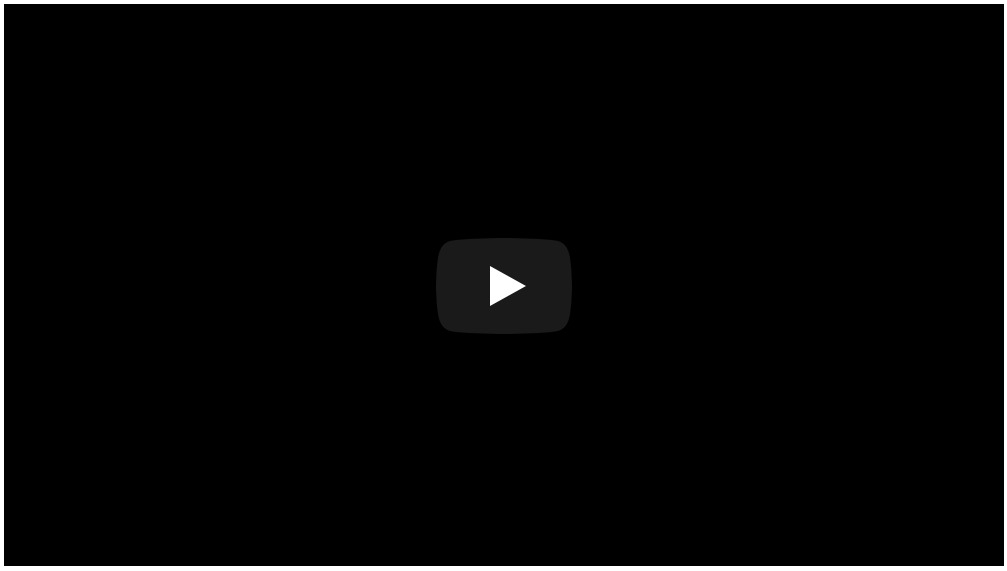
Fitness is often quantifiable and is measured by scientists in the field. However, it is not the absolute fitness of an individual that counts, but rather how it compares to the other organisms in the population. This concept, called relative

fitness, allows researchers to determine which individuals are contributing additional offspring to the next generation and, thus, how the population might evolve.

There are several ways selection can affect population variation:

- stabilizing selection
- directional selection
- diversifying selection
- frequency-dependent selection
- sexual selection

As natural selection influences the allele frequencies in a population, individuals can either become more or less genetically similar and the phenotypes displayed can become more similar or more disparate. In the end, natural selection cannot produce perfect organisms from scratch, it can only generate populations that are better adapted to survive and successfully reproduce in their environments through the aforementioned selections.



Galápagos with David Attenborough: Two hundred years after Charles Darwin set foot on the shores of the Galápagos Islands, David Attenborough travels to this wild and mysterious archipelago. Amongst the flora and fauna of these enchanted volcanic islands, Darwin formulated his groundbreaking theories on evolution. Journey with Attenborough to explore how life on the islands has continued to evolve in biological isolation, and how the ever-changing volcanic landscape has given birth to species and sub-species that exist nowhere else in the world.

19.3B: Stabilizing, Directional, and Diversifying Selection

Stabilizing, directional, and diversifying selection either decrease, shift, or increase the genetic variance of a population.

Learning Objectives

- Contrast stabilizing selection, directional selection, and diversifying selection.

Key Points

- Stabilizing selection results in a decrease of a population ‘s genetic variance when natural selection favors an average phenotype and selects against extreme variations.
- In directional selection, a population’s genetic variance shifts toward a new phenotype when exposed to environmental changes.
- Diversifying or disruptive selection increases genetic variance when natural selection selects for two or more extreme phenotypes that each have specific advantages.
- In diversifying or disruptive selection, average or intermediate phenotypes are often less fit than either extreme phenotype and are unlikely to feature prominently in a population.

Key Terms

- **directional selection**: a mode of natural selection in which a single phenotype is favored, causing the allele frequency to continuously shift in one direction
- **disruptive selection**: (or diversifying selection) a mode of natural selection in which extreme values for a trait are favored over intermediate values
- **stabilizing selection**: a type of natural selection in which genetic diversity decreases as the population stabilizes on a particular trait value

Stabilizing Selection

If natural selection favors an average phenotype by selecting against extreme variation, the population will undergo stabilizing selection. For example, in a population of mice that live in the woods, natural selection will tend to favor individuals that best blend in with the forest floor and are less likely to be spotted by predators. Assuming the ground is a fairly consistent shade of brown, those mice whose fur is most-closely matched to that color will most probably survive and reproduce, passing on their genes for their brown coat. Mice that carry alleles that make them slightly lighter or slightly darker will stand out against the ground and will more probably die from predation. As a result of this stabilizing selection, the population’s genetic variance will decrease.

Stabilizing selection: Stabilizing selection occurs when the population stabilizes on a particular trait value and genetic diversity decreases.

Directional Selection

When the environment changes, populations will often undergo directional selection, which selects for phenotypes at one end of the spectrum of existing variation.

A classic example of this type of selection is the evolution of the peppered moth in eighteenth- and nineteenth-century England. Prior to the Industrial Revolution, the moths were predominately light in color, which allowed them to blend in with the light-colored trees and lichens in their environment. As soot began spewing from factories, the trees darkened and the light-colored moths became easier for predatory birds to spot.

Directional selection: Directional selection occurs when a single phenotype is favored, causing the allele frequency to continuously shift in one direction.

Over time, the frequency of the melanic form of the moth increased because their darker coloration provided camouflage against the sooty tree; they had a higher survival rate in habitats affected by air pollution. Similarly, the hypothetical mouse

population may evolve to take on a different coloration if their forest floor habitat changed. The result of this type of selection is a shift in the population's genetic variance toward the new, fit phenotype.

Figure $19.3B.1$: **The Evolution of the Peppered Moth**: Typica and carbonaria morphs resting on the same tree.The light-colored typica (below the bark's scar) is nearly invisible on this pollution-free tree, camouflaging it from predators.

Diversifying (or Disruptive) Selection

Sometimes natural selection can select for two or more distinct phenotypes that each have their advantages. In these cases, the intermediate phenotypes are often less fit than their extreme counterparts. Known as diversifying or disruptive selection, this is seen in many populations of animals that have multiple male mating strategies, such as lobsters. Large, dominant alpha males obtain mates by brute force, while small males can sneak in for furtive copulations with the females in an alpha male's territory. In this case, both the alpha males and the "sneaking" males will be selected for, but medium-sized males, which cannot overtake the alpha males and are too big to sneak copulations, are selected against.

image

Diversifying (or disruptive) selection: Diversifying selection occurs when extreme values for a trait are favored over the intermediate values.This type of selection often drives speciation.

Diversifying selection can also occur when environmental changes favor individuals on either end of the phenotypic spectrum. Imagine a population of mice living at the beach where there is light-colored sand interspersed with patches of tall grass. In this scenario, light-colored mice that blend in with the sand would be favored, as well as dark-colored mice that can hide in the grass. Medium-colored mice, on the other hand, would not blend in with either the grass or the sand and, thus, would more probably be eaten by predators. The result of this type of selection is increased genetic variance as the population becomes more diverse.

Comparing Types of Natural Selection

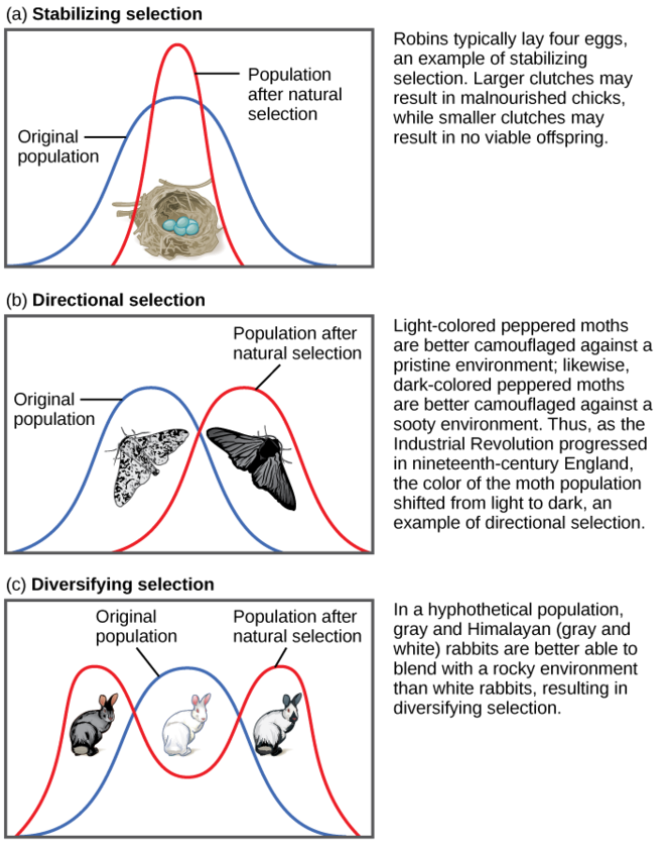


Figure 19.3B.1: **Types of natural selection**: Different types of natural selection can impact the distribution of phenotypes within a population.In (a) stabilizing selection, an average phenotype is favored.In (b) directional selection, a change in the environment shifts the spectrum of phenotypes observed.In (c) diversifying selection, two or more extreme phenotypes are selected for, while the average phenotype is selected against.

19.3C: Frequency-Dependent Selection

In frequency-dependent selection, phenotypes that are either common or rare are favored through natural selection.

Learning Objectives

- Describe frequency-dependent selection

Key Points

- Negative frequency -dependent selection selects for rare phenotypes in a population and increases a population's genetic variance.
- Positive frequency-dependent selection selects for common phenotypes in a population and decreases genetic variance.
- In the example of male side-blotched lizards, populations of each color pattern increase or decrease at various stages depending on their frequency; this ensures that both common and rare phenotypes continue to be cyclically present.
- Infectious agents such as microbes can exhibit negative frequency-dependent selection; as a host population becomes immune to a common strain of the microbe, less common strains of the microbe are automatically favored.
- Variation in color pattern mimicry by the scarlet kingsnake is dependent on the prevalence of the eastern coral snake, the model for this mimicry, in a particular geographical region. The more prevalent the coral snake is in a region, the more common and variable the scarlet kingsnake's color pattern will be, making this an example of positive frequency-dependent selection.

Key Terms

- **frequency-dependent selection**: the term given to an evolutionary process where the fitness of a phenotype is dependent on its frequency relative to other phenotypes in a given population
- **polygynous**: having more than one female as mate

Frequency-dependent Selection

Another type of selection, called frequency-dependent selection, favors phenotypes that are either common (positive frequency-dependent selection) or rare (negative frequency-dependent selection).

Negative Frequency-dependent Selection

An interesting example of this type of selection is seen in a unique group of lizards of the Pacific Northwest. Male common side-blotched lizards come in three throat-color patterns: orange, blue, and yellow. Each of these forms has a different reproductive strategy: orange males are the strongest and can fight other males for access to their females; blue males are medium-sized and form strong pair bonds with their mates; and yellow males are the smallest and look a bit like female, allowing them to sneak copulations. Like a game of rock-paper-scissors, orange beats blue, blue beats yellow, and yellow beats orange in the competition for females. The big, strong orange males can fight off the blue males to mate with the blue's pair-bonded females; the blue males are successful at guarding their mates against yellow sneaker males; and the yellow males can sneak copulations from the potential mates of the large, polygynous orange males.

Figure 19.3C.1: **Frequency-dependent selection in side-blotched lizards**: A yellow-throated side-blotched lizard is smaller than either the blue-throated or orange-throated males and appears a bit like the females of the species, allowing it to sneak copulations. Frequency-dependent selection allows for both common and rare phenotypes of the population to appear in a frequency-aided cycle.

In this scenario, orange males will be favored by natural selection when the population is dominated by blue males, blue males will thrive when the population is mostly yellow males, and yellow males will be selected for when orange males are the most populous. As a result, populations of side-blotched lizards cycle in the distribution of these phenotypes. In one generation, orange might be predominant and then yellow males will begin to rise in frequency. Once yellow males make up a majority of the population, blue males will be selected for.Finally, when blue males become common, orange males will once again be favored.

An example of negative frequency-dependent selection can also be seen in the interaction between the human immune system and various infectious microbes such as pathogenic bacteria or viruses. As a particular human population is infected by a common strain of microbe, the majority of individuals in the population become immune to it. This then selects for rarer strains of the microbe which can still infect the population because of genome mutations; these strains have greater evolutionary fitness because they are less common.

Positive Frequency-dependent Selection

An example of positive frequency-dependent selection is the mimicry of the warning coloration of dangerous species of animals by other species that are harmless. The scarlet kingsnake, a harmless species, mimics the coloration of the eastern coral snake, a venomous species typically found in the same geographical region. Predators learn to avoid both species of snake due to the similar coloration, and as a result the scarlet kingsnake becomes more common, and its coloration phenotype becomes more variable due to relaxed selection. This phenotype is therefore more “fit” as the population of species that possess it (both dangerous and harmless) becomes more numerous. In geographic areas where the coral snake is less common, the pattern becomes less advantageous to the kingsnake, and much less variable in its expression, presumably because predators in these regions are not “educated” to avoid the pattern.

Figure 19.3C.1: **Lampropeltis elapsoides, the scarlet kingsnake**: The scarlet kingsnake mimics the coloration of the poisonous eastern coral snake. Positive frequency-dependent selection reinforces the common phenotype because predators avoid the distinct coloration.

Figure 19.3C.1: **Micrurus fulvius, the eastern coral snake**: The eastern coral snake is poisonous.

Negative frequency-dependent selection serves to increase the population's genetic variance by selecting for rare phenotypes, whereas positive frequency-dependent selection usually decreases genetic variance by selecting for common phenotypes.

19.3D: Sexual Selection

Sexual selection, the selection pressure on males and females to obtain matings, can result in traits designed to maximize sexual success.

Learning Objectives

- Discuss the effects of sexual dimorphism on the reproductive potential of an organism

Key Points

- Sexual selection often results in the development of secondary sexual characteristics, which help to maximize a species ' reproductive success, but do not provide any survival benefits.
- The handicap principle states that only the best males survive the risks from traits that may actually be detrimental to a species; therefore, they are more fit as mating partners.
- In the good genes hypothesis, females will choose males that show off impressive traits to ensure they pass on genetic superiority to their offspring.
- Sexual dimorphisms, obvious morphological differences between the sexes of a species, arise when there is more variance in the reproductive success of either males or females.

Key Terms

- **sexual dimorphism**: a physical difference between male and female individuals of the same species
- **sexual selection**: a type of natural selection, where members of the sexes acquire distinct forms because members choose mates with particular features or because competition for mates with certain traits succeed
- **handicap principle**: a theory that suggests that animals of greater biological fitness signal this status through a behavior or morphology that effectively lowers their chances of survival

Sexual Selection

The selection pressures on males and females to obtain matings is known as sexual selection. Sexual selection takes two major forms: intersexual selection (also known as 'mate choice' or 'female choice') in which males compete with each other to be chosen by females; and intrasexual selection (also known as 'male–male competition') in which members of the less limited sex (typically males) compete aggressively among themselves for access to the limiting sex. The limiting sex is the sex which has the higher parental investment, which therefore faces the most pressure to make a good mate decision.

Figure $19.3D.1$: **Sexual selection in elk**: This male elk has large antlers to compete with rival males for available females (intrasexual competition).Tn addition, the many points on his antlers represent health and longevity, and therefore he may be more desirable to females (intersexual selection).

Sexual Dimorphism

Males and females of certain species are often quite different from one another in ways beyond the reproductive organs. Males are often larger, for example, and display many elaborate colors and adornments, such as the peacock's tail, while females tend to be smaller and duller in decoration. These differences are called sexual dimorphisms and arise from the variation in male reproductive success.

Females almost always mate, while mating is not guaranteed for males. The bigger, stronger, or more decorated males usually obtain the vast majority of the total matings, while other males receive none. This can occur because the males are better at fighting off other males, or because females will choose to mate with the bigger or more decorated males. In either case, this variation in reproductive success generates a strong selection pressure among males to obtain those matings, resulting in the evolution of bigger body size and elaborate ornaments in order to increase their chances of mating. Females, on the other hand, tend to get a handful of selected matings; therefore, they are more likely to select more desirable males.

Figure 19.3D.1: **Sexual dimorphism**: Morphological differences between males and females of the same species is known as sexual dimorphism.These differences can be observed in (a) peacocks and peahens, (b) Argiope appensa spiders (the female spider is the large one), and (c) wood ducks.

Sexual dimorphism varies widely among species; some species are even sex-role reversed. In such cases, females tend to have a greater variation in their reproductive success than males and are, correspondingly, selected for the bigger body size and elaborate traits usually characteristic of males.

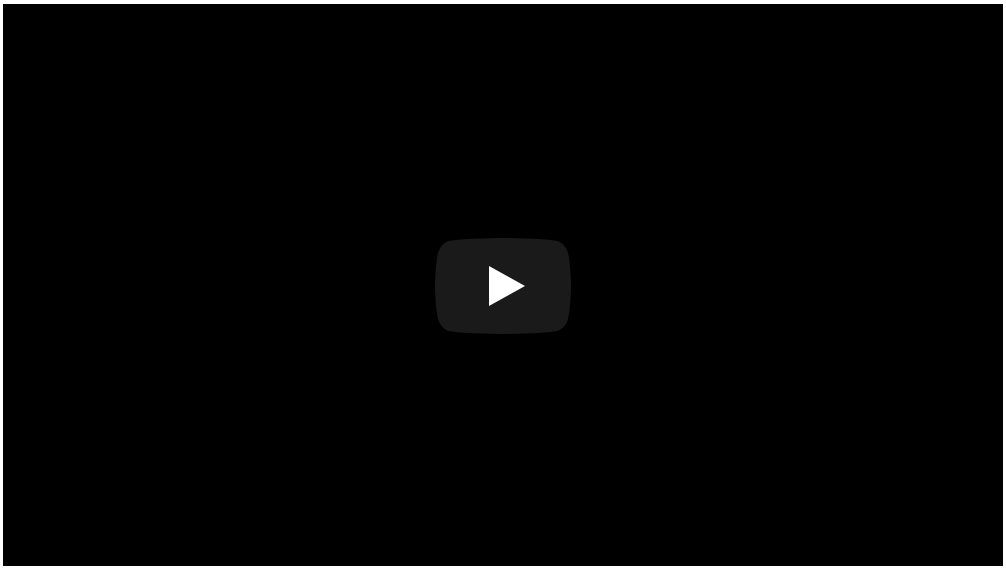
The Handicap Principle

Sexual selection can be so strong that it selects for traits that are actually detrimental to the individual's survival, even though they maximize its reproductive success. For example, while the male peacock's tail is beautiful and the male with the largest, most colorful tail will more probably win the female, it is not a practical appendage. In addition to being more visible to predators, it makes the males slower in their attempted escapes. There is some evidence that this risk, in fact, is why females like the big tails in the first place. Because large tails carry risk, only the best males survive that risk and therefore the bigger the tail, the more fit the male. This idea is known as the handicap principle.

Figure 19.3D.1: **A male bird of paradise**: This male bird of paradise carries an extremely long tail as the result of sexual selection.The tail is flamboyant and detrimental to the bird's own survival, but it increases his reproductive success.This may be an example of the handicap principle.

The Good Genes Hypothesis

The good genes hypothesis states that males develop these impressive ornaments to show off their efficient metabolism or their ability to fight disease. Females then choose males with the most impressive traits because it signals their genetic superiority, which they will then pass on to their offspring. Though it might be argued that females should not be so selective because it will likely reduce their number of offspring, if better males father more fit offspring, it may be beneficial. Fewer, healthier offspring may increase the chances of survival more than many, weaker offspring.



BBC Planet Earth – Birds of Paradise mating dance: Extraordinary Courtship displays from these weird and wonderful creatures. From episode 1 “Pole to Pole”. This is an example of the extreme behaviors that arise from intense sexual selection pressure.

19.3E: No Perfect Organism

Natural selection cannot create novel, perfect species because it only selects on existing variations in a population.

Learning Objectives

- Explain the limitations encountered in natural selection

Key Points

- Natural selection is limited by a population ‘s existing genetic variation.
- Natural selection is limited through linkage disequilibrium, where alleles that are physically proximate on the chromosome are passed on together at greater frequencies.
- In a polymorphic population, two phenotypes may be maintained in the population despite the higher fitness of one morph if the intermediate phenotype is detrimental.
- Evolution is not purposefully adaptive; it is the result of various selection forces working together to influence genetic and phenotypical variances within a population.

Key Terms

- **linkage disequilibrium**: a non-random association of two or more alleles at two or more loci; normally caused by an interaction between genes
- **genetic hitchhiking**: changes in the frequency of an allele because of linkage with a positively or negatively selected allele at another locus
- **polymorphism**: the regular existence of two or more different genotypes within a given species or population

No Perfect Organism

Natural selection is a driving force in evolution and can generate populations that are adapted to survive and successfully reproduce in their environments. However, natural selection cannot produce the perfect organism. Natural selection can only select on existing variation in the population; it cannot create anything from scratch. Therefore, the process of evolution is limited by a population’s existing genetic variance, the physical proximity of alleles, non-beneficial intermediate morphs in a polymorphic population, and non-adaptive evolutionary forces.

Natural Selection Acts on Individuals, not Alleles

Natural selection is also limited because it acts on the phenotypes of individuals, not alleles. Some alleles may be more likely to be passed on with alleles that confer a beneficial phenotype because of their physical proximity on the chromosomes. Alleles that are carried together are in linkage disequilibrium. When a neutral allele is linked to beneficial allele, consequently meaning that it has a selective advantage, the allele frequency can increase in the population through genetic hitchhiking (also called genetic draft).

Any given individual may carry some beneficial alleles and some unfavorable alleles. Natural selection acts on the net effect of these alleles and corresponding fitness of the phenotype. As a result, good alleles can be lost if they are carried by individuals that also have several overwhelmingly bad alleles; similarly, bad alleles can be kept if they are carried by individuals that have enough good alleles to result in an overall fitness benefit.

Polymorphism

Furthermore, natural selection can be constrained by the relationships between different polymorphisms. One morph may confer a higher fitness than another, but may not increase in frequency because the intermediate morph is detrimental.

Figure 19.3E.1:

morphism in the grove snail: Color and pattern morphs of the grove snail, Cepaea nemoralis.The polymorphism, two or more different genotypes exist within a given species, in grove snails seems to have several causes, including predation by thrushes.

For example, consider a hypothetical population of mice that live in the desert. Some are light-colored and blend in with the sand, while others are dark and blend in with the patches of black rock. The dark-colored mice may be more fit than the light-colored mice, and according to the principles of natural selection the frequency of light-colored mice is expected to decrease over time. However, the intermediate phenotype of a medium-colored coat is very bad for the mice: these cannot blend in with either the sand or the rock and will more vulnerable to predators. As a result, the frequency of a dark-colored mice would not increase because the intermediate morphs are less fit than either light-colored or dark-colored mice. This a common example of disruptive selection.

Not all Evolution is Adaptive

Finally, it is important to understand that not all evolution is adaptive. While natural selection selects the fittest individuals and often results in a more fit population overall, other forces of evolution, including genetic drift and gene flow, often do the opposite by introducing deleterious alleles to the population's gene pool. Evolution has no purpose. It is not changing a population into a preconceived ideal. It is simply the sum of various forces and their influence on the genetic and phenotypic variance of a population.

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CHAPTER OVERVIEW

2: THE CHEMICAL FOUNDATION OF LIFE

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2.1B: ATOMIC NUMBER AND MASS NUMBER
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2.2C: WATER'S HIGH HEAT CAPACITY
2.2D: WATER'S SOLVENT PROPERTIES
2.2E: WATER'S COHESIVE AND ADHESIVE PROPERTIES
2.2F: PH, BUFFERS, ACIDS, AND BASES
2.3: CARBON

2.3A: THE CHEMICAL BASIS FOR LIFE
2.3B: HYDROCARBONS
2.3C: ORGANIC ISOMERS
2.3D: ORGANIC ENANTIOMERS
2.3E: ORGANIC MOLECULES AND FUNCTIONAL GROUPS
2.4: SYNTHESIS OF BIOLOGICAL MACROMOLECULES

2.4A: TYPES OF BIOLOGICAL MACROMOLECULES
Biological macromolecules, the large molecules necessary for life, include carbohydrates, lipids, nucleic acids, and proteins.

2.4B: DEHYDRATION SYNTHESIS
2.4C: HYDROLYSIS

Atoms, Isotopes, Ions, and Molecules

- 2.1A: Overview of Atomic Structure
- 2.1B: Atomic Number and Mass Number
- 2.1C: Isotopes
- 2.1D: The Periodic Table
- 2.1E: Electron Shells and the Bohr Model
- 2.1F: Electron Orbitals

Electron orbitals are three-dimensional representations of the space in which an electron is likely to be found.

- 2.1G: Chemical Reactions and Molecules
- 2.1H: Ions and Ionic Bonds
- 2.1I: Covalent Bonds and Other Bonds and Interactions
- 2.1J: Hydrogen Bonding and Van der Waals Forces

2.1A: Overview of Atomic Structure

Atoms are made up of particles called protons, neutrons, and electrons, which are responsible for the mass and charge of atoms.

Learning Objectives

- Discuss the electronic and structural properties of an atom

Key Points

- An atom is composed of two regions: the nucleus, which is in the center of the atom and contains protons and neutrons, and the outer region of the atom, which holds its electrons in orbit around the nucleus.
- Protons and neutrons have approximately the same mass, about 1.67 × 10-24 grams, which scientists define as one atomic mass unit (amu) or one Dalton.
- Each electron has a negative charge (-1) equal to the positive charge of a proton (+1).
- Neutrons are uncharged particles found within the nucleus.

Key Terms

- **atom**: The smallest possible amount of matter which still retains its identity as a chemical element, consisting of a nucleus surrounded by electrons.
- **proton**: Positively charged subatomic particle forming part of the nucleus of an atom and determining the atomic number of an element. It weighs 1 amu.
- **neutron**: A subatomic particle forming part of the nucleus of an atom. It has no charge. It is equal in mass to a proton or it weighs 1 amu.

An atom is the smallest unit of matter that retains all of the chemical properties of an element. Atoms combine to form molecules, which then interact to form solids, gases, or liquids. For example, water is composed of hydrogen and oxygen atoms that have combined to form water molecules. Many biological processes are devoted to breaking down molecules into their component atoms so they can be reassembled into a more useful molecule.

Atomic Particles

Atoms consist of three basic particles: protons, electrons, and neutrons. The nucleus (center) of the atom contains the protons (positively charged) and the neutrons (no charge). The outermost regions of the atom are called electron shells and contain the electrons (negatively charged). Atoms have different properties based on the arrangement and number of their basic particles.

The hydrogen atom (H) contains only one proton, one electron, and no neutrons. This can be determined using the atomic number and the mass number of the element (see the concept on atomic numbers and mass numbers).

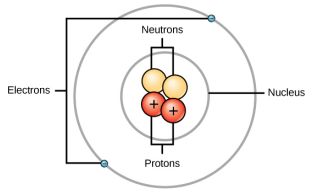


Figure 2.1A.1: **Structure of an atom**: Elements, such as helium, depicted here, are made up of atoms. Atoms are made up of protons and neutrons located within the nucleus, with electrons in orbitals surrounding the nucleus.

Atomic Mass

Protons and neutrons have approximately the same mass, about 1.67×10^{-24} grams. Scientists define this amount of mass as one atomic mass unit (amu) or one Dalton. Although similar in mass, protons are positively charged, while neutrons have no charge. Therefore, the number of neutrons in an atom contributes significantly to its mass, but not to its charge.

Electrons are much smaller in mass than protons, weighing only 9.11×10^{-28} grams, or about 1/1800 of an atomic mass unit. Therefore, they do not contribute much to an element's overall atomic mass. When considering atomic mass, it is customary to ignore the mass of any electrons and calculate the atom's mass based on the number of protons and neutrons alone.

Electrons contribute greatly to the atom's charge, as each electron has a negative charge equal to the positive charge of a proton. Scientists define these charges as "+1" and "-1. " In an uncharged, neutral atom, the number of electrons orbiting the nucleus is equal to the number of protons inside the nucleus. In these atoms, the positive and negative charges cancel each other out, leading to an atom with no net charge.

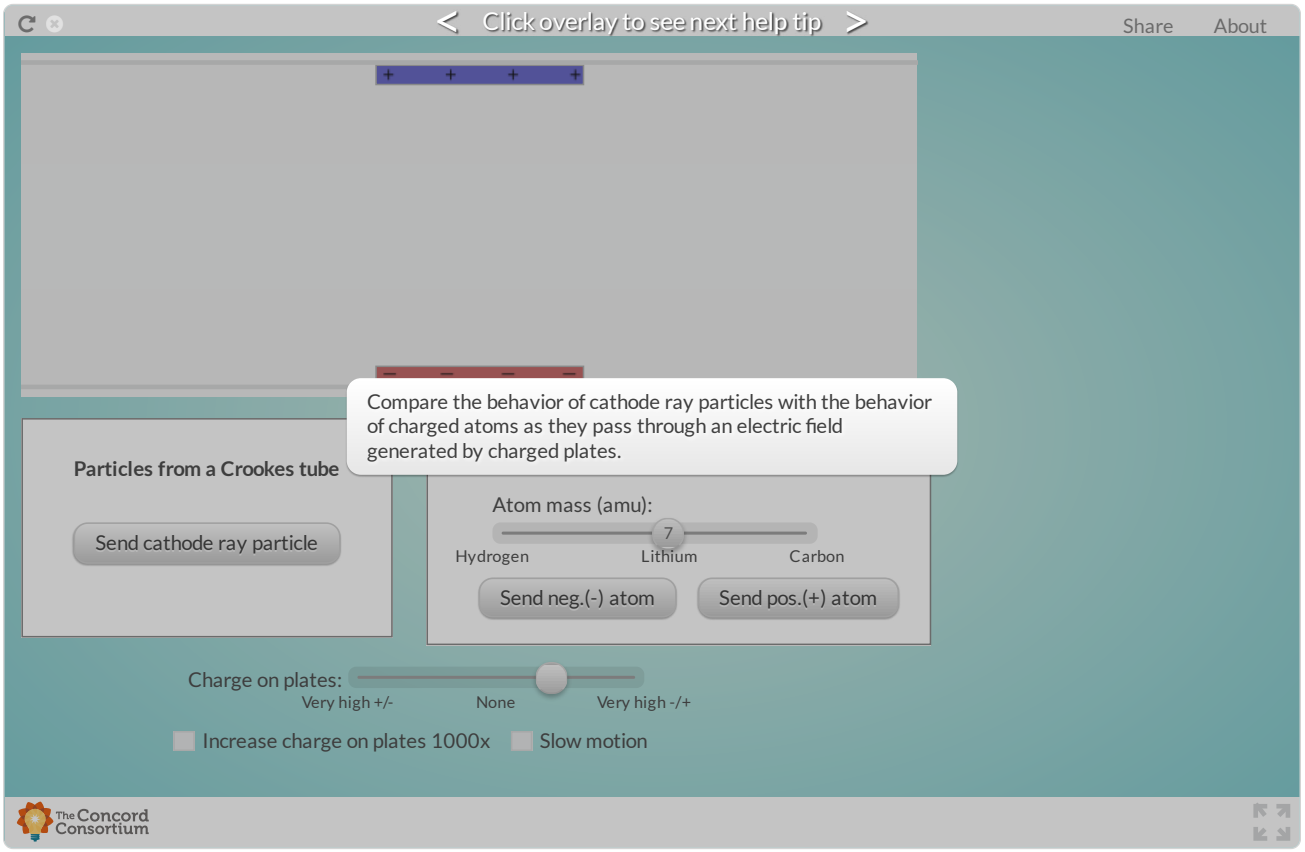
Protons, Neutrons, and Electrons			
	Charge	**Mass (amu)**	**Location**
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	–1	0	orbitals

Figure 2.1A.1: **Protons, neutrons, and electrons**: Both protons and neutrons have a mass of 1 amu and are found in the nucleus. However, protons have a charge of +1, and neutrons are uncharged. Electrons have a mass of approximately 0 amu, orbit the nucleus, and have a charge of -1.

Exploring Electron Properties: Compare the behavior of electrons to that of other charged particles to discover properties of electrons such as charge and mass.

Volume of Atoms

Accounting for the sizes of protons, neutrons, and electrons, most of the volume of an atom—greater than 99 percent—is, in fact, empty space. Despite all this empty space, solid objects do not just pass through one another. The electrons that surround all atoms are negatively charged and cause atoms to repel one another, preventing atoms from occupying the same space. These intermolecular forces prevent you from falling through an object like your chair.



Interactive: Build an Atom: Build an atom out of protons, neutrons, and electrons, and see how the element, charge, and mass change. Then play a game to test your ideas!

2.1B: Atomic Number and Mass Number

The atomic number is the number of protons in an element, while the mass number is the number of protons plus the number of neutrons.

Learning Objectives

- Determine the relationship between the mass number of an atom, its atomic number, its atomic mass, and its number of subatomic particles

Key Points

- Neutral atoms of each element contain an equal number of protons and electrons.
- The number of protons determines an element's atomic number and is used to distinguish one element from another.
- The number of neutrons is variable, resulting in isotopes, which are different forms of the same atom that vary only in the number of neutrons they possess.
- Together, the number of protons and the number of neutrons determine an element's mass number.
- Since an element's isotopes have slightly different mass numbers, the atomic mass is calculated by obtaining the mean of the mass numbers for its isotopes.

Key Terms

- **mass number**: The sum of the number of protons and the number of neutrons in an atom.
- **atomic number**: The number of protons in an atom.
- **atomic mass**: The average mass of an atom, taking into account all its naturally occurring isotopes.

Atomic Number

Neutral atoms of an element contain an equal number of protons and electrons. The number of protons determines an element's atomic number (Z) and distinguishes one element from another. For example, carbon's atomic number (Z) is 6 because it has 6 protons. The number of neutrons can vary to produce isotopes, which are atoms of the same element that have different numbers of neutrons. The number of electrons can also be different in atoms of the same element, thus producing ions (charged atoms). For instance, iron, Fe, can exist in its neutral state, or in the +2 and +3 ionic states.

Mass Number

An element's mass number (A) is the sum of the number of protons and the number of neutrons. The small contribution of mass from electrons is disregarded in calculating the mass number. This approximation of mass can be used to easily calculate how many neutrons an element has by simply subtracting the number of protons from the mass number. Protons and neutrons both weigh about one atomic mass unit or amu. Isotopes of the same element will have the same atomic number but different mass numbers.

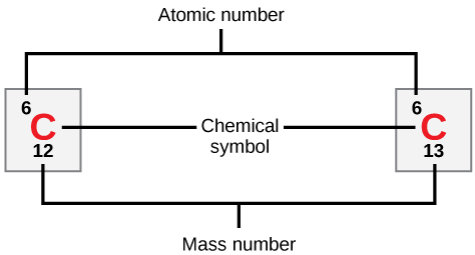


Figure 2.1*B*.1: **Atomic number, chemical symbol, and mass number**: Carbon has an atomic number of six, and two stable isotopes with mass numbers of twelve and thirteen, respectively. Its average atomic mass is 12.11.

Scientists determine the atomic mass by calculating the mean of the mass numbers for its naturally-occurring isotopes. Often, the resulting number contains a decimal. For example, the atomic mass of chlorine (Cl) is 35.45 amu because chlorine is composed of several isotopes, some (the majority) with an atomic mass of 35 amu (17 protons and 18 neutrons) and some with an atomic mass of 37 amu (17 protons and 20 neutrons).

Given an atomic number (Z) and mass number (A), you can find the number of protons, neutrons, and electrons in a neutral atom. For example, a lithium atom (Z=3, A=7 amu) contains three protons (found from Z), three electrons (as the number of protons is equal to the number of electrons in an atom), and four neutrons (7 – 3 = 4).

2.1C: Isotopes

Isotopes are various forms of an element that have the same number of protons, but a different number of neutrons.

Learning Objectives

- Discuss the properties of isotopes and their use in radiometric dating

Key Points

- Isotopes are atoms of the same element that contain an identical number of protons, but a different number of neutrons.
- Despite having different numbers of neutrons, isotopes of the same element have very similar physical properties.
- Some isotopes are unstable and will undergo radioactive decay to become other elements.
- The predictable half-life of different decaying isotopes allows scientists to date material based on its isotopic composition, such as with Carbon-14 dating.

Key Terms

- **isotope**: Any of two or more forms of an element where the atoms have the same number of protons, but a different number of neutrons within their nuclei.
- **half-life**: The time it takes for half of the original concentration of an isotope to decay back to its more stable form.
- **radioactive isotopes**: an atom with an unstable nucleus, characterized by excess energy available that undergoes radioactive decay and creates most commonly gamma rays, alpha or beta particles.
- **radiocarbon dating**: Determining the age of an object by comparing the ratio of the 14C concentration found in it to the amount of 14C in the atmosphere.

What is an Isotope?

Isotopes are various forms of an element that have the same number of protons but a different number of neutrons. Some elements, such as carbon, potassium, and uranium, have multiple naturally-occurring isotopes. Isotopes are defined first by their element and then by the sum of the protons and neutrons present.

- Carbon-12 (or ^{12}C) contains six protons, six neutrons, and six electrons; therefore, it has a mass number of 12 amu (six protons and six neutrons).
- Carbon-14 (or ^{14}C) contains six protons, eight neutrons, and six electrons; its atomic mass is 14 amu (six protons and eight neutrons).

While the mass of individual isotopes is different, their physical and chemical properties remain mostly unchanged.

Isotopes do differ in their stability. Carbon-12 (^{12}C) is the most abundant of the carbon isotopes, accounting for 98.89% of carbon on Earth. Carbon-14 (^{14}C) is unstable and only occurs in trace amounts. Unstable isotopes most commonly emit alpha particles (He^{2+}) and electrons. Neutrons, protons, and positrons can also be emitted and electrons can be captured to attain a more stable atomic configuration (lower level of potential energy) through a process called radioactive decay. The new atoms created may be in a high energy state and emit gamma rays which lowers the energy but alone does not change the atom into another isotope. These atoms are called radioactive isotopes or radioisotopes.

Radiocarbon Dating

Carbon is normally present in the atmosphere in the form of gaseous compounds like carbon dioxide and methane. Carbon-14 (^{14}C) is a naturally-occurring radioisotope that is created from atmospheric ^{14}N (nitrogen) by the addition of a neutron and the loss of a proton, which is caused by cosmic rays. This is a continuous process so more ^{14}C is always being created in the atmosphere. Once produced, the ^{14}C often combines with the oxygen in the atmosphere to form carbon dioxide. Carbon dioxide produced in this way diffuses in the atmosphere, is dissolved in the ocean, and is incorporated by plants via photosynthesis. Animals eat the plants and, ultimately, the radiocarbon is distributed throughout the biosphere.

In living organisms, the relative amount of ^{14}C in their body is approximately equal to the concentration of ^{14}C in the atmosphere. When an organism dies, it is no longer ingesting ^{14}C, so the ratio between ^{14}C and ^{12}C will decline as ^{14}C

gradually decays back to ^{14}N. This slow process, which is called beta decay, releases energy through the emission of electrons from the nucleus or positrons.

After approximately 5,730 years, half of the starting concentration of ^{14}C will have been converted back to ^{14}N. This is referred to as its half-life, or the time it takes for half of the original concentration of an isotope to decay back to its more stable form. Because the half-life of ^{14}C is long, it is used to date formerly-living objects such as old bones or wood. Comparing the ratio of the ^{14}C concentration found in an object to the amount of ^{14}C in the atmosphere, the amount of the isotope that has not yet decayed can be determined. On the basis of this amount, the age of the material can be accurately calculated, as long as the material is believed to be less than 50,000 years old. This technique is called radiocarbon dating, or carbon dating for short.

Figure 2.1C.1: **Application of carbon dating**: The age of carbon-containing remains less than 50,000 years old, such as this pygmy mammoth, can be determined using carbon dating.

Other elements have isotopes with different half lives. For example, ^{40}K (potassium-40) has a half-life of 1.25 billion years, and ^{235}U (uranium-235) has a half-life of about 700 million years. Scientists often use these other radioactive elements to date objects that are older than 50,000 years (the limit of carbon dating). Through the use of radiometric dating, scientists can study the age of fossils or other remains of extinct organisms.

2.1D: The Periodic Table

Everything in the universe is made of one or more elements. The periodic table is a means of organizing the various elements according to similar physical and chemical properties.

Learning Objectives

- Discuss the organization of the periodic table

Key Points

- All matter is made from atoms of one or more elements. Living creatures consist mainly of carbon, hydrogen, oxygen, and nitrogen (CHON).
- Combining elements creates compounds that may have emergent properties.
- The periodic table is a listing of the elements according to increasing atomic number that is further organized into columns based on similar physical and chemical properties and electron configuration.
- As one moves down a column or across a row, there are some general trends for the properties of the elements.
- The periodic table continues to expand today as heavier and heavier elements are created in laboratories around the world.

Key Terms

- **element**: Pure chemical substances consisting of only one type of atom with a defined set of chemical and physical properties.
- **emergent properties**: Properties found in compound structures that are different from those of the individual components and would not be predicted based on the properties of the individual components.
- **periodic table**: A tabular chart of the chemical elements according to their atomic numbers so that elements with similar properties are in the same column.

Matter and Elements

Matter comprises all of the physical objects in the universe, those that take up space and have mass. All matter is composed of atoms of one or more elements, pure substances with specific chemical and physical properties. There are 98 elements that naturally occur on earth, yet living systems use a relatively small number of these. Living creatures are composed mainly of just four elements: carbon, hydrogen, oxygen, and nitrogen (often remembered by the acronym CHON). As elements are bonded together they form compounds that often have new emergent properties that are different from the properties of the individual elements. Life is an example of an emergent property that arises from the specific collection of molecules found in cells.

Elements of the human body arranged by percent of total mass: There are 25 elements believed to play an active role in human health. Carbon, hydrogen, oxygen, and nitrogen make up approximately 96% of the mass in a human body.

The Periodic Table

The different elements are organized and displayed in the periodic table. Devised by Russian chemist Dmitri Mendeleev (1834–1907) in 1869, the table groups elements that, although unique, share certain chemical properties with other elements. In the periodic table the elements are organized and displayed according to their atomic number and are arranged in a series of rows (periods) and columns (groups) based on shared chemical and physical properties. If you look at a periodic table, you will see the groups numbered at the top of each column from left to right starting with 1 and ending with 18. In addition to providing the atomic number for each element, the periodic table also displays the element's atomic mass. Looking at carbon, for example, its symbol (C) and name appear, as well as its atomic number of six (in the upper left-hand corner) and its atomic mass of 12.11.

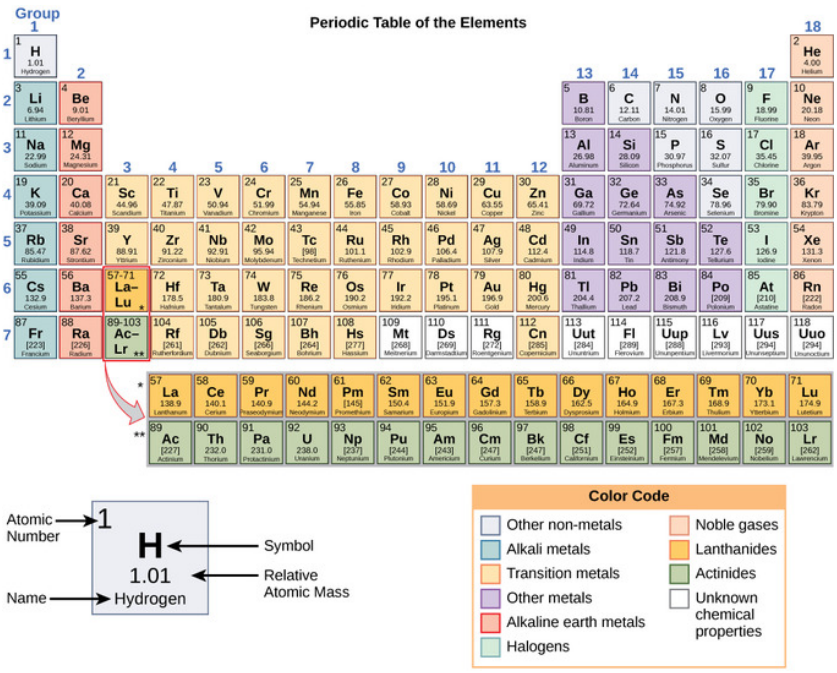


Figure 2.1D.1: **The periodic table**: The periodic table shows the atomic mass and atomic number of each element. The atomic number appears above the symbol for the element and the approximate atomic mass appears below it.

The arrangement of the periodic table allows the elements to be grouped according to their chemical properties. Within the main group elements (Groups 1-2, 13-18), there are some general trends that we can observe. The further down a given group, the elements have an increased metallic character: they are good conductors of both heat and electricity, solids at room temperature, and shiny in appearance. Moving from left to right across a period, the elements have greater non-metallic character. These elements are insulators, poor heat conductors, and can exist in different phases at room temperature (brittle solid, liquid, or gas). The elements at the boundary between the metallic elements (grey elements) and nonmetal elements (green elements) are metalloid in character (pink elements). They have low electrical conductivity that increases with temperature. They also share properties with both the metals and the nonmetals.

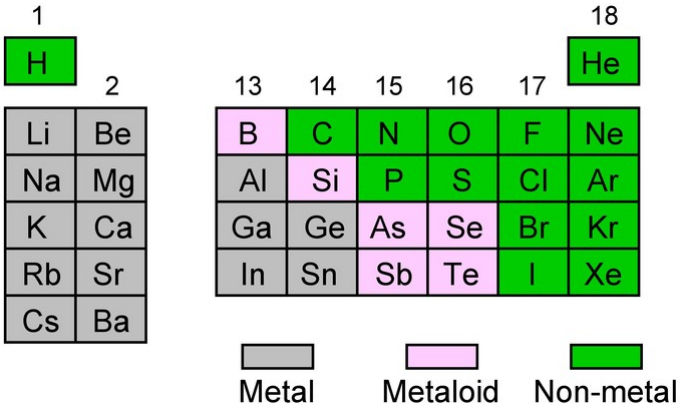


Figure 2.1D.1: **The main group elements**: Within the p-block at the boundary between the metallic elements (grey elements) and nonmetal elements (green elements) there is positioned boron and silicon that are metalloid in character (pink elements), i.e., they have low electrical conductivity that increases with temperature.

Today, the periodic table continues to expand as heavier and heavier elements are synthesized in laboratories. These large elements are extremely unstable and, as such, are very difficult to detect; but their continued creation is an ongoing challenge undertaken by scientists around the world.

2.1E: Electron Shells and the Bohr Model

Niels Bohr proposed an early model of the atom as a central nucleus containing protons and neutrons being orbited by electrons in shells.

Learning Objectives

- Construct an atom according to the Bohr model

Key Points

- In the Bohr model of the atom, the nucleus contains the majority of the mass of the atom in its protons and neutrons.
- Orbiting the positively-charged core are the negatively charged electrons, which contribute little in terms of mass, but are electrically equivalent to the protons in the nucleus.
- In most cases, electrons fill the lower- energy orbitals first, followed by the next higher energy orbital until it is full, and so on until all electrons have been placed.
- Atoms tend to be most stable with a full outer shell (one which, after the first, contains 8 electrons), leading to what is commonly called the ” octet rule “.
- The properties of an element are determined by its outermost electrons, or those in the highest energy orbital.
- Atoms that do not have full outer shells will tend to gain or lose electrons, resulting in a full outer shell and, therefore, stability.

Key Terms

- **octet rule**: A rule stating that atoms lose, gain, or share electrons in order to have a full valence shell of 8 electrons. (Hydrogen is excluded because it can hold a maximum of 2 electrons in its valence shell.)
- **electron shell**: The collective states of all electrons in an atom having the same principal quantum number (visualized as an orbit in which the electrons move).

Electron Shells and the Bohr Model

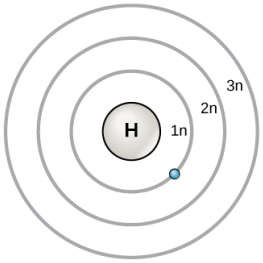


Figure 2.1E.1: **Orbitals in the Bohr model**: The Bohr model was developed by Niels Bohr in 1913. In this model, electrons exist within principal shells. An electron normally exists in the lowest energy shell available, which is the one closest to the nucleus. Energy from a photon of light can bump it up to a higher energy shell, but this situation is unstable and the electron quickly decays back to the ground state. In the process, a photon of light is released.

As previously discussed, there is a connection between the number of protons in an element, the atomic number that distinguishes one element from another, and the number of electrons it has. In all electrically-neutral atoms, the number of electrons is the same as the number of protons. Each element, when electrically neutral, has a number of electrons equal to its atomic number.

An early model of the atom was developed in 1913 by Danish scientist Niels Bohr (1885–1962). The Bohr model shows the atom as a central nucleus containing protons and neutrons with the electrons in circular orbitals at specific distances from the nucleus. These orbits form electron shells or energy levels, which are a way of visualizing the number of electrons in the various shells. These energy levels are designated by a number and the symbol “n.” For example, 1n represents the first energy level located closest to the nucleus.

Electrons fill orbit shells in a consistent order. Under standard conditions, atoms fill the inner shells (closer to the nucleus) first, often resulting in a variable number of electrons in the outermost shell. The innermost shell has a maximum of two electrons, but the next two electron shells can each have a maximum of eight electrons. This is known as the octet rule which states that, with the exception of the innermost shell, atoms are more stable energetically when they have eight electrons in their valence shell, the outermost electron shell. Examples of some neutral atoms and their electron configurations are shown in. As shown, helium has a complete outer electron shell, with two electrons filling its first and only shell. Similarly, neon has a complete outer 2n shell containing eight electrons. In contrast, chlorine and sodium have seven and one electrons in their outer shells, respectively. Theoretically, they would be more energetically stable if they followed the octet rule and had eight.

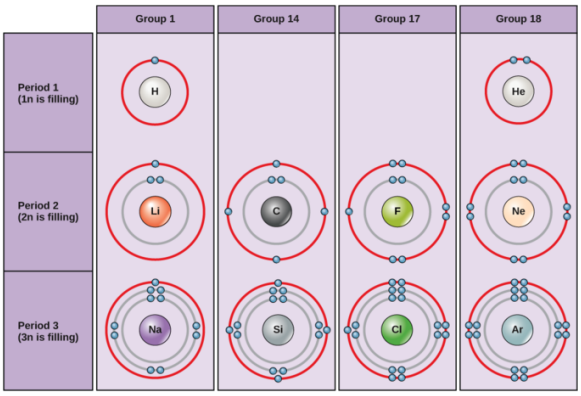


Figure 2.1E.1: **Bohr diagrams**: Bohr diagrams indicate how many electrons fill each principal shell. Group 18 elements (helium, neon, and argon are shown) have a full outer, or valence, shell. A full valence shell is the most stable electron configuration. Elements in other groups have partially-filled valence shells and gain or lose electrons to achieve a stable electron configuration.

An atom may gain or lose electrons to achieve a full valence shell, the most stable electron configuration. The periodic table is arranged in columns and rows based on the number of electrons and where these electrons are located, providing a tool to understand how electrons are distributed in the outer shell of an atom. As shown in, the group 18 atoms helium (He), neon (Ne), and argon (Ar) all have filled outer electron shells, making it unnecessary for them to gain or lose electrons to attain stability; they are highly stable as single atoms. Their non-reactivity has resulted in their being named the inert gases (or noble gases). In comparison, the group 1 elements, including hydrogen (H), lithium (Li), and sodium (Na), all have one electron in their outermost shells. This means that they can achieve a stable configuration and a filled outer shell by donating or losing an electron. As a result of losing a negatively-charged electron, they become positively-charged ions. When an atom loses an electron to become a positively-charged ion, this is indicated by a plus sign after the element symbol; for example, Na^+. Group 17 elements, including fluorine and chlorine, have seven electrons in their outermost shells; they tend to fill this shell by gaining an electron from other atoms, making them negatively-charged ions. When an atom gains an electron to become a negatively-charged ion this is indicated by a minus sign after the element symbol; for example, F-. Thus, the columns of the periodic table represent the potential shared state of these elements' outer electron shells that is responsible for their similar chemical characteristics.

2.1F: Electron Orbitals

Electron orbitals are three-dimensional representations of the space in which an electron is likely to be found.

Learning Objectives

- Distinguish between electron orbitals in the Bohr model versus the quantum mechanical orbitals

Key Points

- The Bohr model of the atom does not accurately reflect how electrons are spatially distributed around the nucleus as they do not circle the nucleus like the earth orbits the sun.
- The electron orbitals are the result of mathematical equations from quantum mechanics known as wave functions and can predict within a certain level of probability where an electron might be at any given time.
- The number and type of orbitals increases with increasing atomic number, filling in various electron shells.
- The area where an electron is most likely to be found is called its orbital.

Key Terms

- **electron shell**: The collective states of all electrons in an atom having the same principal quantum number (visualized as an orbit in which the electrons move).
- **orbital**: A specification of the energy and probability density of an electron at any point in an atom or molecule.

Although useful to explain the reactivity and chemical bonding of certain elements, the Bohr model of the atom does not accurately reflect how electrons are spatially distributed surrounding the nucleus. They do not circle the nucleus like the earth orbits the sun, but are rather found in electron orbitals. These relatively complex shapes result from the fact that electrons behave not just like particles, but also like waves. Mathematical equations from quantum mechanics known as wave functions can predict within a certain level of probability where an electron might be at any given time. The area where an electron is most likely to be found is called its orbital.

First Electron Shell

The closest orbital to the nucleus, called the 1s orbital, can hold up to two electrons. This orbital is equivalent to the innermost electron shell of the Bohr model of the atom. It is called the 1s orbital because it is spherical around the nucleus. The 1s orbital is always filled before any other orbital. Hydrogen has one electron; therefore, it has only one spot within the 1s orbital occupied. This is designated as $1s^1$, where the superscripted 1 refers to the one electron within the 1s orbital. Helium has two electrons; therefore, it can completely fill the 1s orbital with its two electrons. This is designated as $1s^2$, referring to the two electrons of helium in the 1s orbital. On the periodic table, hydrogen and helium are the only two elements in the first row (period); this is because they are the sole elements to have electrons only in their first shell, the 1s orbital.

Second Electron Shell

The second electron shell may contain eight electrons. This shell contains another spherical s orbital and three “dumbbell” shaped p orbitals, each of which can hold two electrons. After the 1s orbital is filled, the second electron shell is filled, first filling its 2s orbital and then its three p orbitals. When filling the p orbitals, each takes a single electron; once each p orbital has an electron, a second may be added. Lithium (Li) contains three electrons that occupy the first and second shells. Two electrons fill the 1s orbital, and the third electron then fills the 2s orbital. Its electron configuration is $1s^2 2s^1$. Neon (Ne), on the other hand, has a total of ten electrons: two are in its innermost 1s orbital, and eight fill its second shell (two each in the 2s and three p orbitals). Thus, it is an inert gas and energetically stable: it rarely forms a chemical bond with other atoms.

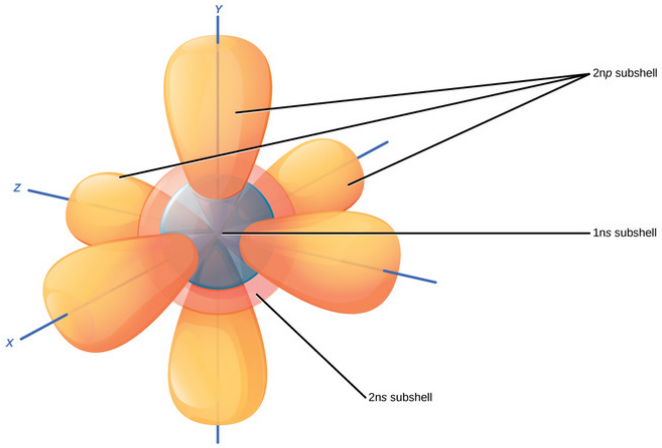


Figure 2.1F. 1: **Diagram of the S and P orbitals**: The s subshells are shaped like spheres. Both the 1n and 2n principal shells have an s orbital, but the size of the sphere is larger in the 2n orbital. Each sphere is a single orbital. p subshells are made up of three dumbbell-shaped orbitals. Principal shell 2n has a p subshell, but shell 1 does not.

Third Electron Shell

Larger elements have additional orbitals, making up the third electron shell. Subshells d and f have more complex shapes and contain five and seven orbitals, respectively. Principal shell 3n has s, p, and d subshells and can hold 18 electrons. Principal shell 4n has s, p, d, and f orbitals and can hold 32 electrons. Moving away from the nucleus, the number of electrons and orbitals found in the energy levels increases. Progressing from one atom to the next in the periodic table, the electron structure can be worked out by fitting an extra electron into the next available orbital. While the concepts of electron shells and orbitals are closely related, orbitals provide a more accurate depiction of the electron configuration of an atom because the orbital model specifies the different shapes and special orientations of all the places that electrons may occupy.

2.1G: Chemical Reactions and Molecules

Chemical reactions occur when two or more atoms bond together to form molecules or when bonded atoms are broken apart.

Learning Objectives

- Describe the properties of chemical reactions and compounds

Key Points

- Atoms form chemical bonds with other atoms thereby obtaining the electrons they need to attain a stable electron configuration.
- The substances used in the beginning of a chemical reaction are called the reactants and the substances found at the end of the reaction are known as the products.
- Some reactions are reversible and will reach a relative balance between reactants and products: a state called equilibrium.
- An arrow is typically drawn between the reactants and products to indicate the direction of the chemical reaction.

Key Terms

- **reactant**: Any of the participants present at the start of a chemical reaction.
- **molecule**: The smallest particle of a specific compound that retains the chemical properties of that compound; two or more atoms held together by chemical bonds.
- **reaction**: A transformation in which one or more substances is converted into another by combination or decomposition

Chemical Reactions and Molecules

According to the octet rule, elements are most stable when their outermost shell is filled with electrons. This is because it is energetically favorable for atoms to be in that configuration. However, since not all elements have enough electrons to fill their outermost shells, atoms form chemical bonds with other atoms, which helps them obtain the electrons they need to attain a stable electron configuration. When two or more atoms chemically bond with each other, the resultant chemical structure is a molecule. The familiar water molecule, H_2O, consists of two hydrogen atoms and one oxygen atom, which bond together to form water. Atoms can form molecules by donating, accepting, or sharing electrons to fill their outer shells.

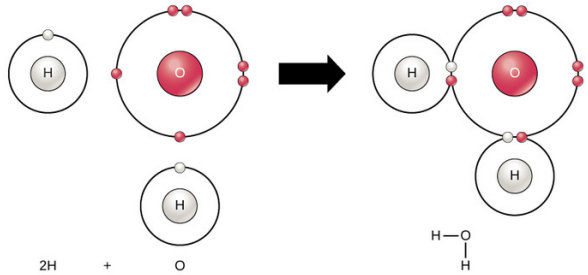


Figure 2.1G.1: **Atoms bond to form molecules**: Two or more atoms may bond with each other to form a molecule. When two hydrogens and an oxygen share electrons via covalent bonds, a water molecule is formed.

Chemical reactions occur when two or more atoms bond together to form molecules or when bonded atoms are broken apart. The substances used in the beginning of a chemical reaction are called the reactants (usually found on the left side of a chemical equation), and the substances found at the end of the reaction are known as the products (usually found on the right side of a chemical equation). An arrow is typically drawn between the reactants and products to indicate the direction of the chemical reaction. For the creation of the water molecule shown above, the chemical equation would be:

$2H2+O2→2H2O2H2+O2→2H2O$

An example of a simple chemical reaction is the breaking down of hydrogen peroxide molecules, each of which consists of two hydrogen atoms bonded to two oxygen atoms (H_2O_2). The reactant hydrogen peroxide is broken down into water (H_2O), and oxygen, which consists of two bonded oxygen atoms (O_2). In the equation below, the reaction includes two hydrogen peroxide molecules and two water molecules. This is an example of a balanced chemical equation, wherein the number of atoms of each element is the same on each side of the equation. According to the law of conservation of matter, the number of atoms before and after a chemical reaction should be equal, such that no atoms are, under normal circumstances, created or destroyed.

$2H2O2→2H2O+O22H2O2→2H2O+O2$

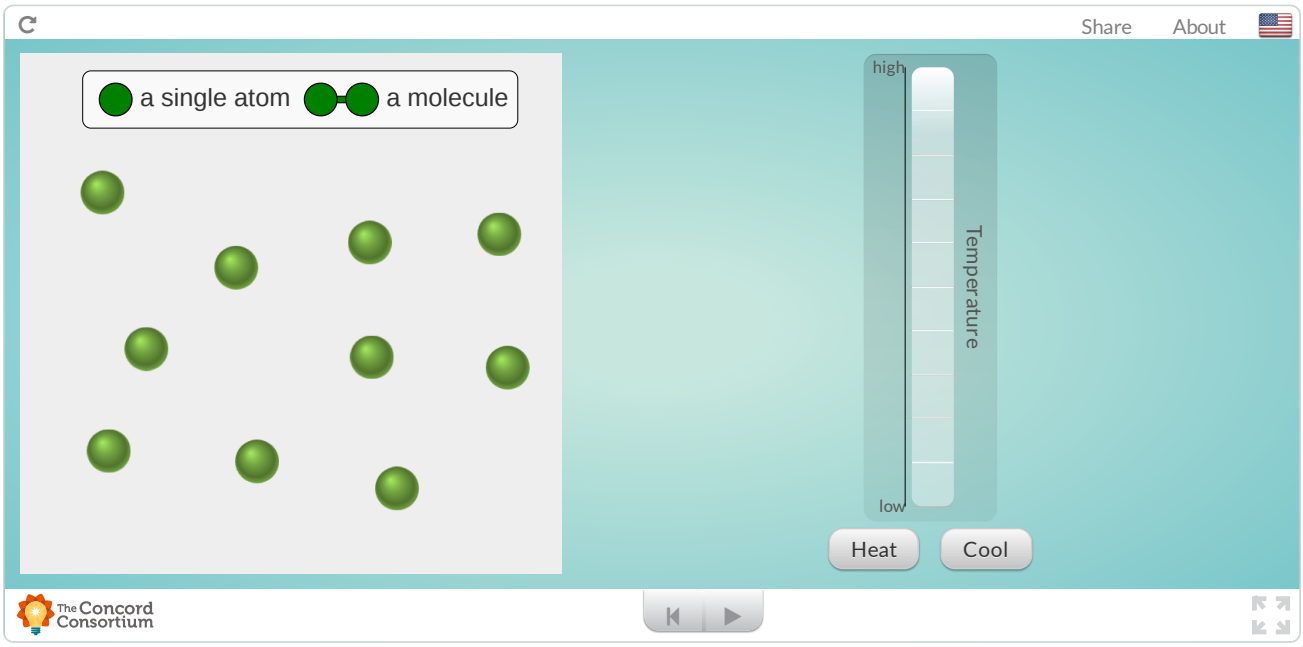
Even though all of the reactants and products of this reaction are molecules (each atom remains bonded to at least one other atom), in this reaction only hydrogen peroxide and water are representative of a subclass of molecules known as compounds: they contain atoms of more than one type of element. Molecular oxygen, on the other hand, consists of two doubly bonded oxygen atoms and is not classified as a compound but as an element.

Irreversible and Reversible Reactions

Some chemical reactions, such as the one shown above, can proceed in one direction until the reactants are all used up. The equations that describe these reactions contain a unidirectional arrow and are irreversible. Reversible reactions are those that can go in either direction. In reversible reactions, reactants are turned into products, but when the concentration of product goes beyond a certain threshold, some of these products will be converted back into reactants; at this point, the designations of products and reactants are reversed. This back and forth continues until a certain relative balance between reactants and products occurs: a state called equilibrium. These situations of reversible reactions are often denoted by a chemical equation with a double headed arrow pointing towards both the reactants and products.

For example, in human blood, excess hydrogen ions (H^+) bind to bicarbonate ions (HCO_3^-) forming an equilibrium state with carbonic acid (H_2CO_3). If carbonic acid were added to this system, some of it would be converted to bicarbonate and hydrogen ions.

In biological reactions, however, equilibrium is rarely obtained because the concentrations of the reactants or products or both are constantly changing, often with a product of one reaction being a reactant for another. To return to the example of excess hydrogen ions in the blood, the formation of carbonic acid will be the major direction of the reaction. However, the carbonic acid can also leave the body as carbon dioxide gas (via exhalation) instead of being converted back to bicarbonate ion, thus driving the reaction to the right by the chemical law known as law of mass action. These reactions are important for maintaining the homeostasis of our blood.



Interactive: What is a Chemical Reaction?: Explore reactions in which chemical bonds are formed and broken with this model. Press run, then try heating and cooling the atoms to see how temperature affects the balance between bond formation and breaking.

H: Ions and Ionic Bonds

Ionic bonds are attractions between oppositely charged atoms or groups of atoms where electrons are donated and accepted.

Learning Objectives

- Predict whether a given element will more likely form a cation or an anion

Key Points

- Ions form from elements when they gain or lose an electron causing the number of protons to be unequal to the number of electrons, resulting in a net charge.
- If there are more electrons than protons (from an element gaining one or more electrons), the ion is negatively charged and called an anion.
- If there are more protons than electrons (via loss of electrons), the ion is positively charged and is called a cation.
- Ionic bonds result from the interaction between a positively charged cation and a negatively charged anion.

Key Terms

- **ion**: An atom, or group of atoms, bearing an electrical charge, such as the sodium and chlorine atoms in a salt solution.
- **ionic bond**: A strong chemical bond caused by the electrostatic attraction between two oppositely charged ions.

Ions and Ionic Bonds

Some atoms are more stable when they gain or lose an electron (or possibly two) and form ions. This results in a full outermost electron shell and makes them energetically more stable. Now, because the number of electrons does not equal the number of protons, each ion has a net charge. Cations are positive ions that are formed by losing electrons (as the number of protons is now greater than the number of electrons). Negative ions are formed by gaining electrons and are called anions (wherein there are more electrons than protons in a molecule). Anions are designated by their elemental name being altered to end in “-ide”. For example, the anion of chlorine is called chloride, and the anion of sulfur is called sulfide.

This movement of electrons from one element to another is referred to as electron transfer. As illustrated, sodium (Na) only has one electron in its outer electron shell. It takes less energy for sodium to donate that one electron than it does to accept seven more electrons to fill the outer shell. When sodium loses an electron, it will have 11 protons, 11 neutrons, and only 10 electrons. This leaves it with an overall charge of +1 since there are now more protons than electrons. It is now referred to as a sodium ion. Chlorine (Cl) in its lowest energy state (called the ground state) has seven electrons in its outer shell. Again, it is more energy efficient for chlorine to gain one electron than to lose seven. Therefore, it tends to gain an electron to create an ion with 17 protons, 17 neutrons, and 18 electrons. This gives it a net charge of -1 since there are now more electrons than protons. It is now referred to as a chloride ion. In this example, sodium will donate its one electron to empty its shell, and chlorine will accept that electron to fill its shell. Both ions now satisfy the octet rule and have complete outer shells. These transactions can normally only take place simultaneously; in order for a sodium atom to lose an electron, it must be in the presence of a suitable recipient like a chlorine atom.

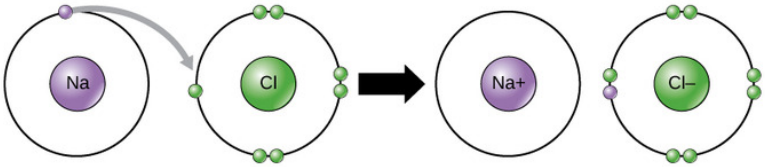


Figure 2.1H.1: **Electron Transfer Between Na and Cl**: In the formation of an ionic compound, metals lose electrons and nonmetals gain electrons to achieve an octet. In this example, sodium loses one electron to empty its shell and chlorine accepts that electron to fill its shell.

Ionic bonds are formed between ions with opposite charges. For instance, positively charged sodium ions and negatively charged chloride ions bond together to form sodium chloride, or table salt, a crystalline molecule with zero net charge. The

ve force holding the two atoms together is called the electromagnetic force and is responsible for the attraction n oppositely charged ions.

Certain salts are referred to in physiology as electrolytes (including sodium, potassium, and calcium). Electrolytes are ions necessary for nerve impulse conduction, muscle contractions, and water balance. Many sports drinks and dietary supplements provide these ions to replace those lost from the body via sweating during exercise.

2.1I: Covalent Bonds and Other Bonds and Interactions

Covalent bonds result from a sharing of electrons between two atoms and hold most biomolecules together.

Learning Objectives

- Compare the relative strength of different types of bonding interactions

Key Points

- A polar covalent bond arises when two atoms of different electronegativity share two electrons unequally.
- A non-polar covalent bond is one in which the electrons are shared equally between two atoms.
- Hydrogen bonds and Van Der Waals are responsible for the folding of proteins, the binding of ligands to proteins, and many other processes between molecules.

Key Terms

- **hydrogen bond**: A weak bond in which a hydrogen atom in one molecule is attracted to an electronegative atom (usually nitrogen or oxygen) in the same or different molecule.
- **covalent bond**: A type of chemical bond where two atoms are connected to each other by the sharing of two or more electrons.
- **dipole**: Any object (such as a magnet, polar molecule or antenna), that is oppositely charged at two points (or poles).

Covalent Bonds and Other Bonds and Interactions

The octet rule can be satisfied by the sharing of electrons between atoms to form covalent bonds. These bonds are stronger and much more common than are ionic bonds in the molecules of living organisms. Covalent bonds are commonly found in carbon-based organic molecules, such as DNA and proteins. Covalent bonds are also found in inorganic molecules such as H_2O, CO_2, and O_2. One, two, or three pairs of electrons may be shared between two atoms, making single, double, and triple bonds, respectively. The more covalent bonds between two atoms, the stronger their connection. Thus, triple bonds are the strongest.

The strength of different levels of covalent bonding is one of the main reasons living organisms have a difficult time in acquiring nitrogen for use in constructing nitrogenous molecules, even though molecular nitrogen, N_2, is the most abundant gas in the atmosphere. Molecular nitrogen consists of two nitrogen atoms triple bonded to each other. The resulting strong triple bond makes it difficult for living systems to break apart this nitrogen in order to use it as constituents of biomolecules, such as proteins, DNA, and RNA.

The formation of water molecules is an example of covalent bonding. The hydrogen and oxygen atoms that combine to form water molecules are bound together by covalent bonds. The electron from the hydrogen splits its time between the incomplete outer shell of the hydrogen atom and the incomplete outer shell of the oxygen atom. In return, the oxygen atom shares one of its electrons with the hydrogen atom, creating a two-electron single covalent bond. To completely fill the outer shell of oxygen, which has six electrons in its outer shell, two electrons (one from each hydrogen atom) are needed. Each hydrogen atom needs only a single electron to fill its outer shell, hence the well-known formula H_2O. The electrons that are shared between the two elements fill the outer shell of each, making both elements more stable.

Polar Covalent Bonds

There are two types of covalent bonds: polar and nonpolar. In a polar covalent bond, the electrons are unequally shared by the atoms because they are more attracted to one nucleus than the other. The relative attraction of an atom to an electron is known as its electronegativity: atoms that are more attracted to an electron are considered to be more electronegative. Because of the unequal distribution of electrons between the atoms of different elements, a slightly positive (δ+) or slightly negative (δ-) charge develops. This partial charge is known as a dipole; this is an important property of water and accounts for many of its characteristics. The dipole in water occurs because oxygen has a higher electronegativity than hydrogen, which means that the shared electrons spend more time in the vicinity of the oxygen nucleus than they do near the nucleus of the hydrogen atoms.

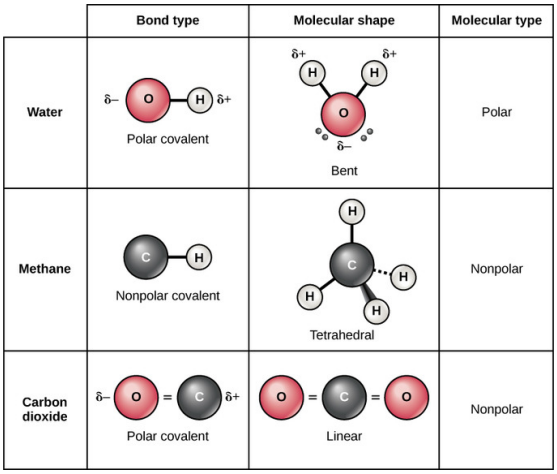


Figure 2.1*I*.1: **Polar and Nonpolar Covalent Bonds**: Whether a molecule is polar or nonpolar depends both on bond type and molecular shape. Both water and carbon dioxide have polar covalent bonds, but carbon dioxide is linear, so the partial charges on the molecule cancel each other out.

Nonpolar Covalent Bonds

Nonpolar covalent bonds form between two atoms of the same element or between different elements that share electrons equally. For example, molecular oxygen (O_2) is nonpolar because the electrons will be equally distributed between the two oxygen atoms. The four bonds of methane are also considered to be nonpolar

because the electronegativies of carbon and hydrogen are nearly identical.

Hydrogen Bonds and Van Der Waals Interactions

Not all bonds are ionic or covalent; weaker bonds can also form between molecules. Two types of weak bonds that frequently occur are hydrogen bonds and van der Waals interactions. Without these two types of bonds, life as we know it would not exist.

Hydrogen bonds provide many of the critical, life-sustaining properties of water and also stabilize the structures of proteins and DNA, the building block of cells. When polar covalent bonds containing hydrogen are formed, the hydrogen atom in that bond has a slightly positive charge (δ+) because the shared electrons are pulled more strongly toward the other element and away from the hydrogen atom. Because the hydrogen has a slightly positive charge, it's attracted to neighboring negative charges. The weak interaction between the δ+ charge of a hydrogen atom from one molecule and the δ- charge of a more electronegative atom is called a hydrogen bond. Individual hydrogen bonds are weak and easily broken; however, they occur in very large numbers in water and in organic polymers, and the additive force can be very strong. For example, hydrogen bonds are responsible for zipping together the DNA double helix.

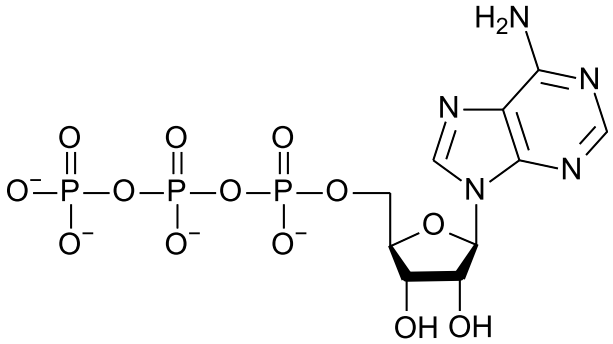
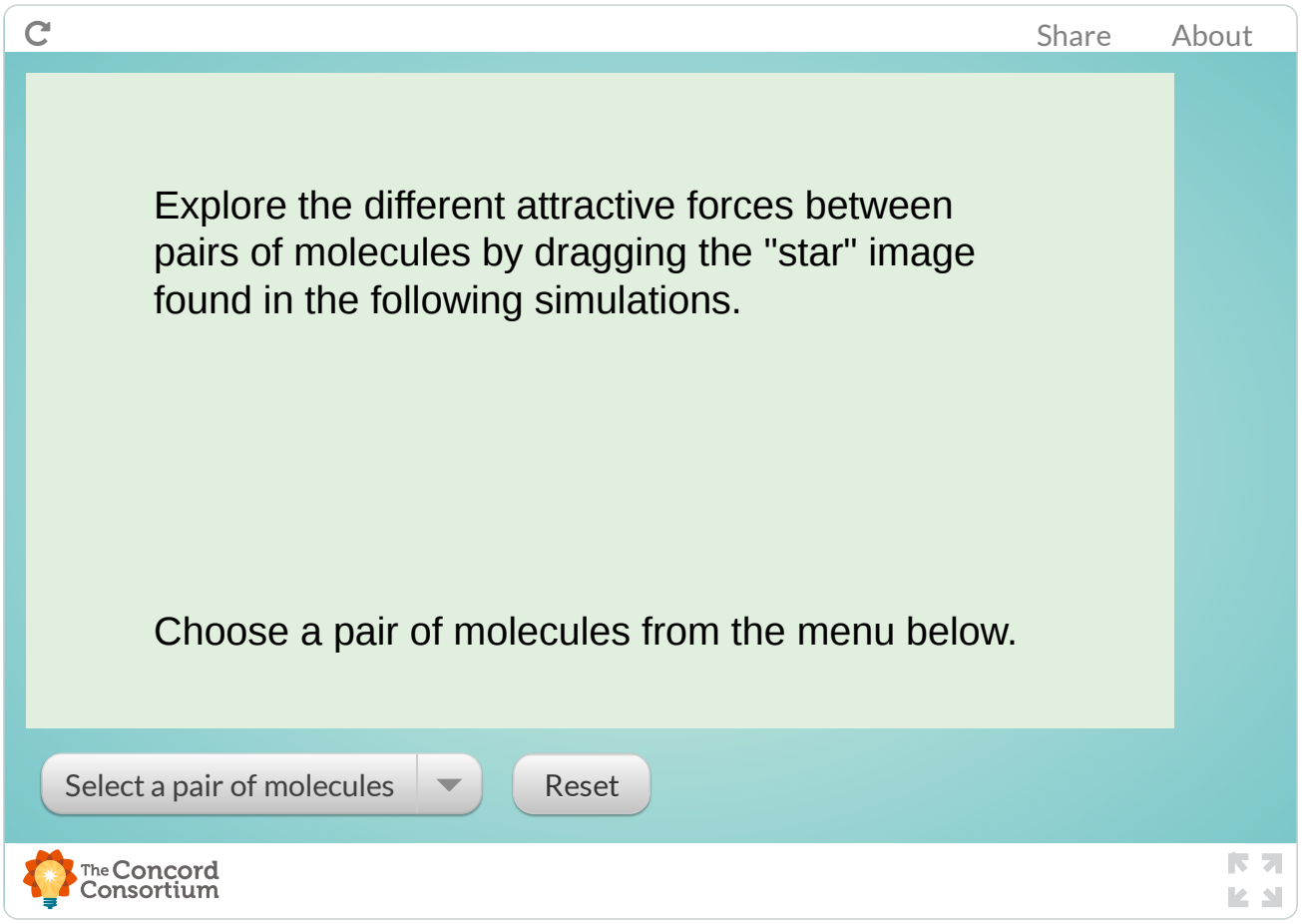


Figure 2.1*I*.1: **Adenosine Triphosphate, ATP**: Adenosine Triphosphate, or ATP, is the most commonly used cofactor in nature. Its biosynthesis involves the fixation of nitrogen to provide feedstocks that eventually produce the carbon-nitrogen bonds it contains.

Like hydrogen bonds, van der Waals interactions are weak interactions between molecules. Van der Waals attractions can occur between any two or more molecules and are dependent on slight fluctuations of the electron densities, which can lead to slight temporary dipoles around a molecule. For these attractions to happen, the molecules need to be very close to one another. These bonds, along with hydrogen bonds, help form the three-dimensional structures of the proteins in our cells that are required for their proper function.



Interactions between different types of molecules: In this interactive, you can explore how different types of molecules interact with each other based on their bonds.

2.1J: Hydrogen Bonding and Van der Waals Forces

Hydrogen bonds and van der Waals interactions are two types of weak bonds that are necessary to the basic building blocks of life.

Learning Objectives

- Describe how hydrogen bonds and van der Waals interactions occur

Key Points

- Hydrogen bonds provide many of the critical, life-sustaining properties of water and also stabilize the structures of proteins and DNA, the building block of cells.
- Hydrogen bonds occur in inorganic molecules, such as water, and organic molecules, such as DNA and proteins.
- Van der Waals attractions can occur between any two or more molecules and are dependent on slight fluctuations of the electron densities.
- While hydrogen bonds and van der Waals interactions are weak individually, they are strong combined in vast numbers.

Key Terms

- **van der Waals interactions**: A weak force of attraction between electrically neutral molecules that collide with or pass very close to each other. The van der Waals force is caused by temporary attractions between electron-rich regions of one molecule and electron-poor regions of another.
- **electronegativity**: The tendency of an atom or molecule to draw electrons towards itself, form dipoles, and thus form bonds.
- **hydrogen bond**: The attraction between a partially positively-charged hydrogen atom attached to a highly electronegative atom (such as nitrogen, oxygen, or fluorine) and another nearby electronegative atom.

Ionic and covalent bonds between elements require energy to break. Ionic bonds are not as strong as covalent, which determines their behavior in biological systems. However, not all bonds are ionic or covalent bonds. Weaker bonds can also form between molecules. Two weak bonds that occur frequently are hydrogen bonds and van der Waals interactions.

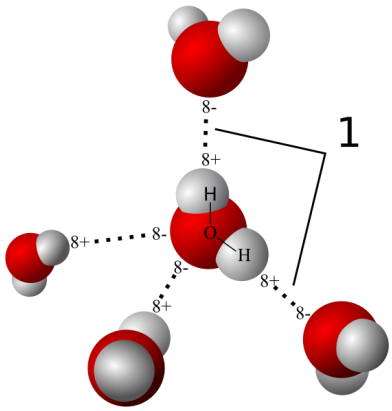


Figure 2.1*J*. 1: **Hydrogen bonds between water molecules**: The slightly negative oxygen side of the water molecule and the slightly positive hydrogen side of the water molecule are attracted to each other and form a hydrogen bond.

Hydrogen Bonding

Hydrogen bonds provide many of the critical, life-sustaining properties of water and also stabilize the structures of proteins and DNA, the building block of cells. When polar covalent bonds containing hydrogen form, the hydrogen in that bond has a slightly positive charge because hydrogen's one electron is pulled more strongly toward the other element and away from the hydrogen. Because the hydrogen is slightly positive, it will be attracted to neighboring negative charges. When this happens, an interaction occurs between the $\delta^+$of the hydrogen from one molecule and the $\delta-$ charge on the more electronegative atoms of another molecule, usually oxygen or nitrogen, or within the same molecule. This interaction is called a hydrogen bond. This type of bond is common and occurs regularly between water molecules. Individual hydrogen bonds are weak and easily broken; however, they occur in very large numbers in water and in organic polymers, creating a major force in combination. Hydrogen bonds are also responsible for zipping together the DNA double helix.

Applications for Hydrogen Bonds

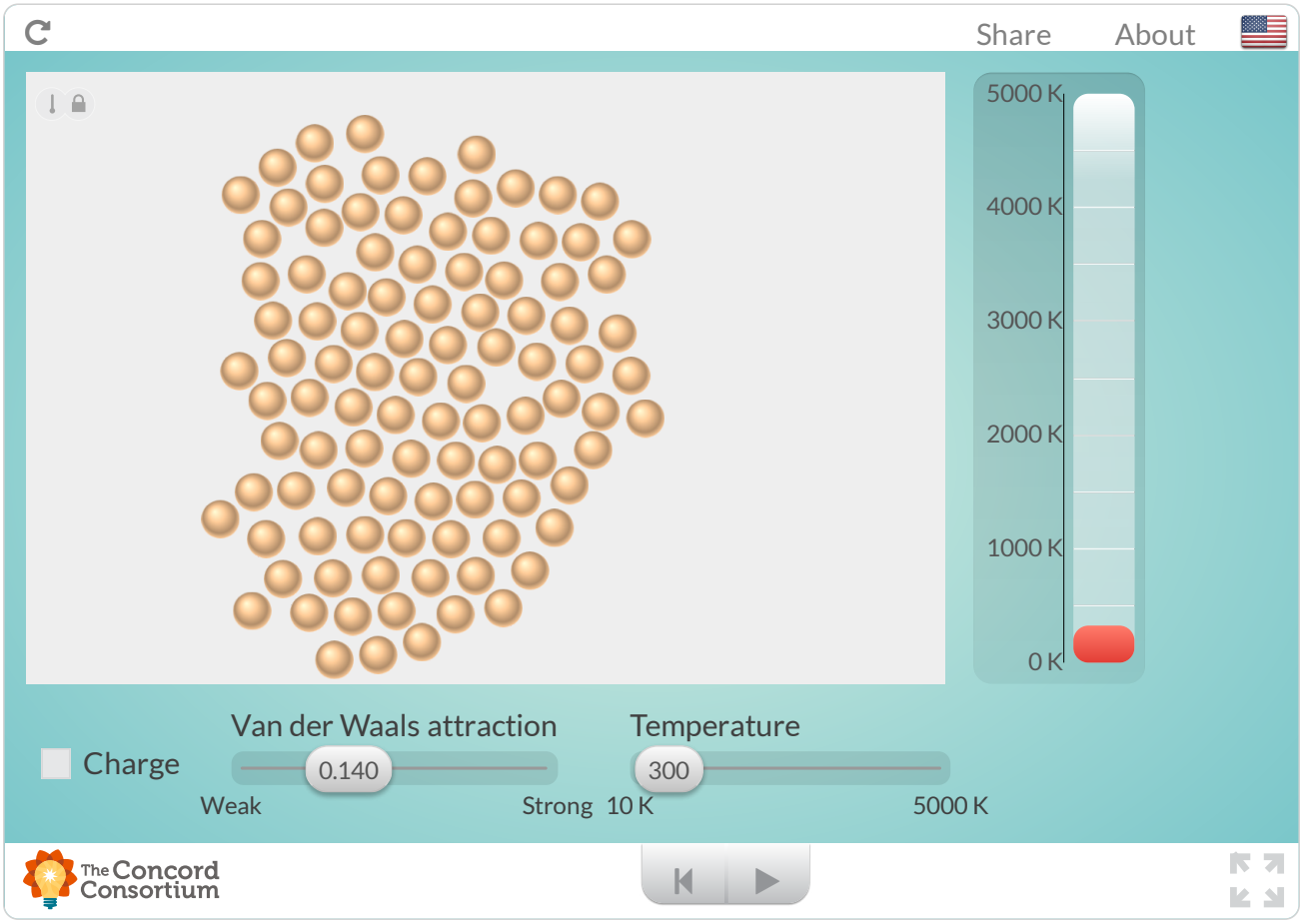
Hydrogen bonds occur in inorganic molecules, such as water, and organic molecules, such as DNA and proteins. The two complementary strands of DNA are held together by hydrogen bonds between complementary nucleotides (A&T, C&G). Hydrogen bonding in water contributes to its unique properties, including its high boiling point (100 °C) and surface tension.

Figure 2.1.J. 1: Water droplets on a leaf: The hydrogen bonds formed between water molecules in water droplets are stronger than the other intermolecular forces between the water molecules and the leaf, contributing to high surface tension and distinct water droplets.

In biology, intramolecular hydrogen bonding is partly responsible for the secondary, tertiary, and quaternary structures of proteins and nucleic acids. The hydrogen bonds help the proteins and nucleic acids form and maintain specific shapes.

Van der Waals Interactions

Like hydrogen bonds, van der Waals interactions are weak attractions or interactions between molecules. Van der Waals attractions can occur between any two or more molecules and are dependent on slight fluctuations of the electron densities, which are not always symmetrical around an atom. For these attractions to happen, the molecules need to be very close to one another. These bonds—along with ionic, covalent, and hydrogen bonds—contribute to the three-dimensional structure of proteins that is necessary for their proper function.



Van der Waals attraction: Explore how Van der Waals attractions and temperature affect intermolecular interactions.

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2.2: Water

Topic hierarchy

- 2.2A: Water’s Polarity
- 2.2B: Water’s States: Gas, Liquid, and Solid

The orientation of hydrogen bonds as water changes states dictates the properties of water in its gaseous, liquid, and solid forms.

- 2.2C: Water’s Heat of Vaporization

Evaporation of water requires a substantial amount of energy due to the high heat of vaporization of water.

- 2.2D: Water’s Solvent Properties
- 2.2E: Water’s Cohesive and Adhesive Properties
- 2.2F: pH, Buffers, Acids, and Bases

2.2A: Water's Polarity

Water's polarity is responsible for many of its properties including its attractiveness to other molecules.

Learning Objectives

- Describe the actions that occur due to water's polarity

Key Points

- The difference in electronegativities between oxygen and hydrogen atoms creates partial negative and positive charges, respectively, on the atoms.
- Water molecules attract or are attracted to other polar molecules.
- Molecules that do not dissolve in water are known as hydrophobic (water fearing) molecules.

Key Terms

- **hydrophilic**: having an affinity for water; able to absorb, or be wetted by water
- **hydrophobic**: lacking an affinity for water; unable to absorb, or be wetted by water
- **polarity**: The intermolecular forces between the slightly positively-charged end of one molecule to the negative end of another or the same molecule.

One of water's important properties is that it is composed of polar molecules. The two hydrogen atoms and one oxygen atom within water molecules (H_2O) form polar covalent bonds. While there is no net charge to a water molecule, the polarity of water creates a slightly positive charge on hydrogen and a slightly negative charge on oxygen, contributing to water's properties of attraction. Water's charges are generated because oxygen is more electronegative, or electron loving, than hydrogen. Thus, it is more likely that a shared electron would be found near the oxygen nucleus than the hydrogen nucleus. Since water is a nonlinear, or bent, molecule, the difference in electronegativities between the oxygen and hydrogen atoms generates the partial negative charge near the oxygen and partial positive charges near both hydrogens.

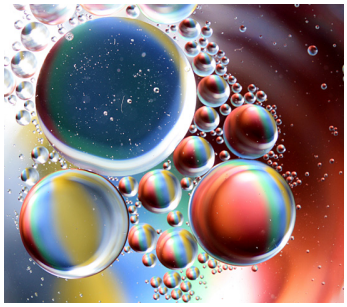
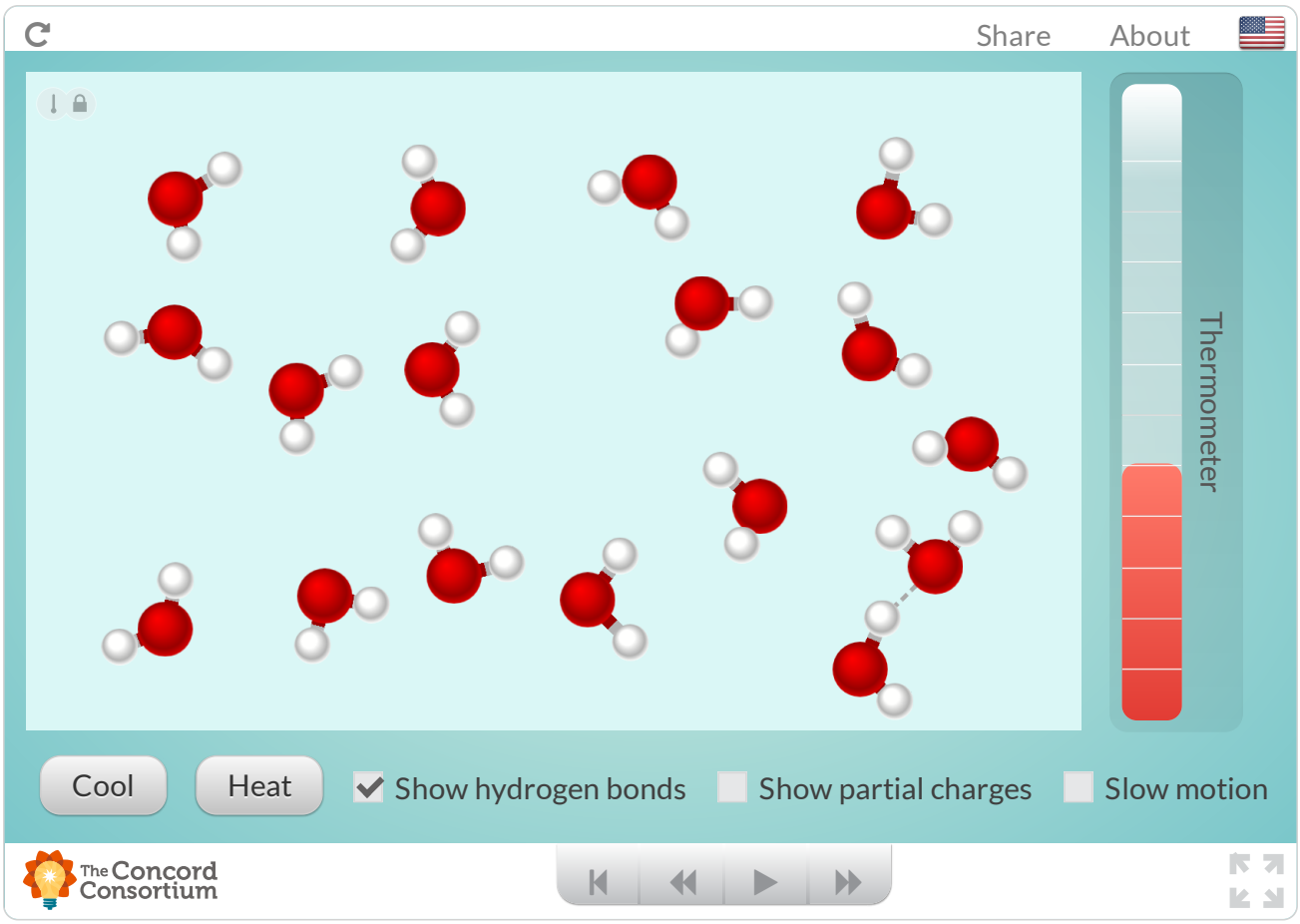


Figure 2.2*A*.1: **Nonpolar Molecules**: Oil and water do not mix. As this macro image of oil and water shows, oil does not dissolve in water but forms droplets instead. This is due to it being a nonpolar compound.

As a result of water's polarity, each water molecule attracts other water molecules because of the opposite charges between them, forming hydrogen bonds. Water also attracts, or is attracted to, other polar molecules and ions, including many biomolecules, such as sugars, nucleic acids, and some amino acids. A polar substance that interacts readily with or dissolves in water is referred to as hydrophilic (hydro- = "water"; -philic = "loving"). In contrast, nonpolar molecules, such as oils and fats, do not interact well with water, as shown in. These molecules separate from it rather than dissolve in it, as we see in salad dressings containing oil and vinegar (an acidic water solution). These nonpolar compounds are called hydrophobic (hydro- = "water"; -phobic = "fearing").



Hydrogen bonds: This interactive shows the interaction of the hydrogen bonds among water molecules.

: Water's States: Gas, Liquid, and Solid

LEARNING OBJECTIVES

- Explain the biological significance of ice's ability to float on water

Water's States: Gas, Liquid, and Solid

The formation of hydrogen bonds is an important quality of liquid water that is crucial to life as we know it. As water molecules make hydrogen bonds with each other, water takes on some unique chemical characteristics compared to other liquids, and since living things have a high water content, understanding these chemical features is key to understanding life. In liquid water, hydrogen bonds are constantly formed and broken as the water molecules slide past each other. The breaking of these bonds is caused by the motion (kinetic energy) of the water molecules due to the heat contained in the system. When the heat is raised as water is boiled, the higher kinetic energy of the water molecules causes the hydrogen bonds to break completely and allows water molecules to escape into the air as gas (steam or water vapor). On the other hand, when the temperature of water is reduced and water freezes, the water molecules form a crystalline structure maintained by hydrogen bonding (there is not enough energy to break the hydrogen bonds). This makes ice less dense than liquid water, a phenomenon not seen in the solidification of other liquids.

***Phases of matter**: See what happens to intermolecular bonds during phase changes in this interactive.*

Water's lower density in its solid form is due to the way hydrogen bonds are oriented as it freezes: the water molecules are pushed farther apart compared to liquid water. With most other liquids, solidification when the temperature drops includes the lowering of kinetic energy between molecules, allowing them to pack even more tightly than in liquid form and giving the solid a greater density than the liquid.

The low density of ice, an anomaly, causes it to float at the surface of liquid water, such as an iceberg or the ice cubes in a glass of water. In lakes and ponds, ice forms on the surface of the water creating an insulating barrier that protects the animals and plant life in the pond from freezing. Without this layer of insulating ice, plants and animals living in the pond would freeze in the solid block of ice and could not survive. The detrimental effect of freezing on living organisms is caused by the expansion of ice relative to liquid water. The ice crystals that form upon freezing rupture the delicate membranes essential for the function of living cells, irreversibly damaging them. Cells can only survive freezing if the water in them is temporarily replaced by another liquid like glycerol.

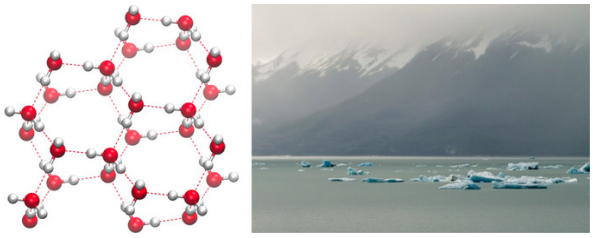


Figure 2.2B.1: **Ice Density**: Hydrogen bonding makes ice less dense than liquid water. The (a) lattice structure of ice makes it less dense than the freely flowing molecules of liquid water, enabling it to (b) float on water.

Key Points

- As water is boiled, kinetic energy causes the hydrogen bonds to break completely and allows water molecules to escape into the air as gas (steam or water vapor).
- When water freezes, water molecules form a crystalline structure maintained by hydrogen bonding.
- Solid water, or ice, is less dense than liquid water.
- Ice is less dense than water because the orientation of hydrogen bonds causes molecules to push farther apart, which lowers the density.
- For other liquids, solidification when the temperature drops includes the lowering of kinetic energy, which allows molecules to pack more tightly and makes the solid denser than its liquid form.
- Because ice is less dense than water, it is able to float at the surface of water.

Key Terms

- **density**: A measure of the amount of matter contained by a given volume.

2.2C: Water's Heat of Vaporization

LEARNING OBJECTIVES

- Explain how heat of vaporization is related to the boiling point of water

Water in its liquid form has an unusually high boiling point temperature, a value close to 100°C. As a result of the network of hydrogen bonding present between water molecules, a high input of energy is required to transform one gram of liquid water into water vapor, an energy requirement called the heat of vaporization. Water has a heat of vaporization value of 40.65 kJ/mol. A considerable amount of heat energy (586 calories) is required to accomplish this change in water. This process occurs on the surface of water. As liquid water heats up, hydrogen bonding makes it difficult to separate the water molecules from each other, which is required for it to enter its gaseous phase (steam). As a result, water acts as a heat sink, or heat reservoir, and requires much more heat to boil than does a liquid such as ethanol (grain alcohol), whose hydrogen bonding with other ethanol molecules is weaker than water's hydrogen bonding. Eventually, as water reaches its boiling point of 100° Celsius (212° Fahrenheit), the heat is able to break the hydrogen bonds between the water molecules, and the kinetic energy (motion) between the water molecules allows them to escape from the liquid as a gas. Even when below its boiling point, water's individual molecules acquire enough energy from each other such that some surface water molecules can escape and vaporize; this process is known as evaporation.

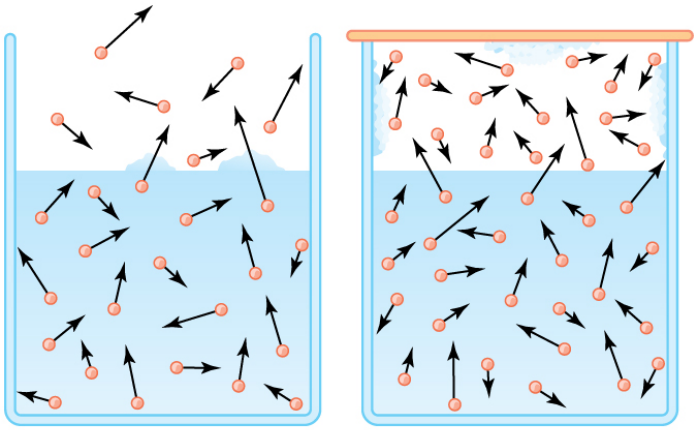


Figure 2.2C.1: **Humidity, Evaporation, and Boiling**: (a) Because of the distribution of speeds and kinetic energies, some water molecules can break away to the vapor phase even at temperatures below the ordinary boiling point. (b) If the container is sealed, evaporation will continue until there is enough vapor density for the condensation rate to equal the evaporation rate. This vapor density and the partial pressure it creates are the saturation values. They increase with temperature and are independent of the presence of other gases, such as air. They depend only on the vapor pressure of water.

The fact that hydrogen bonds need to be broken for water to evaporate means that a substantial amount of energy is used in the process. As the water evaporates, energy is taken up by the process, cooling the environment where the evaporation is taking place. In many living organisms, including humans, the evaporation of sweat, which is 90 percent water, allows the organism to cool so that homeostasis of body temperature can be maintained.

Key Points

- The dissociation of liquid water molecules, which changes the substance to a gas, requires a lot of energy.
- The boiling point of water is the temperature in which there is enough energy to break the hydrogen bonds between water molecules.
- Water is converted from its liquid form to its gaseous form (steam) when the heat of vaporization is reached.
- Evaporation of sweat (mostly water) removes heat from the surface of skin, cooling the body.

Key Terms

- **heat of vaporization**: The energy required to transform a given quantity of a substance from a liquid into a gas at a given pressure (often atmospheric pressure).

2.2C: Water's High Heat Capacity

Water is able to absorb a high amount of heat before increasing in temperature, allowing humans to maintain body temperature.

Learning Objectives

- Explain the biological significance of water's high specific heat

Key Points

- Water has the highest heat capacity of all liquids.
- Oceans cool slower than the land due to the high heat capacity of water.
- To change the temperature of 1 gram of water by 1 degree Celsius, it takes 1.00 calorie.

Key Terms

- **heat capacity**: The capability of a substance to absorb heat energy
- **specific heat**: the amount of heat, in calories, needed to raise the temperature of 1 gram of water by 1 degree Celsius

Water's High Heat Capacity

The capability for a molecule to absorb heat energy is called heat capacity, which can be calculated by the equation shown in the figure. Water's high heat capacity is a property caused by hydrogen bonding among water molecules. When heat is absorbed, hydrogen bonds are broken and water molecules can move freely. When the temperature of water decreases, the hydrogen bonds are formed and release a considerable amount of energy. Water has the highest specific heat capacity of any liquid. Specific heat is defined as the amount of heat one gram of a substance must absorb or lose to change its temperature by one degree Celsius. For water, this amount is one calorie, or 4.184 Joules. As a result, it takes water a long time to heat and a long time to cool. In fact, the specific heat capacity of water is about five times more than that of sand. This explains why the land cools faster than the sea.

$C=Q\Delta T.C=Q\Delta T.$

The resistance to sudden temperature changes makes water an excellent habitat, allowing organisms to survive without experiencing wide temperature fluctuation. Furthermore, because many organisms are mainly composed of water, the property of high heat capacity allows highly regulated internal body temperatures. For example, the temperature of your body does not drastically drop to the same temperature as the outside temperature while you are skiing or playing in the snow. Due to its high heat capacity, water is used by warm blooded animals to more evenly disperse heat in their bodies; it acts in a similar manner to a car's cooling system, transporting heat from warm places to cool places, causing the body to maintain a more even temperature.

2.2D: Water's Solvent Properties

Water's polarity makes it an excellent solvent for other polar molecules and ions.

Learning Objectives

- Explain why some molecules do not dissolve in water.

Key Points

- Water dissociates salts by separating the cations and anions and forming new interactions between the water and ions.
- Water dissolves many biomolecules, because they are polar and therefore hydrophilic.

Key Terms

- **dissociation**: The process by which a compound or complex body breaks up into simpler constituents such as atoms or ions, usually reversibly.
- **hydration shell**: The term given to a solvation shell (a structure composed of a chemical that acts as a solvent and surrounds a solute species) with a water solvent; also referred to as a hydration sphere.

Water's Solvent Properties

Water, which not only dissolves many compounds but also dissolves more substances than any other liquid, is considered the universal solvent. A polar molecule with partially-positive and negative charges, it readily dissolves ions and polar molecules. Water is therefore referred to as a solvent: a substance capable of dissolving other polar molecules and ionic compounds. The charges associated with these molecules form hydrogen bonds with water, surrounding the particle with water molecules. This is referred to as a sphere of hydration, or a hydration shell, and serves to keep the particles separated or dispersed in the water.

When ionic compounds are added to water, individual ions interact with the polar regions of the water molecules during the dissociation process, disrupting their ionic bonds. Dissociation occurs when atoms or groups of atoms break off from molecules and form ions. Consider table salt (NaCl, or sodium chloride): when NaCl crystals are added to water, the molecules of NaCl dissociate into Na^+ and Cl^-ions, and spheres of hydration form around the ions. The positively-charged sodium ion is surrounded by the partially-negative charge of the water molecule's oxygen; the negatively-charged chloride ion is surrounded by the partially-positive charge of the hydrogen in the water molecule.

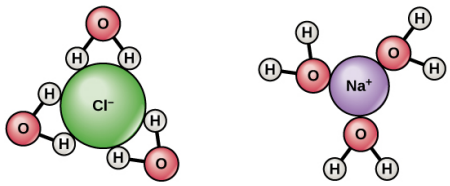


Figure 2.2D.1: **Dissociation of NaCl in water**: When table salt (NaCl) is mixed in water, spheres of hydration form around the ions.

Since many biomolecules are either polar or charged, water readily dissolves these hydrophilic compounds. Water is a poor solvent, however, for hydrophobic molecules such as lipids. Nonpolar molecules experience hydrophobic interactions in water: the water changes its hydrogen bonding patterns around the hydrophobic molecules to produce a cage-like structure called a clathrate. This change in the hydrogen-bonding pattern of the water solvent causes the system's overall entropy to greatly decrease, as the molecules become more ordered than in liquid water. Thermodynamically, such a large decrease in entropy is not spontaneous, and the hydrophobic molecule will not dissolve.

: Water's Cohesive and Adhesive Properties

Cohesion allows substances to withstand rupture when placed under stress while adhesion is the attraction between water and other molecules.

Learning Objectives

- Describe the cohesive and adhesive properties of water.

Key Points

- Cohesion holds hydrogen bonds together to create surface tension on water.
- Since water is attracted to other molecules, adhesive forces pull the water toward other molecules.
- Water is transported in plants through both cohesive and adhesive forces; these forces pull water and the dissolved minerals from the roots to the leaves and other parts of the plant.

Key Terms

- **adhesion**: The ability of a substance to stick to an unlike substance; attraction between unlike molecules
- **cohesion**: Various intermolecular forces that hold solids and liquids together; attraction between like molecules

Water's Cohesive and Adhesive Properties

Have you ever filled a glass of water to the very top and then slowly added a few more drops? Before it overflows, the water forms a dome-like shape above the rim of the glass. This water can stay above the glass because of the property of cohesion. In cohesion, water molecules are attracted to each other (because of hydrogen bonding), keeping the molecules together at the liquid-gas (water-air) interface, although there is no more room in the glass.

Cohesion allows for the development of surface tension, the capacity of a substance to withstand being ruptured when placed under tension or stress. This is also why water forms droplets when placed on a dry surface rather than being flattened out by gravity. When a small scrap of paper is placed onto the droplet of water, the paper floats on top of the water droplet even though paper is denser (the mass per unit volume) than the water. Cohesion and surface tension keep the hydrogen bonds of water molecules intact and support the item floating on the top. It's even possible to "float" a needle on top of a glass of water if it is placed gently without breaking the surface tension.

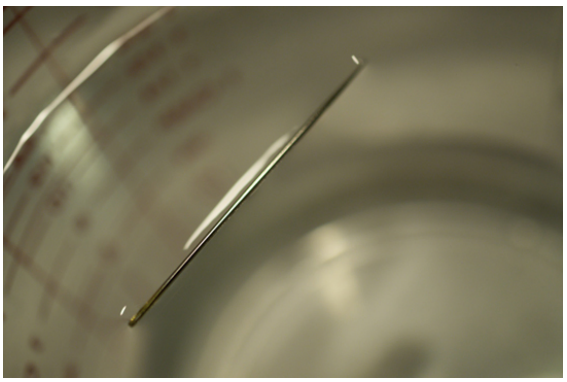


Figure $2.2E.1$: **Surface Tension**: The weight of the needle is pulling the surface downward; at the same time, the surface tension is pulling it up, suspending it on the surface of the water and keeping it from sinking. Notice the indentation in the water around the needle.

These cohesive forces are related to water's property of adhesion, or the attraction between water molecules and other molecules. This attraction is sometimes stronger than water's cohesive forces, especially when the water is exposed to charged surfaces such as those found on the inside of thin glass tubes known as capillary tubes. Adhesion is observed when water "climbs" up the tube placed in a glass of water: notice that the water appears to be higher on the sides of the tube than in the middle. This is because the water molecules are attracted to the charged glass walls of the capillary more than they are to each other and therefore adhere to it. This type of adhesion is called capillary action.

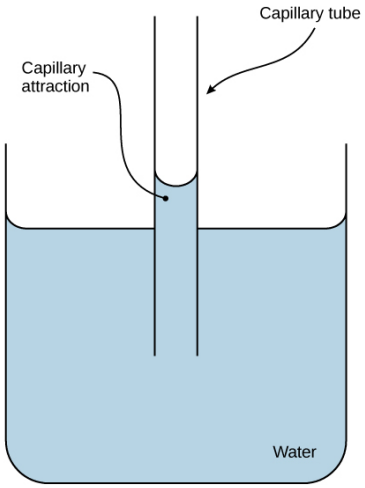


Figure 2.2E. 1: **Adhesion**: Capillary action in a glass tube is caused by the adhesive forces exerted by the internal surface of the glass exceeding the cohesive forces between the water molecules themselves.

Why are cohesive and adhesive forces important for life? Cohesive and adhesive forces are important for the transport of water from the roots to the leaves in plants. These forces create a "pull" on the water column. This pull results from the tendency of water molecules being evaporated on the surface of the plant to stay connected to water molecules below them, and so they are pulled along. Plants use this natural phenomenon to help transport water from their roots to their leaves. Without these properties of water, plants would be unable to receive the water and the dissolved minerals they require. In another example, insects such as the water strider use the surface tension of water to stay afloat on the surface layer of water and even mate there.

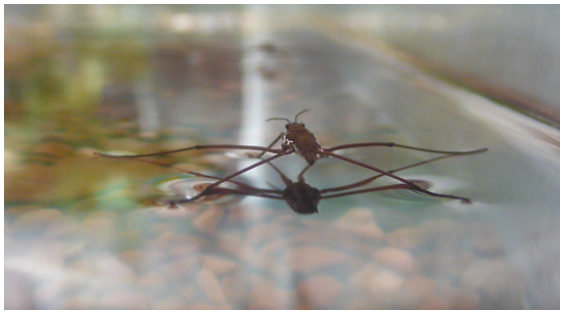


Figure 2.2E. 1: **Cohesion & Adhesion**: Water's cohesive and adhesive properties allow this water strider (Gerris sp.) to stay afloat.

2.2F: pH, Buffers, Acids, and Bases

Acids dissociate into H^+ and lower pH, while bases dissociate into OH^- and raise pH; buffers can absorb these excess ions to maintain pH.

Learning Objectives

- Explain the composition of buffer solutions and how they maintain a steady pH

Key Points

- A basic solution will have a pH above 7.0, while an acidic solution will have a pH below 7.0.
- Buffers are solutions that contain a weak acid and its a conjugate base; as such, they can absorb excess H^+ ions or OH^- ions, thereby maintaining an overall steady pH in the solution.
- pH is equal to the negative logarithm of the concentration of H^+ ions in solution: $pH = – \log[H^+]$.

Key Terms

- **alkaline**: having a pH greater than 7; basic
- **acidic**: having a pH less than 7
- **buffer**: a solution composed of a weak acid and its conjugate base that can be used to stabilize the pH of a solution

Self-Ionization of Water

Hydrogen ions are spontaneously generated in pure water by the dissociation (ionization) of a small percentage of water molecules into equal numbers of hydrogen (H^+) ions and hydroxide (OH^-) ions. The hydroxide ions remain in solution because of their hydrogen bonds with other water molecules; the hydrogen ions, consisting of naked protons, are immediately attracted to un-ionized water molecules and form hydronium ions (H_30^+). By convention, scientists refer to hydrogen ions and their concentration as if they were free in this state in liquid water.

$2H2O \leftrightharpoons H3O^+ + OH^- 2H2O \leftrightharpoons H3O^+ + OH^-$

The concentration of hydrogen ions dissociating from pure water is 1×10^{-7} moles H^+ ions per liter of water. The pH is calculated as the negative of the base 10 logarithm of this concentration:

$pH = -\log[H^+]$

The negative log of 1×10^{-7} is equal to 7.0, which is also known as neutral pH. Human cells and blood each maintain near-neutral pH.

pH Scale

The pH of a solution indicates its acidity or basicity (alkalinity). The pH scale is an inverse logarithm that ranges from 0 to 14: anything below 7.0 (ranging from 0.0 to 6.9) is acidic, and anything above 7.0 (from 7.1 to 14.0) is basic (or alkaline). Extremes in pH in either direction from 7.0 are usually considered inhospitable to life. The pH in cells (6.8) and the blood (7.4) are both very close to neutral, whereas the environment in the stomach is highly acidic, with a pH of 1 to 2.

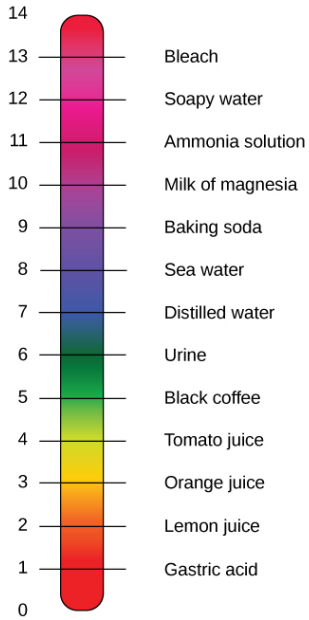


Figure 2.2F.1: **The pH scale**: The pH scale measures the concentration of hydrogen ions (H^+) in a solution.

Non-neutral pH readings result from dissolving acids or bases in water. Using the negative logarithm to generate positive integers, high concentrations of hydrogen ions yield a low pH, and low concentrations a high pH.

An acid is a substance that increases the concentration of hydrogen ions (H^+) in a solution, usually by dissociating one of its hydrogen atoms. A base provides either hydroxide ions (OH^-) or other negatively-charged ions that react with hydrogen ions in solution, thereby reducing the concentration of H^+ and raising the pH.

Strong Acids and Strong Bases

The stronger the acid, the more readily it donates H^+. For example, hydrochloric acid (HCl) is highly acidic and completely dissociates into hydrogen and chloride ions, whereas the acids in tomato juice or vinegar do not completely dissociate and are considered weak acids; conversely, strong bases readily donate OH^- and/or react with hydrogen ions. Sodium hydroxide (NaOH) and many household cleaners are highly basic and give up OH^- rapidly when placed in water; the OH^- ions react with H^+ in solution, creating new water molecules and lowering the amount of free H^+ in the system, thereby raising the overall pH. An example of a weak basic solution is seawater, which has a pH near 8.0, close enough to neutral that well-adapted marine organisms thrive in this alkaline environment.

Buffers

How can organisms whose bodies require a near-neutral pH ingest acidic and basic substances (a human drinking orange juice, for example) and survive? Buffers are the key. Buffers usually consist of a weak acid and its conjugate base; this enables them to readily absorb excess H^+ or OH^-, keeping the system's pH within a narrow range.

Maintaining a constant blood pH is critical to a person's well-being. The buffer that maintains the pH of human blood involves carbonic acid (H_2CO_3), bicarbonate ion (HCO_3^-), and carbon dioxide (CO_2). When bicarbonate ions combine with free hydrogen ions and become carbonic acid, hydrogen ions are removed, moderating pH changes. Similarly, excess carbonic acid can be converted into carbon dioxide gas and exhaled through the lungs; this prevents too many free hydrogen ions from building up in the blood and dangerously reducing its pH; likewise, if too much OH^- is introduced into the system, carbonic acid will combine with it to create bicarbonate, lowering the pH. Without this buffer system, the body's pH would fluctuate enough to jeopardize survival.

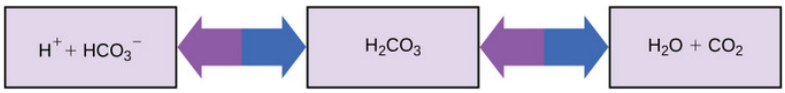


Figure 2.2F.1: Buffers in the body: This diagram shows the body's buffering of blood pH levels: the blue arrows show the process of raising pH as more CO2 is made; the purple arrows indicate the reverse process, lowering pH as more bicarbonate is created.

Antacids, which combat excess stomach acid, are another example of buffers. Many over-the-counter medications work similarly to blood buffers, often with at least one ion (usually carbonate) capable of absorbing hydrogen and moderating pH, bringing relief to those that suffer “heartburn” from stomach acid after eating.

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2.3: Carbon

Topic hierarchy

- 2.3A: The Chemical Basis for Life
- 2.3B: Hydrocarbons
- 2.3C: Organic Isomers
- 2.3D: Organic Enantiomers
- 2.3E: Organic Molecules and Functional Groups

2.3A: The Chemical Basis for Life

Carbon is the most important element to living things because it can form many different kinds of bonds and form essential compounds.

Learning Objectives

- Explain the properties of carbon that allow it to serve as a building block for biomolecules

Key Points

- All living things contain carbon in some form.
- Carbon is the primary component of macromolecules, including proteins, lipids, nucleic acids, and carbohydrates.
- Carbon's molecular structure allows it to bond in many different ways and with many different elements.
- The carbon cycle shows how carbon moves through the living and non-living parts of the environment.

Key Terms

- **octet rule**: A rule stating that atoms lose, gain, or share electrons in order to have a full valence shell of 8 electrons (has some exceptions).
- **carbon cycle**: the physical cycle of carbon through the earth's biosphere, geosphere, hydrosphere, and atmosphere; includes such processes as photosynthesis, decomposition, respiration and carbonification
- **macromolecule**: a very large molecule, especially used in reference to large biological polymers (e.g., nucleic acids and proteins)

Carbon is the fourth most abundant element in the universe and is the building block of life on earth. On earth, carbon circulates through the land, ocean, and atmosphere, creating what is known as the Carbon Cycle. This global carbon cycle can be divided further into two separate cycles: the geological carbon cycles takes place over millions of years, whereas the biological or physical carbon cycle takes place from days to thousands of years. In a nonliving environment, carbon can exist as carbon dioxide (CO_2), carbonate rocks, coal, petroleum, natural gas, and dead organic matter. Plants and algae convert carbon dioxide to organic matter through the process of photosynthesis, the energy of light.

Figure 2.3A.1: **Carbon is present in all life**: All living things contain carbon in some form, and carbon is the primary component of macromolecules, including proteins, lipids, nucleic acids, and carbohydrates. Carbon exists in many forms in this leaf, including in the cellulose to form the leaf's structure and in chlorophyll, the pigment which makes the leaf green.

Carbon is Important to Life

In its metabolism of food and respiration, an animal consumes glucose ($C_6H_{12}O_6$), which combines with oxygen (O_2) to produce carbon dioxide (CO_2), water (H_2O), and energy, which is given off as heat. The animal has no need for the carbon dioxide and releases it into the atmosphere. A plant, on the other hand, uses the opposite reaction of an animal through photosynthesis. It intakes carbon dioxide, water, and energy from sunlight to make its own glucose and oxygen gas. The glucose is used for chemical energy, which the plant metabolizes in a similar way to an animal. The plant then emits the remaining oxygen into the environment.

Cells are made of many complex molecules called macromolecules, which include proteins, nucleic acids (RNA and DNA), carbohydrates, and lipids. The macromolecules are a subset of organic molecules (any carbon-containing liquid,

solid, or gas) that are especially important for life. The fundamental component for all of these macromolecules is carbon. The carbon atom has unique properties that allow it to form covalent bonds to as many as four different atoms, making this versatile element ideal to serve as the basic structural component, or “backbone,” of the macromolecules.

Structure of Carbon

Individual carbon atoms have an incomplete outermost electron shell. With an atomic number of 6 (six electrons and six protons), the first two electrons fill the inner shell, leaving four in the second shell. Therefore, carbon atoms can form four covalent bonds with other atoms to satisfy the octet rule. The methane molecule provides an example: it has the chemical formula CH_4. Each of its four hydrogen atoms forms a single covalent bond with the carbon atom by sharing a pair of electrons. This results in a filled outermost shell.

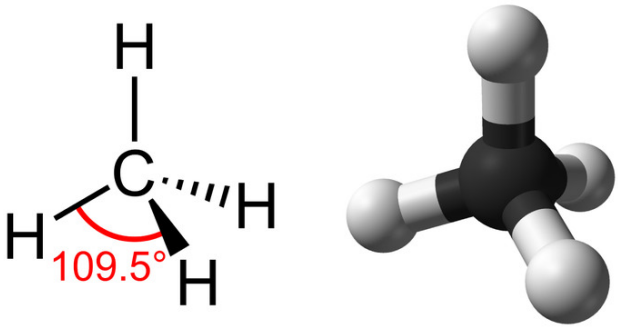

Figure 2.3A.1: **Structure of Methane**: Methane has a tetrahedral geometry, with each of the four hydrogen atoms spaced 109.5° apart.

2.3B: Hydrocarbons

Hydrocarbons are important molecules that can form chains and rings due to the bonding patterns of carbon atoms.

Learning Objectives

- Discuss the role of hydrocarbons in biomacromolecules

Key Points

- Hydrocarbons are molecules that contain only carbon and hydrogen.
- Due to carbon's unique bonding patterns, hydrocarbons can have single, double, or triple bonds between the carbon atoms.
- The names of hydrocarbons with single bonds end in "-ane," those with double bonds end in "-ene," and those with triple bonds end in "-yne".
- The bonding of hydrocarbons allows them to form rings or chains.

Key Terms

- **covalent bond**: A type of chemical bond where two atoms are connected to each other by the sharing of two or more electrons.
- **aliphatic**: Of a class of organic compounds in which the carbon atoms are arranged in an open chain.
- **aromatic**: Having a closed ring of alternate single and double bonds with delocalized electrons.

Hydrocarbons

Hydrocarbons are organic molecules consisting entirely of carbon and hydrogen, such as methane (CH_4). Hydrocarbons are often used as fuels: the propane in a gas grill or the butane in a lighter. The many covalent bonds between the atoms in hydrocarbons store a great amount of energy, which is released when these molecules are burned (oxidized). Methane, an excellent fuel, is the simplest hydrocarbon molecule, with a central carbon atom bonded to four different hydrogen atoms. The geometry of the methane molecule, where the atoms reside in three dimensions, is determined by the shape of its electron orbitals. The carbon and the four hydrogen atoms form a shape known as a tetrahedron, with four triangular faces; for this reason, methane is described as having tetrahedral geometry.

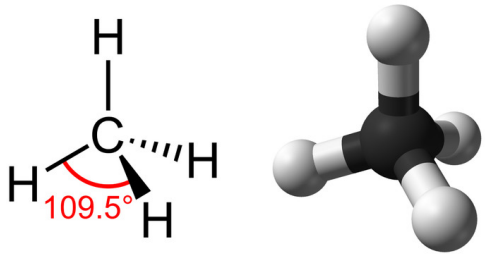

Figure 2.3B.1: **Methane**: Methane has a tetrahedral geometry, with each of the four hydrogen atoms spaced 109.5° apart.

As the backbone of the large molecules of living things, hydrocarbons may exist as linear carbon chains, carbon rings, or combinations of both. Furthermore, individual carbon-to-carbon bonds may be single, double, or triple covalent bonds; each type of bond affects the geometry of the molecule in a specific way. This three-dimensional shape or conformation of the large molecules of life (macromolecules) is critical to how they function.

Hydrocarbon Chains

Hydrocarbon chains are formed by successive bonds between carbon atoms and may be branched or unbranched. The overall geometry of the molecule is altered by the different geometries of single, double, and triple covalent bonds. The hydrocarbons ethane, ethene, and ethyne serve as examples of how different carbon-to-carbon bonds affect the geometry of the molecule. The names of all three molecules start with the prefix "eth-," which is the prefix for two carbon hydrocarbons. The suffixes "-ane," "-ene," and "-yne" refer to the presence of single, double, or triple carbon-carbon bonds, respectively. Thus, propane, propene, and propyne follow the same pattern with three carbon molecules, butane, butene, and butyne for four carbon molecules, and so on. Double and triple bonds change the geometry of the molecule:

single bonds allow rotation along the axis of the bond, whereas double bonds lead to a planar configuration and triple bonds to a linear one. These geometries have a significant impact on the shape a particular molecule can assume.

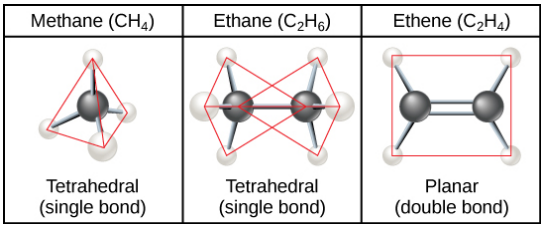

Figure 2.3B.1: **Hydrocarbon Chains**: When carbon forms single bonds with other atoms, the shape is tetrahedral. When two carbon atoms form a double bond, the shape is planar, or flat. Single bonds, like those found in ethane, are able to rotate. Double bonds, like those found in ethene cannot rotate, so the atoms on either side are locked in place.

Hydrocarbon Rings

The hydrocarbons discussed so far have been aliphatic hydrocarbons, which consist of linear chains of carbon atoms. Another type of hydrocarbon, aromatic hydrocarbons, consists of closed rings of carbon atoms. Ring structures are found in hydrocarbons, sometimes with the presence of double bonds, which can be seen by comparing the structure of cyclohexane to benzene. The benzene ring is present in many biological molecules including some amino acids and most steroids, which includes cholesterol and the hormones estrogen and testosterone. The benzene ring is also found in the herbicide 2,4-D. Benzene is a natural component of crude oil and has been classified as a carcinogen. Some hydrocarbons have both aliphatic and aromatic portions; beta-carotene is an example of such a hydrocarbon.

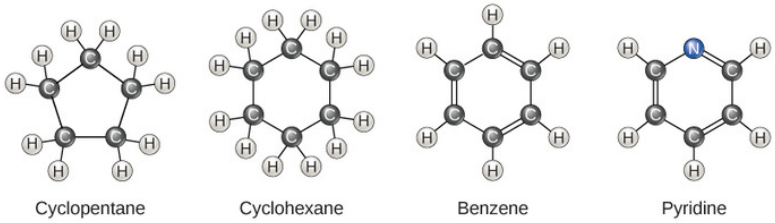

Figure 2.3B.1: **Hydrocarbon Rings**: Carbon can form five-and six membered rings. Single or double bonds may connect the carbons in the ring, and nitrogen may be substituted for carbon.